Kinsus Interconnect Technology (3189 TT)

Target price: **TWD1,580.00** (from TWD1,068.00)

Share price (6 Oct): **TWD1,070.00** | Up/downside: **+47.7%**

5 4 3 2 **1**

Buy (unchanged)

3Q26 results preview: several favourable trends ahead

- Rising high-end ABF substrate likely to drive blended ASP increase
- Multicore and optics likely to be the next growth drivers
- Reaffirming our Buy (1) call; raising 12M TP to TWD1,580

Sheng Cheng
(886) 2 8758 6253
sheng.cheng@daiwacm-cathay.com.tw

Thomas Kim
(886) 2 8758 6249
thomas.kim@daiwacm-cathay.com.tw

What's new: We believe an improving product mix, driven by rising Nvidia's CPU contribution, to be a key near-term share price catalyst.

What's the impact: **Near-term update.** After Kinsus released its August preliminary results on 23 September, we note that we had underestimated the 3Q26E gross margin expansion on product optimisation and price hike scale. The results imply a GM of 33% in August (vs. 26% in 2Q26). We therefore lift our 3Q-4Q26E gross margin to 33.2-35.1% (from 28.9-30.3%) to reflect an improving product mix, including greater CPU substrate contribution. We believe the gross margin of high-end chipset substrate may be c.50%, supporting margin expansion. We also expect a c.10% QoQ price hike for both BT and ABF substrate in 3Q26, above Kinsus' guidance of a high-single-digit% QoQ, making these substrates the other growth driver. We see further gross margin expansion in 2027-28 on the ongoing price hike trend and increasing high-end ABF substrate product mix.

Business outlook. We believe several favourable trends lie ahead. First, we see a rising contribution from high-end GPU and CPU ABF substrates. We believe Nvidia's CPU currently accounts for 5-10% of total revenue, with Kinsus holding a 70-80% market share. We also expect Kinsus to acquire a teens% market share in AMD's Venice, which should gradually ramp up from 4Q26. These will likely become key growth drivers of the ABF substrate segment in 2027, in our view. We believe the ramp-up progress for GPU substrate for these two clients is slightly slower than that for CPU. Second, we believe Kinsus will benefit from the rising BT substrate demand for optical transceivers. In our view, migration to on-board optics (OBO) and NPO/CPO will require ABF substrate beneath the optics engine (OE) in OBO; the OE and IC will need a bigger ABF substrate as we move to the NPO/CPO generation. As such, we expect the total ABF substrate demand to increase with rising adoption of optics in AI servers. Lastly, we expect the adoption of multiple-core ABF from 2028 to improve the rigidity and transmission performance before the official launch of ceramic and glass-core substrate, which we expect to increase the ABF TAM as well.

What we recommend: We raise our 2026-28E earnings by 19-29% to reflect the better-than-expected gross margin in August. We reaffirm our Buy (1) call and raise our 12M TP to TWD1,580, based on an unchanged target PER of 36x (high end of 3-year-trading range of 10-36x), applied to the 1-year-forward EPS forecast. Key downside risks: weaker-than-expected demand for memory, PCs, gaming GPUs or AI servers.

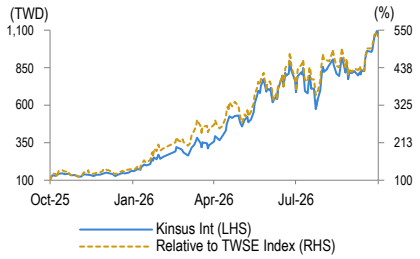
How we differ: Our 2026-28E EPS are 27-57% above the Bloomberg consensus, likely on our more bullish view on its gross margin expansion.

Forecast revisions (%)

Year to 31 Dec	26E	27E	28E
Revenue change	1.3	-	2.2
Net profit change	29.1	28.7	18.8
Core EPS (FD) change	18.6	18.3	9.2

Source: Daiwa forecasts

Share price performance



12-month range	111.18-1,095.00
Market cap (USDbn)	17.18
3m avg daily turnover (USDm)	428.04
Shares outstanding (m)	510
Major shareholder	Asus Investment Co Ltd (13.2%)

Financial summary (TWD)

Year to 31 Dec	26E	27E	28E
Revenue (m)	54,062	88,448	125,676
Operating profit (m)	9,142	24,773	43,198
Net profit (m)	7,558	22,325	39,673
Core EPS (fully-diluted)	14.824	43.787	77.810
EPS change (%)	322.9	195.4	77.7
Daiwa vs Cons. EPS (%)	29.1	57.1	46.7
PER (x)	72.2	24.4	13.8
Dividend yield (%)	0.3	0.4	0.4
DPS	3.5	4.5	4.5
PBR (x)	11.6	8.1	5.3
EV/EBITDA (x)	30.7	15.1	8.8
ROE (%)	15.2	32.4	40.0

Source: FactSet, Daiwa forecasts

See important disclosures, including any required research certifications, beginning on page 6

Flash

Kinsus: revenue and earnings forecast revisions

	2026E			2027E			2028E		
(TWDm)	Previous	New	Consensus	Previous	New	Consensus	Previous	New	Consensus
Revenue	53,385	54,062	53,912	88,484	88,448	80,794	122,920	125,676	120,243
Diff (%)		1.3%	0.3%		0.0%	9.5%		2.2%	4.5%
Gross Margin (%)	27.0%	29.65%	27.3%	35.6%	40.5%	33.7%	43.2%	46.37%	38.5%
Operating profit	7,578	9,142	7,907	20,449	24,773	18,170	38,330	43,198	32,345
Op Margin (%)	14.2%	16.9%	14.7%	23.1%	28.0%	22.5%	31.2%	34.4%	26.9%
Net profit	5,855	7,558	5,857	17,344	22,325	14,212	33,395	39,673	27,051
EPS	12.50	14.82	11.49	37.02	43.79	27.87	71.29	77.81	53.06
Diff (%)		18.6%	29.1%		18.3%	57.1%		9.2%	46.7%

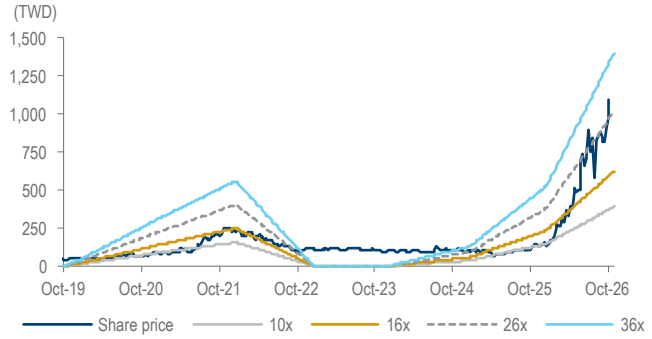
Source: Bloomberg, Daiwa forecasts

Kinsus: quarterly and annual P&L statement

	2026E				2027E				2025	2026E	2027E	2028E
(TWDm)	1Q	2Q	3QE	4QE	1QE	2QE	3QE	4QE				
Net revenue	**11,105**	**12,496**	**14,539**	**15,922**	**18,046**	**21,099**	**23,941**	**25,361**	**39,351**	**54,062**	**88,448**	**125,676**
COGS	(8,755)	(9,235)	(9,716)	(10,326)	(11,304)	(12,750)	(14,067)	(14,480)	(31,037)	(38,032)	(52,601)	(67,397)
Gross profit	**2,349**	**3,261**	**4,824**	**5,596**	**6,743**	**8,349**	**9,874**	**10,881**	**8,314**	**16,030**	**35,847**	**58,279**
Operating expenses	1,459	1,582	1,841	2,006	2,274	2,637	2,993	3,170	5,645	6,888	11,074	15,081
Operating profit	**891**	**1,679**	**2,983**	**3,590**	**4,469**	**5,712**	**6,881**	**7,711**	**2,670**	**9,142**	**24,773**	**43,198**
Non-operating profit	74	106	88	90	89	84	90	92	417	358	354	359
Pre-tax profit	**964**	**1,784**	**3,071**	**3,680**	**4,558**	**5,796**	**6,971**	**7,803**	**3,087**	**9,499**	**25,127**	**43,557**
Income taxes	(89)	(101)	(174)	(208)	(258)	(328)	(394)	(441)	(370)	(571)	(1,420)	(2,462)
Net profit	**549**	**1,311**	**2,586**	**3,112**	**3,986**	**5,132**	**6,230**	**6,978**	**1,596**	**7,558**	**22,325**	**39,673**
Net EPS (TWD)	**1.17**	**2.57**	**5.07**	**6.10**	**7.82**	**10.06**	**12.22**	**13.69**	**3.5**	**14.8**	**43.8**	**77.8**
Operating Ratios												
Gross margin	21.2%	26.1%	33.2%	35.1%	37.4%	39.6%	41.2%	42.9%	21.1%	29.7%	40.5%	46.4%
Operating margin	8.0%	13.4%	20.5%	22.5%	24.8%	27.1%	28.7%	30.4%	6.8%	16.9%	28.0%	34.4%
Pre-tax margin	8.7%	14.3%	21.1%	23.1%	25.3%	27.5%	29.1%	30.8%	7.8%	17.6%	28.4%	34.7%
Net margin	4.9%	10.5%	17.8%	19.5%	22.1%	24.3%	26.0%	27.5%	4.1%	14.0%	25.2%	31.6%
YoY (%)												
Net revenue	29%	31%	40%	47%	63%	69%	65%	59%	29%	37%	64%	42%
Gross profit	21%	62%	146%	134%	187%	156%	105%	94%	-4%	93%	124%	63%
Operating profit	78%	171%	375%	290%	402%	240%	131%	115%	144%	242%	171%	74%
Pre-tax profit	38%	188%	331%	249%	373%	225%	127%	112%	93%	208%	165%	73%
Net profit	99%	288%	662%	384%	626%	292%	141%	124%	3164%	374%	195%	78%
QoQ (%)												
Net revenue	3%	13%	16%	10%	13%	17%	13%	6%				
Gross profit	-2%	39%	48%	16%	20%	24%	18%	10%				
Operating profit	-3%	88%	78%	20%	24%	28%	20%	12%				
Pre-tax profit	-9%	85%	72%	20%	24%	27%	20%	12%				
Net profit	-14%	139%	97%	20%	28%	29%	21%	12%				

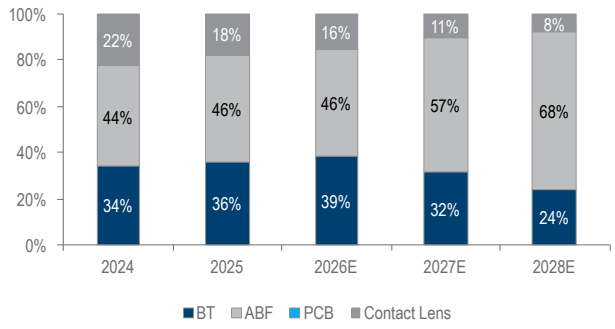
Source: Company, Daiwa forecasts

Kinsus: 1-year forward PER bands



Source: TEJ, Daiwa forecasts

Kinsus: revenue breakdown by product



Source: Company, Daiwa forecasts

Financial summary

Key assumptions

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Global PC shipment (mn)	361	301	260	263	287	244	244	244
Regular server shipmnet (mn)	14	15	12	14	16	18	22	26
Global smartphone shipment (mn)	1,655	1,437	1,380	1,437	1,429	1,249	1,255	1,300

Profit and loss (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
ABF	10,191	16,526	11,978	13,319	17,998	24,558	48,820	82,562
BT	17,565	18,796	8,065	10,398	14,314	21,234	30,530	33,105
Other Revenue	5,580	6,336	6,790	6,817	7,039	8,271	9,098	10,008
Total Revenue	**33,336**	**41,658**	**26,832**	**30,535**	**39,351**	**54,062**	**88,448**	**125,676**
Other income	0	0	0	0	0	0	0	0
COGS	(22,583)	(25,771)	(20,075)	(21,867)	(31,037)	(38,032)	(52,601)	(67,397)
SG&A	(2,879)	(3,532)	(3,301)	(5,010)	(3,433)	(4,241)	(6,960)	(9,312)
Other op.expenses	(2,413)	(2,484)	(2,397)	(2,612)	(2,212)	(2,647)	(4,114)	(5,769)
Operating profit	**5,456**	**9,851**	**1,042**	**1,095**	**2,670**	**9,142**	**24,773**	**43,198**
Net-interest inc./(exp.)	(68)	(172)	(332)	(394)	(391)	(336)	(356)	(350)
Assoc/forex/extraord./others	253	607	716	902	809	694	710	709
Pre-tax profit	**5,641**	**10,287**	**1,426**	**1,603**	**3,087**	**9,499**	**25,127**	**43,557**
Tax	(672)	(2,158)	(256)	(272)	(370)	(571)	(1,420)	(2,462)
Min. int./pref. div./others	(633)	(957)	(1,123)	(1,282)	(1,121)	(1,370)	(1,381)	(1,422)
Net profit (reported)	**4,336**	**7,172**	**48**	**49**	**1,596**	**7,558**	**22,325**	**39,673**
Net profit (adjusted)	**4,336**	**7,172**	**48**	**49**	**1,596**	**7,558**	**22,325**	**39,673**
EPS (reported)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
EPS (adjusted)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
EPS (adjusted fully-diluted)(TWD)	**9.622**	**15.901**	**0.105**	**0.108**	**3.505**	**14.824**	**43.787**	**77.810**
DPS (TWD)	**4.500**	**6.476**	**1.000**	**1.000**	**1.518**	**3.500**	**4.500**	**4.500**
EBIT	**5,456**	**9,851**	**1,042**	**1,095**	**2,670**	**9,142**	**24,773**	**43,198**
EBITDA	**9,833**	**14,958**	**6,394**	**7,228**	**9,914**	**17,313**	**33,761**	**53,085**

Cash flow (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Profit before tax	5,641	10,287	1,426	1,603	3,087	9,499	25,127	43,557
Depreciation and amortisation	4,377	5,107	5,351	6,133	7,244	8,171	8,988	9,887
Tax paid	(672)	(2,158)	(256)	(272)	(370)	(571)	(1,420)	(2,462)
Change in working capital	(1,347)	(176)	1,528	(906)	(1,959)	(3,452)	(2,982)	(3,707)
Other operational CF items	2,575	2,901	(1,793)	1,147	91	(634)	0	0
Cash flow from operations	**10,574**	**15,961**	**6,256**	**7,705**	**8,093**	**13,013**	**29,713**	**47,274**
Capex	(13,384)	(17,094)	(10,136)	(10,408)	(6,117)	(7,832)	(5,483)	(3,838)
Net (acquisitions)/disposals	0	0	0	0	0	0	0	0
Other investing CF items	489	559	(2,351)	1,301	(1,752)	(2,374)	(44)	(85)
Cash flow from investing	**(12,895)**	**(16,535)**	**(12,486)**	**(9,107)**	**(7,869)**	**(10,206)**	**(5,526)**	**(3,922)**
Change in debt	6,544	1,903	6,043	1,624	(1,684)	(8,891)	0	(45)
Net share issues/(repurchases)	0	0	0	0	0	0	0	0
Dividends paid	(451)	(2,443)	(3,438)	(454)	(457)	(774)	(1,785)	(2,294)
Other financing CF items	(94)	2,411	2,730	(173)	(1,130)	8,625	0	0
Cash flow from financing	**6,000**	**1,870**	**5,335**	**996**	**(3,270)**	**(1,041)**	**(1,785)**	**(2,340)**
Forex effect/others	(12)	56	(88)	71	(39)	51	39	35
Change in cash	**3,667**	**1,352**	**(983)**	**(335)**	**(3,084)**	**1,817**	**22,441**	**41,047**
Free cash flow	(2,809)	(1,133)	(3,880)	(2,703)	1,976	5,181	24,230	43,436

Source: FactSet, Daiwa forecasts

Financial summary continued …

Balance sheet (TWDm)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & short-term investment	15,968	17,923	21,013	16,676	16,336	21,188	43,628	84,676
Inventory	3,466	3,481	2,612	2,981	4,828	6,237	8,747	10,924
Accounts receivable	5,679	5,050	4,299	5,284	6,804	10,300	16,405	22,824
Other current assets	1,312	1,647	1,445	4,735	970	1,080	1,080	1,080
Total current assets	**26,424**	**28,100**	**29,369**	**29,677**	**28,938**	**38,804**	**69,860**	**119,503**
Fixed assets	31,586	44,101	48,209	49,658	50,897	51,643	48,138	42,089
Goodwill & intangibles	0	0	0	0	0	0	0	0
Other non-current assets	376	432	489	100	138	434	478	562
Total assets	**58,386**	**72,633**	**78,067**	**79,435**	**79,973**	**90,882**	**118,476**	**162,154**
Short-term debt	1,100	377	1,409	2,452	3,053	915	915	869
Accounts payable	10,150	10,798	8,457	7,521	10,030	14,004	19,638	24,526
Other current liabilities	2,733	4,199	4,393	7,018	7,023	5,571	5,571	5,571
Total current liabilities	**13,983**	**15,374**	**14,259**	**16,990**	**20,106**	**20,490**	**26,124**	**30,966**
Long-term debt	9,387	10,770	15,280	13,779	11,197	6,219	6,219	6,219
Other non-current liabilities	2,027	7,674	9,369	8,788	7,297	6,318	6,318	6,318
Total liabilities	**25,397**	**33,818**	**38,908**	**39,557**	**38,600**	**33,027**	**38,661**	**43,503**
Share capital	4,508	4,528	4,544	4,566	4,568	5,269	5,269	5,269
Reserves/R.E./others	28,481	34,288	34,615	35,312	36,805	52,585	74,546	113,382
Shareholders' equity	**32,989**	**38,816**	**39,159**	**39,878**	**41,373**	**57,854**	**79,815**	**118,651**
Minority interests	0	0	0	0	0	0	0	0
Total equity & liabilities	**58,386**	**72,633**	**78,067**	**79,435**	**79,973**	**90,882**	**118,476**	**162,154**
EV	540,075	538,779	541,231	545,110	543,470	531,501	509,061	467,968
Net debt/(cash)	**(5,481)**	**(6,776)**	**(4,325)**	**(445)**	**(2,085)**	**(14,054)**	**(36,495)**	**(77,588)**
BVPS (TWD)	**64.478**	**75.804**	**69.748**	**69.314**	**71.656**	**92.357**	**131.414**	**202.420**

Key ratios (%)

Year to 31 Dec	2021	2022	2023	2024	2025	2026E	2027E	2028E
Sales (YoY)	23.0	25.0	(35.6)	13.8	28.9	37.4	63.6	42.1
EBITDA (YoY)	70.8	52.1	(57.3)	13.1	37.1	74.6	95.0	57.2
Operating profit (YoY)	307.0	80.6	(89.4)	5.1	143.7	242.4	171.0	74.4
Net profit (YoY)	700.2	65.4	(99.3)	2.9	3,164.4	373.6	195.4	77.7
Core EPS (fully-diluted) (YoY)	698.1	65.3	(99.3)	2.5	3,153.4	322.9	195.4	77.7
Gross-profit margin	32.3	38.1	25.2	28.4	21.1	29.7	40.5	46.4
EBITDA margin	29.5	35.9	23.8	23.7	25.2	32.0	38.2	42.2
Operating-profit margin	16.4	23.6	3.9	3.6	6.8	16.9	28.0	34.4
Net profit margin	13.0	17.2	0.2	0.2	4.1	14.0	25.2	31.6
ROAE	13.9	20.0	0.1	0.1	3.9	15.2	32.4	40.0
ROAA	8.6	10.9	0.1	0.1	2.0	8.8	21.3	28.3
ROCE	14.0	21.1	2.0	2.0	4.8	15.2	32.6	40.6
ROIC	19.9	26.1	2.6	2.4	6.0	20.7	53.7	96.6
Net debt to equity	net cash	net cash	2.5	4.6	4.8	net cash	net cash	net cash
Effective tax rate	11.9	21.0	17.9	17.0	12.0	6.0	5.7	5.7
Accounts receivable (days)	55.2	47.0	63.6	57.3	56.1	57.7	55.1	57.0
Current ratio (x)	1.9	1.8	2.1	1.7	1.4	1.9	2.7	3.9
Net interest cover (x)	84.0	61.0	5.3	5.1	8.9	29.3	71.6	125.5
Net dividend payout	52.6	41.9	909.2	909.2	43.2	23.5	10.3	5.8
Free cash flow yield	n.a.	n.a.	n.a.	n.a.	0.4	0.9	4.4	8.0

Source: FactSet, Daiwa forecasts

Company profile

Kinsus was founded in 2000 as a subsidiary of Asustek. The company mainly focus on niche ball grid array (BGA) substrates to avoid price competition among mainstream products like graphics card and chipsets, and it has become top-3 flip chip manufacturer in global. Besides, its manufacturing sites are based in Taiwan and mainland China.

ESG analysis

ESG risks

Risks		Management	Analyst comments
G	Executive/board quality	2	Kinsus's leadership team comprises experienced professionals dedicated to steering the company towards sustainable growth. This diverse board composition ensures a comprehensive range of expertise and perspectives, contributing to Kinsus's strategic decision-making and corporate governamce.
	Capital management	2	Kinsus maintains a robust financial framework, ensuring the availability of resources necessary for operational efficiency and strategic investments. The company's financial health is regularly disclosed through annual reports, providing transparency to stakeholders.
	Related party & transaction	1	The company adheres to stringent corporate governance policies to manage related party transactions, ensuring transparency and compliance with regulatory standards. The company discloses such transactions in its financial reports, maintaining accountability and stakeholder trust. For instance, the company's annual general meeting documents provide details on related party transactions, ensuring transparency and accountability.
S	Supply chain management	1	Kinsus is committed to sustainable supply chain practices, emphasizing environmental, social, and governance considerations. The company has established comprehensive ESG policies, including commitments to labor standards, environmental health and safety, ethics, and conflict-free minerals. These policies guide the company's interactions with suppliers and partners, promoting sustainability throughout the supply chain.
E	Data security	3	The company recognizes the importance of information security in its operations. As part of its corporate governance framework, Kinsus is likely to implement standard data protection practices to safeguard sensitive information and ensure compliance with relevant regulations. Stakeholders are encouraged to consult the company's official communications or contact the company directly for detailed information on data security policies.

Note: Management score represents a company's ability to manage/benefit from certain ESG topics. The scores range from 1 to 3, with 1 being the strongest.

Update Date: 31 Jan 2026

Source: Daiwa, Company

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Buy*	73.96%
Hold**	19.58%
Sell***	6.46%

Source: Daiwa

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Daiwa Capital Markets Hong Kong Limited

data

6 October 2026

MEMO

WinWay Technology (6515 TT)

Share price (6 Oct): **TWD6,130.00** 12-mth rating: **Buy (1)** **Information Technology: Taiwan**

Limited near-term catalysts before 2027 capacity unlock

Leon Chen
(886) 2 8758 6247
leon.chen@daiwacm-cathay.com.tw

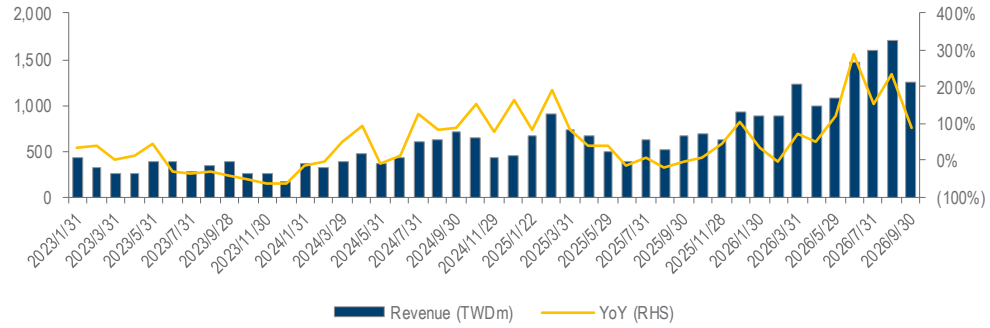
Stacy Lin
(886) 2 8758 6252
stacy.lin@daiwacm-cathay.com.tw

Summary: Winway reported September revenue of TWD1.25bn (-26.7% MoM; +87.7% YoY), brining 3Q26 revenue to a record high of TWD4.56bn (+29.3% QoQ; +152.6% YoY). The 3Q26 top-line result beat market consensus by 6.7% but missed Daiwa's estimate by 9.1%. Following August's exceptionally strong top line (TWD1.71bn), we had factored in a higher shipment run-rate for September. However, the miss stems primarily from the lumpiness of probe-card deliveries (high concentration to a single customer) and customer qualification timelines, rather than order cancellations or share loss.

We forecast 4Q26 revenue to track flattish to slightly down QoQ, reflecting lower probe-card shipment volumes compared to 3Q26, alongside typical year-end pull-in deceleration from customers. On profitability, we maintain our view that 2Q26 marked the trough for gross margin this year. As the revenue contribution from lower-margin MEMS probe cards declines, a more favourable product mix should drive a steady QoQ gross margin recovery, which may drive gross margin to reach 40% earlier than market expectations.

Although we remain positive on Winway's structural growth profile, near-term catalysts appear muted. This reflects the revenue lumpiness in probe cards, while additional socket capacity will not come online until March 2027—pushing the corresponding revenue recognition into 2Q27. We will revisit our financial model upon the company's 3Q26 results release. We currently have a Buy (1) rating on Winway with a 12-month TP of TWD10,000.

Winway monthly revenue trend



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Rating	Percentage of total
Buy*	73.96%
Hold**	19.58%
Sell***	6.46%

Source: Daiwa

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- There may be a difference between bid price etc. and ask price etc. of OTC derivatives handled by our company.
- Before engaging in any trading, please thoroughly confirm accounting and tax treatments regarding your trading in financial instruments with such experts as certified public accountants.

* The amount of the trading commission cannot be stated here in advance because it will be determined between our company and you based on current market conditions and the content of each transaction etc.

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King Slide

Maintain Buy: Sep sales -31% MoM from high bases; Aug earnings remain solid

Maintain Rating: BUY | PO: 14,888 TWD | Price: 13,100 TWD

Sep sales -31% MoM from high bases; 3Q26 sales in-line

King Slide's Sep sales of NT$4.46bn declined 31% MoM (+196% YoY), after the strong sales in the past few months (up 17-46% MoM in Apr-Jul and up 1% MoM in Aug). 3Q26 sales of NT$17.34bn (+60% QoQ, +297% YoY) were 0.5% above BofAe of NT$17.26bn and 0.9% below consensus of NT$17.49bn.

We are not surprised with the MoM sales drop in Sep as (1) the company's MoM sales trend has been volatile (Exhibit 1 and Exhibit 2), (2) sales bases in Apr-Aug were at record levels, (3) contribution from Nvidia Rubin GPU projects should be more meaningful in 4Q26 after hardware components ramp-up, and (4) partial general server orders/demand likely were pushed out due to key component shortage (e.g. memory, CPU).

Solid Aug EPS NT$40.07; pre-tax/net margin 74.6%/59.1%

In addition to Sep sales, on the same day (6 Oct), King Slide also announced its Aug unaudited key financial data as required by local regulations (Exhibit 3). Aug sales were NT$6,466mn (+344% YoY), while pre-tax profit was NT$4,826mn (+216% YoY) and net income was NT$3,819mn (+212% YoY). EPS came in at NT$40.07 (+212% YoY). Pre-tax margin and net margin were 74.6% and 59.1%, respectively.

Compared to July unaudited figures announced on 14 Aug (see our 17 Aug report), including sales of NT$6,407mn, pre-tax margin of 93.4%, net margin of 74.7%, and EPS of NT$50.24, we believe the margin and EPS differences mainly reflected FX impact under similar sales scale.

Combining Jul and Aug unaudited results, sales totaled NT$12.87bn and EPS reached NT$90.31. Pre-tax margin and net margin were 84.0% and 66.9%, respectively. Jul + Aug sales achieved 75%/74% of BofAe/consensus 3Q26 sales. Pre-tax profit, net income, and EPS all achieved 79% of BofAe (79.4% pre-tax margin, 62.9% net margin, and NT$113.87 EPS) and 77-78% of consensus (80.3% pre-tax margin, 63.1% net margin, and NT$117.62 EPS). We also noted that Jul + Aug EPS of NT$90.31 already surpassed EPS of NT$74.38 in 2Q26 with similar pre-tax/net margin (84.0%/66.9% in Jul + Aug, vs. 84.2%/65.4% in 2Q26).

Maintain Buy; expect a neutral-to-positive market reaction

We expect a neutral-to-positive market reaction in the near term in view of significant MoM sales decline in Sep but also still solid Aug unaudited data. We maintain Buy on King Slide as the server rail market leader with broad and diversified client/project base. We expect the company to post 3Q26 results after market close on 12 Nov (Thu).

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06 October 2026

Equity

Doris Kao >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3722
doris.kao@bofa.com

Robert Cheng >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3731
robert.cheng@bofa.com

Katherine Zhu >>
Research Analyst
Merrill Lynch (Hong Kong)
+852 3508 3374
kexin.zhu@bofa.com

Carol Hu >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3721
carol.hu@bofa.com

Stock Data

Price	13,100 TWD
Price Objective	14,888 TWD
Date Established	7-Aug-2026
Investment Opinion	C-1-7
52-Week Range	2,875 TWD-15,085 TWD
Mrkt Val / Shares Out (mn)	39,309 USD / 95.3
Market Value (mn)	1,248,392 TWD
Average Daily Value (mn)	201.14 USD
Free Float	67.5%
BofA Ticker / Exchange	KGSLF / TAI
Bloomberg / Reuters	2059 TT / 2059.TW
ROE (2026E)	77.4%
Net Dbt to Eqty (Dec-2025A)	-84.2%

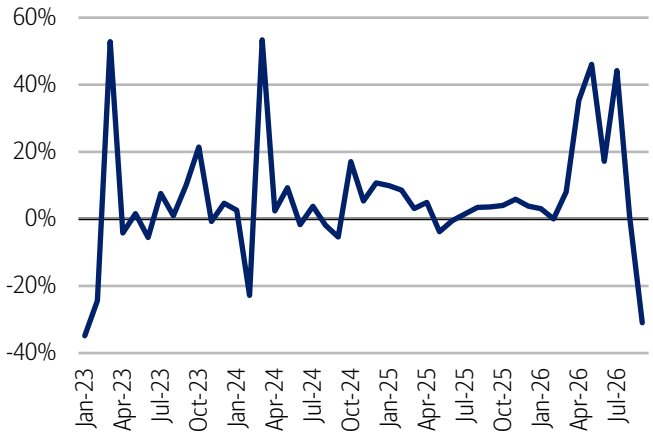
Acronyms

CPU: Central Processing Unit

FX: Foreign Exchange

GPU: Graphic Processing Unit

Exhibit 1: King Slide's sales MoM trend has been volatile in AI era
King Slide monthly sales MoM trend, Jan'23-Sep'26



Source: BofA Global Research, company data. AI: artificial intelligence

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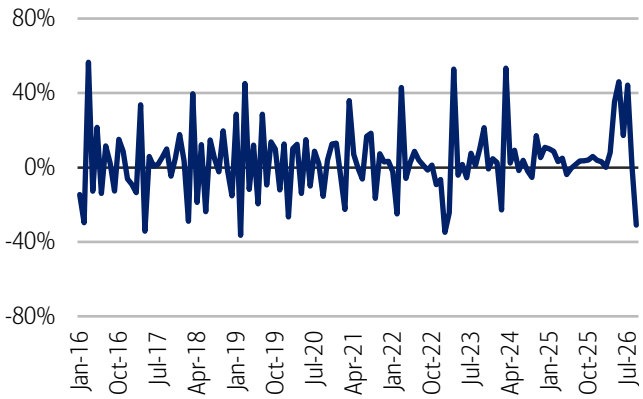
Exhibit 2: King Slide's sales MoM trend has been volatile in the past decade
King Slide monthly sales MoM trend, Jan'16-Sep'26



Source: BofA Global Research, company data

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Exhibit 3: Jul + Aug sales achieved 75%/74% of BofA/consensus 3Q26 estimate and EPS achieved 79%/77% of BofA/consensus 3Q26 estimate
King Slide Jul'26 and Aug'26 unaudited key financial data vs. 3Q26 BofAe/consensus

NT$mn	Jul'26	YoY %	Aug'26	YoY %	Jul + Aug Subtotal	3Q26 BofAe	Jul + Aug Run rate	3Q26 Consensus	Jul + Aug Run rate
Sales	6,407	336%	6,466	344%	12,873	17,260	75%	17,494	74%
Pretax profit	5,985	321%	4,826	216%	10,811	13,702	79%	14,049	77%
Net income	4,787	321%	3,819	212%	8,606	10,852	79%	11,038	78%
EPS (NT$)	50.24	321%	40.07	212%	90.31	113.87	79%	117.62	77%
Margin %									
Pre-tax margin	93.4%		74.6%		84.0%	79.4%		80.3%	
Net margin	74.7%		59.1%		66.9%	62.9%		63.1%	

Source: BofA Global Research estimates, Bloomberg, company data

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Price objective basis & risk

King Slide (KGSLF)

Our PO of NT$14,888 is based on 32x 2027E EPS. We use 32x P/E multiple (2.5SD above P/E average since listing) in view of strong AI server market demand, leading position in server rail, solid/high margin profiles, and double-digit sales/EPS CAGR in 2026-28E. Its early step into server market since 2001 with proven track record, much larger scale than peer, and 3,600+ patents as strong moats for high/above-peer margins. Its above-30x P/E is a similar case to other component sub-sector leaders in Taiwan.

Upside risks to our PO are: (1) even stronger AI server demand, (2) margin upside thanks to larger scale and/or better efficiency, (3) better kitchen slides market demand, (4) faster capacity expansion, (5) favorable FX movement (mainly USD/NTD) for GM and/or non-operating items.

Downside risks to our PO are: (1) softer AI server demand due to GPU/components shortage, (2) GM pressure due to higher raw material cost (e.g. steel, aluminum), (3) softer kitchen slides market demand under macro uncertainty/weakness, (4) delayed product certification, (5) slower capacity expansion, (6) unfavorable FX movement (mainly USD/NTD) to weigh on GM and/or non-operating items.

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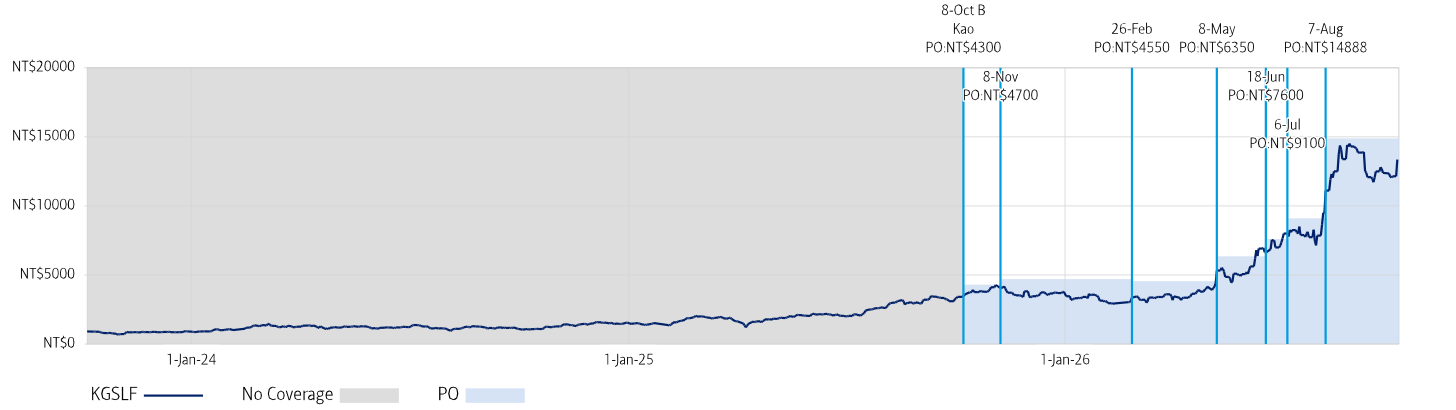
King Slide (KGSLF) Price Chart



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Hold	82	19.95%	Hold	46	56.10%
Sell	74	18.00%	Sell	31	41.89%

Equity Investment Rating Distribution: Global Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
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Hold	806	22.35%	Hold	563	69.85%
Sell	744	20.63%	Sell	446	59.95%

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October 6, 2026 11:43 AM GMT

Accton Technology Corporation | Asia Pacific

September Sales -1% MoM/ +66% YoY

In this report, we focus on Accton's monthly sales, which we believe could be a catalyst for share price movement.

Details:

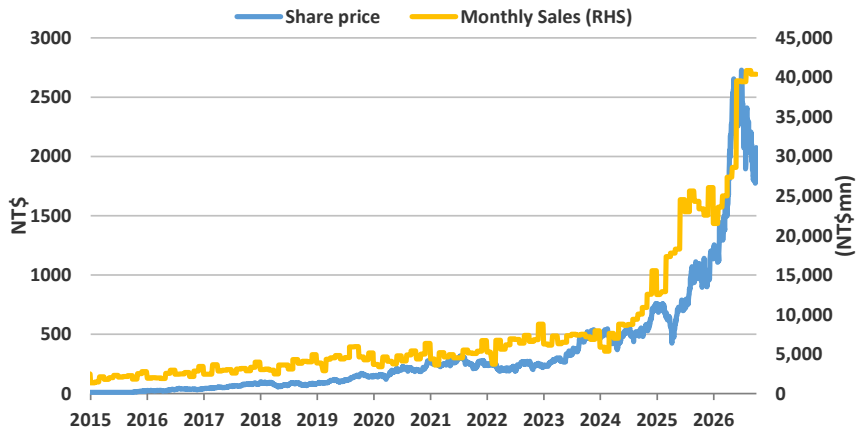
- September sales were NT$40,399mn (-1% MoM/+66% YoY).
- 3Q26 revenue was NT$120,744mn (+26% QoQ/+66% YoY), in line with MSe of NT$120,883mn (+27% QoQ/+66% YoY) and consensus of NT$117.390mn (+23% QoQ/+61% YoY) by 3%.

Our view:

- We believe the consignment arrangement of DRAM, PCB, etc., from August might have capped the upside on revenue growth, but it should be a favorable factor for gross margin, emerging from August/September and likely extending into 4Q26.
- Driven by the network switch business and its major customer's AI accelerator module ramp, we expect revenue to continue growing sequentially in 4Q26, with the momentum in 2027 strengthened by potential new customers and new products.

- Stay OW with a PT of NT$3,500, based on 30x 2027e P/E.

Exhibit 1: Accton's monthly sales vs. its share price since 2015



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MORGAN STANLEY TAIWAN LIMITED+

Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

Accton Technology Corporation (2345.TW, 2345 TT)
Top Pick

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$3,500.00
Up/downside to price target (%)	71
Shr price, close (Oct 6, 2026)	NT$2,050.00
52-Week Range	NT$2,825.00-884.00
Sh out, dil, curr (mn)	559
Mkt cap, curr (mn)	NT$1,145,761
EV, curr (mn)	NT$1,085,110
Avg daily trading value (mn)	NT$8,214

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	47.13	83.77	115.77	145.77
Prior EPS (NT$)**	-	-	-	-
EPS (NT$)§	45.42	86.65	118.19	165.38
Revenue, net (NT$ mn)	248,320	413,479	541,272	667,428
EBITDA (NT$ mn)	33,894	62,436	85,653	106,950
ModelWare net inc (NT$ mn)	26,342	46,817	64,707	81,472
P/E	25.1	24.5	17.7	14.1
P/BV	11.5	12.3	8.0	5.6
RNOA (%)	622.3	(1,479.8)	4,110.1	1,435.9
ROE (%)	72.6	81.2	69.2	56.9
EV/EBITDA	17.8	16.9	11.8	8.9
Div yld (%)	0.9	1.0	1.3	1.7
FCF yld ratio (%)**	4.5	3.7	5.3	6.8
Leverage (EOP) (%)	(101.7)	(96.2)	(94.8)	(94.4)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
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Valuation Methodology and Risks

Accton Technology Corporation (2345.TW)

Base case, P/E, which we believe better captures the company's earnings growth momentum. We assign a target P/E of 30x to our 2027 EPS estimate, which we view as justified by an operating profit CAGR of 48% for 2025-28, based on our estimates. This implies a PEG of around 0.6, vs. its average of 0.7 over 2019-25, when the stock was trading at a P/E of 22x for an operating profit CAGR of 32%.

Risks to Upside

- Stronger data center capex
- Faster network switch spec migration
- InfiniBand losing share in AI networks
- Stronger custom AI accelerator growth
- Stronger ODM direct business model adoption

Risks to Downside

- Weaker data center capex
- Slower network switch spec migration
- InfiniBand taking more share in AI networks
- Softer custom AI accelerator growth
- Softer ODM direct business model adoption

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Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

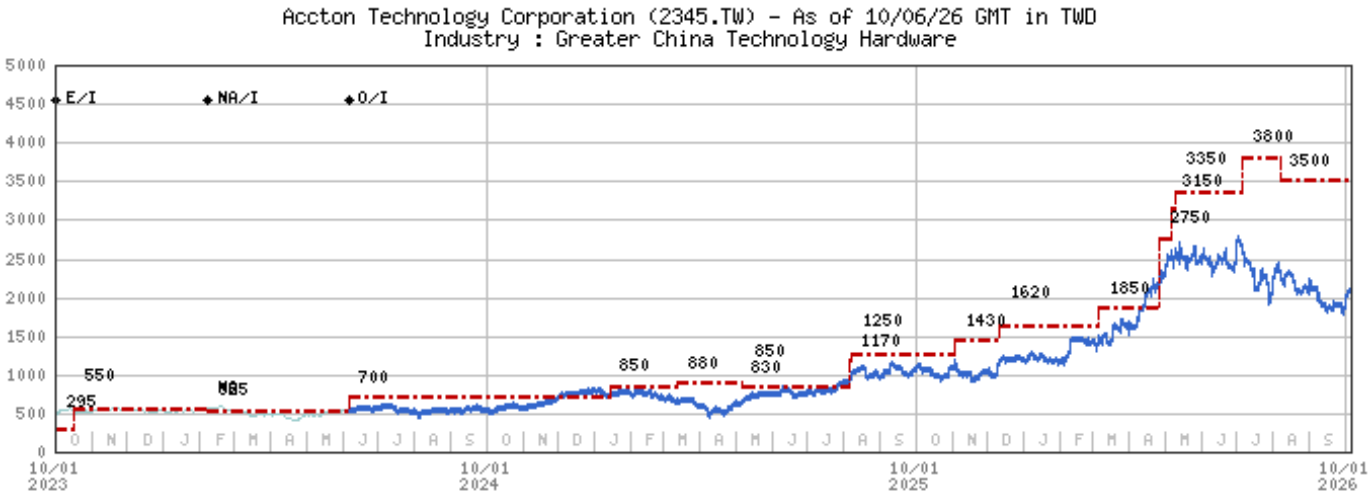
Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : O/I; 11/12/21 : E/I; 2/7/24 : NA/I; 6/6/24 : O/I

Price Target History: 7/28/21 : 336; 11/12/21 : 265; 1/10/22 : 280; 3/7/22 : 270; 5/13/22 : 235; 6/7/22 : 248; 8/12/22 : 275; 5/22/23 : 285; 8/10/23 : 295; 10/17/23 : 550; 2/7/24 : NA; 2/8/24 : 525; 6/6/24 : 700; 1/14/25 : 850; 3/13/25 : 880; 5/6/25 : 830; 5/9/25 : 850; 8/6/25 : 1170; 8/8/25 : 1250; 11/4/25 : 1430; 12/12/25 : 1620; 3/6/26 : 1850; 4/27/26 : 2750; 5/7/26 : 3150; 5/11/26 : 3350; 7/6/26 : 3800; 8/7/26 : 3500

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/06/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$39.36
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$23.90
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,045.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,855.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.43
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$52.30
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$5.00
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$38.14
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$64.10
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77

Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.26
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$180.00
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,079.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.80
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,050.00
Advantech (2395.TW)	O (01/20/2021)	NT$738.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,195.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,685.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$37.45
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,530.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,325.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$885.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,350.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,050.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,415.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$339.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.10
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,100.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$299.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.05
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$1,000.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$206.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.90
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,995.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$5.02
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$380.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,260.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$256.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.10
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,920.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.40
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,865.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,505.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$90.00
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$333.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,310.00

Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$567.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,290.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$625.00
Zhen Ding (4958.TW)	++	NT$574.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$551.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,755.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

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UPDATE

October 6, 2026 12:55 AM GMT

Bizlink | Asia Pacific

September Sales +5% MoM/ +39% YoY

In this report, we focus on Bizlink's monthly sales, which we believe could be a catalyst for its share price.

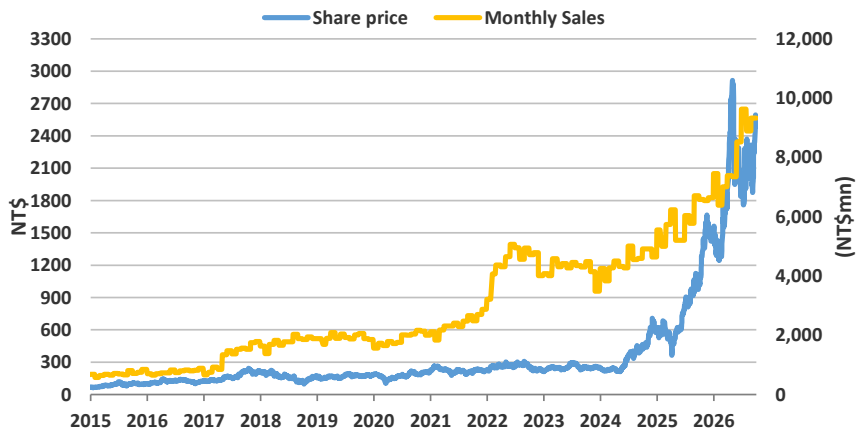
Details:

- September sales came in at NT$9,332mn (+5% MoM/+39% YoY)
- 3Q26 revenue reached NT$27,854mn (+20% QoQ/+50% YoY), slightly short of MSe of NT$28,521mn (+23% QoQ/+55% YoY) by 2% while exceeding consensus of NT$27,423mn (+18% QoQ/+49% YoY) by 2%.

Our view:

- We think 3Q26 revenue came in largely in-line with market expectations.
- Looking into 2H26, we expect HPC to remain the main growth driver, as power interconnects should continue to see momentum from AI/general servers. Meanwhile, data interconnects should grow along with new ASIC model ramp-up and GPU server demand. Moreover, semi cap equipment revenue growth should also sustain.
- We believe Bizlink has allocated increasing resources toward the optical interconnect space, including the collaboration on ALC with Credo, mini VCSEL optical module with ams-OSRAM and shuffle box with Corning. We think that these opportunities could help offset a potential slowdown in its AEC business as AEC solutions reach their scalability limits at higher data rates, likely from 3.2T.
- Stay OW.

Exhibit 1: Bizlink's monthly sales vs. its share price since 2015



Note: Past performance is no guarantee of future results. Results shown do not include transaction costs. Source: Company data, TEJ, Morgan Stanley Research.

MORGAN STANLEY TAIWAN LIMITED+

Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

Bizlink (3665.TW, 3665 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$3,665.00
Up/downside to price target (%)	47
Shr price, close (Oct 5, 2026)	NT$2,485.00
52-Week Range	NT$3,010.00-1,000.00
Sh out, dil, curr (mn)	193
Mkt cap, curr (mn)	NT$480,515
EV, curr (mn)	NT$486,091
Avg daily trading value (mn)	NT$6,712

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	46.57	69.77	120.77	143.77
Prior EPS (NT$)**	-	-	-	-
EPS (NT$)§	47.71	68.93	119.70	154.48
Revenue, net (NT$ mn)	71,247	101,953	151,516	174,996
EBITDA (NT$ mn)	15,154	21,153	34,384	41,510
ModelWare net inc (NT$ mn)	9,005	13,491	23,353	27,800
P/E	32.6	35.6	20.6	17.3
P/BV	6.4	8.6	6.4	5.1
RNOA (%)	24.2	26.2	39.8	41.7
ROE (%)	24.9	28.9	40.8	36.5
EV/EBITDA	20.1	22.8	13.7	11.0
Div yld (%)	0.8	0.6	0.9	1.6
FCF yld ratio (%)**	(0.8)	1.5	2.9	5.1
Leverage (EOP) (%)	(0.9)	(8.3)	(19.1)	(32.5)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

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AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
Bizlink 3665.TW			
September 2026 monthly sales	06 Oct 2026	Unchanged	In-line

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Bizlink (3665.TW)

Base case, derived from 30x 2027e P/E. We think our target P/E multiple is justified by an earnings CAGR of 46% seen over 2025-28e, which implies a PEG of around 0.7. This is in line with the average of 0.7 in 2019-24, when the stock was trading at an average 13x P/E for a 19% earnings CAGR.

Risks to Upside

- Faster and wider AEC adoption
- Data center and semi equipment capex increases that drive stronger HPC and semi-cap business
- Stronger recovery for automotive business

Risks to Downside

- Lower-than-expected AEC adoption
- Intensifying competition
- Softer-than-expected AI capex
- Slower-than-expected growth in the semi capital equipment market
- Continued weakness in the auto industry

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(as of September 30, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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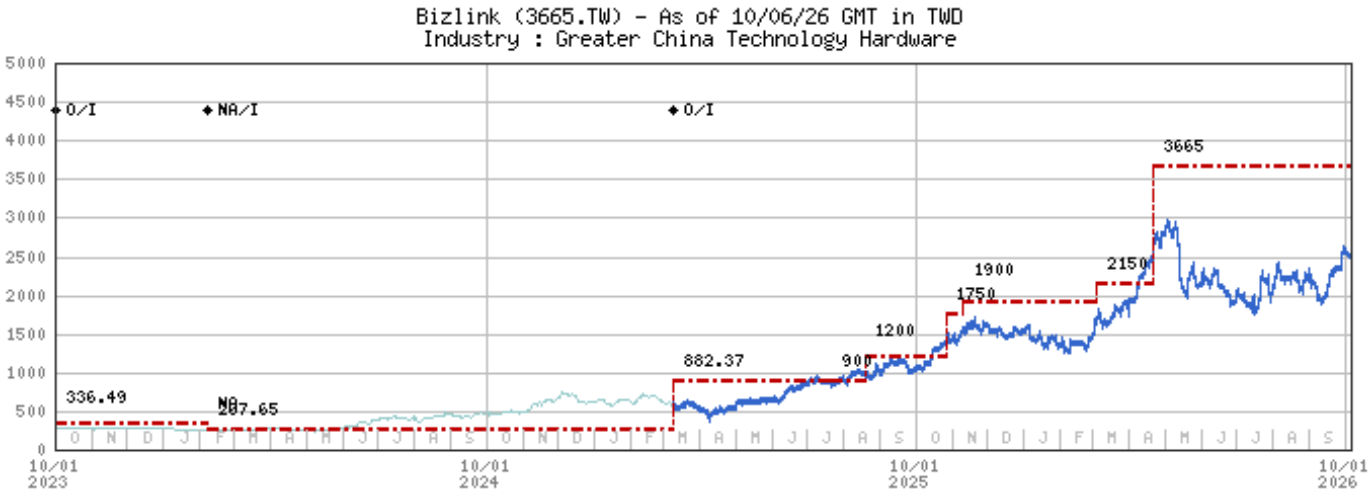
Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : NA/I; 8/18/22 : E/I; 2/20/23 : O/I; 2/7/24 : NA/I; 3/10/25 : O/I

Price Target History: 8/18/22 : 336.49; 11/11/22 : 293.23; 2/20/23 : 336.49; 2/7/24 : NA; 2/8/24 : 267.65; 3/10/25 : 882.37; 7/21/25 : 900; 8/19/25 : 1200; 10/27/25 : 1750; 11/10/25 : 1900; 3/4/26 : 2150; 4/22/26 : 3665

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -·- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) ▬
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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/05/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$38.64
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.76
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,055.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,730.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.46
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$53.35
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.91
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$37.16
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$61.95
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$23.88

Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$179.10
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,111.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.62
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,075.00
Advantech (2395.TW)	O (01/20/2021)	NT$740.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,180.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,585.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$39.05
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,550.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,485.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$890.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,325.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,005.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$146.50
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$68.20
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,450.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$345.50
Innolux (3481.TW)	E (04/07/2025)	NT$52.30
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,355.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$298.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.40
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$995.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$207.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.80
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,700.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.86
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$388.50
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,215.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$254.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.30
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$3,000.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.92
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,875.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,495.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$88.70
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$331.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,325.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$606.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50

Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,150.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$626.00
Zhen Ding (4958.TW)	++	NT$576.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$570.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,780.00

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* Historical prices are not split adjusted.

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October 6, 2026 11:48 AM GMT

Chroma Ate Inc. | Asia Pacific

September Sales +22% MoM/ +206% YoY

In this report, we focus on Chroma's monthly sales, which we believe could be a catalyst for share price movement.

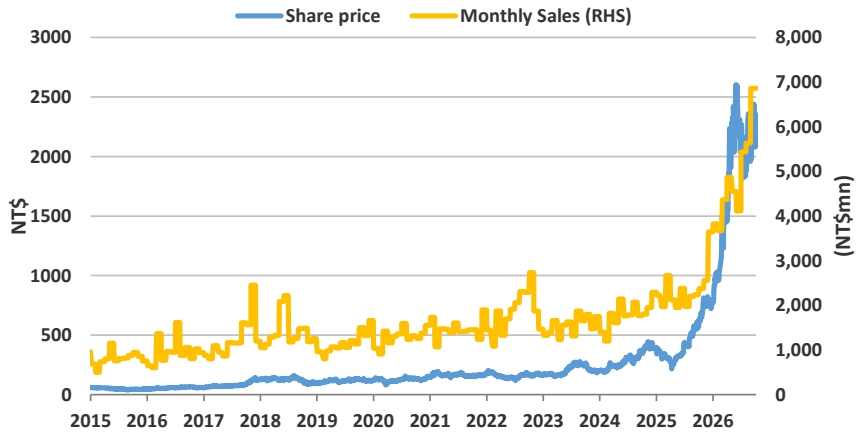
Details:

- September revenue was NT$6,860mn (22% MoM/+206% YoY), a record high.
- 3Q26 revenue came in at NT$17,926mn (+32% QoQ/+180% YoY), significantly higher than our estimate at NT$14,772mn (+9% QoQ/+130% YoY) and consensus at NT$14,555mn (+8% QoQ/+127% YoY).

Our view:

- We believe that the strength in 3Q26 was mainly driven by the semiconductor and photonics business, especially SLT, metrology, transceiver/CPO testing, etc.
- We think that Chroma might start to ship FT handlers and ovens in small volume from late 2026/early 2027, which represent new business opportunities for the coming years.
- Stay OW.

Exhibit 1: Chroma's monthly sales vs. its share price since 2015



Source: Company data, TEJ, Morgan Stanley Research. Note: Past performance is no guarantee of future results. Results shown do not include transaction costs.

MORGAN STANLEY TAIWAN LIMITED+

Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

Chroma Ate Inc. (2360.TW, 2360 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$2,800.00
Up/downside to price target (%)	19
Shr price, close (Oct 6, 2026)	NT$2,350.00
52-Week Range	NT$2,795.00-599.00
Sh out, dil, curr (mn)	425
Mkt cap, curr (mn)	US$31,512
EV, curr (mn)	US$31,169
Avg daily trading value (mn)	US$182

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	27.49	43.97	62.77	74.77
EPS (NT$)§	26.88	44.69	66.84	86.80
Prior EPS (NT$)**	-	-	-	-
Revenue, net (NT$ mn)	28,311	55,154	76,117	89,224
EBITDA (NT$ mn)	10,038	22,684	32,742	38,930
ModelWare net inc (NT $ mn)	11,692	18,697	26,691	31,797
P/E	28.2	53.4	37.4	31.4
P/BV	10.1	22.4	16.9	13.7
RNOA (%)	49.4	80.3	95.7	98.3
ROE (%)	45.9	57.4	59.9	53.9
EV/EBITDA	31.8	43.3	29.7	24.7
Div yld (%)	1.2	0.8	1.3	1.9
FCF yld ratio (%)**	2.4	1.5	2.3	3.0
Leverage (EOP) (%)	(33.4)	(39.0)	(46.3)	(52.8)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

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AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
Chroma Ate Inc. 2360.TW			
September 2026 sales	06 Oct 2026	Unchanged	↑ Modest upside surprise

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Chroma Ate Inc. (2360.TW)

Base case, 45x 2027e P/E. We use P/E as our valuation methodology because we think it better captures Chroma's earnings growth momentum. The stock traded at an average valuation of ~19x P/E during 2014-25 with an operating profit CAGR of 20%, implying a PEG of 1.0. We think our target multiple of 45x is justified by the 61% operating profit CAGR we project for 2025-28.

Risks to Upside

- Better-than-expected macro conditions
- Stronger semi and photonics and power testing demand
- Margin improvement via better product mix
- Better opex control

Risks to Downside

- Macro slowdown
- Softer-than-expected AI capex
- Slower progress in the advanced packaging capacity expansion
- Weaker consumer electronics and automobile related demand
- Margin contraction arising from lack of revenue scale and increasing opex

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Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

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Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

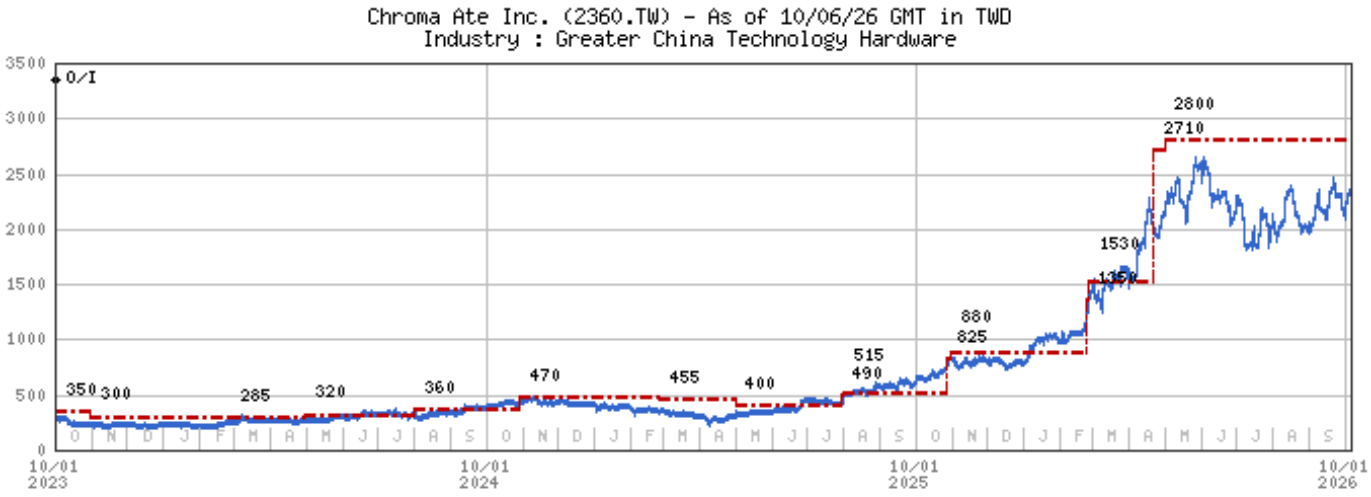
Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : O/I; 10/5/21 : O/I

Price Target History: 7/31/20 : 200; 10/5/21 : 220; 2/23/22 : 230; 2/24/23 : 215; 4/27/23 : 225; 7/7/23 : 320; 8/1/23 : 350; 10/31/23 : 300; 2/26/24 : 285; 4/30/24 : 320; 8/1/24 : 360; 10/30/24 : 470; 2/25/25 : 455; 5/1/25 : 400; 7/30/25 : 490; 8/1/25 : 515; 10/28/25 : 825; 10/31/25 : 880; 2/23/26 : 1350; 2/26/26 : 1530; 4/21/26 : 2710; 5/1/26 : 2800

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)

Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/06/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$39.36
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$23.90
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,045.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,855.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.43
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$52.30
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$5.00
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$38.14
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$64.10
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43

Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.26
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$180.00
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,079.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.80
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,050.00
Advantech (2395.TW)	O (01/20/2021)	NT$738.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,195.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,685.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$37.45
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,530.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,325.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$885.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,350.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,050.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,415.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$339.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.10
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,100.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$299.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.05
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$1,000.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$206.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.90
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,995.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$5.02
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$380.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,260.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$256.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.10
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,920.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.40
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,865.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,505.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$90.00
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$333.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,310.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$567.00

Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,290.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$625.00
Zhen Ding (4958.TW)	++	NT$574.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$551.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,755.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

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UPDATE

October 6, 2026 11:47 AM GMT

Hiwin Technologies Corp. | Asia Pacific

September Sales Flat MoM/ +30% YoY

In this report, we focus on Hiwin's monthly sales, which we believe could be a catalyst for its share price.

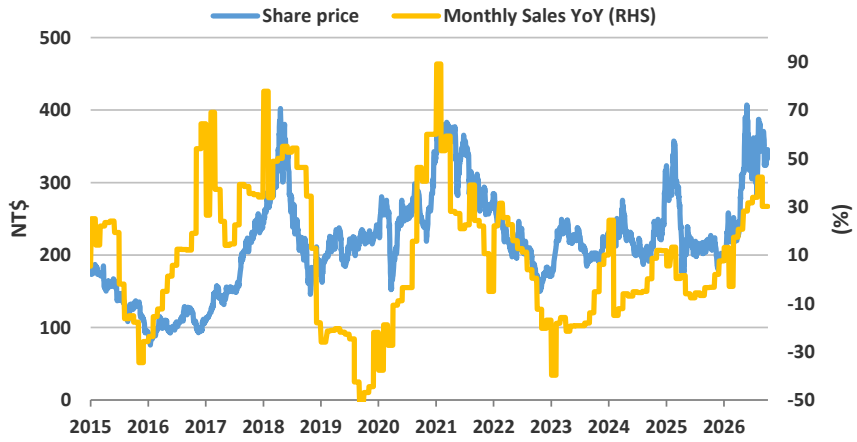
Details:

- September sales were NT$2,720mn (flat MoM/+30% YoY).
- 3Q26 revenue came in at NT$8,110mn (+8% QoQ/+35% YoY), largely in line with our estimate at NT$7,996mn (+6% QoQ/+33% YoY) and the consensus of NT$7,964mn (+6% QoQ/+33% YoY).

Our view:

- On top of the sequential growth on the top line, we believe that Hiwin should also further expand its gross margin on a QoQ basis in 3Q26, owing to the higher utilization and the tailwinds of price hikes.
- We think the momentum should remain favorable in the near term.
- We stay constructive on the stock, with current valuation at 29x 2027e P/E vs. a peak cycle level of 35-40x.

Exhibit 1: Hiwin's monthly sales YoY vs. its share price since 2015



Source: Past performance is no guarantee of future results. Results shown do not include transaction costs. Source: Company data, TEJ, Morgan Stanley Research.

MORGAN STANLEY TAIWAN LIMITED+

Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

Hiwin Technologies Corp. (2049.TW, 2049 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$450.00
Up/downside to price target (%)	33
Shr price, close (Oct 6, 2026)	NT$339.50
52-Week Range	NT$426.00-180.50
Sh out, dil, curr (mn)	354
Mkt cap, curr (mn)	US$3,787
EV, curr (mn)	US$3,776
Avg daily trading value (mn)	US$45

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	4.31	9.77	11.77	13.77
Prior EPS (NT$)**	-	-	-	-
Revenue, net (NT$ mn)	24,263	29,992	34,688	39,123
EBITDA (NT$ mn)	4,001	6,485	7,488	8,431
ModelWare net inc (NT $ mn)	1,526	3,457	4,165	4,873
P/E	45.0	34.7	28.8	24.6
P/BV	1.4	2.2	2.0	1.9
RNOA (%)	3.6	8.7	11.1	12.7
ROE (%)	4.1	9.2	10.4	11.2
EV/EBITDA	12.6	17.9	15.3	13.3
Div yld (%)	1.2	0.6	0.9	1.1
FCF yld ratio (%)**	2.0	3.7	2.6	2.9

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
e = Morgan Stanley Research estimates

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AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
Hiwin Technologies Corp. 2049.TW			
September 2026 Sales	06 Oct 2026	Unchanged	In-line

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Hiwin Technologies Corp. (2049.TW)

Base case, P/E. We think this valuation methodology better captures Hiwin's earnings growth momentum, as its valuation tends to move with the business cycle. We apply a target multiple of 38x to our 2027 EPS estimate. We view this multiple as justified by the 47% operating profit CAGR we project for 2025-28.

Risks to Upside

- Stronger industrial automation demand pickup.
- Market share gains.
- Earlier-than-expected revenue contribution from humanoids.

Risks to Downside

- Weaker industrial automation demand.
- More pricing pressure.
- Market share loss.
- More broad-based impact on the global economy from geopolitical risks.

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(as of September 30, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
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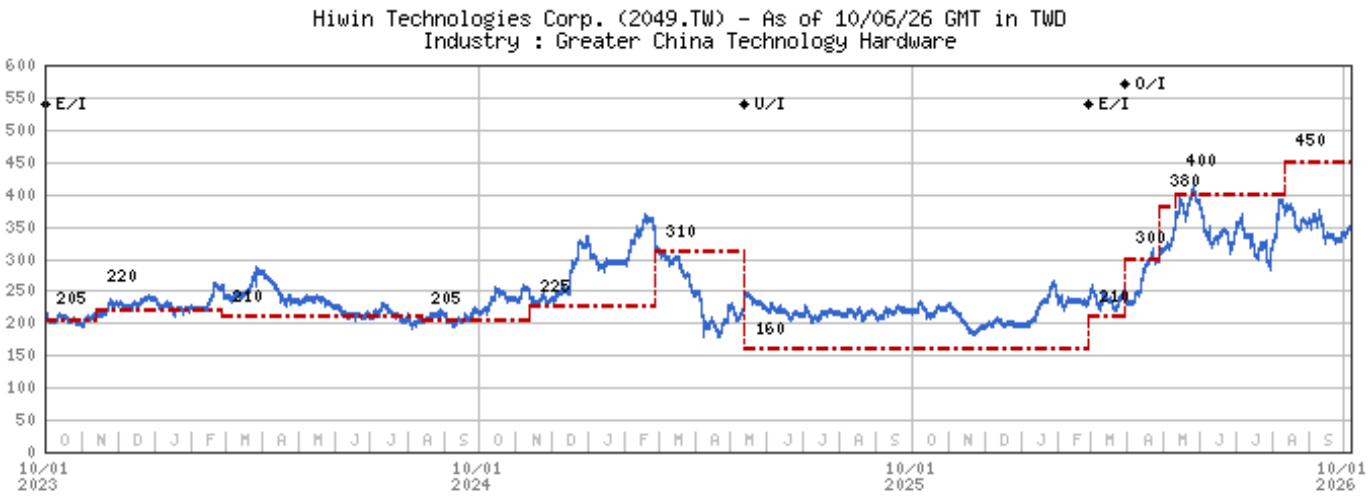
Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : O/I; 8/11/22 : E/I; 2/21/23 : O/I; 8/11/23 : E/I; 5/12/25 : U/I; 2/27/26 : E/I; 3/30/26 : O/I

Price Target History: 8/5/21 : 480.04; 10/5/21 : 479.6; 2/28/22 : 341.16; 3/24/22 : 301.61; 7/20/22 : 295; 8/11/22 : 230; 11/21/22 : 200; 2/21/23 : 295; 2/27/23 : 283; 5/10/23 : 280; 8/11/23 : 205; 11/13/23 : 220; 2/27/24 : 210; 8/13/24 : 205; 11/12/24 : 225; 2/27/25 : 310; 5/12/25 : 160; 2/27/26 : 210; 3/30/26 : 300; 4/28/26 : 380; 5/11/26 : 400; 8/12/26 : 450

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)

Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/06/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$39.36
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$23.90
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,045.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,855.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.43
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$52.30
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$5.00
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$38.14
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$64.10
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43

Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.26
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$180.00
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,079.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.80
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,050.00
Advantech (2395.TW)	O (01/20/2021)	NT$738.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,195.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,685.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$37.45
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,530.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,325.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$885.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,350.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,050.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,415.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$339.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.10
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,100.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$299.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.05
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$1,000.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$206.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.90
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,995.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$5.02
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$380.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,260.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$256.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.10
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,920.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.40
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,865.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,505.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$90.00
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$333.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,310.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$567.00

Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,290.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$625.00
Zhen Ding (4958.TW)	++	NT$574.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$551.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,755.00

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ASIA PACIFIC INSIGHT

October 6, 2026 12:44 AM GMT

Greater China Technology Hardware

Integrated Power Solutions Driving Secular Demand

Increasing power consumption, along with the deployment of BBU/CBU/800/±400VDC/SST in AI clusters, could push the technology barrier higher and present growth opportunities for players with superior capabilities. Fuel cells could also emerge as a new opportunity amid power constraints.

Secular demand for power amid AI capex: AI-related power consumption will continue to increase, and we believe this will lead to secular demand for power-related components, such as power supply units (PSUs), battery backup units (BBUs), capacitor bank units (CBUs), coolant distribution units (CDUs), and so on. Moreover, the requirement for efficiency will inevitably necessitate the adoption of 800/±400 voltage direct current (VDC) in the power rack/center/block configuration (initially from late 2026 and broadly in 2028, we estimate), eventually extending to the deployment of solid state transformers (SST) in the longer term – all pointing to a more centralized power delivery architecture in the future. This trend of aggregation could raise the technology barriers amid the fusion of white/grey space, favoring players with better capabilities and execution. Overall, we forecast the datacenter PSU market size to grow at a 75% three-year CAGR to reach US$33bn in 2028.

Fuel cells to improve flexibility and resilience of AI power delivery: Despite the strong AI capex outlook (MSe global cloud capex +97%/20% YoY in 2026/2027), power has become a bottleneck, indicated by MS's forecast power shortfall of 5GW in 2026, scaling to a total 72GW through 2029 (see report). Fuel cells, among other alternative energy sources, could emerge as a good solution to pick up some slack and enhance efficiency through active micro grid management. Those who are able to offer comprehensive solutions from power sources to power delivery could be increasingly more competitive in the market.

Assuming coverage of Delta and Lite-On:

- Delta (OW, PT NT$2,500, based on 30x 2027/2028 P/E) should maintain its dominant position in the industry with its integrated power/thermal related solutions while fuel cells could add incremental growth opportunities in the longer term. We think a 30x P/E target multiple is justified by our estimated 63% earnings CAGR in 2025-28, driven by a 44% revenue CAGR and operating margin expansion to 22.1% in 2028 vs 15.1% in 2025,
- Lite-On (EW, PT NT$315, based on 17x 2027/2028 P/E), the second-largest player in AI power-related products, should also ride the tailwinds despite its slower progress in HVDC power racks and liquid cooling solutions. We think a 17x P/E target multiple is justified by our estimated 47% earnings CAGR in 2025-28 (29% revenue CAGR and 6.8ppt OPM expansion to 16.8% in 2028).

MORGAN STANLEY TAIWAN LIMITED+
Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

MORGAN STANLEY MUFG SECURITIES CO., LTD.+
Kazuo Yoshikawa, CFA
Equity Analyst
Kazuo.Yoshikawa@morganstanleymufg.com +81 3 6836-8408

MORGAN STANLEY ASIA LIMITED+
Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

MORGAN STANLEY TAIWAN LIMITED+
Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY MUFG SECURITIES CO., LTD.+
Hidetaka Suzuki
Research Associate
Hidetaka.Suzuki@morganstanleymufg.com +81 3 6836-8402

GREATER CHINA TECHNOLOGY HARDWARE

Asia Pacific Industry View	**In-Line**

WHAT'S CHANGED

Delta Electronics Inc. (2308.TW)	**From**	**To**
Price Target	NT$2,700.00	**NT$2,500.00**
Lite-On Technology (2301.TW)	**From**	**To**
Price Target	NT$176.00	**NT$315.00**

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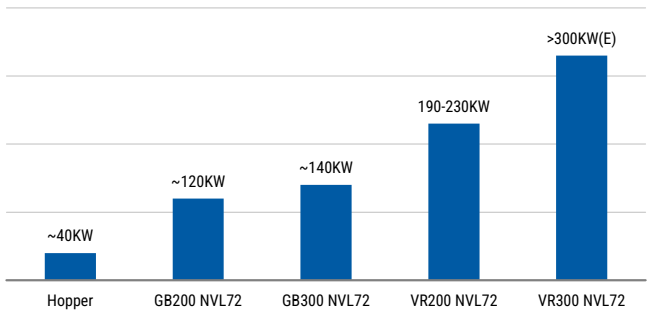
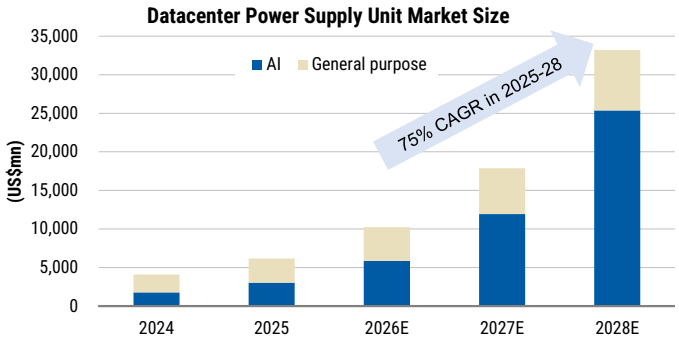
The Story in Charts

Exhibit 1: AI servers continuing to see higher TDP...



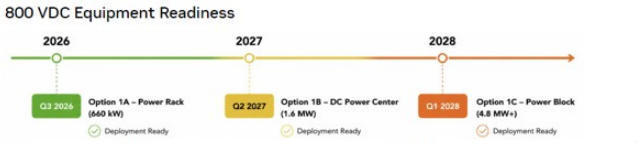
Source: Company data, Morgan Stanley Research estimates

Exhibit 2: ...which drives data center power supply demand



Source: Company data, Morgan Stanley Research estimates

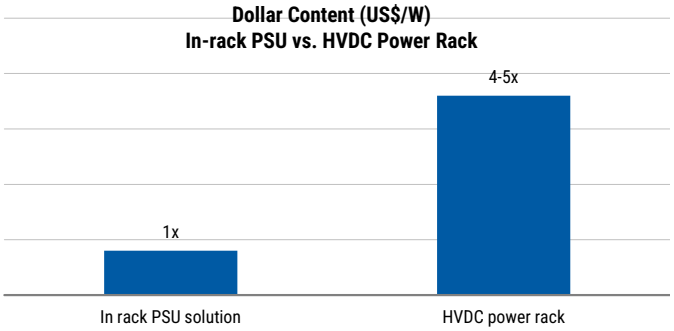
Exhibit 3: HVDC to see early adoption from late 2026...



Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

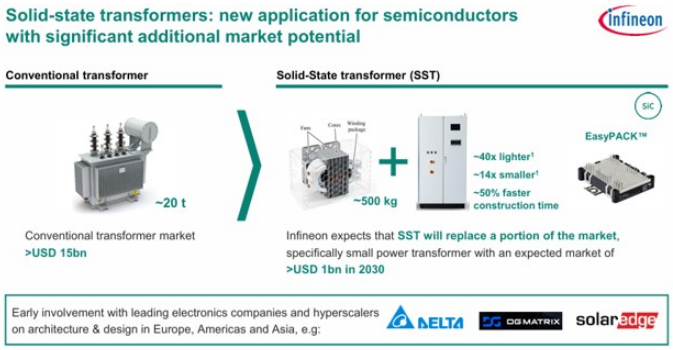
Source: Nvidia

Exhibit 4: ...with a much higher dollar content



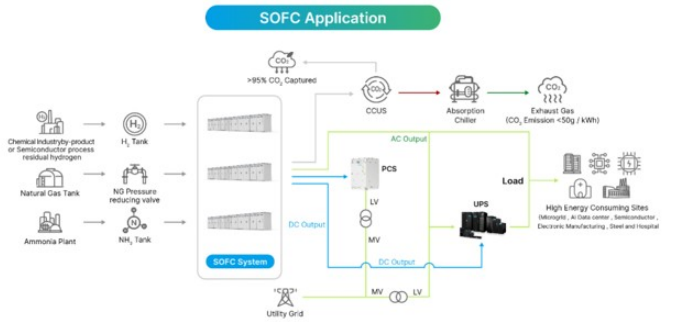
Source: Morgan Stanley Research

Exhibit 5: SST is also in the pipeline for more centralized power delivery



Source: Infineon

Exhibit 6: Fuel cells are a good option for better resilience in power architecture



Source: Delta

Executive Summary

AI cluster performance evolving to fulfill demand

The rapid adoption of artificial intelligence is driving a significant increase in global data center compute demand. Advances in foundation models, reasoning capabilities, agentic AI, and enterprise AI deployment are increasing the computational resources required for both training and inference. To support this growth, hyperscalers and model developers continue to deploy larger and more powerful AI clusters, driving demand for greater compute, memory, and networking capacity.

As AI infrastructure scales, cluster performance is increasingly determined by the efficiency of the entire system rather than the capabilities of individual processors alone. Compute, networking, cooling, and power infrastructure must all evolve together to support higher rack densities and larger deployments. While semiconductor innovation remains critical, supporting infrastructure is becoming an equally important factor in enabling future AI growth.

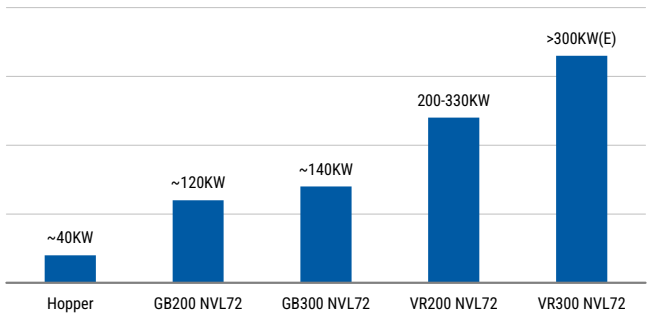
AI cluster performance is fundamentally a power problem

As AI clusters continue to scale in size and performance, power consumption is becoming one of the most important constraints on future infrastructure deployment. The power challenge is no longer limited to GPUs alone, as modern AI clusters comprise tightly integrated compute, networking, memory, storage, and cooling systems, all of which require increasing amounts of power as performance requirements rise. Consequently, the next phase of AI infrastructure investment should be viewed not only as a semiconductor upgrade cycle but also as a broad-based power infrastructure upgrade cycle.

Server power consumption has increased significantly across successive AI hardware generations. Nvidia's architecture roadmap illustrates the industry's migration from relatively modest server-level power budgets under the Ampere and Hopper generations toward rack-scale systems under Blackwell and future Rubin architectures. Estimated rack-level power consumption has increased from roughly 40kW for Hopper systems to approximately 120kW for GB200 NVL72 and 140kW for GB300 NVL72, with future Vera Rubin systems expected to approach 200-300kW per rack with current Vera Rubin systems reaching up to 330kW per rack while future Vera Rubin generations are expected to surpass these levels. As compute density continues to increase, power delivery infrastructure is becoming a critical enabler of system performance.

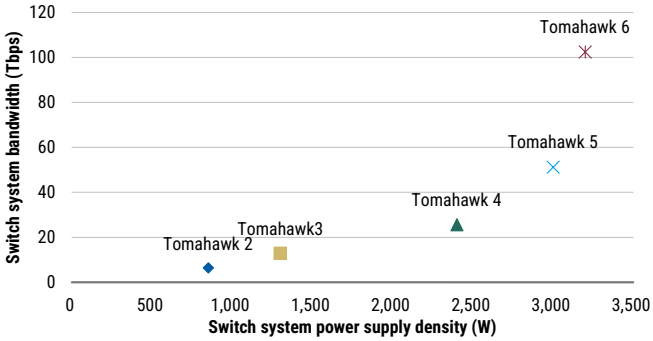
The increase in power consumption is not limited to compute hardware. Networking systems are also becoming increasingly power intensive as switch bandwidth scales from 100G architectures toward 800G and 1.6T generations. Despite substantial process-node improvements, switch ASIC power consumption has continued to rise due to increasing complexity and higher SerDes speeds. As a result, networking equipment now requires significantly larger power budgets than previous generations, creating additional demand for higher-capacity power supplies throughout the data center.

Exhibit 7: TDP roadmap for Nvidia racks



Source: Morgan Stanley Research

Exhibit 8: Power supply density for Tomahawks



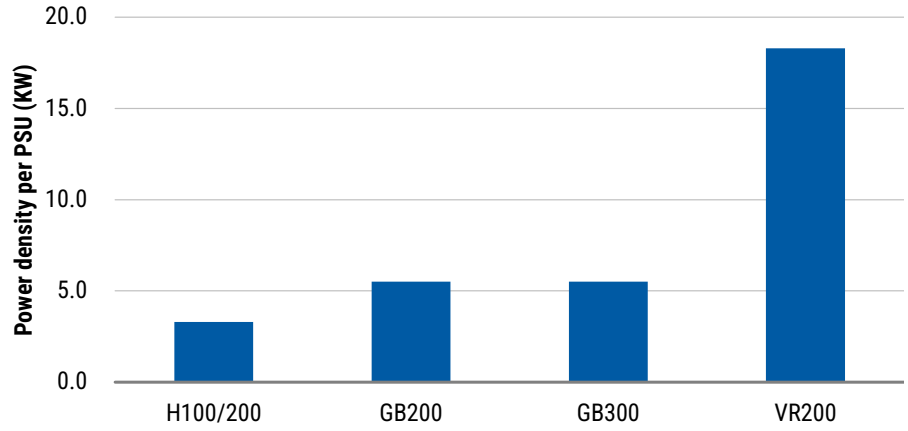
Source: Morgan Stanley Research

The growth in power consumption is driving a corresponding upgrade cycle for power-supply units (PSUs). Earlier AI server generations typically relied on multiple 3-3.3KW power supplies installed directly within individual server chassis. As the industry transitions toward rack-scale architectures, power conversion is increasingly centralized into dedicated power shelves capable of supporting substantially larger power loads. In Blackwell systems, power shelves utilize 5.5KW power supplies, while future Rubin-generation platforms are expected to increase PSU ratings to approximately 18KW, representing a significant increase in both PSU content and power density.

Delivering higher power within a similar footprint creates significant engineering challenges. Increasing PSU power density requires advances across multiple areas, including higher-performance power semiconductors such as silicon carbide (SiC) and gallium nitride (GaN), more efficient power-conversion topologies, smaller and more capable magnetic components, improved thermal-management solutions, and greater system integration. As output power increases, thermal constraints become increasingly important, making heat dissipation, airflow optimization, and mechanical design critical determinants of PSU performance.

Consequently, on top of the volume driven growth, the power-supply industry is benefiting from the technology driven dynamics, where efficiency, power density, and thermal performance are becoming key competitive differentiators. As AI rack power requirements continue to increase, PSU innovation will become an increasingly important component of the broader AI infrastructure investment cycle.

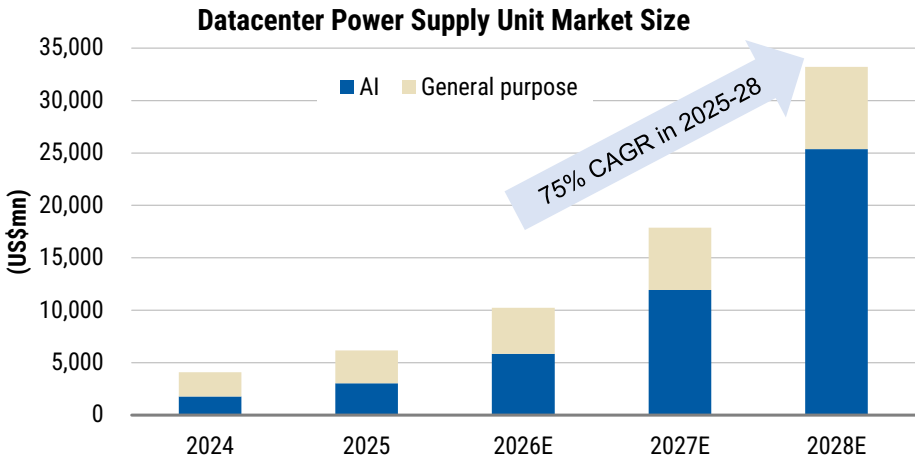
Exhibit 9: Power density per PSU trend in Nvidia server systems



Source: Nvidia, Morgan Stanley Research

Overall, we estimate the global datacenter power supply market will grow at a three-year CAGR of 75% to reach US$33bn in 2028 from US$6bn in 2025, driven mainly by server shipment growth, increasing power consumption per rack, and the adoption of HVDC power racks from 2027.

Exhibit 10: Global datacenter power supply market forecast



Source: Company data, Morgan Stanley Research estimates

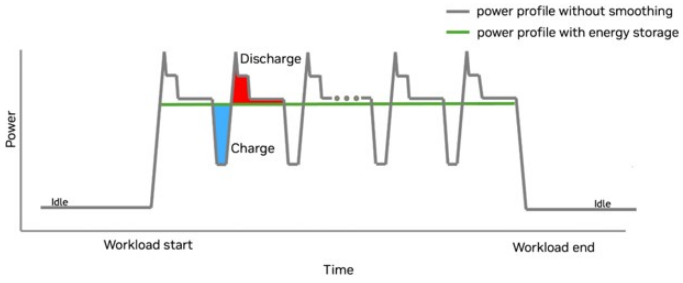
Complementary hardware for more resilient power delivery

As AI clusters become more power intensive, maintaining stable and resilient power delivery is emerging as a critical design priority. Traditional centralized power architectures are increasingly being complemented by localized energy-storage and power-conditioning technologies that can respond rapidly to dynamic workload demands.

Battery Backup Units (BBUs) represent one of the most important additions to modern AI power architectures. Positioned close to the computing workload, BBUs provide localized energy storage capable of responding significantly faster than facility-level backup systems. Their role extends beyond conventional backup power functionality. BBUs can

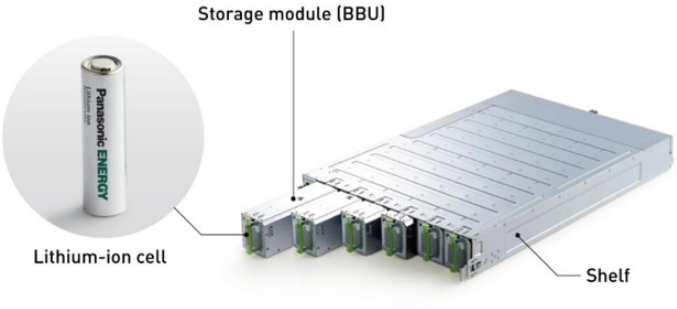
absorb transient fluctuations, smooth power ramps, and improve overall rack-level power stability. By providing local buffering capacity, BBUs help reduce the burden on upstream power distribution infrastructure while improving overall system resilience.

Exhibit 11: Concept of peak shaving for AI server power loads



Source: Nvidia

Exhibit 12: BBU



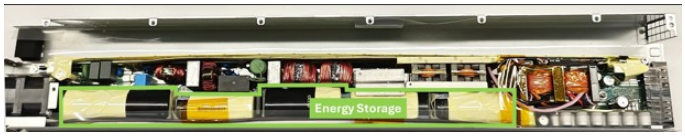
Source: Panasonic

Capacitor Backup Units (CBUs) address an even shorter-duration segment of the power stability challenge. Utilizing supercapacitors and other high-power-density storage technologies, CBUs can deliver and absorb power within milliseconds, making them particularly effective for managing rapid GPU power spikes. These systems act as shock absorbers within the electrical architecture, preventing sudden load changes from propagating throughout the broader power network. As AI rack power densities continue increasing, capacitor-based energy storage is becoming an attractive solution for improving power quality without requiring substantial oversizing of facility infrastructure.

Exhibit 13: CBU

Source: Panasonic

Exhibit 14: GB300 power shelf with integrated capacitors



Source: Nvidia

Together, BBUs and CBUs provide complementary benefits. BBUs are designed to handle longer-duration events, while CBUs are optimized for extremely fast transient response. The integration of these technologies enables operators to improve power quality, reduce stranded infrastructure capacity, enhance system reliability, and increase effective utilization of expensive electrical assets. As rack densities continue to rise, localized energy-storage solutions are likely to become a standard component of AI data center design rather than an optional enhancement.

The adoption of BBU and CBU architectures may also enable future increases in rack density. As operators attempt to push toward several hundred kilowatts per rack, localized power-buffering solutions can help mitigate transient peaks and reduce stress on grid-facing infrastructure. This allows operators to deploy higher-density computing resources without proportionately increasing upstream electrical capacity.

Exhibit 15: Comparison of CBU and BBU

Dimension	CBU	BBU
Primary storage medium	EDLC supercapacitors, LICs or high-capacitance	Typically lithium-ion battery cells managed by a BMS
Primary strength	High power, fast charge/discharge and frequent	Higher stored energy and longer sustained
Main function	Transient smoothing, peak shaving, voltage	Outage backup, source-transfer bridging and orderly workload
Response profile	Well suited to millisecond-class load support and	Fast switchover, but generally optimized for sustained support rather
Typical duration	Milliseconds to seconds; selected systems can extend toward tens of seconds or more	Commonly tens of seconds to several minutes, depending on battery and rack design
Power versus energy density	Higher power capability but lower energy storage per unit volume than a battery	Higher energy density, making it more economical for longer events
Cycle life	Eaton specifies more than one million cycles and up to 20 years for certain supercapacitor modules	Battery life is more sensitive to depth of discharge, temperature and repeated cycling; BMS supervision is therefore essential.
State management	Requires cell balancing, voltage/temperature monitoring and state-of-health tracking, but does not use a conventional battery fuel	Requires BMS functions including charge control, state of charge, state of health and cell protection
Best-fit use case	Repetitive AI power spikes, inrush events, PSU load leveling and bus stabilization	Power continuity through outages, brownouts or generator/source-transfer intervals

Source: Morgan Stanley Research

HVDC emerging as a necessity, with SST in the pipeline

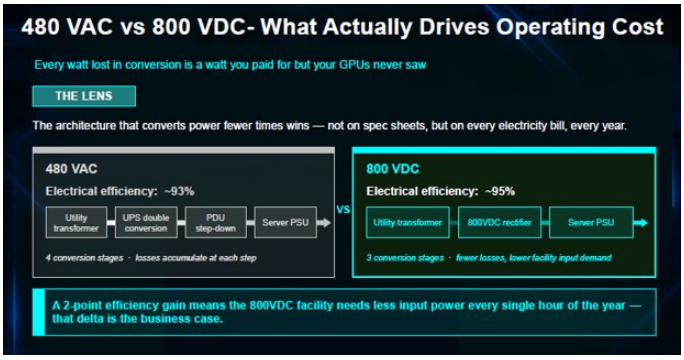
800VDC and ±400VDC progressing simultaneously

The rapid increase in AI rack power density is creating growing challenges for conventional power-distribution architectures. As rack-level power consumption rises from tens of kilowatts to several hundreds of KWs, electrical losses, cable size, thermal management requirements, and overall system complexity increase significantly. Under traditional low-voltage power architectures, delivering larger amounts of power requires proportionally higher current, which increases resistive losses and places greater stress on cables, connectors, busbars, and power-conversion equipment.

At the same time, every power-conversion stage introduces efficiency losses and generates additional heat. As AI deployments scale into multi-megawatt and eventually gigawatt-level campuses, even small efficiency improvements can translate into meaningful reductions in operating costs, cooling requirements, and infrastructure investment. Consequently, the industry is increasingly focused on power architectures that can reduce conversion stages, lower current levels, and improve overall power-delivery efficiency.

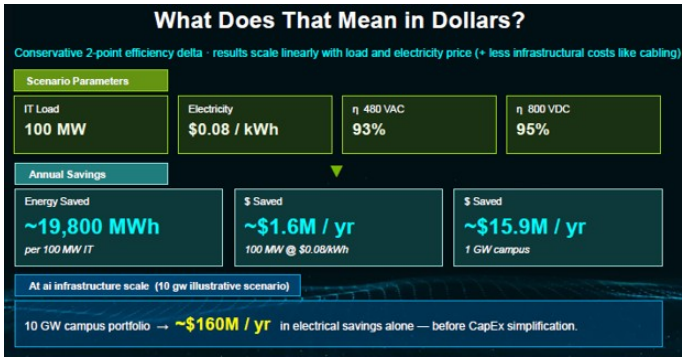
Against this backdrop, high-voltage direct current (HVDC) is emerging as one of the most important architectural upgrades for next-generation AI infrastructure. By increasing distribution voltage, HVDC allows the same amount of power to be delivered at a lower current, reducing conduction losses and improving system efficiency. Higher-voltage architectures also improve scalability for ultra-high-density rack deployments by alleviating some of the physical and thermal constraints associated with conventional power-distribution systems.

Exhibit 16: Better power efficiency for 800VDC architecture



Source: Delta

Exhibit 17: Better efficiency leads to cost savings



Source: Delta

Nvidia's roadmap illustrates the industry's gradual migration toward higher-voltage power architectures. Earlier AI server generations primarily utilized conventional power-delivery schemes centered around rack-level 48V distribution. As rack power requirements increased with Blackwell-generation systems, industry participants began evaluating higher-voltage architectures capable of supporting substantially greater power density. Looking ahead, the industry is increasingly converging around 800VDC as a potential next-generation standard for large-scale AI deployments, particularly as rack-level power consumption approaches several hundred kilowatts. The transition reflects the need to improve power-distribution efficiency while supporting the continued growth of AI computing infrastructure.

We believe that the adoption of the 800VDC power rack could start from late 4Q26, though it should be small scope deployment through 2027, since the in-rack power shelf is the reference design in the Vera Rubin NVL72 rack. Broader adoption should happen more likely from late 2027 and 2028, as the Vera Rubin Ultra rack further pushes up the power consumption.

Exhibit 18: HVDC deployment roadmap

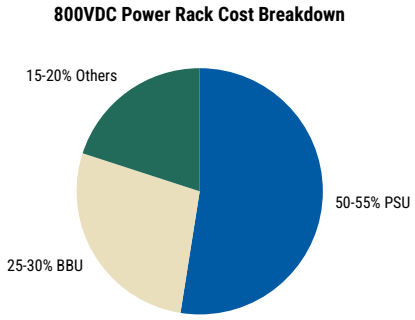
Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

Source: Nvidia

The 800VDC power rack BOM is primarily driven by the PSU, which accounts for 50–55% of total cost, followed by the BBU at 25–30%, with other components comprising the remaining 15–20%. This cost structure reflects the broader functionality integrated into an HVDC power rack, combining power conversion, backup power and supporting components within a centralized architecture.

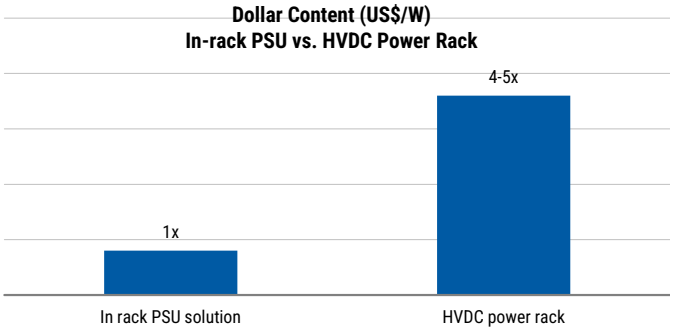
For power supply players, one of the most direct implications is that an HVDC power rack carries around 4–5x higher dollar content per watt compared with a conventional in-rack PSU solution, since the former integrates more components into a robust system and power supply players usually ship the entire rack instead of just the components.

Exhibit 19: BOM cost breakdown for an 800VDC power rack



Source: Morgan Stanley Research

Exhibit 20: Higher dollar content for HVDC power racks



Source: Morgan Stanley Research

However, the industry has not yet fully converged on a single architecture. While Nvidia's roadmap increasingly points toward 800VDC, several hyperscalers and members of the Open Compute Project (OCP) ecosystem have expressed interest in ±400VDC architectures. Both approaches deliver similar total voltage levels and pursue many of the same objectives, including lower current, higher efficiency, and improved scalability. The debate largely centers on implementation trade-offs rather than end goals.

Supporters of 800VDC generally highlight the advantages of a simpler single-bus architecture, potentially lower system complexity, and closer alignment with existing industrial high-voltage DC equipment. The architecture avoids the need to maintain balanced positive and negative voltage rails and may simplify certain aspects of power conversion and system integration. These characteristics make 800VDC particularly attractive in environments where operational simplicity and interoperability with existing electrical infrastructure are key priorities.

Advocates of ±400VDC, meanwhile, emphasize the advantages of a bipolar architecture. By creating positive and negative voltage rails around a neutral reference point, ±400VDC can potentially reduce insulation requirements, simplify certain fault-management scenarios, and provide improved flexibility for load balancing and future system expansion. Many hyperscalers also view the bipolar approach as being more compatible with long-term visions for highly distributed power systems incorporating energy storage, renewable energy, and microgrid functionality.

Importantly, the ultimate beneficiaries may be largely independent of which standard gains wider adoption. With the exception of certain power-semiconductor and protection-system requirements, many component suppliers across the power ecosystem remain

broadly agnostic between 800VDC and ±400VDC implementations. Higher-voltage architectures in either form require similar upgrades in power conversion, power management, cooling, connectors, magnetics, and infrastructure equipment. As a result, the industry's migration toward HVDC matters significantly more than the eventual selection of any single voltage standard.

Exhibit 21: Comparison of 800VDC and ±400VDC

Metric	±400VDC	800VDC
Voltage to ground	400V relative to ground	800V relative to ground
Voltage differential	800V	800V
Power distribution efficiency	High	Slightly higher
Scalability	Good	Excellent
Insulation requirements	Lower	Higher
Maintenance safety	Better	More challenging
System architecture simplicity	Dual-rail system	Single-bus system
Grounding Complexity	More complicated	Simpler
Electrical footprint	Small	Smaller
Ability to reduce conversion stages	Moderate	Highest
Long-distance distribution capability	Excellent	Excellent
Maturity of ecosystem	Better today	Early but rapidly developing
Industry players	OCP, hyperscalers	Nvidia
Long term industry direction	Transitional architecture	Likely end-state architecture

Source: Morgan Stanley Research

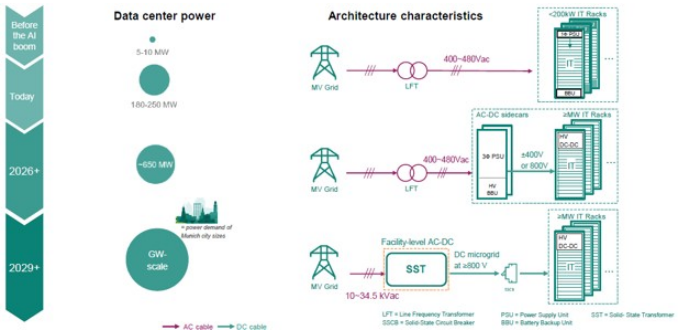
Solid-state transformers to follow after 800VDC/±400VDC matures

Looking further ahead, solid-state transformers (SSTs) represent a potential next-generation evolution of power infrastructure. Unlike traditional transformers that operate primarily as passive devices, SSTs leverage power electronics to provide voltage conversion, power conditioning, and bidirectional energy management within a single platform. This enables greater controllability and flexibility while supporting increasingly digitalized power networks.

Although SSTs remain at an early stage of commercialization and face challenges related to economics, reliability, and large-scale deployment, the technology aligns well with long-term data center trends. As AI campuses evolve into utility-scale power consumers incorporating energy storage, distributed generation, and advanced power management, SSTs could eventually become a key enabling technology.

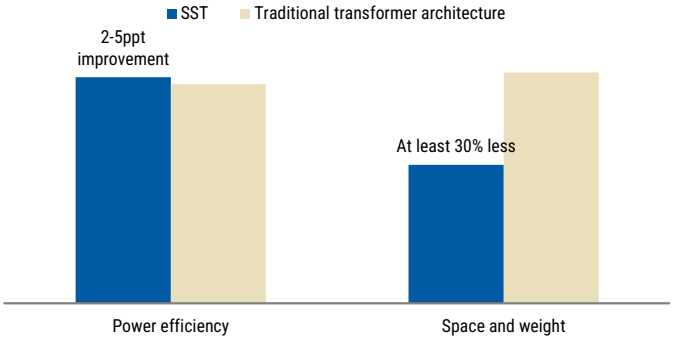
Therefore, while HVDC is expected to drive the next phase of power architecture upgrades, SSTs represent a longer-term technology roadmap that could further transform how future data centers distribute and manage electricity.

Exhibit 22: SST enabling more efficient power delivery architecture



Source: Infineon

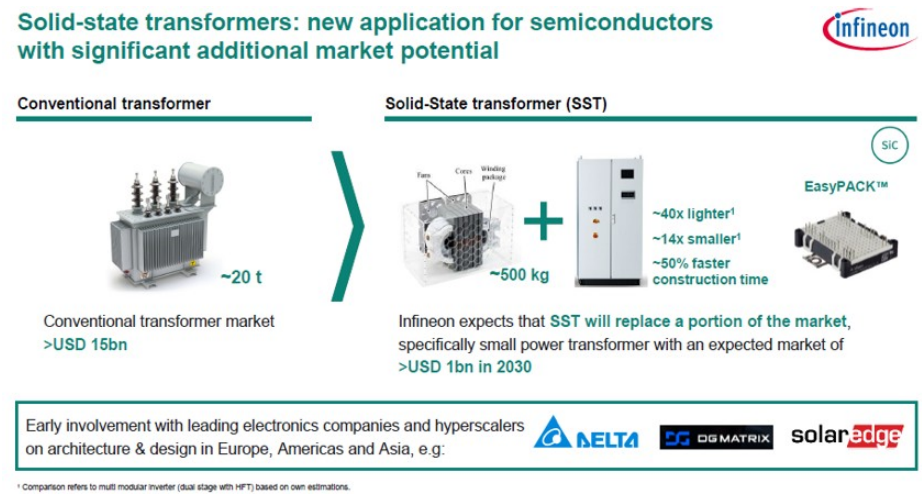
Exhibit 23: SST offers better power efficiency with at least 30% less space/weight



Source: Morgan Stanley Research

According to Infineon, SST could represent an emerging >US$1bn market opportunity by 2030, as the technology begins to replace a portion of the >US$15bn conventional transformer market, particularly in smaller-power applications. While SST penetration is expected to remain relatively modest versus the overall transformer market, the opportunity is significant for relevant players in the longer term.

Exhibit 24: SST to see initial adoption in 2030

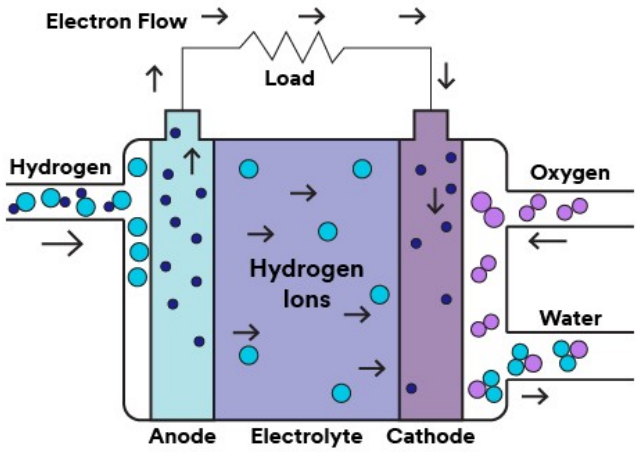


Source: Infineon

Fuel cells for better flexibility and resilience

Fuel cells generate electricity through an electrochemical reaction rather than combustion. Hydrogen, natural gas, biogas, or other fuels are supplied to the fuel cell stack, where chemical energy is converted directly into electricity, heat, and in some cases water. Unlike batteries, which store energy, fuel cells continuously generate power as long as fuel is supplied.

Exhibit 25: Fuel cell diagram



Source: 3M

Fuel cells can offer several benefits, including:

- **High energy efficiency:** Direct electrochemical conversion enables higher efficiency than conventional combustion-based power generation, reducing fuel consumption and operating costs.
- **Low carbon emissions**: Fuel cells produce minimal direct emissions and can support long-term decarbonization goals when powered by low-carbon fuels such as hydrogen.
- **Grid independence and energy security:** Provides reliable onsite power generation, reducing dependence on grid availability and improving energy resilience.
- **Long-duration energy storage enabler:** Hydrogen-based fuel-cell systems can complement renewable energy by storing energy over extended periods, supporting long-duration and seasonal energy storage.
- **Fuel flexibility:** Certain fuel-cell technologies can operate on hydrogen, natural gas, biogas, ammonia-derived hydrogen, and other hydrogen-rich fuels, allowing deployment using existing fuel infrastructure.
- **Low maintenance and quiet operation:** Limited moving parts reduce maintenance requirements while providing quieter operation than conventional combustion-based generators.
- **Scalable deployment:** Can be deployed across a wide range of power levels, from distributed backup systems to utility-scale installations, making fuel cells suitable for large AI data center campuses

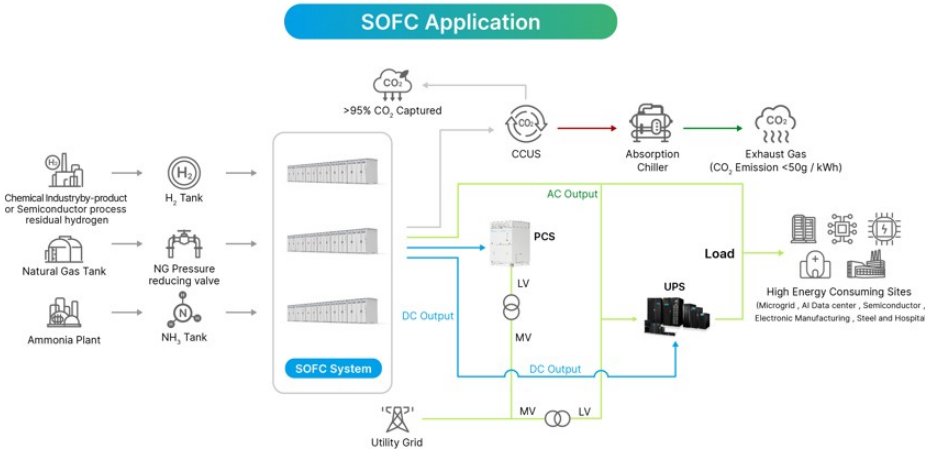
However, the following challenging are hurdles for wider deployment:

- **High system cost:** Fuel cells remain more expensive than conventional power-generation technologies, limiting large-scale adoption.
- **Limited hydrogen infrastructure:** Hydrogen production, transportation, storage, and refueling networks remain underdeveloped in most regions.
- **Durability requirements:** Long-term reliability and stack lifetime remain key challenges, particularly for large-scale and mission-critical deployments.
- **Regulatory and safety considerations:** Hydrogen handling requires specialized infrastructure and compliance with stringent safety regulations, increasing deployment complexity.
- **Lack of standardization:** Multiple fuel-cell technologies compete across different applications, creating uncertainty around long-term technology standards and adoption paths.

Fuel cells have the potential to fundamentally reshape future data center power architectures. Traditionally, data centers rely on grid electricity as the primary power source and diesel generators for backup power. However, as AI-driven workloads significantly increase power demand and power availability becomes a key deployment constraint, hyperscalers are increasingly evaluating fuel cells not only as a backup-power solution but also as a potential primary power source. This could accelerate the transition from a grid-centric architecture toward one incorporating distributed onsite generation, enabling data center deployment in regions where utility power capacity is limited.

Over the longer term, fuel cells could become a core component of AI data center microgrids that integrate renewable generation, battery storage, hydrogen storage, and intelligent energy-management systems. Such an architecture would improve scalability, resilience, and energy independence while reducing reliance on centralized utility infrastructure. In addition, because fuel cells inherently generate DC power, they align well with the industry's migration toward HVDC-based power distribution architectures. This could simplify power delivery by reducing conversion stages between generation and IT equipment, improving overall system efficiency and supporting next-generation high-density AI deployment

Exhibit 26: Fuel cell in datacenter power delivery architecture



Source: Delta

Stock picks

Delta (2308.TW, OW, PT NT$2,500)

Dominant supplier in the AI power supply offerings: Delta has been one of the most competitive players in power related components and systems, as its offerings spanning from the core power supply units, the complementary BBU/CBU for AI servers, the highly anticipated HVDC power rack, grey space power facilities to air/cooling solutions. We estimate that it has ~60% market share in the datacenter power supply market and is among the leading suppliers of the 800VDC/±400VDC power rack to Nvidia/hyperscalers in 2H26 (wider adoption likely from late 2027). Longer term, when the HVDC becomes more mature, the centralized power delivery architecture could migrate into SSTs, where Delta already has small volume shipment now. With the solid technology, close customer relationships and strong execution, we believe Delta should stay competitive even amid the fusion of the white space and grey space.

Extending from power delivery to power source: In addition to the offerings in the power delivery, Delta signed a licensing agreement with Ceres Power (CRW.GB) for its solid oxide fuel cell (SOFC) technology in early 2024. This could extend its business scope into the power source area. which we believe could add more value to its customers as it could help mitigate the power constraints for datacenter build and enhance the operational efficiency through active micro grid management.

Solid near term momentum: Despite the slower revenue growth owing to the component supply tightness in 3Q26, we expects its 4Q26 revenue to grow by double digit % QoQ on alleviated component supply and solid demand. 3Q26 GM should be flat to slightly up QoQ and 4Q26 might see further expansion on the tailwinds price hikes.

OW with PT of NT$2,500: We see power/thermal as one of the major secular AI related themes in the downstream tech space and Delta is well positioned to benefit, potentially driving a 63% earnings CAGR in 2025-28, in our estimate. We thus assume the coverage of Delta with an OW rating and the PT of NT$2,500, based on 30x 2027/2028 P/E. Our target multiple implies a PEG of 0.5 vs. the average PEG of 1.2 in 2020-25, when the stock was trading at an average 12M forward P/E of 22x with the earnings CAGR of 19%.

Lite-On (2301.TW, EW, PT NT$315)

Rising tide lifting all boats: Lite-On is the second largest power supply vendor globally and has been engaging with hyperscalers regarding the ASIC and Nvidia GPU servers. Looking ahead, it expects to expand its share in a couple of hyperscalers from late 2026. It should have over 50% of the total revenue generated from the cloud segment for PSUs, BBUs, and to a lesser extent, other mechanical components like liquid cooling, chassis/rack, etc. Moreover, it also has the HVDC power rack solutions ready and is qualifying with potential customers. The cloud shall remain as the major growth driver in the coming years.

Aspiration in the optical communication space: Lite-On has always had an opto business (10-20% of total revenue in 2026) in both visible and invisible light applications. Post the strategic investment at Dense Light (21% stake), an InP epi wafer and chip process manufacturer, it plans to leverage its expertise in packaging and provides the optical

modules and engine for the data interconnects. While the end applications looks promising, we think it might take some time to see more meaningful revenue contribution in Lite-On.

Cloud momentum extending into the coming quarters: We expect its 4Q26 revenue to grow by double digit QoQ driven mainly by the could segment and GM should normalize at ~25% in 2H26 vs. 27.2% in 2Q26 on some inventory write-off reversal and high margin product shipment delayed from the previous quarter.

EW with PT of NT$315: We think Lite-On will grow along with the AI capex cycle, driven by its power-related offerings (47% earnings CAGR in 2025-28, in our estimate), though execution and progress with new customers is worth monitoring. We assume coverage of Lite-On with an EW rating and a PT of NT$315, based on 17x 2027/2028 P/E. Our target multiple implies a PEG of 0.4 vs. the average PEG of 1.6 in 2020-25, when the stock was trading at an average 12M forward P/E of 14x on an earnings CAGR of 9%.

Panasonic Holdings (EW, PT ¥4,700, covered by Kazuo Yoshikawa)

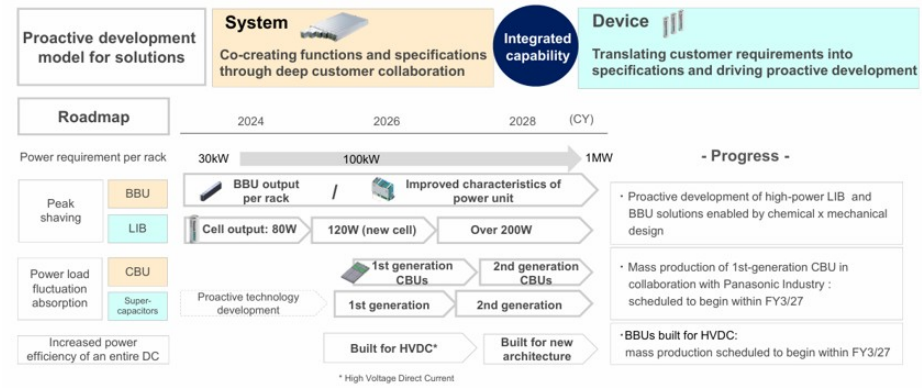
Panasonic has an approximately 80% share of the BBU market and is well positioned to benefit from rising AI data center power demand: The company expects data center energy storage revenue to grow from ¥322bn in FY3/26 to approximately ¥550bn in FY3/27 and around ¥1trn in FY3/29. Importantly, all of the business underpinning its FY3/29 revenue target has already been awarded, meaning agreements have been reached with customers on product development and orders.

BBU: Panasonic has begun limited mass production of new high-capacity BBU cells above 200W, with products adopted by a major GPU vendor and ramping from June-July. Discussions for next-generation server platforms are in the final stages, and the company expects to maintain its approximately 80% market share. It also sees scope for higher pricing as rack-level power consumption rises.

The company is also close to securing awards for BBUs used in dedicated power racks. As its medium-term plan includes only already-awarded business, additional design wins could provide upside.

CBU: CBUs are emerging as a new growth opportunity as GPU and ASIC power consumption increases. Panasonic is working with power-supply vendors to propose integrated BBU+CBU solutions and is receiving inquiries from US and Chinese hyperscalers, including customers that were not previously BBU users.

Exhibit 27: Panasonic Holdings: Strengthen proposal and development capabilities for the next generation



Source: Company data

Risks to our positive view

Risks to our positive view on power-related demand include:

- Component supply tightness leading to lower shipment of products
- Shortfall in hardware shipments, yield issues, electricity availability, etc. slowing down datacenter builds
- Lower-than-expected penetration of new power delivery architectures like 800VDC/±400VDC, SST, etc.
- More intense competition leading to margin erosion among most players
- Hyperscalers prove unable to finance aggressive capex through internal and external funding.

AI Cluster Performance Evolving to Fulfill Demand

AI adoption driving exponential growth in compute demand

AI is becoming a mainstream computing workload

Over the past three years, generative AI has evolved from a research application into a core technology platform. Consumer applications such as ChatGPT, Claude, Gemini, and Copilot now serve hundreds of millions of users globally, while enterprises are increasingly embedding AI into software development, customer service, document analysis, enterprise search, healthcare, financial research, and industrial automation.

Historically, software demand scaled roughly with the number of users. AI demand is fundamentally different because every user interaction consumes computational resources in real time. Unlike traditional cloud applications, where the bulk of computation occurs during software development, AI applications continuously consume compute whenever a model generates a response. Consequently, rising AI adoption directly translates into higher demand for data center compute resources.

Model complexity continues to increase

The AI industry has historically improved model performance through scaling laws: larger models trained on larger datasets using more compute generally produce better outcomes. While efficiency improvements continue, leading AI developers remain committed to scaling model capabilities.

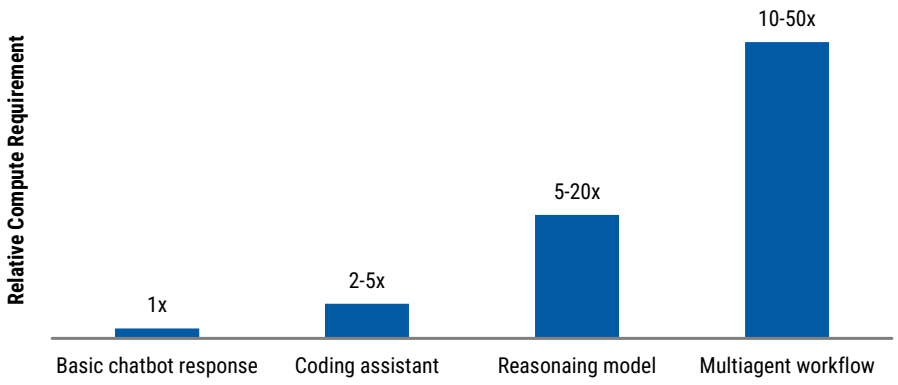
More importantly, the industry is moving beyond conventional language models toward significantly more compute-intensive architectures.

Reasoning models

Traditional LLMs generate answers directly from a prompt. New reasoning models perform multiple internal computational steps before producing a response. Instead of generating a few hundred output tokens, advanced reasoning systems may generate thousands or even tens of thousands of internal reasoning tokens while evaluating different solution paths. This dramatically increases inference workloads.

The industry is increasingly shifting from "parameter scaling" toward "test-time compute scaling," where additional compute is consumed during inference to improve answer quality. From an infrastructure perspective, this is highly significant because inference workloads occur continuously rather than once during model training.

Exhibit 28: Increasing demand for compute power from reasoning models



Source: Morgan Stanley Research estimates

Agentic AI

The next wave of AI applications involves autonomous agents capable of planning, executing, verifying, and refining tasks.

Instead of generating a single answer, an AI agent may search multiple information sources, run code, validate results, generate follow-up tasks, and produce a final response. Each step requires additional inference cycles.

A workflow that previously required one model invocation may now require dozens of model calls, substantially increasing compute consumption per user.

Multimodal AI

AI models are increasingly processing images, video, audio, sensor data, and text simultaneously.

Video understanding is particularly compute intensive because each second of video contains vastly more information than a text prompt.

As multimodal AI adoption grows, compute demand per query rises significantly, creating another structural driver of accelerator demand.

Training demand remains robust

Although inference is becoming the largest source of AI compute demand, training workloads continue to expand. Frontier model development increasingly requires clusters containing tens of thousands of accelerators operating simultaneously.

Training runs are becoming larger because developers are pursuing larger parameter counts, more training data, longer training durations, multimodal datasets and reinforcement learning and reasoning optimization

Consequently, each new generation of foundation models typically consumes substantially more compute than the previous generation.

Importantly, training infrastructure sets the baseline for future inference deployment. Every major model developer must maintain access to large-scale training clusters to remain competitive, supporting continued demand for high-end AI accelerators.

More compute requests to be fulfilled by better chip performance

Rising compute demand cannot be addressed simply by adding more servers. Modern AI workloads increasingly encounter bottlenecks related to memory bandwidth, interconnect performance, power consumption, and cooling. As a result, each new generation of AI hardware delivers improvements across multiple dimensions.

Compute performance

The latest AI accelerators provide several-fold performance improvements versus prior generations. For example, NVIDIA's Blackwell architecture was specifically designed to support both large-scale training and high-volume inference, with significant gains in AI throughput compared with previous Hopper-based systems. Performance improvements are necessary because reasoning and agentic workloads require substantially more operations per query.

Memory capacity and bandwidth

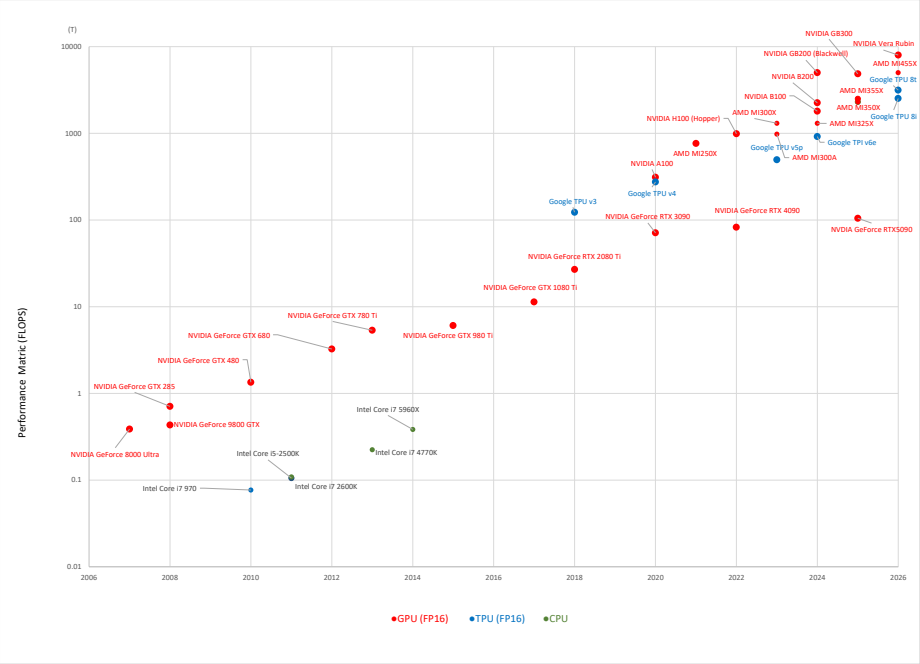
AI models are increasingly constrained by memory rather than raw compute, because longer models, longer context windows, reasoning models and agentic workflows all require more memory. As model complexity increases, modern high-end accelerators usually incorporate hundreds of gigabytes of High Bandwidth Memory (HBM), providing memory bandwidth measured in multiple terabytes per second.

Interconnect performance

As AI clusters scale from hundreds to thousands of accelerators, communication between processors becomes increasingly important. A large AI training cluster effectively behaves like a single massive computer. Consequently, networking technologies such as NVLink, InfiniBand, Ethernet, etc., have become critical enablers of AI scaling.

Many AI workloads can no longer be accelerated simply by increasing GPU performance; cluster interconnect bandwidth must also improve.

Exhibit 29: Compute power for HPCs continuing to evolve



Source: Company data, Morgan Stanley Research

AI Cluster Performance Is Fundamentally a Power Problem

As AI clusters become larger and more powerful, every component within the cluster must deliver higher throughput, lower latency, and greater bandwidth. Achieving these improvements inevitably requires more electrical power.

Therefore, the power challenge is no longer limited to GPUs. Modern AI clusters are highly integrated systems comprising compute servers, networking fabrics, memory subsystems, storage infrastructure, and cooling equipment. As cluster performance increases, power consumption rises across the entire infrastructure stack.

In our view, the next phase of AI infrastructure investment should be viewed not simply as a semiconductor upgrade cycle, but as a broad-based power consumption cycle driven by the need for ever-higher AI cluster performance.

Server systems seeing increasing TDP

Nvidia's AI-server power roadmap is best understood as a transition from chip-defined, air-cooled nodes to power-, cooling- and fabric-constrained rack-scale computers: the Ampere-era A100 SXM carried a disclosed 400W standard TDP—up to 500W in custom thermal designs—with 40/80GB HBM2/HBM2e and 600GB/s NVLink, while the eight-GPU, dual-AMD-EPYC DGX A100 was a conventional 6U server with a 6.5KW maximum input rating, meaning GPU silicon represented only roughly half of the node envelope once CPUs, NVSwitches, memory, NICs, drives, fans and conversion losses were included.

Hopper stepped the H100 SXM to a disclosed 700W TDP, 80GB HBM3, roughly 3TB/s memory bandwidth and 900GB/s fourth-generation NVLink, while H200 retained essentially the same Hopper compute architecture and power profile but substituted 141GB of faster HBM3e at 4.8TB/s—making H200 primarily a memory-capacity/bandwidth upgrade rather than a compute-TDP step-up—and Nvidia rates the eight-GPU, dual-Xeon DGX H100/H200 at 10.2KW maximum input, versus 6.5KW for DGX A100;

The architectural discontinuity arrives with GB200. Rather than treating an eight-GPU server as the scale-up unit and using InfiniBand/Ethernet primarily between nodes, Grace Blackwell NVL72 makes the entire rack the scale-up domain, integrating 18 one-U compute trays—each with two 72-core Grace Arm CPUs and four Blackwell GPUs—plus nine NVLink switch trays, management switches, , a 50–51V DC busbar, copper NVLink cartridges and liquid manifolds, so that 36 CPUs and 72 GPUs operate as one low-latency NVLink-connected system. Based on the same compute die, Nvidia upgraded the HBM content to further improve the performance of GB300 chip. Overall, the rack level power consumption is estimated at ~120KW for the GB200 NVL72 and ~140KW for the GB300 NVL72.

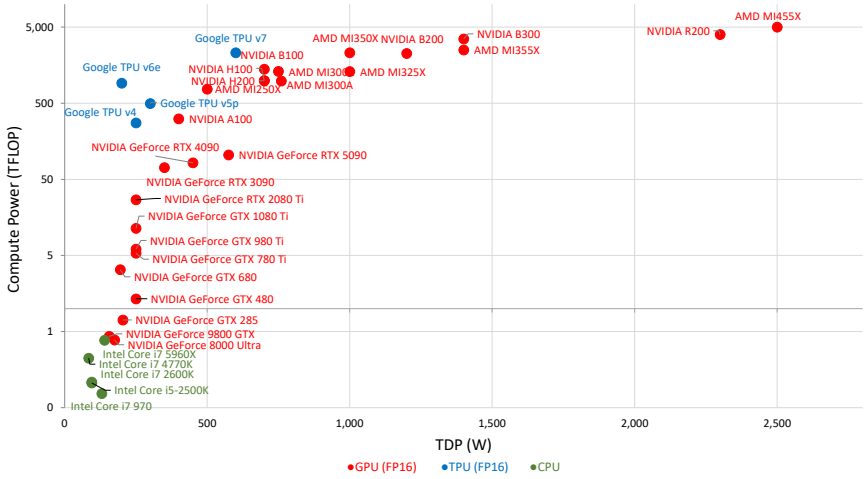
Vera Rubin—often called VR200/R200 in industry roadmaps—pushes the same rack-level philosophy further. Nvidia's disclosed architecture combines 72 dual-die Rubin GPUs with 36 new Vera CPUs, replacing Grace's 72 Arm Neoverse cores with 88 custom Olympus cores, raising CPU memory capacity and bandwidth to as much as 1.5TB LPDDR5X and 1.2TB/s, and doubling coherent CPU–GPU NVLink-C2C bandwidth to 1.8TB/s; Rubin itself moves from HBM3e to 288GB HBM4 at up to 22TB/s, introduces a third-

generation Transformer Engine and raises sixth-generation NVLink scale-up bandwidth to 3.6TB/s per GPU, while ConnectX-9 provides 1.6Tbps/GPU scale-out connectivity and BlueField-4 increasingly offloads storage, security and infrastructure processing.

Nvidia has not published a simple Rubin chip TDP in its primary datasheet, but our supply chain checks indicate roughly 1.8kW at an efficiency-oriented operating point and up to approximately 2.3kW at the performance mode, which is consistently corroborated by system vendors, with ASUS specifying 187kW Max-Q and 227kW Max-P for its fully liquid-cooled Vera Rubin NVL72.

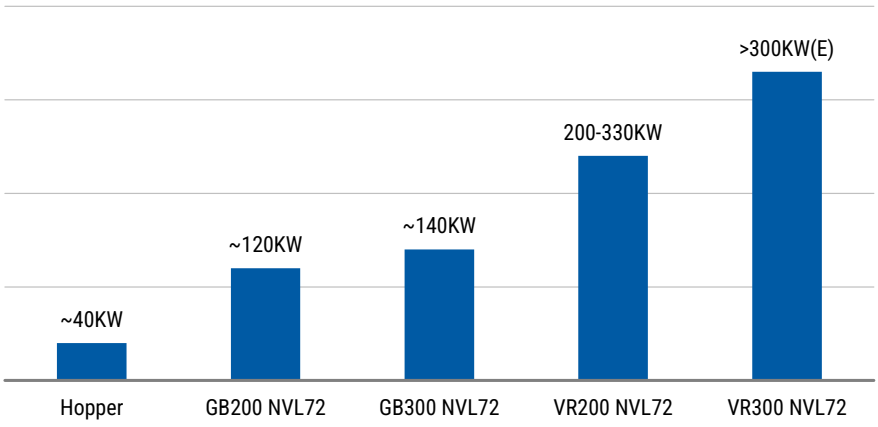
Looking ahead, we expect this trend to extend into the future along with the technology advancement and more critical workloads.

Exhibit 30: Compute power vs. TDP for HPCs



Source: Company data, Morgan Stanley Research

Exhibit 31: TDP roadmap for Nvidia racks



Source: Company data, Morgan Stanley Research

Networking consuming more power as well

The network switch system is another key devices within the datacenter and has been evolving over the past years, driven primarily by advances in the switch ASIC. From Tomahawk 1 (2014) to Tomahawk 6 (2025), the process node shrank from 28nm to 3nm, while aggregate switching bandwidth increased from 3.2 Tbps to 102.4 Tbps, representing a 32× increase. Over the same period, SerDes speed improved from 25 Gbps NRZ to 200 Gbps PAM4, enabling maximum port speeds to scale from 100G to 1.6T.

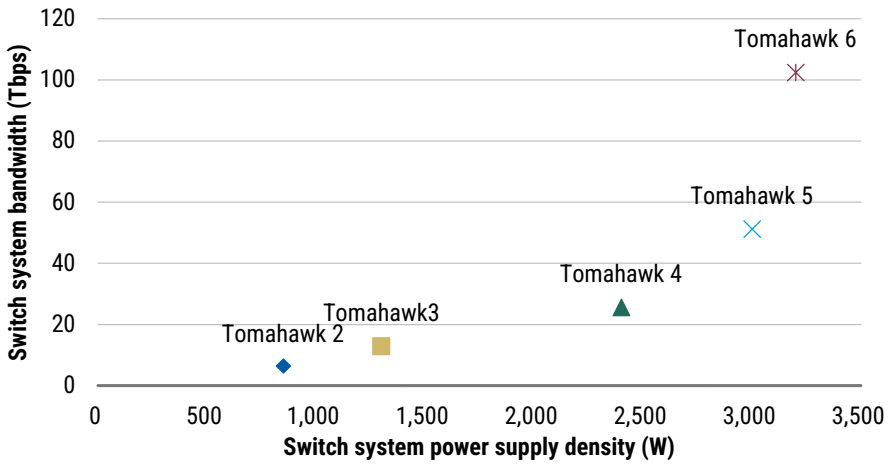
While process scaling significantly improved switching efficiency, it did not prevent ASIC power from rising. ASIC TDP increased from 150 W to 800 W, reflecting the growing complexity of higher radix switches and faster SerDes interfaces.

The impact of ASIC scaling is evident at the system level. As switch bandwidth increased from 100G-class systems to 800G and 1.6T platforms, power delivery requirements grew dramatically. Representative PSU ratings increased from 850W in Tomahawk 2 systems to 3,200W in Tomahawk 6 systems, nearly a 4× increase.

The rise in PSU power is driven not only by the higher ASIC TDP, but also by the growing power consumption of high-speed optics, increased cooling requirements, and larger overall system power budgets. As systems evolved from 100G to 1.6T ports, switch vendors were forced to add more fan redundancy, larger heatsinks, and in some cases larger chassis footprints. For example, Tomahawk 6 systems have expanded to 4U form factors with 7+1 fan redundancy to accommodate the thermal load associated with 102.4 Tbps switching capacity.

Alongside the increasing power consumption of the server racks, network switches are also witnessing higher power budgets as well.

Exhibit 32: Power supply density for different network switch generations



Source: Company data, Morgan Stanley Research

Power supply growing on both volume and specs

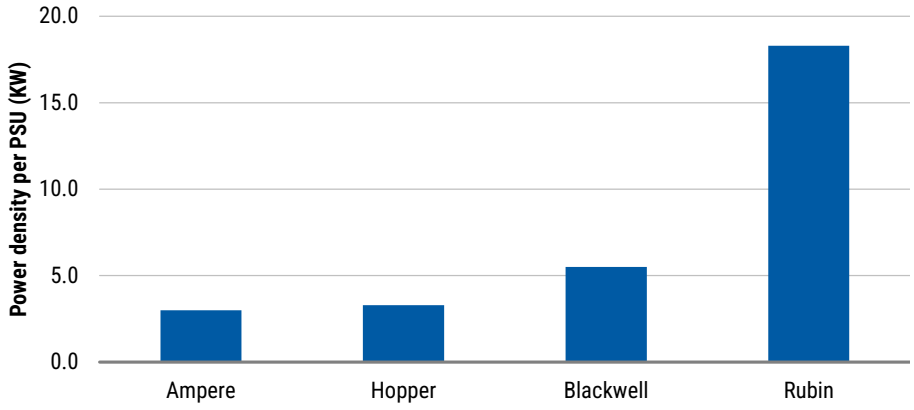
PSU spec upgrade trend

In response to the increasing power consumption of the AI servers, the power supply architecture has been migrating as well. in the A100 and H100/200 single system servers, 6 power supply units are placed directly inside the chassis, with each capable to deliver 3.0-3.3KW of power.

When Nvidia make a broad based move to the rack level server solution from the Blackwell generation, the power supply units were no longer placed directly within the chassis housing the HPCs that consume power; instead, power supply units are separated and grouped into power shelves. In the GB200/300 systems, there are 8 power shelves sitting on the same rack as the compute trays and switch trays, each enclosing six 5.5KW power supply units.

For the upcoming Vera Rubin rack, the power supply unit will be upgraded to 18.3KW of output to comprise a power shelf of 110KW, and each power shelf will take 3U of space on the rack vs. 1U in the Blackwell generation.

Exhibit 33: Power density for PSUs in Nvidia server systems



Source: Company data, Morgan Stanley Research

Challenges increasing the power density of PSUs

To come up with power supply unit with higher power density, manufacturers might need to solve the following potential hurdles:

Better power semiconductors (SiC and GaN)

The biggest improvement in PSU power density comes from replacing traditional silicon power chips with newer materials such as silicon carbide (SiC) and gallium nitride (GaN). These devices waste less energy during power conversion and can operate at much higher speeds. As a result, many supporting components inside the PSU—such as transformers and inductors—can be made significantly smaller. This allows PSU manufacturers to fit more output power into the same physical size while also improving efficiency. In many ways, SiC and GaN have become the foundation for the latest generation of high-density AI server power supplies.

More efficient power conversion designs

Modern PSUs use more sophisticated circuit designs, i.e., bridgeless PFC vs. conventional one, that reduce energy losses during power conversion. A conventional boost PFC includes a diode bridge and therefore incurs multiple semiconductor conduction drops. A bridgeless totem-pole PFC reduces the number of devices in the active current path and is now a preferred architecture for dense server PSUs. Instead of simply converting electricity from one voltage to another, newer architectures are designed to minimize wasted energy throughout the process. Higher efficiency means less heat is generated, which directly enables higher power density. This is particularly important in AI servers, where even small efficiency improvements can translate into meaningful reductions in heat generation and cooling requirements.

Smaller and smarter magnetic components

Transformers and inductors are often among the largest components inside a PSU. A major area of innovation is reducing their size without sacrificing performance. Manufacturers are increasingly using planar magnetics, where copper traces on circuit boards replace traditional wire windings. These designs are thinner, easier to cool, and more space-efficient. Some suppliers are also combining multiple magnetic functions into a single component, further reducing overall volume. As PSU power levels increase from 3 KW to 5.5 KW and beyond, shrinking magnetic components becomes one of the most important contributors to power-density improvement.

Improved thermal management

As power output increases, heat becomes the primary obstacle to higher density. Even highly efficient PSUs still generate a significant amount of heat. Therefore, manufacturers are investing heavily in better cooling solutions, including optimized airflow paths, advanced heatsinks, vapor chambers, and liquid-cooling technologies. The goal is to remove heat more effectively so that components can operate safely even when packed more tightly together. In many modern PSU designs, thermal engineering is just as important as electrical engineering in determining achievable power density.

Higher component integration

Another way to increase power density is to reduce the amount of unused space inside the PSU. Manufacturers are integrating more functions into fewer components and using three-dimensional layouts that make better use of available volume. Components are stacked, arranged vertically, or combined into integrated modules to maximize space utilization. This approach is similar to how smartphones have become more powerful without becoming larger—the same principle is increasingly being applied to server power supplies.

Higher efficiency creates more thermal headroom

Efficiency improvements provide a double benefit. First, less electricity is wasted. Second, less heat is generated. For example, a 5.5 kW PSU operating at 97.5% efficiency generates roughly 30% less heat than one operating at 96.5% efficiency. While a one-percentage-point improvement may appear small, it can remove tens of watts of heat from the system. This allows engineers to either increase output power within the same form factor or reduce cooling requirements while maintaining existing power levels.

Reducing the size of energy storage components

Server PSUs contain capacitors that store energy and maintain power delivery during short interruptions. Historically, these capacitors occupied a significant amount of space. Newer designs use smarter control techniques that allow manufacturers to achieve the same performance with smaller capacitor banks. Reducing capacitor volume frees up valuable internal space that can be used for higher power output or additional functionality.

For example, in Infineon's 3.3 KW example, the calculated capacitance for a conventional 10ms hold-up implementation was approximately 2.5 mF; allowing deeper discharge through a baby-boost stage reduced the requirement to about 900 µF. The 12 KW design similarly uses a bidirectional buck-boost buffer to achieve 20ms hold-up at full load without the capacitor volume that a passive solution would require. The penalty is additional semiconductors, an inductor, control software and failure modes, but the net volumetric benefit can be substantial.

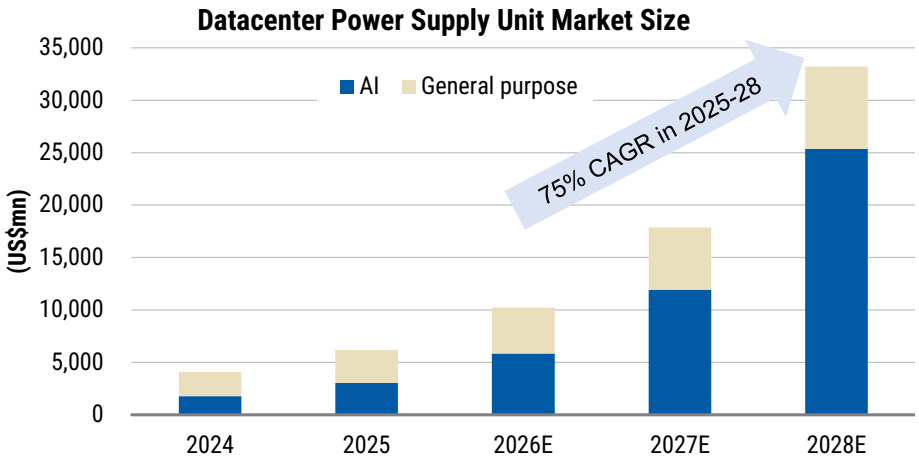
Advanced mechanical design

Power density is ultimately measured in watts per unit volume, making physical packaging extremely important. Manufacturers continuously optimize component placement, airflow channels, connector design, and enclosure structures to eliminate unused space. The best PSU designs are often not defined by a single breakthrough component but rather by hundreds of small mechanical optimizations that collectively improve volumetric efficiency.

Datacenter PSU market size to grow at a 75% CAGR in 2025-28

Overall, we estimate the global datacenter power supply market will grow at a three-year CAGR of 75% to reach US$33bn in 2028 from US$6bn in 2025, driven mainly by server shipment growth, increasing power consumption per rack, and the adoption of HVDC power racks from 2027.

Exhibit 34: Global datacenter power supply market forecast



Source: Company data, Morgan Stanley Research estimates

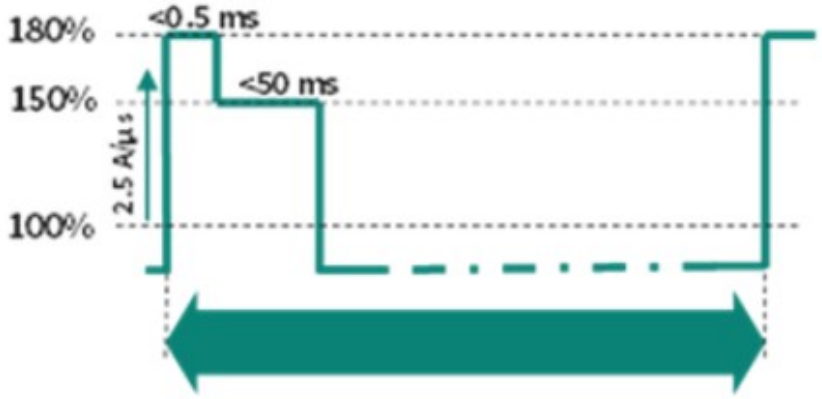
Complementary Hardware for More Resilient Power Delivery

Transient power spikes at AI servers

AI servers have a much more dynamic power profile than conventional enterprise servers because GPU utilization changes sharply throughout a workload. During large-model training, thousands of GPUs typically operate in a bulk-synchronous manner, and power consumption approaches the accelerator's design limit during tensor-compute phases, then falls during collective communications, check pointing, data movement, or periods in which GPUs wait for slower devices to synchronize. Inference can create a similar pattern as request arrivals, batching, context length and token-generation activity change rapidly.

Because the GPUs perform these transitions together, the power variation of one accelerator is multiplied across the server, rack and cluster rather than being averaged out by unrelated workloads, so the magnitude of the challenge is also rising with system density. Power spikes are therefore not simply the result of higher average consumption; they reflect the combination of higher accelerator power, more GPUs per system and increasingly synchronized workload behavior.

Exhibit 35: Transient power spikes at AI servers



Source: Infineon

System performance and reliability potentially compromised

The key issue is that power delivery systems are designed to supply a relatively stable amount of power, while AI workloads can change power demand almost instantaneously. When the load changes faster than the power-delivery system can react, a temporary mismatch occurs between power demanded by the GPUs and power supplied by the PSU, creating electrical instability throughout the server.

Sudden increases in GPU demand lead to voltage drops

When GPUs transition from a low-utilization state (communication, synchronization, checkpointing, etc.) to a full-compute state, their current demand can increase within microseconds. However, the PSU cannot instantly increase its output because its control loop requires time to detect the load change and adjust power conversion.

During this response period, the additional energy must come from capacitors located throughout the power chain. As these capacitors discharge, the voltage on the DC bus begins to fall, creating voltage drop.

If the drop becomes sufficiently large, several operational issues can occur:

- GPU throttling: The GPU reduces clock speed or performance to remain within safe operating limits.
- Computation errors: Voltage instability can affect timing margins in high-speed processors and memory.
- Unexpected resets or shutdowns: Protection circuits may trigger if voltage falls below specified thresholds.
- Training interruption: Large AI training jobs running across thousands of GPUs may fail if even a small number of nodes experience instability.

In a large AI cluster, a single server failure can force the entire training job to restart from the latest checkpoint, resulting in significant productivity loss.

Sudden decreases in GPU demand results in voltage spike

The opposite problem occurs when GPUs rapidly reduce power consumption.

For example, after a compute-intensive phase, GPUs may enter a communication phase where power consumption drops sharply. The PSU, however, is still delivering power based on the previous load condition. The excess energy has nowhere to go immediately and accumulates in the DC bus, causing a temporary voltage spike.

This can create:

- Overvoltage stress on components
- Protection-circuit activation
- Accelerated degradation of capacitors and voltage regulators
- Higher risk of component failure over time

Although individual spikes may be small, AI servers can experience these transitions continuously during training, causing repeated stress throughout the power-delivery chain.

Excessive current surges throughout the system

Power spikes are not only a voltage problem—they also create large current excursions.

A 1KW increase in power demand at 12V requires roughly 83A of additional current. In modern AI servers, where multiple GPUs can change utilization simultaneously, current swings can reach hundreds of amperes within a very short period.

These current surges lead to:

- Higher conductor losses
- Increased connector heating
- Greater stress on voltage-regulator modules (VRMs)
- Larger thermal fluctuations in power components
- As GPU power ratings rise to over 1KW, the magnitude of these transient currents becomes increasingly difficult to manage.

Thermal cycling and reliability degradation

Perhaps the most overlooked issue is the impact on long-term reliability.

Every power excursion creates a corresponding temperature excursion. When power rises, semiconductor junction temperatures increase; when power falls, temperatures decrease. This repeated heating and cooling cycle occurs thousands or even millions of times over a system's lifetime.

The result is accelerated wear on:

- Electrolytic capacitors
- Power MOSFETs
- Solder joints
- PCB interconnects
- Connectors and busbars

Consequently, even if average power remains within specification, transient power spikes can shorten component lifetime and increase failure rates.

Infrastructure overdesign and lower efficiency

To prevent instability, operators often provision the power infrastructure for the worst-case peak rather than average consumption.

For example:

- A rack averaging 80KW may need infrastructure sized for 120-140KW transient peaks.
- PSUs must be oversized to accommodate short-duration excursions.
- UPS systems must reserve capacity for sudden load increases.
- Cooling systems must handle peak thermal output.

This creates a significant amount of stranded capacity, where expensive electrical and cooling infrastructure remains underutilized most of the time. Furthermore, PSUs and UPS systems typically operate most efficiently near their optimal loading point; oversized systems therefore tend to operate at lower efficiency, increasing operating costs.

Conventional power delivery systems not designed to deal with power volatility

In the conventional power delivery architecture, the UPS protects the data hall against utility outages and power-quality disturbances, while the PSU converts upstream power into the tightly regulated DC output required by the server. Both remain essential in the modern AI server clusters, but neither was originally optimized to absorb large, repetitive and synchronized AI load steps economically.

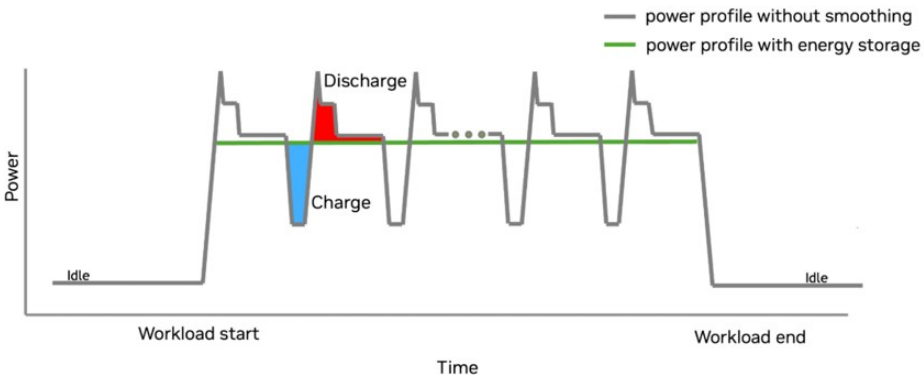
The PSU can use its internal capacitors to cover the initial transient, but capacitor volume is limited; once that stored energy is depleted, the PSU control loop must raise input current, potentially producing current overshoot and transferring the disturbance upstream.

The UPS is located farther from the accelerator and sees the aggregated load of many PSUs. A transient originating near the GPU can therefore occur on microsecond-to-millisecond timescales—including permitted GPU electrical-design-power excursions around 50 ms—before the complete upstream conversion chain can rebalance power.

This is a response time and location mismatch, rather than evidence that UPSs or PSUs are fundamentally incapable of supporting AI. New AI-tolerant UPS platforms are being designed for faster and larger load steps, but many legacy systems may still experience voltage excursions, nuisance alarms, lower partial-load efficiency or unwanted bypass transfers under steep synchronized loads.

To complement the conventional power delivery architecture, more AI server clusters are adopting BBUs or CBUs to shave peak loads and mitigate the above-mentioned issues.

Exhibit 36: Concept of peak shaving for AI server power loads



Source: Nvidia

Battery backup units seeing increasing importance

What's a BBU?

A BBU is an integrated battery, power-conversion and control system that supplies DC power to server loads when the normal power source is interrupted or temporarily insufficient. In an OCP-style rack, the BBU modules are housed in a dedicated shelf located near the PSU power shelf and connect directly to the common DC busbar. Under normal conditions, the PSUs power the servers and maintain the BBU's charge; when the bus

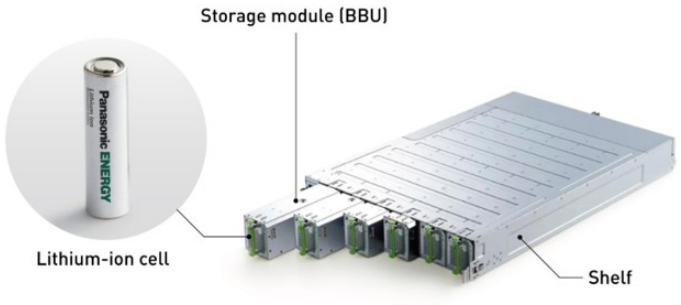
voltage falls below a programmed threshold, the BBU's bidirectional converter changes from charging or standby into discharge mode and supports the rack. Analog Devices' ORV3 reference design detects a bus voltage below 48.5V for more than 2ms and is required to ramp to full support within another 2ms, while maintaining the bus above the specified minimum voltage.

Key components within a BBU

- **Battery-cell pack:** The cells store the energy used during discharge. Lithium-ion cylindrical cells are commonly arranged in series and parallel to obtain the required voltage, power and runtime; the ORV3 reference module uses an interchangeable 11-series, 6-parallel battery pack. The cell must provide not only sufficient energy but also high discharge power, low internal resistance and predictable performance over its qualified life.
- **Bidirectional DC/DC converter:** The converter controls energy flow between the battery pack and the rack DC bus. It steps voltage in the appropriate direction during charging and discharging and regulates the output so that the battery's changing voltage does not destabilize the server bus. Multiphase designs divide the large current across several parallel power stages; TI's 12 KW design uses six 2 KW phases and reports peak efficiency above 99%, including above 99.5% when battery and bus voltages are close.
- **Battery management system:** The BMS measures individual cell voltages, current and temperature, balances cells and estimates state of charge and state of health. It also protects against overcharge, excessive discharge, overheating and short circuit. Accurate state-of-health information is important because the operator must know whether an apparently charged BBU can still deliver its required rack power and runtime after years of standby operation and periodic discharge events.
- **Power-path and protection circuitry:** Fuses, current sensors, disconnect switches, protection MOSFETs and hot-swap circuits isolate faults and prevent uncontrolled current flow. Typical protections include overvoltage, undervoltage, overcurrent and overtemperature shutdown. Hot-swap capability allows a failed or end-of-life module to be replaced without intentionally shutting down the entire rack, while active current sharing ensures that parallel BBU modules carry an appropriate portion of the load.
- **Control, monitoring and communications:** A module controller manages operating modes, converter control, fault handling, diagnostics and communication with the BMS and rack controller. Interfaces such as PMBus, Modbus, CAN, RS-485 or Ethernet can expose battery voltage, current, temperature, state of charge, state of health, fan speed and fault status to data-center management systems. Analog Devices' reference platform uses separate controllers for battery monitoring and overall BBU logic, communications and housekeeping.
- **Thermal-management system:** Fans, temperature sensors, airflow paths and mechanical spacing around cells remove heat during high-rate charging and discharging. Thermal management is both a performance and a safety requirement: excessive temperature accelerates battery ageing and can restrict available discharge power. Analog Devices' ORV3 design therefore incorporates fan control, airflow simulation and cell-to-cell spacing, while Murata's commercial 3 KW BBU includes internal forced-air cooling and overtemperature protection.

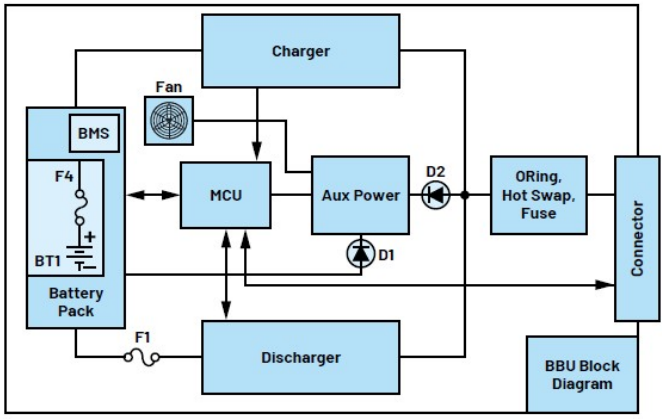
- **Shelf, backplane and mechanical integration:** The shelf distributes DC power and communications across several modules and provides retention, service access and cooling. Modularity allows power and runtime to be scaled by adding units and enables redundancy such as the ORV3 5+1 configuration. Murata specifies a 2U, hot-swappable 3 KW module, while Eaton's 19-inch BBU shelf accepts six 2.75 kW batteries and scales runtime according to the application requirement

Exhibit 37: BBU



Source: Panasonic

Exhibit 38: BBU diagram



Source: ADI

Key functions of BBUs

Backup power and outage ride-through

The BBU's fundamental function is to maintain rack operation during a brief loss of upstream power. The ride-through interval gives the data center time to restore the primary source, transfer to another source, preserve critical data, migrate workloads or perform an orderly shutdown. Rather than attempting long-duration operation, rack BBUs are generally optimized for high power over a relatively short period; the ORV3 example specifies 3KW per module for four minutes, and Murata guarantees a 240-second hold-up period at full-rated load under stated age and temperature conditions.

For AI systems, this protection is particularly valuable because a rack interruption can affect a synchronized workload extending well beyond the failed server. Local ride-through prevents a momentary upstream disturbance from immediately removing GPU, CPU, networking and storage power from the rack. Infineon consequently describes the server BBU as an integrated backup supply intended to prevent data loss and maintain AI workloads during outages or grid failures.

Peak shaving and load smoothing

A BBU can be operated as an active energy buffer. as it charges during lower-demand periods and discharges when server consumption temporarily exceeds a defined rack or PSU limit. This allows a relatively stable upstream power source to support a more volatile downstream AI workload. The economic benefit is that PSU, UPS and electrical capacity can be designed closer to sustained demand rather than every short theoretical peak, although the achievable benefit depends on the frequency, duration and predictability of those peaks.

The important limitation is battery cycling. A BBU designed primarily for emergency standby may not be optimized for continuous, repetitive peak shaving, so frequent cycling affects heat generation, state of charge, usable life and replacement schedules. Very short and highly repetitive transients may therefore be better supported by a CBU or hybrid battery-capacitor system, leaving the battery to cover longer events. Renesas correspondingly distinguishes distributed BBUs from hybrid systems that combine batteries with capacitors for fast transient response and longer-duration backup.

Power stabilization, redundancy and fault isolation

A rack BBU adds another independent source to the local power path. If a power shelf, upstream feed or AC source is interrupted, the BBU can sustain the rack while the fault is isolated or another source assumes the load. A modular shelf further improves resilience because one BBU failure need not remove the entire backup function; the remaining modules can continue operating within the available redundancy and power limits.

Distributed deployment also narrows the fault domain. A defective rack BBU affects a much smaller set of compute equipment than a failure within a centralized backup system serving a large room or data hall. The tradeoff is a much larger population of batteries, converters, controllers and fans that must be monitored, tested and eventually replaced. Digital state-of-health monitoring, fault reporting and hot-swappable construction are therefore central to the operating model rather than optional features.

BBUs vs. UPSs

While some of the functions that BBUs offer look similar to what traditional UPSs can do, we think BBUs and UPSs should be viewed as complementary rather than substitutive technologies, as they address different challenges within the datacenter power architecture.

A UPS is designed to provide facility- or row-level protection against utility outages, voltage sags, surges, and other power-quality disturbances, ensuring continuous AC power delivery to a broad range of IT equipment. In contrast, a BBU operates much closer to the AI workload, supplying DC power directly at the rack level and responding to localized power events that occur faster than the upstream power infrastructure can efficiently manage.

As AI racks become denser and more power intensive, the UPS continues to serve as the first line of defense against external power disruptions, while the BBU acts as a second layer of protection by stabilizing the rack DC bus, supporting transient load spikes, and providing short-duration ride-through capability. Together, they create a hierarchical power protection architecture in which the UPS ensures overall system availability and power quality, while the BBU enhances rack-level resilience and power-delivery performance. Rather than replacing the UPS, BBUs help reduce the burden on centralized backup systems, improve power utilization efficiency, and enable the reliable operation of next generation AI infrastructure with increasingly dynamic power profiles

Exhibit 39: Comparison of BBUs and UPSs

Dimension	BBU	UPS
Location	Inside the server rack, in a BBU shelf, or in an adjacent rack/sidecar close to the compute	Typically deployed at equipment room, row or facility level and protects multiple racks or other critical loads
Power path	Usually DC-coupled to the rack bus through a bidirectional DC/DC converter	Takes AC input and delivers AC output; the server PSU subsequently converts AC again for the server
Primary objective	Short rack-local ride-through, DC-bus support, modular redundancy and—when designed for it—transient or peak-load assistance	Broad power conditioning and uninterrupted AC supply during utility sags, surges, frequency disturbances and outages
Response	Monitors the local bus and can inject power within a few milliseconds; the cited ORV3 reference requires full support within 2ms after a qualifying 2ms event	An online double-conversion UPS already powers the load through its inverter and therefore has zero transfer time between utility and battery operation
Runtime	Generally optimized for short-duration, high-power support; OCP's reference example is four minutes at full load	Runtime varies widely with battery-bank sizing and system design; its broader function is to maintain protected AC loads until another source or operating response becomes available
Scalability	Added rack-by-rack or module-by-module; power and runtime can be matched to each compute deployment	Scaled centrally in larger KW blocks and requires capacity planning for the aggregate protected load
Fault domain	A module or shelf fault is normally localized to one rack or a limited group of racks	A centralized system protects a much broader load domain, so its redundancy and maintenance design are correspondingly critical

Source: Morgan Stanley Research

Capacitor backup unit (CBU) another complementary power device

What's a CBU?

A CBU is an integrated energy storage subsystem positioned close to the AI load, typically in a dedicated rack shelf, capacitor tray or adjacent power sidecar, and connected to the server rack's DC power path. Its principal storage medium may be an electric double-layer capacitor (EDLC or supercapacitor), lithium-ion capacitor (LIC), or a bank of high-capacitance electrolytic capacitors, depending on the required response time, energy duration, footprint and cost. Dedicated CBU products increasingly use EDLCs or LICs because their low internal resistance supports rapid, repetitive charge and discharge.

The CBU sits electrically alongside the PSU. Under steady-state conditions, the PSU supplies the AI rack and maintains the CBU at its target state of charge. When GPU demand temporarily exceeds the desired PSU output, the CBU discharges through a controlled power converter into the DC bus; when GPU demand falls, the converter reverses direction and recharges the capacitors at a controlled rate. This prevents the full GPU load step from passing through the PSU and upstream equipment. Infineon notes that a separate CBU must be compatible with, and potentially communicate with, the PSU shelves to coordinate rack power.

An alternative to deploying a standalone CBU shelf is to integrate the energy-storage capacitors directly into the power shelf, effectively combining power conversion and transient-energy buffering within the same subsystem. NVIDIA's GB300 NVL72 architecture is a good example: its enhanced power shelves incorporate a substantial capacitor bank alongside the PSU modules, with NVIDIA allocating approximately 65 joules of stored energy per GPU.

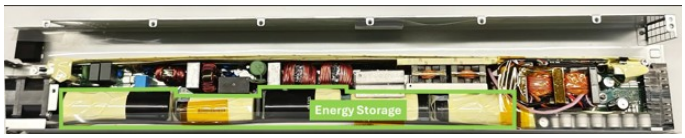
NVIDIA reported that this approach reduced peak upstream power demand by roughly 30% in a measured Megatron LLM workload. Compared with a separate CBU, integrating capacitors into the power shelf can save valuable rack space, shorten the electrical path between the energy buffer and the load, and simplify coordination between the PSU and

energy-storage system, since charging, discharging, and PSU output can be managed as one integrated power architecture. The trade-off is that capacitor capacity becomes more tightly coupled to the power-shelf design, potentially reducing flexibility in independently scaling or servicing the energy-storage system. Nevertheless, as AI racks become increasingly power dense, integrating transient-energy storage directly into the power shelf could become an attractive alternative to standalone CBUs, particularly where rack space and fast transient response are key design priorities.

Exhibit 40: CBU

Source: Panasonic

Exhibit 41: GB300 power shelf with integrated capacitors



Source: Nvidia

Key components within a CBU

- **Capacitor bank:** The storage section contains many capacitor cells arranged in series to reach the required voltage and in parallel to provide the required power and energy. Individual EDLC cells commonly operate at only a few volts, so high-voltage systems require long series strings; Abracon cites approximately 2.7 V as a typical supercapacitor cell rating, while TI's NVIDIA-oriented reference architecture demonstrates an 800 V CBU using EDLC cells. LICs can provide more stored energy than conventional EDLCs, while EDLCs generally emphasize lower resistance, high peak power and very high cycling endurance. Delta, for example, offers a power-capacitance shelf using fast-charge/discharge LICs, rated to provide 15 seconds of hold-up at 20 KW.
- **Cell-balancing system:** Because cell capacitance, leakage current and internal resistance vary across a series string, the voltage does not divide perfectly equally. A balancing circuit prevents an individual cell from exceeding its rated voltage during charging or being reverse-biased during deep discharge. Passive balancing uses resistors to bleed excess charge, while active approaches use MOSFETs, amplifiers or switched converters to redistribute energy more efficiently; the design tradeoff is between cost, balancing speed, component count and standby loss.
- **Bidirectional DC/DC converter:** This is the active interface between the capacitor bank and the rack DC bus. During a GPU power spike, it converts the capacitor's changing voltage into a regulated bus contribution; during lower demand, it reverses power flow to recharge the bank. The converter also allows more of the capacitors' stored energy to be used than would be possible with a direct connection, because capacitor voltage declines materially during discharge.

Infineon specifically identifies a capacitor bank plus bidirectional converter as the main CBU structure, while Skeleton Technologies notes that converter-based integration provides greater energy utilization and more precise power-flow control than a direct bus connection.

- **Controller and management software:** The controller determines when the CBU should charge, remain idle or discharge, based on bus voltage, PSU output, rack power limits and capacitor state. It may also calculate state of health, manage balancing, regulate recharge current and coordinate multiple CBU modules.
- **Protection and switching:** A high-power capacitor bank can release very large fault current, so fuses, contactors, pre-charge circuits and overvoltage, overcurrent and overtemperature protection remain essential. Pre-charge limits inrush current when the CBU is connected to the DC bus, while contactors or solid-state switches isolate the bank during a fault or service event. Skeleton's reference system integrates switchgear, fuse-fault detection, pre-charge/discharge contactors and protection against high-current and overheating events; industry discussions also highlight the need for faster DC fault isolation as CBU voltage rises.
- **Monitoring and communications:** Voltage, current, temperature, equivalent series resistance and state-of-health measurements allow operators to identify degradation before the CBU can no longer meet its specified peak-power requirement. Panasonic emphasizes voltage, temperature and ESR monitoring for predictive maintenance, while Skeleton provides single-cell voltage sensing, multiple temperature sensors and communication bus to its master controller.
- **Thermal and mechanical system:** Although supercapacitors are efficient, high repetitive current produces resistive heating, particularly in cells, busbars, converters and connectors. The shelf therefore requires designed airflow, temperature sensing and, at higher power levels, potentially liquid cooling. Skeleton's cabinet uses controlled fan cooling and also offers liquid-cooled configurations, while the Open Compute Project presentation from Panasonic identifies lower-resistance capacitors and liquid cooling as development priorities for future higher-output CBUs

Main functions of CBU

Transient power stabilization

One of the most fundamental roles of a CBU is to stabilize the rack DC power bus when GPU power consumption changes very rapidly. AI workloads frequently alternate between compute-intensive phases and lower-power communication, synchronization, or idle phases. Because many GPUs operate in parallel, these transitions can create very large changes in rack power within milliseconds.

When GPU demand suddenly increases, the PSU cannot instantaneously increase its output. The CBU therefore discharges stored energy into the DC bus during this short response gap, helping prevent voltage drop and excessive current excursions. Conversely, when GPU demand suddenly falls, the CBU can absorb excess energy while the PSU reduces its output, helping limit voltage overshoot.

The key objective here is power quality and system stability, rather than reducing the overall power capacity required by the rack. In other words, the CBU acts as a fast

electrical shock absorber between the highly dynamic GPU load and the slower PSU. This becomes increasingly important as AI racks move above 100 KW and synchronized GPU workloads create larger absolute power swings.

Peak shaving

Another major function is peak shaving, whose objective is to limit the maximum power that the PSU and upstream infrastructure need to supply.

The benefit is potentially significant because the PSU, connectors, protection devices, and portions of the upstream power infrastructure no longer need to be oversized solely for short-duration GPU peaks. It can also allow PSUs to operate closer to their optimal loading range, improving utilization and potentially efficiency.

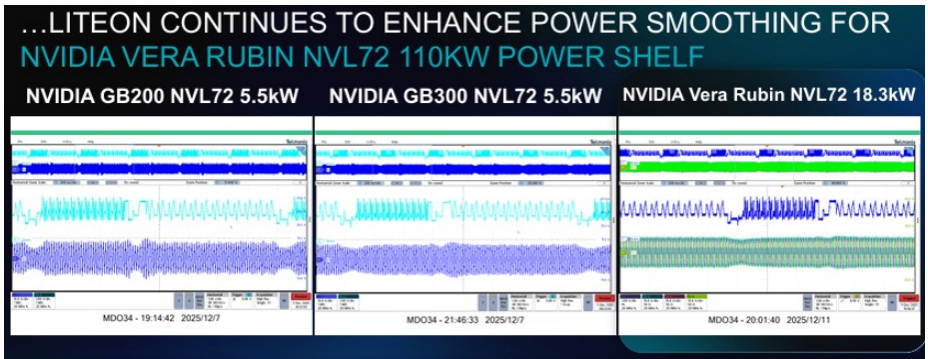
Short duration ride-through

The third function is ride-through, where the CBU temporarily powers the rack when the normal power source is interrupted altogether. This is fundamentally different from the first two functions because the issue is no longer a mismatch between GPU demand and PSU response—the upstream power source itself has temporarily disappeared.

A sufficiently sized CBU can maintain the DC bus during events such as a brief AC interruption, PSU switchover, source-transfer event, or temporary power-shelf failure. Depending on the amount and type of capacitance installed, this support can range from milliseconds to several seconds. For example, some LIC-based capacitance shelves are designed to provide multi-second hold-up at high rack power.

However, ride-through is generally a secondary function for a CBU. Capacitors have high power density but relatively low energy density, meaning they can deliver a large amount of power very quickly but cannot economically sustain that power for long periods. As the required backup duration increases from seconds toward minutes, a battery-based BBU generally becomes more suitable.

Exhibit 42: Capacitors help smooth power delivery



	Ipk-max	Ipk-min	Max/Min Ratio
NVIDIA GB200 NVL72	15.9A	8.3A	91.57%
NVIDIA GB300 NVL72	13.3A	8.6A	54.65%
NVIDIA Vera Rubin NVL72	**12.8A**	**10.3A**	**24.27%**

Source: Lite On

CBUs vs. BBUs

The two technologies should consequently be regarded as complementary rather than directly substitutive. A CBU handles the high-power, short-duration and high-cycle portion of the load profile, while a BBU supplies the larger amount of energy required when the upstream source remains unavailable. Some describe BBUs and CBUs as coexisting in the same power sidecar, and hybrid architectures can allocate fast transients to capacitors while preserving battery charge and cycle life for longer outages.

The CBU's major advantages are rapid bidirectional response, long cycle life, high efficiency and the ability to absorb repeated GPU peaks without the same electrochemical wear mechanism as a conventional battery. It can also recharge quickly between workload cycles and may require less routine replacement. However, the CBU has materially lower energy density, its voltage falls continuously during discharge, and large banks require balancing, high-current protection, cooling and valuable rack or sidecar space.

Exhibit 43: Comparison of CBUs and BBUs

Dimension	CBU	BBU
Primary storage medium	EDLC supercapacitors, LICs or high-capacitance	Typically lithium-ion battery cells managed by a BMS
Primary strength	High power, fast charge/discharge and frequent	Higher stored energy and longer sustained
Main function	Transient smoothing, peak shaving, voltage	Outage backup, source-transfer bridging and orderly workload
Response profile	Well suited to millisecond-class load support and	Fast switchover, but generally optimized for sustained support rather
Typical duration	Milliseconds to seconds; selected systems can extend toward tens of seconds or more	Commonly tens of seconds to several minutes, depending on battery and rack design
Power versus energy density	Higher power capability but lower energy storage per unit volume than a battery	Higher energy density, making it more economical for longer events
Cycle life	Eaton specifies more than one million cycles and up to 20 years for certain supercapacitor modules	Battery life is more sensitive to depth of discharge, temperature and repeated cycling; BMS supervision is therefore essential.
State management	Requires cell balancing, voltage/temperature monitoring and state-of-health tracking, but does not use a conventional battery fuel	Requires BMS functions including charge control, state of charge, state of health and cell protection
Best-fit use case	Repetitive AI power spikes, inrush events, PSU load leveling and bus stabilization	Power continuity through outages, brownouts or generator/source-transfer intervals

Source: Morgan Stanley Research

HVDC Emerging, with SST in the Pipeline

HVDC coming along with the increasing power consumption of AI clusters

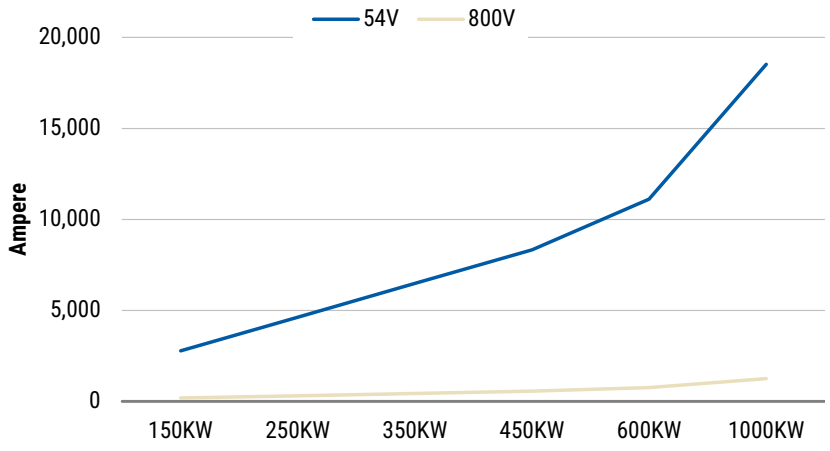
Why do we need HVDC?

Low voltage turns higher rack power into an extreme-current problem

The fundamental relationship is that at a given rack load, increasing the bulk-distribution voltage from 54 V to 800 V reduces current by approximately 14.8 times. A balanced ±400 V architecture also presents 800 V between its +400 V and –400 V conductors, so its bulk-transfer current is broadly equivalent to a 0–800 V architecture.

This current differential has an outsized mechanical consequence. For the same conductor resistance, I^2R loss at 54V would theoretically be approximately 219 times that at 800 V, though real systems avoid this level of loss by adding copper, parallel conductors and cooling. The result is not a 219-fold difference in actual operating loss, but a trade-off in which low-voltage designs consume increasing amounts of busbar cross-section, connector area, rack volume and cooling capacity to remain within voltage-drop and temperature limits.

Exhibit 44: Transmission current for different power scales: 54V vs 800V



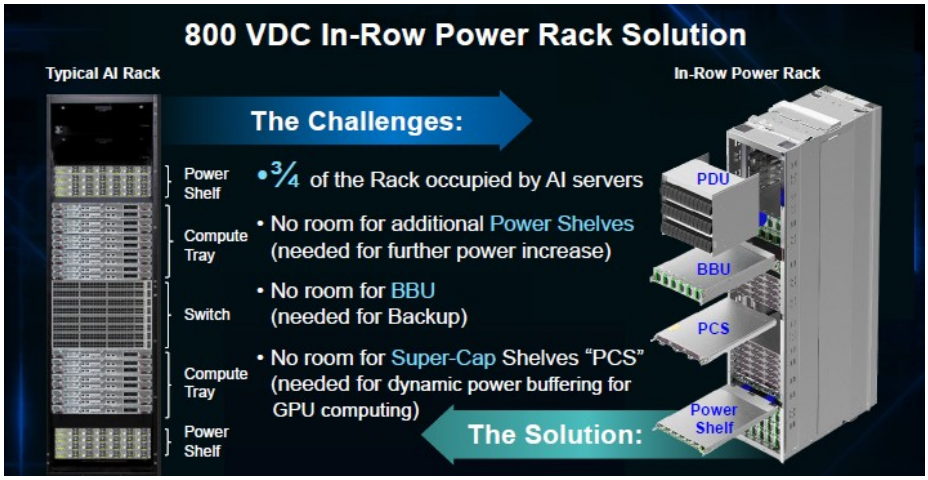
Source: Delta

Copper, connectors and thermal management become system constraints

At several thousand amperes, distribution cannot rely on a single ordinary conductor or connector. Designers must use broad laminated busbars, multiple parallel contacts, tightly controlled joint resistance and temperature monitoring. A small increase in contact resistance produces disproportionately high heat at these currents, raising the risk of hot spots, connector ageing and nuisance protection events. OCP's 100KW HVDC-to-LVDC shelf illustrates where the current problem moves: only two 62.5 A HVDC inputs are required on the ±400V side, while the 49–50V output approaches 2,000A and therefore uses a substantial bolted output connection with dedicated thermistors.

The same issue appears in power-shelf packaging. Delta notes that a typical AI rack may already allocate roughly three-quarters of its volume to servers, leaving insufficient room for additional power shelves, BBUs or supercapacitor shelves as power rises. Moving AC/DC conversion, protection and storage into an adjacent power rack therefore releases compute space and separates the thermal design of power electronics from that of GPUs. Microsoft's Mt Diablo rationale similarly states that disaggregated power can enable up to 35% more accelerators in the IT rack and allows power shelf count to be sized independently for each workload configuration

Exhibit 45: Not enough room for power devices on the traditional AI server rack



Source: Delta

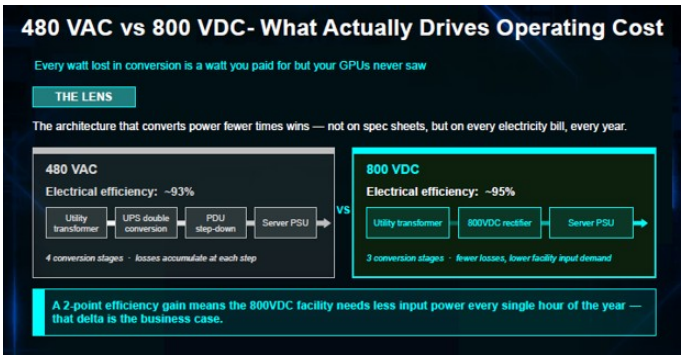
HVDC enables denser, fewer and more efficient power stages

As AI rack power scales from today's GB200/GB300 systems to Vera Rubin and future multi-hundred-kilowatt racks, power distribution efficiency becomes increasingly important. Our checks indicate that a conventional 480VAC architecture delivers an end-to-end electrical efficiency of roughly 93%, while an 800VDC architecture improves efficiency to about 95% by eliminating one power conversion stage and simplifying the power path from grid to rack.

The higher-voltage 800VDC distribution also reduces current levels for the same power load, lowering resistive (I^2R) losses in cables and distribution equipment and enabling more efficient delivery to increasingly power-dense AI racks.

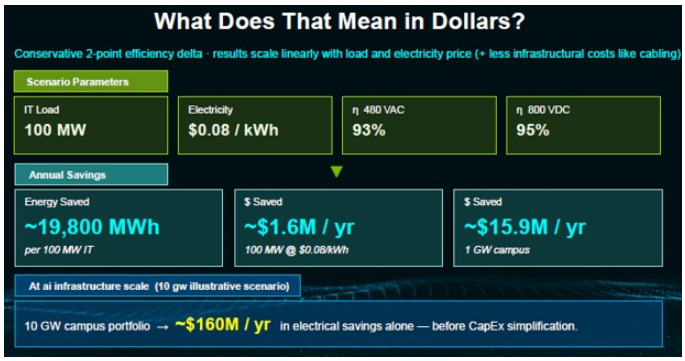
While a 2ppts of efficiency improvement may appear modest, the savings become substantial at hyperscale AI data center deployments. Delta's example shows that for a 100MW IT load, improving facility efficiency from 93% to 95% reduces annual energy consumption by approximately 19,800MWh, translating into roughly US$1.6 million of annual electricity cost savings assuming a power price of US$0.08/kWh. At a 1GW AI campus, the savings scale to nearly US$16 million per year, and at emerging multi-gigawatt AI factory deployments, the annual benefit can exceed US$100 million. Consequently, the transition from conventional AC distribution and 54VDC rack power architectures toward 800VDC distribution is not only a technical necessity for supporting future 500kW-to-1MW AI racks, but also a compelling economic proposition that lowers operating expenses and improves total cost of ownership through reduced energy losses and simplified electrical infrastructure.

Exhibit 46: Better power efficiency for 800VDC architecture



Source: Delta

Exhibit 47: Better efficiency leads to cost savings



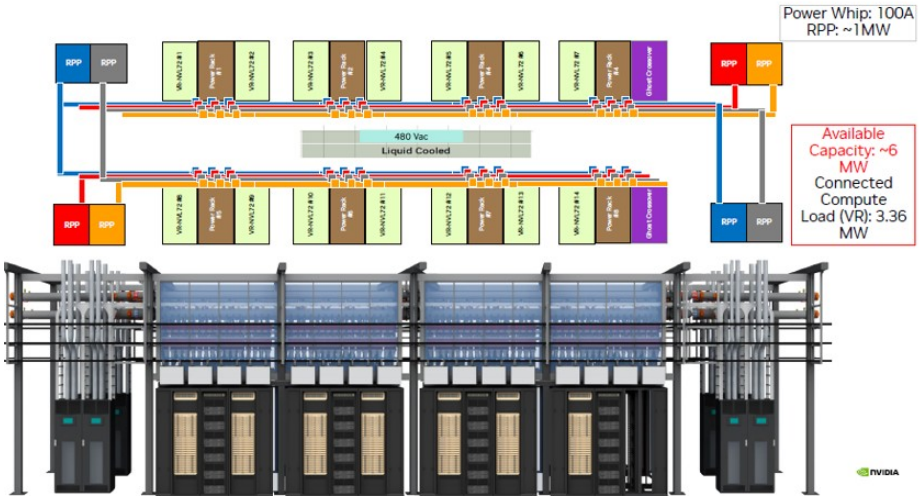
Source: Delta

What's the deployment roadmap for HVDC?

The deployment roadmap for 800VDC is best understood as a multi-stage evolution rather than a single-step transition. Both Nvidia and major hyperscalers recognize that existing 480VAC infrastructure cannot efficiently support the rapid increase in rack power. As rack densities increase, conventional power whips, busbars, connectors, and PSU architectures increasingly become bottlenecks, driving the need for higher voltage distribution

The first phase of deployment focuses on introducing 800VDC power racks within existing AC-based data centers. In this architecture, the facility remains largely unchanged, but a dedicated power rack converts 415-480VAC into 800VDC before distributing power locally to AI racks. This approach minimizes disruption to existing infrastructure while allowing operators to deploy Vera Rubin-class systems that exceed the practical limits of traditional power shelves. Nvidia refers to this as the row-level power rack architecture, which serves as the industry's bridge solution between legacy AC facilities and future native DC environments.

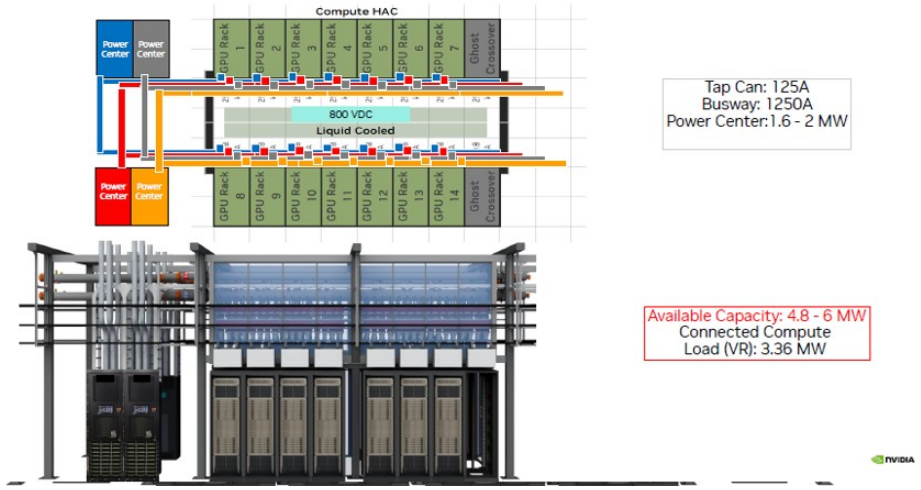
Exhibit 48: Phase I - 800VDC power rack



Source: Nvidia

The second phase expands 800VDC beyond individual rows into data hall-level power centers. Instead of deploying a dedicated power rack for every few compute cabinets, centralized rectifier or power-center systems distribute 800VDC through DC busways across an entire hall. Nvidia's roadmap identifies 1.6-2MW power centers and 1250A DC busway architectures as the next step, improving power density while reducing equipment duplication. This stage also aligns well with the deployment strategies being discussed by hyperscalers and the OCP community because it preserves existing medium-voltage and low-voltage infrastructure while moving a significant portion of distribution into DC

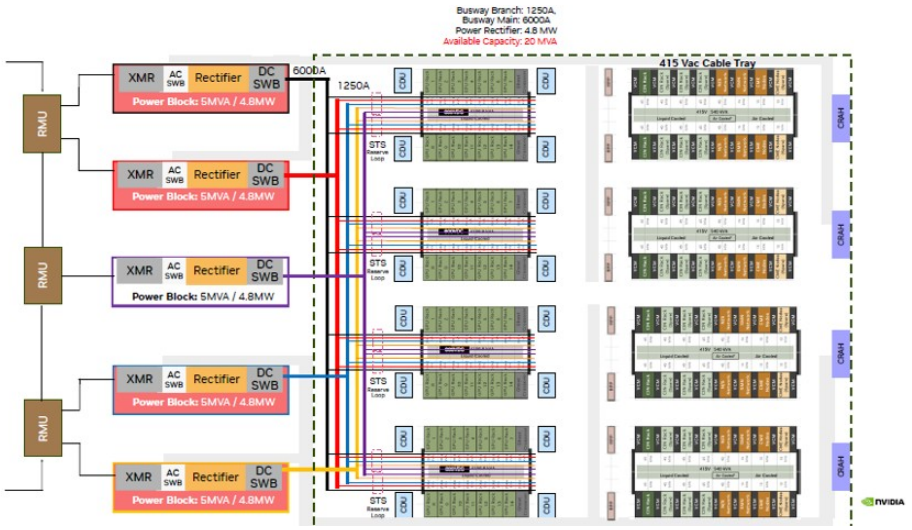
Exhibit 49: Phase II – 800VDC power center



Source: Nvidia

The third phase is the transition toward facility-wide 800VDC distribution, where AC is converted to DC near the utility entrance or medium-voltage substation and distributed throughout the data center via high-capacity DC switchboards and busways. Nvidia describes this as the DC power block architecture, utilizing approximately 4.8MW power blocks, 6000A DC busways, and integrated redundancy schemes. At this stage, the architecture begins to resemble an "AI factory," where power distribution is simplified, energy storage can be integrated directly onto the DC bus, and multiple megawatts of compute capacity can be deployed as modular building blocks

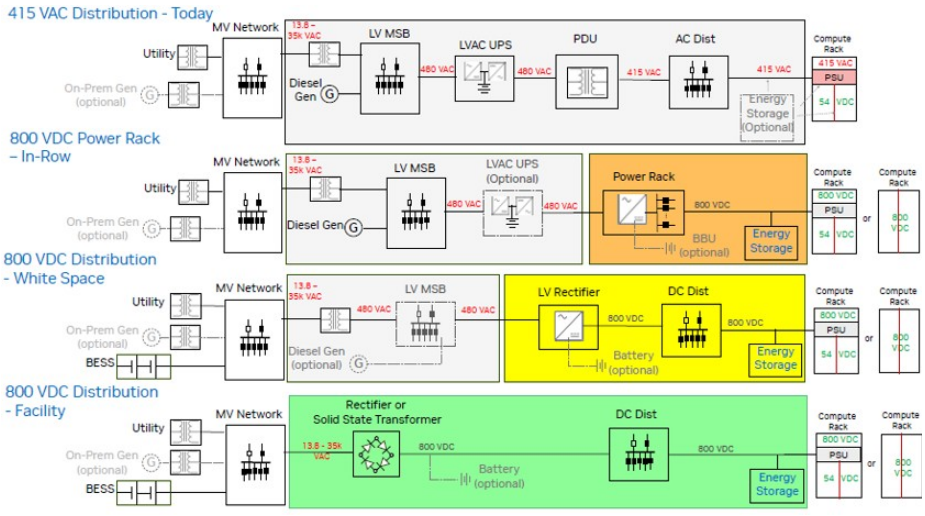
Exhibit 50: Phase III – power block



Source: Nvidia

The final stage is the emergence of native 800VDC compute racks, which Nvidia identifies as the long-term destination. Today, even in 800VDC deployments, power is generally stepped down to 54V before entering the servers. Future systems, however, are expected to accept 800VDC directly within the rack and potentially at the server level, eliminating another conversion stage and further reducing power losses. Nvidia's public roadmap shows a progression from current 54V architectures, to 800VDC-fed PSUs, and ultimately to 1MW-class native 800VDC racks capable of supporting next-generation AI accelerators

Exhibit 51: 800VDC native server racks



Source: Nvidia

In sum, deployment is likely to occur gradually because the ecosystem remains under development. NVIDIA notes that several enabling technologies—including DC protection devices, busways, connectors, grounding standards, arc-flash modeling, and solid-state transformers— are still undergoing certification and standardization. The roadmap presented by NVIDIA suggests a phased commercialization timeline, with 800VDC power racks reaching deployment readiness in 2026, hall-level power centers in 2027, and facility-scale DC power blocks becoming deployment-ready around 2028. Industry

collaboration among NVIDIA, Microsoft, Google, and OCP participants is focused on ensuring interoperability and regulatory acceptance before broader hyperscale adoption.

Exhibit 52: HVDC deployment roadmap

Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

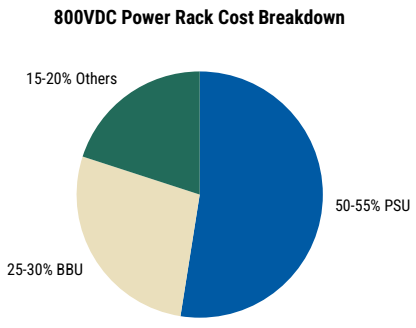
Source: Nvidia

800VDC represents 4-5x dollar content vs. in-rack power supply architecture

The 800VDC power rack BOM is primarily driven by the PSU, which accounts for 50–55% of the total cost, followed by the BBU at 25–30%, with other components comprising the remaining 15–20%. This cost structure reflects the broader functionality integrated into an HVDC power rack, combining power conversion, backup power and supporting components within a centralized architecture.

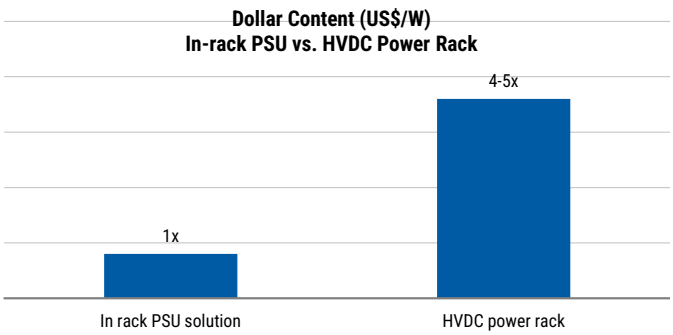
For power supply players, one of the most implications is that an HVDC power rack carries around 4–5x higher dollar content per watt compared with a conventional in-rack PSU solution, since the former integrates more components into a robust system and power supply players usually ship the entire rack instead of just the components.

Exhibit 53: BOM cost breakdown for an 800VDC power rack



Source: Morgan Stanley Research

Exhibit 54: Higher dollar content for HVDC power racks



Source: Morgan Stanley Research

800VDC vs. ±400VDC?

The debate between ±400VDC and 800VDC centers on balancing near-term practicality against long-term scalability. While both architectures deliver an identical 800V differential voltage and therefore achieve similar reductions in current, copper loss, and cable size versus traditional 54V rack power systems, they differ significantly in implementation.

Under a ±400VDC architecture, each conductor operates at only 400V relative to ground, resulting in less stringent insulation, creepage, clearance, and protection requirements. This makes the architecture easier to deploy, maintain, and certify, which explains why many hyperscalers and OCP members view ±400VDC as an attractive migration path toward high-density AI racks.

By contrast, NVIDIA and several power infrastructure vendors are increasingly advocating for a single-bus 800VDC architecture because it simplifies overall power distribution, reduces conversion stages, and is better suited for future 500kW to 1MW rack deployments.

We believe that ±400VDC appears well positioned as a near-term bridge architecture due to its lower deployment risk and more mature ecosystem, whereas 800VDC is increasingly viewed as the long-term end-state architecture for AI factories as rack power scales toward the megawatt era.

Exhibit 55: Comparison of 800VDC and ±400VDC

Metric	±400VDC	800VDC
Voltage to ground	400V relative to ground	800V relative to ground
Voltage differential	800V	800V
Power distribution efficiency	High	Slightly higher
Scalability	Good	Excellent
Insulation requirements	Lower	Higher
Maintenance safety	Better	More challenging
System architecture simplicity	Dual-rail system	Single-bus system
Grounding Complexity	More complicated	Simpler
Electrical footprint	Small	Smaller
Ability to reduce conversion stages	Moderate	Highest
Long-distance distribution capability	Excellent	Excellent
Maturity of ecosystem	Better today	Early but rapidly developing
Industry players	OCP, hyperscalers	Nvidia
Long term industry direction	Transitional architecture	Likely end-state architecture

Source: Morgan Stanley Research

From a component supplier perspective, the transition toward HVDC is far more important than which architecture ultimately wins, as most hardware content increases under either scenario. PSU vendors, BBU suppliers, busbar manufacturers, connector providers, power-whip suppliers, magnetics vendors, and rack-level DC-DC converter manufacturers are largely architecture-agnostic, since both ±400VDC and 800VDC require a similar shift toward high-voltage power distribution, centralized power shelves, and rack-level conversion.

Our understanding is that the primary exception is the power semiconductor industry, where the choice of architecture could materially influence device content and technology adoption. A ±400VDC system can often be supported using existing 650V-900V MOSFET and SiC platforms, benefiting suppliers with exposure to lower-voltage power devices. By contrast, 800VDC architectures increasingly require 1200V-class SiC MOSFETs, higher-

voltage gate drivers, hot-swap controllers, isolation components, and advanced protection circuitry, driving greater semiconductor content per watt and raising technology barriers.

Solid state transformer potentially being the holy grail in AI datacenter power delivery

What is a solid-state transformer ?

A datacenter solid state transformer (SST) integrates power electronics to replace or augment traditional line-frequency transformers, and it's typically built around three major functional stages. The first is the medium-voltage AC-to-DC rectifier stage, which uses high-voltage semiconductor switches—often silicon carbide (SiC) or gallium nitride (GaN) devices due to their superior switching speed and thermal performance—to convert incoming grid-level medium voltage AC into a regulated DC bus. The second is the isolated DC-DC conversion stage, usually implemented as a dual active bridge (DAB) or LLC resonant converter, which provides galvanic isolation through a high-frequency transformer (operating in the kHz to tens-of-kHz range rather than 50/60 Hz), dramatically shrinking the magnetic core and transformer size compared to conventional units while stepping voltage down toward datacenter distribution levels. The third is the low-voltage DC-AC or DC-DC output stage, which conditions power into the form needed by the facility—either regulated DC for direct DC-bus distribution to IT racks or inverted AC for legacy PDU/UPS infrastructure. Supporting these power stages are the control and sensing subsystems (digital controllers, gate drivers, voltage/current sensors) that manage switching, fault protection, and grid synchronization; thermal management components (heat sinks, cold plates, or liquid cooling loops) to handle the concentrated losses in the semiconductor devices; and increasingly, integrated energy storage interfaces or DC microgrid ports that allow the SST to natively support battery buffering, renewable integration, and finer-grained power quality control than a passive transformer ever could

A Solid-State Transformer (SST) is a power-electronics-based transformer that replaces the traditional 50/60Hz iron-core transformer with a combination of power semiconductors, control electronics, and a medium-frequency transformer. Unlike conventional transformers that perform only voltage conversion, SSTs actively process power through multiple stages, enabling both AC and DC inputs/outputs, bidirectional power flow, voltage regulation, reactive power compensation, and intelligent grid control

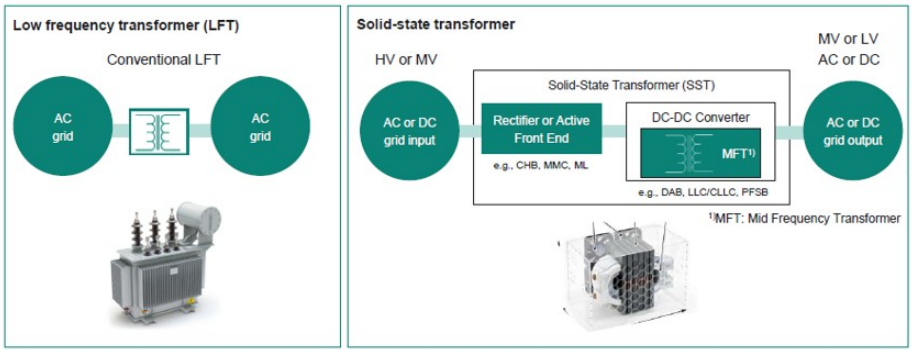
A SST is typically built around three major functional stages:

- The first is the medium-voltage AC-to-DC rectifier stage, which uses high-voltage semiconductor switches—often silicon carbide (SiC) or gallium nitride (GaN) devices due to their superior switching speed and thermal performance—to convert incoming grid-level medium voltage AC into a regulated DC bus.
- The second is the isolated DC-DC conversion stage, usually implemented as a dual active bridge (DAB) or LLC resonant converter, which provides galvanic isolation through a high-frequency transformer (operating in the KHz to tens-of-LHz range rather than 50/60 Hz), dramatically shrinking the magnetic core and transformer size compared to conventional units while stepping voltage down toward datacenter distribution levels.
- The third is the low-voltage DC-AC or DC-DC output stage, which conditions

power into the form needed by the facility—either regulated DC for direct DC-bus distribution to IT racks or inverted AC for legacy PDU/UPS infrastructure.

Supporting these power stages are the control and sensing subsystems (digital controllers, gate drivers, voltage/current sensors) that manage switching, fault protection, and grid synchronization; thermal management components (heat sinks, cold plates, or liquid cooling loops) to handle the concentrated losses in the semiconductor devices; and increasingly, integrated energy storage interfaces or DC microgrid ports that allow the SST to natively support battery buffering, renewable integration, and finer-grained power quality control than a passive transformer ever could.

Exhibit 56: Solid state vs. traditional transformers



Source: Infineon

Why do we need SSTs?

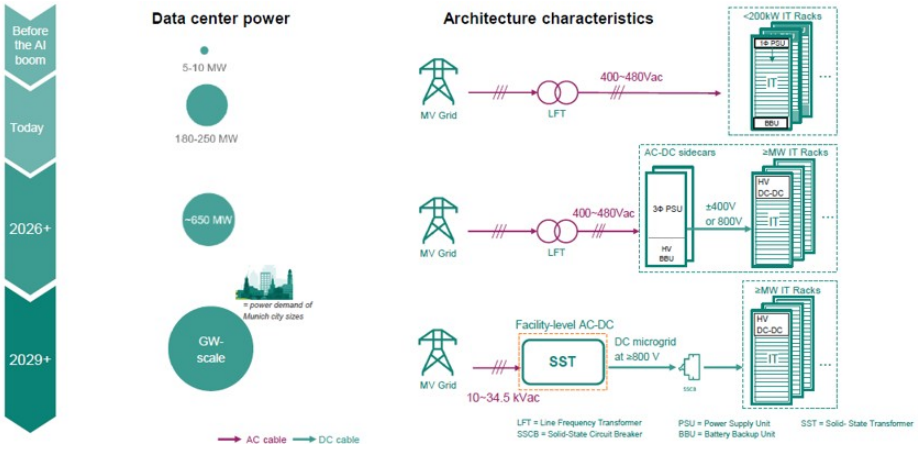
Improving efficiency in an era of gigawatt data centers

One of the strongest drivers for SST adoption is the rapid increase in AI data center power demand, which has progressed from 5-10MW facilities in the pre-AI era to campuses exceeding 650MW today, with gigawatt-scale facilities expected beyond 2029.

Traditional architectures use medium-voltage AC distribution, followed by low-frequency transformers and multiple power conversion stages before power reaches IT racks. SSTs allow facilities to convert 10-34.5KV medium-voltage power directly into ±400V or 800V DC distribution networks. This reduces conversion losses and improves overall grid-to-chip efficiency.

The efficiency advantage could be at the order of about 2-5ppt vs, the traditional transformer architecture, but at a 1GW AI campus, this magnitude of improvement could translate into roughly 20-50MW of continuous power savings. Over the lifetime of a facility, this can result in substantial reductions in electricity cost and carbon emissions. SSTs are therefore becoming increasingly attractive for hyperscaler operators seeking lower total cost of ownership (TCO).

Exhibit 57: SST enabling more efficient power delivery architecture



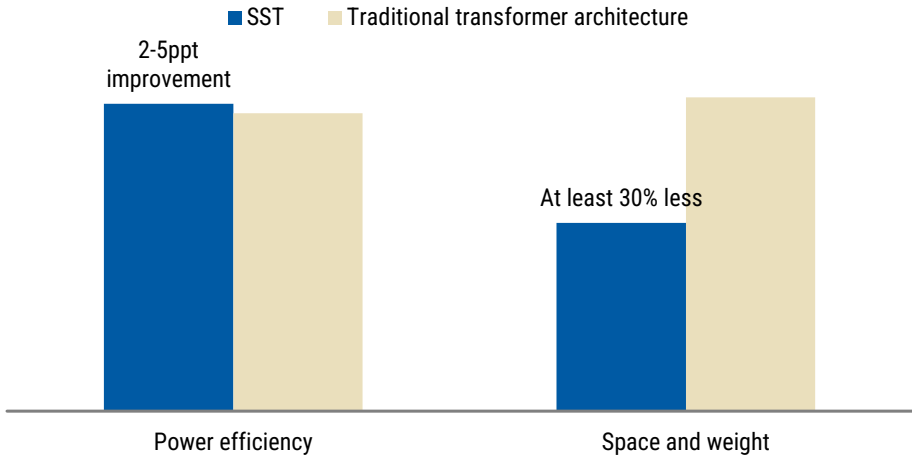
Source: Infineon

Higher power density and smaller footprint

Because SSTs use medium-frequency transformers instead of bulky 50/60Hz transformers, they can achieve significantly higher power density. Infineon indicates that SST systems can reduce both footprint and weight by at least 30% compared with conventional transformer-based installations. They also accelerate site energization and facilitate modular deployment

This becomes particularly valuable in urban data centers, utility substations, EV charging hubs, and energy storage sites where space constraints directly affect project economics.

Exhibit 58: SST offers better power efficiency with at least 30% less space/weight



Source: Infineon, Delta Electronics, Morgan Stanley Research

Supporting distributed energy resources

As renewable energy penetration increases, future grids will require seamless integration of solar generation, battery storage, fuel cells, gas turbines, and other distributed energy resources.

SSTs provide intelligent power routing, bidirectional operation, and dynamic voltage control. This allows them to function as energy hubs that connect renewable generation,

storage, and loads through a common DC infrastructure. Such capabilities are difficult or impossible to achieve using passive transformers alone.

Improved grid intelligence

Unlike conventional transformers, SSTs can actively regulate voltage and frequency, provide reactive power compensation, support grid-forming functionality, and improve overall power quality. These capabilities make SSTs an important enabler for future smart grids, microgrids, and virtual power plants.

What's the deployment status now?

Despite decades of research, SSTs remain at an early commercial deployment stage. While we started to see the SST models being unveiled by relevant suppliers and there start to be some pilot adoption, we believe that the 800VDC power rack/center/block architecture shall be deployed first, and SSTs will take more time, as more fundamental design change in the datacenter and all hardware inside will involved. We believe wider adoption probably won't happen before 2030.

Exhibit 59: SST from Delta

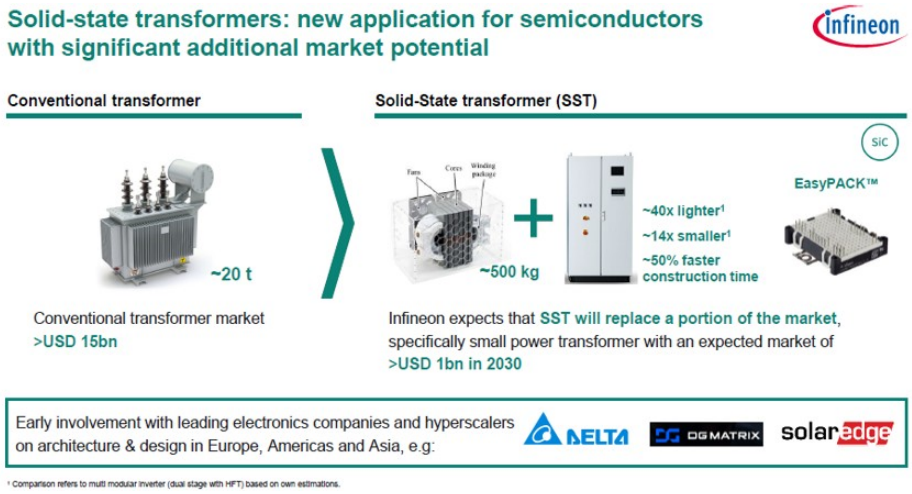
Source: Delta

Exhibit 60: SST from Eaton

Source: Eaton

According to Infineon, SST could represent a >US$1bn market opportunity by 2030, as the technology begins to replace a portion of the >US$15bn conventional transformer market, particularly in smaller-power applications. While SST penetration is expected to remain relatively modest versus the overall transformer market, the opportunity is significant for relevant players in the longer term.

Exhibit 61: SST to see initial adoption in 2030



Source: Infineon

Key hurdles for the wider adoption of SSTs

- **High system cost remains the primary barrier:** Despite their technical advantages, SSTs remain significantly more expensive than conventional transformers. Traditional transformers consist primarily of steel cores, copper windings, insulation materials, and cooling systems, all of which benefit from mature manufacturing processes and economies of scale. In contrast, SSTs require large numbers of high-voltage power semiconductors, gate drivers, digital controllers, sensors, protection systems, and sophisticated thermal management solutions. The extensive use of silicon carbide (SiC) devices, which remain substantially more expensive than conventional silicon-based components, further increases the overall system cost. Although SSTs can deliver efficiency gains, improved controllability, and reduced footprint, the economic benefit is often insufficient to offset the substantial upfront capital expenditure in many utility applications. Consequently, SST deployment is currently limited to applications where efficiency, power density, or system flexibility carry exceptionally high economic value, such as AI data centers, megawatt EV charging stations, and grid-scale battery storage systems.
- **Long term reliability has yet to be proven:** Electric utilities and large infrastructure operators place an extremely high value on reliability because transformers are expected to operate continuously for several decades with minimal maintenance. Conventional transformers routinely achieve service lifetimes exceeding 30 to 40 years and have accumulated more than a century of operating history. SSTs, however, are fundamentally different devices that incorporate thousands of semiconductor junctions, complex control electronics, communication networks, cooling systems, and software algorithms. While each individual technology may be mature, the long-term reliability of the complete SST system remains largely unproven at scale. Operators therefore remain cautious about replacing a passive asset with an active power-electronic system whose failure modes are more complex and less understood. For critical infrastructure such as substations, industrial facilities, or hyperscale data centers, reliability concerns often outweigh potential efficiency improvements.

- **DC protection technology ss still evolving:** One of the most significant technical challenges associated with SST deployment is fault protection, particularly in medium-voltage and high-voltage DC networks. Unlike AC systems, where current naturally crosses zero multiple times each cycle and facilitates fault interruption, DC currents continue flowing continuously during fault conditions. As a result, fault currents can rise extremely quickly, making them more difficult to detect and interrupt safely. Future SST-based power architectures therefore require advanced protection devices such as solid-state circuit breakers (SSCBs), which can react far faster than conventional mechanical breakers. However, DC protection technologies are still evolving, and there is currently no universally accepted approach equivalent to the mature protection schemes available in AC grids. The absence of standardized protection architectures increases deployment risk and remains one of the key obstacles to the wider adoption of SST-based DC distribution systems.
- **System complexity creates engineering, qualification and maintenance challenges:** Compared with traditional transformers, SSTs are considerably more complex systems that require the integration of multiple high-performance subsystems. A complete SST includes medium-voltage power converters, isolated DC/DC stages, high-frequency transformers, semiconductor modules, digital controllers, communication interfaces, and advanced protection mechanisms. As voltage and power levels increase, the number of converter modules and semiconductor devices required can rise significantly. Although the introduction of higher-voltage SiC devices can reduce subsystem count and simplify design, SST architectures remain substantially more complicated than conventional transformers. This complexity increases development costs, certification requirements, maintenance needs, and operational risks, all of which slow adoption among traditionally conservative utility customers.
- **Industry standards and ecosystems are still developing:** The transformer industry benefits from decades of established standards, testing procedures, maintenance practices, and supply chain infrastructure. SSTs, by comparison, represent an emerging technology with limited deployment history and evolving standards. Many utilities remain uncertain regarding qualification procedures, maintenance requirements, spare-parts strategies, and long-term asset management. Furthermore, widespread SST adoption will require parallel development of supporting technologies such as DC protection systems, DC switchgear, energy management software, and medium-voltage DC distribution networks. Until the broader ecosystem matures, many customers will continue to view SSTs as a promising but relatively high-risk alternative to proven transformer technologies.

Fuel Cells for Better Flexibility and Resilience

What are fuel cells?

Fuel cells are electrochemical energy-conversion devices that generate electricity by combining hydrogen and oxygen without combustion. Unlike conventional power generation technologies, which convert fuel into heat and then into electricity through multiple intermediate steps, fuel cells produce electricity directly through an electrochemical reaction, resulting in higher efficiency and significantly lower emissions.

When powered by hydrogen, the primary byproducts are water and heat, making fuel cells one of the cleanest forms of power generation available today.

Fuel cells can be deployed across a wide range of applications, including passenger vehicles, heavy-duty trucks, buses, trains, ships, stationary power generation, backup power systems, and increasingly data centers. As governments and industries pursue decarbonization while seeking reliable and dispatchable power sources.

Major fuel-cell technologies include:

- SOFC (Solid Oxide Fuel Cell): Preferred for stationary power and data centers.
- PEMFC (Proton Exchange Membrane Fuel Cell): Dominant in transportation and backup power.
- MCFC (Molten Carbonate Fuel Cell): Large-scale distributed generation.
- PAFC (Phosphoric Acid Fuel Cell): Commercial combined heat and power.
- DMFC (Direct Methanol Fuel Cell): Portable applications

Exhibit 62: Fuel cell offering - Delta

Source: Delta

Exhibit 63: Fuel cell offering - Bloom Energy

Source: Bloom Energy

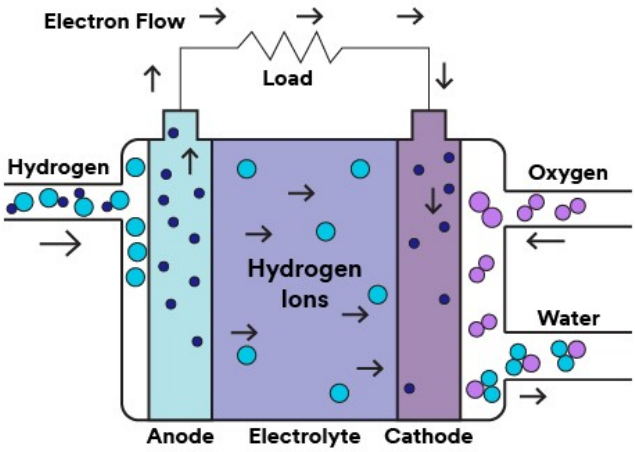
How do fuel cells work?

While there are several fuel-cell architectures, the fundamental operating principle is largely the same: fuel is supplied to the anode, oxygen is supplied to the cathode, and an electrolyte in between allows charged ions to move while forcing electrons to flow through an external circuit. This flow of electrons generates usable electricity, while the

reaction also produces heat and water as byproducts. Unlike conventional generators, which rely on burning fuel to drive a turbine or engine, fuel cells produce electricity directly through electrochemical reactions, allowing them to achieve significantly higher efficiency and lower emissions.

Using solid oxide fuel cell (SOFC) technology as an example, air is supplied to the cathode side of the cell, where oxygen molecules gain electrons and form oxygen ions (O^{2-}). These oxygen ions migrate through a ceramic electrolyte toward the anode side of the cell. At the same time, hydrogen or another hydrogen-rich fuel is supplied to the anode. When the oxygen ions reach the anode, they react with the fuel to produce water, heat, and electrons. The generated electrons cannot pass through the electrolyte and therefore travel through an external electrical circuit before returning to the cathode, creating a continuous flow of electricity. As long as fuel and air continue to be supplied, the fuel cell can generate power continuously.

Exhibit 64: Fuel cell diagram



Source: 3M

What benefits can fuel cells offer?

- **High energy efficiency:** Fuel cells convert the chemical energy of hydrogen directly into electricity through an electrochemical reaction, avoiding the multiple energy-conversion steps associated with combustion-based power generation. This allows fuel cells to achieve electrical efficiencies of over 60%, with some systems exceeding 80% total efficiency when waste heat is recovered through combined heat-and-power configurations, compared with roughly 35-45% for conventional gas-fired power generation. The efficiency advantage translates directly into lower fuel consumption, reduced operating costs, and improved energy utilization over the lifetime of the system.
- **Low carbon emissions:** When powered by hydrogen, fuel cells produce only water and heat as byproducts, generating no direct carbon emissions at the point of use. Even when operating on hydrogen derived from natural gas, fuel cells typically produce significantly lower greenhouse gas emissions than conventional combustion-based power systems. As green hydrogen production scales, fuel cells offer a potential pathway toward near-zero-carbon power generation for transportation, industrial facilities, and distributed energy systems.

- **Grid independence and energy security:** Fuel cells can provide reliable on-site power generation independent of the electrical grid, improving energy security and resilience. This makes them attractive for mission-critical applications such as data centers, hospitals, telecommunications infrastructure, defense installations, and industrial facilities that require uninterrupted power. In regions facing grid congestion or reliability concerns, fuel cells can also serve as a distributed power source located directly at the point of consumption.
- **Long duration energy storage enabler:** Fuel cells play an important role in the emerging hydrogen economy by enabling excess renewable energy from solar and wind farms to be stored as hydrogen and later converted back into electricity. Unlike batteries, which become increasingly expensive for longer storage durations, hydrogen storage can economically provide energy over days, weeks, or even seasonal periods. This capability could become increasingly valuable as renewable energy penetration rises worldwide.
- **Fuel flexibility:** Certain fuel-cell technologies, particularly Solid Oxide Fuel Cells (SOFCs), can operate on a variety of fuels including hydrogen, natural gas, biogas, ammonia-derived hydrogen, and other hydrogen-rich fuels. This flexibility allows users to deploy fuel-cell systems today using existing fuel infrastructure while maintaining a pathway toward pure hydrogen operation as hydrogen supply chains mature.
- **Quiet operation and reduced maintenance:** Because fuel cells contain relatively few moving parts compared with combustion engines and turbine-based generators, they operate quietly and often require less routine maintenance. The absence of combustion also reduces mechanical wear and vibration, which can improve system reliability and lower lifecycle operating costs, particularly in stationary power applications.
- **Scalability across multiple applications:** Fuel cells can be deployed across a wide range of power levels, from portable power systems measured in watts to utility-scale installations generating multiple megawatts. This scalability allows the technology to serve residential buildings, commercial facilities, industrial plants, transportation platforms, microgrids, and large-scale data centers, making fuel cells one of the most versatile clean-energy technologies available today.

Limitations and constraints of fuel cells

- **High system cost:** Despite significant technological progress, fuel cells remain more expensive than many conventional power-generation technologies. The cost challenge stems from complex stack manufacturing, specialized materials, balance-of-plant components, and relatively low production volumes compared with mature technologies such as internal combustion engines, gas turbines, and batteries. While costs have declined over the past decade, achieving widespread commercial adoption will likely require further manufacturing scale, supply-chain development, and technological improvements.
- **Limited hydrogen infrastructure:** One of the biggest challenges facing fuel-cell deployment is the lack of hydrogen production, transportation, storage, and refueling infrastructure. Unlike electricity, which is already widely distributed through established grids, hydrogen supply networks remain in the early stages of

development in most regions. This infrastructure gap creates a chicken-and-egg challenge, where end users are reluctant to adopt fuel cell technologies without reliable hydrogen availability, while infrastructure developers are hesitant to invest until demand reaches sufficient scale.

- **Durability and lifetime requirements:** Although fuel-cell durability has improved substantially, long-term reliability remains a key focus area, particularly for commercial transportation and utility-scale installations. Continuous exposure to heat, humidity, contaminants, and repeated operating cycles can gradually degrade fuel cell performance over time. Customers in sectors such as transportation, power generation, and industrial infrastructure often require operating lifetimes measured in tens of thousands of hours, creating significant engineering challenges for system developers.
- **Regulatory and safety considerations:** Hydrogen is a highly flammable gas and requires specialized handling, storage, and transportation systems. Although hydrogen technologies have well-established safety standards, new fueling stations, pipelines, and storage facilities must comply with stringent regulations. Developing these standards and obtaining approvals can extend deployment timelines and increase project costs.
- **Technology fragmentation and lack of standardization:** The fuel cell industry encompasses multiple technology platforms, including PEM, SOFC, MCFC, and other architectures, each optimized for different applications. While this diversity encourages innovation, it can also slow industry standardization and create uncertainty for customers evaluating long-term technology choices. Unlike the battery industry, which is increasingly converging around a few dominant chemistries, the fuel cell sector remains relatively fragmented.

Implications to datacenter power architecture

Transition from backup power to primary power source

Traditionally, data centers rely on grid electricity for primary power and diesel generators for backup power during outages. Fuel cells are increasingly being evaluated not only as a replacement for diesel backup generators but also as a potential primary power source. As AI-driven workloads substantially increase power density and overall electricity demand, hyperscalers are exploring fuel cells as a means of deploying computing capacity in locations where grid power is constrained or unavailable. This could fundamentally shift data-center power architecture from a grid-centric design to one that incorporates distributed on-site generation.

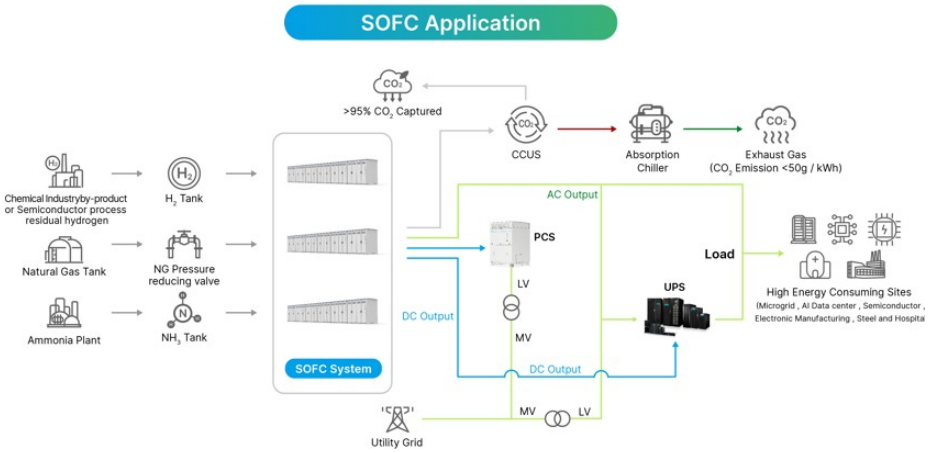
Potential long term evolution toward microgrid-based datacenters

In the longer term, fuel cells could become a core component of data-center microgrids combining renewable generation, battery storage, hydrogen storage, and intelligent energy management systems. Rather than functioning as passive consumers of grid electricity, future AI-focused data centers may increasingly operate as self-contained energy ecosystems capable of managing their own generation, storage, and load balancing. Such an architecture could provide greater scalability, resilience, and energy independence than traditional grid-connected designs

Potential simplification of power distribution architecture

Fuel cells inherently generate DC power, which aligns more closely with the power requirements of modern servers and IT equipment. Current data centers typically involve multiple AC/DC and DC/DC conversion stages between the utility grid and the processor, each introducing efficiency losses. Over time, fuel cell adoption could encourage a greater migration toward direct DC power distribution architectures, reducing conversion losses and improving overall facility efficiency. Moreover, its DC output fits well with the shift into HVDC power delivery architecture for datacenters.

Exhibit 65: Fuel cells in datacenter power delivery architecture



Source: Delta

Risk Reward – Delta Electronics Inc. (2308.TW)

Strong combination: Power + cooling for AI infrastructure

PRICE TARGET NT$2,500.00

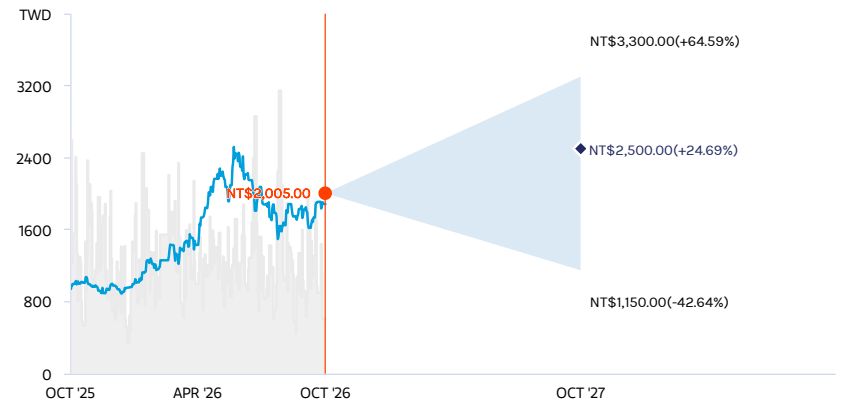
Base case: P/E multiple of 30x based on average 2027-28 EPS. We believe our target multiple is justified by the 63% earnings CAGR we estimate for 2025-28, implying a PEG of 0.5, versus the stock's ~23x P/E and 23% earnings CAGR during 2022-25, which implied a PEG of 1.0.

Consensus Price Target Distribution NT$2,200.00 NT$2,621.29 MS PT NT$4,120.00

Source: Refinitiv, Morgan Stanley Research

◆ Mean ◆ Morgan Stanley Estimates

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We view AI server power supplies and power management/cooling solutions for data centers as key growth drivers for the coming decade.
- Our expectation for a re-rating reflects expanding gross/operating margins and increasing exposure to AI-related businesses.
- Our 2026-28 earnings estimates reflect strength in AI server power supplies and liquid cooling.
- Our price target implies a P/E of 30x based on average 2027-28 EPS.

Consensus Rating Distribution

100%	Overweight
0%	Equal-weight
0%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth:	*Positive*
Self-help:	*Positive*
Technology Diffusion:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$3,300.00

40x average 2027-28e P/E

Strong sales growth with significant margin expansion: We assume a greater-than-expected contribution from higher-margin businesses such as power electronics and liquid cooling as well as greater synergies from acquired companies.

BASE CASE NT$2,500.00

30x average 2027-28e P/E

Strong YoY revenue growth in 2026-28, coupled with operating leverage: We expect steady growth in AI server power supply and data center infrastructure to be the key drivers. We see favorable mix and scale merits driving profit to grow more than double in 2027 vs. 2025.

BEAR CASE NT$1,150.00

14x average 2027-28e P/E

Moderate sales growth with slight margin improvement: We assume further profit drag from legacy products, smaller profit contribution from higher-margin businesses, higher opex and/or less meaningful contribution from acquired companies.

Risk Reward – Delta Electronics Inc. (2308.TW)

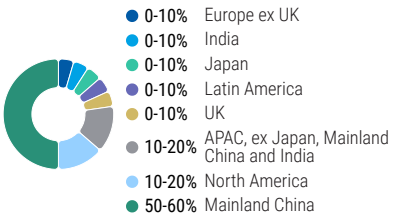
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Power Electronics YoY (%)	25.0	48.9	52.6	51.8
Automation YoY (%)	4.8	10.9	5.2	5.2
Infrastructure YoY (%)	81.6	44.6	51.0	48.2

INVESTMENT DRIVERS

- Transition to EV-related business and data infrastructure
- Operating margin recovery
- FINI holdings

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster-than-expected global economic growth
- Falling raw material prices
- Margin improvement

RISKS TO DOWNSIDE

- Global economic slowdown
- Ongoing component supply tightness
- Rising raw material prices
- Margin contraction as a result of higher operating expenses or faster ASP erosion

OWNERSHIP POSITIONING

Inst. Owners, % Active 64.7%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	746,914	775,166 / 774,709	807,877
EBITDA (NT$, mn)	163,606	169,213 / 175,311	191,720
Net income (NT$, mn)	96,767	103,307 / 105,731	111,870
EPS (NT$)	37.25	39.77 / 40.50	43.07

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Financial Summary – Delta

Exhibit 66: Delta: Financial Summary

Consolidated Income Statements

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Net sales	**421,148**	**554,885**	**775,166**	**1,131,877**	**1,659,520**
COGS	(284,567)	(364,729)	(494,374)	(707,257)	(1,028,632)
Gross profit	136,580	190,157	280,792	424,620	630,887
Operating expenses	(88,928)	(106,224)	(142,491)	(191,135)	(263,905)
Operating income	**47,652**	**83,932**	**138,300**	**233,484**	**366,983**
Non-operating income	**3,664**	**3,934**	**10,365**	**7,100**	**7,300**
Interest income	1,859	1,501	1,687	2,100	2,300
Investment income	(162)	(200)	372	-	-
Exchange gain	1,041	644	2,091	-	-
Other	926	1,989	6,215	5,000	5,000
Pre-tax income	**51,316**	**87,866**	**148,666**	**240,584**	**374,283**
Income tax	(10,925)	(19,930)	(32,925)	(52,189)	(81,172)
Minority interests	(5,163)	(7,828)	(12,433)	(20,158)	(31,363)
Net income	**35,229**	**60,108**	**103,307**	**168,237**	**261,747**
Modelware EPS (NT$)	**13.56**	**23.14**	**39.77**	**64.77**	**100.77**

Consolidated Cash Flow Statements

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Operating CF	**72,895**	**98,474**	**116,226**	**177,240**	**263,652**
Net Profits	35,229	60,108	103,307	168,237	261,747
Depreciation & Amort.	25,016	27,839	30,912	33,985	37,058
Change in WC	(2,486)	(24,190)	(55,485)	(86,295)	(125,847)
Other adjustments	15,137	34,717	37,491	61,313	90,694
Investing CF	**(40,360)**	**(53,214)**	**(70,000)**	**(70,000)**	**(60,000)**
Change of FA (Capex)	(33,435)	(45,950)	(70,000)	(70,000)	(60,000)
Change of L/T invest.	(4,739)	-	-	-	-
Other adjustments	(1,644)	(3,517)	-	-	-
Financing CF	**(11,118)**	**(9,401)**	**36,107**	**(12,414)**	**(54,050)**
Increase in L/T debt	8,552	(11,308)	12,530	2,978	3,829
Increase in S/T debt	(1,417)	446	29	212	272
Issuance of stock	-	-	-	-	-
Cash dividends	(18,696)	(20,313)	(30,131)	(51,787)	(84,335)
Other adjustments	444	21,774	53,680	36,183	26,183
FX adjust.	8,586	(2,146)	-	-	-
Net change in cash	**30,003**	**33,713**	**82,333**	**94,826**	**149,602**

Consolidated Balance Sheets

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Cash	**117,459**	**151,172**	**233,505**	**328,332**	**477,934**
Mkt securities	4,333	3,941	3,941	3,941	3,941
A/R	89,921	121,717	170,036	248,282	364,023
Inventory	83,856	101,478	142,217	205,395	298,726
Others	15,355	17,473	24,410	35,642	52,257
Current Assets	**310,925**	**395,781**	**574,109**	**821,592**	**1,196,881**
Long-term investments	4,740	4,876	4,876	4,876	4,876
Fixed assets	115,710	142,040	127,820	127,652	124,410
Other assets	100,523	96,922	96,922	96,922	96,922
Total Assets	**531,898**	**639,619**	**803,727**	**1,051,042**	**1,423,089**
S/T borrowings	1,918	2,364	2,393	2,605	2,877
AP/NP	69,223	94,452	128,025	183,154	266,379
Other ST liabilities	81,142	112,849	157,649	230,195	337,504
Total current liabilities	**152,283**	**209,665**	**288,067**	**415,954**	**606,759**
L/T debt	32,209	21,143	33,673	36,650	40,480
Other LT liabilities	67,840	83,422	83,422	83,422	83,422
Total Liabilities	**252,332**	**314,230**	**405,162**	**536,026**	**730,661**
Common shares	25,975	25,975	25,975	25,975	25,975
Retained Earnings	102,429	140,925	214,101	330,552	507,964
Other SH equity	151,161	158,489	158,489	158,489	158,489
Shareholders' equity	**279,565**	**325,389**	**398,565**	**515,016**	**692,428**
Total Liab./SE	**531,898**	**639,619**	**803,727**	**1,051,042**	**1,423,089**

Consolidated Financial Ratios

	2024	2025	2026E	2027E	2028E
Margins (%)					
Gross margin	32.4	34.3	36.2	37.5	38.0
Operating margin	11.3	15.1	17.8	20.6	22.1
Pretax margin	12.2	15.8	19.2	21.3	22.6
Net margin	8.4	10.8	13.3	14.9	15.8
YoY growth (%)					
Sales	5	32	40	46	47
Operating profits	16	76	65	69	57
Pretax profits	5	71	69	62	56
Net profits	5	71	72	63	56
Others					
Cash div payout (%)	52%	50%	50%	50%	50%
Adjusted cash div (NT$)	7.0	11.6	19.9	32.5	50.5
Yield (%)	0%	1%	1%	2%	3%
Net Debt/Equity (%)	-31%	-40%	-51%	-57%	-63%
Liabilities/Equity (%)	90%	97%	102%	104%	106%
Liabilities/Assets (%)	47%	49%	50%	51%	51%
ROAE (%)	13%	20%	29%	37%	43%
ROAA (%)	7%	10%	14%	18%	21%
AR/NR days	73	70	69	67	67
Inventory days	103	93	90	90	89
AP/NP days	79	82	82	80	80
Cash conversion	97	80	77	77	77

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 67: Delta: Quarterly Financials

Delta (2308.TW) (NT$ mn)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E
Net sales	**159,353**	**183,256**	**201,379**	**231,177**	**231,791**	**265,526**	**302,930**	**331,630**	**554,885**	**775,166**	**1,131,877**
COGS	(100,392)	(117,948)	(129,404)	(146,630)	(146,821)	(166,230)	(188,191)	(206,015)	(364,729)	(494,374)	(707,257)
Gross profit	**58,961**	**65,308**	**71,976**	**84,547**	**84,970**	**99,295**	**114,739**	**125,615**	**190,157**	**280,792**	**424,620**
Operating expenses	(30,544)	(34,738)	(36,983)	(40,226)	(40,422)	(45,405)	(50,589)	(54,719)	(106,224)	(142,491)	(191,135)
SG&A	(16,886)	(18,956)	(20,208)	(21,963)	(22,067)	(24,959)	(27,870)	(30,178)	(57,482)	(78,014)	(105,074)
R&D exp.	(13,657)	(15,782)	(16,775)	(18,263)	(18,356)	(20,445)	(22,720)	(24,541)	(48,743)	(64,477)	(86,061)
Operating income	**28,417**	**30,570**	**34,992**	**44,321**	**44,548**	**53,891**	**64,149**	**70,896**	**83,932**	**138,300**	**233,484**
Non-operating income	**2,676**	**4,430**	**1,630**	**1,630**	**1,775**	**1,775**	**1,775**	**1,775**	**3,934**	**10,365**	**7,100**
Interest income	428	398	430	430	525	525	525	525	1,501	1,687	2,100
Investment income	487	(115)	-	-	-	-	-	-	(200)	372	-
Disposal of investment	-	-	-	-	-	-	-	-	-	-	-
Exchange gain	1,688	403	-	-	-	-	-	-	644	2,091	-
Other	72	3,743	1,200	1,200	1,250	1,250	1,250	1,250	1,989	6,215	5,000
Pre-tax income	**31,093**	**35,000**	**36,622**	**45,951**	**46,323**	**55,666**	**65,924**	**72,671**	**87,866**	**148,666**	**240,584**
Income tax	(7,258)	(7,831)	(7,910)	(9,925)	(10,423)	(11,968)	(14,174)	(15,624)	(19,930)	(32,925)	(52,189)
Minority interest	(3,279)	(2,033)	(3,158)	(3,963)	(3,841)	(4,676)	(5,537)	(6,104)	(7,828)	(12,433)	(20,158)
Net income	**20,556**	**25,136**	**25,554**	**32,063**	**32,059**	**39,022**	**46,213**	**50,943**	**60,108**	**103,307**	**168,237**
EPS (NT$)	**7.91**	**9.68**	**9.84**	**12.34**	**12.34**	**15.02**	**17.79**	**19.61**	**23.14**	**39.77**	**64.77**
Margins (%)											
Gross Margin	37.0	35.6	35.7	36.6	36.7	37.4	37.9	37.9	34.3	36.2	37.5
Operating Margin	17.8	16.7	17.4	19.2	19.2	20.3	21.2	21.4	15.1	17.8	20.6
Pretax Margin	19.5	19.1	18.2	19.9	20.0	21.0	21.8	21.9	15.8	19.2	21.3
Net Margin	12.9	13.7	12.7	13.9	13.8	14.7	15.3	15.4	10.8	13.3	14.9
QoQ Growth (%)											
Sales	(1)	15	10	15	0	15	14	9			
Gross Profit	5	11	10	17	1	17	16	9			
Operating Profit	8	8	14	27	1	21	19	11			
Pretax Profit	21	13	5	25	1	20	18	10			
Net Profit	19	22	2	25	(0)	22	18	10			
YoY Growth (%)											
Sales	34	48	34	43	45	45	50	43	32	40	46
Gross Profit	56	48	37	51	44	52	59	49	39	48	51
Operating Profit	102	64	41	68	57	76	83	60	76	65	69
Pretax Profit	99	79	36	79	49	59	80	58	71	69	62
Net Profit	101	80	37	85	56	55	81	59	71	72	63

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

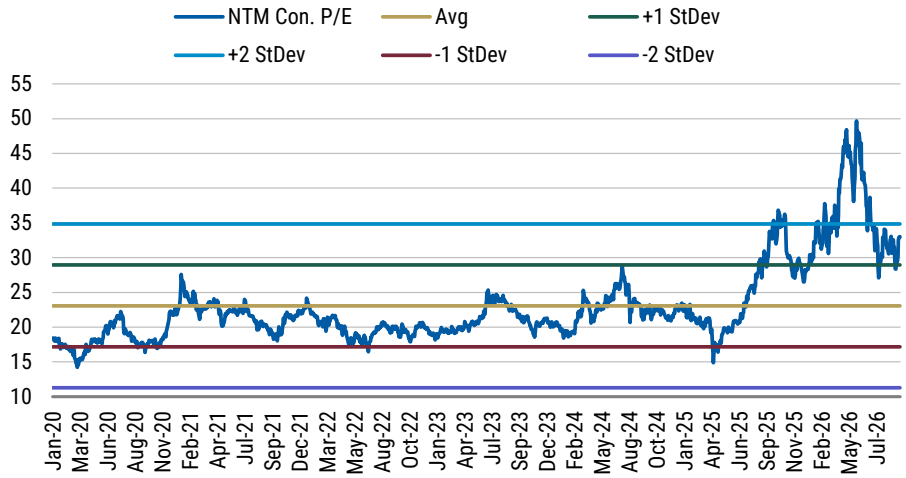
Valuation Methodology – Delta

Our price target (base case scenario value) is NT$2,500, derived from 30x 2027-28e P/E: We use price/earnings as our valuation methodology because we believe it appropriately captures Delta's earnings growth potential. We believe our target P/E multiple of 30x is justified by an earnings CAGR of 63% over 2025-28, which implies a PEG of 0.5. This is lower than the average of 1.0 in 2022-25, when the stock was trading at an average P/E of 23x for a 23% earnings CAGR.

Bull case scenario value – NT$3,300: We apply a 40x P/E multiple to our average base case estimate for 2027-2028. This scenario assumes a greater-than-expected contribution from higher-margin businesses such as power electronics and liquid cooling as well as greater synergies from acquired companies.

Bear case scenario value – NT$1,150: We apply a 14x P/E multiple to our average base case estimate for 2027-2028. This scenario assumes further profit drag from legacy products, smaller profit contribution from higher-margin businesses, higher opex and/or less meaningful contribution from acquired companies.

Exhibit 68: Delta: NTM P/E chart



Source: FactSet consensus estimates, Morgan Stanley Research

Why Does Delta Trade at a Significant Valuation Discount to Delta Thailand?

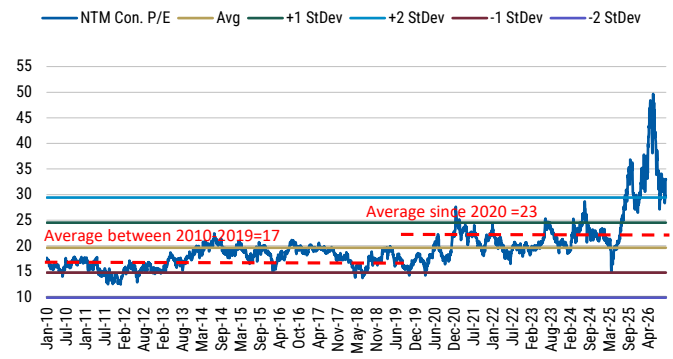
Discussion of the valuation gap between Delta Electronics (2308.TW, currently 33x 12M forward consensus P/E) and Delta Electronics Thailand (DELTA.TH, 71x currently) has been ongoing for several years. If we look at their valuations before 2020, Delta was actually trading at a premium, with an average 17x 12M forward P/E in 2010-2020 vs. Delta Thailand's 13x. This made sense, as Delta is much more competitive in the global arena and Delta Thailand was set up as a subsidiary focused on manufacturing.

The situation flipped after 2020. Though Delta has been re-rated to trade at an average 23x 12M forward P/E since 2020, Delta Thailand has seen its valuation surge to an average of 60x over the same period. This cannot be explained by moves in their respective markets, either, as TWSE went from 14x forward P/E in 2010-20 to 16x after 2020, while the SET went from 13x to 16x.

We think this could be attributable to a few factors:

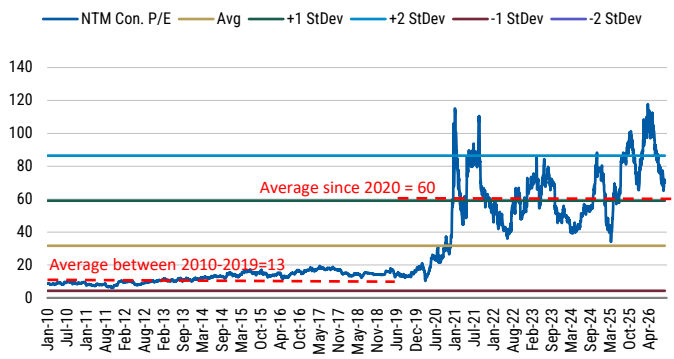
- **Different positioning within the Delta group:** Delta group has increased its stake in Delta Thailand to over 60% from about 20% in 2019 through a tender offer. Since then, there has been much more collaboration between Delta group and Delta Thailand. As Thailand emerged as one of the major manufacturing hubs within the Delta group, its earnings growth (28% in 2020-25) outpaced that of Delta Taiwan (19%).
- **Scarcity in the Thai stock market:** Delta Thailand is one of the relatively few large-cap Thai-listed companies that offer direct exposure to AI/data-center power and EV-related themes, whereas there are many such alternatives in Taiwan.
- **Limited free float:** As the Delta group owns a ~62% stake in Delta Thailand and there have been a few quite stable long-term shareholders, we think the effective free float is probably 20% or less.

Exhibit 69: Delta NTM P/E



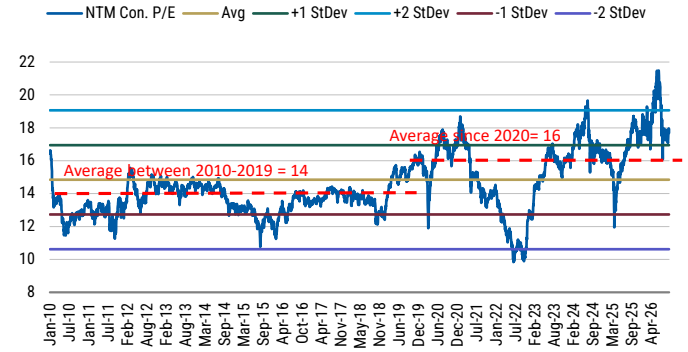
Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 70: DET NTM P/E



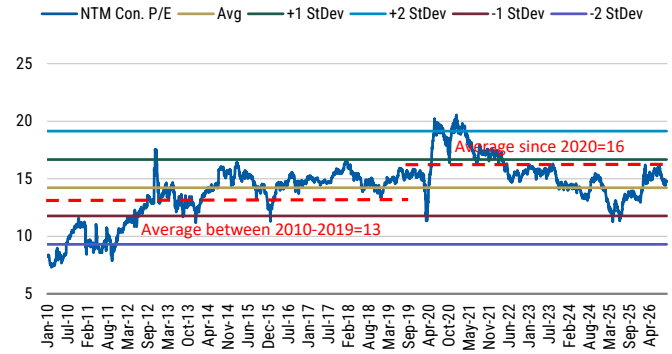
Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 71: TWSE NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

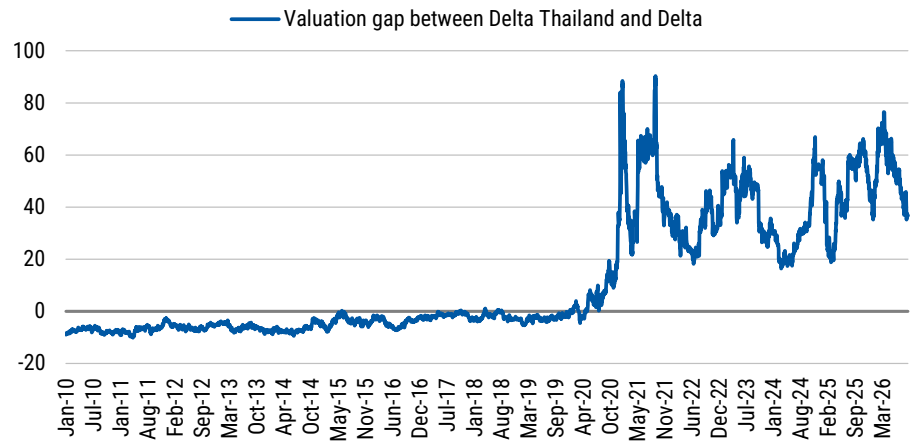
Exhibit 72: SET Index NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 73:

Valuation gap between Delta Thailand and Delta based on 12M forward P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Global Footprint – Delta

Comprehensive Global Operations

Aggressive capacity expansion plans

The company has been continuously diversifying its production base away from China owing to the trade turmoil back in 2018-2019, and then by supply chain disruptions during Covid. Management explicitly set a target of lowering China production to 60% of total capacity (from >70% previously) for the first time back in 2020. New capacity was added in Thailand, India, and Taiwan during 2020-2021. It then saw demand surge from various sectors, including EVs, automation and telecom, during 2022-23, followed by AI-related power suppliers and cooling solutions emerging in 2024. In order to grasp the emerging opportunities, relevant plant build-outs were planned accordingly, on top of the ongoing geographical diversification. It expanded its presence—production sites, R&D, and sales teams—in Europe and the US to better cope with operational disruptions and capture domestic demand, in addition to its existing setups in Taiwan, Thailand, and India. There was then a substantial increase in capex of NT$12.2bn in 2025 (vs. NT$5.6-6bn in 2023-24) as the company further accelerated its expansion plan, mainly on AI-related power/cooling demand and increasing capacity tightness, specifically in Thailand and the US. Delta continued to expand in Thailand by adding three new plants, with another three under construction for power and liquid cooling system offerings. In view of the capacity tightness reserved for AI-related business, it also started to prioritize margin-accretive business—i.e., de-emphasize business with net margins below 10%—which further optimized product mix.

Looking ahead into 2026 and beyond, expansion is likely to remain aggressive in order to fulfill substantial power and liquid cooling turnkey solution demand for large-scale data centers, while management also indicated that the new capacity added at the end of 2025 has already been fully taken. With capacity staying tight for AI power and cooling offerings, the company will continue to expand actively in China, Thailand, the US, and Tainan, Taiwan. It also plans to add mass production lines in Taoyuan, scheduled to be ready by 2028, in addition to the pilot run lines in Tainan. Against this backdrop, the capex budget has been revised up several times and is now roughly NT$50bn for 2026.

Exhibit 74: Delta's global footprint

Source: Company data, Morgan Stanley Research. Note: Data as of 2024.

Financial Analysis – Delta

44% Revenue CAGR in 2025-28e

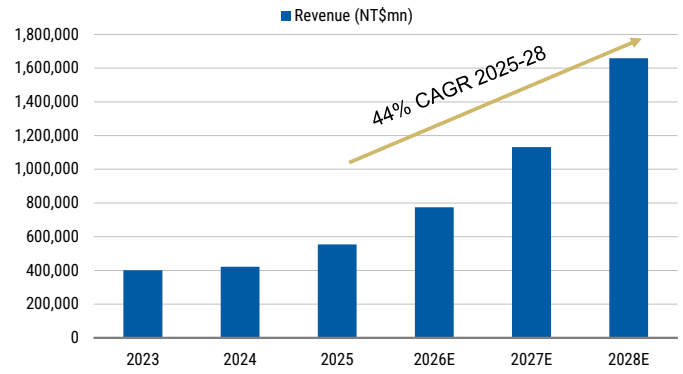
Delta's revenue growth was moderate in 2020 at merely 5%, making it the only year of the Covid disruption (2019-22) that did not see double-digit growth. Growth picked up to 11% in 2021, underpinned by surging demand for cloud, networking, and work-from-home infrastructure amid the pandemic, while the next uplift to 22% growth in 2022 was mainly fueled by increasing demand for EV- and data center-related components (server power supplies, networking switches, energy storage systems, etc). Revenue momentum subsequently moderated in 2023, growing just 4%, as the EV component business was impacted by the UAW strike in the US and automation demand in China stayed subdued, despite sustained demand for AI server power supplies and modules. Growth in 2024 was at a similar level, as strength in data center offerings were not enough to offset weakness in its legacy businesses.

Revenue surged 32% in 2025 to a record NT$554,885mn, driven by AI server rack power and cooling solutions, along with further spec upgrades and value growth.

We expect revenue to expand at a 44% CAGR over 2025-28, supported by ongoing AI server shipments, spec upgrades, 800 VDC power solutions, and turnkey liquid cooling offerings. As the upgrade to 800V HVDC power architecture is likely to take years given the complex design and high safety requirements, we expect AI server power solution upgrades to be separated into a few stages – from power shelves in AI server racks to standalone side-car power racks, from 50V DC to 800V DC output, from 72kW per power shelf to >250kW, etc, as well as power wattage increases for next-generation GPU rack designs.

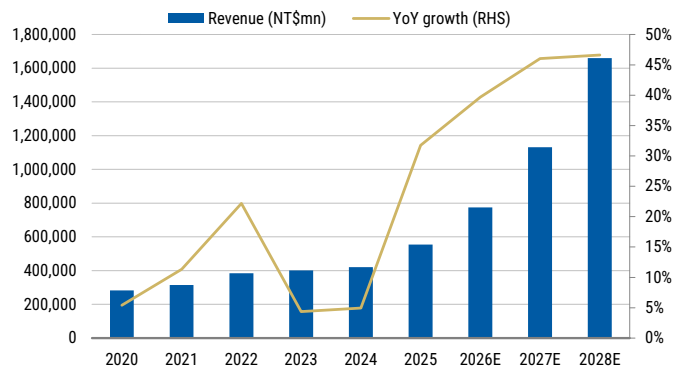
By segment, Power Electronics has been the primary revenue driver, contributing at least 50% over the past years despite the company separating the Mobility segment from Power Electronics since 2024 so as to further distinguish revenue generated from EVs/ autos. We expect the segment to remain the key revenue contributor, supported by strong demand from AI server-related power solutions, and estimate it will reach 58% of revenue by 2028 vs. 50% in 2025. Infrastructure, the second-largest revenue contributor historically, is expected to grow along with the Power Electronics segment, fueled by demand for data center power-related solutions (i.e. UPSs, CDUs, sidecars, etc) as well as energy infrastructure-related offerings (i.e. SSTs (solid-state transformers), fuel cells, modular power solutions, micro grid management, etc.). We model the segment's revenue contribution to reach 35% by 2028 from 24% in 2024 and 33% in 2025. For Automation and Mobility, we expect their revenue contributions to decline sequentially throughout the years, with Automation dropping from 10% in 2025 to 4% in 2028 and Mobility falling from 7% in 2025 to 2% in 2028.

Exhibit 75: Delta: Revenue CAGR, 2025-28e



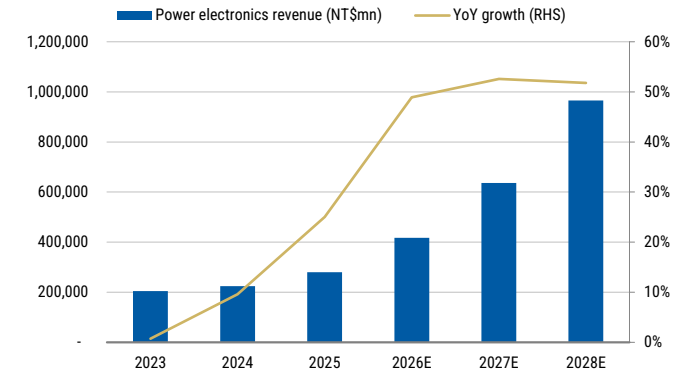
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 76: Delta: Annual revenue growth



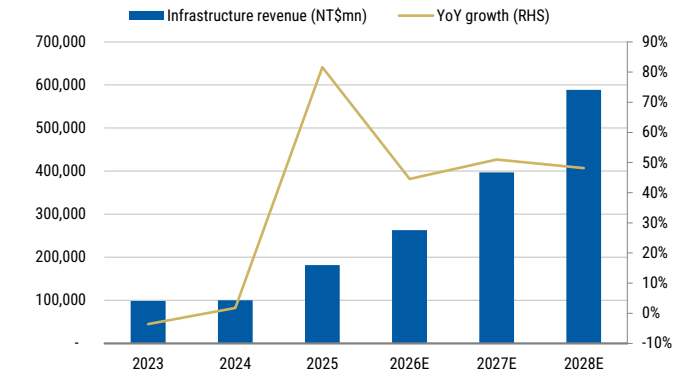
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 77: Delta: Power electronics revenue



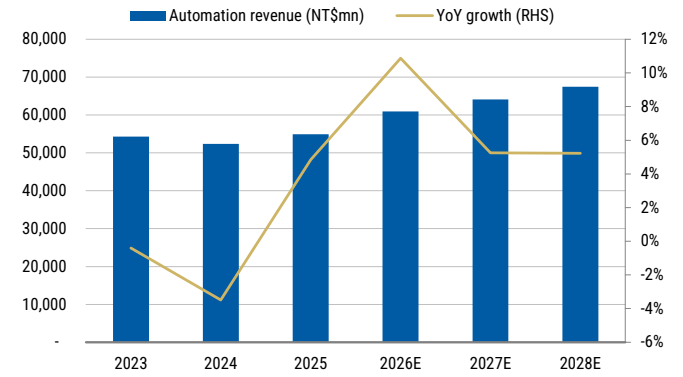
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 78: Delta: Infrastructure revenue



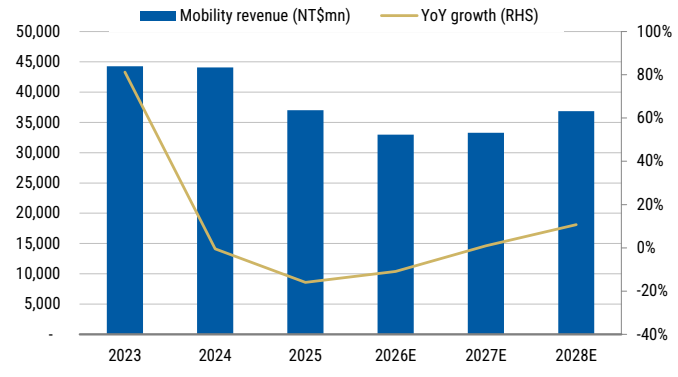
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 79: Delta: Automation revenue



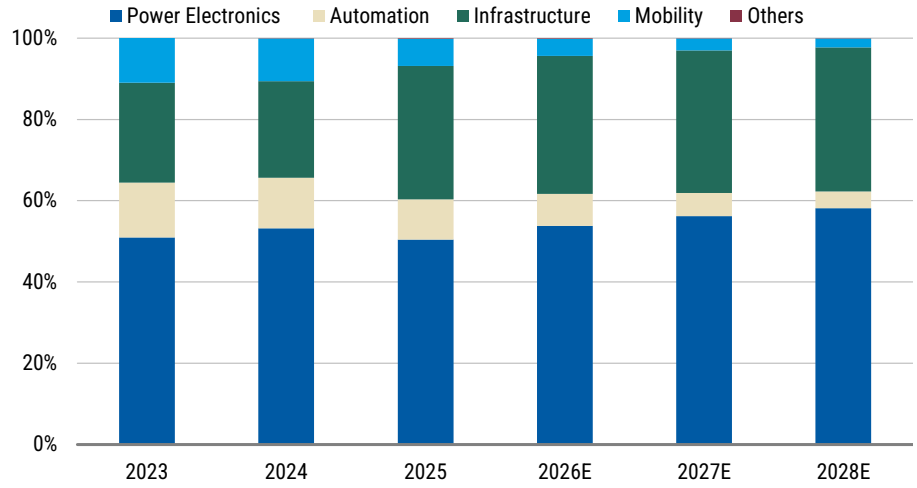
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 80: Delta: Mobility revenue



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 81:
Delta: Revenue breakdown by segment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

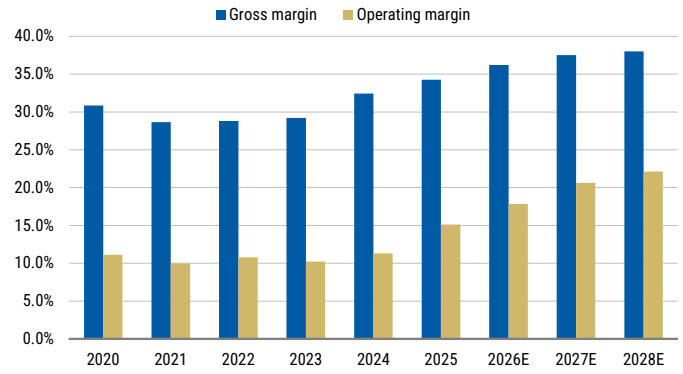
Continuous margin expansion on favorable product mix

Delta's gross margin stayed largely between 28-29% during 2021-23, as revenue was mainly driven by relatively lower-margin products, including EV- and consumer/IT-related products, while server-related offerings had yet to scale up. 2024 served as an inflection point, with the product mix changing structurally, supported by robust demand for AI data center-related offerings, while its legacy businesses stayed relatively weak. Gross margin thus grew by 3.2ppt YoY during the year, from 29.2% in 2023 to 32.4% in 2024. Increasing AI computing needs further increased power consumption and cooling requirements per rack, boosting demand for high-density AI power and cooling products along with related system-level solutions in 2025, which lifted gross margin by another 1.8ppt YoY to 34.3% for the year.

The opex ratio was largely stable, fluctuating around 18-19% between 2020 and 2023. It then increased to 19% in 2023 and 21.1% in 2024, mainly due to operating deleverage, higher R&D costs, and ongoing capacity expansion. The ratio declined back to 19.1% in 2025 on the back of an enlarged revenue base, despite R&D-related investments remaining elevated.

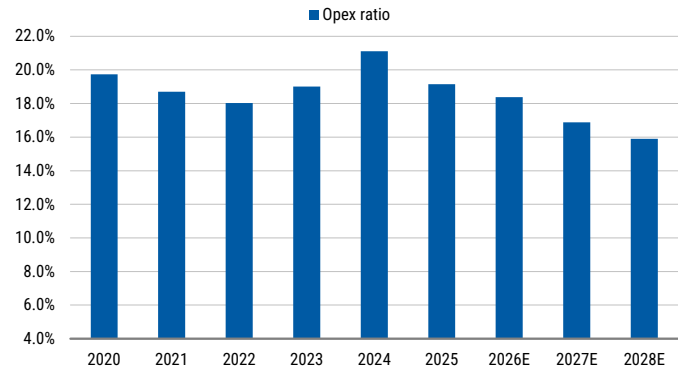
Looking ahead, we expect gross margin to continue expanding on the back of increasing revenue contribution from AI-related power supplies. We thus pencil in a 2.0ppt gross margin expansion to 36.2% in 2026, followed by further increases of 1.3ppt and 0.5ppt YoY in 2027 and 2028, respectively, reaching 38.0% by 2028. On the opex ratio, we expect Delta to continue benefiting from operating leverage in the coming years, and thus estimate operating margin to improve more substantially than gross margin, expanding by 2.7ppt YoY in 2026 to 17.8%, followed by further expansions of 2.8ppt and 1.5ppt in 2027 and 2028, reaching 20.6% and 22.1%, respectively.

Exhibit 82: Delta: Margin trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 83: Delta: Opex ratio



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Net profit to grow at a 63% CAGR in 2025-28

Delta's operating profit grew sequentially throughout 2020-22, supported by gross margin expansion and a lower opex ratio, from NT$31,426mn in 2020 to NT$41,439mn in 2022. It then saw a slight 1% YoY decline in 2023 due to higher operating expenses as the company further stepped up its R&D investments, despite a 0.4ppt gross margin expansion in 2023. Operating margin also edged down accordingly by 0.6ppt to 10.2% during the year.

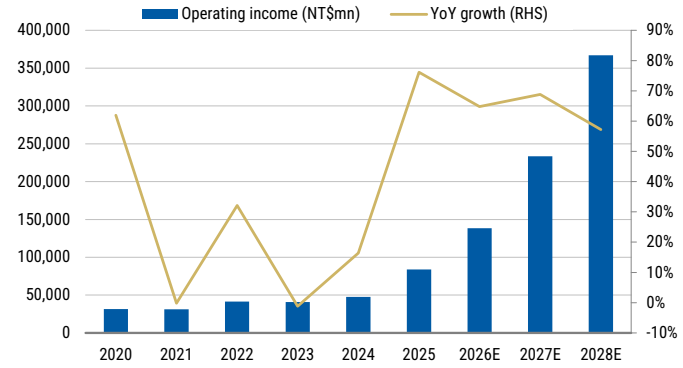
Operating profit returned to a growth trajectory from 2024 onward, growing 16% YoY in 2024 and 76% YoY in 2025, as AI-related power offerings started to scale up. Operating margin expanded faster than gross margin, supported by operating leverage and disciplined cost controls, increasing by 1.1ppt in 2024 to 11.3% and another 3.8ppt to 15.1% in 2025.

With the tax rate staying largely stable at ~20% during 2020-23 before starting to edge up in 2024 and reaching 23% in 2025, Delta's net profit growth was largely aligned with its operating profit growth trajectory. 2023 was an exception, as net profit still managed to grow 2% YoY despite a 1% YoY decline in operating profit, mainly due to higher FX gains.

The net margin trend has been largely aligned with that of operating margin in recent years. After dropping from 9.0% in 2020 to 8.5% in 2021, as lower gross margin was offset by a lower opex ratio, net margin hovered around the 8.3-8.5% level during 2021-24. Net margin was largely flat YoY in 2024 despite a 1.1ppt expansion in operating margin, mainly due to lower non-operating income and a slightly higher tax rate. We then saw a surge in 2025, with net margin expanding 2.5ppt YoY to 10.8%, benefiting from a larger contribution from higher-margin AI power offerings.

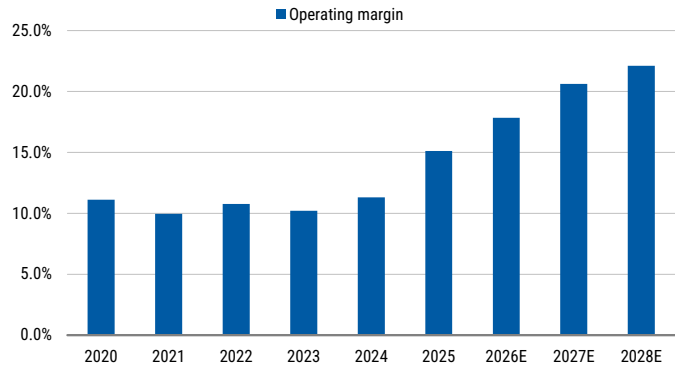
Looking ahead, we forecast net profit to grow 72% YoY in 2026e, followed by 63% YoY growth in 2027e and 56% YoY growth in 2028e, implying a 63% CAGR in 2025-28e, assuming a stable tax rate of 22%. Net margin is likely to rise accordingly, from 10.8% in 2025 to 13.3% in 2026, followed by further expansion to 14.9% in 2027 and 15.8% in 2028.

Exhibit 84: Delta: Operating income trend



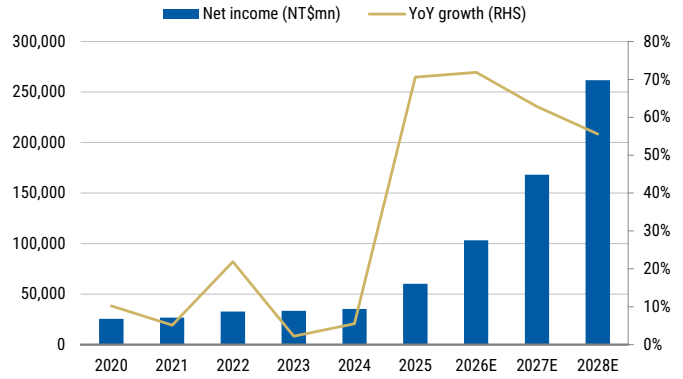
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 85: Delta: Operating margin



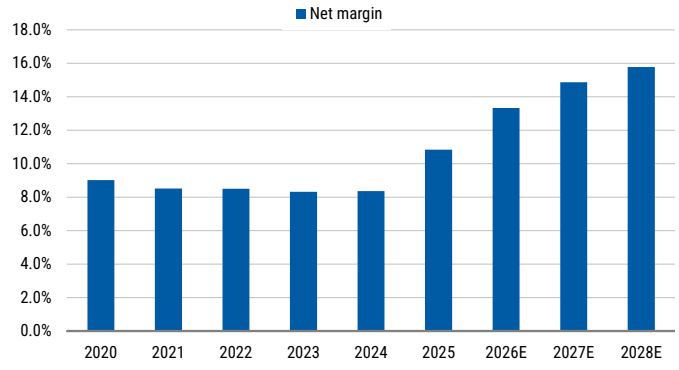
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 86: Delta: Net profit trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 87: Delta: Net margin



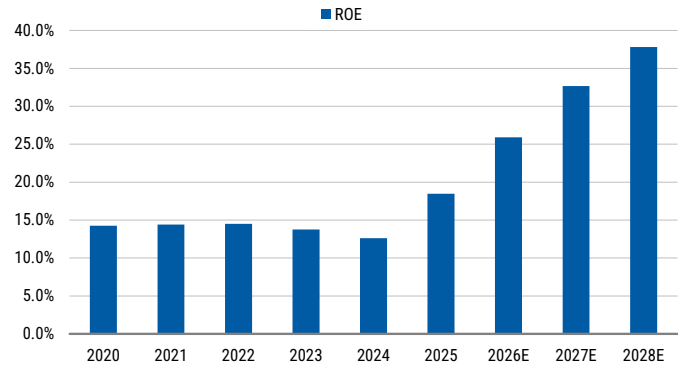
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

ROE to rise alongside margin expansion

ROE rose gradually but steadily during 2020-22, from 14.3% to 14.5%, mainly driven by improving asset turnover, offset by lower net margin. ROE edged down to 13.8% in 2023 and fell further to 12.6% in 2024 on the back of lower asset turnover, while net margin and financial leverage remained at similar levels. It then rebounded to 18.5% in 2025, supported by both improving asset turnover and net margin expansion, which was fueled by an increasingly favorable product mix and operating leverage.

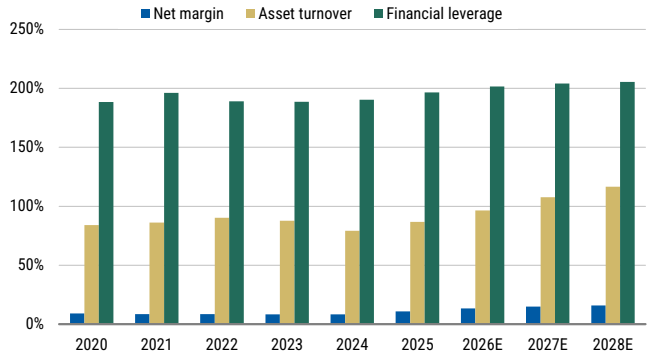
Looking ahead, we expect ROE to continue rising, reaching 25.9% in 2026, 32.7% in 2027, and 37.8% in 2028, primarily driven by continued margin expansion and improving asset turnover. Net margin is projected to expand from 10.8% in 2025 to 15.8% in 2028, supported by gross margin improvement—from 34.3% to 38.0%—and a declining operating expense ratio from 19.1% in 2025 to 15.9% in 2028.

Exhibit 88: Delta: ROE trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 89: Delta: DuPont analysis



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

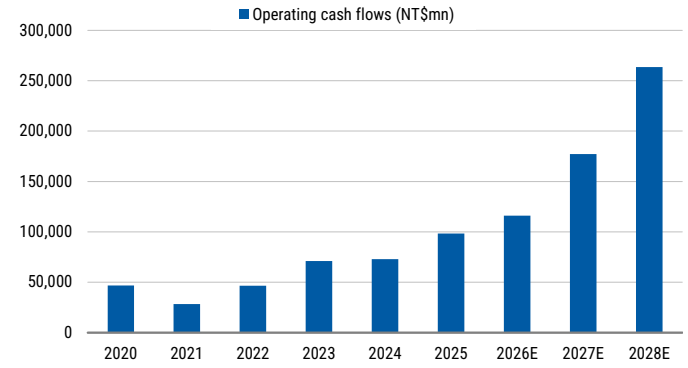
Improving balance sheet and continuous free cash flow generation

Delta has generated positive operating cash flow in recent years. Operating cash flow declined in 2021, mainly due to higher inventory buildup amid supply chain disruptions during the year. Free cash flow generation has also remained positive, while fluctuating with capital expenditures over the years. Free cash flow was exceptionally low at NT $5,292mn in 2021, as the company started to step up capex mainly for geographic capacity diversification and greater production flexibility—adding new capacity in Thailand, India, Tainan/Zhongli in Taiwan, and Wuhu, Wujiang, and Chongqing in Mainland China. Both absolute capex and the capex-to-sales ratio have trended upward since 2022, as Delta continued to add capacity for EV-related offerings and power and cooling solutions during 2023-24, with further acceleration in 2025 due to capacity tightness for AI power and cooling offerings, especially in Thailand and the US.

Looking ahead, the company aims to continue adding capacity worldwide to fulfill substantial demand for power and liquid-cooling turnkey solutions for large-scale data centers operated by major hyperscalers beyond 2027. Capacity remains tight for AI power and cooling offerings, so it continues to expand actively in China, Thailand, the US, and Tainan. Management also plans to add mass-production lines in Taoyuan, Taiwan, scheduled to be ready by 2028, in addition to the pilot-run lines in Tainan. Accordingly, we now estimate capex to rise to NT$70bn in 2026 from ~NT$46bn in 2025 and stay at a similar level in 2027 before edging down slightly to NT$60bn in 2028.

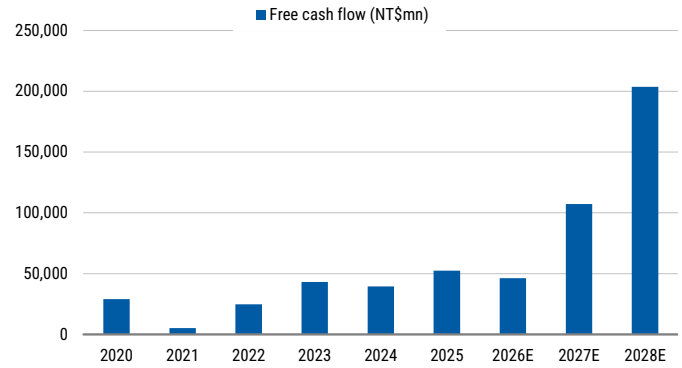
Delta's net cash position has been rising since 2021, with its net cash-to-equity ratio improving from just 2% in 2021 to 40% in 2025. We expect this ratio to continue rising, reaching 51% in 2026, 57% in 2027, and 63% in 2028, on strong free cash flow generation despite a higher total debt level amid ongoing capacity expansion.

Exhibit 90: Delta: Positive operating cash flow



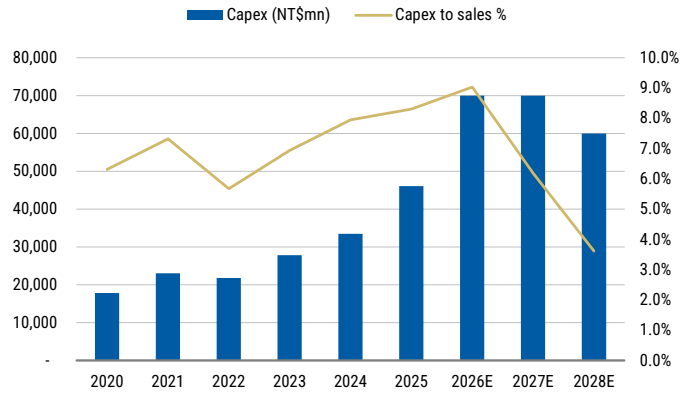
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 91: Delta: Improving free cash flow generation



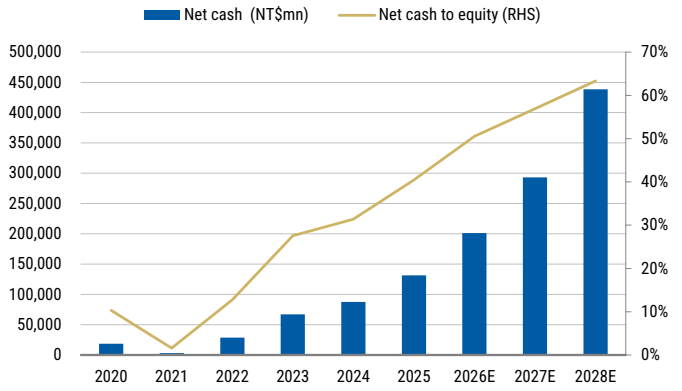
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 92: Delta: Capex and capex-to-sales ratio



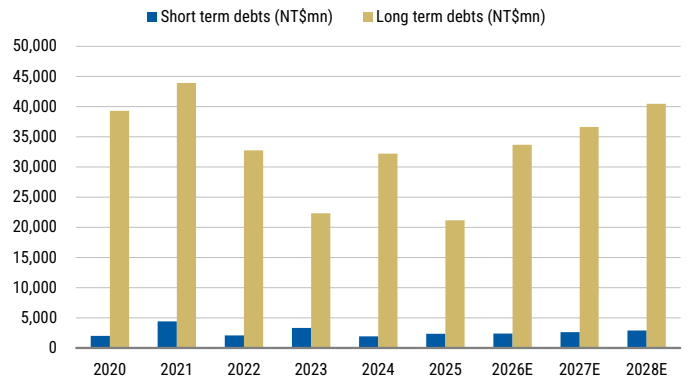
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 93: Delta: Net cash position



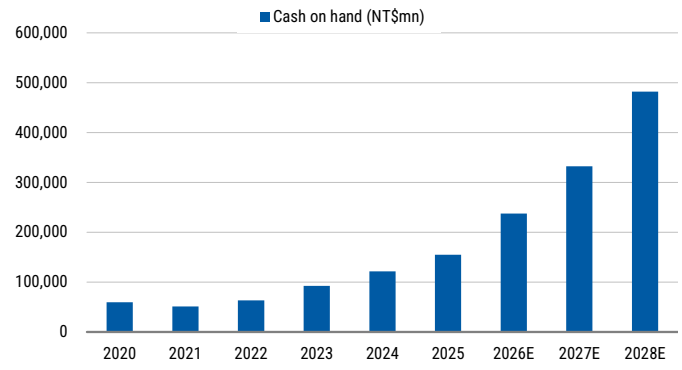
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 94: Delta: Debt position



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 95: Delta: Cash on hand



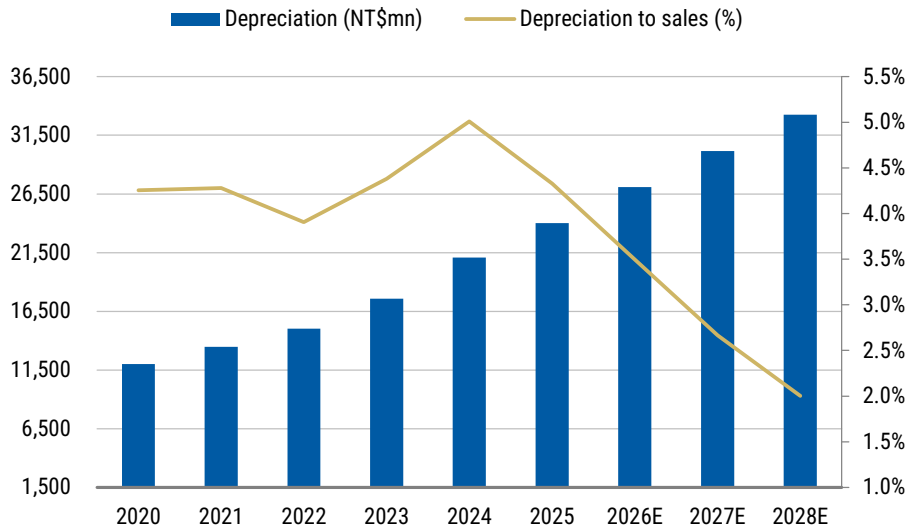
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

After edging down slightly in 2022 from a relatively high 7.3% in 2021, Delta's capex-to-sales ratio has been rising alongside its aggressive capacity expansion plans since 2022, despite an expanding revenue base. Looking ahead, as we expect capex to increase further in 2026, we project a capex-to-sales ratio of 9.0% for the year on robust revenue growth, before declining to 6.2% in 2027 and 3.6% in 2028.

The depreciation-to-sales ratio hovered around the 4-5% level during 2020-25. We

estimate the ratio will decline sequentially as the revenue base grows faster than capex and depreciation, from 4.3% in 2025 to 3.5% in 2026, 2.7% in 2027, and 2.0% in 2028.

Exhibit 96: Delta: Depreciation and depreciation to sales

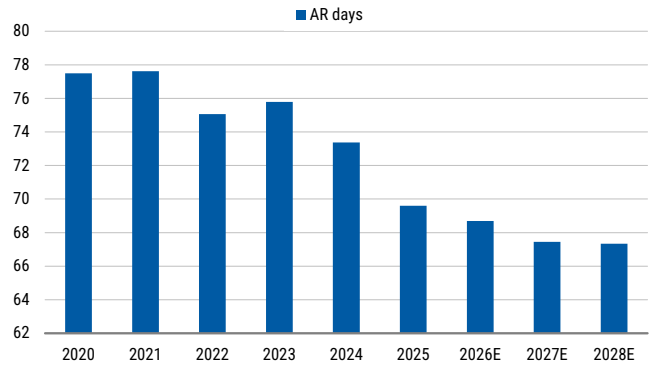


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Delta's cash conversion cycle (CCC) rose from 77 days in 2020 to a peak of 101 days in 2023 before improving to 80 days in 2025. By 2025, inventory days had increased from 80 days in 2020 to 93 days, while AP days had risen to 82 days after falling from 81 days in 2020 to 75 days in 2023. This was partly offset by improving AR days, which declined from 77 days in 2020 to 70 days in 2025.

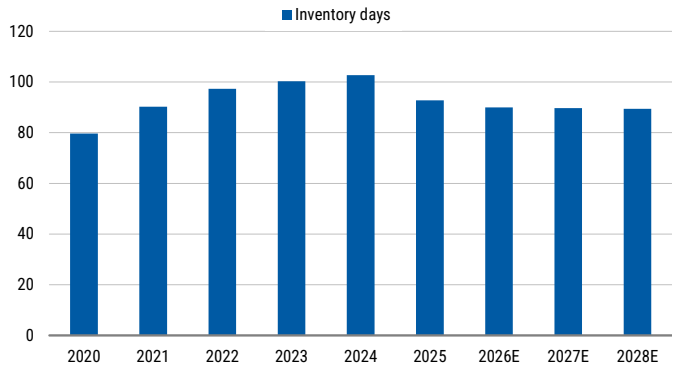
Looking ahead, we expect the overall cash conversion cycle to improve further and remain at 77 days throughout 2026-28, mainly supported by improving AR days and a modest decline in inventory days.

Exhibit 97: Delta: AR days



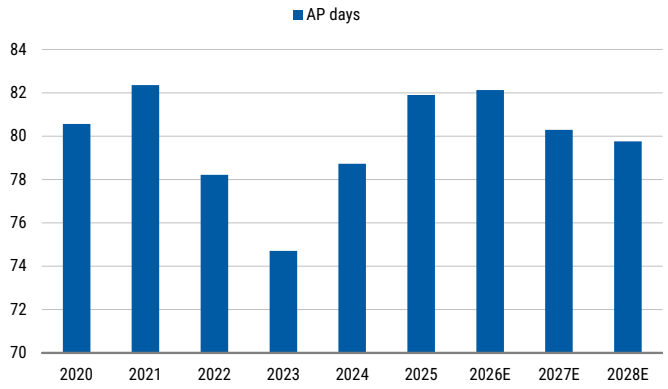
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 98: Delta: Inventory days



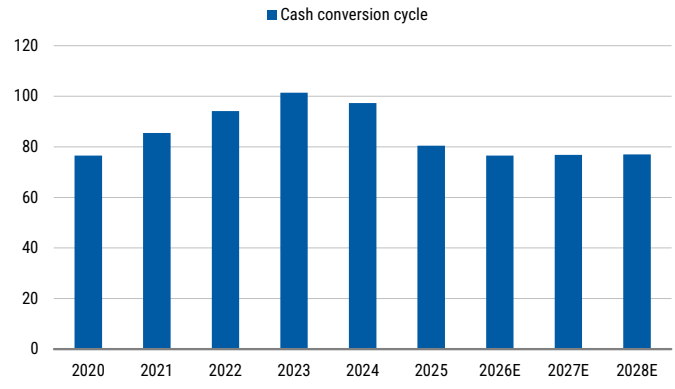
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 99: Delta: AP days



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 100: Delta: Cash conversion cycle



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Company Background – Delta

Company snapshot

Delta Electronics is a global provider of power and thermal management technologies, with a focus on energy-efficient products and integrated solutions. Founded in 1971, the company serves a broad range of applications spanning components, power and system products, thermal management, mobility, industrial and building automation, data centers, telecom, and energy infrastructure, across industries including automotive, healthcare, data centers, and power grids. By combining its capabilities in power electronics, digital technologies, and system integration, Delta aims to support customers in improving operational efficiency and advancing smart manufacturing, smart cities, and more sustainable infrastructure.

The company is headquartered in Taipei, Taiwan. Its global network consists of 293 operations, including 165 sales offices, 55 manufacturing sites, and 73 R&D centers, supporting customers worldwide.

Exhibit 101: Delta: Key milestones

Year	Key Milestones
1971	Delta Electronics, Inc. was established
1976	Achieved >US$1mn of revenue
1977	Moved plant to Taoyuan Guishan Industrial Park
1983	- Mass production of switching power supplies
1988	- Listed on the Taiwan Stock Exchange (code 2308) - Mass production of brushless DC fans - Invested in the establishment of Delta Electronics (Thailand) PCL, with an original stake of 6%
1989	- Established first Japan office in Tokyo - Established Delta Power Electronics laboratory in Raleigh, US
1992	- Inauguration of the first plant in China (Dongguan) - Established Delta Greentech (China) Co., Ltd. in Shanghai
1995	Delta Electronics (Thailand) PCL was listed on Thailand Stock Exchange
1999	- Inaugurated its new Taipei HQ - Established the Delta Power Electronics Center (DPEC) in Shanghai
2000	Established the Delta Power Electronics Science and Education Development program
2001	Inaugurated its Wuhu plant in China
2003	Acquired ASCOM Energy Systems and its plant in Slovakia
2006	Inaugurated Delta's Tainan Plant I
2008	Inaugurated its Rudrapur plant in India and Wuhu plant in China
2010	Inaugurated Chenzhou plant in China
2011	Inaugurated Shanghai R&D and Operations Center and Taoyuan R&D Center in Taiwan
2012	Mr. Yancey Hai was elected Chairman and Mr. Ping Cheng was appointed as CEO
2015	- Acquired Eltek ASA, a leading provider of telecom power, industrial power, and data center power solutions - Inaugurated Delta Americas' headquarters green building and Taoyuan Plant 5
2017	- Announced organizational adjustment to focus on Power Electronics, Automation, and Infrastructure as main business categories - Acquired a controlling stake in VIVOTEK
2019	- Acquired a controlling stake in Delta Electronics (Thailand) Public Company Limited - Acquired lighting-solutions brand Amerlux
2020	Accquired Trihedral, a Canadian SCADA and IIoT software company, to strengthen industrial automation and system-integration businesses
2021	Acquired March Networks, a global video-surveillance provider, and Universal Instruments, a leader in precision automation solutions for electronics manufacturing
2022	- Inaugurated a new facility at the Automotive Campus in Helmond, the Netherlands - Purchased land and a factory in Plano, Texas, United States
2023	Acquired TB&C, a leading provider of automotive high-voltage hybrid components
2024	- Mr. Ping Cheng selected as Chairman while concurrently serves as CEO - Adjusted business categories into Power Electronics, Mobility, Automation, and Infrastructure
2025	- Inaugurated new North East Asia HQ in Tokyo - Acquired ACT Genomics and Noda RF Technologies

Source: Company data, Morgan Stanley Research

Product offerings

Delta's key offerings can be categorized into four key areas: Power Electronics, Infrastructure, Mobility, and Automation.

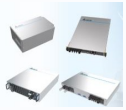
1. Power Electronics

Delta operates in switching power supplies, thermal management, and passive components, supplying products for ICT, cloud computing, EVs, and industrial and commercial applications. The company frames its business around energy efficiency and

sustainability and now also offers grid-to-chip power and thermal management solutions for data centers.

- Components: Includes essential passive and magnetic components used in portable devices, AI cloud computing, automotive electronics, and IoT.
- Power and System: Contains power products and systems for AI, cloud computing, IoT, client devices, and industrial, medical, lighting, e-mobility, and electrical-appliance applications.
- Fans and Thermal Management: This mainly covers air and liquid cooling solutions for AI cloud and edge computing, EVs, as well as industrial, commercial, and residential sectors. Key products includes DC and EC fans, thermal modules, liquid cooling cold plates, and coolant distribution units, etc.

Exhibit 102: Delta: Key offerings for the power electronics segment

POWER ELECTRONICS							
Components							
Inductors	RF Inductors	Standard Power Module (DC/DC Converter & AC/DC Module)	Transformers	Networking	EMI Filters	Solenoids	Current Sensing Resistor
				Wire-Wound Common Mode Choke			
Power and System							
Embedded Power	Adapters	Industrial Power	Medical power	Industrial Battery Charging	USB Socket Outlets	High Voltage Power	Hydrogen Power
				Modular Charging System			
Fans and Thermal Management							
Chip Level Cooling Solution	DC Brushless Fans and Blowers	Thermal Module	Cold Plate Loop	In-rack CDU	Rack Manifold	Electical Pump	In-row CDU
Communication Equipment Cooling	Energy Infrastructure and Industrial Automation Cooling	Portable Device and Equipment Cooling	Infrastructure Ventilation	Automotive Thermal			
DC Brushless Fans and Blowers	DC Brushless Fans and Blowers	Laptop cooling	DC Brushless Fans and Blowers				

Source: Company data, Morgan Stanley Research

2. Infrastructure

Delta supplies ICT and energy infrastructure to support demand related to AI, smart cities, and IoT. Its ICT infrastructure solutions target data centers and telecom networks, while its energy infrastructure offerings integrate AIoT, power conversion, and microgrid control across energy generation, storage, and management.

- ICT Infrastructure: It mainly provides infrastructure for AI computing and 5G

communications under this category, ranging from telecom cell sites to data centers. The portfolio includes power, air and liquid cooling, networking equipment, and management software.

- Energy Infrastructure: The company offers renewable energy, AIoT-based energy management, and power conversion technologies for low-carbon transformation. Solutions include solar inverters, wind power converters, energy storage, EV charging, and management platforms for smart grids. It also supplies semiconductor high-voltage power supplies, automated testing instruments, and variable-frequency drives for industrial processes.
- Display and Visualization: Delta's display technologies serve control rooms, public information systems, projection mapping, and exhibition or event spaces. Offerings range from large-scale and 8K projection systems to integrated video-wall solutions and can be customized for mission-critical operations.

Exhibit 103: Delta: Key offerings for the infrastructure segment

INFRASTRUCTURE							
ICT Infrastructure							
Uninterruptible Power Supplies (UPS)	Battery Backup System (BBU)	800 VDC In-Row Power	Power Distribution Unit (PDU)	Rack-Mount Remote Power Panel (rRPP)	Liquid and Air Cooling	Modular Rack & Rack Mount Accessories	Lithium-ion UPS Battery System
Ethernet Switches	Indoor/Outdoor Telecom Power Systems	Renewable Hybrid Power Systems	Small Cell Power Systems	Li-ion Battery Backup Systems	Campus/Enterprise Networks		
Energy Infrastructure							
EV Charger	Power Conversion System (PCS)	All-in-One ESS solution	Distributed ESS Solutions	Solar Inverter	Wind Power Converter	Solid State Transformer Platforms	Medium Voltage Drives
High Voltage Power	DC EV Charger ATS	Power Supply ATS	Solid Oxide Fuel Cell (SOFC)	Hydrogen Energy Solution	Active Power Filter (APF) / Static Var Generator (SVG)		
Display and Visualization							
Business and Education Multimedia Projector	High Performance Projector	DLP, LED and LCD Video Wall	Wireless Collaboration and Signage	Vivitek			
				vivitek			

Source: Company data, Morgan Stanley Research

3. Mobility

Delta applies its power electronics and motor technologies to smart mobility, supplying

power systems, drive technologies, and integrated powertrain solutions. It works with automakers to develop electric vehicles, positioning the business as part of the low-carbon transition and sustainable urban mobility.

- EV Powertrain System: The company's powertrain portfolio for EV makers includes On-board Charging Modules (OBCM), DC/DC converters, power electronics combos, and traction inverters, along with X-in-One integrated solutions.

Exhibit 104: Delta: Key offerings for the mobility segment

MOBILITY						
EV Powertrain System						
EV Power Electronics	Traction Products	x-EV Power Management	In-Vehicle Network and AI Driving Computing	Automotive Thermal	Low Voltage Powertrain	EV Charging

Source: Company data, Morgan Stanley Research

4. Automation

Delta provides automation solutions for industrial and building applications through integrated software and hardware. Its industrial automation offerings support smart production and operational efficiency, while its building automation solutions target energy-efficient, secure, and sustainable spaces.

- Industrial Automation: This segment provides automation solutions across industries by utilizing AI, big data, IoT, and digitized manufacturing models. Relevant offerings support diverse production needs, including customized, small-volume and high-variety production, as well as mass production.
- Building Automation: Delta's Building Automation Solutions use an IoT architecture and integrate technologies from five brands: Amerlux lighting, Delta Controls and LOYTEC IP-based building control, and March Networks and VIVOTEK surveillance. The management platform enables operators to manage energy use, occupant comfort, and sustainability goals through a single interface.

Exhibit 105: Delta: Key offerings for the automation segment

AUTOMATION							
Industrial Automation							
Inverters - AC Motor Drives	Drive System for Motion Control Solutions	Motors	Servo Systems - AC Servo Motors and Drives	CNC Solution	PC-Based Motion Control Solution	Linear Motion	Embedded Motion Controller
Industrial Ethernet Solution	Temperature Controllers	High Voltage Power	SCARA Robot	Electric Servo Press			
Building Automation							
Delta Controls	LOYTEC	Indoor Air Quality Monitor	LED Lighting - Industrial/ Road/Fishing/Commercial	Smart Surveillance - VIVOTEK	Healthy Lighting		

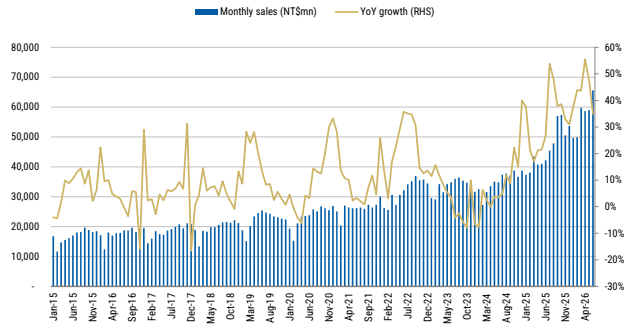
Source: Company data, Morgan Stanley Research

Monthly sales vs. share price

Delta's monthly sales show that seasonality has been an important factor over the years. Following a substantial increase in monthly sales since 2024, traditional seasonal patterns have become less pronounced as revenue from AI-related offerings has started to outweigh them, while liquid-cooling-related offerings have added a new layer of monthly sales volatility. The correlation between the company's monthly sales and share price movements has remained high, at ~87% since 2023.

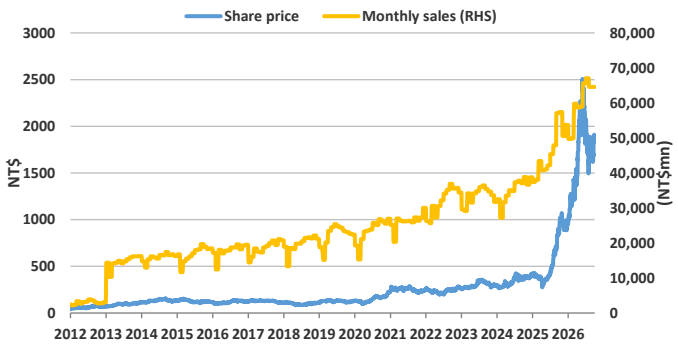
The company reported August monthly sales of NT$64,586mn (-4% MoM/+35% YoY). We saw the Power Electronics segment decline sequentially from a high base, offset by growth in Automation and slight growth in the Infrastructure segment.

Exhibit 106: Delta: Monthly sales vs. YoY sales growth



Source: TEJ, Morgan Stanley Research

Exhibit 107: Delta: Monthly sales vs. share price



Source: TEJ, Morgan Stanley Research

Key management & shareholding structure

Delta's board currently consists of eight directors and five independent non-executive directors. Its founder, Chung-Hua Cheng (Bruce Cheng), and his sons, Ping Cheng and An Cheng, together owned a 7% stake as of March 2026. Excluding the founder and his family, the management team held a stake of ~0.1% as of March 2026.

The FINI holding percentage as of July 14, 2026, was 63%, versus a previous peak of 78% in February 2011 and a historical average (since 2010) of 69%.

Exhibit 108: Delta: Shareholding structure

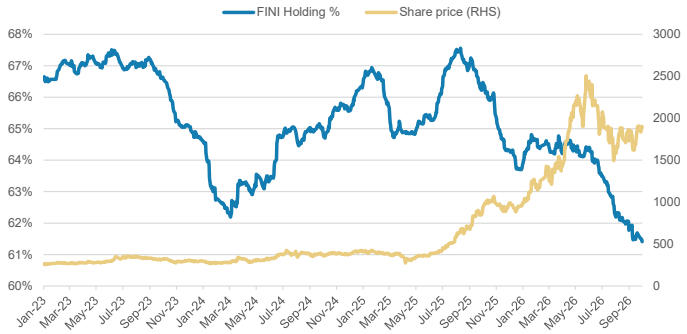
Shareholders	%
Dejico International LTD	10.30%
Deltron Holding LTD	8.40%
New Labor Retirement Pension Fund	3.75%
Chung-Hua Cheng (Bruce CH Cheng)*	3.15%
Government of Singapore-GOS-EFM C	2.28%
Ping Cheng*	2.14%
An Cheng (Victor Cheng)*	1.94%
Yuanta/P-shares Taiwan Top 50 ETF	1.43%
Labor Retirement Reserve Fund (The Old Fund)	1.42%
Labor Insurance Fund	1.32%
Total	**36.13%**

Source: Company data, Morgan Stanley Research. Note: Data as of March 2026. (*) are shown as insiders.

Exhibit 109: Delta: FINI holdings vs. share price since 2010

Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 110: Delta: FINI holdings vs. share price since 2023



Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 111:

Delta: Management profiles

Name	Title	Education	Work Experience
Bruce C.H. Cheng	Founder and Honorary Chairman Chairman, Delta Electronics Foundation	Bachelor of Electrical Engineering — National Cheng Kung University	- 2012–present — Honorary Chairman, Delta Electronics - 1990–present — Founded and chairs the Delta Electronics Foundation - 1971–2012 — Founder and Chairman, Delta Electronics - Before 1971 — Worked at TRW and Air Asia Co.
Ping Cheng	Chairman and CEO	Bachelor of Business Administration — California State University, East Bay	- Joined Delta in 1988 - 2024–present — Chairman - 2012–present — CEO of Delta Electronics - 2010–2012 — Corporate Senior Vice President and President of China Region - 2008–2010 — Senior Vice President & General Manager of Power Supply Business Group - 1998–2008 — Deputy General Manager of Component Business Group (Delta China); Executive Vice President of China Region
Mark Ko	Vice Chairman	Bachelor of Control Engineering — National Chiao Tung University	- Joined Delta in 1988 - 2012–present — Vice Chairman of Delta Electronics - 2004–2012 — President and Chief Operating Officer of Delta Group - 1994–2004 — General Manager of Power System Business Group
Simon Chang	Board Member President and COO	EMBA — Chung Yuan Christian University	- Joined Delta in 1981 - 2017–present — Board Member; President and Chief Operating Officer, Delta Electronics - Before 2017 — Senior Vice President and General Manager of Industrial Automation Business Group - From 1995 — Director of Industrial Automation Division
Victor Cheng	Board Member CEO of Delta Electronics (Thailand) PCL.	-Master of Electrical Engineering — Santa Clara University -Bachelor of Electrical Engineering — Santa Clara University	- Joined Delta in 1993 - 2024–present — Board Member of Delta; CEO of Delta Electronics (Thailand) PCL. - 2023–2024 — Executive Vice President of Infrastructure Business - 2017–2023 — General Manager of Information and Communications Technology Business Group - 2014–2017 — General Manager of Power System Business Group - 2002–2019 — President, then Chairman of Delta Networks - 1999–2002 — Head of Video Display Business Unit - 1993–1999 — International Sales Director to Assistant General Manager of Video Display Business Unit
Shan-Shan Guo	Board Member Chief Brand Officer Vice Chairman, Delta Electronics Foundation	- Master of Marketing and Communications — University of Michigan - Bachelor of Foreign Languages and Literatures — National Taiwan University	- Joined Delta in 2010 - 2011–present — Chief Brand Officer of Delta
Johnson Lee	Executive Vice President Infrastructure Business (IFB)	- EMBA — National Taiwan University - Bachelor of Industrial Engineering and Systems Management — Feng Chia University	- Joined Delta in 1985 - 2024–present — Executive Vice President of Infrastructure Business - 2012–2017 — Chief Operating Officer of Delta Electronics - 2004–2012 — General Manager of Power Supplies Business Group - Earlier — Procurement manager; General Manager of materials management
Ted Shyy	Executive Vice President Power Electronics Business (PEB)	Master in Materials Science and Engineering — National Cheng Kung University	- Joined Delta in 1988 - 2024–present — Executive Vice President of Power Electronics Business - 2014–present — Chief Executive Officer of Cyntec (concurrent role) - 2011–2014 — General Manager of Delta Components Business Group - 2003–2014 — General Manager of Cyntec - 1991–2003 — Rotation in Research and Development, production and marketing, under Cyntec, a joint venture between Delta and Japan's SUSUMU
James Tang	Executive Vice President Mobility Business (EVB)	- EMBA — National Taiwan University - Bachelor of Management Science — National Cheng Kung University	- Joined Delta in 1988 - 2024–present — Executive Vice President of Mobility Business; continues as General Manager of Electric Vehicle Solutions Business Group - 2017–present — General Manager of Electric Vehicle Solutions Business Group - 2008–2016 — General Manager across Power Supplies, Consumer and Lighting Solutions, and Embedded Power Supplies business groups - 1988–2008 — Procurement; Materials Manager; Director of Personal Computer Business Group
Jimmy Yiin	Executive Vice President Global Business Operations (GBO)	Master of Business Administration — National Taiwan University and Fudan University	- Joined Delta in 1992 - 2023–present — Executive Vice President of Global Business Operations - 2019–2022 — General Manager of Power System Business Group - 2014–2019 — Vice Chairman of Delta Greentech (China); General Manager of Delta Electronics (Shanghai) - 2008–2013 — General Manager of Delta Electronics (Jiangsu) - 1999–2007 — Purchasing manager/materials section chief of Delta Dongguan and Delta Networks
Rock Huang	Executive Vice President Global Manufacturing	Master of Industrial Engineering and Management — National Taipei University of Technology	- Joined Delta in 1987; ~40 years of experience in the manufacturing field - 2026–present — Executive Vice President of Global Manufacturing - 2020–2026 — President of Global Manufacturing - 2019–2020 — President of Delta China - 2018–2019 — Executive Vice President of Delta China - 2007–2018 — General Manager of Southern China, leading Dongguan team

Source: Company data, Morgan Stanley Research

Risk Reward – Lite-On Technology (2301.TW)

Stronger ASIC server power supply but fair risk-reward

PRICE TARGET NT$315.00

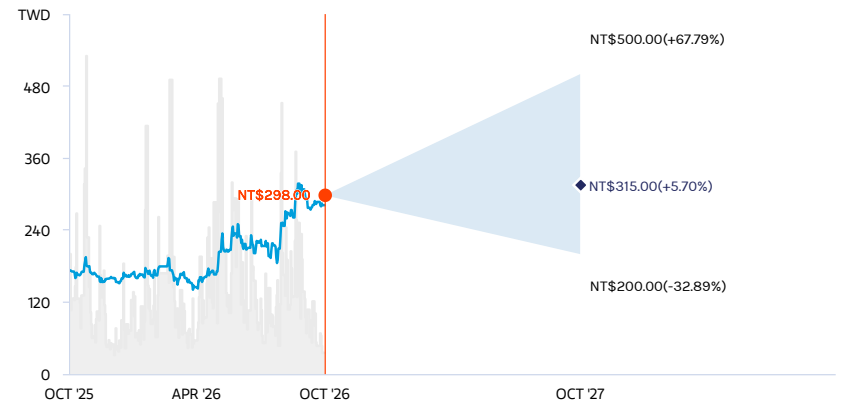
Base case: P/E multiple of 17x based on average 2027-28 EPS. We believe our target multiple is justified by the 47% earnings CAGR we estimate for 2025-28, implying a PEG of 0.4, versus the stock's 16x P/E and 2% earnings CAGR during 2022-25, which implied a PEG of 7.2.

Consensus Price Target Distribution NT$176.00 NT$300.50 MS PT NT$400.00

Source: Refinitiv, Morgan Stanley Research

◆ Mean ◆ Morgan Stanley Estimates

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- We expect Lite-On Tech to benefit from strong AI server power demand in 2026-27, with the company gaining more traction on ASIC platforms.
- We are mindful of weakness in consumer/IT demand amid rising input costs, which could weigh on Lite-On Tech's profit in 2026-27.
- Liquid-cooling sidecar CDU system shipments may take more time to ramp up.

Consensus Rating Distribution

77% Overweight
23% Equal-weight
0% Underweight
● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian:	*Negative*
New Data Era:	*Positive*
Technology Diffusion:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$500.00

27x average 2027-28e P/E

Better-than-expected margin expansion: Operating efficiency and accelerating production automation drive margin expansion, while new-business revenue comes in earlier than expected. Growth in cloud computing and automotive electronics is stronger than expected.

BASE CASE NT$315.00

17x average 2027-28e P/E

AI server projects progress, but some business uncertainties remain: We expect margin improvement from a better product mix as AI server power supply projects ramp up. However, we are mindful of weakness in consumer/IT demand amid rising input costs.

BEAR CASE NT$200.00

11x average 2027-28e P/E

Operating margin contraction: Operating efficiency improves less than expected, while pricing competition intensifies amid macro volatility; automotive electronics take longer to ramp up.

Risk Reward – Lite-On Technology (2301.TW)

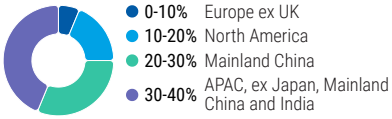
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Opto-electronics OPM (%)	8.3	7.9	7.5	7.5
Information Technologies OPM (%)	9.2	7.7	7.8	8.6
Opto-electronics revenue YoY (%)	0.2	8.0	1.7	2.3
Information Technologies revenue YoY (%)	7.7	(5.2)	0.5	(0.2)

INVESTMENT DRIVERS

- LED adoption (outdoor lighting/auto)
- Margin expansion
- Raw material prices, particularly for steel and copper

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger optical business growth
- Stronger data center/server demand
- PC/NB demand growth
- Downtrend in raw material prices

RISKS TO DOWNSIDE

- Macro demand weakness
- Weaker data center/server demand
- PC/NB demand slowdown
- Uptrend in raw material prices

OWNERSHIP POSITIONING

Inst. Owners, % Active 65.2%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	210,118	224,829 / 219,960	229,691
EBITDA (NT$, mn)	31,349	33,549 / 34,893	38,115
Net income (NT$, mn)	23,541	25,187 / 24,910	26,835
EPS (NT$)	10.30	11.07 / 10.88	11.79

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Financial Summary – Lite-On

Exhibit 112: Financial Summary – Lite-On

Consolidated Income Statement

NT$ mn	2024	2025	2026E	2027E	2028E
Net sales	**137,134**	**166,085**	**224,829**	**297,964**	**357,670**
COGS	-107,504	-128,046	-168,703	-219,712	-259,785
Gross profit	**29,630**	**38,039**	**56,126**	**78,252**	**97,885**
Operating expenses	-16,648	-21,301	-26,044	-33,189	-37,645
- Promotion	-5,184	-7,629	-6,902	-8,782	-10,247
- ADM	-3,990	-4,965	-7,384	-9,340	-10,162
- R&D	-7,476	-8,721	-11,787	-15,037	-17,200
Operating income	**12,981**	**16,738**	**30,082**	**45,063**	**60,240**
Non-operating income	2,653	2,975	2,876	2,500	2,500
Interest income	1,491	1,119	799	800	800
Investment income	2,808	-956	1,436	1,200	1,200
Disposal of investment	75	-537	-77	0	0
Exchange gain	2,808	1,091	1,390	0	0
Other	-4,528	2,258	-672	500	500
Pre-tax income	**15,635**	**19,713**	**32,958**	**47,563**	**62,740**
Income tax	-3,674	-4,601	-7,767	-11,227	-14,793
Minority	19	-4	3	0	0
Reported net income	**11,942**	**15,116**	**25,187**	**36,337**	**47,947**
EPS (NT$)	**5.21**	**6.64**	**11.07**	**15.97**	**21.07**

Consolidated Cash Flow Statement

NT$ mn	2024	2025	2026E	2027E	2028E
Cash flow from Operations	**12,411**	**14,533**	**34,885**	**43,140**	**51,224**
Net profits	11,942	15,116	25,187	36,337	47,947
Depreciation + Amortization	4,047	3,668	3,994	4,065	4,137
Inv. Gain/Loss (Equity)	-15	-86	0	0	0
Disposal of Investment	9	568	0	0	0
Working Capital Change	-3,886	-2,099	-12,395	-12,092	-10,109
Other adjustments	313	-2,635	18,098	14,830	9,248
Cash flow from Investing	**-4,843**	**-7,395**	**-13,287**	**-8,993**	**-11,919**
Capex	-3,283	-6,795	-13,287	-8,993	-11,919
Change of LT investment	-2,234	-43	0	0	0
Change of ST investment	1,453	-417	0	0	0
Other adjustments	-779	-140	0	0	0
Cash flow from financing	**-3,363**	**-23,099**	**-19,866**	**-28,659**	**-37,818**
Increase in L/T debt	0	0	0	0	0
Increase in S/T debt	7,584	-10,116	0	0	0
Cash Dividend Paid	-10,372	-10,297	-19,866	-28,660	-37,818
Issuance of stock	0	0	0	0	0
Other adjustments	-574	-2,686	0	0	0
Exchange rate adjustment	3,736	-3,748	0	0	0
Net change in cash	**7,942**	**-19,709**	**1,732**	**5,487**	**1,487**

Consolidated Balance Sheet

NT$ mn	2024	2025	2026E	2027E	2028E
Cash	100,684	80,974	82,706	88,193	89,680
Mkt Securities	933	1,009	1,009	1,009	1,009
AR/NR	37,182	42,318	60,960	80,790	96,978
Inventory	26,817	34,262	42,083	54,807	64,804
Others	3,501	5,458	5,458	5,458	5,458
Current Assets	**169,117**	**164,021**	**192,216**	**230,257**	**257,928**
Long-term investments	8,195	7,748	7,748	7,748	7,748
Fixed assets	18,776	24,752	34,495	39,873	48,104
Deferred Asset	1,628	2,246	2,246	2,246	2,246
Intangible Assets	3,156	3,060	2,610	2,160	1,710
Other assets	3,811	3,860	3,860	3,860	3,860
Total Assets	**204,683**	**205,687**	**243,175**	**286,144**	**321,597**
S/T borrowings	30,187	20,029	20,029	20,029	20,029
AP/NP	43,126	53,608	67,676	88,138	104,213
Other ST liabilities	34,556	35,960	54,059	68,888	78,137
Total Current Liabilities	**107,869**	**109,597**	**141,764**	**177,056**	**202,379**
LT debt(include CB)	3,000	3,000	3,000	3,000	3,000
Other LT liabilities	2,903	3,111	3,111	3,111	3,111
Total Liabilities	**113,772**	**115,709**	**147,875**	**183,167**	**208,491**
Common shares	23,473	23,168	23,168	23,168	23,168
Retained Earnings	46,277	42,393	47,714	55,390	65,519
Other shareholders' equity	21,161	24,418	24,418	24,418	24,418
Total Equity	**90,911**	**89,979**	**95,300**	**102,976**	**113,105**
Total Liab. & Shldr's Equity	**204,683**	**205,687**	**243,175**	**286,143**	**321,596**

Consolidated Financial Ratios

	2024	2025	2026E	2027E	2028E
Margins					
Gross margin	21.6%	22.9%	25.0%	26.3%	27.4%
Operating margin	9.5%	10.1%	13.4%	15.1%	16.8%
Pretax margin	11.4%	11.9%	14.7%	16.0%	17.5%
Net margin	8.7%	9.1%	11.2%	12.2%	13.4%
YoY growth					
Sales	-7.6%	21.1%	35.4%	32.5%	20.0%
Operating profits	-10.6%	28.9%	79.7%	49.8%	33.7%
Pretax profits	-14.9%	26.1%	67.2%	44.3%	31.9%
Net profits	-18.0%	26.6%	66.6%	44.3%	32.0%
EPS	-18.1%	27.5%	66.6%	44.3%	32.0%
Net Debt/Equity	-75.3%	-65.5%	-63.7%	-64.3%	-59.8%
Liabilities/Equity	125.1%	128.6%	155.2%	177.9%	184.3%
Liabilities/Assets	55.6%	56.3%	60.8%	64.0%	64.8%
ROAE	13.6%	16.7%	27.2%	36.7%	44.4%
ROAA	6.1%	7.4%	11.2%	13.7%	15.8%
AR/NR Turnover (days)	92.2	87.4	83.8	86.8	90.7
AP/NP Turnover (days)	142.7	137.9	131.2	129.4	135.1
Inventory Turnover (days)	89.3	87.1	82.6	80.5	84.0
Cash conversion cycle (days)	38.9	36.5	35.2	37.9	39.6

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 113: Lite-on: Quarterly Financials

NT$ mn	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	2025	2026E	2027E
Net sales	**36,417**	**40,421**	**44,891**	**44,356**	**43,407**	**52,703**	**59,526**	**69,193**	**166,085**	**224,829**	**297,964**
COGS	-28,195	-31,479	-33,648	-34,724	-34,004	-38,355	-44,867	-51,477	-128,046	-168,703	-219,712
Gross profit	**8,222**	**8,942**	**11,243**	**9,633**	**9,403**	**14,348**	**14,659**	**17,716**	**38,039**	**56,126**	**78,252**
Operating expenses	-4,546	-5,215	-6,560	-4,980	-5,304	-6,146	-6,851	-7,743	-21,301	-26,044	-33,189
- Promotion	-1,532	-2,051	-2,535	-1,512	-1,500	-1,623	-1,786	-1,993	-7,629	-6,902	-8,782
- ADM	-998	-1,048	-1,792	-1,127	-1,457	-1,783	-1,964	-2,180	-4,965	-7,384	-9,340
- R&D	-2,023	-2,110	-2,244	-2,345	-2,341	-2,787	-3,095	-3,563	-8,721	-11,787	-15,037
Operating income	**3,675**	**3,727**	**4,683**	**4,653**	**4,099**	**8,203**	**7,807**	**9,973**	**16,738**	**30,082**	**45,063**
Non-operating income	994	536	886	560	871	1,166	445	395	2,975	2,876	2,500
Net interest income	424	323	184	188	214	195	195	195	1,119	799	800
Net investment income	596	-3,235	1,389	293	99	1,237	50	50	-956	1,436	1,200
Net gain of investment dispos	137	-702	23	5	1	-78	0	0	-537	-77	0
Exchange gain	379	2,540	-751	-122	284	1,106	0	0	1,091	1,390	0
Others	-542	1,609	41	195	272	-1,295	200	150	2,258	-672	500
Pre-tax income	4,669	4,262	5,569	5,213	4,970	9,368	8,252	10,368	19,713	32,958	47,563
Income tax	-1,214	-1,108	-924	-1,355	-1,195	-2,248	-1,939	-2,385	-4,601	-7,767	-11,227
Net income	**3,456**	**3,155**	**4,645**	**3,860**	**3,775**	**7,116**	**6,313**	**7,983**	**15,116**	**25,187**	**36,337**
EPS (NT$)	**1.51**	**1.39**	**2.05**	**1.70**	**1.66**	**3.14**	**2.77**	**3.51**	**6.64**	**11.07**	**15.97**
Margins											
Gross margin	22.6%	22.1%	25.0%	21.7%	21.7%	27.2%	24.6%	25.6%	22.9%	25.0%	26.3%
Operating margin	10.1%	9.2%	10.4%	10.5%	9.4%	15.6%	13.1%	14.4%	10.1%	13.4%	15.1%
Pre-tax margin	12.8%	10.5%	12.4%	11.8%	11.4%	17.8%	13.9%	15.0%	11.9%	14.7%	16.0%
Net margin	9.5%	7.8%	10.3%	8.7%	8.7%	13.5%	10.6%	11.5%	9.1%	11.2%	12.2%
QoQ Growth											
Sales	-4.9%	11.0%	11.1%	-1.2%	-2.1%	21.4%	12.9%	16.2%			
Gross profit	1.0%	8.8%	25.7%	-14.3%	-2.4%	52.6%	2.2%	20.9%			
Operating profit	5.1%	1.4%	25.7%	-0.6%	-11.9%	100.1%	-4.8%	27.7%			
Pre-tax profit	17.1%	-8.7%	30.7%	-6.4%	-4.7%	88.5%	-11.9%	25.6%			
Net profit	13.3%	-8.7%	47.2%	-16.9%	-2.2%	88.5%	-11.3%	26.5%			
YoY Growth											
Sales	26.6%	21.4%	22.1%	15.8%	19.2%	30.4%	32.6%	56.0%	21.1%	35.4%	32.5%
Gross profit	40.6%	21.0%	36.3%	18.3%	14.4%	60.5%	30.4%	83.9%	28.4%	47.5%	39.4%
Operating profit	64.8%	13.6%	19.3%	33.1%	11.5%	120.1%	66.7%	114.3%	28.9%	79.7%	49.8%
Pre-tax profit	49.2%	4.4%	25.6%	30.7%	6.4%	119.8%	48.2%	98.9%	26.1%	67.2%	44.3%
Net profit	44.7%	1.3%	37.0%	26.6%	9.3%	125.5%	35.9%	106.8%	26.6%	66.6%	44.3%

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

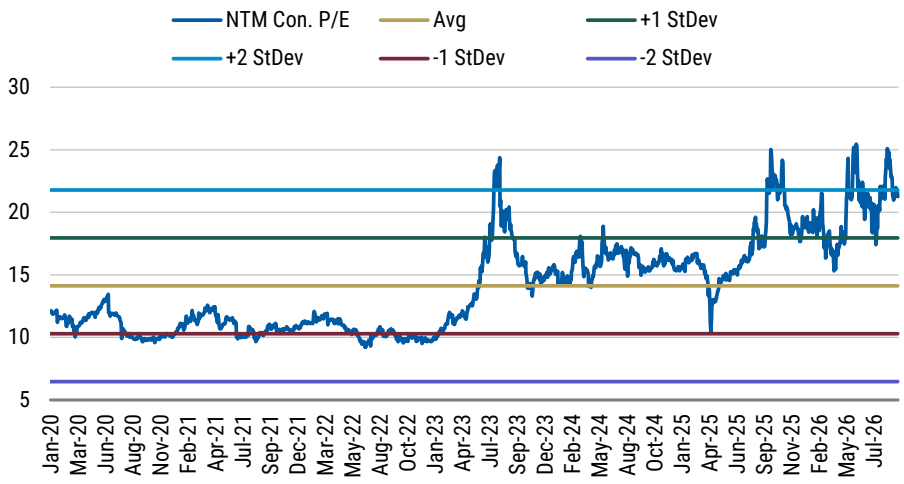
Valuation Methodology – Lite-On

Our price target (base case scenario value) is NT$315, derived from 17x 2027-28e P/E: We use price/earnings as our valuation methodology because we believe it appropriately captures Lite-On's earnings growth potential. We believe our target P/E multiple of 17x is justified by an earnings CAGR of 47% over 2025-28, implying a PEG of 0.4. This is lower than the average PEG of 7.2 in 2022-25, when the stock was trading at an average P/E of 16x for a 2% earnings CAGR.

Bull case scenario value – NT$500: We apply a 27x P/E multiple to our average base case estimate for 2027-28. This scenario assumes better-than-expected margin expansion, driven by improved operating efficiency and accelerating production automation, while new business revenue comes in earlier than expected. Growth in cloud computing and automotive electronics is stronger than expected.

Bear case scenario value – NT$200: We apply an 11x P/E multiple to our average base case estimate for 2027-28. This scenario assumes weaker-than-expected improvement in operating efficiency, while pricing competition intensifies amid macro volatility; automotive electronics take longer to ramp up.

Exhibit 114: Lite-on: NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Global Footprint – Lite-On

More capacity expansion to come

Lite-On's production footprint spans the globe, with 15 sites in total, although it remains heavily weighted toward China. The company has been expanding aggressively, both to meet strong demand for AI power supplies and to support its strategy of geographic risk diversification and proximity to customers.

Looking at its expansion plans in recent years, the company has continued to demonstrate its commitment to diversifying production away from China and increasing proximity to customers. It completed its Thailand plant expansion and announced the second-phase expansion of its Vietnam facility in 2021. It then established automotive electronics and cloud power production lines in Dallas, Texas, in 2H22, along with the official launch of the second phase of its Hai Phong plant in Vietnam, which focuses on networking and consumer products. In 2023, it began expanding its second Kaohsiung plant and announced additional capacity expansion in Vietnam. In 2024, it committed a further US $120mn to establish a wholly owned subsidiary in Vietnam and began progressively shifting production of high-end power products to that facility. In 2025, the company aimed to double capacity as demand for AI server power supplies and battery backup units (BBUs) surged, focusing primarily on capacity expansion in the US, Kaohsiung, and Vietnam.

For 2026, the company expects capacity allocation of ~40% in China, 27% in Vietnam, 20% in Taiwan, and 10% in the US, with the remaining non-China footprint (i.e., Mexico, Brazil, and Thailand) accounting for ~3% of total capacity. It has also announced investments of US$919mn and US$149mn in its US and Vietnam subsidiaries, respectively, to continue expanding power-related capacity, mainly for PSUs. The capex budget for 2026 has also been raised to ~NT$13bn, versus ~NT$7bn in 2025.

Exhibit 115: Lite-on's global footprint

Source: Lite-on.

Financial Analysis – Lite-On

29% revenue CAGR in 2025-28

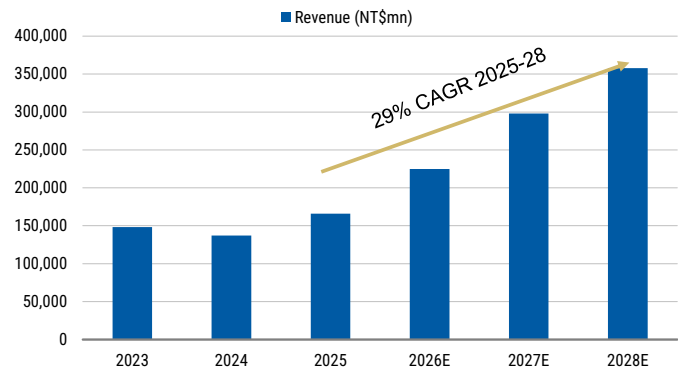
Lite-On's revenue growth was relatively moderate during 2020-22, with revenue expanding at a CAGR of 5% over the period. The company benefited from surging cloud computing demand, which drove increasing demand for power supplies, along with rising demand from EV/automotive electronics, although growth was offset by elevated component costs and supply-chain disruptions during COVID-19. Revenue then fell by 14% in 2023 and 8% in 2024 due to a post-pandemic demand pullback across the consumer electronics market and PC/NB model transitions, coupled with softening demand from EV-related businesses. The IT segment declined 20%/11% YoY, while the opto-electronics segment declined 11%/9% YoY in 2023/2024, respectively.

Revenue rebounded in 2025, growing 21% YoY, as the company began to benefit from surging demand for AI/cloud server power supplies. Contributions from ramping high-voltage/AI power infrastructure products began to increase, coupled with a partial recovery in the IT segment, mainly fueled by power adapter shipments for game consoles and gaming PCs/NBs, as well as related power offerings for low-Earth-orbit applications.

Looking ahead, we estimate revenue to expand at a 29% CAGR in 2025-28e, driven by the AI server power demand boom, while we remain mindful of weakness in consumer/IT demand amid elevated component costs.

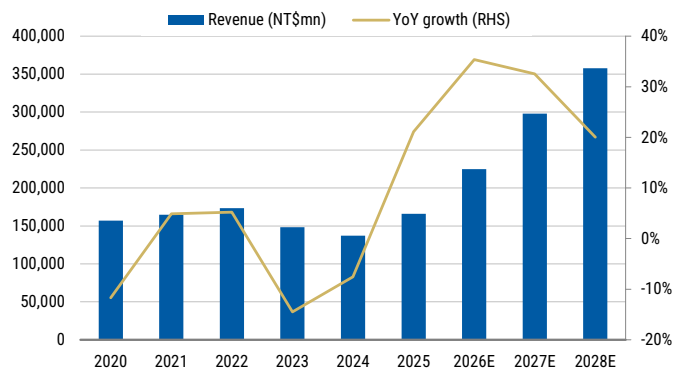
By product segment, IT-related products were the primary revenue driver in past years, while 2025 marked a crossover point, with the contribution from the Cloud & AIoT segment surpassing that of the IT segment, mainly supported by AI power-related offerings. IT revenue contribution declined from 72% in 2020 to 38% in 2025, and we estimate it will reach 17% by 2028. For the Cloud & AIoT segment, revenue contribution grew substantially from 27% in 2021 (0% in 2020) to 45% in 2025; we estimate it will account for 74% of total revenue by 2028, on the back of surging AI power-related demand. For opto-electronics, we expect revenue contribution to decline from 17% in 2025 to 9% in 2028.

Exhibit 116: Lite-on: Revenue CAGR, 2025-28e



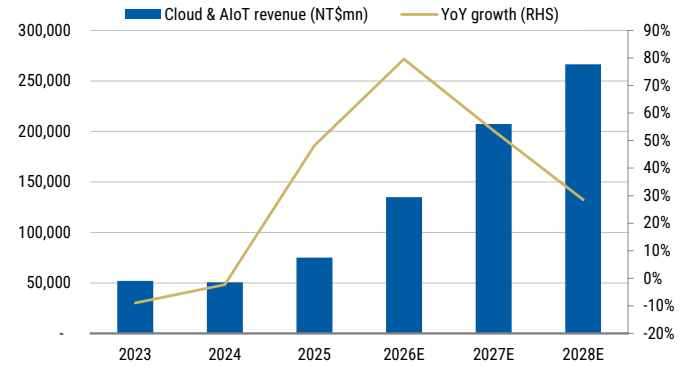
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 117: Lite-on: Annual revenue growth



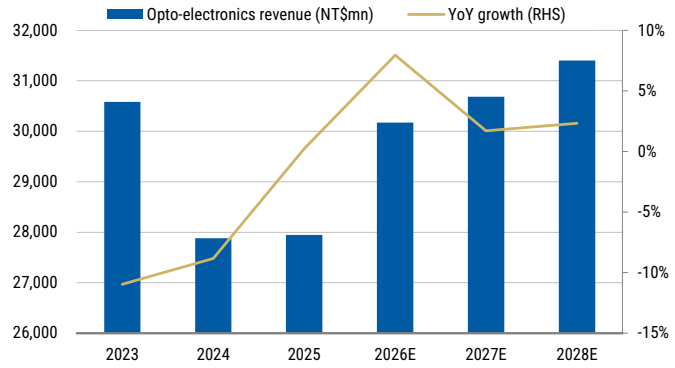
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 118: Lite-on: Cloud & AIoT revenue



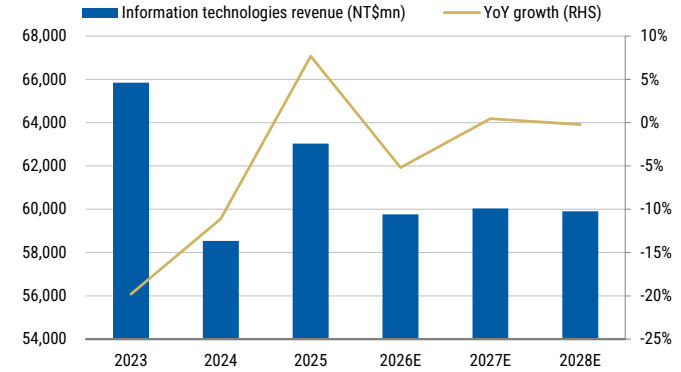
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 119: Lite-on: Opto-electronics revenue



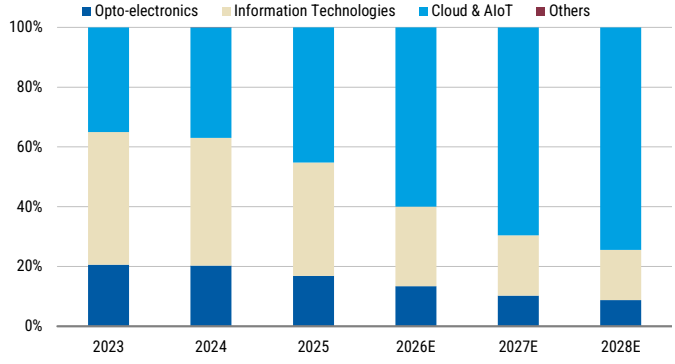
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 120: Lite-on: IT revenue



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 121: Lite-on: Revenue breakdown by segment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

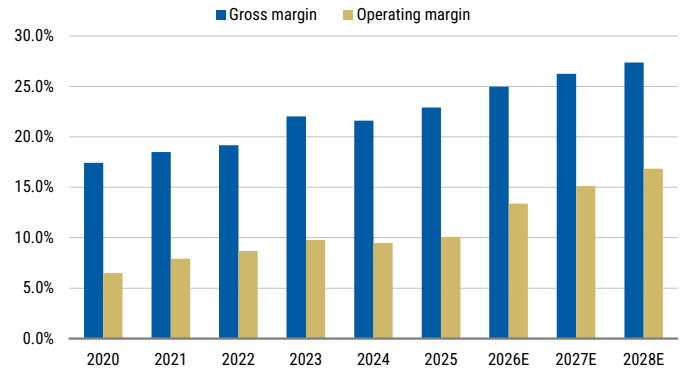
Continuing margin expansion

Lite-On's gross margin rose steadily over the years, from 17.4% in 2020 to 22.0% in 2023, thanks to a continually improving product mix. Gross margin dipped to 21.6% in 2024, mainly due to a smaller revenue base, while contributions from AI/cloud had yet to scale and margins were weighed down by learning-curve costs at the new Vietnam site. It then rose to 22.9% in 2025 as the AI server power business began to scale alongside rising BBU (battery backup unit) shipments, resulting in a more favorable product mix.

The opex ratio was largely stable at 10.5-11.0% during 2020-22. It then increased to 12.2% in 2023 and 12.1% in 2024, mainly due to operating deleverage and higher R&D costs. The opex ratio edged up further to 12.8% in 2025, mainly due to ongoing investment in R&D to support new business development in AI server power supplies, BBUs, liquid cooling, etc., along with a one-off tariff prepayment.

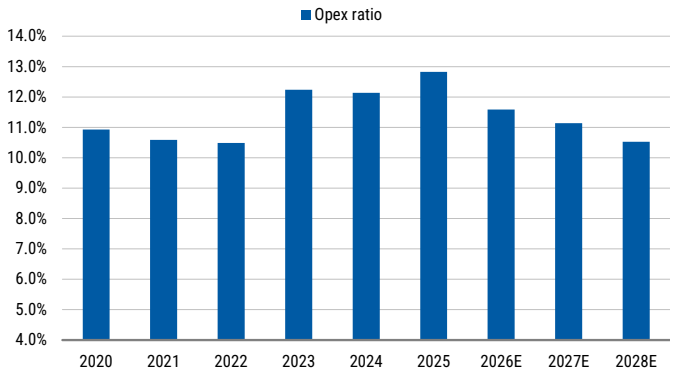
Looking ahead, we expect margins to continue expanding on a favorable product mix. We now pencil in a 2.1ppt gross margin improvement in 2026 to 25.0%, followed by expansions of 1.3ppt and 1.1ppt in 2027 and 2028, to 26.3% and 27.4%, respectively. For the opex ratio, we expect it to decline sequentially as the revenue base expands. Accordingly, we estimate operating margin to improve by 3.3ppt to 13.4% in 2026, followed by expansions of 1.7ppt in both 2027 and 2028, reaching 15.1% and 16.8%.

Exhibit 122: Lite-on: Margin trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 123: Lite-on: Opex ratio



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Net profit to grow at a 47% CAGR in 2025-28

Operating profit grew steadily alongside gross margin expansion during 2020-22, from NT $10,206mn to NT$15,073mn. It then fell 4% YoY in 2023 and a further 11% YoY in 2024 due to higher operating expenses, mainly from R&D costs, despite gross margin expansion in 2023. Operating margin increased by 1.1ppt YoY to 9.8% in 2023 amid a smaller revenue base, before declining by 0.3ppt YoY to 9.5% in 2024 on lower gross margin and an elevated opex ratio.

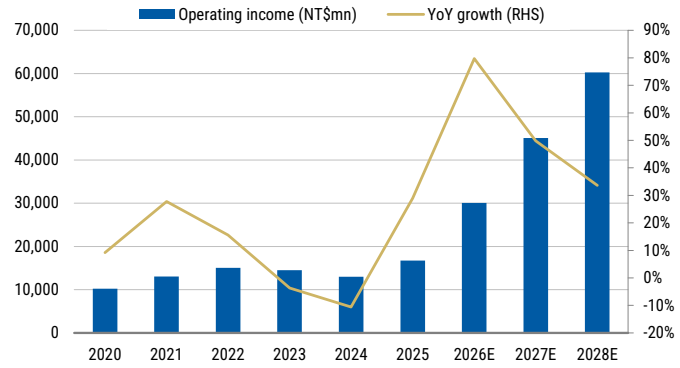
Operating profit rebounded by 29% YoY to NT$16,738mn in 2025, supported by both an expanding revenue base and strengthening gross margin. Operating margin also improved, increasing by 0.6ppt YoY to 10.1%, as the recovery in gross margin was partly offset by higher opex.

With the tax rate remaining relatively stable at 20-24%, Lite-On's net profit growth has generally tracked its operating profit growth. An exception occurred in 2023, when net profit grew 3% YoY despite a 4% decline in operating profit, mainly due to higher FX gains during the year.

Net margin has broadly tracked operating margin in recent years. It reached a record high of 9.8% in 2023 on improving gross margin, despite a higher opex ratio. It then declined to 8.7% in 2024 on gross margin erosion before rebounding to 9.1% in 2025, benefiting from increasing demand for AI power supplies.

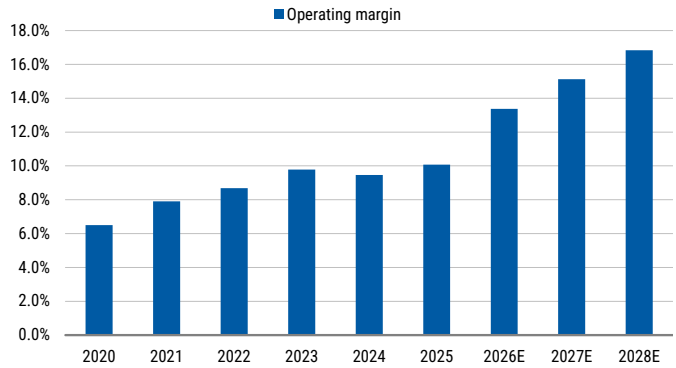
Looking ahead, we forecast net profit to grow 67% YoY in 2026e, followed by 44% YoY growth in 2027e and 32% YoY growth in 2028e, implying a 47% CAGR in 2025-28e. Net margin is likely to rise to 11.2% in 2026, 12.2% in 2027, and 13.4% in 2028.

Exhibit 124: Lite-on: 35% CAGR for operating profit in 2024-27



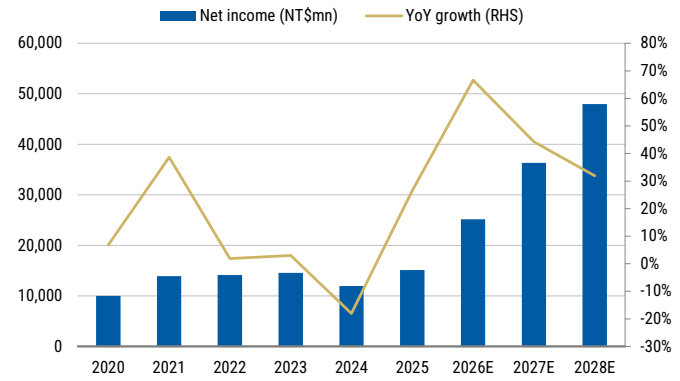
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 125: Lite-on: Operating margin



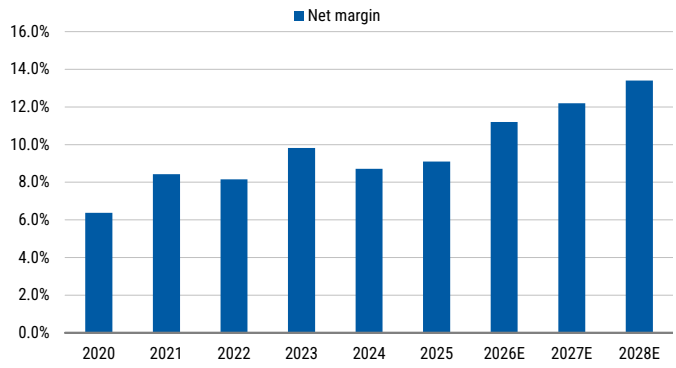
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 126: Lite-on: Net profit trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 127: Lite-on: Net margin



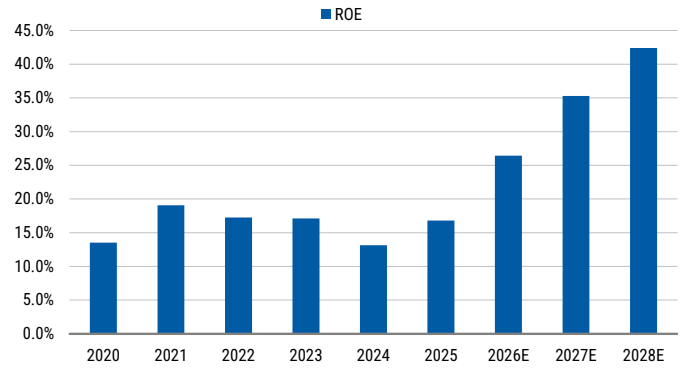
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

ROE to rise

ROE movements were highly correlated with changes in net margin during 2020-22, as asset turnover and financial leverage were largely stable over the period. ROE edged down slightly in 2023 despite higher net margin, which increased from 8.2% to 9.8%, due to lower asset turnover and financial leverage. ROE fell further to a trough of 13.1% in 2024 as net margin and asset turnover declined further. It then rebounded to 16.8% in 2025, primarily supported by net margin expansion, fueled by a favorable product mix, along with an improvement in asset turnover back to the 80% level.

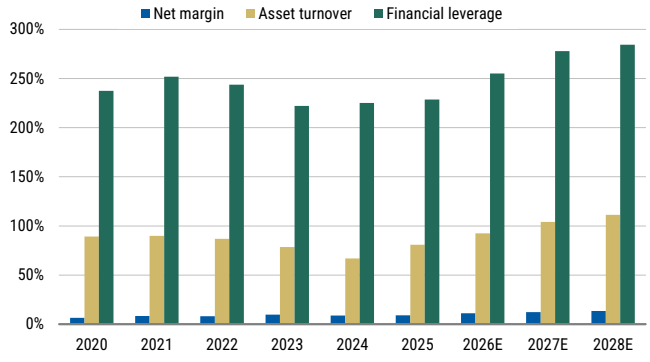
Looking ahead, we expect ROE to continue rising, reaching 26.4% in 2026, 35.3% in 2027, and 42.4% in 2028, primarily driven by continued margin expansion and improving asset turnover. Net margin is projected to increase from 9.1% in 2025 to 13.4% by 2028, supported by gross margin improvement from 22.9% to 27.4% and a decline in the operating expense ratio from 12.8% in 2025 to 10.5% in 2028.

Exhibit 128: ROE trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 129: DuPont analysis



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Improving balance sheet and continuous free cash flow generation

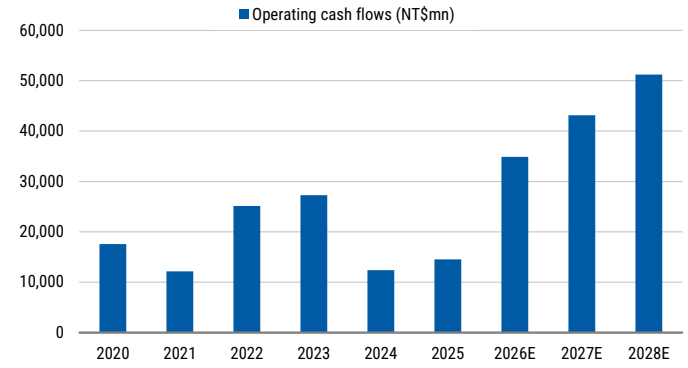
Lite-On has consistently generated positive operating cash flow in recent years, although cash flow dropped notably in 2021, 2024, and 2025, mainly due to higher working-capital-related cash outflows. The company has also consistently generated positive free cash flow, with capital expenditures generally at the NT$3-4bn level over the years. An exception occurred in 2025, when capital expenditure rose to NT$6.7bn, mainly for power-related capacity expansion in Vietnam, Taiwan, and the US.

As the company has announced investments of US$919mn and US$149mn in its US and Vietnam subsidiaries, respectively, alongside continued expansion of power-related capacity, mainly for PSUs, we now estimate capex to rise to ~NT$13bn in 2026, versus ~NT $7bn in 2025. Capex should then decline to NT$9bn in 2027 as investment levels normalize.

Lite-On's net cash position has remained relatively healthy, with its net cash-to-equity ratio improving from 65% in 2020 to 80% in 2023. It then declined to 75% in 2024 and 66% in 2025, as total debt increased by NT$7.7bn in 2024, while cash on hand declined in 2025, likely due to increasing capital investment.

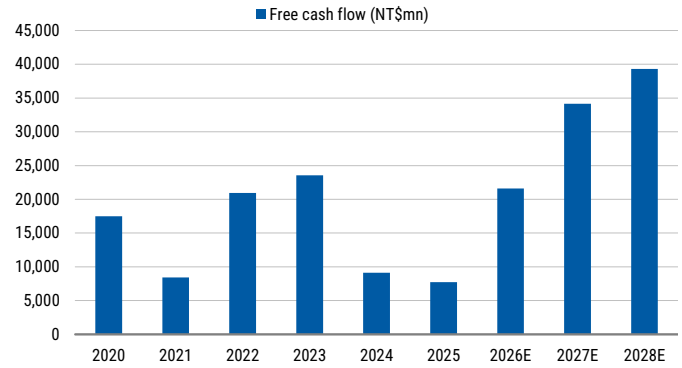
Looking ahead, we assume the total debt position will remain stable despite ongoing capex investment. The net cash position should continue to grow sequentially on the back of positive free cash flow generation.

Exhibit 130: Lite-on: Positive operating cash flow



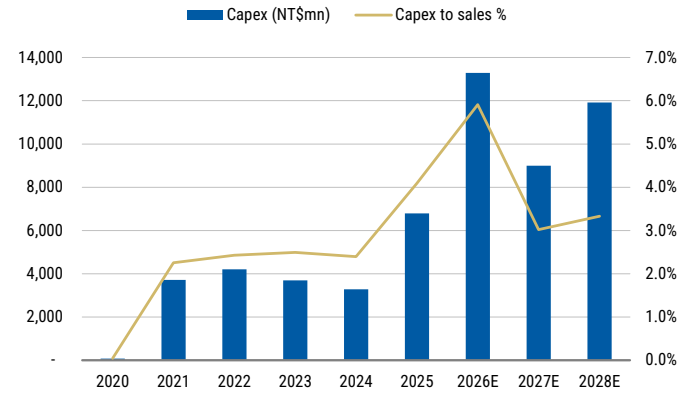
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 131: Lite-on: Improving free cash flow generation



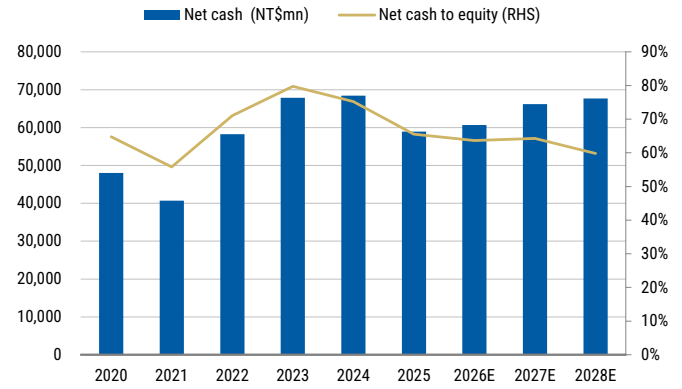
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 132: Lite-on: Capex and capex-to-sales ratio



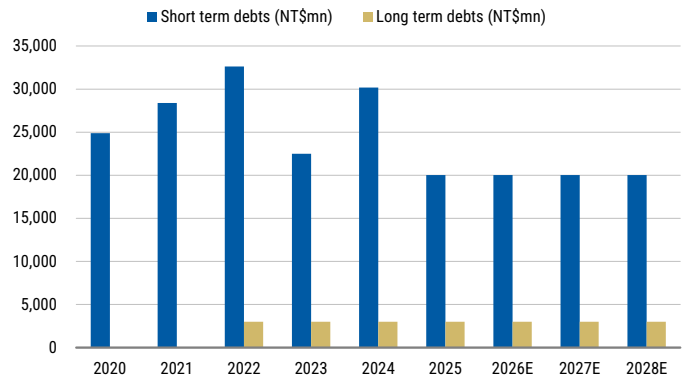
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 133: Lite-on: Net cash position



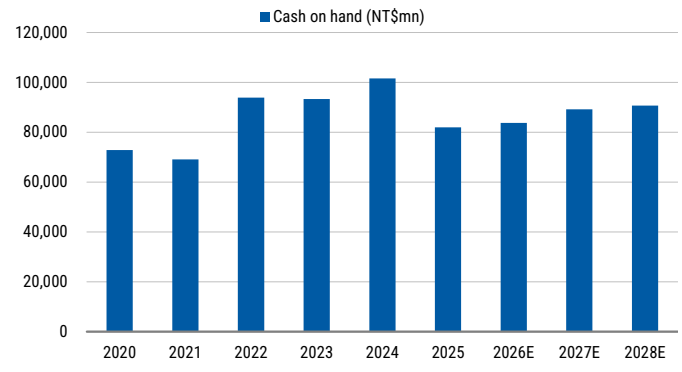
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 134: Lite-on: Debt position



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 135: Lite-on: Cash on hand

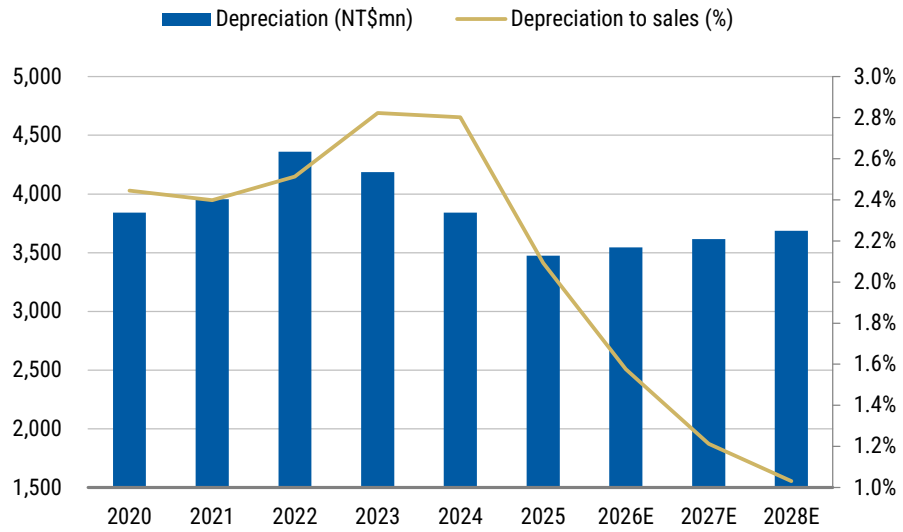


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Lite-On's average capex-to-sales ratio was around 2.3-2.4% during 2021-24, then rose to 4.1% in 2025 as the company began aggressively expanding global capacity, despite a larger revenue base during the year. Looking ahead, we project the capex-to-sales ratio to rise to 5.9% in 2026 amid continued production-site development, before declining to 3.0-3.3% in 2027-28 as investments normalize.

The depreciation-to-sales ratio remained at 2.4-2.5% during 2020-22 before rising to 2.8% in 2023, in line with higher capex during the year. The ratio then declined to 2.1% in 2025 on a larger revenue base. Looking ahead, we expect the ratio to decline further as the revenue base grows faster than capex and depreciation, falling to 1.6% in 2026 and 1.0% in 2028.

Exhibit 136: Lite-on: Depreciation and depreciation to sales

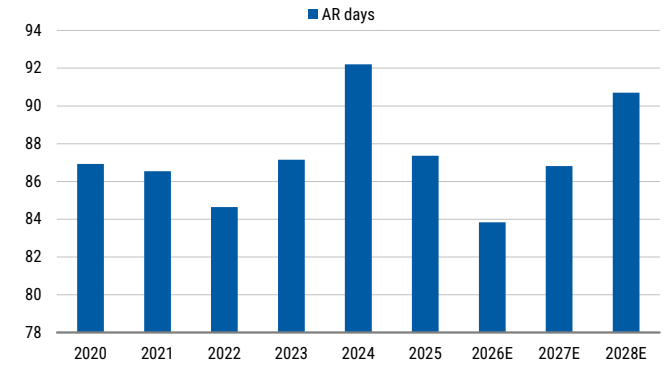


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Lite-On's cash conversion cycle (CCC) fluctuated within a relatively narrow band of roughly 35-44 days between 2020 and 2025, reflecting generally stable but modestly shifting working-capital efficiency. CCC was at a relatively low level of 35 days in 2020, supported by lower inventory days and higher AP days, before rising steadily to 44 days in 2022 as both inventory days (78) and AR days (85) increased, while AP days were slightly lower (119). From 2023 onward, the cycle improved meaningfully, falling to 36 days as AP days extended sharply from 119 in 2022 to 135. This improvement partially reversed in 2024, with CCC ticking up to 39 days as inventory and AR days both increased to 89 and 92 days, respectively, even as AP days extended further to 143. CCC then eased again to 37 days in 2025 as AP days moderated slightly to 138, alongside stable inventory and receivables levels.

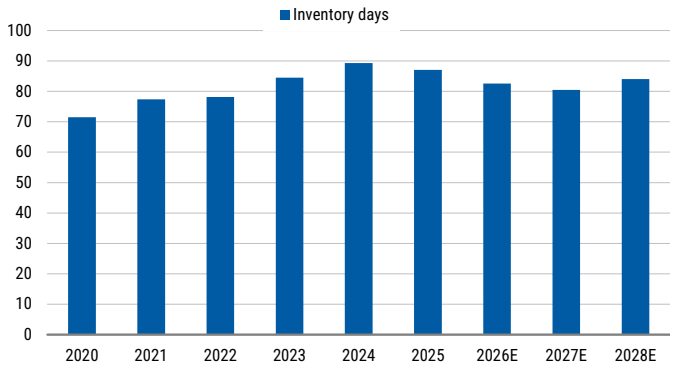
Looking ahead, we expect the overall cash conversion cycle to edge up slightly from 37 days in 2025 to 40 days in 2028, as increasing AR days and lower AP days are partially offset by improving inventory days.

Exhibit 137: Lite-on: AR days



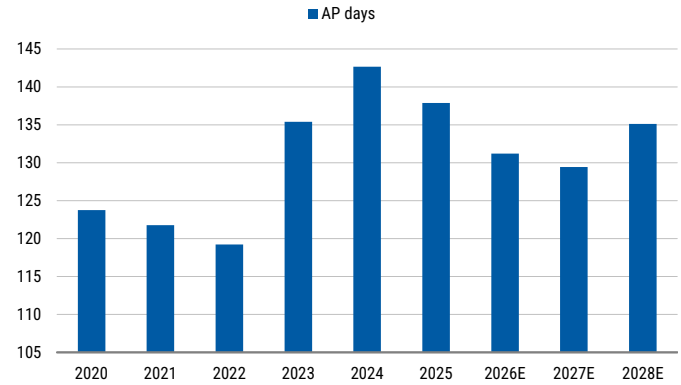
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 138: Lite-on: Inventory days



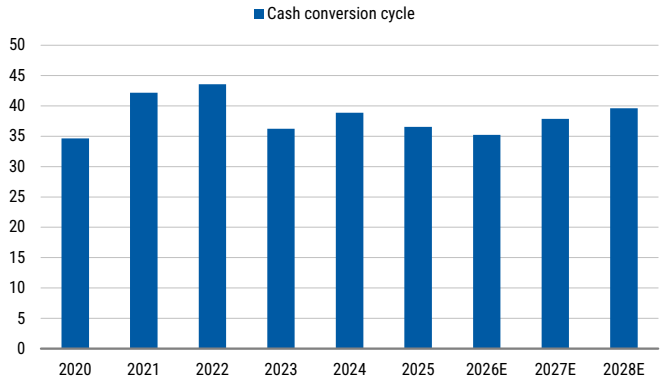
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 139: Lite-on: AP days



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 140: Lite-on: Cash conversion cycle



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Company Background – Lite-On

Company snapshot

Founded in 1975, Lite-On Technology is a Taiwan-based manufacturer of optoelectronic components, power supplies, and information technology products. The company produced Taiwan's first LEDs and became the first technology company listed on the Taiwan Stock Exchange in 1983. It mainly operates through four business segments: Information Products (power supplies, servers, networking equipment, keyboards, and chassis), Optoelectronics (LED packaging, LED lighting, automotive electronic components, and camera modules), Storage (storage components), and other supporting businesses. Additional product lines include energy management solutions, such as data center power supply systems and cloud server chassis; networking and surveillance products; and automotive electronics, including ADAS and EV charging solutions.

Lite-On is headquartered in Taipei, Taiwan. The company operates manufacturing sites, R&D centers, and sales offices across Asia (Taiwan, China, Japan, and Singapore), the US, and Europe (the UK, the Netherlands, and Germany).

Exhibit 141: Lite-on: Key milestones

Year	Key Milestones
1975	Established LITEON's first LED factory in Zhonghe, Taiwan
1983	Became the first electronics company listed on the Taiwan Stock Exchange (Ticker: 2301)
1985	Began manufacturing power supply units.
1989	Opened first overseas plant in Bangkok, Thailand, for optoelectronic products.
1990	• Set up a switching power supply plant in Penang, Malaysia. • Silitek acquired Maxi Switch, the No. 3 U.S. keyboard maker.
1991	Acquired NPE's switching power supply plant in the UK.
1994	Established a subsidiary in California, USA.
1999	Spun off the Optoelectronics business unit as LITEON IT Corp.
2002	LITEON Electronics, Silitek, LITEON Technology, and GVC merged into LITEON Technology Corporation (Ticker: 2301)
2003	Moved global headquarters to Neihu Science Park.
2006	• Acquired BenQ's Optical Disk Drive business, becoming a top-3 global ODD maker. • Partnered with Philips to enter the automotive disc drive market. • Acquired 100% of LITEON Automotive Corp., expanding into automotive electronics.
2007	• Acquired 100% of Li Shin International via stock swap, strengthening power supply resources for LCD TVs and other products. • Completed tender offer for Perlos, the world's largest handset casing maker.
2009	Opened East China Regional HQ in Wujin, Changzhou.
2010	• Invested in Epyon Power, a supplier of fast-charging technology. • Completed the world's first Optoelectronic Semiconductor Carbon Footprint and Type III Environmental Product Declarations.
2011	Completed the world's first Product Category Rules for Circuit Boards.
2013	Acquired Power Innovations (US), strengthening cloud and power management capabilities.
2014	Selected for the Carbon Disclosure Project's Climate Disclosure Leadership Index.
2016	Opened a new operations center in Kaohsiung's Nanzih Export Processing Zone.
2018	Divested the mobile camera module and mechanical business to focus resources.
2019	Acquired 100% of LOJ via tender offer.
2020	• Launched succession plan: Tom Soong named as Chairman, Anson Chiu named as President. • Completed sale of the SSD business to KIOXIA Holdings.
2021	• Launched an Employee Stock Ownership Trust (ESOT). • Joined the O-RAN Alliance and launched 5G small cells.
2022	• Began reconstructing the Zhonghe office into the "LITEON Zhonghe Digitech Center." • Inaugurated Phase II of the LITEON Vietnam plant in Haiphong.
2023	Opened its first corporate store, 'LITEON Collection,' in Neihu Science Park.
2024	• Took a 19.99% strategic stake in COSEL. • Broke ground on a new low-carbon building in Kaohsiung.
2025	• Returned to COMPUTEX with next-gen AI solutions, winning the Sustainable Design Award (Bronze). • Reopened its founding Zhonghe headquarters after a three-year reconstruction.

Source: Company data, Morgan Stanley Research

Exhibit 142: Lite-on: Global operation locations

Source: Company data

Exhibit 143: Lite-on: Global footprint

Headquarter	
Taipei City, Taiwan	
Sales Offices	
Milpitas, California, U.S.A.	
Plano, Texas, U.S.A.	
Tokyo, Japan	
Shanghai, China	
Singapore	
Hoofddorp, NL	
Cramlington, Northumberland, UK	
Wetzlar, DE	
Subsidiary	**Entity name**
American Fork, Utah, U.S.A.	Power Innovations International

Source: Company data, Morgan Stanley Research

Product offerings

Lite-On's key offerings can be categorized into six key areas: Optoelectronic Semiconductors, Energy Management, Consumer Electronics, Networking and Surveillance, Automotive Electronics, and Roadway Intellimation System.

1. Optoelectronic semiconductors

Its product portfolio spans a broad technological range, encompassing illumination and sensing applications, low- to high-power devices, standard and custom-made solutions, and components ranging from LEDs to lasers and from visible light to infrared and ultraviolet spectra.

Exhibit 144: Lite-on: Offerings for optoelectronic semiconductors

Optoelectronic Semiconductors						
Visible, UV & Automotive LEDs				Couplers	Optical Sensors	IR Emitters & Detectors
Illumination / UV-C / Automotive				Isolation & Signal Transfer	Ambient / Proximity / Gesture	Infrared Sensing
Automotive LED	SMD LED	Flash LED	UV LED	Photocouplers	Optical Sensors	IR Emitters & Detectors

Source: Company data, Morgan Stanley Research

2. Energy management

The product range serves the communications, industrial automation, smart home, medical, office, and automotive electronics industries. It offers a comprehensive suite of platinum- to titanium-efficiency-grade power supplies (CRPS, GPU, and SLIM series) suitable for both standard and AI servers, supporting AC or DC input with high power density, high reliability, and modular scalability. Power-shelf and battery-backup-unit (BBU) solutions, along with rack-cooling and liquid-cooling technologies for data centers, cloud infrastructure, and AI/machine-learning platforms, are also included in this segment.

Exhibit 145: Lite-on: Offerings for energy management

Energy Management			
Data Center Power Supply System	**Cloud Server Chassis**	**Power Supply Module**	**Magnetic Components**
CRPS / GPU / SLIM, Platinum-Titanium	Open Rack Standard V3	PoE / Industrial / LED Driver	Transformers & Inductors
			Includes components apply for cloud infrastructure, industrial power supply, white appliances, electric vehicle and consumer electric, etc
Server Power Networking Power Open Compute Project	Server Rack Server Chassis	POE Injector Industrial Power Adapter LED Driver	Magnetic Components

Source: Company data, Morgan Stanley Research

3. Consumer electronics

This segment is committed to advancing home entertainment and office solutions through feature-rich product design intended to enhance the overall user experience. Offerings include office/gaming solutions, laptop keyboard modules and touchpads, laptop power adapters, USB Type-C/PD chargers, etc.

Exhibit 146: Lite-on: Offerings for consumer electronics

Consumer Electronics		
3C Accessories	**Power Supply Module**	
Office / Gaming / Laptop peripherals	PC & Networking Power	
Office Solutions Gaming Solutions Laptop Keyboard Module Touchpad Laptop Power Adapters USB Type C/PD Charger	PC Power Supply	Networking Adapter

Source: Company data, Morgan Stanley Research

4. Networking and surveillance

The product portfolio covers networking solutions, video surveillance, AIoT solutions, and server-related offerings. The solutions are designed to be user-friendly and adaptable to emerging trends in digital intelligence.

Exhibit 147: Lite-on: Offerings for networking and surveillance

Networking and Surveillance		
Video Surveillance	**Access Control & Alarms**	**Networking Solutions**
Home / Building / Module	Smart Hub / Access / Alarm	5G / WLAN / Switch / Module
Home Security Camera Building Surveillance Camera Camera Module	Smart Hub Access Control Alarm Kits	5G Total Solution Wireless LAN Wired Switch Wireless Module

Source: Company data, Morgan Stanley Research

5. Automotive electronics

Lite-On is able to leverage its core competencies in optoelectronics and power management, along with more than four decades of expertise in automotive electronics. This segment is dedicated to advancing vehicle-charging technology, ADAS (Advanced Driver Assistance Systems) solutions, and automotive lighting modules through high-end engineering. AC charging solutions are offered across multiple current ratings (32/40A, 48A, 80A, and an EU-specific variant), while DC charging solutions include a 30kW fast-charging offering.

Exhibit 148: Lite-on: Offerings for automotive electronics

Automotive Electronics		
ADAS Solutions	**AC Charging Solutions**	**DC Charging Solutions**
Vision & Lighting	32/40A - 48A - 80A - EU	Fast Charging
Automotive Vision Solutions Vehicle Lighting Solutions	IC 32/40A IC 48A IC 80A IC EU	DC 30kW

Source: Company data, Morgan Stanley Research

6. Roadway Intellimation System

This segment is mainly operated under the LEOTEK brand, a subsidiary of the LITEON Technology Group with dual headquarters in Silicon Valley and Taipei. It has served road infrastructure across more than 30 countries—including North America, Europe, the Middle East, and the Asia-Pacific region—since 1992. The segment comprises intelligent roadway lighting, traffic signal systems, and AI-driven predictive maintenance solutions, all developed to support smarter and more efficient road infrastructure.

Exhibit 149: Lite-on: Offerings for Roadway Intellimation System

Roadway Intellimation System		
Roadway Lighting Intellimation	**Intelligent Traffic Signal System**	**AI Predictive Maintenance Solution**
Smart LED Streetlights	AI-driven Signal Control	Condition Monitoring
		Includes AI warning and maintenance management, traffic flow data analysis and computation, traffic-adaptive signal control, and priority passage for emergency vehicles, etc
Roadway Lighting Intellimation	Intelligent Traffic Signal System	AI Predictive Maintenance Solution

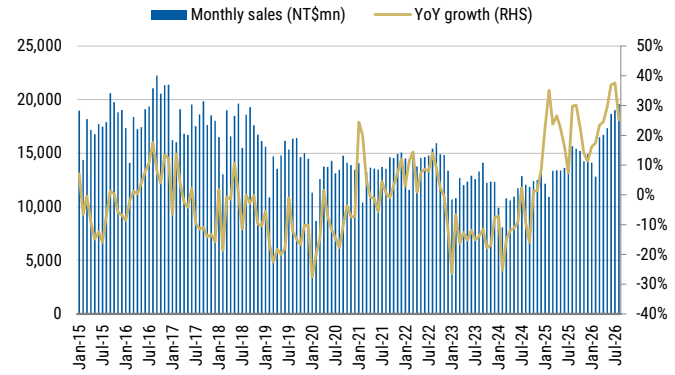
Source: Company data, Morgan Stanley Research

Monthly sales vs. share price

Lite-On's monthly sales have hovered around the NT$10-20bn level since 2013. Notably, we see a high correlation of ~76% between the company's monthly sales and its share price movements since 2023.

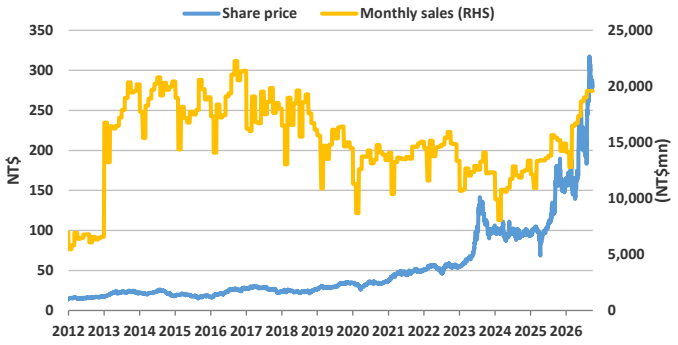
The company reported August monthly sales of NT$19,592mn (+3% MoM/+25% YoY).

Exhibit 150: Lite-on: Monthly sales vs. sales YoY growth



Source: TEJ, Morgan Stanley Research

Exhibit 151: Lite-on: Monthly sales vs. share price



Source: TEJ, Morgan Stanley Research

Key management & shareholding structure

Lite-On's board currently consists of five directors and four independent non-executive directors.

Its founder, Raymond Soong, and current Chairman, Tom Soong, together owned a stake of ~5% as of June 2026. Excluding the founder and Chairman, the management team owned a stake of ~0.1% as of June 2026.

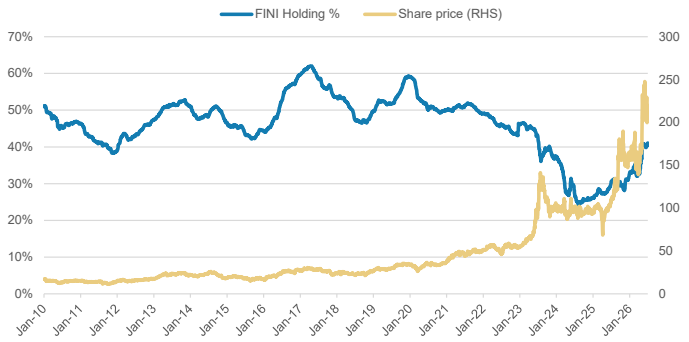
The FINI holding percentage as of July 7, 2026, was 41%, versus a previous peak of 62% in April 2017 and a historical average (since 2010) of 46%.

Exhibit 152: Lite-on: Shareholding structure

Shareholders	Ownership
Cathay MSCI Taiwan ESG Sustainability High Dividend Yield ETF	5.6%
Yuanta/P-shares Taiwan Dividend Plus ETF	5.5%
Ta-Rong Investment Co., Ltd	3.7%
Raymond Soong*	3.4%
Fuh Hwa Taiwan Technology Dividend Highlight ETF	3.3%
Labor Pension fund	2.6%
Ming-Hsing Investment Co., Ltd	2.6%
Ta-Sung Investment Co., Ltd*	2.1%
Chunghwa Post Co., Ltd	2.0%
Lite-On Technology Corporation (Treasury shares)*	1.7%
Total	**32.6%**

Source: Company data, Morgan Stanley Research. Note: Data as of August 2025. (*) are shown as insiders.

Exhibit 153: Lite-on: FINI holdings vs. share price since 2010



Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 154: Lite-on: FINI holdings vs. share price since 2023

Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 155:

Lite-on: Management profiles

Name	Title	Education	Work Experience
Tom Soong	Chairman and Chief Sustainability Officer, LITEON Technology	Electrical Engineering, University of Southern California, USA; International Business Program Attendance, NTU-FUDAN EMBA	- 20+ years at LITEON - Special Assistant to Chairman & Vice Chairman, LITEON - CEO of LITEON Smart Life and Applications Business Group - General Manager, LITEON Shanghai Operational Center
Anson Chiu	President and Director, LITEON Technology	Bachelor of Industrial Management, Lunghwa University of Science and Technology	- Joined LITEON Technology in 1987 - 35+ years in global supply chain and business development' - CEO, LITEON Power Conversion Business Group - General Manager, LITEON Power Conversion Business Group - Director, NB Business Unit, Power Conversion Business Group
Raymond Soong	Founder and Director, LITEON Technology and LITEON Cultural Foundation	Fellow, Industrial Technology Research Institute; Honorary Doctor of Management, National Chiao Tung University; Honorary Doctor of Management, Taipei University of Technology	- Chairman, LITEON Group - Chairman & Founder, LITEON Cultural Foundation - Chief Engineer, Texas Instruments Taiwan Ltd. - Board of Councilors member, Doctorate College of Technology, USC
Harvey Chang	Director Representative, Lite-On Technology Corp.	MBA, The Wharton School, University of Pennsylvania	- Independent Director, LITEON Technology - Chairman, IC Broadcasting - Chairman, TVBS and Catchplay Media Holdings Ltd. - President & CEO, Taiwan Mobile - Senior VP & CFO, TSMC
Sarina Lin	Director Representative, Lite-On; Advisor to Chairman's Office, Lite-On	Doctor of Laws, National Taiwan University	- Chief Legal Officer, Lite-On - Special Assistant to Chairman and Attorney, MediaTek Inc. - Director, MediaTek Foundation - SVP, CHRO & Chief Legal Officer, MStar Semiconductor - Director & CEO, MStar Semiconductor Charity Foundation - Director and Supervisor, MStar Semiconductor
Albert Hsueh	Independent Director, LITEON Technology;	MBA, Bloomsburg University, Pennsylvania, U.S.A.	- Chairman, PricewaterhouseCoopers Taiwan - Director, Corporate Governance Association in Taiwan - Professor, National Taiwan University and Tsinghua University, College of Technology Management
Mike Yang	Independent Director, Lite-On Technology Corp.	Master of Science, Arizona State University	- Associate Vice President, ADI Corp. Procurement Dept. - Director, LIONS Taiwan Technology Inc. - Director, Quanta Culture & Education Foundation - Cloud computing and global technology leadership
MK Lu	Independent Director, Lite-On Technology Corp.	Honorary Doctor, Tatung University; Honorary Doctor of Engineering, National Chiao Tung University; Department of Electrical Engineering, Tatung University	- Chairman, Tatung Company - Chairman, Sino-American Silicon Products Inc. - Chairman, Actron Technology Corporation - President, Lite-On Semiconductor Corp. - President, Vishay Lite-On Power Semiconductor Corp. - Vice President, Silitek Corp.
Jesse Ding	Independent Director, Lite-On Technology Corp.	PhD program, The Wharton School, University of Pennsylvania; MBA, University of Detroit, USA	- Chairman, EnTie Commercial Bank - General Manager, EnTie Commercial Bank - General Manager, TAIPEI FUBON COMMERCIAL BANK CO., LTD - Financial accounting and leadership decision-making

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Panasonic Holdings (6752.T)

Our target price is based on P/E 19.5x our FY3/28 EPS forecast.

We apply: 36x to EPS generated by AI infrastructure-related businesses, 5x to EPS attributable to IRA subsidies in the automotive battery business, and 13.5x to EPS from all other businesses.

Risks to Upside

- Acceleration of restructuring and exit/scale-down of unprofitable businesses
- Stronger than expected growth of datacenter BBU, automotive battery business

Risks to Downside

- Deteriorating visibility for energy business
- Expectations for business portfolio revisions sink

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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(as of September 30, 2026)

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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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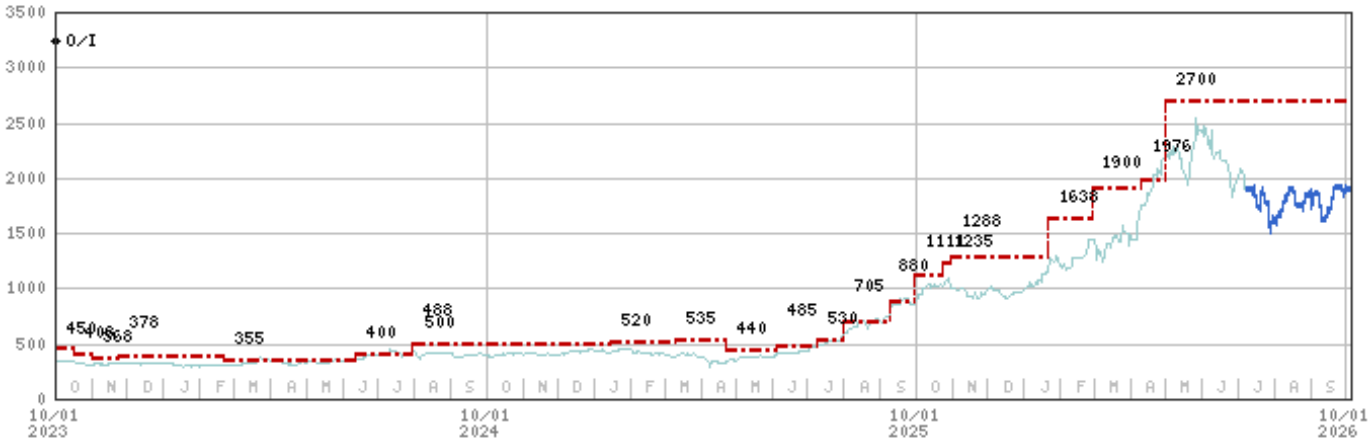
Stock Price, Price Target and Rating History (See Rating Definitions)

Delta Electronics Inc. (2308.TW) - As of 10/05/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : O/I

Price Target History: 9/9/21 : 345; 10/29/21 : 325; 2/25/22 : 303; 7/29/22 : 312; 9/21/22 : 326; 10/6/22 : 350; 11/23/22 : 368; 8/2/23 : 450; 10/17/23 : 408; 11/1/23 : 368; 11/23/23 : 378; 2/21/24 : 355; 6/12/24 : 400; 7/30/24 : 488; 8/1/24 : 500; 1/15/25 : 520; 3/11/25 : 535; 4/23/25 : 440; 6/4/25 : 485; 7/9/25 : 530; 8/1/25 : 705; 9/8/25 : 880; 9/30/25 : 1111; 10/24/25 : 1235; 10/30/25 : 1288; 1/21/26 : 1638; 2/27/26 : 1900; 4/10/26 : 1976; 5/1/26 : 2700

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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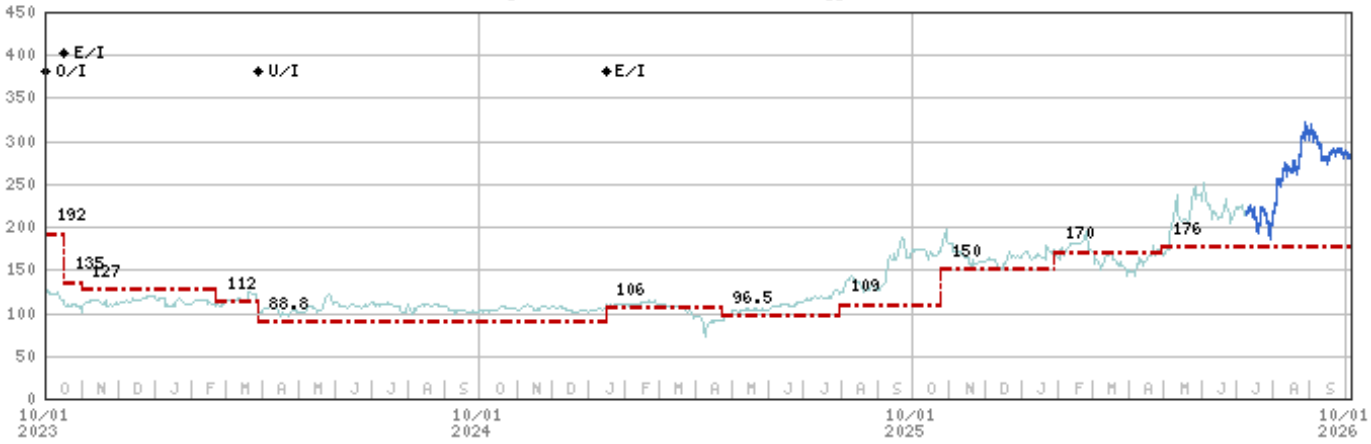
Lite-On Technology (2301.TW) - As of 10/05/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : E/I; 7/31/23 : O/I; 10/17/23 : E/I; 3/28/24 : U/I; 1/15/25 : E/I

Price Target History: 4/30/21 : 62; 7/19/22 : 65; 2/21/23 : 61.2; 7/31/23 : 192; 10/17/23 : 135; 10/31/23 : 127; 2/21/24 : 112; 3/28/24 : 88.8; 1/15/25 : 106; 4/23/25 : 96.5; 7/31/25 : 109; 10/24/25 : 150; 1/28/26 : 170; 4/29/26 : 176

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
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Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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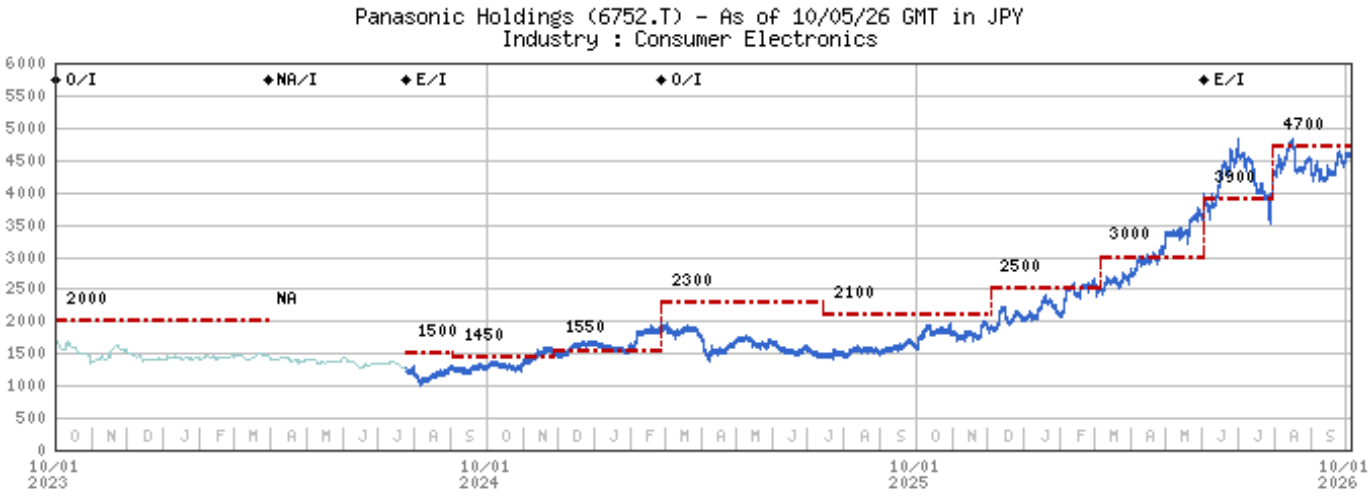

Stock Rating History: 10/1/21 : O/I; 3/29/24 : NA/I; 7/25/24 : E/I; 2/26/25 : O/I; 6/3/26 : E/I

Price Target History: 9/22/21 : 1900; 3/1/22 : 1700; 6/13/22 : 1500; 12/1/22 : 1600; 2/20/23 : 1500; 6/7/23 : 2000; 3/29/24 : NA; 7/25/24 : 1500; 9/3/24 : 1450; 11/28/24 : 1550; 2/26/25 : 2300; 7/14/25 : 2100; 12/3/25 : 2500; 3/6/26 : 3000; 6/3/26 : 3900; 7/31/26 : 4700

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/05/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$38.64
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.76
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,055.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,730.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.46
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$53.35
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.91
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$37.16
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$61.95
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43

Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$23.88
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$179.10
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,111.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.62
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,075.00
Advantech (2395.TW)	O (01/20/2021)	NT$740.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,180.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,585.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$39.05
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,550.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,485.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$890.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,325.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,005.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$146.50
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$68.20
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,450.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$345.50
Innolux (3481.TW)	E (04/07/2025)	NT$52.30
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,355.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$298.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.40
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$995.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$207.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.80
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,700.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.86
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$388.50
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,215.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$254.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.30
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$3,000.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.92
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,875.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,495.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$88.70
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$331.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,325.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$606.00

Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,150.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$626.00
Zhen Ding (4958.TW)	++	NT$576.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$570.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,780.00

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ASIA PACIFIC INSIGHT

October 6, 2026 12:44 AM GMT

Greater China Technology Hardware

Integrated Power Solutions Driving Secular Demand

Increasing power consumption, along with the deployment of BBU/CBU/800/±400VDC/SST in AI clusters, could push the technology barrier higher and present growth opportunities for players with superior capabilities. Fuel cells could also emerge as a new opportunity amid power constraints.

Secular demand for power amid AI capex: AI-related power consumption will continue to increase, and we believe this will lead to secular demand for power-related components, such as power supply units (PSUs), battery backup units (BBUs), capacitor bank units (CBUs), coolant distribution units (CDUs), and so on. Moreover, the requirement for efficiency will inevitably necessitate the adoption of 800/±400 voltage direct current (VDC) in the power rack/center/block configuration (initially from late 2026 and broadly in 2028, we estimate), eventually extending to the deployment of solid state transformers (SST) in the longer term – all pointing to a more centralized power delivery architecture in the future. This trend of aggregation could raise the technology barriers amid the fusion of white/grey space, favoring players with better capabilities and execution. Overall, we forecast the datacenter PSU market size to grow at a 75% three-year CAGR to reach US$33bn in 2028.

Fuel cells to improve flexibility and resilience of AI power delivery: Despite the strong AI capex outlook (MSe global cloud capex +97%/20% YoY in 2026/2027), power has become a bottleneck, indicated by MS's forecast power shortfall of 5GW in 2026, scaling to a total 72GW through 2029 (see report). Fuel cells, among other alternative energy sources, could emerge as a good solution to pick up some slack and enhance efficiency through active micro grid management. Those who are able to offer comprehensive solutions from power sources to power delivery could be increasingly more competitive in the market.

Assuming coverage of Delta and Lite-On:

- Delta (OW, PT NT$2,500, based on 30x 2027/2028 P/E) should maintain its dominant position in the industry with its integrated power/thermal related solutions while fuel cells could add incremental growth opportunities in the longer term. We think a 30x P/E target multiple is justified by our estimated 63% earnings CAGR in 2025-28, driven by a 44% revenue CAGR and operating margin expansion to 22.1% in 2028 vs 15.1% in 2025,
- Lite-On (EW, PT NT$315, based on 17x 2027/2028 P/E), the second-largest player in AI power-related products, should also ride the tailwinds despite its slower progress in HVDC power racks and liquid cooling solutions. We think a 17x P/E target multiple is justified by our estimated 47% earnings CAGR in 2025-28 (29% revenue CAGR and 6.8ppt OPM expansion to 16.8% in 2028).

MORGAN STANLEY TAIWAN LIMITED+
Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

MORGAN STANLEY MUFG SECURITIES CO., LTD.+
Kazuo Yoshikawa, CFA
Equity Analyst
Kazuo.Yoshikawa@morganstanleymufg.com +81 3 6836-8408

MORGAN STANLEY ASIA LIMITED+
Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

MORGAN STANLEY TAIWAN LIMITED+
Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY MUFG SECURITIES CO., LTD.+
Hidetaka Suzuki
Research Associate
Hidetaka.Suzuki@morganstanleymufg.com +81 3 6836-8402

GREATER CHINA TECHNOLOGY HARDWARE

Asia Pacific Industry View	**In-Line**

WHAT'S CHANGED

Delta Electronics Inc. (2308.TW)	**From**	**To**
Price Target	NT$2,700.00	**NT$2,500.00**
Lite-On Technology (2301.TW)	**From**	**To**
Price Target	NT$176.00	**NT$315.00**

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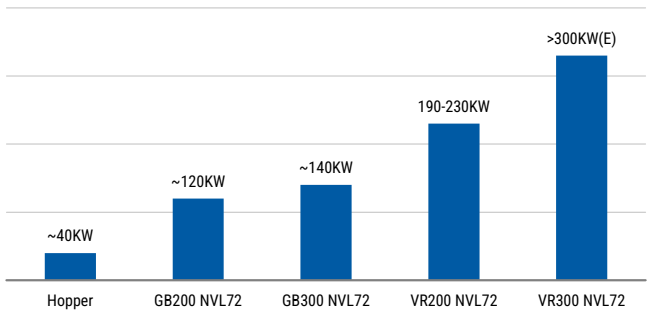
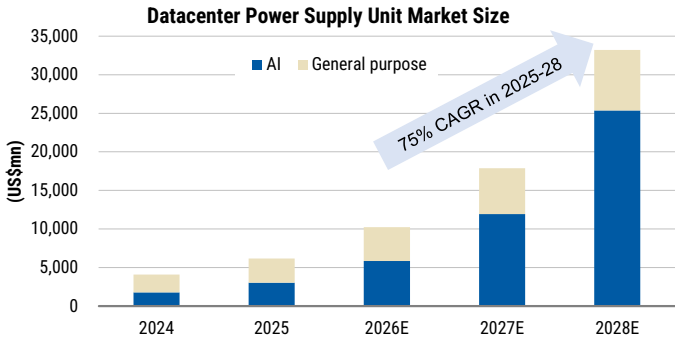
The Story in Charts

Exhibit 1: AI servers continuing to see higher TDP...



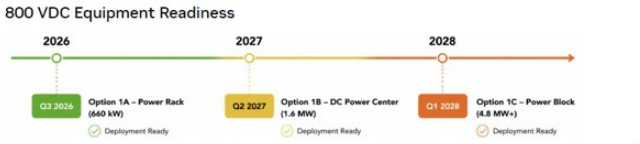
Source: Company data, Morgan Stanley Research estimates

Exhibit 2: ...which drives data center power supply demand



Source: Company data, Morgan Stanley Research estimates

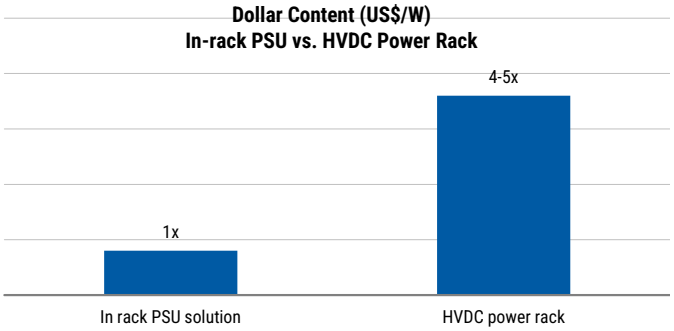
Exhibit 3: HVDC to see early adoption from late 2026...



Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

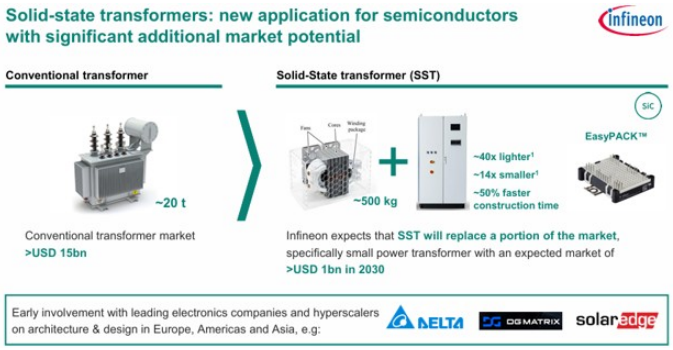
Source: Nvidia

Exhibit 4: ...with a much higher dollar content



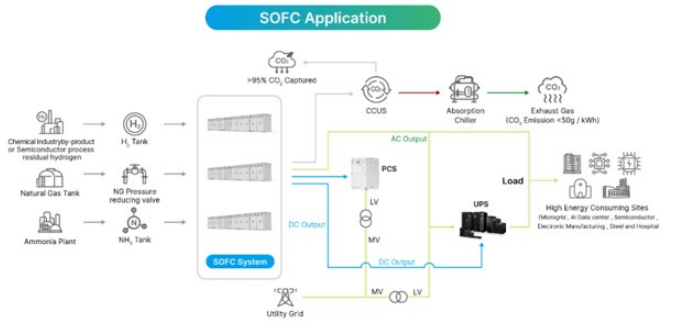
Source: Morgan Stanley Research

Exhibit 5: SST is also in the pipeline for more centralized power delivery



Source: Infineon

Exhibit 6: Fuel cells are a good option for better resilience in power architecture



Source: Delta

Executive Summary

AI cluster performance evolving to fulfill demand

The rapid adoption of artificial intelligence is driving a significant increase in global data center compute demand. Advances in foundation models, reasoning capabilities, agentic AI, and enterprise AI deployment are increasing the computational resources required for both training and inference. To support this growth, hyperscalers and model developers continue to deploy larger and more powerful AI clusters, driving demand for greater compute, memory, and networking capacity.

As AI infrastructure scales, cluster performance is increasingly determined by the efficiency of the entire system rather than the capabilities of individual processors alone. Compute, networking, cooling, and power infrastructure must all evolve together to support higher rack densities and larger deployments. While semiconductor innovation remains critical, supporting infrastructure is becoming an equally important factor in enabling future AI growth.

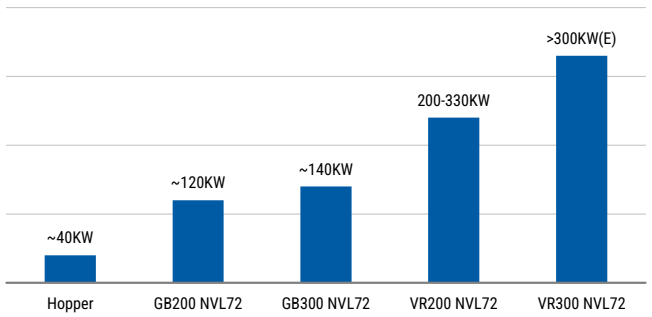
AI cluster performance is fundamentally a power problem

As AI clusters continue to scale in size and performance, power consumption is becoming one of the most important constraints on future infrastructure deployment. The power challenge is no longer limited to GPUs alone, as modern AI clusters comprise tightly integrated compute, networking, memory, storage, and cooling systems, all of which require increasing amounts of power as performance requirements rise. Consequently, the next phase of AI infrastructure investment should be viewed not only as a semiconductor upgrade cycle but also as a broad-based power infrastructure upgrade cycle.

Server power consumption has increased significantly across successive AI hardware generations. Nvidia's architecture roadmap illustrates the industry's migration from relatively modest server-level power budgets under the Ampere and Hopper generations toward rack-scale systems under Blackwell and future Rubin architectures. Estimated rack-level power consumption has increased from roughly 40kW for Hopper systems to approximately 120kW for GB200 NVL72 and 140kW for GB300 NVL72, with future Vera Rubin systems expected to approach 200-300kW per rack with current Vera Rubin systems reaching up to 330kW per rack while future Vera Rubin generations are expected to surpass these levels. As compute density continues to increase, power delivery infrastructure is becoming a critical enabler of system performance.

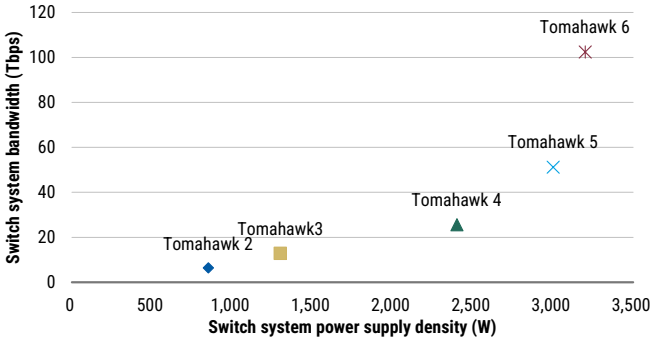
The increase in power consumption is not limited to compute hardware. Networking systems are also becoming increasingly power intensive as switch bandwidth scales from 100G architectures toward 800G and 1.6T generations. Despite substantial process-node improvements, switch ASIC power consumption has continued to rise due to increasing complexity and higher SerDes speeds. As a result, networking equipment now requires significantly larger power budgets than previous generations, creating additional demand for higher-capacity power supplies throughout the data center.

Exhibit 7: TDP roadmap for Nvidia racks



Source: Morgan Stanley Research

Exhibit 8: Power supply density for Tomahawks



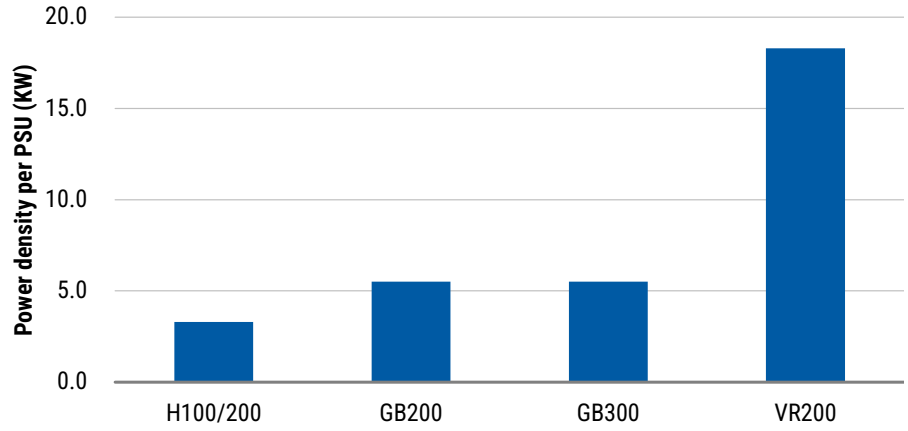
Source: Morgan Stanley Research

The growth in power consumption is driving a corresponding upgrade cycle for power-supply units (PSUs). Earlier AI server generations typically relied on multiple 3-3.3KW power supplies installed directly within individual server chassis. As the industry transitions toward rack-scale architectures, power conversion is increasingly centralized into dedicated power shelves capable of supporting substantially larger power loads. In Blackwell systems, power shelves utilize 5.5KW power supplies, while future Rubin-generation platforms are expected to increase PSU ratings to approximately 18KW, representing a significant increase in both PSU content and power density.

Delivering higher power within a similar footprint creates significant engineering challenges. Increasing PSU power density requires advances across multiple areas, including higher-performance power semiconductors such as silicon carbide (SiC) and gallium nitride (GaN), more efficient power-conversion topologies, smaller and more capable magnetic components, improved thermal-management solutions, and greater system integration. As output power increases, thermal constraints become increasingly important, making heat dissipation, airflow optimization, and mechanical design critical determinants of PSU performance.

Consequently, on top of the volume driven growth, the power-supply industry is benefiting from the technology driven dynamics, where efficiency, power density, and thermal performance are becoming key competitive differentiators. As AI rack power requirements continue to increase, PSU innovation will become an increasingly important component of the broader AI infrastructure investment cycle.

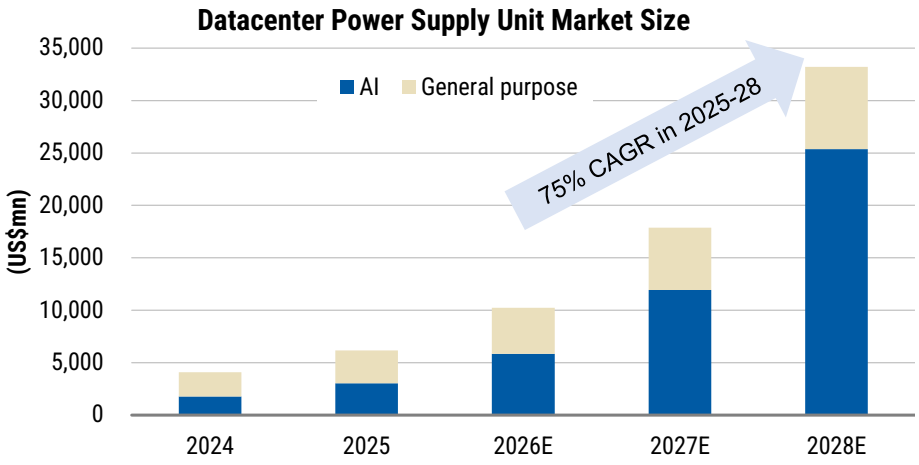
Exhibit 9: Power density per PSU trend in Nvidia server systems



Source: Nvidia, Morgan Stanley Research

Overall, we estimate the global datacenter power supply market will grow at a three-year CAGR of 75% to reach US$33bn in 2028 from US$6bn in 2025, driven mainly by server shipment growth, increasing power consumption per rack, and the adoption of HVDC power racks from 2027.

Exhibit 10: Global datacenter power supply market forecast



Source: Company data, Morgan Stanley Research estimates

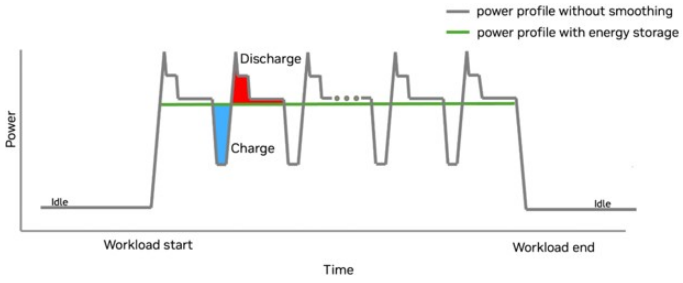
Complementary hardware for more resilient power delivery

As AI clusters become more power intensive, maintaining stable and resilient power delivery is emerging as a critical design priority. Traditional centralized power architectures are increasingly being complemented by localized energy-storage and power-conditioning technologies that can respond rapidly to dynamic workload demands.

Battery Backup Units (BBUs) represent one of the most important additions to modern AI power architectures. Positioned close to the computing workload, BBUs provide localized energy storage capable of responding significantly faster than facility-level backup systems. Their role extends beyond conventional backup power functionality. BBUs can

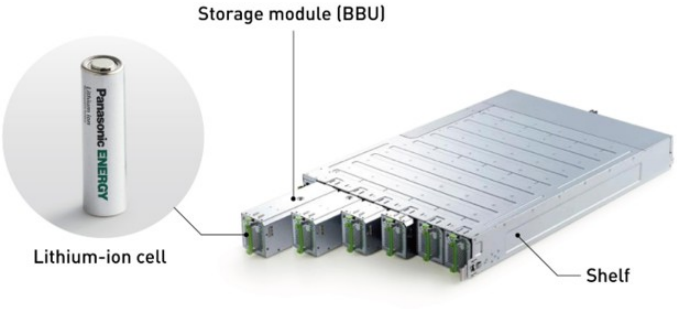
absorb transient fluctuations, smooth power ramps, and improve overall rack-level power stability. By providing local buffering capacity, BBUs help reduce the burden on upstream power distribution infrastructure while improving overall system resilience.

Exhibit 11: Concept of peak shaving for AI server power loads



Source: Nvidia

Exhibit 12: BBU



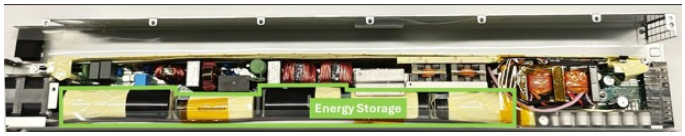
Source: Panasonic

Capacitor Backup Units (CBUs) address an even shorter-duration segment of the power stability challenge. Utilizing supercapacitors and other high-power-density storage technologies, CBUs can deliver and absorb power within milliseconds, making them particularly effective for managing rapid GPU power spikes. These systems act as shock absorbers within the electrical architecture, preventing sudden load changes from propagating throughout the broader power network. As AI rack power densities continue increasing, capacitor-based energy storage is becoming an attractive solution for improving power quality without requiring substantial oversizing of facility infrastructure.

Exhibit 13: CBU

Source: Panasonic

Exhibit 14: GB300 power shelf with integrated capacitors



Source: Nvidia

Together, BBUs and CBUs provide complementary benefits. BBUs are designed to handle longer-duration events, while CBUs are optimized for extremely fast transient response. The integration of these technologies enables operators to improve power quality, reduce stranded infrastructure capacity, enhance system reliability, and increase effective utilization of expensive electrical assets. As rack densities continue to rise, localized energy-storage solutions are likely to become a standard component of AI data center design rather than an optional enhancement.

The adoption of BBU and CBU architectures may also enable future increases in rack density. As operators attempt to push toward several hundred kilowatts per rack, localized power-buffering solutions can help mitigate transient peaks and reduce stress on grid-facing infrastructure. This allows operators to deploy higher-density computing resources without proportionately increasing upstream electrical capacity.

Exhibit 15: Comparison of CBU and BBU

Dimension	CBU	BBU
Primary storage medium	EDLC supercapacitors, LICs or high-capacitance	Typically lithium-ion battery cells managed by a BMS
Primary strength	High power, fast charge/discharge and frequent	Higher stored energy and longer sustained
Main function	Transient smoothing, peak shaving, voltage	Outage backup, source-transfer bridging and orderly workload
Response profile	Well suited to millisecond-class load support and	Fast switchover, but generally optimized for sustained support rather
Typical duration	Milliseconds to seconds; selected systems can extend toward tens of seconds or more	Commonly tens of seconds to several minutes, depending on battery and rack design
Power versus energy density	Higher power capability but lower energy storage per unit volume than a battery	Higher energy density, making it more economical for longer events
Cycle life	Eaton specifies more than one million cycles and up to 20 years for certain supercapacitor modules	Battery life is more sensitive to depth of discharge, temperature and repeated cycling; BMS supervision is therefore essential.
State management	Requires cell balancing, voltage/temperature monitoring and state-of-health tracking, but does not use a conventional battery fuel	Requires BMS functions including charge control, state of charge, state of health and cell protection
Best-fit use case	Repetitive AI power spikes, inrush events, PSU load leveling and bus stabilization	Power continuity through outages, brownouts or generator/source-transfer intervals

Source: Morgan Stanley Research

HVDC emerging as a necessity, with SST in the pipeline

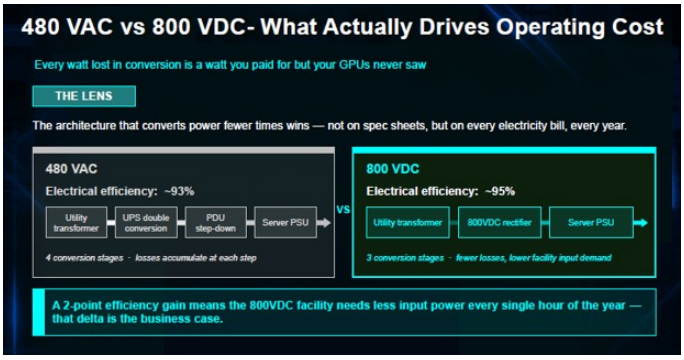
800VDC and ±400VDC progressing simultaneously

The rapid increase in AI rack power density is creating growing challenges for conventional power-distribution architectures. As rack-level power consumption rises from tens of kilowatts to several hundreds of KWs, electrical losses, cable size, thermal management requirements, and overall system complexity increase significantly. Under traditional low-voltage power architectures, delivering larger amounts of power requires proportionally higher current, which increases resistive losses and places greater stress on cables, connectors, busbars, and power-conversion equipment.

At the same time, every power-conversion stage introduces efficiency losses and generates additional heat. As AI deployments scale into multi-megawatt and eventually gigawatt-level campuses, even small efficiency improvements can translate into meaningful reductions in operating costs, cooling requirements, and infrastructure investment. Consequently, the industry is increasingly focused on power architectures that can reduce conversion stages, lower current levels, and improve overall power-delivery efficiency.

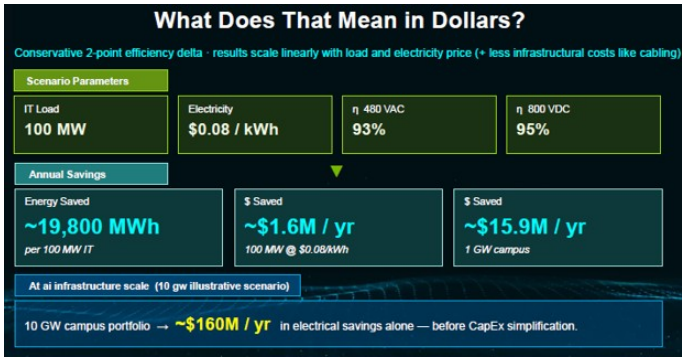
Against this backdrop, high-voltage direct current (HVDC) is emerging as one of the most important architectural upgrades for next-generation AI infrastructure. By increasing distribution voltage, HVDC allows the same amount of power to be delivered at a lower current, reducing conduction losses and improving system efficiency. Higher-voltage architectures also improve scalability for ultra-high-density rack deployments by alleviating some of the physical and thermal constraints associated with conventional power-distribution systems.

Exhibit 16: Better power efficiency for 800VDC architecture



Source: Delta

Exhibit 17: Better efficiency leads to cost savings



Source: Delta

Nvidia's roadmap illustrates the industry's gradual migration toward higher-voltage power architectures. Earlier AI server generations primarily utilized conventional power-delivery schemes centered around rack-level 48V distribution. As rack power requirements increased with Blackwell-generation systems, industry participants began evaluating higher-voltage architectures capable of supporting substantially greater power density. Looking ahead, the industry is increasingly converging around 800VDC as a potential next-generation standard for large-scale AI deployments, particularly as rack-level power consumption approaches several hundred kilowatts. The transition reflects the need to improve power-distribution efficiency while supporting the continued growth of AI computing infrastructure.

We believe that the adoption of the 800VDC power rack could start from late 4Q26, though it should be small scope deployment through 2027, since the in-rack power shelf is the reference design in the Vera Rubin NVL72 rack. Broader adoption should happen more likely from late 2027 and 2028, as the Vera Rubin Ultra rack further pushes up the power consumption.

Exhibit 18: HVDC deployment roadmap

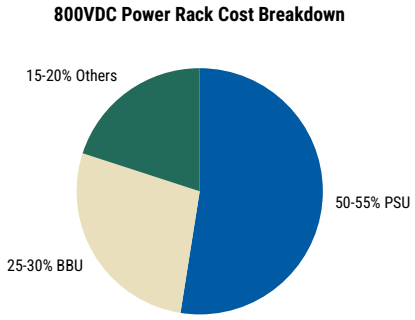
Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

Source: Nvidia

The 800VDC power rack BOM is primarily driven by the PSU, which accounts for 50–55% of total cost, followed by the BBU at 25–30%, with other components comprising the remaining 15–20%. This cost structure reflects the broader functionality integrated into an HVDC power rack, combining power conversion, backup power and supporting components within a centralized architecture.

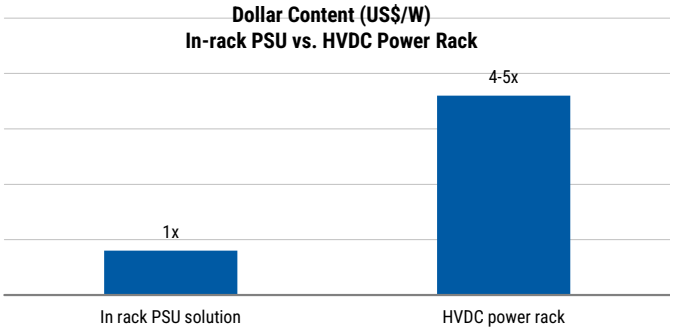
For power supply players, one of the most direct implications is that an HVDC power rack carries around 4–5x higher dollar content per watt compared with a conventional in-rack PSU solution, since the former integrates more components into a robust system and power supply players usually ship the entire rack instead of just the components.

Exhibit 19: BOM cost breakdown for an 800VDC power rack



Source: Morgan Stanley Research

Exhibit 20: Higher dollar content for HVDC power racks



Source: Morgan Stanley Research

However, the industry has not yet fully converged on a single architecture. While Nvidia's roadmap increasingly points toward 800VDC, several hyperscalers and members of the Open Compute Project (OCP) ecosystem have expressed interest in ±400VDC architectures. Both approaches deliver similar total voltage levels and pursue many of the same objectives, including lower current, higher efficiency, and improved scalability. The debate largely centers on implementation trade-offs rather than end goals.

Supporters of 800VDC generally highlight the advantages of a simpler single-bus architecture, potentially lower system complexity, and closer alignment with existing industrial high-voltage DC equipment. The architecture avoids the need to maintain balanced positive and negative voltage rails and may simplify certain aspects of power conversion and system integration. These characteristics make 800VDC particularly attractive in environments where operational simplicity and interoperability with existing electrical infrastructure are key priorities.

Advocates of ±400VDC, meanwhile, emphasize the advantages of a bipolar architecture. By creating positive and negative voltage rails around a neutral reference point, ±400VDC can potentially reduce insulation requirements, simplify certain fault-management scenarios, and provide improved flexibility for load balancing and future system expansion. Many hyperscalers also view the bipolar approach as being more compatible with long-term visions for highly distributed power systems incorporating energy storage, renewable energy, and microgrid functionality.

Importantly, the ultimate beneficiaries may be largely independent of which standard gains wider adoption. With the exception of certain power-semiconductor and protection-system requirements, many component suppliers across the power ecosystem remain

broadly agnostic between 800VDC and ±400VDC implementations. Higher-voltage architectures in either form require similar upgrades in power conversion, power management, cooling, connectors, magnetics, and infrastructure equipment. As a result, the industry's migration toward HVDC matters significantly more than the eventual selection of any single voltage standard.

Exhibit 21: Comparison of 800VDC and ±400VDC

Metric	±400VDC	800VDC
Voltage to ground	400V relative to ground	800V relative to ground
Voltage differential	800V	800V
Power distribution efficiency	High	Slightly higher
Scalability	Good	Excellent
Insulation requirements	Lower	Higher
Maintenance safety	Better	More challenging
System architecture simplicity	Dual-rail system	Single-bus system
Grounding Complexity	More complicated	Simpler
Electrical footprint	Small	Smaller
Ability to reduce conversion stages	Moderate	Highest
Long-distance distribution capability	Excellent	Excellent
Maturity of ecosystem	Better today	Early but rapidly developing
Industry players	OCP, hyperscalers	Nvidia
Long term industry direction	Transitional architecture	Likely end-state architecture

Source: Morgan Stanley Research

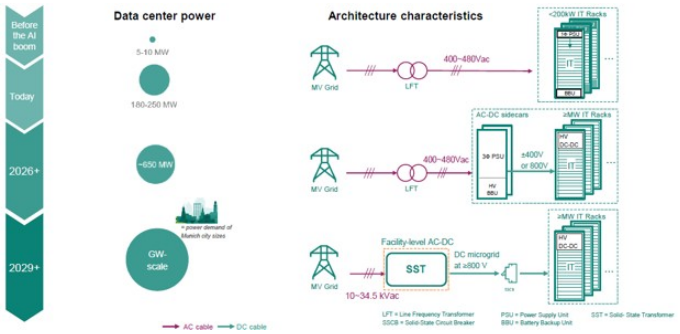
Solid-state transformers to follow after 800VDC/±400VDC matures

Looking further ahead, solid-state transformers (SSTs) represent a potential next-generation evolution of power infrastructure. Unlike traditional transformers that operate primarily as passive devices, SSTs leverage power electronics to provide voltage conversion, power conditioning, and bidirectional energy management within a single platform. This enables greater controllability and flexibility while supporting increasingly digitalized power networks.

Although SSTs remain at an early stage of commercialization and face challenges related to economics, reliability, and large-scale deployment, the technology aligns well with long-term data center trends. As AI campuses evolve into utility-scale power consumers incorporating energy storage, distributed generation, and advanced power management, SSTs could eventually become a key enabling technology.

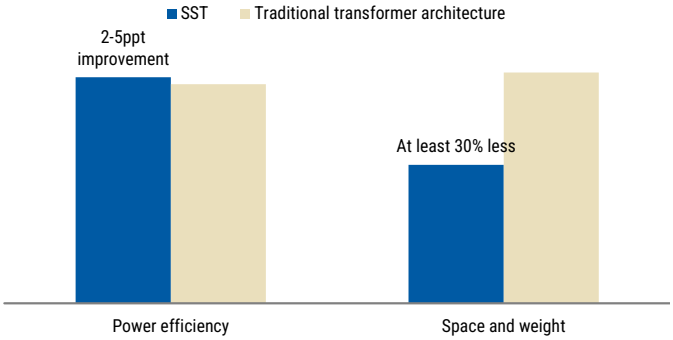
Therefore, while HVDC is expected to drive the next phase of power architecture upgrades, SSTs represent a longer-term technology roadmap that could further transform how future data centers distribute and manage electricity.

Exhibit 22: SST enabling more efficient power delivery architecture



Source: Infineon

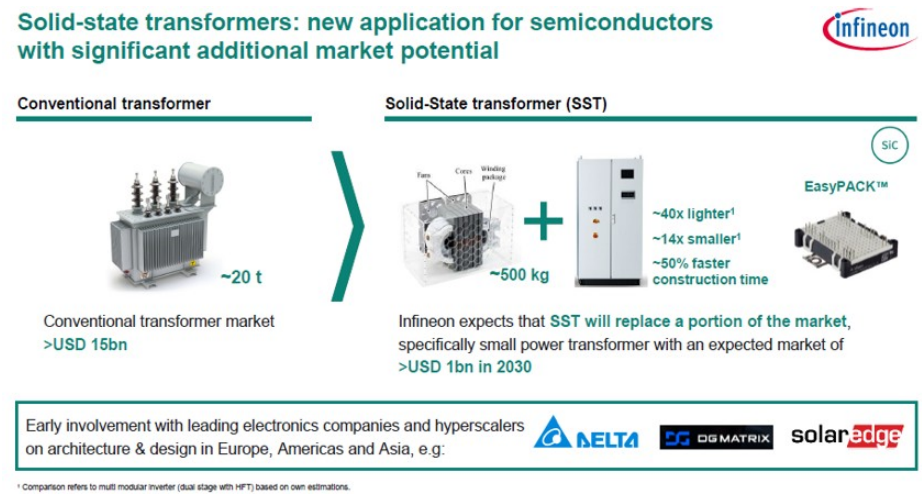
Exhibit 23: SST offers better power efficiency with at least 30% less space/weight



Source: Morgan Stanley Research

According to Infineon, SST could represent an emerging >US$1bn market opportunity by 2030, as the technology begins to replace a portion of the >US$15bn conventional transformer market, particularly in smaller-power applications. While SST penetration is expected to remain relatively modest versus the overall transformer market, the opportunity is significant for relevant players in the longer term.

Exhibit 24: SST to see initial adoption in 2030

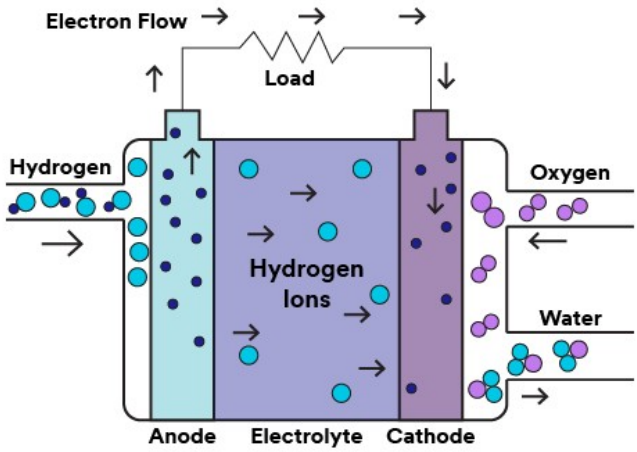


Source: Infineon

Fuel cells for better flexibility and resilience

Fuel cells generate electricity through an electrochemical reaction rather than combustion. Hydrogen, natural gas, biogas, or other fuels are supplied to the fuel cell stack, where chemical energy is converted directly into electricity, heat, and in some cases water. Unlike batteries, which store energy, fuel cells continuously generate power as long as fuel is supplied.

Exhibit 25: Fuel cell diagram



Source: 3M

Fuel cells can offer several benefits, including:

- **High energy efficiency:** Direct electrochemical conversion enables higher efficiency than conventional combustion-based power generation, reducing fuel consumption and operating costs.
- **Low carbon emissions**: Fuel cells produce minimal direct emissions and can support long-term decarbonization goals when powered by low-carbon fuels such as hydrogen.
- **Grid independence and energy security:** Provides reliable onsite power generation, reducing dependence on grid availability and improving energy resilience.
- **Long-duration energy storage enabler:** Hydrogen-based fuel-cell systems can complement renewable energy by storing energy over extended periods, supporting long-duration and seasonal energy storage.
- **Fuel flexibility:** Certain fuel-cell technologies can operate on hydrogen, natural gas, biogas, ammonia-derived hydrogen, and other hydrogen-rich fuels, allowing deployment using existing fuel infrastructure.
- **Low maintenance and quiet operation:** Limited moving parts reduce maintenance requirements while providing quieter operation than conventional combustion-based generators.
- **Scalable deployment:** Can be deployed across a wide range of power levels, from distributed backup systems to utility-scale installations, making fuel cells suitable for large AI data center campuses

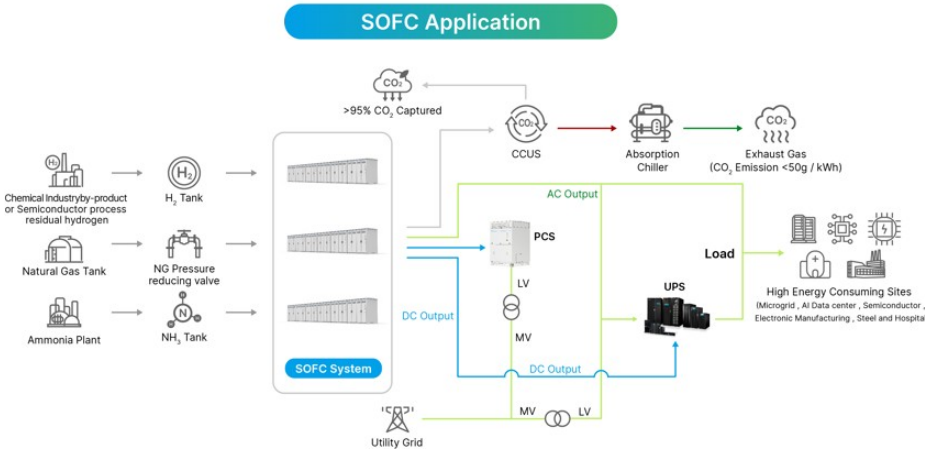
However, the following challenging are hurdles for wider deployment:

- **High system cost:** Fuel cells remain more expensive than conventional power-generation technologies, limiting large-scale adoption.
- **Limited hydrogen infrastructure:** Hydrogen production, transportation, storage, and refueling networks remain underdeveloped in most regions.
- **Durability requirements:** Long-term reliability and stack lifetime remain key challenges, particularly for large-scale and mission-critical deployments.
- **Regulatory and safety considerations:** Hydrogen handling requires specialized infrastructure and compliance with stringent safety regulations, increasing deployment complexity.
- **Lack of standardization:** Multiple fuel-cell technologies compete across different applications, creating uncertainty around long-term technology standards and adoption paths.

Fuel cells have the potential to fundamentally reshape future data center power architectures. Traditionally, data centers rely on grid electricity as the primary power source and diesel generators for backup power. However, as AI-driven workloads significantly increase power demand and power availability becomes a key deployment constraint, hyperscalers are increasingly evaluating fuel cells not only as a backup-power solution but also as a potential primary power source. This could accelerate the transition from a grid-centric architecture toward one incorporating distributed onsite generation, enabling data center deployment in regions where utility power capacity is limited.

Over the longer term, fuel cells could become a core component of AI data center microgrids that integrate renewable generation, battery storage, hydrogen storage, and intelligent energy-management systems. Such an architecture would improve scalability, resilience, and energy independence while reducing reliance on centralized utility infrastructure. In addition, because fuel cells inherently generate DC power, they align well with the industry's migration toward HVDC-based power distribution architectures. This could simplify power delivery by reducing conversion stages between generation and IT equipment, improving overall system efficiency and supporting next-generation high-density AI deployment

Exhibit 26: Fuel cell in datacenter power delivery architecture



Source: Delta

Stock picks

Delta (2308.TW, OW, PT NT$2,500)

Dominant supplier in the AI power supply offerings: Delta has been one of the most competitive players in power related components and systems, as its offerings spanning from the core power supply units, the complementary BBU/CBU for AI servers, the highly anticipated HVDC power rack, grey space power facilities to air/cooling solutions. We estimate that it has ~60% market share in the datacenter power supply market and is among the leading suppliers of the 800VDC/±400VDC power rack to Nvidia/hyperscalers in 2H26 (wider adoption likely from late 2027). Longer term, when the HVDC becomes more mature, the centralized power delivery architecture could migrate into SSTs, where Delta already has small volume shipment now. With the solid technology, close customer relationships and strong execution, we believe Delta should stay competitive even amid the fusion of the white space and grey space.

Extending from power delivery to power source: In addition to the offerings in the power delivery, Delta signed a licensing agreement with Ceres Power (CRW.GB) for its solid oxide fuel cell (SOFC) technology in early 2024. This could extend its business scope into the power source area. which we believe could add more value to its customers as it could help mitigate the power constraints for datacenter build and enhance the operational efficiency through active micro grid management.

Solid near term momentum: Despite the slower revenue growth owing to the component supply tightness in 3Q26, we expects its 4Q26 revenue to grow by double digit % QoQ on alleviated component supply and solid demand. 3Q26 GM should be flat to slightly up QoQ and 4Q26 might see further expansion on the tailwinds price hikes.

OW with PT of NT$2,500: We see power/thermal as one of the major secular AI related themes in the downstream tech space and Delta is well positioned to benefit, potentially driving a 63% earnings CAGR in 2025-28, in our estimate. We thus assume the coverage of Delta with an OW rating and the PT of NT$2,500, based on 30x 2027/2028 P/E. Our target multiple implies a PEG of 0.5 vs. the average PEG of 1.2 in 2020-25, when the stock was trading at an average 12M forward P/E of 22x with the earnings CAGR of 19%.

Lite-On (2301.TW, EW, PT NT$315)

Rising tide lifting all boats: Lite-On is the second largest power supply vendor globally and has been engaging with hyperscalers regarding the ASIC and Nvidia GPU servers. Looking ahead, it expects to expand its share in a couple of hyperscalers from late 2026. It should have over 50% of the total revenue generated from the cloud segment for PSUs, BBUs, and to a lesser extent, other mechanical components like liquid cooling, chassis/rack, etc. Moreover, it also has the HVDC power rack solutions ready and is qualifying with potential customers. The cloud shall remain as the major growth driver in the coming years.

Aspiration in the optical communication space: Lite-On has always had an opto business (10-20% of total revenue in 2026) in both visible and invisible light applications. Post the strategic investment at Dense Light (21% stake), an InP epi wafer and chip process manufacturer, it plans to leverage its expertise in packaging and provides the optical

modules and engine for the data interconnects. While the end applications looks promising, we think it might take some time to see more meaningful revenue contribution in Lite-On.

Cloud momentum extending into the coming quarters: We expect its 4Q26 revenue to grow by double digit QoQ driven mainly by the could segment and GM should normalize at ~25% in 2H26 vs. 27.2% in 2Q26 on some inventory write-off reversal and high margin product shipment delayed from the previous quarter.

EW with PT of NT$315: We think Lite-On will grow along with the AI capex cycle, driven by its power-related offerings (47% earnings CAGR in 2025-28, in our estimate), though execution and progress with new customers is worth monitoring. We assume coverage of Lite-On with an EW rating and a PT of NT$315, based on 17x 2027/2028 P/E. Our target multiple implies a PEG of 0.4 vs. the average PEG of 1.6 in 2020-25, when the stock was trading at an average 12M forward P/E of 14x on an earnings CAGR of 9%.

Panasonic Holdings (EW, PT ¥4,700, covered by Kazuo Yoshikawa)

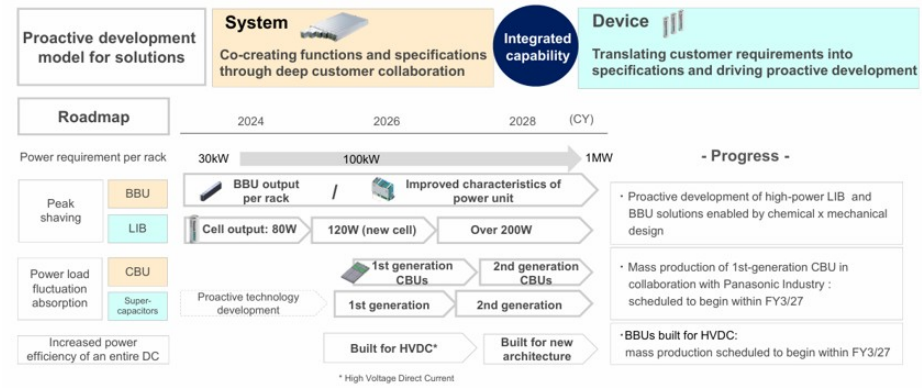
Panasonic has an approximately 80% share of the BBU market and is well positioned to benefit from rising AI data center power demand: The company expects data center energy storage revenue to grow from ¥322bn in FY3/26 to approximately ¥550bn in FY3/27 and around ¥1trn in FY3/29. Importantly, all of the business underpinning its FY3/29 revenue target has already been awarded, meaning agreements have been reached with customers on product development and orders.

BBU: Panasonic has begun limited mass production of new high-capacity BBU cells above 200W, with products adopted by a major GPU vendor and ramping from June-July. Discussions for next-generation server platforms are in the final stages, and the company expects to maintain its approximately 80% market share. It also sees scope for higher pricing as rack-level power consumption rises.

The company is also close to securing awards for BBUs used in dedicated power racks. As its medium-term plan includes only already-awarded business, additional design wins could provide upside.

CBU: CBUs are emerging as a new growth opportunity as GPU and ASIC power consumption increases. Panasonic is working with power-supply vendors to propose integrated BBU+CBU solutions and is receiving inquiries from US and Chinese hyperscalers, including customers that were not previously BBU users.

Exhibit 27: Panasonic Holdings: Strengthen proposal and development capabilities for the next generation



Source: Company data

Risks to our positive view

Risks to our positive view on power-related demand include:

- Component supply tightness leading to lower shipment of products
- Shortfall in hardware shipments, yield issues, electricity availability, etc. slowing down datacenter builds
- Lower-than-expected penetration of new power delivery architectures like 800VDC/±400VDC, SST, etc.
- More intense competition leading to margin erosion among most players
- Hyperscalers prove unable to finance aggressive capex through internal and external funding.

AI Cluster Performance Evolving to Fulfill Demand

AI adoption driving exponential growth in compute demand

AI is becoming a mainstream computing workload

Over the past three years, generative AI has evolved from a research application into a core technology platform. Consumer applications such as ChatGPT, Claude, Gemini, and Copilot now serve hundreds of millions of users globally, while enterprises are increasingly embedding AI into software development, customer service, document analysis, enterprise search, healthcare, financial research, and industrial automation.

Historically, software demand scaled roughly with the number of users. AI demand is fundamentally different because every user interaction consumes computational resources in real time. Unlike traditional cloud applications, where the bulk of computation occurs during software development, AI applications continuously consume compute whenever a model generates a response. Consequently, rising AI adoption directly translates into higher demand for data center compute resources.

Model complexity continues to increase

The AI industry has historically improved model performance through scaling laws: larger models trained on larger datasets using more compute generally produce better outcomes. While efficiency improvements continue, leading AI developers remain committed to scaling model capabilities.

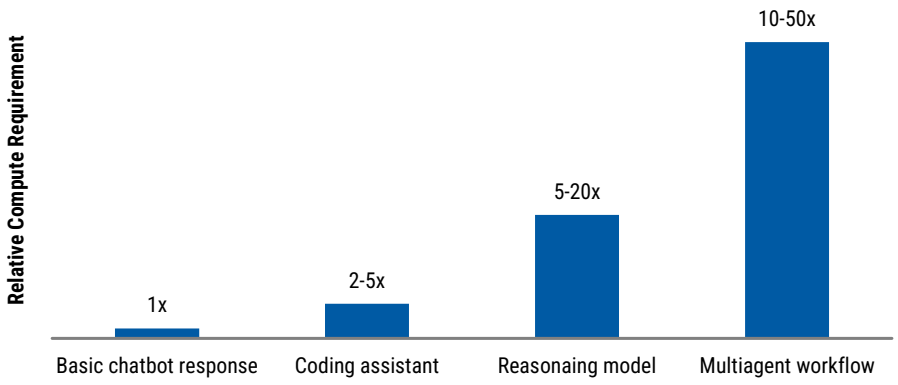
More importantly, the industry is moving beyond conventional language models toward significantly more compute-intensive architectures.

Reasoning models

Traditional LLMs generate answers directly from a prompt. New reasoning models perform multiple internal computational steps before producing a response. Instead of generating a few hundred output tokens, advanced reasoning systems may generate thousands or even tens of thousands of internal reasoning tokens while evaluating different solution paths. This dramatically increases inference workloads.

The industry is increasingly shifting from "parameter scaling" toward "test-time compute scaling," where additional compute is consumed during inference to improve answer quality. From an infrastructure perspective, this is highly significant because inference workloads occur continuously rather than once during model training.

Exhibit 28: Increasing demand for compute power from reasoning models



Source: Morgan Stanley Research estimates

Agentic AI

The next wave of AI applications involves autonomous agents capable of planning, executing, verifying, and refining tasks.

Instead of generating a single answer, an AI agent may search multiple information sources, run code, validate results, generate follow-up tasks, and produce a final response. Each step requires additional inference cycles.

A workflow that previously required one model invocation may now require dozens of model calls, substantially increasing compute consumption per user.

Multimodal AI

AI models are increasingly processing images, video, audio, sensor data, and text simultaneously.

Video understanding is particularly compute intensive because each second of video contains vastly more information than a text prompt.

As multimodal AI adoption grows, compute demand per query rises significantly, creating another structural driver of accelerator demand.

Training demand remains robust

Although inference is becoming the largest source of AI compute demand, training workloads continue to expand. Frontier model development increasingly requires clusters containing tens of thousands of accelerators operating simultaneously.

Training runs are becoming larger because developers are pursuing larger parameter counts, more training data, longer training durations, multimodal datasets and reinforcement learning and reasoning optimization

Consequently, each new generation of foundation models typically consumes substantially more compute than the previous generation.

Importantly, training infrastructure sets the baseline for future inference deployment. Every major model developer must maintain access to large-scale training clusters to remain competitive, supporting continued demand for high-end AI accelerators.

More compute requests to be fulfilled by better chip performance

Rising compute demand cannot be addressed simply by adding more servers. Modern AI workloads increasingly encounter bottlenecks related to memory bandwidth, interconnect performance, power consumption, and cooling. As a result, each new generation of AI hardware delivers improvements across multiple dimensions.

Compute performance

The latest AI accelerators provide several-fold performance improvements versus prior generations. For example, NVIDIA's Blackwell architecture was specifically designed to support both large-scale training and high-volume inference, with significant gains in AI throughput compared with previous Hopper-based systems. Performance improvements are necessary because reasoning and agentic workloads require substantially more operations per query.

Memory capacity and bandwidth

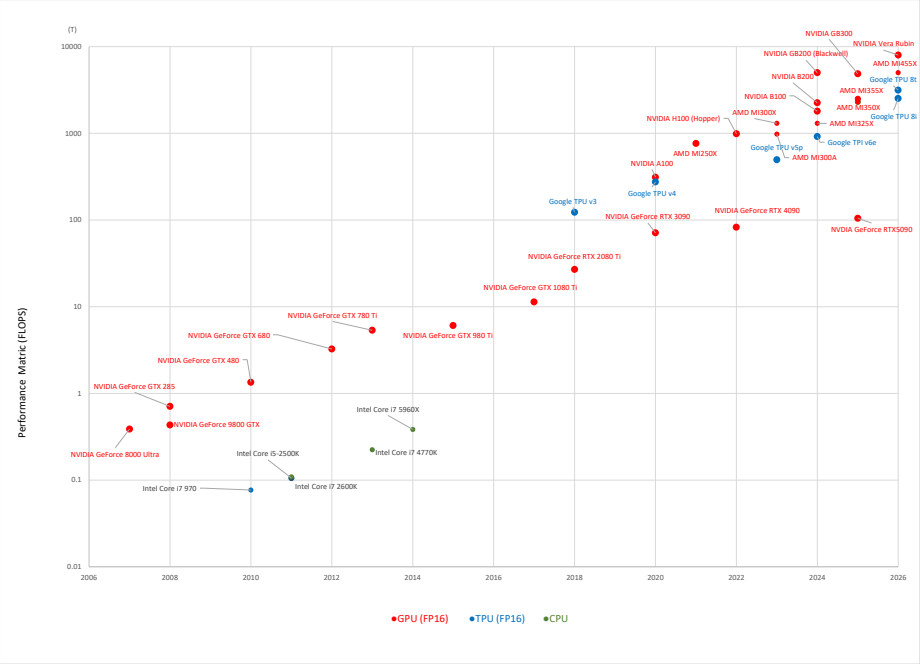
AI models are increasingly constrained by memory rather than raw compute, because longer models, longer context windows, reasoning models and agentic workflows all require more memory. As model complexity increases, modern high-end accelerators usually incorporate hundreds of gigabytes of High Bandwidth Memory (HBM), providing memory bandwidth measured in multiple terabytes per second.

Interconnect performance

As AI clusters scale from hundreds to thousands of accelerators, communication between processors becomes increasingly important. A large AI training cluster effectively behaves like a single massive computer. Consequently, networking technologies such as NVLink, InfiniBand, Ethernet, etc., have become critical enablers of AI scaling.

Many AI workloads can no longer be accelerated simply by increasing GPU performance; cluster interconnect bandwidth must also improve.

Exhibit 29: Compute power for HPCs continuing to evolve



Source: Company data, Morgan Stanley Research

AI Cluster Performance Is Fundamentally a Power Problem

As AI clusters become larger and more powerful, every component within the cluster must deliver higher throughput, lower latency, and greater bandwidth. Achieving these improvements inevitably requires more electrical power.

Therefore, the power challenge is no longer limited to GPUs. Modern AI clusters are highly integrated systems comprising compute servers, networking fabrics, memory subsystems, storage infrastructure, and cooling equipment. As cluster performance increases, power consumption rises across the entire infrastructure stack.

In our view, the next phase of AI infrastructure investment should be viewed not simply as a semiconductor upgrade cycle, but as a broad-based power consumption cycle driven by the need for ever-higher AI cluster performance.

Server systems seeing increasing TDP

Nvidia's AI-server power roadmap is best understood as a transition from chip-defined, air-cooled nodes to power-, cooling- and fabric-constrained rack-scale computers: the Ampere-era A100 SXM carried a disclosed 400W standard TDP—up to 500W in custom thermal designs—with 40/80GB HBM2/HBM2e and 600GB/s NVLink, while the eight-GPU, dual-AMD-EPYC DGX A100 was a conventional 6U server with a 6.5KW maximum input rating, meaning GPU silicon represented only roughly half of the node envelope once CPUs, NVSwitches, memory, NICs, drives, fans and conversion losses were included.

Hopper stepped the H100 SXM to a disclosed 700W TDP, 80GB HBM3, roughly 3TB/s memory bandwidth and 900GB/s fourth-generation NVLink, while H200 retained essentially the same Hopper compute architecture and power profile but substituted 141GB of faster HBM3e at 4.8TB/s—making H200 primarily a memory-capacity/bandwidth upgrade rather than a compute-TDP step-up—and Nvidia rates the eight-GPU, dual-Xeon DGX H100/H200 at 10.2KW maximum input, versus 6.5KW for DGX A100;

The architectural discontinuity arrives with GB200. Rather than treating an eight-GPU server as the scale-up unit and using InfiniBand/Ethernet primarily between nodes, Grace Blackwell NVL72 makes the entire rack the scale-up domain, integrating 18 one-U compute trays—each with two 72-core Grace Arm CPUs and four Blackwell GPUs—plus nine NVLink switch trays, management switches, , a 50–51V DC busbar, copper NVLink cartridges and liquid manifolds, so that 36 CPUs and 72 GPUs operate as one low-latency NVLink-connected system. Based on the same compute die, Nvidia upgraded the HBM content to further improve the performance of GB300 chip. Overall, the rack level power consumption is estimated at ~120KW for the GB200 NVL72 and ~140KW for the GB300 NVL72.

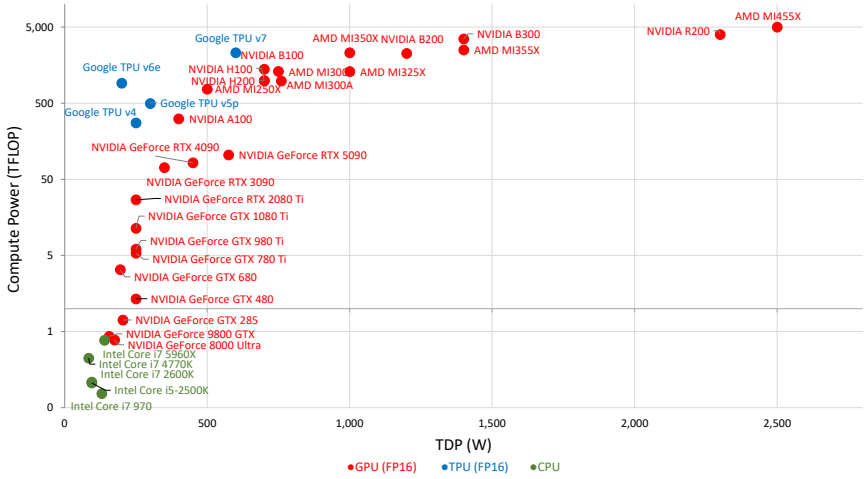
Vera Rubin—often called VR200/R200 in industry roadmaps—pushes the same rack-level philosophy further. Nvidia's disclosed architecture combines 72 dual-die Rubin GPUs with 36 new Vera CPUs, replacing Grace's 72 Arm Neoverse cores with 88 custom Olympus cores, raising CPU memory capacity and bandwidth to as much as 1.5TB LPDDR5X and 1.2TB/s, and doubling coherent CPU–GPU NVLink-C2C bandwidth to 1.8TB/s; Rubin itself moves from HBM3e to 288GB HBM4 at up to 22TB/s, introduces a third-

generation Transformer Engine and raises sixth-generation NVLink scale-up bandwidth to 3.6TB/s per GPU, while ConnectX-9 provides 1.6Tbps/GPU scale-out connectivity and BlueField-4 increasingly offloads storage, security and infrastructure processing.

Nvidia has not published a simple Rubin chip TDP in its primary datasheet, but our supply chain checks indicate roughly 1.8kW at an efficiency-oriented operating point and up to approximately 2.3kW at the performance mode, which is consistently corroborated by system vendors, with ASUS specifying 187kW Max-Q and 227kW Max-P for its fully liquid-cooled Vera Rubin NVL72.

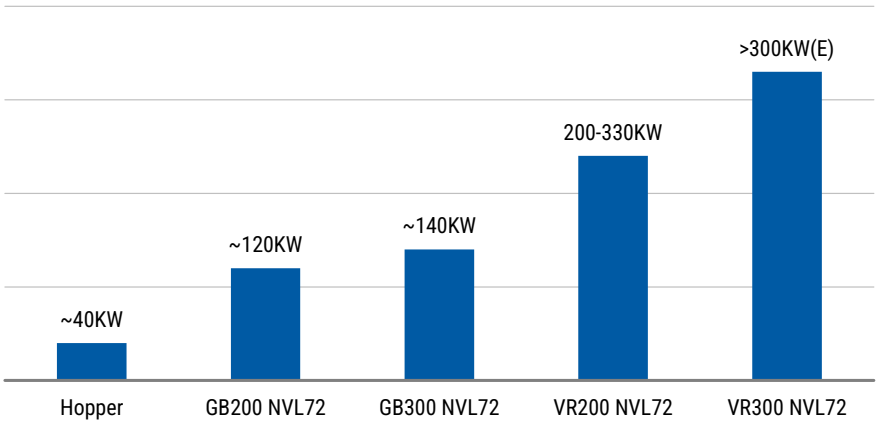
Looking ahead, we expect this trend to extend into the future along with the technology advancement and more critical workloads.

Exhibit 30: Compute power vs. TDP for HPCs



Source: Company data, Morgan Stanley Research

Exhibit 31: TDP roadmap for Nvidia racks



Source: Company data, Morgan Stanley Research

Networking consuming more power as well

The network switch system is another key devices within the datacenter and has been evolving over the past years, driven primarily by advances in the switch ASIC. From Tomahawk 1 (2014) to Tomahawk 6 (2025), the process node shrank from 28nm to 3nm, while aggregate switching bandwidth increased from 3.2 Tbps to 102.4 Tbps, representing a 32× increase. Over the same period, SerDes speed improved from 25 Gbps NRZ to 200 Gbps PAM4, enabling maximum port speeds to scale from 100G to 1.6T.

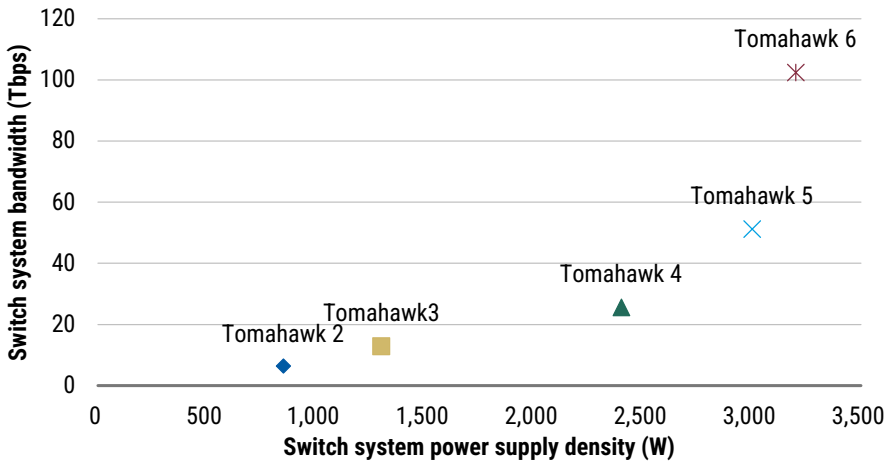
While process scaling significantly improved switching efficiency, it did not prevent ASIC power from rising. ASIC TDP increased from 150 W to 800 W, reflecting the growing complexity of higher radix switches and faster SerDes interfaces.

The impact of ASIC scaling is evident at the system level. As switch bandwidth increased from 100G-class systems to 800G and 1.6T platforms, power delivery requirements grew dramatically. Representative PSU ratings increased from 850W in Tomahawk 2 systems to 3,200W in Tomahawk 6 systems, nearly a 4× increase.

The rise in PSU power is driven not only by the higher ASIC TDP, but also by the growing power consumption of high-speed optics, increased cooling requirements, and larger overall system power budgets. As systems evolved from 100G to 1.6T ports, switch vendors were forced to add more fan redundancy, larger heatsinks, and in some cases larger chassis footprints. For example, Tomahawk 6 systems have expanded to 4U form factors with 7+1 fan redundancy to accommodate the thermal load associated with 102.4 Tbps switching capacity.

Alongside the increasing power consumption of the server racks, network switches are also witnessing higher power budgets as well.

Exhibit 32: Power supply density for different network switch generations



Source: Company data, Morgan Stanley Research

Power supply growing on both volume and specs

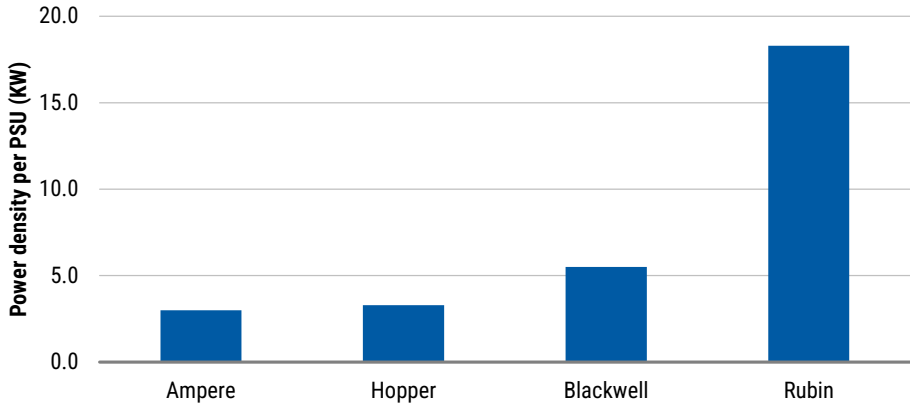
PSU spec upgrade trend

In response to the increasing power consumption of the AI servers, the power supply architecture has been migrating as well. in the A100 and H100/200 single system servers, 6 power supply units are placed directly inside the chassis, with each capable to deliver 3.0-3.3KW of power.

When Nvidia make a broad based move to the rack level server solution from the Blackwell generation, the power supply units were no longer placed directly within the chassis housing the HPCs that consume power; instead, power supply units are separated and grouped into power shelves. In the GB200/300 systems, there are 8 power shelves sitting on the same rack as the compute trays and switch trays, each enclosing six 5.5KW power supply units.

For the upcoming Vera Rubin rack, the power supply unit will be upgraded to 18.3KW of output to comprise a power shelf of 110KW, and each power shelf will take 3U of space on the rack vs. 1U in the Blackwell generation.

Exhibit 33: Power density for PSUs in Nvidia server systems



Source: Company data, Morgan Stanley Research

Challenges increasing the power density of PSUs

To come up with power supply unit with higher power density, manufacturers might need to solve the following potential hurdles:

Better power semiconductors (SiC and GaN)

The biggest improvement in PSU power density comes from replacing traditional silicon power chips with newer materials such as silicon carbide (SiC) and gallium nitride (GaN). These devices waste less energy during power conversion and can operate at much higher speeds. As a result, many supporting components inside the PSU—such as transformers and inductors—can be made significantly smaller. This allows PSU manufacturers to fit more output power into the same physical size while also improving efficiency. In many ways, SiC and GaN have become the foundation for the latest generation of high-density AI server power supplies.

More efficient power conversion designs

Modern PSUs use more sophisticated circuit designs, i.e., bridgeless PFC vs. conventional one, that reduce energy losses during power conversion. A conventional boost PFC includes a diode bridge and therefore incurs multiple semiconductor conduction drops. A bridgeless totem-pole PFC reduces the number of devices in the active current path and is now a preferred architecture for dense server PSUs. Instead of simply converting electricity from one voltage to another, newer architectures are designed to minimize wasted energy throughout the process. Higher efficiency means less heat is generated, which directly enables higher power density. This is particularly important in AI servers, where even small efficiency improvements can translate into meaningful reductions in heat generation and cooling requirements.

Smaller and smarter magnetic components

Transformers and inductors are often among the largest components inside a PSU. A major area of innovation is reducing their size without sacrificing performance. Manufacturers are increasingly using planar magnetics, where copper traces on circuit boards replace traditional wire windings. These designs are thinner, easier to cool, and more space-efficient. Some suppliers are also combining multiple magnetic functions into a single component, further reducing overall volume. As PSU power levels increase from 3 KW to 5.5 KW and beyond, shrinking magnetic components becomes one of the most important contributors to power-density improvement.

Improved thermal management

As power output increases, heat becomes the primary obstacle to higher density. Even highly efficient PSUs still generate a significant amount of heat. Therefore, manufacturers are investing heavily in better cooling solutions, including optimized airflow paths, advanced heatsinks, vapor chambers, and liquid-cooling technologies. The goal is to remove heat more effectively so that components can operate safely even when packed more tightly together. In many modern PSU designs, thermal engineering is just as important as electrical engineering in determining achievable power density.

Higher component integration

Another way to increase power density is to reduce the amount of unused space inside the PSU. Manufacturers are integrating more functions into fewer components and using three-dimensional layouts that make better use of available volume. Components are stacked, arranged vertically, or combined into integrated modules to maximize space utilization. This approach is similar to how smartphones have become more powerful without becoming larger—the same principle is increasingly being applied to server power supplies.

Higher efficiency creates more thermal headroom

Efficiency improvements provide a double benefit. First, less electricity is wasted. Second, less heat is generated. For example, a 5.5 kW PSU operating at 97.5% efficiency generates roughly 30% less heat than one operating at 96.5% efficiency. While a one-percentage-point improvement may appear small, it can remove tens of watts of heat from the system. This allows engineers to either increase output power within the same form factor or reduce cooling requirements while maintaining existing power levels.

Reducing the size of energy storage components

Server PSUs contain capacitors that store energy and maintain power delivery during short interruptions. Historically, these capacitors occupied a significant amount of space. Newer designs use smarter control techniques that allow manufacturers to achieve the same performance with smaller capacitor banks. Reducing capacitor volume frees up valuable internal space that can be used for higher power output or additional functionality.

For example, in Infineon's 3.3 KW example, the calculated capacitance for a conventional 10ms hold-up implementation was approximately 2.5 mF; allowing deeper discharge through a baby-boost stage reduced the requirement to about 900 µF. The 12 KW design similarly uses a bidirectional buck-boost buffer to achieve 20ms hold-up at full load without the capacitor volume that a passive solution would require. The penalty is additional semiconductors, an inductor, control software and failure modes, but the net volumetric benefit can be substantial.

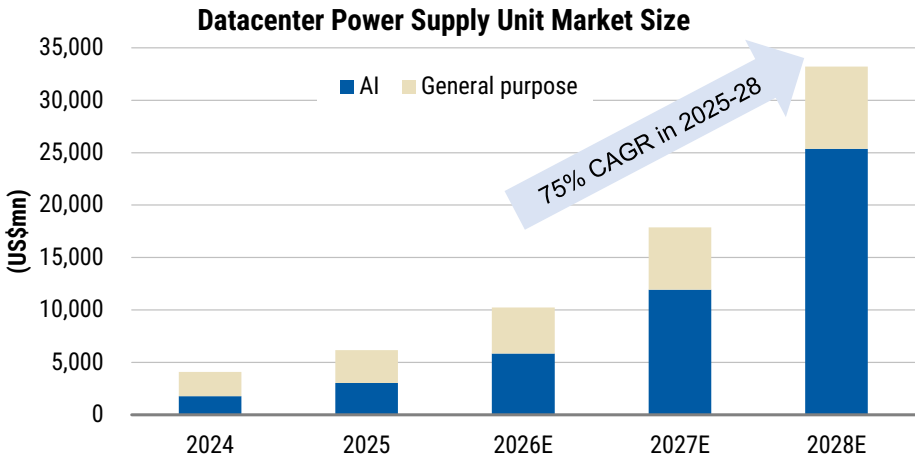
Advanced mechanical design

Power density is ultimately measured in watts per unit volume, making physical packaging extremely important. Manufacturers continuously optimize component placement, airflow channels, connector design, and enclosure structures to eliminate unused space. The best PSU designs are often not defined by a single breakthrough component but rather by hundreds of small mechanical optimizations that collectively improve volumetric efficiency.

Datacenter PSU market size to grow at a 75% CAGR in 2025-28

Overall, we estimate the global datacenter power supply market will grow at a three-year CAGR of 75% to reach US$33bn in 2028 from US$6bn in 2025, driven mainly by server shipment growth, increasing power consumption per rack, and the adoption of HVDC power racks from 2027.

Exhibit 34: Global datacenter power supply market forecast



Source: Company data, Morgan Stanley Research estimates

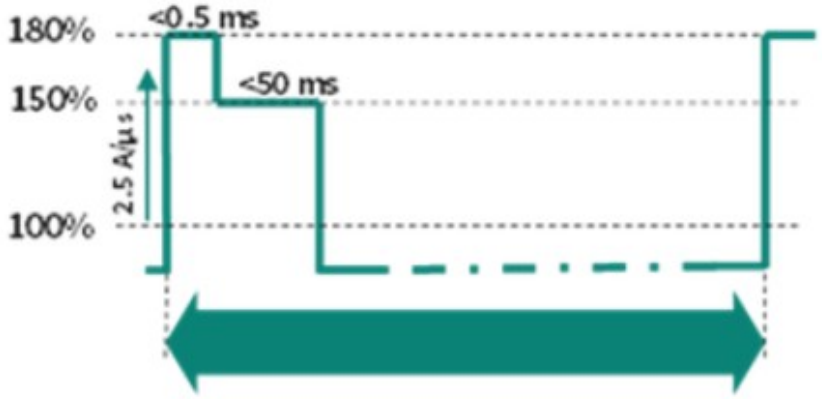
Complementary Hardware for More Resilient Power Delivery

Transient power spikes at AI servers

AI servers have a much more dynamic power profile than conventional enterprise servers because GPU utilization changes sharply throughout a workload. During large-model training, thousands of GPUs typically operate in a bulk-synchronous manner, and power consumption approaches the accelerator's design limit during tensor-compute phases, then falls during collective communications, check pointing, data movement, or periods in which GPUs wait for slower devices to synchronize. Inference can create a similar pattern as request arrivals, batching, context length and token-generation activity change rapidly.

Because the GPUs perform these transitions together, the power variation of one accelerator is multiplied across the server, rack and cluster rather than being averaged out by unrelated workloads, so the magnitude of the challenge is also rising with system density. Power spikes are therefore not simply the result of higher average consumption; they reflect the combination of higher accelerator power, more GPUs per system and increasingly synchronized workload behavior.

Exhibit 35: Transient power spikes at AI servers



Source: Infineon

System performance and reliability potentially compromised

The key issue is that power delivery systems are designed to supply a relatively stable amount of power, while AI workloads can change power demand almost instantaneously. When the load changes faster than the power-delivery system can react, a temporary mismatch occurs between power demanded by the GPUs and power supplied by the PSU, creating electrical instability throughout the server.

Sudden increases in GPU demand lead to voltage drops

When GPUs transition from a low-utilization state (communication, synchronization, checkpointing, etc.) to a full-compute state, their current demand can increase within microseconds. However, the PSU cannot instantly increase its output because its control loop requires time to detect the load change and adjust power conversion.

During this response period, the additional energy must come from capacitors located throughout the power chain. As these capacitors discharge, the voltage on the DC bus begins to fall, creating voltage drop.

If the drop becomes sufficiently large, several operational issues can occur:

- GPU throttling: The GPU reduces clock speed or performance to remain within safe operating limits.
- Computation errors: Voltage instability can affect timing margins in high-speed processors and memory.
- Unexpected resets or shutdowns: Protection circuits may trigger if voltage falls below specified thresholds.
- Training interruption: Large AI training jobs running across thousands of GPUs may fail if even a small number of nodes experience instability.

In a large AI cluster, a single server failure can force the entire training job to restart from the latest checkpoint, resulting in significant productivity loss.

Sudden decreases in GPU demand results in voltage spike

The opposite problem occurs when GPUs rapidly reduce power consumption.

For example, after a compute-intensive phase, GPUs may enter a communication phase where power consumption drops sharply. The PSU, however, is still delivering power based on the previous load condition. The excess energy has nowhere to go immediately and accumulates in the DC bus, causing a temporary voltage spike.

This can create:

- Overvoltage stress on components
- Protection-circuit activation
- Accelerated degradation of capacitors and voltage regulators
- Higher risk of component failure over time

Although individual spikes may be small, AI servers can experience these transitions continuously during training, causing repeated stress throughout the power-delivery chain.

Excessive current surges throughout the system

Power spikes are not only a voltage problem—they also create large current excursions.

A 1KW increase in power demand at 12V requires roughly 83A of additional current. In modern AI servers, where multiple GPUs can change utilization simultaneously, current swings can reach hundreds of amperes within a very short period.

These current surges lead to:

- Higher conductor losses
- Increased connector heating
- Greater stress on voltage-regulator modules (VRMs)
- Larger thermal fluctuations in power components
- As GPU power ratings rise to over 1KW, the magnitude of these transient currents becomes increasingly difficult to manage.

Thermal cycling and reliability degradation

Perhaps the most overlooked issue is the impact on long-term reliability.

Every power excursion creates a corresponding temperature excursion. When power rises, semiconductor junction temperatures increase; when power falls, temperatures decrease. This repeated heating and cooling cycle occurs thousands or even millions of times over a system's lifetime.

The result is accelerated wear on:

- Electrolytic capacitors
- Power MOSFETs
- Solder joints
- PCB interconnects
- Connectors and busbars

Consequently, even if average power remains within specification, transient power spikes can shorten component lifetime and increase failure rates.

Infrastructure overdesign and lower efficiency

To prevent instability, operators often provision the power infrastructure for the worst-case peak rather than average consumption.

For example:

- A rack averaging 80KW may need infrastructure sized for 120-140KW transient peaks.
- PSUs must be oversized to accommodate short-duration excursions.
- UPS systems must reserve capacity for sudden load increases.
- Cooling systems must handle peak thermal output.

This creates a significant amount of stranded capacity, where expensive electrical and cooling infrastructure remains underutilized most of the time. Furthermore, PSUs and UPS systems typically operate most efficiently near their optimal loading point; oversized systems therefore tend to operate at lower efficiency, increasing operating costs.

Conventional power delivery systems not designed to deal with power volatility

In the conventional power delivery architecture, the UPS protects the data hall against utility outages and power-quality disturbances, while the PSU converts upstream power into the tightly regulated DC output required by the server. Both remain essential in the modern AI server clusters, but neither was originally optimized to absorb large, repetitive and synchronized AI load steps economically.

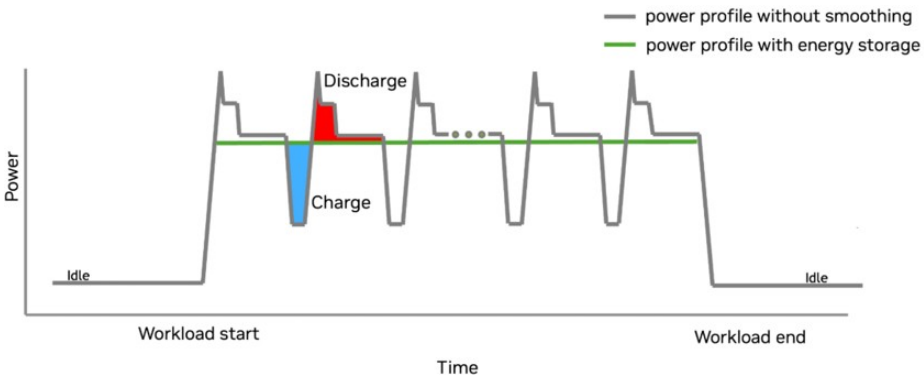
The PSU can use its internal capacitors to cover the initial transient, but capacitor volume is limited; once that stored energy is depleted, the PSU control loop must raise input current, potentially producing current overshoot and transferring the disturbance upstream.

The UPS is located farther from the accelerator and sees the aggregated load of many PSUs. A transient originating near the GPU can therefore occur on microsecond-to-millisecond timescales—including permitted GPU electrical-design-power excursions around 50 ms—before the complete upstream conversion chain can rebalance power.

This is a response time and location mismatch, rather than evidence that UPSs or PSUs are fundamentally incapable of supporting AI. New AI-tolerant UPS platforms are being designed for faster and larger load steps, but many legacy systems may still experience voltage excursions, nuisance alarms, lower partial-load efficiency or unwanted bypass transfers under steep synchronized loads.

To complement the conventional power delivery architecture, more AI server clusters are adopting BBUs or CBUs to shave peak loads and mitigate the above-mentioned issues.

Exhibit 36: Concept of peak shaving for AI server power loads



Source: Nvidia

Battery backup units seeing increasing importance

What's a BBU?

A BBU is an integrated battery, power-conversion and control system that supplies DC power to server loads when the normal power source is interrupted or temporarily insufficient. In an OCP-style rack, the BBU modules are housed in a dedicated shelf located near the PSU power shelf and connect directly to the common DC busbar. Under normal conditions, the PSUs power the servers and maintain the BBU's charge; when the bus

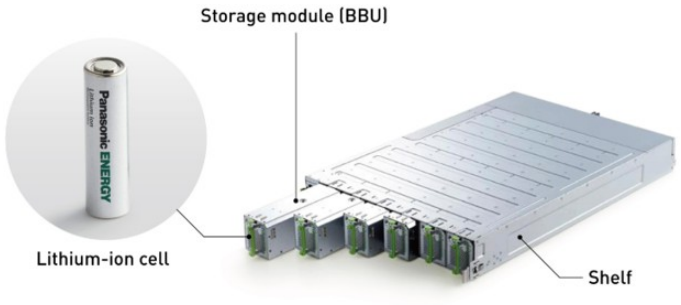
voltage falls below a programmed threshold, the BBU's bidirectional converter changes from charging or standby into discharge mode and supports the rack. Analog Devices' ORV3 reference design detects a bus voltage below 48.5V for more than 2ms and is required to ramp to full support within another 2ms, while maintaining the bus above the specified minimum voltage.

Key components within a BBU

- **Battery-cell pack:** The cells store the energy used during discharge. Lithium-ion cylindrical cells are commonly arranged in series and parallel to obtain the required voltage, power and runtime; the ORV3 reference module uses an interchangeable 11-series, 6-parallel battery pack. The cell must provide not only sufficient energy but also high discharge power, low internal resistance and predictable performance over its qualified life.
- **Bidirectional DC/DC converter:** The converter controls energy flow between the battery pack and the rack DC bus. It steps voltage in the appropriate direction during charging and discharging and regulates the output so that the battery's changing voltage does not destabilize the server bus. Multiphase designs divide the large current across several parallel power stages; TI's 12 KW design uses six 2 KW phases and reports peak efficiency above 99%, including above 99.5% when battery and bus voltages are close.
- **Battery management system:** The BMS measures individual cell voltages, current and temperature, balances cells and estimates state of charge and state of health. It also protects against overcharge, excessive discharge, overheating and short circuit. Accurate state-of-health information is important because the operator must know whether an apparently charged BBU can still deliver its required rack power and runtime after years of standby operation and periodic discharge events.
- **Power-path and protection circuitry:** Fuses, current sensors, disconnect switches, protection MOSFETs and hot-swap circuits isolate faults and prevent uncontrolled current flow. Typical protections include overvoltage, undervoltage, overcurrent and overtemperature shutdown. Hot-swap capability allows a failed or end-of-life module to be replaced without intentionally shutting down the entire rack, while active current sharing ensures that parallel BBU modules carry an appropriate portion of the load.
- **Control, monitoring and communications:** A module controller manages operating modes, converter control, fault handling, diagnostics and communication with the BMS and rack controller. Interfaces such as PMBus, Modbus, CAN, RS-485 or Ethernet can expose battery voltage, current, temperature, state of charge, state of health, fan speed and fault status to data-center management systems. Analog Devices' reference platform uses separate controllers for battery monitoring and overall BBU logic, communications and housekeeping.
- **Thermal-management system:** Fans, temperature sensors, airflow paths and mechanical spacing around cells remove heat during high-rate charging and discharging. Thermal management is both a performance and a safety requirement: excessive temperature accelerates battery ageing and can restrict available discharge power. Analog Devices' ORV3 design therefore incorporates fan control, airflow simulation and cell-to-cell spacing, while Murata's commercial 3 KW BBU includes internal forced-air cooling and overtemperature protection.

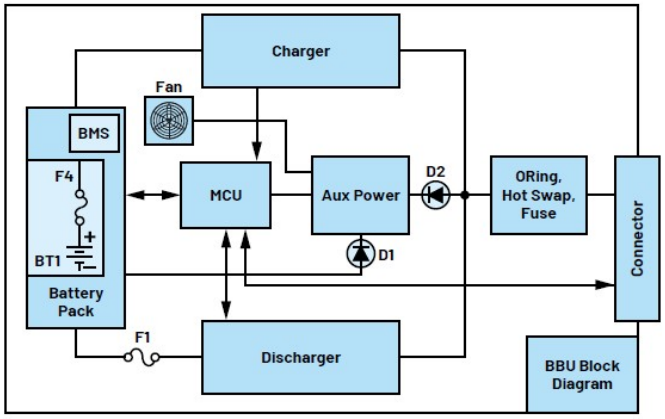
- **Shelf, backplane and mechanical integration:** The shelf distributes DC power and communications across several modules and provides retention, service access and cooling. Modularity allows power and runtime to be scaled by adding units and enables redundancy such as the ORV3 5+1 configuration. Murata specifies a 2U, hot-swappable 3 KW module, while Eaton's 19-inch BBU shelf accepts six 2.75 kW batteries and scales runtime according to the application requirement

Exhibit 37: BBU



Source: Panasonic

Exhibit 38: BBU diagram



Source: ADI

Key functions of BBUs

Backup power and outage ride-through

The BBU's fundamental function is to maintain rack operation during a brief loss of upstream power. The ride-through interval gives the data center time to restore the primary source, transfer to another source, preserve critical data, migrate workloads or perform an orderly shutdown. Rather than attempting long-duration operation, rack BBUs are generally optimized for high power over a relatively short period; the ORV3 example specifies 3KW per module for four minutes, and Murata guarantees a 240-second hold-up period at full-rated load under stated age and temperature conditions.

For AI systems, this protection is particularly valuable because a rack interruption can affect a synchronized workload extending well beyond the failed server. Local ride-through prevents a momentary upstream disturbance from immediately removing GPU, CPU, networking and storage power from the rack. Infineon consequently describes the server BBU as an integrated backup supply intended to prevent data loss and maintain AI workloads during outages or grid failures.

Peak shaving and load smoothing

A BBU can be operated as an active energy buffer. as it charges during lower-demand periods and discharges when server consumption temporarily exceeds a defined rack or PSU limit. This allows a relatively stable upstream power source to support a more volatile downstream AI workload. The economic benefit is that PSU, UPS and electrical capacity can be designed closer to sustained demand rather than every short theoretical peak, although the achievable benefit depends on the frequency, duration and predictability of those peaks.

The important limitation is battery cycling. A BBU designed primarily for emergency standby may not be optimized for continuous, repetitive peak shaving, so frequent cycling affects heat generation, state of charge, usable life and replacement schedules. Very short and highly repetitive transients may therefore be better supported by a CBU or hybrid battery-capacitor system, leaving the battery to cover longer events. Renesas correspondingly distinguishes distributed BBUs from hybrid systems that combine batteries with capacitors for fast transient response and longer-duration backup.

Power stabilization, redundancy and fault isolation

A rack BBU adds another independent source to the local power path. If a power shelf, upstream feed or AC source is interrupted, the BBU can sustain the rack while the fault is isolated or another source assumes the load. A modular shelf further improves resilience because one BBU failure need not remove the entire backup function; the remaining modules can continue operating within the available redundancy and power limits.

Distributed deployment also narrows the fault domain. A defective rack BBU affects a much smaller set of compute equipment than a failure within a centralized backup system serving a large room or data hall. The tradeoff is a much larger population of batteries, converters, controllers and fans that must be monitored, tested and eventually replaced. Digital state-of-health monitoring, fault reporting and hot-swappable construction are therefore central to the operating model rather than optional features.

BBUs vs. UPSs

While some of the functions that BBUs offer look similar to what traditional UPSs can do, we think BBUs and UPSs should be viewed as complementary rather than substitutive technologies, as they address different challenges within the datacenter power architecture.

A UPS is designed to provide facility- or row-level protection against utility outages, voltage sags, surges, and other power-quality disturbances, ensuring continuous AC power delivery to a broad range of IT equipment. In contrast, a BBU operates much closer to the AI workload, supplying DC power directly at the rack level and responding to localized power events that occur faster than the upstream power infrastructure can efficiently manage.

As AI racks become denser and more power intensive, the UPS continues to serve as the first line of defense against external power disruptions, while the BBU acts as a second layer of protection by stabilizing the rack DC bus, supporting transient load spikes, and providing short-duration ride-through capability. Together, they create a hierarchical power protection architecture in which the UPS ensures overall system availability and power quality, while the BBU enhances rack-level resilience and power-delivery performance. Rather than replacing the UPS, BBUs help reduce the burden on centralized backup systems, improve power utilization efficiency, and enable the reliable operation of next generation AI infrastructure with increasingly dynamic power profiles

Exhibit 39: Comparison of BBUs and UPSs

Dimension	BBU	UPS
Location	Inside the server rack, in a BBU shelf, or in an adjacent rack/sidecar close to the compute	Typically deployed at equipment room, row or facility level and protects multiple racks or other critical loads
Power path	Usually DC-coupled to the rack bus through a bidirectional DC/DC converter	Takes AC input and delivers AC output; the server PSU subsequently converts AC again for the server
Primary objective	Short rack-local ride-through, DC-bus support, modular redundancy and—when designed for it—transient or peak-load assistance	Broad power conditioning and uninterrupted AC supply during utility sags, surges, frequency disturbances and outages
Response	Monitors the local bus and can inject power within a few milliseconds; the cited ORV3 reference requires full support within 2ms after a qualifying 2ms event	An online double-conversion UPS already powers the load through its inverter and therefore has zero transfer time between utility and battery operation
Runtime	Generally optimized for short-duration, high-power support; OCP's reference example is four minutes at full load	Runtime varies widely with battery-bank sizing and system design; its broader function is to maintain protected AC loads until another source or operating response becomes available
Scalability	Added rack-by-rack or module-by-module; power and runtime can be matched to each compute deployment	Scaled centrally in larger KW blocks and requires capacity planning for the aggregate protected load
Fault domain	A module or shelf fault is normally localized to one rack or a limited group of racks	A centralized system protects a much broader load domain, so its redundancy and maintenance design are correspondingly critical

Source: Morgan Stanley Research

Capacitor backup unit (CBU) another complementary power device

What's a CBU?

A CBU is an integrated energy storage subsystem positioned close to the AI load, typically in a dedicated rack shelf, capacitor tray or adjacent power sidecar, and connected to the server rack's DC power path. Its principal storage medium may be an electric double-layer capacitor (EDLC or supercapacitor), lithium-ion capacitor (LIC), or a bank of high-capacitance electrolytic capacitors, depending on the required response time, energy duration, footprint and cost. Dedicated CBU products increasingly use EDLCs or LICs because their low internal resistance supports rapid, repetitive charge and discharge.

The CBU sits electrically alongside the PSU. Under steady-state conditions, the PSU supplies the AI rack and maintains the CBU at its target state of charge. When GPU demand temporarily exceeds the desired PSU output, the CBU discharges through a controlled power converter into the DC bus; when GPU demand falls, the converter reverses direction and recharges the capacitors at a controlled rate. This prevents the full GPU load step from passing through the PSU and upstream equipment. Infineon notes that a separate CBU must be compatible with, and potentially communicate with, the PSU shelves to coordinate rack power.

An alternative to deploying a standalone CBU shelf is to integrate the energy-storage capacitors directly into the power shelf, effectively combining power conversion and transient-energy buffering within the same subsystem. NVIDIA's GB300 NVL72 architecture is a good example: its enhanced power shelves incorporate a substantial capacitor bank alongside the PSU modules, with NVIDIA allocating approximately 65 joules of stored energy per GPU.

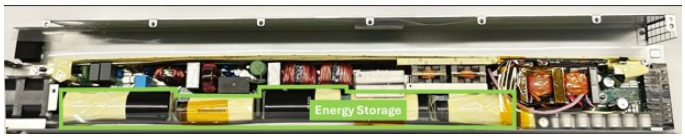
NVIDIA reported that this approach reduced peak upstream power demand by roughly 30% in a measured Megatron LLM workload. Compared with a separate CBU, integrating capacitors into the power shelf can save valuable rack space, shorten the electrical path between the energy buffer and the load, and simplify coordination between the PSU and

energy-storage system, since charging, discharging, and PSU output can be managed as one integrated power architecture. The trade-off is that capacitor capacity becomes more tightly coupled to the power-shelf design, potentially reducing flexibility in independently scaling or servicing the energy-storage system. Nevertheless, as AI racks become increasingly power dense, integrating transient-energy storage directly into the power shelf could become an attractive alternative to standalone CBUs, particularly where rack space and fast transient response are key design priorities.

Exhibit 40: CBU

Source: Panasonic

Exhibit 41: GB300 power shelf with integrated capacitors



Source: Nvidia

Key components within a CBU

- **Capacitor bank:** The storage section contains many capacitor cells arranged in series to reach the required voltage and in parallel to provide the required power and energy. Individual EDLC cells commonly operate at only a few volts, so high-voltage systems require long series strings; Abracon cites approximately 2.7 V as a typical supercapacitor cell rating, while TI's NVIDIA-oriented reference architecture demonstrates an 800 V CBU using EDLC cells. LICs can provide more stored energy than conventional EDLCs, while EDLCs generally emphasize lower resistance, high peak power and very high cycling endurance. Delta, for example, offers a power-capacitance shelf using fast-charge/discharge LICs, rated to provide 15 seconds of hold-up at 20 KW.
- **Cell-balancing system:** Because cell capacitance, leakage current and internal resistance vary across a series string, the voltage does not divide perfectly equally. A balancing circuit prevents an individual cell from exceeding its rated voltage during charging or being reverse-biased during deep discharge. Passive balancing uses resistors to bleed excess charge, while active approaches use MOSFETs, amplifiers or switched converters to redistribute energy more efficiently; the design tradeoff is between cost, balancing speed, component count and standby loss.
- **Bidirectional DC/DC converter:** This is the active interface between the capacitor bank and the rack DC bus. During a GPU power spike, it converts the capacitor's changing voltage into a regulated bus contribution; during lower demand, it reverses power flow to recharge the bank. The converter also allows more of the capacitors' stored energy to be used than would be possible with a direct connection, because capacitor voltage declines materially during discharge.

Infineon specifically identifies a capacitor bank plus bidirectional converter as the main CBU structure, while Skeleton Technologies notes that converter-based integration provides greater energy utilization and more precise power-flow control than a direct bus connection.

- **Controller and management software:** The controller determines when the CBU should charge, remain idle or discharge, based on bus voltage, PSU output, rack power limits and capacitor state. It may also calculate state of health, manage balancing, regulate recharge current and coordinate multiple CBU modules.
- **Protection and switching:** A high-power capacitor bank can release very large fault current, so fuses, contactors, pre-charge circuits and overvoltage, overcurrent and overtemperature protection remain essential. Pre-charge limits inrush current when the CBU is connected to the DC bus, while contactors or solid-state switches isolate the bank during a fault or service event. Skeleton's reference system integrates switchgear, fuse-fault detection, pre-charge/discharge contactors and protection against high-current and overheating events; industry discussions also highlight the need for faster DC fault isolation as CBU voltage rises.
- **Monitoring and communications:** Voltage, current, temperature, equivalent series resistance and state-of-health measurements allow operators to identify degradation before the CBU can no longer meet its specified peak-power requirement. Panasonic emphasizes voltage, temperature and ESR monitoring for predictive maintenance, while Skeleton provides single-cell voltage sensing, multiple temperature sensors and communication bus to its master controller.
- **Thermal and mechanical system:** Although supercapacitors are efficient, high repetitive current produces resistive heating, particularly in cells, busbars, converters and connectors. The shelf therefore requires designed airflow, temperature sensing and, at higher power levels, potentially liquid cooling. Skeleton's cabinet uses controlled fan cooling and also offers liquid-cooled configurations, while the Open Compute Project presentation from Panasonic identifies lower-resistance capacitors and liquid cooling as development priorities for future higher-output CBUs

Main functions of CBU

Transient power stabilization

One of the most fundamental roles of a CBU is to stabilize the rack DC power bus when GPU power consumption changes very rapidly. AI workloads frequently alternate between compute-intensive phases and lower-power communication, synchronization, or idle phases. Because many GPUs operate in parallel, these transitions can create very large changes in rack power within milliseconds.

When GPU demand suddenly increases, the PSU cannot instantaneously increase its output. The CBU therefore discharges stored energy into the DC bus during this short response gap, helping prevent voltage drop and excessive current excursions. Conversely, when GPU demand suddenly falls, the CBU can absorb excess energy while the PSU reduces its output, helping limit voltage overshoot.

The key objective here is power quality and system stability, rather than reducing the overall power capacity required by the rack. In other words, the CBU acts as a fast

electrical shock absorber between the highly dynamic GPU load and the slower PSU. This becomes increasingly important as AI racks move above 100 KW and synchronized GPU workloads create larger absolute power swings.

Peak shaving

Another major function is peak shaving, whose objective is to limit the maximum power that the PSU and upstream infrastructure need to supply.

The benefit is potentially significant because the PSU, connectors, protection devices, and portions of the upstream power infrastructure no longer need to be oversized solely for short-duration GPU peaks. It can also allow PSUs to operate closer to their optimal loading range, improving utilization and potentially efficiency.

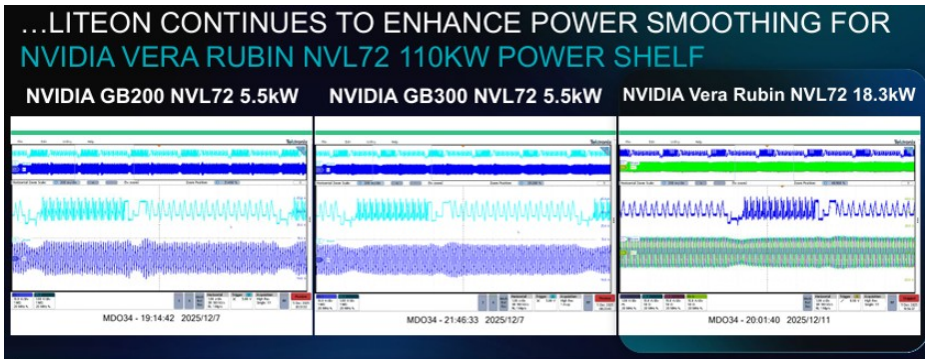
Short duration ride-through

The third function is ride-through, where the CBU temporarily powers the rack when the normal power source is interrupted altogether. This is fundamentally different from the first two functions because the issue is no longer a mismatch between GPU demand and PSU response—the upstream power source itself has temporarily disappeared.

A sufficiently sized CBU can maintain the DC bus during events such as a brief AC interruption, PSU switchover, source-transfer event, or temporary power-shelf failure. Depending on the amount and type of capacitance installed, this support can range from milliseconds to several seconds. For example, some LIC-based capacitance shelves are designed to provide multi-second hold-up at high rack power.

However, ride-through is generally a secondary function for a CBU. Capacitors have high power density but relatively low energy density, meaning they can deliver a large amount of power very quickly but cannot economically sustain that power for long periods. As the required backup duration increases from seconds toward minutes, a battery-based BBU generally becomes more suitable.

Exhibit 42: Capacitors help smooth power delivery



	Ipk-max	Ipk-min	Max/Min Ratio
NVIDIA GB200 NVL72	15.9A	8.3A	91.57%
NVIDIA GB300 NVL72	13.3A	8.6A	54.65%
NVIDIA Vera Rubin NVL72	**12.8A**	**10.3A**	**24.27%**

Source: Lite On

CBUs vs. BBUs

The two technologies should consequently be regarded as complementary rather than directly substitutive. A CBU handles the high-power, short-duration and high-cycle portion of the load profile, while a BBU supplies the larger amount of energy required when the upstream source remains unavailable. Some describe BBUs and CBUs as coexisting in the same power sidecar, and hybrid architectures can allocate fast transients to capacitors while preserving battery charge and cycle life for longer outages.

The CBU's major advantages are rapid bidirectional response, long cycle life, high efficiency and the ability to absorb repeated GPU peaks without the same electrochemical wear mechanism as a conventional battery. It can also recharge quickly between workload cycles and may require less routine replacement. However, the CBU has materially lower energy density, its voltage falls continuously during discharge, and large banks require balancing, high-current protection, cooling and valuable rack or sidecar space.

Exhibit 43: Comparison of CBUs and BBUs

Dimension	CBU	BBU
Primary storage medium	EDLC supercapacitors, LICs or high-capacitance	Typically lithium-ion battery cells managed by a BMS
Primary strength	High power, fast charge/discharge and frequent	Higher stored energy and longer sustained
Main function	Transient smoothing, peak shaving, voltage	Outage backup, source-transfer bridging and orderly workload
Response profile	Well suited to millisecond-class load support and	Fast switchover, but generally optimized for sustained support rather
Typical duration	Milliseconds to seconds; selected systems can extend toward tens of seconds or more	Commonly tens of seconds to several minutes, depending on battery and rack design
Power versus energy density	Higher power capability but lower energy storage per unit volume than a battery	Higher energy density, making it more economical for longer events
Cycle life	Eaton specifies more than one million cycles and up to 20 years for certain supercapacitor modules	Battery life is more sensitive to depth of discharge, temperature and repeated cycling; BMS supervision is therefore essential.
State management	Requires cell balancing, voltage/temperature monitoring and state-of-health tracking, but does not use a conventional battery fuel	Requires BMS functions including charge control, state of charge, state of health and cell protection
Best-fit use case	Repetitive AI power spikes, inrush events, PSU load leveling and bus stabilization	Power continuity through outages, brownouts or generator/source-transfer intervals

Source: Morgan Stanley Research

HVDC Emerging, with SST in the Pipeline

HVDC coming along with the increasing power consumption of AI clusters

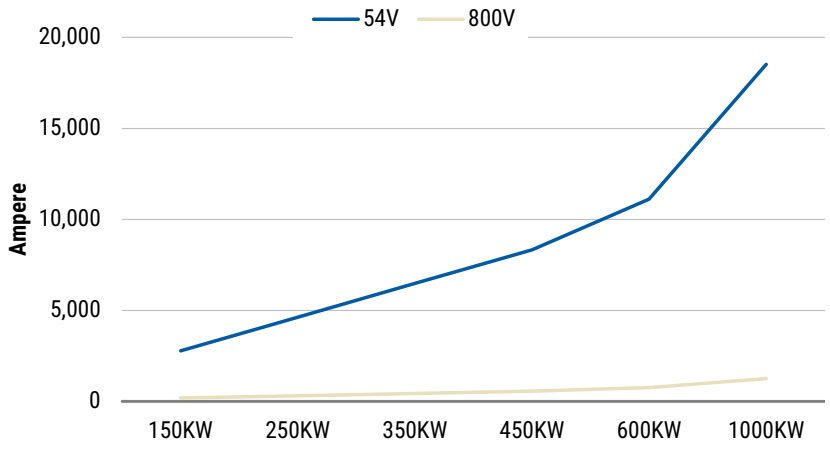
Why do we need HVDC?

Low voltage turns higher rack power into an extreme-current problem

The fundamental relationship is that at a given rack load, increasing the bulk-distribution voltage from 54 V to 800 V reduces current by approximately 14.8 times. A balanced ±400 V architecture also presents 800 V between its +400 V and –400 V conductors, so its bulk-transfer current is broadly equivalent to a 0–800 V architecture.

This current differential has an outsized mechanical consequence. For the same conductor resistance, I^2R loss at 54V would theoretically be approximately 219 times that at 800 V, though real systems avoid this level of loss by adding copper, parallel conductors and cooling. The result is not a 219-fold difference in actual operating loss, but a trade-off in which low-voltage designs consume increasing amounts of busbar cross-section, connector area, rack volume and cooling capacity to remain within voltage-drop and temperature limits.

Exhibit 44: Transmission current for different power scales: 54V vs 800V



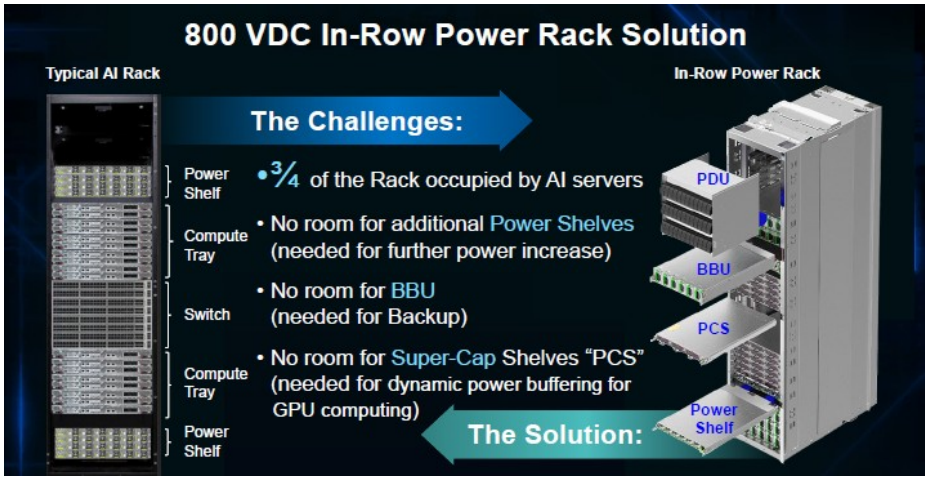
Source: Delta

Copper, connectors and thermal management become system constraints

At several thousand amperes, distribution cannot rely on a single ordinary conductor or connector. Designers must use broad laminated busbars, multiple parallel contacts, tightly controlled joint resistance and temperature monitoring. A small increase in contact resistance produces disproportionately high heat at these currents, raising the risk of hot spots, connector ageing and nuisance protection events. OCP's 100KW HVDC-to-LVDC shelf illustrates where the current problem moves: only two 62.5 A HVDC inputs are required on the ±400V side, while the 49–50V output approaches 2,000A and therefore uses a substantial bolted output connection with dedicated thermistors.

The same issue appears in power-shelf packaging. Delta notes that a typical AI rack may already allocate roughly three-quarters of its volume to servers, leaving insufficient room for additional power shelves, BBUs or supercapacitor shelves as power rises. Moving AC/DC conversion, protection and storage into an adjacent power rack therefore releases compute space and separates the thermal design of power electronics from that of GPUs. Microsoft's Mt Diablo rationale similarly states that disaggregated power can enable up to 35% more accelerators in the IT rack and allows power shelf count to be sized independently for each workload configuration

Exhibit 45: Not enough room for power devices on the traditional AI server rack



Source: Delta

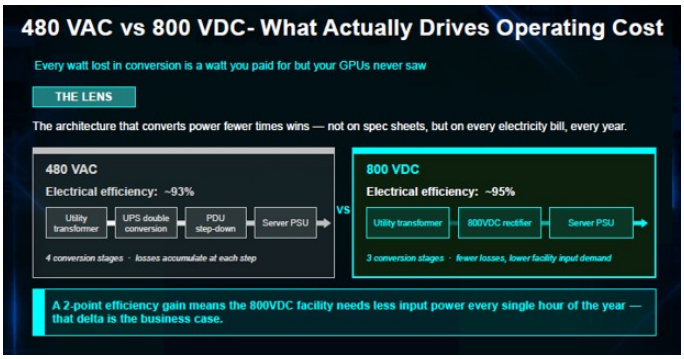
HVDC enables denser, fewer and more efficient power stages

As AI rack power scales from today's GB200/GB300 systems to Vera Rubin and future multi-hundred-kilowatt racks, power distribution efficiency becomes increasingly important. Our checks indicate that a conventional 480VAC architecture delivers an end-to-end electrical efficiency of roughly 93%, while an 800VDC architecture improves efficiency to about 95% by eliminating one power conversion stage and simplifying the power path from grid to rack.

The higher-voltage 800VDC distribution also reduces current levels for the same power load, lowering resistive (I^2R) losses in cables and distribution equipment and enabling more efficient delivery to increasingly power-dense AI racks.

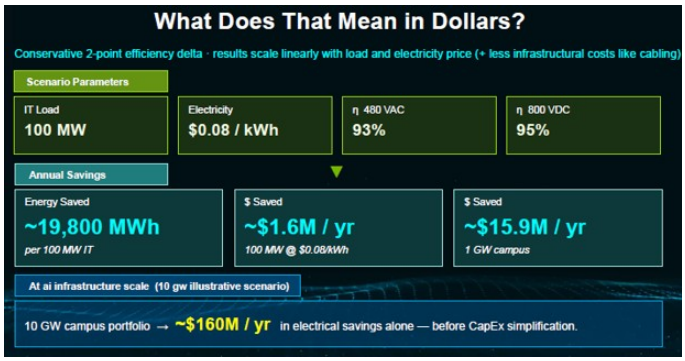
While a 2ppts of efficiency improvement may appear modest, the savings become substantial at hyperscale AI data center deployments. Delta's example shows that for a 100MW IT load, improving facility efficiency from 93% to 95% reduces annual energy consumption by approximately 19,800MWh, translating into roughly US$1.6 million of annual electricity cost savings assuming a power price of US$0.08/kWh. At a 1GW AI campus, the savings scale to nearly US$16 million per year, and at emerging multi-gigawatt AI factory deployments, the annual benefit can exceed US$100 million. Consequently, the transition from conventional AC distribution and 54VDC rack power architectures toward 800VDC distribution is not only a technical necessity for supporting future 500kW-to-1MW AI racks, but also a compelling economic proposition that lowers operating expenses and improves total cost of ownership through reduced energy losses and simplified electrical infrastructure.

Exhibit 46: Better power efficiency for 800VDC architecture



Source: Delta

Exhibit 47: Better efficiency leads to cost savings



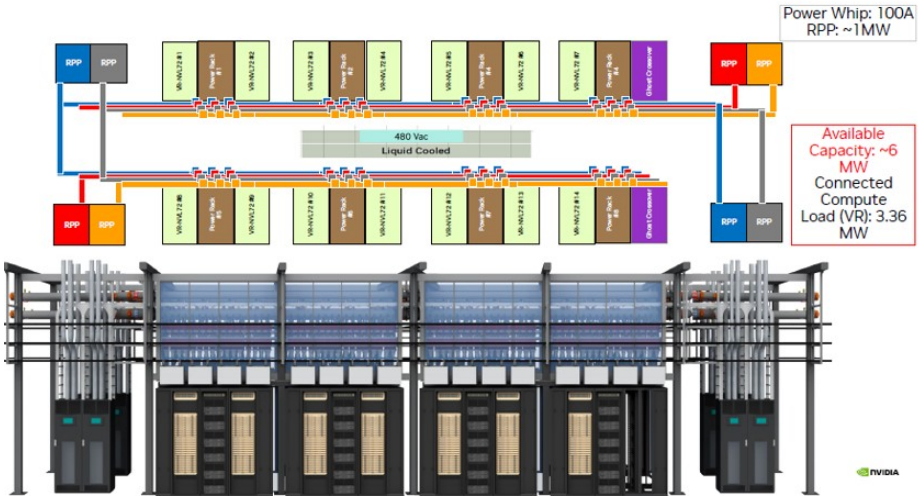
Source: Delta

What's the deployment roadmap for HVDC?

The deployment roadmap for 800VDC is best understood as a multi-stage evolution rather than a single-step transition. Both Nvidia and major hyperscalers recognize that existing 480VAC infrastructure cannot efficiently support the rapid increase in rack power. As rack densities increase, conventional power whips, busbars, connectors, and PSU architectures increasingly become bottlenecks, driving the need for higher voltage distribution

The first phase of deployment focuses on introducing 800VDC power racks within existing AC-based data centers. In this architecture, the facility remains largely unchanged, but a dedicated power rack converts 415-480VAC into 800VDC before distributing power locally to AI racks. This approach minimizes disruption to existing infrastructure while allowing operators to deploy Vera Rubin-class systems that exceed the practical limits of traditional power shelves. Nvidia refers to this as the row-level power rack architecture, which serves as the industry's bridge solution between legacy AC facilities and future native DC environments.

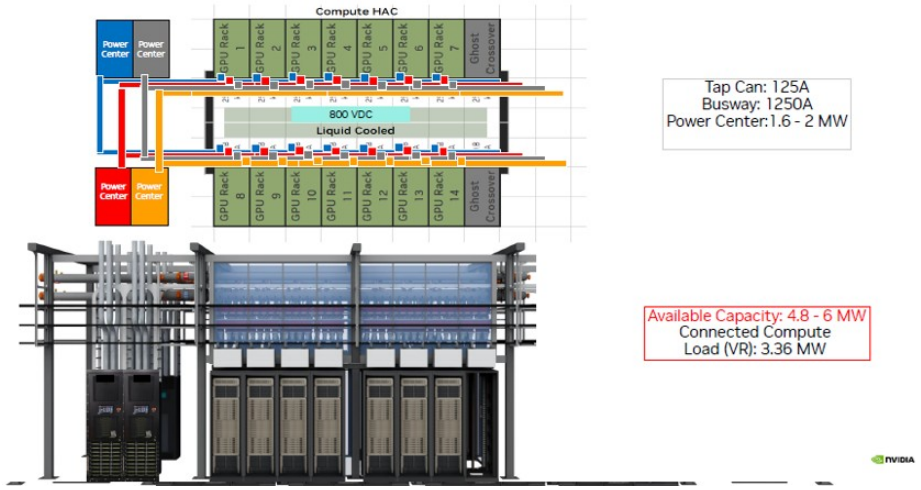
Exhibit 48: Phase I - 800VDC power rack



Source: Nvidia

The second phase expands 800VDC beyond individual rows into data hall-level power centers. Instead of deploying a dedicated power rack for every few compute cabinets, centralized rectifier or power-center systems distribute 800VDC through DC busways across an entire hall. Nvidia's roadmap identifies 1.6-2MW power centers and 1250A DC busway architectures as the next step, improving power density while reducing equipment duplication. This stage also aligns well with the deployment strategies being discussed by hyperscalers and the OCP community because it preserves existing medium-voltage and low-voltage infrastructure while moving a significant portion of distribution into DC

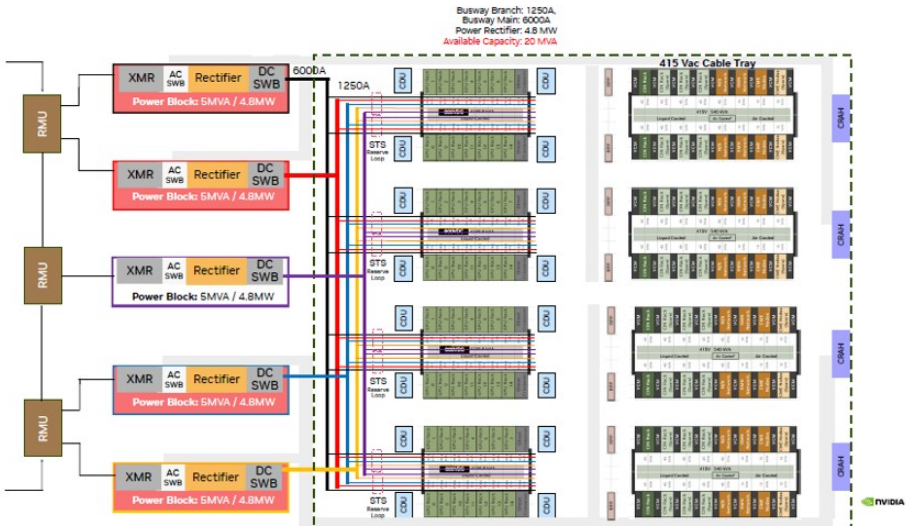
Exhibit 49: Phase II – 800VDC power center



Source: Nvidia

The third phase is the transition toward facility-wide 800VDC distribution, where AC is converted to DC near the utility entrance or medium-voltage substation and distributed throughout the data center via high-capacity DC switchboards and busways. Nvidia describes this as the DC power block architecture, utilizing approximately 4.8MW power blocks, 6000A DC busways, and integrated redundancy schemes. At this stage, the architecture begins to resemble an "AI factory," where power distribution is simplified, energy storage can be integrated directly onto the DC bus, and multiple megawatts of compute capacity can be deployed as modular building blocks

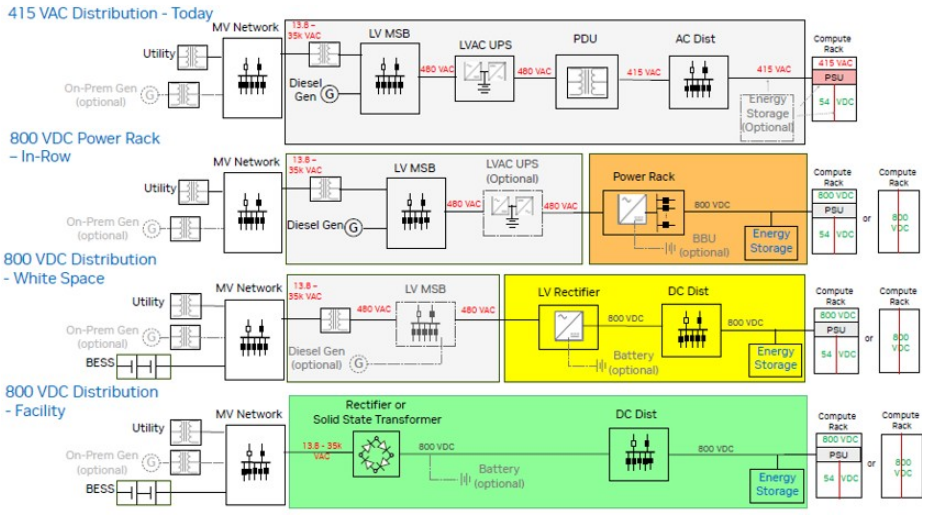
Exhibit 50: Phase III – power block



Source: Nvidia

The final stage is the emergence of native 800VDC compute racks, which Nvidia identifies as the long-term destination. Today, even in 800VDC deployments, power is generally stepped down to 54V before entering the servers. Future systems, however, are expected to accept 800VDC directly within the rack and potentially at the server level, eliminating another conversion stage and further reducing power losses. Nvidia's public roadmap shows a progression from current 54V architectures, to 800VDC-fed PSUs, and ultimately to 1MW-class native 800VDC racks capable of supporting next-generation AI accelerators

Exhibit 51: 800VDC native server racks



Source: Nvidia

In sum, deployment is likely to occur gradually because the ecosystem remains under development. NVIDIA notes that several enabling technologies—including DC protection devices, busways, connectors, grounding standards, arc-flash modeling, and solid-state transformers— are still undergoing certification and standardization. The roadmap presented by NVIDIA suggests a phased commercialization timeline, with 800VDC power racks reaching deployment readiness in 2026, hall-level power centers in 2027, and facility-scale DC power blocks becoming deployment-ready around 2028. Industry

collaboration among NVIDIA, Microsoft, Google, and OCP participants is focused on ensuring interoperability and regulatory acceptance before broader hyperscale adoption.

Exhibit 52: HVDC deployment roadmap

Focus Area	Status	Impact
Grounding & Protection	Architecture converging with UL and OEMs	Foundation for safe and reliable deployment
Equipment Certification	Certification pathways established and progressing	Accelerates deployment readiness
Protection Devices	MCCB available; SSCB ecosystem maturing	Enables fast fault isolation and higher availability
Busway, Connector & Whips	Standardized 125A architecture under development	Supports scalable rack deployment
High-Density Power Conversion	Rectifiers available; SST roadmap advancing	Enables future 800 VDC Compute Product

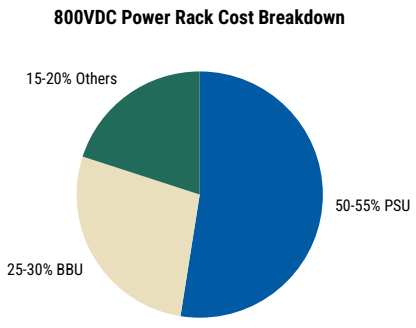
Source: Nvidia

800VDC represents 4-5x dollar content vs. in-rack power supply architecture

The 800VDC power rack BOM is primarily driven by the PSU, which accounts for 50–55% of the total cost, followed by the BBU at 25–30%, with other components comprising the remaining 15–20%. This cost structure reflects the broader functionality integrated into an HVDC power rack, combining power conversion, backup power and supporting components within a centralized architecture.

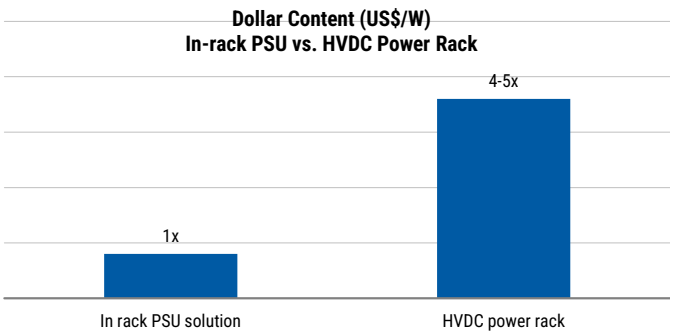
For power supply players, one of the most implications is that an HVDC power rack carries around 4–5x higher dollar content per watt compared with a conventional in-rack PSU solution, since the former integrates more components into a robust system and power supply players usually ship the entire rack instead of just the components.

Exhibit 53: BOM cost breakdown for an 800VDC power rack



Source: Morgan Stanley Research

Exhibit 54: Higher dollar content for HVDC power racks



Source: Morgan Stanley Research

800VDC vs. ±400VDC?

The debate between ±400VDC and 800VDC centers on balancing near-term practicality against long-term scalability. While both architectures deliver an identical 800V differential voltage and therefore achieve similar reductions in current, copper loss, and cable size versus traditional 54V rack power systems, they differ significantly in implementation.

Under a ±400VDC architecture, each conductor operates at only 400V relative to ground, resulting in less stringent insulation, creepage, clearance, and protection requirements. This makes the architecture easier to deploy, maintain, and certify, which explains why many hyperscalers and OCP members view ±400VDC as an attractive migration path toward high-density AI racks.

By contrast, NVIDIA and several power infrastructure vendors are increasingly advocating for a single-bus 800VDC architecture because it simplifies overall power distribution, reduces conversion stages, and is better suited for future 500kW to 1MW rack deployments.

We believe that ±400VDC appears well positioned as a near-term bridge architecture due to its lower deployment risk and more mature ecosystem, whereas 800VDC is increasingly viewed as the long-term end-state architecture for AI factories as rack power scales toward the megawatt era.

Exhibit 55: Comparison of 800VDC and ±400VDC

Metric	±400VDC	800VDC
Voltage to ground	400V relative to ground	800V relative to ground
Voltage differential	800V	800V
Power distribution efficiency	High	Slightly higher
Scalability	Good	Excellent
Insulation requirements	Lower	Higher
Maintenance safety	Better	More challenging
System architecture simplicity	Dual-rail system	Single-bus system
Grounding Complexity	More complicated	Simpler
Electrical footprint	Small	Smaller
Ability to reduce conversion stages	Moderate	Highest
Long-distance distribution capability	Excellent	Excellent
Maturity of ecosystem	Better today	Early but rapidly developing
Industry players	OCP, hyperscalers	Nvidia
Long term industry direction	Transitional architecture	Likely end-state architecture

Source: Morgan Stanley Research

From a component supplier perspective, the transition toward HVDC is far more important than which architecture ultimately wins, as most hardware content increases under either scenario. PSU vendors, BBU suppliers, busbar manufacturers, connector providers, power-whip suppliers, magnetics vendors, and rack-level DC-DC converter manufacturers are largely architecture-agnostic, since both ±400VDC and 800VDC require a similar shift toward high-voltage power distribution, centralized power shelves, and rack-level conversion.

Our understanding is that the primary exception is the power semiconductor industry, where the choice of architecture could materially influence device content and technology adoption. A ±400VDC system can often be supported using existing 650V-900V MOSFET and SiC platforms, benefiting suppliers with exposure to lower-voltage power devices. By contrast, 800VDC architectures increasingly require 1200V-class SiC MOSFETs, higher-

voltage gate drivers, hot-swap controllers, isolation components, and advanced protection circuitry, driving greater semiconductor content per watt and raising technology barriers.

Solid state transformer potentially being the holy grail in AI datacenter power delivery

What is a solid-state transformer ?

A datacenter solid state transformer (SST) integrates power electronics to replace or augment traditional line-frequency transformers, and it's typically built around three major functional stages. The first is the medium-voltage AC-to-DC rectifier stage, which uses high-voltage semiconductor switches—often silicon carbide (SiC) or gallium nitride (GaN) devices due to their superior switching speed and thermal performance—to convert incoming grid-level medium voltage AC into a regulated DC bus. The second is the isolated DC-DC conversion stage, usually implemented as a dual active bridge (DAB) or LLC resonant converter, which provides galvanic isolation through a high-frequency transformer (operating in the kHz to tens-of-kHz range rather than 50/60 Hz), dramatically shrinking the magnetic core and transformer size compared to conventional units while stepping voltage down toward datacenter distribution levels. The third is the low-voltage DC-AC or DC-DC output stage, which conditions power into the form needed by the facility—either regulated DC for direct DC-bus distribution to IT racks or inverted AC for legacy PDU/UPS infrastructure. Supporting these power stages are the control and sensing subsystems (digital controllers, gate drivers, voltage/current sensors) that manage switching, fault protection, and grid synchronization; thermal management components (heat sinks, cold plates, or liquid cooling loops) to handle the concentrated losses in the semiconductor devices; and increasingly, integrated energy storage interfaces or DC microgrid ports that allow the SST to natively support battery buffering, renewable integration, and finer-grained power quality control than a passive transformer ever could

A Solid-State Transformer (SST) is a power-electronics-based transformer that replaces the traditional 50/60Hz iron-core transformer with a combination of power semiconductors, control electronics, and a medium-frequency transformer. Unlike conventional transformers that perform only voltage conversion, SSTs actively process power through multiple stages, enabling both AC and DC inputs/outputs, bidirectional power flow, voltage regulation, reactive power compensation, and intelligent grid control

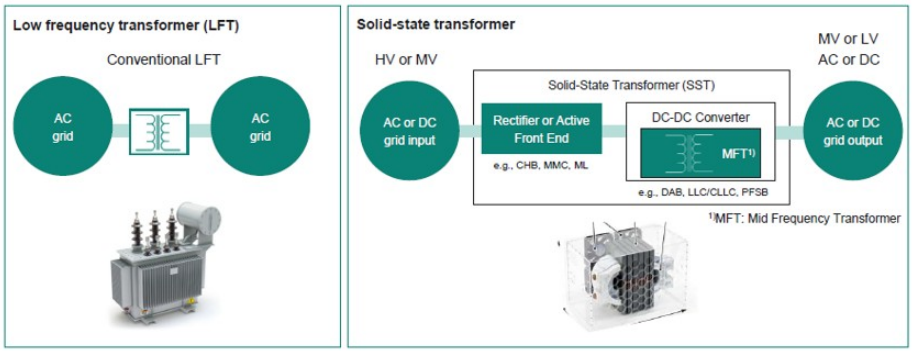
A SST is typically built around three major functional stages:

- The first is the medium-voltage AC-to-DC rectifier stage, which uses high-voltage semiconductor switches—often silicon carbide (SiC) or gallium nitride (GaN) devices due to their superior switching speed and thermal performance—to convert incoming grid-level medium voltage AC into a regulated DC bus.
- The second is the isolated DC-DC conversion stage, usually implemented as a dual active bridge (DAB) or LLC resonant converter, which provides galvanic isolation through a high-frequency transformer (operating in the KHz to tens-of-LHz range rather than 50/60 Hz), dramatically shrinking the magnetic core and transformer size compared to conventional units while stepping voltage down toward datacenter distribution levels.
- The third is the low-voltage DC-AC or DC-DC output stage, which conditions

power into the form needed by the facility—either regulated DC for direct DC-bus distribution to IT racks or inverted AC for legacy PDU/UPS infrastructure.

Supporting these power stages are the control and sensing subsystems (digital controllers, gate drivers, voltage/current sensors) that manage switching, fault protection, and grid synchronization; thermal management components (heat sinks, cold plates, or liquid cooling loops) to handle the concentrated losses in the semiconductor devices; and increasingly, integrated energy storage interfaces or DC microgrid ports that allow the SST to natively support battery buffering, renewable integration, and finer-grained power quality control than a passive transformer ever could.

Exhibit 56: Solid state vs. traditional transformers



Source: Infineon

Why do we need SSTs?

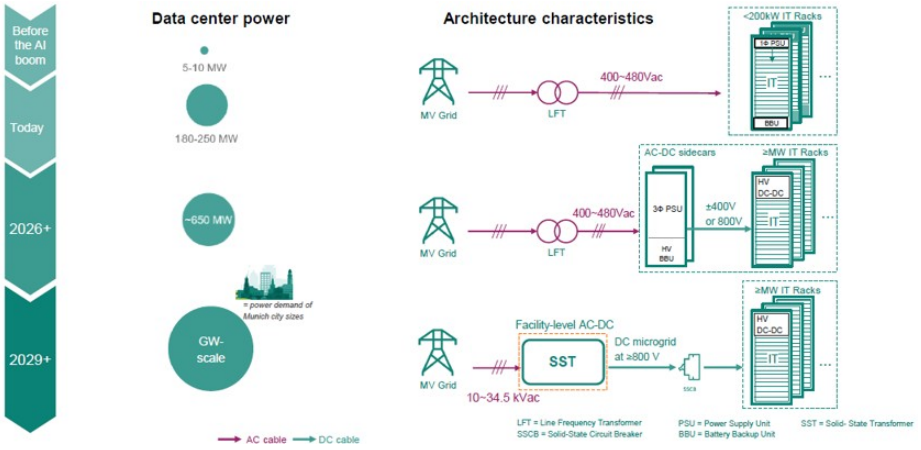
Improving efficiency in an era of gigawatt data centers

One of the strongest drivers for SST adoption is the rapid increase in AI data center power demand, which has progressed from 5-10MW facilities in the pre-AI era to campuses exceeding 650MW today, with gigawatt-scale facilities expected beyond 2029.

Traditional architectures use medium-voltage AC distribution, followed by low-frequency transformers and multiple power conversion stages before power reaches IT racks. SSTs allow facilities to convert 10-34.5KV medium-voltage power directly into ±400V or 800V DC distribution networks. This reduces conversion losses and improves overall grid-to-chip efficiency.

The efficiency advantage could be at the order of about 2-5ppt vs, the traditional transformer architecture, but at a 1GW AI campus, this magnitude of improvement could translate into roughly 20-50MW of continuous power savings. Over the lifetime of a facility, this can result in substantial reductions in electricity cost and carbon emissions. SSTs are therefore becoming increasingly attractive for hyperscaler operators seeking lower total cost of ownership (TCO).

Exhibit 57: SST enabling more efficient power delivery architecture



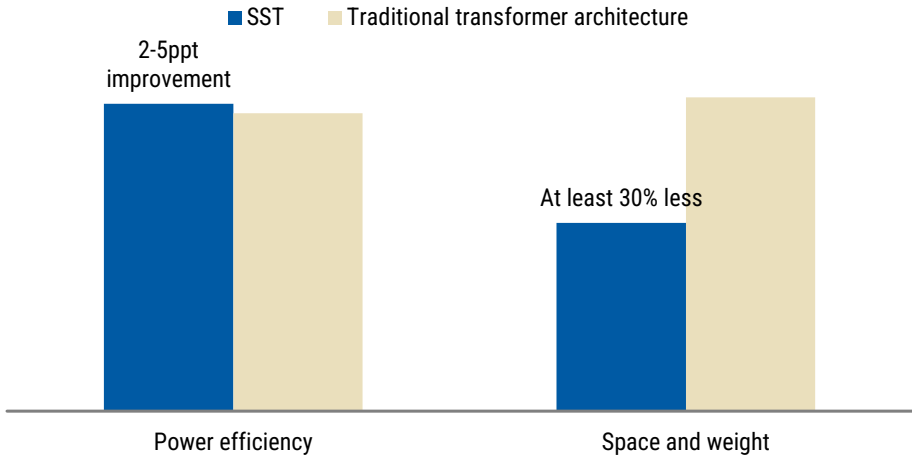
Source: Infineon

Higher power density and smaller footprint

Because SSTs use medium-frequency transformers instead of bulky 50/60Hz transformers, they can achieve significantly higher power density. Infineon indicates that SST systems can reduce both footprint and weight by at least 30% compared with conventional transformer-based installations. They also accelerate site energization and facilitate modular deployment

This becomes particularly valuable in urban data centers, utility substations, EV charging hubs, and energy storage sites where space constraints directly affect project economics.

Exhibit 58: SST offers better power efficiency with at least 30% less space/weight



Source: Infineon, Delta Electronics, Morgan Stanley Research

Supporting distributed energy resources

As renewable energy penetration increases, future grids will require seamless integration of solar generation, battery storage, fuel cells, gas turbines, and other distributed energy resources.

SSTs provide intelligent power routing, bidirectional operation, and dynamic voltage control. This allows them to function as energy hubs that connect renewable generation,

storage, and loads through a common DC infrastructure. Such capabilities are difficult or impossible to achieve using passive transformers alone.

Improved grid intelligence

Unlike conventional transformers, SSTs can actively regulate voltage and frequency, provide reactive power compensation, support grid-forming functionality, and improve overall power quality. These capabilities make SSTs an important enabler for future smart grids, microgrids, and virtual power plants.

What's the deployment status now?

Despite decades of research, SSTs remain at an early commercial deployment stage. While we started to see the SST models being unveiled by relevant suppliers and there start to be some pilot adoption, we believe that the 800VDC power rack/center/block architecture shall be deployed first, and SSTs will take more time, as more fundamental design change in the datacenter and all hardware inside will involved. We believe wider adoption probably won't happen before 2030.

Exhibit 59: SST from Delta

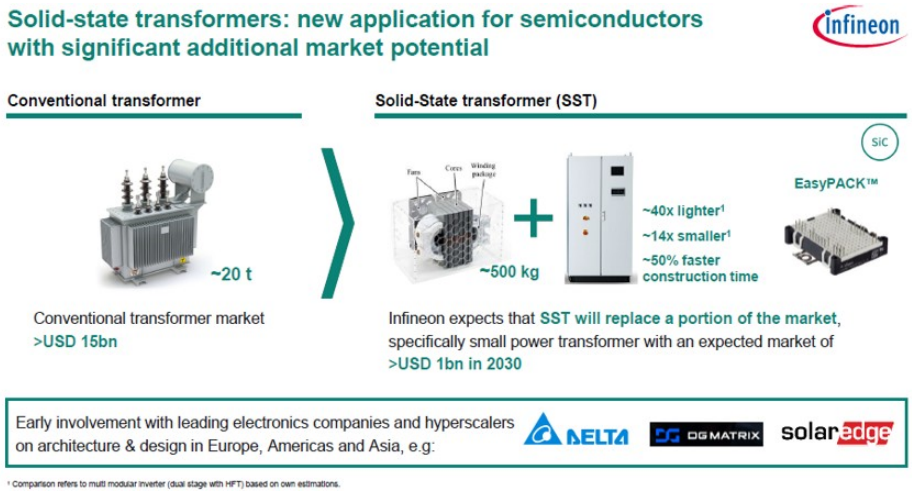
Source: Delta

Exhibit 60: SST from Eaton

Source: Eaton

According to Infineon, SST could represent a >US$1bn market opportunity by 2030, as the technology begins to replace a portion of the >US$15bn conventional transformer market, particularly in smaller-power applications. While SST penetration is expected to remain relatively modest versus the overall transformer market, the opportunity is significant for relevant players in the longer term.

Exhibit 61: SST to see initial adoption in 2030



Source: Infineon

Key hurdles for the wider adoption of SSTs

- **High system cost remains the primary barrier:** Despite their technical advantages, SSTs remain significantly more expensive than conventional transformers. Traditional transformers consist primarily of steel cores, copper windings, insulation materials, and cooling systems, all of which benefit from mature manufacturing processes and economies of scale. In contrast, SSTs require large numbers of high-voltage power semiconductors, gate drivers, digital controllers, sensors, protection systems, and sophisticated thermal management solutions. The extensive use of silicon carbide (SiC) devices, which remain substantially more expensive than conventional silicon-based components, further increases the overall system cost. Although SSTs can deliver efficiency gains, improved controllability, and reduced footprint, the economic benefit is often insufficient to offset the substantial upfront capital expenditure in many utility applications. Consequently, SST deployment is currently limited to applications where efficiency, power density, or system flexibility carry exceptionally high economic value, such as AI data centers, megawatt EV charging stations, and grid-scale battery storage systems.
- **Long term reliability has yet to be proven:** Electric utilities and large infrastructure operators place an extremely high value on reliability because transformers are expected to operate continuously for several decades with minimal maintenance. Conventional transformers routinely achieve service lifetimes exceeding 30 to 40 years and have accumulated more than a century of operating history. SSTs, however, are fundamentally different devices that incorporate thousands of semiconductor junctions, complex control electronics, communication networks, cooling systems, and software algorithms. While each individual technology may be mature, the long-term reliability of the complete SST system remains largely unproven at scale. Operators therefore remain cautious about replacing a passive asset with an active power-electronic system whose failure modes are more complex and less understood. For critical infrastructure such as substations, industrial facilities, or hyperscale data centers, reliability concerns often outweigh potential efficiency improvements.

- **DC protection technology ss still evolving:** One of the most significant technical challenges associated with SST deployment is fault protection, particularly in medium-voltage and high-voltage DC networks. Unlike AC systems, where current naturally crosses zero multiple times each cycle and facilitates fault interruption, DC currents continue flowing continuously during fault conditions. As a result, fault currents can rise extremely quickly, making them more difficult to detect and interrupt safely. Future SST-based power architectures therefore require advanced protection devices such as solid-state circuit breakers (SSCBs), which can react far faster than conventional mechanical breakers. However, DC protection technologies are still evolving, and there is currently no universally accepted approach equivalent to the mature protection schemes available in AC grids. The absence of standardized protection architectures increases deployment risk and remains one of the key obstacles to the wider adoption of SST-based DC distribution systems.
- **System complexity creates engineering, qualification and maintenance challenges:** Compared with traditional transformers, SSTs are considerably more complex systems that require the integration of multiple high-performance subsystems. A complete SST includes medium-voltage power converters, isolated DC/DC stages, high-frequency transformers, semiconductor modules, digital controllers, communication interfaces, and advanced protection mechanisms. As voltage and power levels increase, the number of converter modules and semiconductor devices required can rise significantly. Although the introduction of higher-voltage SiC devices can reduce subsystem count and simplify design, SST architectures remain substantially more complicated than conventional transformers. This complexity increases development costs, certification requirements, maintenance needs, and operational risks, all of which slow adoption among traditionally conservative utility customers.
- **Industry standards and ecosystems are still developing:** The transformer industry benefits from decades of established standards, testing procedures, maintenance practices, and supply chain infrastructure. SSTs, by comparison, represent an emerging technology with limited deployment history and evolving standards. Many utilities remain uncertain regarding qualification procedures, maintenance requirements, spare-parts strategies, and long-term asset management. Furthermore, widespread SST adoption will require parallel development of supporting technologies such as DC protection systems, DC switchgear, energy management software, and medium-voltage DC distribution networks. Until the broader ecosystem matures, many customers will continue to view SSTs as a promising but relatively high-risk alternative to proven transformer technologies.

Fuel Cells for Better Flexibility and Resilience

What are fuel cells?

Fuel cells are electrochemical energy-conversion devices that generate electricity by combining hydrogen and oxygen without combustion. Unlike conventional power generation technologies, which convert fuel into heat and then into electricity through multiple intermediate steps, fuel cells produce electricity directly through an electrochemical reaction, resulting in higher efficiency and significantly lower emissions.

When powered by hydrogen, the primary byproducts are water and heat, making fuel cells one of the cleanest forms of power generation available today.

Fuel cells can be deployed across a wide range of applications, including passenger vehicles, heavy-duty trucks, buses, trains, ships, stationary power generation, backup power systems, and increasingly data centers. As governments and industries pursue decarbonization while seeking reliable and dispatchable power sources.

Major fuel-cell technologies include:

- SOFC (Solid Oxide Fuel Cell): Preferred for stationary power and data centers.
- PEMFC (Proton Exchange Membrane Fuel Cell): Dominant in transportation and backup power.
- MCFC (Molten Carbonate Fuel Cell): Large-scale distributed generation.
- PAFC (Phosphoric Acid Fuel Cell): Commercial combined heat and power.
- DMFC (Direct Methanol Fuel Cell): Portable applications

Exhibit 62: Fuel cell offering - Delta

Source: Delta

Exhibit 63: Fuel cell offering - Bloom Energy

Source: Bloom Energy

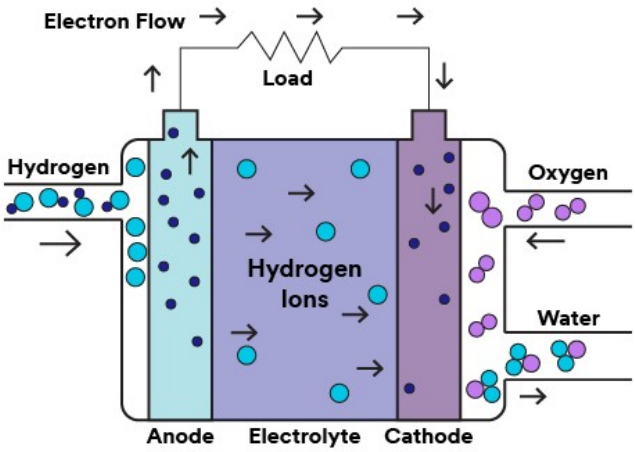
How do fuel cells work?

While there are several fuel-cell architectures, the fundamental operating principle is largely the same: fuel is supplied to the anode, oxygen is supplied to the cathode, and an electrolyte in between allows charged ions to move while forcing electrons to flow through an external circuit. This flow of electrons generates usable electricity, while the

reaction also produces heat and water as byproducts. Unlike conventional generators, which rely on burning fuel to drive a turbine or engine, fuel cells produce electricity directly through electrochemical reactions, allowing them to achieve significantly higher efficiency and lower emissions.

Using solid oxide fuel cell (SOFC) technology as an example, air is supplied to the cathode side of the cell, where oxygen molecules gain electrons and form oxygen ions (O^{2-}). These oxygen ions migrate through a ceramic electrolyte toward the anode side of the cell. At the same time, hydrogen or another hydrogen-rich fuel is supplied to the anode. When the oxygen ions reach the anode, they react with the fuel to produce water, heat, and electrons. The generated electrons cannot pass through the electrolyte and therefore travel through an external electrical circuit before returning to the cathode, creating a continuous flow of electricity. As long as fuel and air continue to be supplied, the fuel cell can generate power continuously.

Exhibit 64: Fuel cell diagram



Source: 3M

What benefits can fuel cells offer?

- **High energy efficiency:** Fuel cells convert the chemical energy of hydrogen directly into electricity through an electrochemical reaction, avoiding the multiple energy-conversion steps associated with combustion-based power generation. This allows fuel cells to achieve electrical efficiencies of over 60%, with some systems exceeding 80% total efficiency when waste heat is recovered through combined heat-and-power configurations, compared with roughly 35-45% for conventional gas-fired power generation. The efficiency advantage translates directly into lower fuel consumption, reduced operating costs, and improved energy utilization over the lifetime of the system.
- **Low carbon emissions:** When powered by hydrogen, fuel cells produce only water and heat as byproducts, generating no direct carbon emissions at the point of use. Even when operating on hydrogen derived from natural gas, fuel cells typically produce significantly lower greenhouse gas emissions than conventional combustion-based power systems. As green hydrogen production scales, fuel cells offer a potential pathway toward near-zero-carbon power generation for transportation, industrial facilities, and distributed energy systems.

- **Grid independence and energy security:** Fuel cells can provide reliable on-site power generation independent of the electrical grid, improving energy security and resilience. This makes them attractive for mission-critical applications such as data centers, hospitals, telecommunications infrastructure, defense installations, and industrial facilities that require uninterrupted power. In regions facing grid congestion or reliability concerns, fuel cells can also serve as a distributed power source located directly at the point of consumption.
- **Long duration energy storage enabler:** Fuel cells play an important role in the emerging hydrogen economy by enabling excess renewable energy from solar and wind farms to be stored as hydrogen and later converted back into electricity. Unlike batteries, which become increasingly expensive for longer storage durations, hydrogen storage can economically provide energy over days, weeks, or even seasonal periods. This capability could become increasingly valuable as renewable energy penetration rises worldwide.
- **Fuel flexibility:** Certain fuel-cell technologies, particularly Solid Oxide Fuel Cells (SOFCs), can operate on a variety of fuels including hydrogen, natural gas, biogas, ammonia-derived hydrogen, and other hydrogen-rich fuels. This flexibility allows users to deploy fuel-cell systems today using existing fuel infrastructure while maintaining a pathway toward pure hydrogen operation as hydrogen supply chains mature.
- **Quiet operation and reduced maintenance:** Because fuel cells contain relatively few moving parts compared with combustion engines and turbine-based generators, they operate quietly and often require less routine maintenance. The absence of combustion also reduces mechanical wear and vibration, which can improve system reliability and lower lifecycle operating costs, particularly in stationary power applications.
- **Scalability across multiple applications:** Fuel cells can be deployed across a wide range of power levels, from portable power systems measured in watts to utility-scale installations generating multiple megawatts. This scalability allows the technology to serve residential buildings, commercial facilities, industrial plants, transportation platforms, microgrids, and large-scale data centers, making fuel cells one of the most versatile clean-energy technologies available today.

Limitations and constraints of fuel cells

- **High system cost:** Despite significant technological progress, fuel cells remain more expensive than many conventional power-generation technologies. The cost challenge stems from complex stack manufacturing, specialized materials, balance-of-plant components, and relatively low production volumes compared with mature technologies such as internal combustion engines, gas turbines, and batteries. While costs have declined over the past decade, achieving widespread commercial adoption will likely require further manufacturing scale, supply-chain development, and technological improvements.
- **Limited hydrogen infrastructure:** One of the biggest challenges facing fuel-cell deployment is the lack of hydrogen production, transportation, storage, and refueling infrastructure. Unlike electricity, which is already widely distributed through established grids, hydrogen supply networks remain in the early stages of

development in most regions. This infrastructure gap creates a chicken-and-egg challenge, where end users are reluctant to adopt fuel cell technologies without reliable hydrogen availability, while infrastructure developers are hesitant to invest until demand reaches sufficient scale.

- **Durability and lifetime requirements:** Although fuel-cell durability has improved substantially, long-term reliability remains a key focus area, particularly for commercial transportation and utility-scale installations. Continuous exposure to heat, humidity, contaminants, and repeated operating cycles can gradually degrade fuel cell performance over time. Customers in sectors such as transportation, power generation, and industrial infrastructure often require operating lifetimes measured in tens of thousands of hours, creating significant engineering challenges for system developers.
- **Regulatory and safety considerations:** Hydrogen is a highly flammable gas and requires specialized handling, storage, and transportation systems. Although hydrogen technologies have well-established safety standards, new fueling stations, pipelines, and storage facilities must comply with stringent regulations. Developing these standards and obtaining approvals can extend deployment timelines and increase project costs.
- **Technology fragmentation and lack of standardization:** The fuel cell industry encompasses multiple technology platforms, including PEM, SOFC, MCFC, and other architectures, each optimized for different applications. While this diversity encourages innovation, it can also slow industry standardization and create uncertainty for customers evaluating long-term technology choices. Unlike the battery industry, which is increasingly converging around a few dominant chemistries, the fuel cell sector remains relatively fragmented.

Implications to datacenter power architecture

Transition from backup power to primary power source

Traditionally, data centers rely on grid electricity for primary power and diesel generators for backup power during outages. Fuel cells are increasingly being evaluated not only as a replacement for diesel backup generators but also as a potential primary power source. As AI-driven workloads substantially increase power density and overall electricity demand, hyperscalers are exploring fuel cells as a means of deploying computing capacity in locations where grid power is constrained or unavailable. This could fundamentally shift data-center power architecture from a grid-centric design to one that incorporates distributed on-site generation.

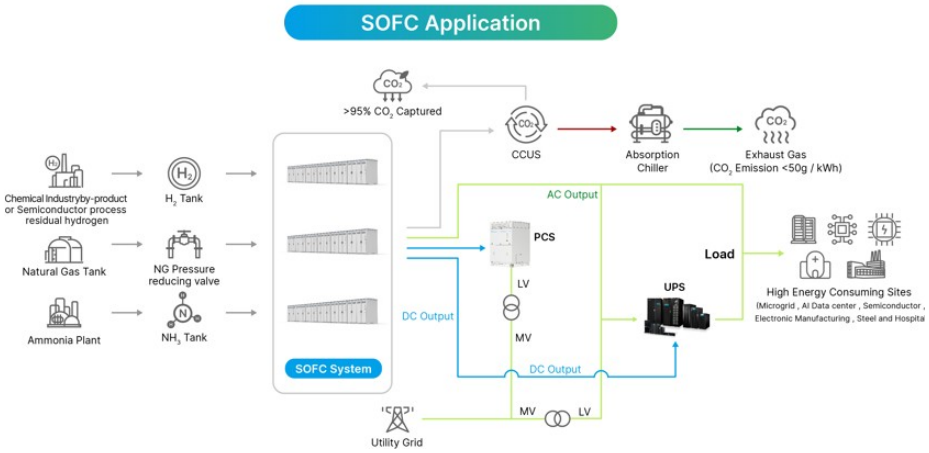
Potential long term evolution toward microgrid-based datacenters

In the longer term, fuel cells could become a core component of data-center microgrids combining renewable generation, battery storage, hydrogen storage, and intelligent energy management systems. Rather than functioning as passive consumers of grid electricity, future AI-focused data centers may increasingly operate as self-contained energy ecosystems capable of managing their own generation, storage, and load balancing. Such an architecture could provide greater scalability, resilience, and energy independence than traditional grid-connected designs

Potential simplification of power distribution architecture

Fuel cells inherently generate DC power, which aligns more closely with the power requirements of modern servers and IT equipment. Current data centers typically involve multiple AC/DC and DC/DC conversion stages between the utility grid and the processor, each introducing efficiency losses. Over time, fuel cell adoption could encourage a greater migration toward direct DC power distribution architectures, reducing conversion losses and improving overall facility efficiency. Moreover, its DC output fits well with the shift into HVDC power delivery architecture for datacenters.

Exhibit 65: Fuel cells in datacenter power delivery architecture



Source: Delta

Risk Reward – Delta Electronics Inc. (2308.TW)

Strong combination: Power + cooling for AI infrastructure

PRICE TARGET NT$2,500.00

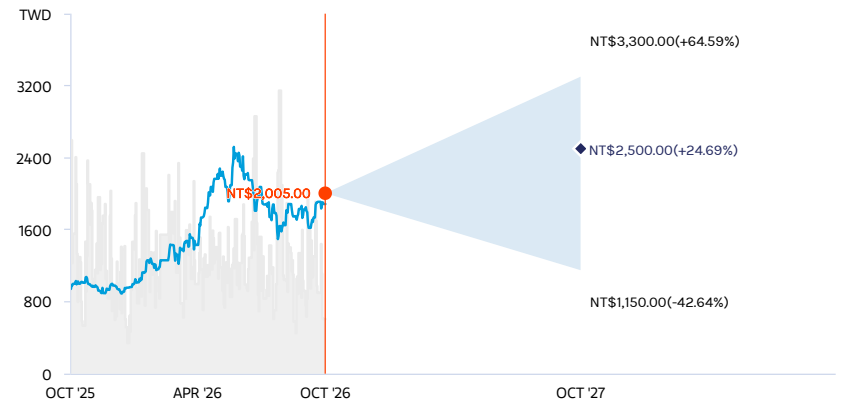
Base case: P/E multiple of 30x based on average 2027-28 EPS. We believe our target multiple is justified by the 63% earnings CAGR we estimate for 2025-28, implying a PEG of 0.5, versus the stock's ~23x P/E and 23% earnings CAGR during 2022-25, which implied a PEG of 1.0.

Consensus Price Target Distribution NT$2,200.00 NT$2,621.29 MS PT NT$4,120.00

Source: Refinitiv, Morgan Stanley Research

◆ Mean ◆ Morgan Stanley Estimates

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We view AI server power supplies and power management/cooling solutions for data centers as key growth drivers for the coming decade.
- Our expectation for a re-rating reflects expanding gross/operating margins and increasing exposure to AI-related businesses.
- Our 2026-28 earnings estimates reflect strength in AI server power supplies and liquid cooling.
- Our price target implies a P/E of 30x based on average 2027-28 EPS.

Consensus Rating Distribution

100%	Overweight
0%	Equal-weight
0%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth:	*Positive*
Self-help:	*Positive*
Technology Diffusion:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$3,300.00

40x average 2027-28e P/E

Strong sales growth with significant margin expansion: We assume a greater-than-expected contribution from higher-margin businesses such as power electronics and liquid cooling as well as greater synergies from acquired companies.

BASE CASE NT$2,500.00

30x average 2027-28e P/E

Strong YoY revenue growth in 2026-28, coupled with operating leverage: We expect steady growth in AI server power supply and data center infrastructure to be the key drivers. We see favorable mix and scale merits driving profit to grow more than double in 2027 vs. 2025.

BEAR CASE NT$1,150.00

14x average 2027-28e P/E

Moderate sales growth with slight margin improvement: We assume further profit drag from legacy products, smaller profit contribution from higher-margin businesses, higher opex and/or less meaningful contribution from acquired companies.

Risk Reward – Delta Electronics Inc. (2308.TW)

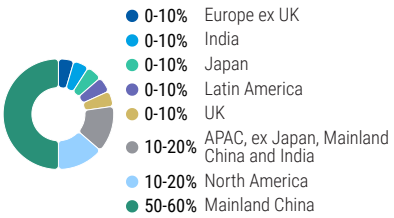
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Power Electronics YoY (%)	25.0	48.9	52.6	51.8
Automation YoY (%)	4.8	10.9	5.2	5.2
Infrastructure YoY (%)	81.6	44.6	51.0	48.2

INVESTMENT DRIVERS

- Transition to EV-related business and data infrastructure
- Operating margin recovery
- FINI holdings

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster-than-expected global economic growth
- Falling raw material prices
- Margin improvement

RISKS TO DOWNSIDE

- Global economic slowdown
- Ongoing component supply tightness
- Rising raw material prices
- Margin contraction as a result of higher operating expenses or faster ASP erosion

OWNERSHIP POSITIONING

Inst. Owners, % Active 64.7%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	746,914	775,166 / 774,709	807,877
EBITDA (NT$, mn)	163,606	169,213 / 175,311	191,720
Net income (NT$, mn)	96,767	103,307 / 105,731	111,870
EPS (NT$)	37.25	39.77 / 40.50	43.07

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Financial Summary – Delta

Exhibit 66: Delta: Financial Summary

Consolidated Income Statements

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Net sales	**421,148**	**554,885**	**775,166**	**1,131,877**	**1,659,520**
COGS	(284,567)	(364,729)	(494,374)	(707,257)	(1,028,632)
Gross profit	136,580	190,157	280,792	424,620	630,887
Operating expenses	(88,928)	(106,224)	(142,491)	(191,135)	(263,905)
Operating income	**47,652**	**83,932**	**138,300**	**233,484**	**366,983**
Non-operating income	**3,664**	**3,934**	**10,365**	**7,100**	**7,300**
Interest income	1,859	1,501	1,687	2,100	2,300
Investment income	(162)	(200)	372	-	-
Exchange gain	1,041	644	2,091	-	-
Other	926	1,989	6,215	5,000	5,000
Pre-tax income	**51,316**	**87,866**	**148,666**	**240,584**	**374,283**
Income tax	(10,925)	(19,930)	(32,925)	(52,189)	(81,172)
Minority interests	(5,163)	(7,828)	(12,433)	(20,158)	(31,363)
Net income	**35,229**	**60,108**	**103,307**	**168,237**	**261,747**
Modelware EPS (NT$)	**13.56**	**23.14**	**39.77**	**64.77**	**100.77**

Consolidated Cash Flow Statements

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Operating CF	**72,895**	**98,474**	**116,226**	**177,240**	**263,652**
Net Profits	35,229	60,108	103,307	168,237	261,747
Depreciation & Amort.	25,016	27,839	30,912	33,985	37,058
Change in WC	(2,486)	(24,190)	(55,485)	(86,295)	(125,847)
Other adjustments	15,137	34,717	37,491	61,313	90,694
Investing CF	**(40,360)**	**(53,214)**	**(70,000)**	**(70,000)**	**(60,000)**
Change of FA (Capex)	(33,435)	(45,950)	(70,000)	(70,000)	(60,000)
Change of L/T invest.	(4,739)	-	-	-	-
Other adjustments	(1,644)	(3,517)	-	-	-
Financing CF	**(11,118)**	**(9,401)**	**36,107**	**(12,414)**	**(54,050)**
Increase in L/T debt	8,552	(11,308)	12,530	2,978	3,829
Increase in S/T debt	(1,417)	446	29	212	272
Issuance of stock	-	-	-	-	-
Cash dividends	(18,696)	(20,313)	(30,131)	(51,787)	(84,335)
Other adjustments	444	21,774	53,680	36,183	26,183
FX adjust.	8,586	(2,146)	-	-	-
Net change in cash	**30,003**	**33,713**	**82,333**	**94,826**	**149,602**

Consolidated Balance Sheets

NT$mn (Year End Dec)	2024	2025	2026E	2027E	2028E
Cash	**117,459**	**151,172**	**233,505**	**328,332**	**477,934**
Mkt securities	4,333	3,941	3,941	3,941	3,941
A/R	89,921	121,717	170,036	248,282	364,023
Inventory	83,856	101,478	142,217	205,395	298,726
Others	15,355	17,473	24,410	35,642	52,257
Current Assets	**310,925**	**395,781**	**574,109**	**821,592**	**1,196,881**
Long-term investments	4,740	4,876	4,876	4,876	4,876
Fixed assets	115,710	142,040	127,820	127,652	124,410
Other assets	100,523	96,922	96,922	96,922	96,922
Total Assets	**531,898**	**639,619**	**803,727**	**1,051,042**	**1,423,089**
S/T borrowings	1,918	2,364	2,393	2,605	2,877
AP/NP	69,223	94,452	128,025	183,154	266,379
Other ST liabilities	81,142	112,849	157,649	230,195	337,504
Total current liabilities	**152,283**	**209,665**	**288,067**	**415,954**	**606,759**
L/T debt	32,209	21,143	33,673	36,650	40,480
Other LT liabilities	67,840	83,422	83,422	83,422	83,422
Total Liabilities	**252,332**	**314,230**	**405,162**	**536,026**	**730,661**
Common shares	25,975	25,975	25,975	25,975	25,975
Retained Earnings	102,429	140,925	214,101	330,552	507,964
Other SH equity	151,161	158,489	158,489	158,489	158,489
Shareholders' equity	**279,565**	**325,389**	**398,565**	**515,016**	**692,428**
Total Liab./SE	**531,898**	**639,619**	**803,727**	**1,051,042**	**1,423,089**

Consolidated Financial Ratios

	2024	2025	2026E	2027E	2028E
Margins (%)					
Gross margin	32.4	34.3	36.2	37.5	38.0
Operating margin	11.3	15.1	17.8	20.6	22.1
Pretax margin	12.2	15.8	19.2	21.3	22.6
Net margin	8.4	10.8	13.3	14.9	15.8
YoY growth (%)					
Sales	5	32	40	46	47
Operating profits	16	76	65	69	57
Pretax profits	5	71	69	62	56
Net profits	5	71	72	63	56
Others					
Cash div payout (%)	52%	50%	50%	50%	50%
Adjusted cash div (NT$)	7.0	11.6	19.9	32.5	50.5
Yield (%)	0%	1%	1%	2%	3%
Net Debt/Equity (%)	-31%	-40%	-51%	-57%	-63%
Liabilities/Equity (%)	90%	97%	102%	104%	106%
Liabilities/Assets (%)	47%	49%	50%	51%	51%
ROAE (%)	13%	20%	29%	37%	43%
ROAA (%)	7%	10%	14%	18%	21%
AR/NR days	73	70	69	67	67
Inventory days	103	93	90	90	89
AP/NP days	79	82	82	80	80
Cash conversion	97	80	77	77	77

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 67: Delta: Quarterly Financials

Delta (2308.TW) (NT$ mn)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E
Net sales	**159,353**	**183,256**	**201,379**	**231,177**	**231,791**	**265,526**	**302,930**	**331,630**	**554,885**	**775,166**	**1,131,877**
COGS	(100,392)	(117,948)	(129,404)	(146,630)	(146,821)	(166,230)	(188,191)	(206,015)	(364,729)	(494,374)	(707,257)
Gross profit	**58,961**	**65,308**	**71,976**	**84,547**	**84,970**	**99,295**	**114,739**	**125,615**	**190,157**	**280,792**	**424,620**
Operating expenses	(30,544)	(34,738)	(36,983)	(40,226)	(40,422)	(45,405)	(50,589)	(54,719)	(106,224)	(142,491)	(191,135)
SG&A	(16,886)	(18,956)	(20,208)	(21,963)	(22,067)	(24,959)	(27,870)	(30,178)	(57,482)	(78,014)	(105,074)
R&D exp.	(13,657)	(15,782)	(16,775)	(18,263)	(18,356)	(20,445)	(22,720)	(24,541)	(48,743)	(64,477)	(86,061)
Operating income	**28,417**	**30,570**	**34,992**	**44,321**	**44,548**	**53,891**	**64,149**	**70,896**	**83,932**	**138,300**	**233,484**
Non-operating income	**2,676**	**4,430**	**1,630**	**1,630**	**1,775**	**1,775**	**1,775**	**1,775**	**3,934**	**10,365**	**7,100**
Interest income	428	398	430	430	525	525	525	525	1,501	1,687	2,100
Investment income	487	(115)	-	-	-	-	-	-	(200)	372	-
Disposal of investment	-	-	-	-	-	-	-	-	-	-	-
Exchange gain	1,688	403	-	-	-	-	-	-	644	2,091	-
Other	72	3,743	1,200	1,200	1,250	1,250	1,250	1,250	1,989	6,215	5,000
Pre-tax income	**31,093**	**35,000**	**36,622**	**45,951**	**46,323**	**55,666**	**65,924**	**72,671**	**87,866**	**148,666**	**240,584**
Income tax	(7,258)	(7,831)	(7,910)	(9,925)	(10,423)	(11,968)	(14,174)	(15,624)	(19,930)	(32,925)	(52,189)
Minority interest	(3,279)	(2,033)	(3,158)	(3,963)	(3,841)	(4,676)	(5,537)	(6,104)	(7,828)	(12,433)	(20,158)
Net income	**20,556**	**25,136**	**25,554**	**32,063**	**32,059**	**39,022**	**46,213**	**50,943**	**60,108**	**103,307**	**168,237**
EPS (NT$)	**7.91**	**9.68**	**9.84**	**12.34**	**12.34**	**15.02**	**17.79**	**19.61**	**23.14**	**39.77**	**64.77**
Margins (%)											
Gross Margin	37.0	35.6	35.7	36.6	36.7	37.4	37.9	37.9	34.3	36.2	37.5
Operating Margin	17.8	16.7	17.4	19.2	19.2	20.3	21.2	21.4	15.1	17.8	20.6
Pretax Margin	19.5	19.1	18.2	19.9	20.0	21.0	21.8	21.9	15.8	19.2	21.3
Net Margin	12.9	13.7	12.7	13.9	13.8	14.7	15.3	15.4	10.8	13.3	14.9
QoQ Growth (%)											
Sales	(1)	15	10	15	0	15	14	9			
Gross Profit	5	11	10	17	1	17	16	9			
Operating Profit	8	8	14	27	1	21	19	11			
Pretax Profit	21	13	5	25	1	20	18	10			
Net Profit	19	22	2	25	(0)	22	18	10			
YoY Growth (%)											
Sales	34	48	34	43	45	45	50	43	32	40	46
Gross Profit	56	48	37	51	44	52	59	49	39	48	51
Operating Profit	102	64	41	68	57	76	83	60	76	65	69
Pretax Profit	99	79	36	79	49	59	80	58	71	69	62
Net Profit	101	80	37	85	56	55	81	59	71	72	63

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

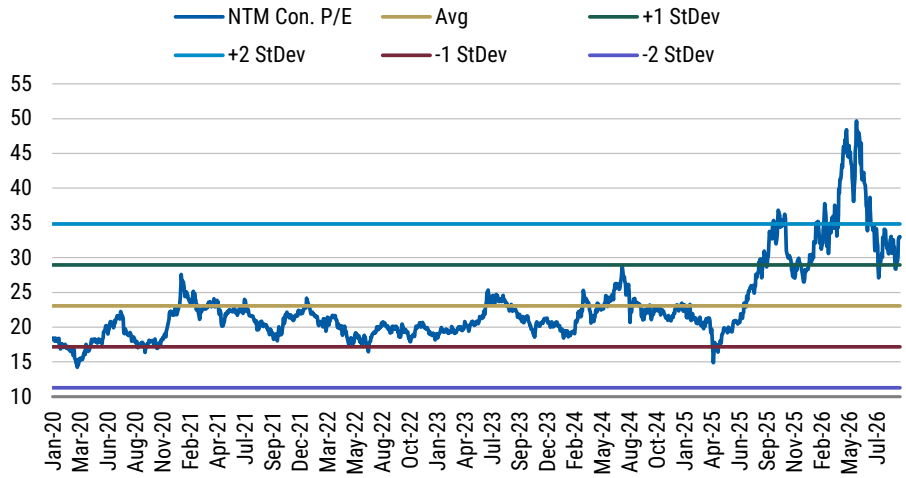
Valuation Methodology – Delta

Our price target (base case scenario value) is NT$2,500, derived from 30x 2027-28e P/E: We use price/earnings as our valuation methodology because we believe it appropriately captures Delta's earnings growth potential. We believe our target P/E multiple of 30x is justified by an earnings CAGR of 63% over 2025-28, which implies a PEG of 0.5. This is lower than the average of 1.0 in 2022-25, when the stock was trading at an average P/E of 23x for a 23% earnings CAGR.

Bull case scenario value – NT$3,300: We apply a 40x P/E multiple to our average base case estimate for 2027-2028. This scenario assumes a greater-than-expected contribution from higher-margin businesses such as power electronics and liquid cooling as well as greater synergies from acquired companies.

Bear case scenario value – NT$1,150: We apply a 14x P/E multiple to our average base case estimate for 2027-2028. This scenario assumes further profit drag from legacy products, smaller profit contribution from higher-margin businesses, higher opex and/or less meaningful contribution from acquired companies.

Exhibit 68: Delta: NTM P/E chart



Source: FactSet consensus estimates, Morgan Stanley Research

Why Does Delta Trade at a Significant Valuation Discount to Delta Thailand?

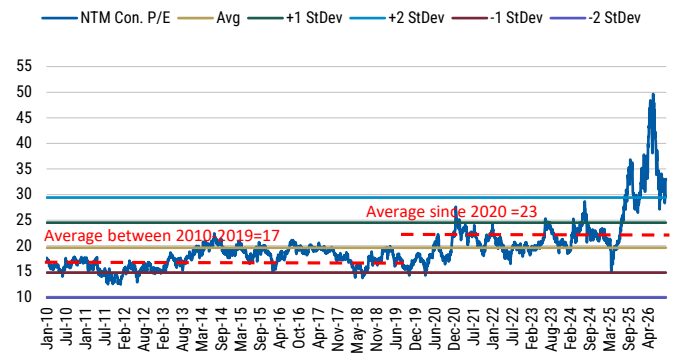
Discussion of the valuation gap between Delta Electronics (2308.TW, currently 33x 12M forward consensus P/E) and Delta Electronics Thailand (DELTA.TH, 71x currently) has been ongoing for several years. If we look at their valuations before 2020, Delta was actually trading at a premium, with an average 17x 12M forward P/E in 2010-2020 vs. Delta Thailand's 13x. This made sense, as Delta is much more competitive in the global arena and Delta Thailand was set up as a subsidiary focused on manufacturing.

The situation flipped after 2020. Though Delta has been re-rated to trade at an average 23x 12M forward P/E since 2020, Delta Thailand has seen its valuation surge to an average of 60x over the same period. This cannot be explained by moves in their respective markets, either, as TWSE went from 14x forward P/E in 2010-20 to 16x after 2020, while the SET went from 13x to 16x.

We think this could be attributable to a few factors:

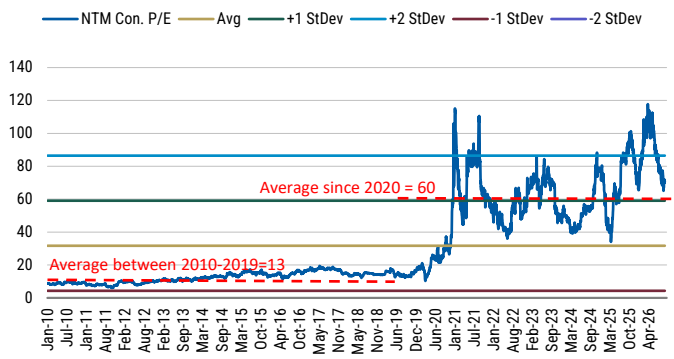
- **Different positioning within the Delta group:** Delta group has increased its stake in Delta Thailand to over 60% from about 20% in 2019 through a tender offer. Since then, there has been much more collaboration between Delta group and Delta Thailand. As Thailand emerged as one of the major manufacturing hubs within the Delta group, its earnings growth (28% in 2020-25) outpaced that of Delta Taiwan (19%).
- **Scarcity in the Thai stock market:** Delta Thailand is one of the relatively few large-cap Thai-listed companies that offer direct exposure to AI/data-center power and EV-related themes, whereas there are many such alternatives in Taiwan.
- **Limited free float:** As the Delta group owns a ~62% stake in Delta Thailand and there have been a few quite stable long-term shareholders, we think the effective free float is probably 20% or less.

Exhibit 69: Delta NTM P/E



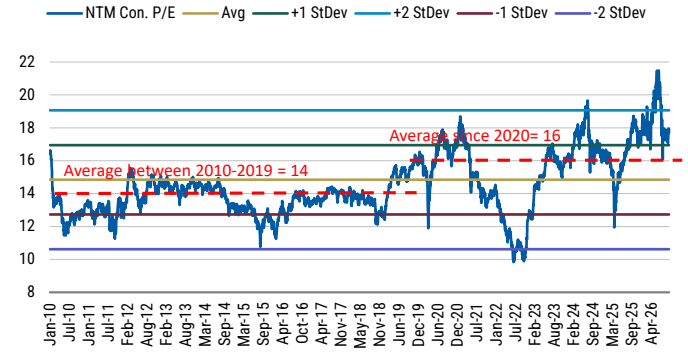
Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 70: DET NTM P/E



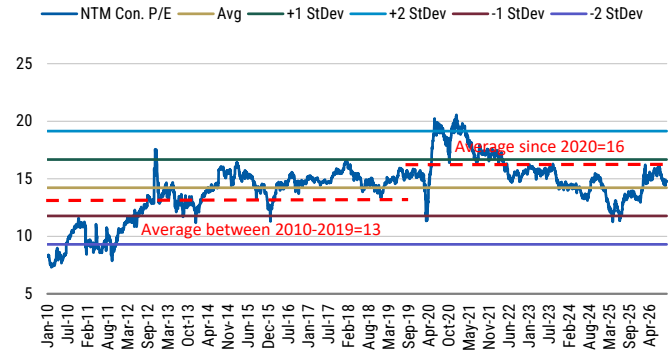
Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 71: TWSE NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

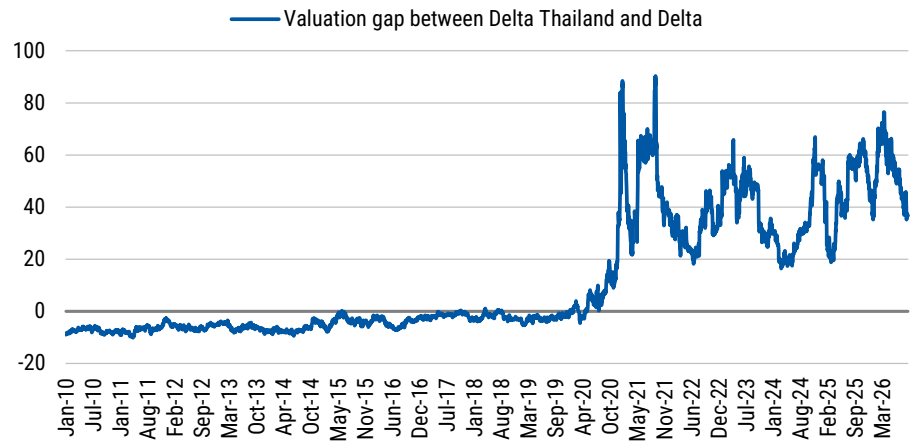
Exhibit 72: SET Index NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Exhibit 73:

Valuation gap between Delta Thailand and Delta based on 12M forward P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Global Footprint – Delta

Comprehensive Global Operations

Aggressive capacity expansion plans

The company has been continuously diversifying its production base away from China owing to the trade turmoil back in 2018-2019, and then by supply chain disruptions during Covid. Management explicitly set a target of lowering China production to 60% of total capacity (from >70% previously) for the first time back in 2020. New capacity was added in Thailand, India, and Taiwan during 2020-2021. It then saw demand surge from various sectors, including EVs, automation and telecom, during 2022-23, followed by AI-related power suppliers and cooling solutions emerging in 2024. In order to grasp the emerging opportunities, relevant plant build-outs were planned accordingly, on top of the ongoing geographical diversification. It expanded its presence—production sites, R&D, and sales teams—in Europe and the US to better cope with operational disruptions and capture domestic demand, in addition to its existing setups in Taiwan, Thailand, and India. There was then a substantial increase in capex of NT$12.2bn in 2025 (vs. NT$5.6-6bn in 2023-24) as the company further accelerated its expansion plan, mainly on AI-related power/cooling demand and increasing capacity tightness, specifically in Thailand and the US. Delta continued to expand in Thailand by adding three new plants, with another three under construction for power and liquid cooling system offerings. In view of the capacity tightness reserved for AI-related business, it also started to prioritize margin-accretive business—i.e., de-emphasize business with net margins below 10%—which further optimized product mix.

Looking ahead into 2026 and beyond, expansion is likely to remain aggressive in order to fulfill substantial power and liquid cooling turnkey solution demand for large-scale data centers, while management also indicated that the new capacity added at the end of 2025 has already been fully taken. With capacity staying tight for AI power and cooling offerings, the company will continue to expand actively in China, Thailand, the US, and Tainan, Taiwan. It also plans to add mass production lines in Taoyuan, scheduled to be ready by 2028, in addition to the pilot run lines in Tainan. Against this backdrop, the capex budget has been revised up several times and is now roughly NT$50bn for 2026.

Exhibit 74: Delta's global footprint

Source: Company data, Morgan Stanley Research. Note: Data as of 2024.

Financial Analysis – Delta

44% Revenue CAGR in 2025-28e

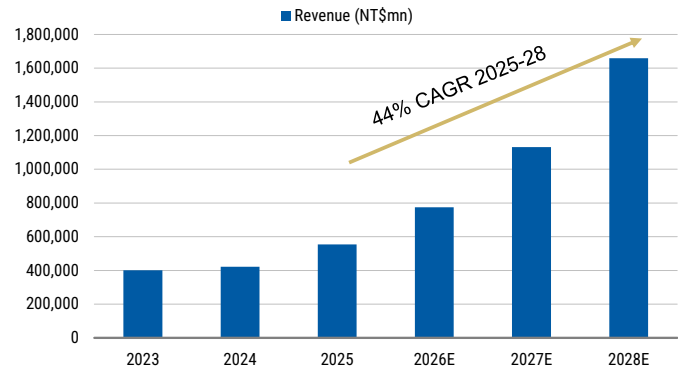
Delta's revenue growth was moderate in 2020 at merely 5%, making it the only year of the Covid disruption (2019-22) that did not see double-digit growth. Growth picked up to 11% in 2021, underpinned by surging demand for cloud, networking, and work-from-home infrastructure amid the pandemic, while the next uplift to 22% growth in 2022 was mainly fueled by increasing demand for EV- and data center-related components (server power supplies, networking switches, energy storage systems, etc). Revenue momentum subsequently moderated in 2023, growing just 4%, as the EV component business was impacted by the UAW strike in the US and automation demand in China stayed subdued, despite sustained demand for AI server power supplies and modules. Growth in 2024 was at a similar level, as strength in data center offerings were not enough to offset weakness in its legacy businesses.

Revenue surged 32% in 2025 to a record NT$554,885mn, driven by AI server rack power and cooling solutions, along with further spec upgrades and value growth.

We expect revenue to expand at a 44% CAGR over 2025-28, supported by ongoing AI server shipments, spec upgrades, 800 VDC power solutions, and turnkey liquid cooling offerings. As the upgrade to 800V HVDC power architecture is likely to take years given the complex design and high safety requirements, we expect AI server power solution upgrades to be separated into a few stages – from power shelves in AI server racks to standalone side-car power racks, from 50V DC to 800V DC output, from 72kW per power shelf to >250kW, etc, as well as power wattage increases for next-generation GPU rack designs.

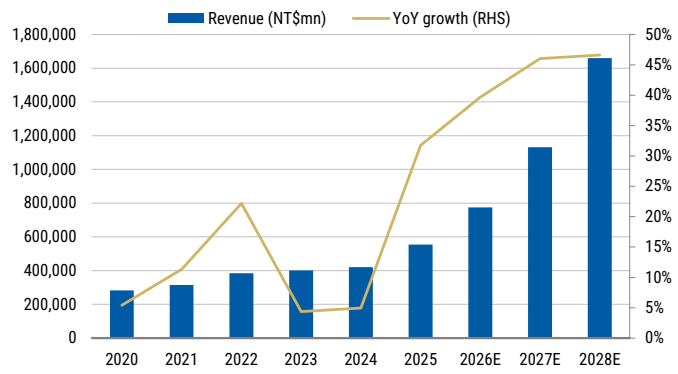
By segment, Power Electronics has been the primary revenue driver, contributing at least 50% over the past years despite the company separating the Mobility segment from Power Electronics since 2024 so as to further distinguish revenue generated from EVs/ autos. We expect the segment to remain the key revenue contributor, supported by strong demand from AI server-related power solutions, and estimate it will reach 58% of revenue by 2028 vs. 50% in 2025. Infrastructure, the second-largest revenue contributor historically, is expected to grow along with the Power Electronics segment, fueled by demand for data center power-related solutions (i.e. UPSs, CDUs, sidecars, etc) as well as energy infrastructure-related offerings (i.e. SSTs (solid-state transformers), fuel cells, modular power solutions, micro grid management, etc.). We model the segment's revenue contribution to reach 35% by 2028 from 24% in 2024 and 33% in 2025. For Automation and Mobility, we expect their revenue contributions to decline sequentially throughout the years, with Automation dropping from 10% in 2025 to 4% in 2028 and Mobility falling from 7% in 2025 to 2% in 2028.

Exhibit 75: Delta: Revenue CAGR, 2025-28e



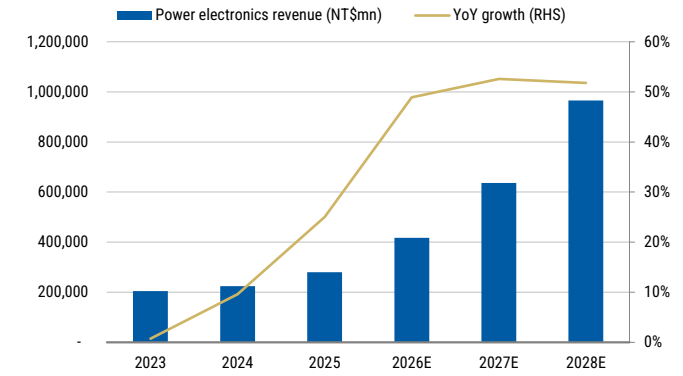
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 76: Delta: Annual revenue growth



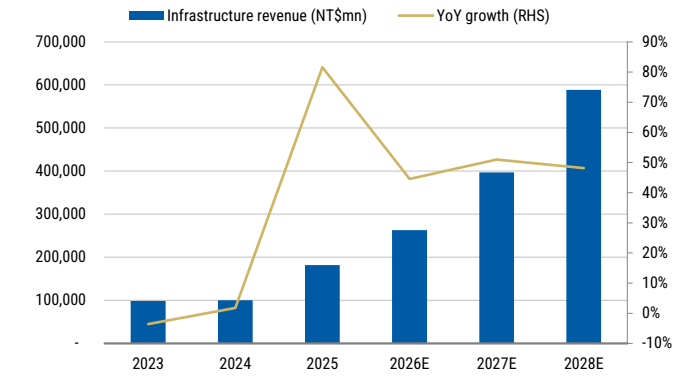
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 77: Delta: Power electronics revenue



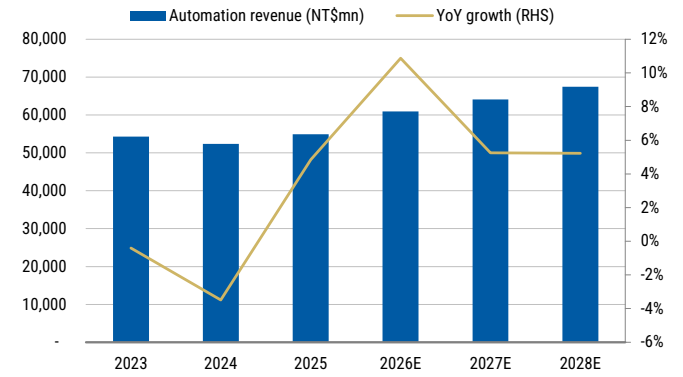
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 78: Delta: Infrastructure revenue



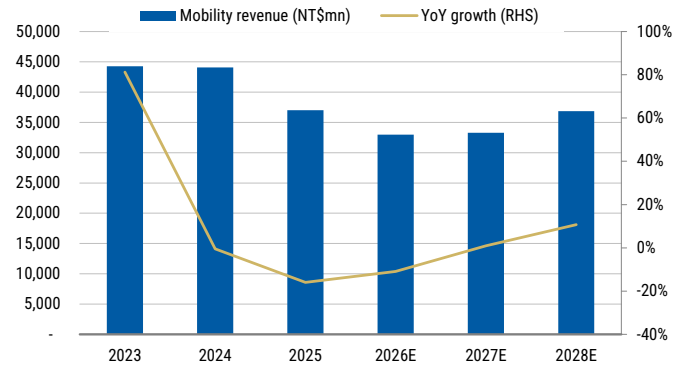
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 79: Delta: Automation revenue



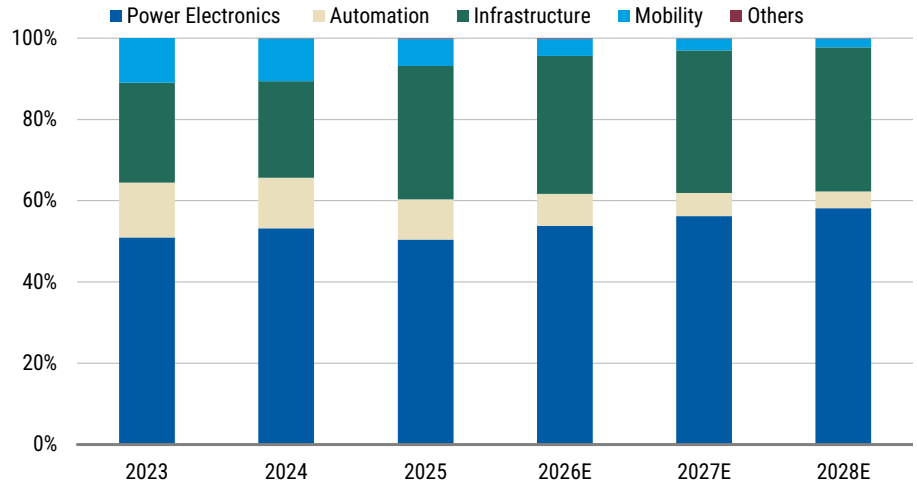
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 80: Delta: Mobility revenue



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 81:
Delta: Revenue breakdown by segment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

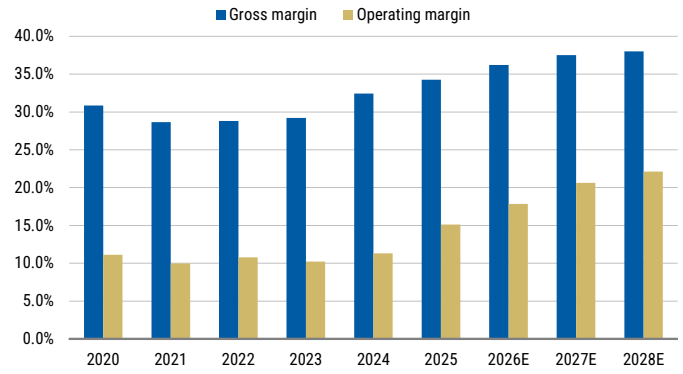
Continuous margin expansion on favorable product mix

Delta's gross margin stayed largely between 28-29% during 2021-23, as revenue was mainly driven by relatively lower-margin products, including EV- and consumer/IT-related products, while server-related offerings had yet to scale up. 2024 served as an inflection point, with the product mix changing structurally, supported by robust demand for AI data center-related offerings, while its legacy businesses stayed relatively weak. Gross margin thus grew by 3.2ppt YoY during the year, from 29.2% in 2023 to 32.4% in 2024. Increasing AI computing needs further increased power consumption and cooling requirements per rack, boosting demand for high-density AI power and cooling products along with related system-level solutions in 2025, which lifted gross margin by another 1.8ppt YoY to 34.3% for the year.

The opex ratio was largely stable, fluctuating around 18-19% between 2020 and 2023. It then increased to 19% in 2023 and 21.1% in 2024, mainly due to operating deleverage, higher R&D costs, and ongoing capacity expansion. The ratio declined back to 19.1% in 2025 on the back of an enlarged revenue base, despite R&D-related investments remaining elevated.

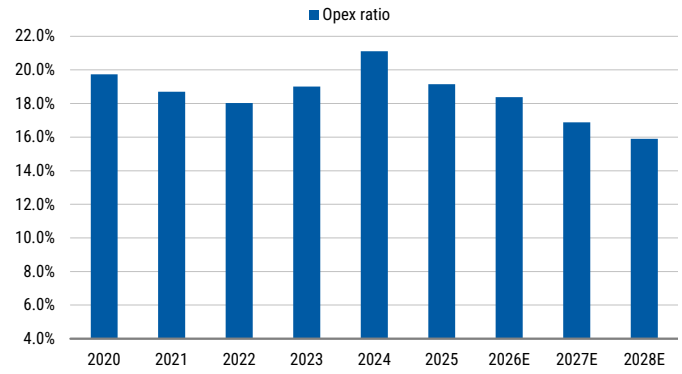
Looking ahead, we expect gross margin to continue expanding on the back of increasing revenue contribution from AI-related power supplies. We thus pencil in a 2.0ppt gross margin expansion to 36.2% in 2026, followed by further increases of 1.3ppt and 0.5ppt YoY in 2027 and 2028, respectively, reaching 38.0% by 2028. On the opex ratio, we expect Delta to continue benefiting from operating leverage in the coming years, and thus estimate operating margin to improve more substantially than gross margin, expanding by 2.7ppt YoY in 2026 to 17.8%, followed by further expansions of 2.8ppt and 1.5ppt in 2027 and 2028, reaching 20.6% and 22.1%, respectively.

Exhibit 82: Delta: Margin trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 83: Delta: Opex ratio



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Net profit to grow at a 63% CAGR in 2025-28

Delta's operating profit grew sequentially throughout 2020-22, supported by gross margin expansion and a lower opex ratio, from NT$31,426mn in 2020 to NT$41,439mn in 2022. It then saw a slight 1% YoY decline in 2023 due to higher operating expenses as the company further stepped up its R&D investments, despite a 0.4ppt gross margin expansion in 2023. Operating margin also edged down accordingly by 0.6ppt to 10.2% during the year.

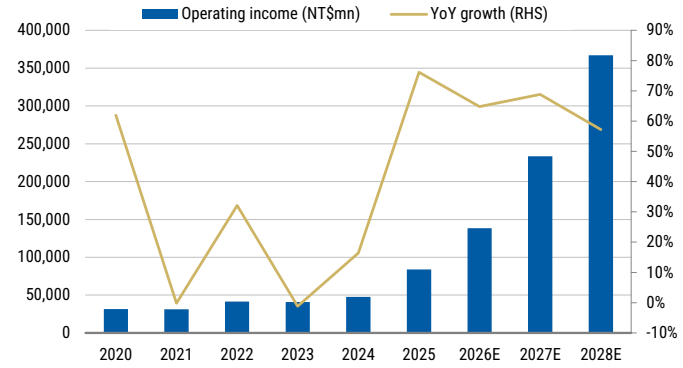
Operating profit returned to a growth trajectory from 2024 onward, growing 16% YoY in 2024 and 76% YoY in 2025, as AI-related power offerings started to scale up. Operating margin expanded faster than gross margin, supported by operating leverage and disciplined cost controls, increasing by 1.1ppt in 2024 to 11.3% and another 3.8ppt to 15.1% in 2025.

With the tax rate staying largely stable at ~20% during 2020-23 before starting to edge up in 2024 and reaching 23% in 2025, Delta's net profit growth was largely aligned with its operating profit growth trajectory. 2023 was an exception, as net profit still managed to grow 2% YoY despite a 1% YoY decline in operating profit, mainly due to higher FX gains.

The net margin trend has been largely aligned with that of operating margin in recent years. After dropping from 9.0% in 2020 to 8.5% in 2021, as lower gross margin was offset by a lower opex ratio, net margin hovered around the 8.3-8.5% level during 2021-24. Net margin was largely flat YoY in 2024 despite a 1.1ppt expansion in operating margin, mainly due to lower non-operating income and a slightly higher tax rate. We then saw a surge in 2025, with net margin expanding 2.5ppt YoY to 10.8%, benefiting from a larger contribution from higher-margin AI power offerings.

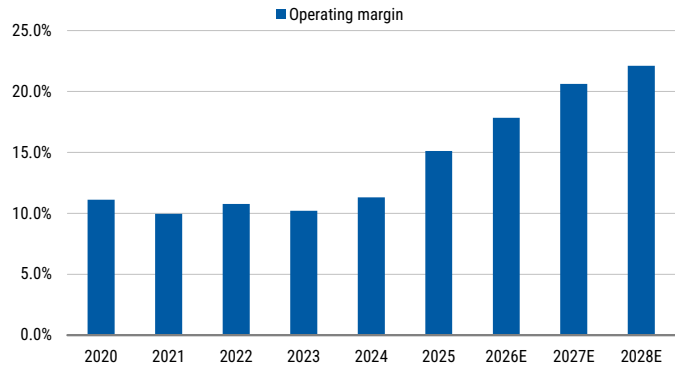
Looking ahead, we forecast net profit to grow 72% YoY in 2026e, followed by 63% YoY growth in 2027e and 56% YoY growth in 2028e, implying a 63% CAGR in 2025-28e, assuming a stable tax rate of 22%. Net margin is likely to rise accordingly, from 10.8% in 2025 to 13.3% in 2026, followed by further expansion to 14.9% in 2027 and 15.8% in 2028.

Exhibit 84: Delta: Operating income trend



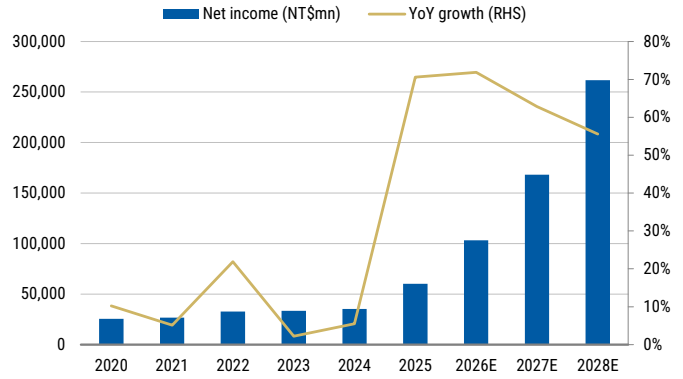
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 85: Delta: Operating margin



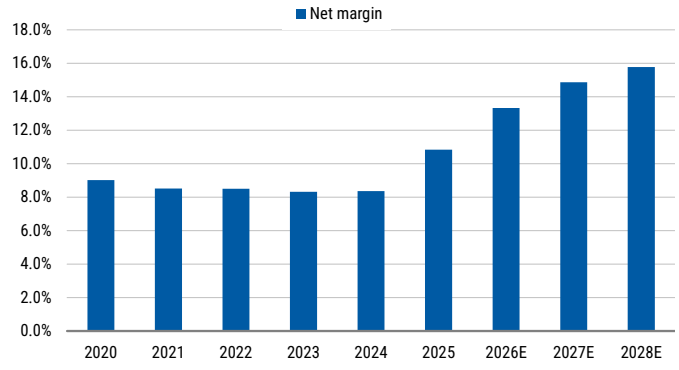
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 86: Delta: Net profit trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 87: Delta: Net margin



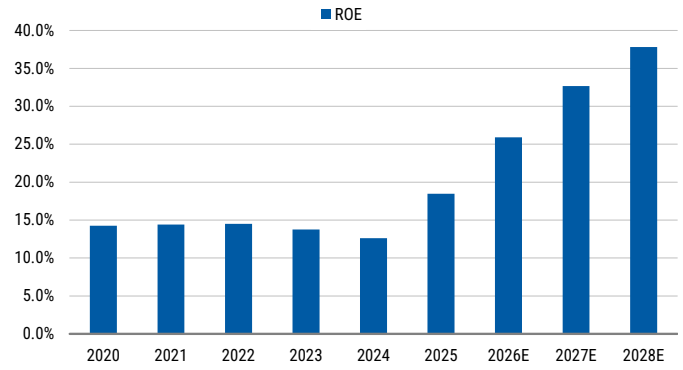
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

ROE to rise alongside margin expansion

ROE rose gradually but steadily during 2020-22, from 14.3% to 14.5%, mainly driven by improving asset turnover, offset by lower net margin. ROE edged down to 13.8% in 2023 and fell further to 12.6% in 2024 on the back of lower asset turnover, while net margin and financial leverage remained at similar levels. It then rebounded to 18.5% in 2025, supported by both improving asset turnover and net margin expansion, which was fueled by an increasingly favorable product mix and operating leverage.

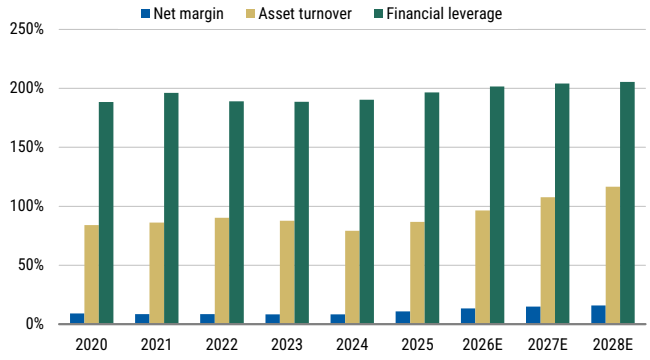
Looking ahead, we expect ROE to continue rising, reaching 25.9% in 2026, 32.7% in 2027, and 37.8% in 2028, primarily driven by continued margin expansion and improving asset turnover. Net margin is projected to expand from 10.8% in 2025 to 15.8% in 2028, supported by gross margin improvement—from 34.3% to 38.0%—and a declining operating expense ratio from 19.1% in 2025 to 15.9% in 2028.

Exhibit 88: Delta: ROE trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 89: Delta: DuPont analysis



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

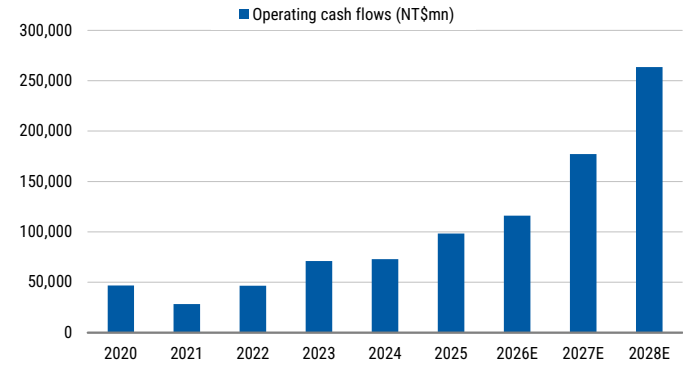
Improving balance sheet and continuous free cash flow generation

Delta has generated positive operating cash flow in recent years. Operating cash flow declined in 2021, mainly due to higher inventory buildup amid supply chain disruptions during the year. Free cash flow generation has also remained positive, while fluctuating with capital expenditures over the years. Free cash flow was exceptionally low at NT $5,292mn in 2021, as the company started to step up capex mainly for geographic capacity diversification and greater production flexibility—adding new capacity in Thailand, India, Tainan/Zhongli in Taiwan, and Wuhu, Wujiang, and Chongqing in Mainland China. Both absolute capex and the capex-to-sales ratio have trended upward since 2022, as Delta continued to add capacity for EV-related offerings and power and cooling solutions during 2023-24, with further acceleration in 2025 due to capacity tightness for AI power and cooling offerings, especially in Thailand and the US.

Looking ahead, the company aims to continue adding capacity worldwide to fulfill substantial demand for power and liquid-cooling turnkey solutions for large-scale data centers operated by major hyperscalers beyond 2027. Capacity remains tight for AI power and cooling offerings, so it continues to expand actively in China, Thailand, the US, and Tainan. Management also plans to add mass-production lines in Taoyuan, Taiwan, scheduled to be ready by 2028, in addition to the pilot-run lines in Tainan. Accordingly, we now estimate capex to rise to NT$70bn in 2026 from ~NT$46bn in 2025 and stay at a similar level in 2027 before edging down slightly to NT$60bn in 2028.

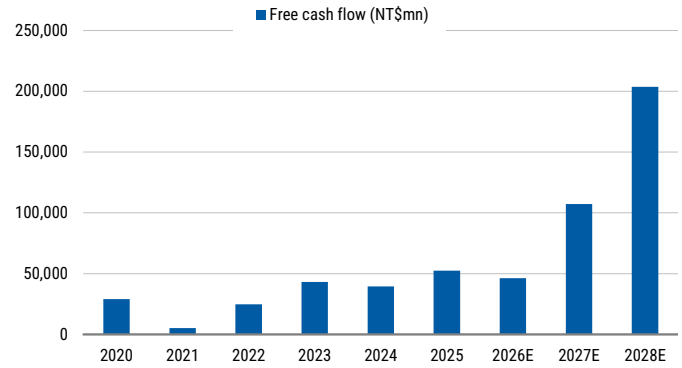
Delta's net cash position has been rising since 2021, with its net cash-to-equity ratio improving from just 2% in 2021 to 40% in 2025. We expect this ratio to continue rising, reaching 51% in 2026, 57% in 2027, and 63% in 2028, on strong free cash flow generation despite a higher total debt level amid ongoing capacity expansion.

Exhibit 90: Delta: Positive operating cash flow



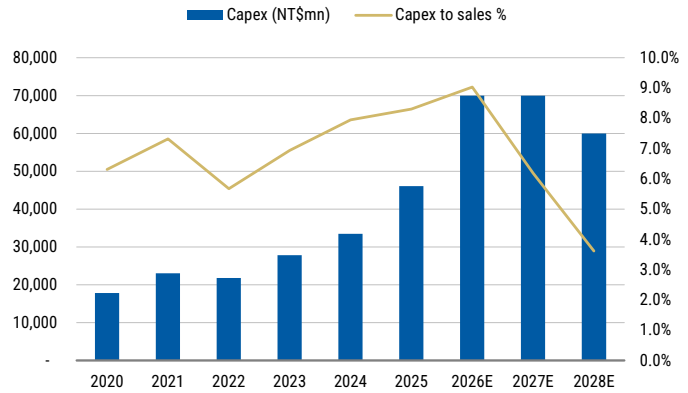
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 91: Delta: Improving free cash flow generation



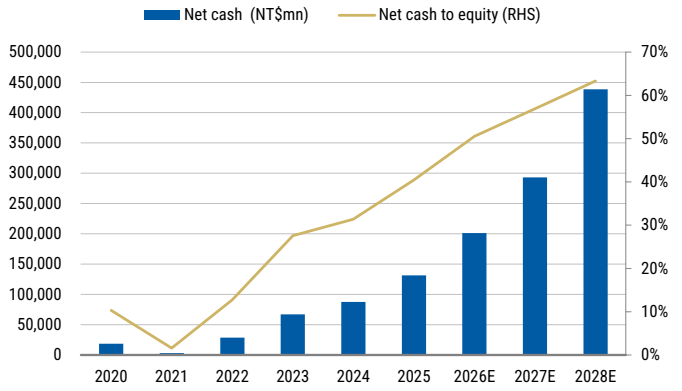
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 92: Delta: Capex and capex-to-sales ratio



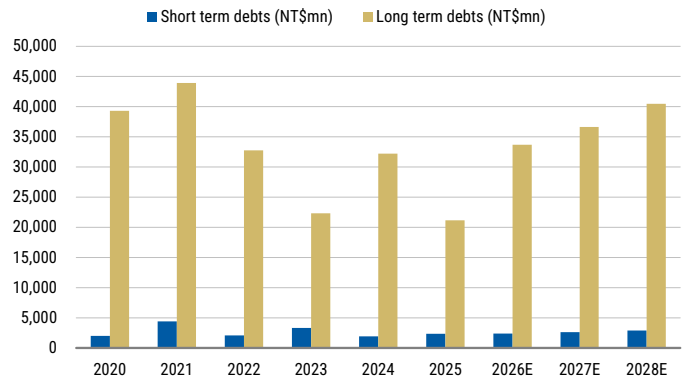
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 93: Delta: Net cash position



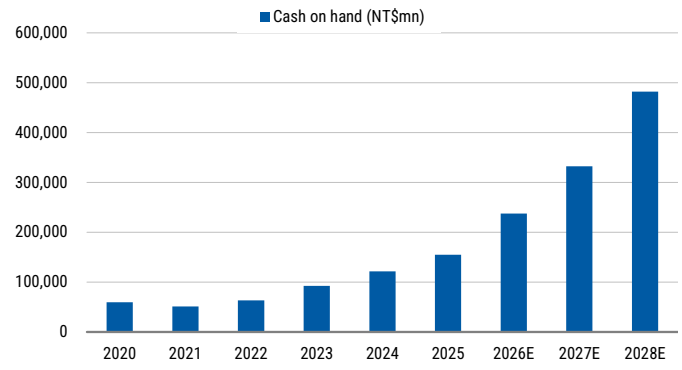
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 94: Delta: Debt position



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 95: Delta: Cash on hand



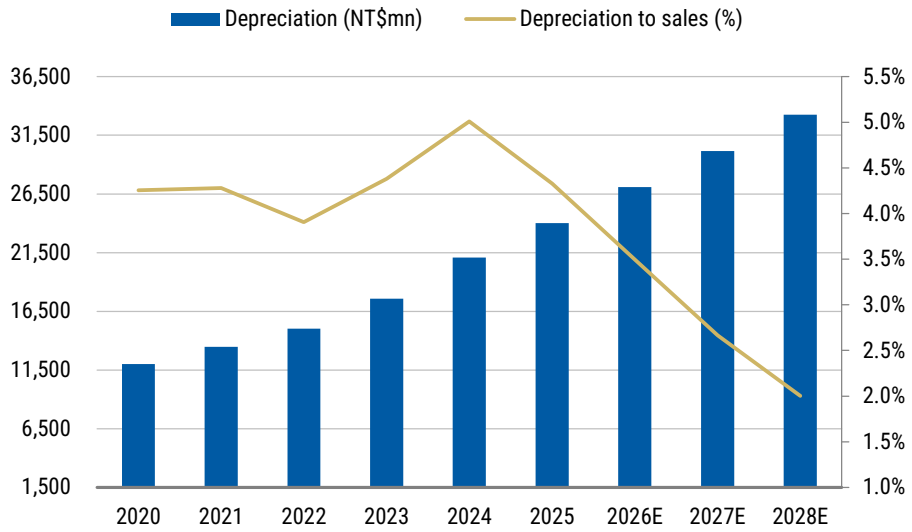
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

After edging down slightly in 2022 from a relatively high 7.3% in 2021, Delta's capex-to-sales ratio has been rising alongside its aggressive capacity expansion plans since 2022, despite an expanding revenue base. Looking ahead, as we expect capex to increase further in 2026, we project a capex-to-sales ratio of 9.0% for the year on robust revenue growth, before declining to 6.2% in 2027 and 3.6% in 2028.

The depreciation-to-sales ratio hovered around the 4-5% level during 2020-25. We

estimate the ratio will decline sequentially as the revenue base grows faster than capex and depreciation, from 4.3% in 2025 to 3.5% in 2026, 2.7% in 2027, and 2.0% in 2028.

Exhibit 96: Delta: Depreciation and depreciation to sales

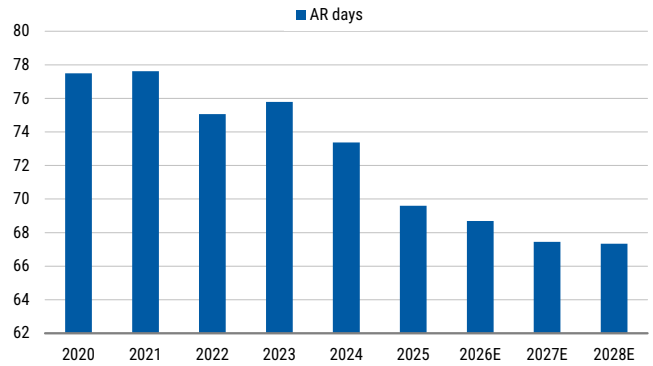


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Delta's cash conversion cycle (CCC) rose from 77 days in 2020 to a peak of 101 days in 2023 before improving to 80 days in 2025. By 2025, inventory days had increased from 80 days in 2020 to 93 days, while AP days had risen to 82 days after falling from 81 days in 2020 to 75 days in 2023. This was partly offset by improving AR days, which declined from 77 days in 2020 to 70 days in 2025.

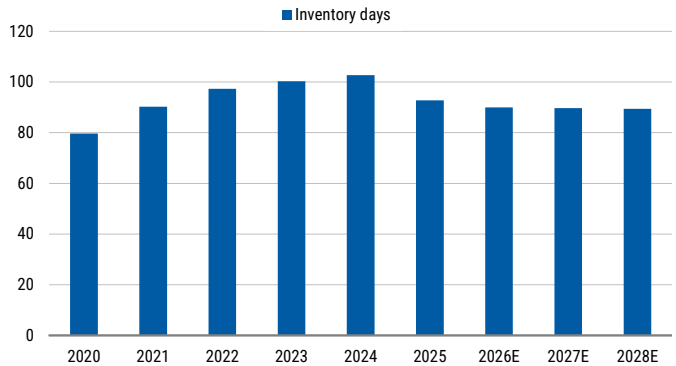
Looking ahead, we expect the overall cash conversion cycle to improve further and remain at 77 days throughout 2026-28, mainly supported by improving AR days and a modest decline in inventory days.

Exhibit 97: Delta: AR days



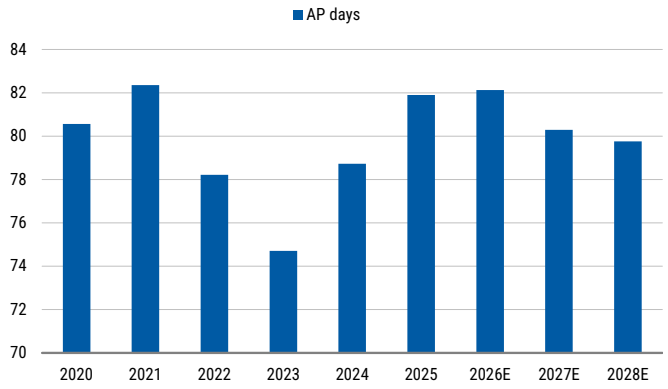
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 98: Delta: Inventory days



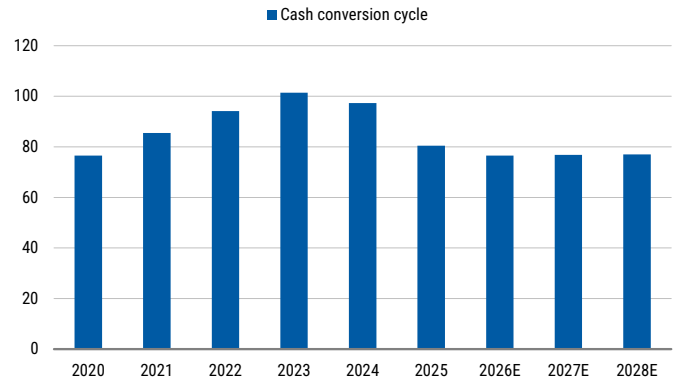
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 99: Delta: AP days



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 100: Delta: Cash conversion cycle



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Company Background – Delta

Company snapshot

Delta Electronics is a global provider of power and thermal management technologies, with a focus on energy-efficient products and integrated solutions. Founded in 1971, the company serves a broad range of applications spanning components, power and system products, thermal management, mobility, industrial and building automation, data centers, telecom, and energy infrastructure, across industries including automotive, healthcare, data centers, and power grids. By combining its capabilities in power electronics, digital technologies, and system integration, Delta aims to support customers in improving operational efficiency and advancing smart manufacturing, smart cities, and more sustainable infrastructure.

The company is headquartered in Taipei, Taiwan. Its global network consists of 293 operations, including 165 sales offices, 55 manufacturing sites, and 73 R&D centers, supporting customers worldwide.

Exhibit 101: Delta: Key milestones

Year	Key Milestones
1971	Delta Electronics, Inc. was established
1976	Achieved >US$1mn of revenue
1977	Moved plant to Taoyuan Guishan Industrial Park
1983	- Mass production of switching power supplies
1988	- Listed on the Taiwan Stock Exchange (code 2308) - Mass production of brushless DC fans - Invested in the establishment of Delta Electronics (Thailand) PCL, with an original stake of 6%
1989	- Established first Japan office in Tokyo - Established Delta Power Electronics laboratory in Raleigh, US
1992	- Inauguration of the first plant in China (Dongguan) - Established Delta Greentech (China) Co., Ltd. in Shanghai
1995	Delta Electronics (Thailand) PCL was listed on Thailand Stock Exchange
1999	- Inaugurated its new Taipei HQ - Established the Delta Power Electronics Center (DPEC) in Shanghai
2000	Established the Delta Power Electronics Science and Education Development program
2001	Inaugurated its Wuhu plant in China
2003	Acquired ASCOM Energy Systems and its plant in Slovakia
2006	Inaugurated Delta's Tainan Plant I
2008	Inaugurated its Rudrapur plant in India and Wuhu plant in China
2010	Inaugurated Chenzhou plant in China
2011	Inaugurated Shanghai R&D and Operations Center and Taoyuan R&D Center in Taiwan
2012	Mr. Yancey Hai was elected Chairman and Mr. Ping Cheng was appointed as CEO
2015	- Acquired Eltek ASA, a leading provider of telecom power, industrial power, and data center power solutions - Inaugurated Delta Americas' headquarters green building and Taoyuan Plant 5
2017	- Announced organizational adjustment to focus on Power Electronics, Automation, and Infrastructure as main business categories - Acquired a controlling stake in VIVOTEK
2019	- Acquired a controlling stake in Delta Electronics (Thailand) Public Company Limited - Acquired lighting-solutions brand Amerlux
2020	Accquired Trihedral, a Canadian SCADA and IIoT software company, to strengthen industrial automation and system-integration businesses
2021	Acquired March Networks, a global video-surveillance provider, and Universal Instruments, a leader in precision automation solutions for electronics manufacturing
2022	- Inaugurated a new facility at the Automotive Campus in Helmond, the Netherlands - Purchased land and a factory in Plano, Texas, United States
2023	Acquired TB&C, a leading provider of automotive high-voltage hybrid components
2024	- Mr. Ping Cheng selected as Chairman while concurrently serves as CEO - Adjusted business categories into Power Electronics, Mobility, Automation, and Infrastructure
2025	- Inaugurated new North East Asia HQ in Tokyo - Acquired ACT Genomics and Noda RF Technologies

Source: Company data, Morgan Stanley Research

Product offerings

Delta's key offerings can be categorized into four key areas: Power Electronics, Infrastructure, Mobility, and Automation.

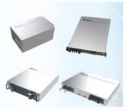
1. Power Electronics

Delta operates in switching power supplies, thermal management, and passive components, supplying products for ICT, cloud computing, EVs, and industrial and commercial applications. The company frames its business around energy efficiency and

sustainability and now also offers grid-to-chip power and thermal management solutions for data centers.

- Components: Includes essential passive and magnetic components used in portable devices, AI cloud computing, automotive electronics, and IoT.
- Power and System: Contains power products and systems for AI, cloud computing, IoT, client devices, and industrial, medical, lighting, e-mobility, and electrical-appliance applications.
- Fans and Thermal Management: This mainly covers air and liquid cooling solutions for AI cloud and edge computing, EVs, as well as industrial, commercial, and residential sectors. Key products includes DC and EC fans, thermal modules, liquid cooling cold plates, and coolant distribution units, etc.

Exhibit 102: Delta: Key offerings for the power electronics segment

POWER ELECTRONICS							
Components							
Inductors	RF Inductors	Standard Power Module (DC/DC Converter & AC/DC Module)	Transformers	Networking	EMI Filters	Solenoids	Current Sensing Resistor
				Wire-Wound Common Mode Choke			
Power and System							
Embedded Power	Adapters	Industrial Power	Medical power	Industrial Battery Charging	USB Socket Outlets	High Voltage Power	Hydrogen Power
				Modular Charging System			
Fans and Thermal Management							
Chip Level Cooling Solution	DC Brushless Fans and Blowers	Thermal Module	Cold Plate Loop	In-rack CDU	Rack Manifold	Electical Pump	In-row CDU
Communication Equipment Cooling	Energy Infrastructure and Industrial Automation Cooling	Portable Device and Equipment Cooling	Infrastructure Ventilation	Automotive Thermal			
DC Brushless Fans and Blowers	DC Brushless Fans and Blowers	Laptop cooling	DC Brushless Fans and Blowers				

Source: Company data, Morgan Stanley Research

2. Infrastructure

Delta supplies ICT and energy infrastructure to support demand related to AI, smart cities, and IoT. Its ICT infrastructure solutions target data centers and telecom networks, while its energy infrastructure offerings integrate AIoT, power conversion, and microgrid control across energy generation, storage, and management.

- ICT Infrastructure: It mainly provides infrastructure for AI computing and 5G

communications under this category, ranging from telecom cell sites to data centers. The portfolio includes power, air and liquid cooling, networking equipment, and management software.

- Energy Infrastructure: The company offers renewable energy, AIoT-based energy management, and power conversion technologies for low-carbon transformation. Solutions include solar inverters, wind power converters, energy storage, EV charging, and management platforms for smart grids. It also supplies semiconductor high-voltage power supplies, automated testing instruments, and variable-frequency drives for industrial processes.
- Display and Visualization: Delta's display technologies serve control rooms, public information systems, projection mapping, and exhibition or event spaces. Offerings range from large-scale and 8K projection systems to integrated video-wall solutions and can be customized for mission-critical operations.

Exhibit 103: Delta: Key offerings for the infrastructure segment

INFRASTRUCTURE							
ICT Infrastructure							
Uninterruptible Power Supplies (UPS)	Battery Backup System (BBU)	800 VDC In-Row Power	Power Distribution Unit (PDU)	Rack-Mount Remote Power Panel (rRPP)	Liquid and Air Cooling	Modular Rack & Rack Mount Accessories	Lithium-ion UPS Battery System
Ethernet Switches	Indoor/Outdoor Telecom Power Systems	Renewable Hybrid Power Systems	Small Cell Power Systems	Li-ion Battery Backup Systems	Campus/Enterprise Networks		
Energy Infrastructure							
EV Charger	Power Conversion System (PCS)	All-in-One ESS solution	Distributed ESS Solutions	Solar Inverter	Wind Power Converter	Solid State Transformer Platforms	Medium Voltage Drives
High Voltage Power	DC EV Charger ATS	Power Supply ATS	Solid Oxide Fuel Cell (SOFC)	Hydrogen Energy Solution	Active Power Filter (APF) / Static Var Generator (SVG)		
Display and Visualization							
Business and Education Multimedia Projector	High Performance Projector	DLP, LED and LCD Video Wall	Wireless Collaboration and Signage	Vivitek			
				vivitek			

Source: Company data, Morgan Stanley Research

3. Mobility

Delta applies its power electronics and motor technologies to smart mobility, supplying

power systems, drive technologies, and integrated powertrain solutions. It works with automakers to develop electric vehicles, positioning the business as part of the low-carbon transition and sustainable urban mobility.

- EV Powertrain System: The company's powertrain portfolio for EV makers includes On-board Charging Modules (OBCM), DC/DC converters, power electronics combos, and traction inverters, along with X-in-One integrated solutions.

Exhibit 104: Delta: Key offerings for the mobility segment

MOBILITY						
EV Powertrain System						
EV Power Electronics	Traction Products	x-EV Power Management	In-Vehicle Network and AI Driving Computing	Automotive Thermal	Low Voltage Powertrain	EV Charging

Source: Company data, Morgan Stanley Research

4. Automation

Delta provides automation solutions for industrial and building applications through integrated software and hardware. Its industrial automation offerings support smart production and operational efficiency, while its building automation solutions target energy-efficient, secure, and sustainable spaces.

- Industrial Automation: This segment provides automation solutions across industries by utilizing AI, big data, IoT, and digitized manufacturing models. Relevant offerings support diverse production needs, including customized, small-volume and high-variety production, as well as mass production.
- Building Automation: Delta's Building Automation Solutions use an IoT architecture and integrate technologies from five brands: Amerlux lighting, Delta Controls and LOYTEC IP-based building control, and March Networks and VIVOTEK surveillance. The management platform enables operators to manage energy use, occupant comfort, and sustainability goals through a single interface.

Exhibit 105: Delta: Key offerings for the automation segment

AUTOMATION							
Industrial Automation							
Inverters - AC Motor Drives	Drive System for Motion Control Solutions	Motors	Servo Systems - AC Servo Motors and Drives	CNC Solution	PC-Based Motion Control Solution	Linear Motion	Embedded Motion Controller
Industrial Ethernet Solution	Temperature Controllers	High Voltage Power	SCARA Robot	Electric Servo Press			
Building Automation							
Delta Controls	LOYTEC	Indoor Air Quality Monitor	LED Lighting - Industrial/ Road/Fishing/Commercial	Smart Surveillance - VIVOTEK	Healthy Lighting		

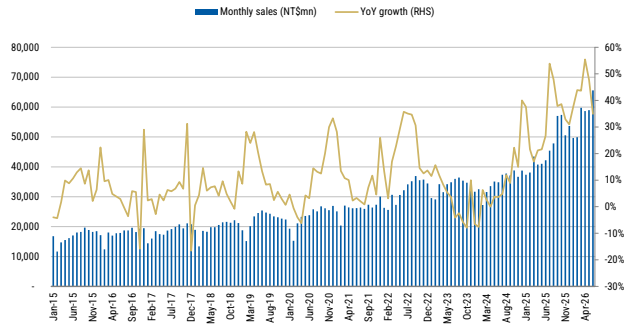
Source: Company data, Morgan Stanley Research

Monthly sales vs. share price

Delta's monthly sales show that seasonality has been an important factor over the years. Following a substantial increase in monthly sales since 2024, traditional seasonal patterns have become less pronounced as revenue from AI-related offerings has started to outweigh them, while liquid-cooling-related offerings have added a new layer of monthly sales volatility. The correlation between the company's monthly sales and share price movements has remained high, at ~87% since 2023.

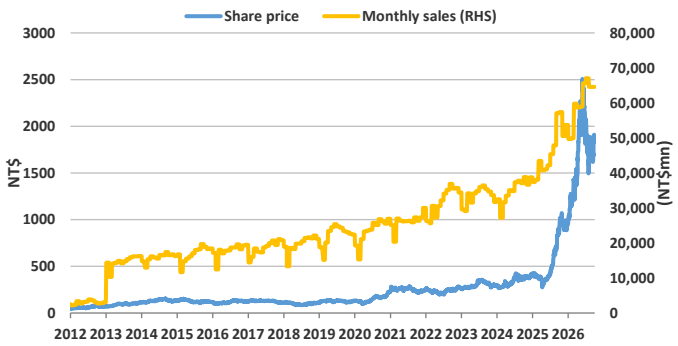
The company reported August monthly sales of NT$64,586mn (-4% MoM/+35% YoY). We saw the Power Electronics segment decline sequentially from a high base, offset by growth in Automation and slight growth in the Infrastructure segment.

Exhibit 106: Delta: Monthly sales vs. YoY sales growth



Source: TEJ, Morgan Stanley Research

Exhibit 107: Delta: Monthly sales vs. share price



Source: TEJ, Morgan Stanley Research

Key management & shareholding structure

Delta's board currently consists of eight directors and five independent non-executive directors. Its founder, Chung-Hua Cheng (Bruce Cheng), and his sons, Ping Cheng and An Cheng, together owned a 7% stake as of March 2026. Excluding the founder and his family, the management team held a stake of ~0.1% as of March 2026.

The FINI holding percentage as of July 14, 2026, was 63%, versus a previous peak of 78% in February 2011 and a historical average (since 2010) of 69%.

Exhibit 108: Delta: Shareholding structure

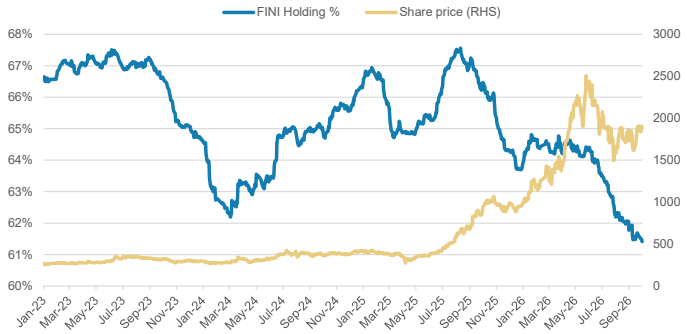
Shareholders	%
Dejico International LTD	10.30%
Deltron Holding LTD	8.40%
New Labor Retirement Pension Fund	3.75%
Chung-Hua Cheng (Bruce CH Cheng)*	3.15%
Government of Singapore-GOS-EFM C	2.28%
Ping Cheng*	2.14%
An Cheng (Victor Cheng)*	1.94%
Yuanta/P-shares Taiwan Top 50 ETF	1.43%
Labor Retirement Reserve Fund (The Old Fund)	1.42%
Labor Insurance Fund	1.32%
Total	**36.13%**

Source: Company data, Morgan Stanley Research. Note: Data as of March 2026. (*) are shown as insiders.

Exhibit 109: Delta: FINI holdings vs. share price since 2010

Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 110: Delta: FINI holdings vs. share price since 2023



Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 111:

Delta: Management profiles

Name	Title	Education	Work Experience
Bruce C.H. Cheng	Founder and Honorary Chairman Chairman, Delta Electronics Foundation	Bachelor of Electrical Engineering — National Cheng Kung University	- 2012–present — Honorary Chairman, Delta Electronics - 1990–present — Founded and chairs the Delta Electronics Foundation - 1971–2012 — Founder and Chairman, Delta Electronics - Before 1971 — Worked at TRW and Air Asia Co.
Ping Cheng	Chairman and CEO	Bachelor of Business Administration — California State University, East Bay	- Joined Delta in 1988 - 2024–present — Chairman - 2012–present — CEO of Delta Electronics - 2010–2012 — Corporate Senior Vice President and President of China Region - 2008–2010 — Senior Vice President & General Manager of Power Supply Business Group - 1998–2008 — Deputy General Manager of Component Business Group (Delta China); Executive Vice President of China Region
Mark Ko	Vice Chairman	Bachelor of Control Engineering — National Chiao Tung University	- Joined Delta in 1988 - 2012–present — Vice Chairman of Delta Electronics - 2004–2012 — President and Chief Operating Officer of Delta Group - 1994–2004 — General Manager of Power System Business Group
Simon Chang	Board Member President and COO	EMBA — Chung Yuan Christian University	- Joined Delta in 1981 - 2017–present — Board Member; President and Chief Operating Officer, Delta Electronics - Before 2017 — Senior Vice President and General Manager of Industrial Automation Business Group - From 1995 — Director of Industrial Automation Division
Victor Cheng	Board Member CEO of Delta Electronics (Thailand) PCL.	-Master of Electrical Engineering — Santa Clara University -Bachelor of Electrical Engineering — Santa Clara University	- Joined Delta in 1993 - 2024–present — Board Member of Delta; CEO of Delta Electronics (Thailand) PCL. - 2023–2024 — Executive Vice President of Infrastructure Business - 2017–2023 — General Manager of Information and Communications Technology Business Group - 2014–2017 — General Manager of Power System Business Group - 2002–2019 — President, then Chairman of Delta Networks - 1999–2002 — Head of Video Display Business Unit - 1993–1999 — International Sales Director to Assistant General Manager of Video Display Business Unit
Shan-Shan Guo	Board Member Chief Brand Officer Vice Chairman, Delta Electronics Foundation	- Master of Marketing and Communications — University of Michigan - Bachelor of Foreign Languages and Literatures — National Taiwan University	- Joined Delta in 2010 - 2011–present — Chief Brand Officer of Delta
Johnson Lee	Executive Vice President Infrastructure Business (IFB)	- EMBA — National Taiwan University - Bachelor of Industrial Engineering and Systems Management — Feng Chia University	- Joined Delta in 1985 - 2024–present — Executive Vice President of Infrastructure Business - 2012–2017 — Chief Operating Officer of Delta Electronics - 2004–2012 — General Manager of Power Supplies Business Group - Earlier — Procurement manager; General Manager of materials management
Ted Shyy	Executive Vice President Power Electronics Business (PEB)	Master in Materials Science and Engineering — National Cheng Kung University	- Joined Delta in 1988 - 2024–present — Executive Vice President of Power Electronics Business - 2014–present — Chief Executive Officer of Cyntec (concurrent role) - 2011–2014 — General Manager of Delta Components Business Group - 2003–2014 — General Manager of Cyntec - 1991–2003 — Rotation in Research and Development, production and marketing, under Cyntec, a joint venture between Delta and Japan's SUSUMU
James Tang	Executive Vice President Mobility Business (EVB)	- EMBA — National Taiwan University - Bachelor of Management Science — National Cheng Kung University	- Joined Delta in 1988 - 2024–present — Executive Vice President of Mobility Business; continues as General Manager of Electric Vehicle Solutions Business Group - 2017–present — General Manager of Electric Vehicle Solutions Business Group - 2008–2016 — General Manager across Power Supplies, Consumer and Lighting Solutions, and Embedded Power Supplies business groups - 1988–2008 — Procurement; Materials Manager; Director of Personal Computer Business Group
Jimmy Yiin	Executive Vice President Global Business Operations (GBO)	Master of Business Administration — National Taiwan University and Fudan University	- Joined Delta in 1992 - 2023–present — Executive Vice President of Global Business Operations - 2019–2022 — General Manager of Power System Business Group - 2014–2019 — Vice Chairman of Delta Greentech (China); General Manager of Delta Electronics (Shanghai) - 2008–2013 — General Manager of Delta Electronics (Jiangsu) - 1999–2007 — Purchasing manager/materials section chief of Delta Dongguan and Delta Networks
Rock Huang	Executive Vice President Global Manufacturing	Master of Industrial Engineering and Management — National Taipei University of Technology	- Joined Delta in 1987; ~40 years of experience in the manufacturing field - 2026–present — Executive Vice President of Global Manufacturing - 2020–2026 — President of Global Manufacturing - 2019–2020 — President of Delta China - 2018–2019 — Executive Vice President of Delta China - 2007–2018 — General Manager of Southern China, leading Dongguan team

Source: Company data, Morgan Stanley Research

Risk Reward – Lite-On Technology (2301.TW)

Stronger ASIC server power supply but fair risk-reward

PRICE TARGET NT$315.00

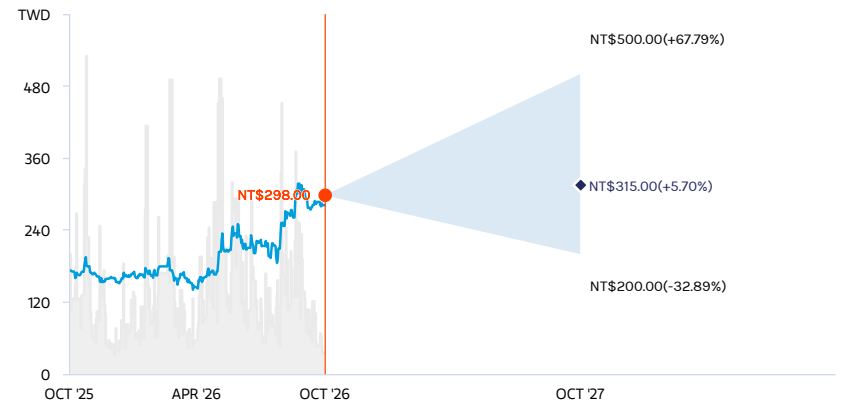
Base case: P/E multiple of 17x based on average 2027-28 EPS. We believe our target multiple is justified by the 47% earnings CAGR we estimate for 2025-28, implying a PEG of 0.4, versus the stock's 16x P/E and 2% earnings CAGR during 2022-25, which implied a PEG of 7.2.

Consensus Price Target Distribution NT$176.00 NT$300.50 MS PT NT$400.00

Source: Refinitiv, Morgan Stanley Research

◆ Mean ◆ Morgan Stanley Estimates

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- We expect Lite-On Tech to benefit from strong AI server power demand in 2026-27, with the company gaining more traction on ASIC platforms.
- We are mindful of weakness in consumer/IT demand amid rising input costs, which could weigh on Lite-On Tech's profit in 2026-27.
- Liquid-cooling sidecar CDU system shipments may take more time to ramp up.

Consensus Rating Distribution

77% Overweight
23% Equal-weight
0% Underweight
● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian:	*Negative*
New Data Era:	*Positive*
Technology Diffusion:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$500.00

27x average 2027-28e P/E

Better-than-expected margin expansion: Operating efficiency and accelerating production automation drive margin expansion, while new-business revenue comes in earlier than expected. Growth in cloud computing and automotive electronics is stronger than expected.

BASE CASE NT$315.00

17x average 2027-28e P/E

AI server projects progress, but some business uncertainties remain: We expect margin improvement from a better product mix as AI server power supply projects ramp up. However, we are mindful of weakness in consumer/IT demand amid rising input costs.

BEAR CASE NT$200.00

11x average 2027-28e P/E

Operating margin contraction: Operating efficiency improves less than expected, while pricing competition intensifies amid macro volatility; automotive electronics take longer to ramp up.

Risk Reward – Lite-On Technology (2301.TW)

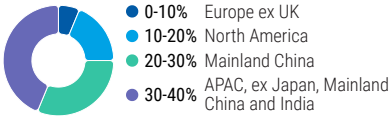
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Opto-electronics OPM (%)	8.3	7.9	7.5	7.5
Information Technologies OPM (%)	9.2	7.7	7.8	8.6
Opto-electronics revenue YoY (%)	0.2	8.0	1.7	2.3
Information Technologies revenue YoY (%)	7.7	(5.2)	0.5	(0.2)

INVESTMENT DRIVERS

- LED adoption (outdoor lighting/auto)
- Margin expansion
- Raw material prices, particularly for steel and copper

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger optical business growth
- Stronger data center/server demand
- PC/NB demand growth
- Downtrend in raw material prices

RISKS TO DOWNSIDE

- Macro demand weakness
- Weaker data center/server demand
- PC/NB demand slowdown
- Uptrend in raw material prices

OWNERSHIP POSITIONING

Inst. Owners, % Active 65.2%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	210,118	224,829 / 219,960	229,691
EBITDA (NT$, mn)	31,349	33,549 / 34,893	38,115
Net income (NT$, mn)	23,541	25,187 / 24,910	26,835
EPS (NT$)	10.30	11.07 / 10.88	11.79

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Financial Summary – Lite-On

Exhibit 112: Financial Summary – Lite-On

Consolidated Income Statement

NT$ mn	2024	2025	2026E	2027E	2028E
Net sales	**137,134**	**166,085**	**224,829**	**297,964**	**357,670**
COGS	-107,504	-128,046	-168,703	-219,712	-259,785
Gross profit	**29,630**	**38,039**	**56,126**	**78,252**	**97,885**
Operating expenses	-16,648	-21,301	-26,044	-33,189	-37,645
- Promotion	-5,184	-7,629	-6,902	-8,782	-10,247
- ADM	-3,990	-4,965	-7,384	-9,340	-10,162
- R&D	-7,476	-8,721	-11,787	-15,037	-17,200
Operating income	**12,981**	**16,738**	**30,082**	**45,063**	**60,240**
Non-operating income	2,653	2,975	2,876	2,500	2,500
Interest income	1,491	1,119	799	800	800
Investment income	2,808	-956	1,436	1,200	1,200
Disposal of investment	75	-537	-77	0	0
Exchange gain	2,808	1,091	1,390	0	0
Other	-4,528	2,258	-672	500	500
Pre-tax income	**15,635**	**19,713**	**32,958**	**47,563**	**62,740**
Income tax	-3,674	-4,601	-7,767	-11,227	-14,793
Minority	19	-4	3	0	0
Reported net income	**11,942**	**15,116**	**25,187**	**36,337**	**47,947**
EPS (NT$)	**5.21**	**6.64**	**11.07**	**15.97**	**21.07**

Consolidated Cash Flow Statement

NT$ mn	2024	2025	2026E	2027E	2028E
Cash flow from Operations	**12,411**	**14,533**	**34,885**	**43,140**	**51,224**
Net profits	11,942	15,116	25,187	36,337	47,947
Depreciation + Amortization	4,047	3,668	3,994	4,065	4,137
Inv. Gain/Loss (Equity)	-15	-86	0	0	0
Disposal of Investment	9	568	0	0	0
Working Capital Change	-3,886	-2,099	-12,395	-12,092	-10,109
Other adjustments	313	-2,635	18,098	14,830	9,248
Cash flow from Investing	**-4,843**	**-7,395**	**-13,287**	**-8,993**	**-11,919**
Capex	-3,283	-6,795	-13,287	-8,993	-11,919
Change of LT investment	-2,234	-43	0	0	0
Change of ST investment	1,453	-417	0	0	0
Other adjustments	-779	-140	0	0	0
Cash flow from financing	**-3,363**	**-23,099**	**-19,866**	**-28,659**	**-37,818**
Increase in L/T debt	0	0	0	0	0
Increase in S/T debt	7,584	-10,116	0	0	0
Cash Dividend Paid	-10,372	-10,297	-19,866	-28,660	-37,818
Issuance of stock	0	0	0	0	0
Other adjustments	-574	-2,686	0	0	0
Exchange rate adjustment	3,736	-3,748	0	0	0
Net change in cash	**7,942**	**-19,709**	**1,732**	**5,487**	**1,487**

Consolidated Balance Sheet

NT$ mn	2024	2025	2026E	2027E	2028E
Cash	100,684	80,974	82,706	88,193	89,680
Mkt Securities	933	1,009	1,009	1,009	1,009
AR/NR	37,182	42,318	60,960	80,790	96,978
Inventory	26,817	34,262	42,083	54,807	64,804
Others	3,501	5,458	5,458	5,458	5,458
Current Assets	**169,117**	**164,021**	**192,216**	**230,257**	**257,928**
Long-term investments	8,195	7,748	7,748	7,748	7,748
Fixed assets	18,776	24,752	34,495	39,873	48,104
Deferred Asset	1,628	2,246	2,246	2,246	2,246
Intangible Assets	3,156	3,060	2,610	2,160	1,710
Other assets	3,811	3,860	3,860	3,860	3,860
Total Assets	**204,683**	**205,687**	**243,175**	**286,144**	**321,597**
S/T borrowings	30,187	20,029	20,029	20,029	20,029
AP/NP	43,126	53,608	67,676	88,138	104,213
Other ST liabilities	34,556	35,960	54,059	68,888	78,137
Total Current Liabilities	**107,869**	**109,597**	**141,764**	**177,056**	**202,379**
LT debt(include CB)	3,000	3,000	3,000	3,000	3,000
Other LT liabilities	2,903	3,111	3,111	3,111	3,111
Total Liabilities	**113,772**	**115,709**	**147,875**	**183,167**	**208,491**
Common shares	23,473	23,168	23,168	23,168	23,168
Retained Earnings	46,277	42,393	47,714	55,390	65,519
Other shareholders' equity	21,161	24,418	24,418	24,418	24,418
Total Equity	**90,911**	**89,979**	**95,300**	**102,976**	**113,105**
Total Liab. & Shldr's Equity	**204,683**	**205,687**	**243,175**	**286,143**	**321,596**

Consolidated Financial Ratios

	2024	2025	2026E	2027E	2028E
Margins					
Gross margin	21.6%	22.9%	25.0%	26.3%	27.4%
Operating margin	9.5%	10.1%	13.4%	15.1%	16.8%
Pretax margin	11.4%	11.9%	14.7%	16.0%	17.5%
Net margin	8.7%	9.1%	11.2%	12.2%	13.4%
YoY growth					
Sales	-7.6%	21.1%	35.4%	32.5%	20.0%
Operating profits	-10.6%	28.9%	79.7%	49.8%	33.7%
Pretax profits	-14.9%	26.1%	67.2%	44.3%	31.9%
Net profits	-18.0%	26.6%	66.6%	44.3%	32.0%
EPS	-18.1%	27.5%	66.6%	44.3%	32.0%
Net Debt/Equity	-75.3%	-65.5%	-63.7%	-64.3%	-59.8%
Liabilities/Equity	125.1%	128.6%	155.2%	177.9%	184.3%
Liabilities/Assets	55.6%	56.3%	60.8%	64.0%	64.8%
ROAE	13.6%	16.7%	27.2%	36.7%	44.4%
ROAA	6.1%	7.4%	11.2%	13.7%	15.8%
AR/NR Turnover (days)	92.2	87.4	83.8	86.8	90.7
AP/NP Turnover (days)	142.7	137.9	131.2	129.4	135.1
Inventory Turnover (days)	89.3	87.1	82.6	80.5	84.0
Cash conversion cycle (days)	38.9	36.5	35.2	37.9	39.6

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 113: Lite-on: Quarterly Financials

NT$ mn	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	2025	2026E	2027E
Net sales	**36,417**	**40,421**	**44,891**	**44,356**	**43,407**	**52,703**	**59,526**	**69,193**	**166,085**	**224,829**	**297,964**
COGS	-28,195	-31,479	-33,648	-34,724	-34,004	-38,355	-44,867	-51,477	-128,046	-168,703	-219,712
Gross profit	**8,222**	**8,942**	**11,243**	**9,633**	**9,403**	**14,348**	**14,659**	**17,716**	**38,039**	**56,126**	**78,252**
Operating expenses	-4,546	-5,215	-6,560	-4,980	-5,304	-6,146	-6,851	-7,743	-21,301	-26,044	-33,189
- Promotion	-1,532	-2,051	-2,535	-1,512	-1,500	-1,623	-1,786	-1,993	-7,629	-6,902	-8,782
- ADM	-998	-1,048	-1,792	-1,127	-1,457	-1,783	-1,964	-2,180	-4,965	-7,384	-9,340
- R&D	-2,023	-2,110	-2,244	-2,345	-2,341	-2,787	-3,095	-3,563	-8,721	-11,787	-15,037
Operating income	**3,675**	**3,727**	**4,683**	**4,653**	**4,099**	**8,203**	**7,807**	**9,973**	**16,738**	**30,082**	**45,063**
Non-operating income	994	536	886	560	871	1,166	445	395	2,975	2,876	2,500
Net interest income	424	323	184	188	214	195	195	195	1,119	799	800
Net investment income	596	-3,235	1,389	293	99	1,237	50	50	-956	1,436	1,200
Net gain of investment dispos	137	-702	23	5	1	-78	0	0	-537	-77	0
Exchange gain	379	2,540	-751	-122	284	1,106	0	0	1,091	1,390	0
Others	-542	1,609	41	195	272	-1,295	200	150	2,258	-672	500
Pre-tax income	4,669	4,262	5,569	5,213	4,970	9,368	8,252	10,368	19,713	32,958	47,563
Income tax	-1,214	-1,108	-924	-1,355	-1,195	-2,248	-1,939	-2,385	-4,601	-7,767	-11,227
Net income	**3,456**	**3,155**	**4,645**	**3,860**	**3,775**	**7,116**	**6,313**	**7,983**	**15,116**	**25,187**	**36,337**
EPS (NT$)	**1.51**	**1.39**	**2.05**	**1.70**	**1.66**	**3.14**	**2.77**	**3.51**	**6.64**	**11.07**	**15.97**
Margins											
Gross margin	22.6%	22.1%	25.0%	21.7%	21.7%	27.2%	24.6%	25.6%	22.9%	25.0%	26.3%
Operating margin	10.1%	9.2%	10.4%	10.5%	9.4%	15.6%	13.1%	14.4%	10.1%	13.4%	15.1%
Pre-tax margin	12.8%	10.5%	12.4%	11.8%	11.4%	17.8%	13.9%	15.0%	11.9%	14.7%	16.0%
Net margin	9.5%	7.8%	10.3%	8.7%	8.7%	13.5%	10.6%	11.5%	9.1%	11.2%	12.2%
QoQ Growth											
Sales	-4.9%	11.0%	11.1%	-1.2%	-2.1%	21.4%	12.9%	16.2%			
Gross profit	1.0%	8.8%	25.7%	-14.3%	-2.4%	52.6%	2.2%	20.9%			
Operating profit	5.1%	1.4%	25.7%	-0.6%	-11.9%	100.1%	-4.8%	27.7%			
Pre-tax profit	17.1%	-8.7%	30.7%	-6.4%	-4.7%	88.5%	-11.9%	25.6%			
Net profit	13.3%	-8.7%	47.2%	-16.9%	-2.2%	88.5%	-11.3%	26.5%			
YoY Growth											
Sales	26.6%	21.4%	22.1%	15.8%	19.2%	30.4%	32.6%	56.0%	21.1%	35.4%	32.5%
Gross profit	40.6%	21.0%	36.3%	18.3%	14.4%	60.5%	30.4%	83.9%	28.4%	47.5%	39.4%
Operating profit	64.8%	13.6%	19.3%	33.1%	11.5%	120.1%	66.7%	114.3%	28.9%	79.7%	49.8%
Pre-tax profit	49.2%	4.4%	25.6%	30.7%	6.4%	119.8%	48.2%	98.9%	26.1%	67.2%	44.3%
Net profit	44.7%	1.3%	37.0%	26.6%	9.3%	125.5%	35.9%	106.8%	26.6%	66.6%	44.3%

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

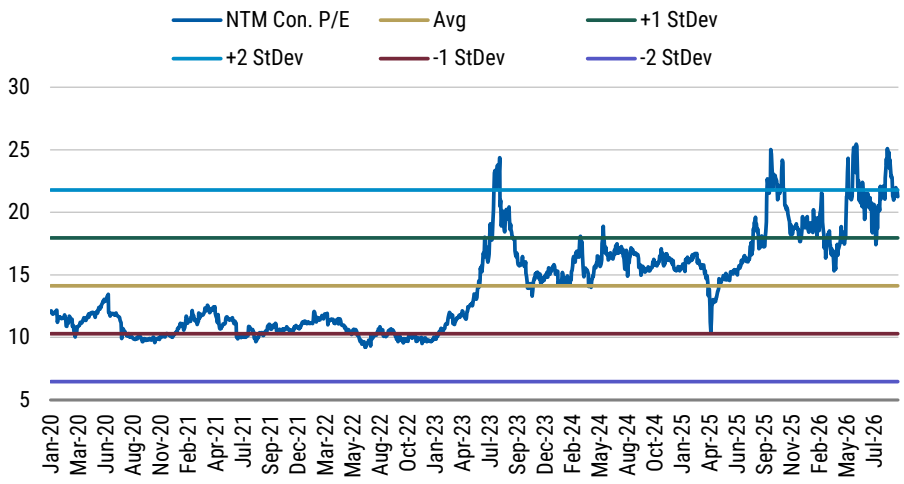
Valuation Methodology – Lite-On

Our price target (base case scenario value) is NT$315, derived from 17x 2027-28e P/E: We use price/earnings as our valuation methodology because we believe it appropriately captures Lite-On's earnings growth potential. We believe our target P/E multiple of 17x is justified by an earnings CAGR of 47% over 2025-28, implying a PEG of 0.4. This is lower than the average PEG of 7.2 in 2022-25, when the stock was trading at an average P/E of 16x for a 2% earnings CAGR.

Bull case scenario value – NT$500: We apply a 27x P/E multiple to our average base case estimate for 2027-28. This scenario assumes better-than-expected margin expansion, driven by improved operating efficiency and accelerating production automation, while new business revenue comes in earlier than expected. Growth in cloud computing and automotive electronics is stronger than expected.

Bear case scenario value – NT$200: We apply an 11x P/E multiple to our average base case estimate for 2027-28. This scenario assumes weaker-than-expected improvement in operating efficiency, while pricing competition intensifies amid macro volatility; automotive electronics take longer to ramp up.

Exhibit 114: Lite-on: NTM P/E



Source: FactSet consensus estimates, Morgan Stanley Research

Global Footprint – Lite-On

More capacity expansion to come

Lite-On's production footprint spans the globe, with 15 sites in total, although it remains heavily weighted toward China. The company has been expanding aggressively, both to meet strong demand for AI power supplies and to support its strategy of geographic risk diversification and proximity to customers.

Looking at its expansion plans in recent years, the company has continued to demonstrate its commitment to diversifying production away from China and increasing proximity to customers. It completed its Thailand plant expansion and announced the second-phase expansion of its Vietnam facility in 2021. It then established automotive electronics and cloud power production lines in Dallas, Texas, in 2H22, along with the official launch of the second phase of its Hai Phong plant in Vietnam, which focuses on networking and consumer products. In 2023, it began expanding its second Kaohsiung plant and announced additional capacity expansion in Vietnam. In 2024, it committed a further US $120mn to establish a wholly owned subsidiary in Vietnam and began progressively shifting production of high-end power products to that facility. In 2025, the company aimed to double capacity as demand for AI server power supplies and battery backup units (BBUs) surged, focusing primarily on capacity expansion in the US, Kaohsiung, and Vietnam.

For 2026, the company expects capacity allocation of ~40% in China, 27% in Vietnam, 20% in Taiwan, and 10% in the US, with the remaining non-China footprint (i.e., Mexico, Brazil, and Thailand) accounting for ~3% of total capacity. It has also announced investments of US$919mn and US$149mn in its US and Vietnam subsidiaries, respectively, to continue expanding power-related capacity, mainly for PSUs. The capex budget for 2026 has also been raised to ~NT$13bn, versus ~NT$7bn in 2025.

Exhibit 115: Lite-on's global footprint

Source: Lite-on.

Financial Analysis – Lite-On

29% revenue CAGR in 2025-28

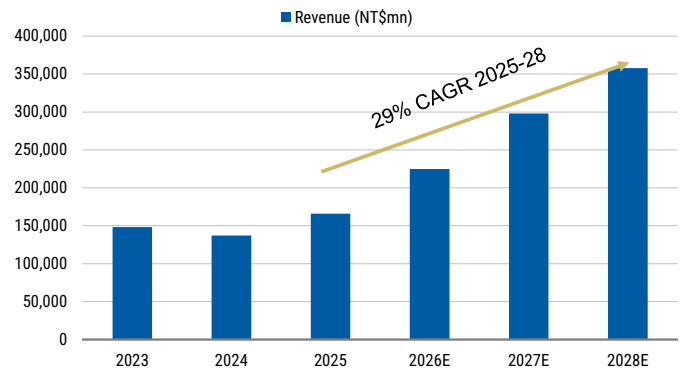
Lite-On's revenue growth was relatively moderate during 2020-22, with revenue expanding at a CAGR of 5% over the period. The company benefited from surging cloud computing demand, which drove increasing demand for power supplies, along with rising demand from EV/automotive electronics, although growth was offset by elevated component costs and supply-chain disruptions during COVID-19. Revenue then fell by 14% in 2023 and 8% in 2024 due to a post-pandemic demand pullback across the consumer electronics market and PC/NB model transitions, coupled with softening demand from EV-related businesses. The IT segment declined 20%/11% YoY, while the opto-electronics segment declined 11%/9% YoY in 2023/2024, respectively.

Revenue rebounded in 2025, growing 21% YoY, as the company began to benefit from surging demand for AI/cloud server power supplies. Contributions from ramping high-voltage/AI power infrastructure products began to increase, coupled with a partial recovery in the IT segment, mainly fueled by power adapter shipments for game consoles and gaming PCs/NBs, as well as related power offerings for low-Earth-orbit applications.

Looking ahead, we estimate revenue to expand at a 29% CAGR in 2025-28e, driven by the AI server power demand boom, while we remain mindful of weakness in consumer/IT demand amid elevated component costs.

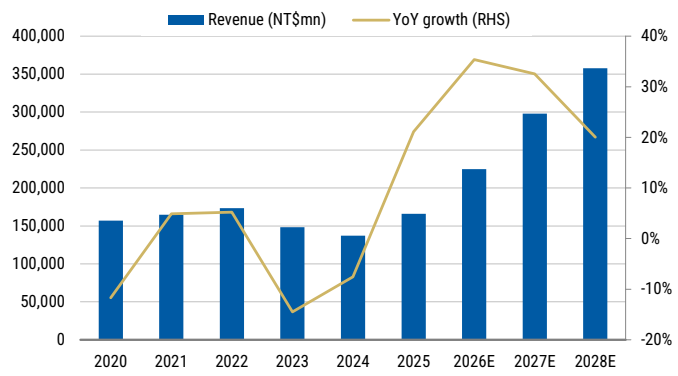
By product segment, IT-related products were the primary revenue driver in past years, while 2025 marked a crossover point, with the contribution from the Cloud & AIoT segment surpassing that of the IT segment, mainly supported by AI power-related offerings. IT revenue contribution declined from 72% in 2020 to 38% in 2025, and we estimate it will reach 17% by 2028. For the Cloud & AIoT segment, revenue contribution grew substantially from 27% in 2021 (0% in 2020) to 45% in 2025; we estimate it will account for 74% of total revenue by 2028, on the back of surging AI power-related demand. For opto-electronics, we expect revenue contribution to decline from 17% in 2025 to 9% in 2028.

Exhibit 116: Lite-on: Revenue CAGR, 2025-28e



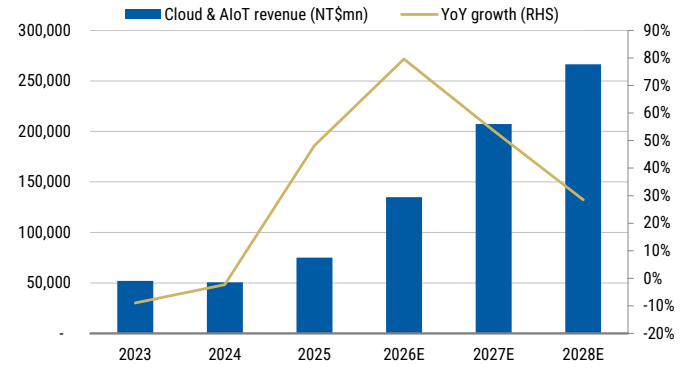
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 117: Lite-on: Annual revenue growth



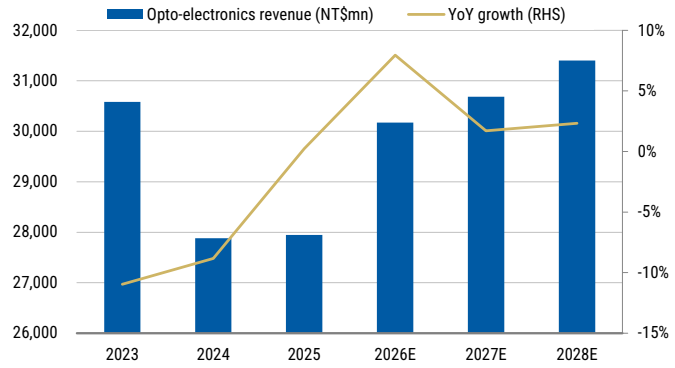
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 118: Lite-on: Cloud & AIoT revenue



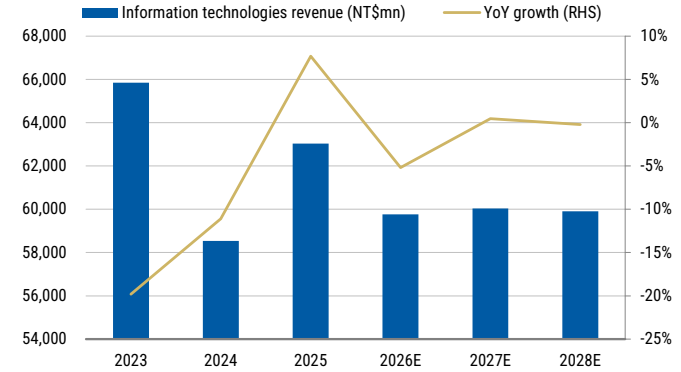
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 119: Lite-on: Opto-electronics revenue



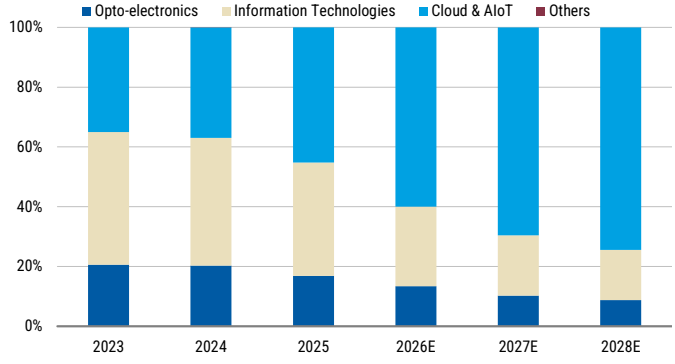
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 120: Lite-on: IT revenue



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 121: Lite-on: Revenue breakdown by segment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

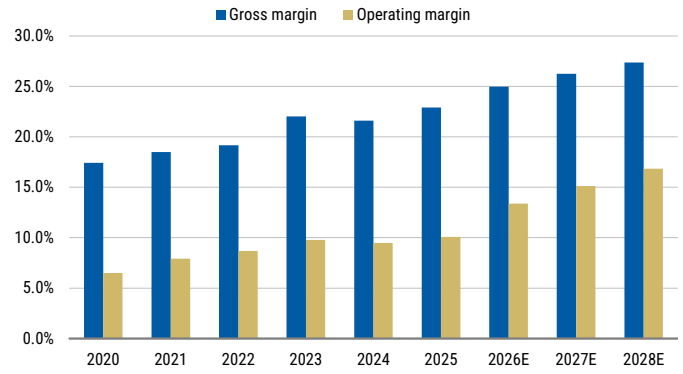
Continuing margin expansion

Lite-On's gross margin rose steadily over the years, from 17.4% in 2020 to 22.0% in 2023, thanks to a continually improving product mix. Gross margin dipped to 21.6% in 2024, mainly due to a smaller revenue base, while contributions from AI/cloud had yet to scale and margins were weighed down by learning-curve costs at the new Vietnam site. It then rose to 22.9% in 2025 as the AI server power business began to scale alongside rising BBU (battery backup unit) shipments, resulting in a more favorable product mix.

The opex ratio was largely stable at 10.5-11.0% during 2020-22. It then increased to 12.2% in 2023 and 12.1% in 2024, mainly due to operating deleverage and higher R&D costs. The opex ratio edged up further to 12.8% in 2025, mainly due to ongoing investment in R&D to support new business development in AI server power supplies, BBUs, liquid cooling, etc., along with a one-off tariff prepayment.

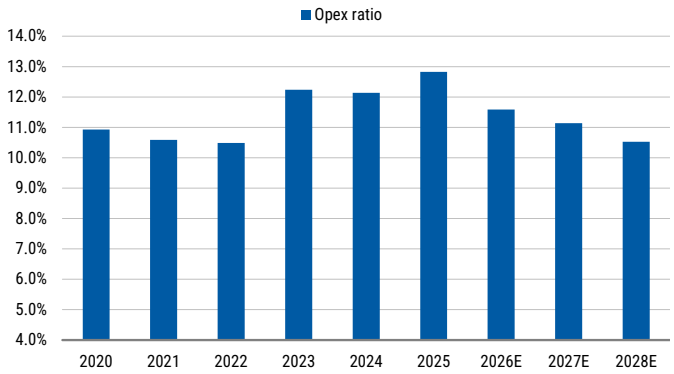
Looking ahead, we expect margins to continue expanding on a favorable product mix. We now pencil in a 2.1ppt gross margin improvement in 2026 to 25.0%, followed by expansions of 1.3ppt and 1.1ppt in 2027 and 2028, to 26.3% and 27.4%, respectively. For the opex ratio, we expect it to decline sequentially as the revenue base expands. Accordingly, we estimate operating margin to improve by 3.3ppt to 13.4% in 2026, followed by expansions of 1.7ppt in both 2027 and 2028, reaching 15.1% and 16.8%.

Exhibit 122: Lite-on: Margin trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 123: Lite-on: Opex ratio



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Net profit to grow at a 47% CAGR in 2025-28

Operating profit grew steadily alongside gross margin expansion during 2020-22, from NT $10,206mn to NT$15,073mn. It then fell 4% YoY in 2023 and a further 11% YoY in 2024 due to higher operating expenses, mainly from R&D costs, despite gross margin expansion in 2023. Operating margin increased by 1.1ppt YoY to 9.8% in 2023 amid a smaller revenue base, before declining by 0.3ppt YoY to 9.5% in 2024 on lower gross margin and an elevated opex ratio.

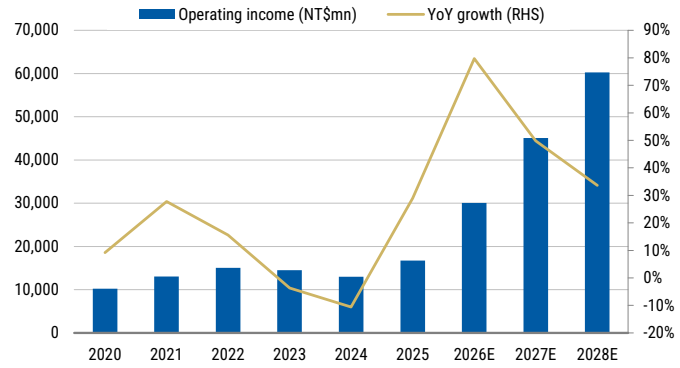
Operating profit rebounded by 29% YoY to NT$16,738mn in 2025, supported by both an expanding revenue base and strengthening gross margin. Operating margin also improved, increasing by 0.6ppt YoY to 10.1%, as the recovery in gross margin was partly offset by higher opex.

With the tax rate remaining relatively stable at 20-24%, Lite-On's net profit growth has generally tracked its operating profit growth. An exception occurred in 2023, when net profit grew 3% YoY despite a 4% decline in operating profit, mainly due to higher FX gains during the year.

Net margin has broadly tracked operating margin in recent years. It reached a record high of 9.8% in 2023 on improving gross margin, despite a higher opex ratio. It then declined to 8.7% in 2024 on gross margin erosion before rebounding to 9.1% in 2025, benefiting from increasing demand for AI power supplies.

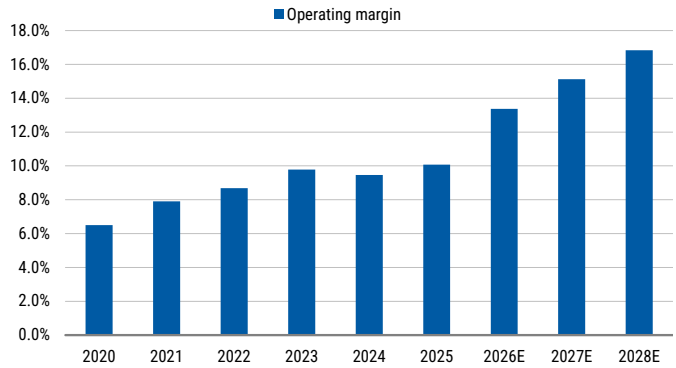
Looking ahead, we forecast net profit to grow 67% YoY in 2026e, followed by 44% YoY growth in 2027e and 32% YoY growth in 2028e, implying a 47% CAGR in 2025-28e. Net margin is likely to rise to 11.2% in 2026, 12.2% in 2027, and 13.4% in 2028.

Exhibit 124: Lite-on: 35% CAGR for operating profit in 2024-27



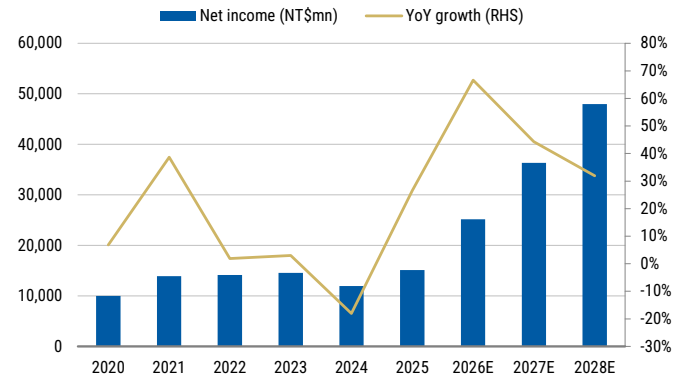
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 125: Lite-on: Operating margin



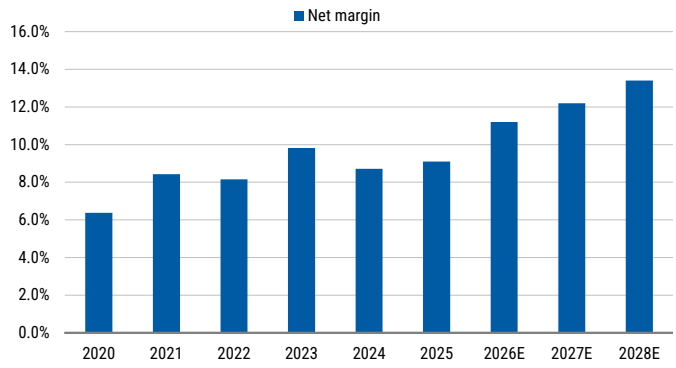
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 126: Lite-on: Net profit trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 127: Lite-on: Net margin



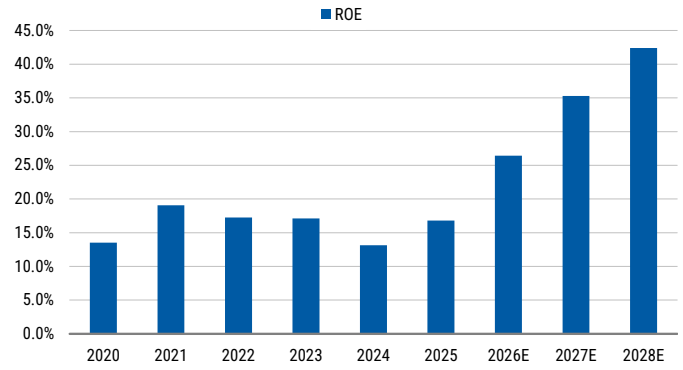
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

ROE to rise

ROE movements were highly correlated with changes in net margin during 2020-22, as asset turnover and financial leverage were largely stable over the period. ROE edged down slightly in 2023 despite higher net margin, which increased from 8.2% to 9.8%, due to lower asset turnover and financial leverage. ROE fell further to a trough of 13.1% in 2024 as net margin and asset turnover declined further. It then rebounded to 16.8% in 2025, primarily supported by net margin expansion, fueled by a favorable product mix, along with an improvement in asset turnover back to the 80% level.

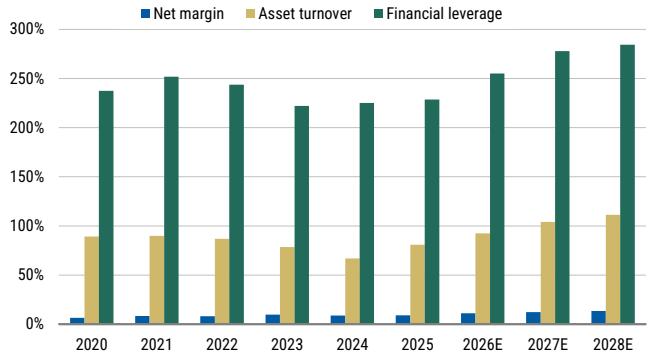
Looking ahead, we expect ROE to continue rising, reaching 26.4% in 2026, 35.3% in 2027, and 42.4% in 2028, primarily driven by continued margin expansion and improving asset turnover. Net margin is projected to increase from 9.1% in 2025 to 13.4% by 2028, supported by gross margin improvement from 22.9% to 27.4% and a decline in the operating expense ratio from 12.8% in 2025 to 10.5% in 2028.

Exhibit 128: ROE trend



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 129: DuPont analysis



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Improving balance sheet and continuous free cash flow generation

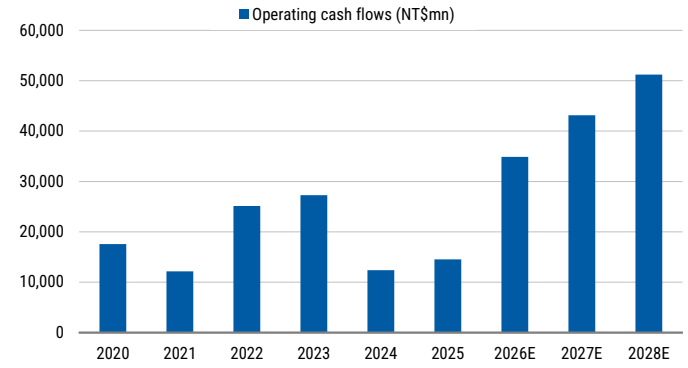
Lite-On has consistently generated positive operating cash flow in recent years, although cash flow dropped notably in 2021, 2024, and 2025, mainly due to higher working-capital-related cash outflows. The company has also consistently generated positive free cash flow, with capital expenditures generally at the NT$3-4bn level over the years. An exception occurred in 2025, when capital expenditure rose to NT$6.7bn, mainly for power-related capacity expansion in Vietnam, Taiwan, and the US.

As the company has announced investments of US$919mn and US$149mn in its US and Vietnam subsidiaries, respectively, alongside continued expansion of power-related capacity, mainly for PSUs, we now estimate capex to rise to ~NT$13bn in 2026, versus ~NT $7bn in 2025. Capex should then decline to NT$9bn in 2027 as investment levels normalize.

Lite-On's net cash position has remained relatively healthy, with its net cash-to-equity ratio improving from 65% in 2020 to 80% in 2023. It then declined to 75% in 2024 and 66% in 2025, as total debt increased by NT$7.7bn in 2024, while cash on hand declined in 2025, likely due to increasing capital investment.

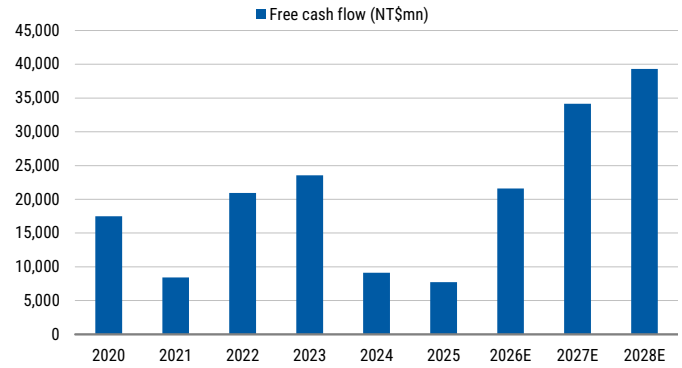
Looking ahead, we assume the total debt position will remain stable despite ongoing capex investment. The net cash position should continue to grow sequentially on the back of positive free cash flow generation.

Exhibit 130: Lite-on: Positive operating cash flow



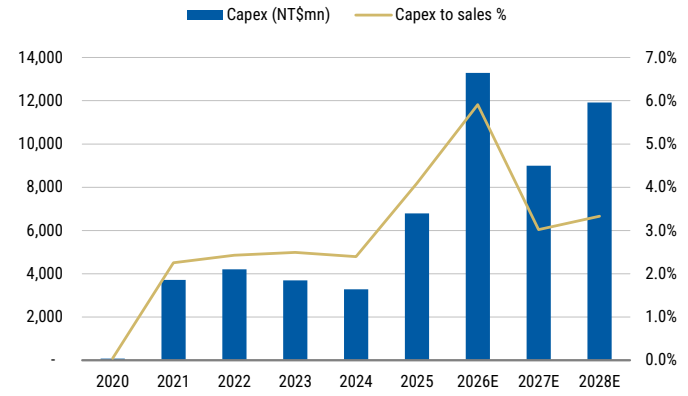
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 131: Lite-on: Improving free cash flow generation



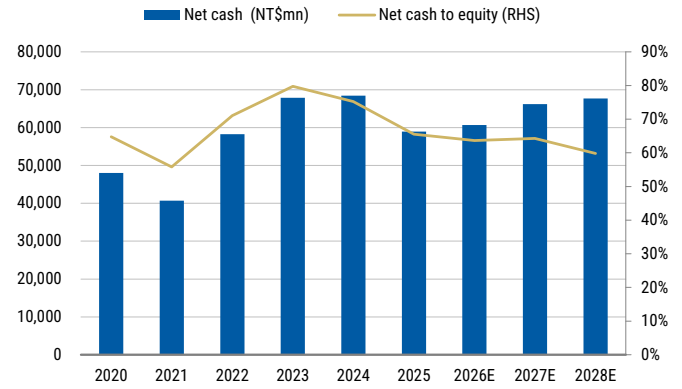
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 132: Lite-on: Capex and capex-to-sales ratio



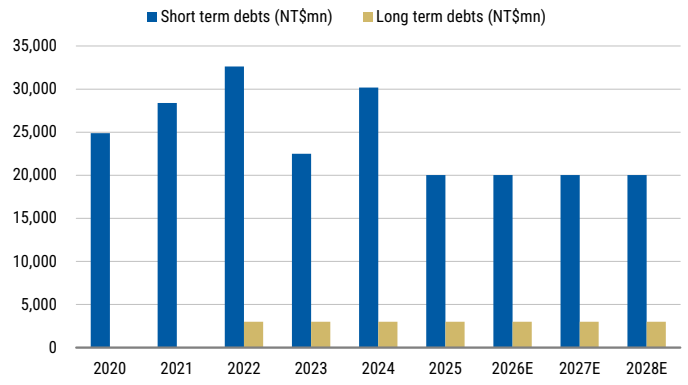
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 133: Lite-on: Net cash position



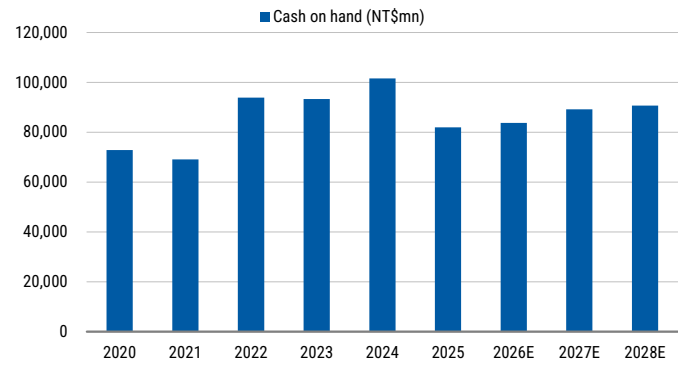
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 134: Lite-on: Debt position



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 135: Lite-on: Cash on hand

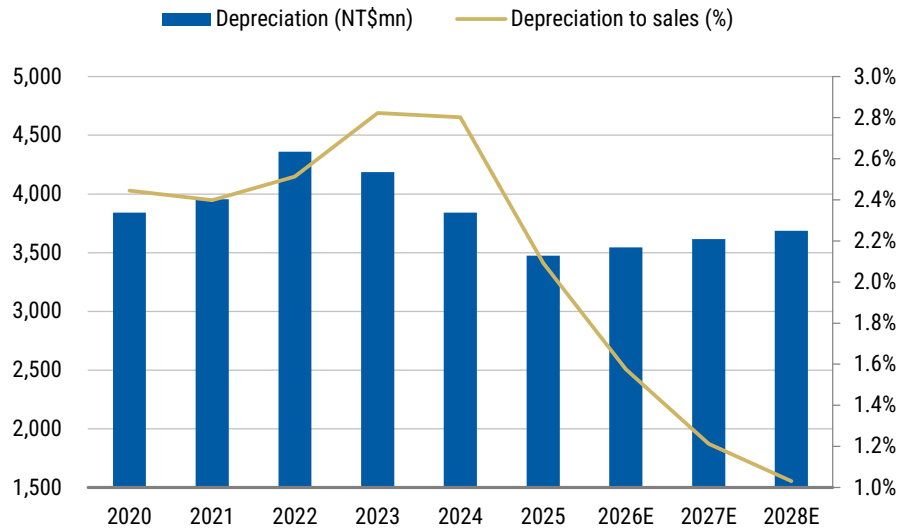


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Lite-On's average capex-to-sales ratio was around 2.3-2.4% during 2021-24, then rose to 4.1% in 2025 as the company began aggressively expanding global capacity, despite a larger revenue base during the year. Looking ahead, we project the capex-to-sales ratio to rise to 5.9% in 2026 amid continued production-site development, before declining to 3.0-3.3% in 2027-28 as investments normalize.

The depreciation-to-sales ratio remained at 2.4-2.5% during 2020-22 before rising to 2.8% in 2023, in line with higher capex during the year. The ratio then declined to 2.1% in 2025 on a larger revenue base. Looking ahead, we expect the ratio to decline further as the revenue base grows faster than capex and depreciation, falling to 1.6% in 2026 and 1.0% in 2028.

Exhibit 136: Lite-on: Depreciation and depreciation to sales

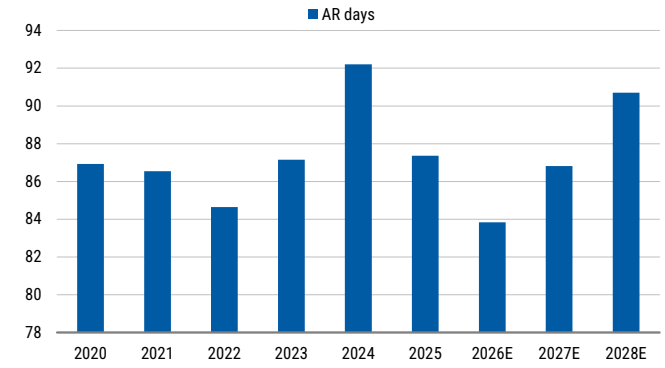


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Lite-On's cash conversion cycle (CCC) fluctuated within a relatively narrow band of roughly 35-44 days between 2020 and 2025, reflecting generally stable but modestly shifting working-capital efficiency. CCC was at a relatively low level of 35 days in 2020, supported by lower inventory days and higher AP days, before rising steadily to 44 days in 2022 as both inventory days (78) and AR days (85) increased, while AP days were slightly lower (119). From 2023 onward, the cycle improved meaningfully, falling to 36 days as AP days extended sharply from 119 in 2022 to 135. This improvement partially reversed in 2024, with CCC ticking up to 39 days as inventory and AR days both increased to 89 and 92 days, respectively, even as AP days extended further to 143. CCC then eased again to 37 days in 2025 as AP days moderated slightly to 138, alongside stable inventory and receivables levels.

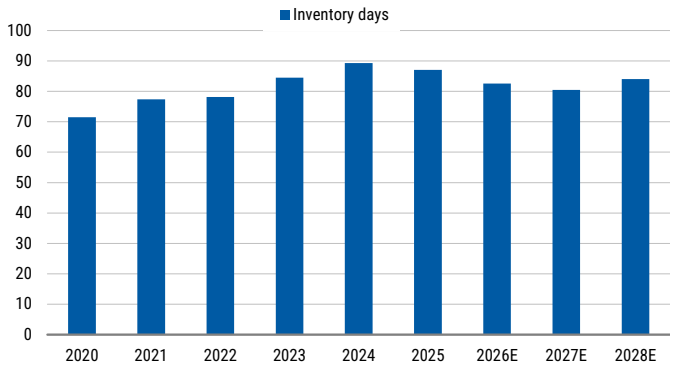
Looking ahead, we expect the overall cash conversion cycle to edge up slightly from 37 days in 2025 to 40 days in 2028, as increasing AR days and lower AP days are partially offset by improving inventory days.

Exhibit 137: Lite-on: AR days



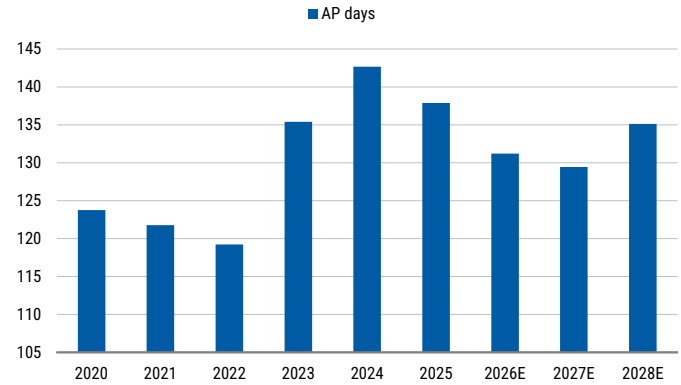
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 138: Lite-on: Inventory days



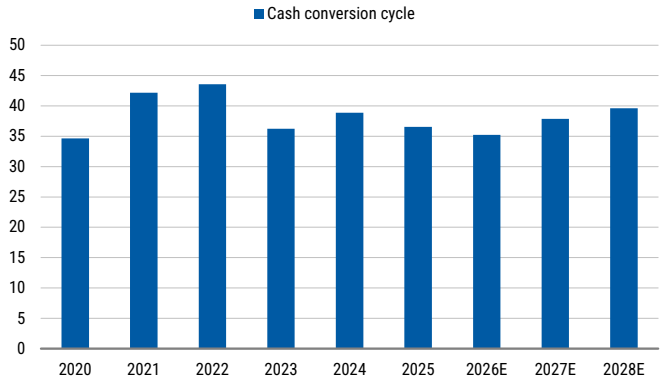
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 139: Lite-on: AP days



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 140: Lite-on: Cash conversion cycle



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Company Background – Lite-On

Company snapshot

Founded in 1975, Lite-On Technology is a Taiwan-based manufacturer of optoelectronic components, power supplies, and information technology products. The company produced Taiwan's first LEDs and became the first technology company listed on the Taiwan Stock Exchange in 1983. It mainly operates through four business segments: Information Products (power supplies, servers, networking equipment, keyboards, and chassis), Optoelectronics (LED packaging, LED lighting, automotive electronic components, and camera modules), Storage (storage components), and other supporting businesses. Additional product lines include energy management solutions, such as data center power supply systems and cloud server chassis; networking and surveillance products; and automotive electronics, including ADAS and EV charging solutions.

Lite-On is headquartered in Taipei, Taiwan. The company operates manufacturing sites, R&D centers, and sales offices across Asia (Taiwan, China, Japan, and Singapore), the US, and Europe (the UK, the Netherlands, and Germany).

Exhibit 141: Lite-on: Key milestones

Year	Key Milestones
1975	Established LITEON's first LED factory in Zhonghe, Taiwan
1983	Became the first electronics company listed on the Taiwan Stock Exchange (Ticker: 2301)
1985	Began manufacturing power supply units.
1989	Opened first overseas plant in Bangkok, Thailand, for optoelectronic products.
1990	• Set up a switching power supply plant in Penang, Malaysia. • Silitek acquired Maxi Switch, the No. 3 U.S. keyboard maker.
1991	Acquired NPE's switching power supply plant in the UK.
1994	Established a subsidiary in California, USA.
1999	Spun off the Optoelectronics business unit as LITEON IT Corp.
2002	LITEON Electronics, Silitek, LITEON Technology, and GVC merged into LITEON Technology Corporation (Ticker: 2301)
2003	Moved global headquarters to Neihu Science Park.
2006	• Acquired BenQ's Optical Disk Drive business, becoming a top-3 global ODD maker. • Partnered with Philips to enter the automotive disc drive market. • Acquired 100% of LITEON Automotive Corp., expanding into automotive electronics.
2007	• Acquired 100% of Li Shin International via stock swap, strengthening power supply resources for LCD TVs and other products. • Completed tender offer for Perlos, the world's largest handset casing maker.
2009	Opened East China Regional HQ in Wujin, Changzhou.
2010	• Invested in Epyon Power, a supplier of fast-charging technology. • Completed the world's first Optoelectronic Semiconductor Carbon Footprint and Type III Environmental Product Declarations.
2011	Completed the world's first Product Category Rules for Circuit Boards.
2013	Acquired Power Innovations (US), strengthening cloud and power management capabilities.
2014	Selected for the Carbon Disclosure Project's Climate Disclosure Leadership Index.
2016	Opened a new operations center in Kaohsiung's Nanzih Export Processing Zone.
2018	Divested the mobile camera module and mechanical business to focus resources.
2019	Acquired 100% of LOJ via tender offer.
2020	• Launched succession plan: Tom Soong named as Chairman, Anson Chiu named as President. • Completed sale of the SSD business to KIOXIA Holdings.
2021	• Launched an Employee Stock Ownership Trust (ESOT). • Joined the O-RAN Alliance and launched 5G small cells.
2022	• Began reconstructing the Zhonghe office into the "LITEON Zhonghe Digitech Center." • Inaugurated Phase II of the LITEON Vietnam plant in Haiphong.
2023	Opened its first corporate store, 'LITEON Collection,' in Neihu Science Park.
2024	• Took a 19.99% strategic stake in COSEL. • Broke ground on a new low-carbon building in Kaohsiung.
2025	• Returned to COMPUTEX with next-gen AI solutions, winning the Sustainable Design Award (Bronze). • Reopened its founding Zhonghe headquarters after a three-year reconstruction.

Source: Company data, Morgan Stanley Research

Exhibit 142: Lite-on: Global operation locations

Source: Company data

Exhibit 143: Lite-on: Global footprint

Headquarter	
Taipei City, Taiwan	
Sales Offices	
Milpitas, California, U.S.A.	
Plano, Texas, U.S.A.	
Tokyo, Japan	
Shanghai, China	
Singapore	
Hoofddorp, NL	
Cramlington, Northumberland, UK	
Wetzlar, DE	
Subsidiary	**Entity name**
American Fork, Utah, U.S.A.	Power Innovations International

Source: Company data, Morgan Stanley Research

Product offerings

Lite-On's key offerings can be categorized into six key areas: Optoelectronic Semiconductors, Energy Management, Consumer Electronics, Networking and Surveillance, Automotive Electronics, and Roadway Intellimation System.

1. Optoelectronic semiconductors

Its product portfolio spans a broad technological range, encompassing illumination and sensing applications, low- to high-power devices, standard and custom-made solutions, and components ranging from LEDs to lasers and from visible light to infrared and ultraviolet spectra.

Exhibit 144: Lite-on: Offerings for optoelectronic semiconductors

Optoelectronic Semiconductors						
Visible, UV & Automotive LEDs				Couplers	Optical Sensors	IR Emitters & Detectors
Illumination / UV-C / Automotive				Isolation & Signal Transfer	Ambient / Proximity / Gesture	Infrared Sensing
Automotive LED	SMD LED	Flash LED	UV LED	Photocouplers	Optical Sensors	IR Emitters & Detectors

Source: Company data, Morgan Stanley Research

2. Energy management

The product range serves the communications, industrial automation, smart home, medical, office, and automotive electronics industries. It offers a comprehensive suite of platinum- to titanium-efficiency-grade power supplies (CRPS, GPU, and SLIM series) suitable for both standard and AI servers, supporting AC or DC input with high power density, high reliability, and modular scalability. Power-shelf and battery-backup-unit (BBU) solutions, along with rack-cooling and liquid-cooling technologies for data centers, cloud infrastructure, and AI/machine-learning platforms, are also included in this segment.

Exhibit 145: Lite-on: Offerings for energy management

Energy Management			
Data Center Power Supply System	**Cloud Server Chassis**	**Power Supply Module**	**Magnetic Components**
CRPS / GPU / SLIM, Platinum-Titanium	Open Rack Standard V3	PoE / Industrial / LED Driver	Transformers & Inductors
			Includes components apply for cloud infrastructure, industrial power supply, white appliances, electric vehicle and consumer electric, etc
Server Power Networking Power Open Compute Project	Server Rack Server Chassis	POE Injector Industrial Power Adapter LED Driver	Magnetic Components

Source: Company data, Morgan Stanley Research

3. Consumer electronics

This segment is committed to advancing home entertainment and office solutions through feature-rich product design intended to enhance the overall user experience. Offerings include office/gaming solutions, laptop keyboard modules and touchpads, laptop power adapters, USB Type-C/PD chargers, etc.

Exhibit 146: Lite-on: Offerings for consumer electronics

Consumer Electronics		
3C Accessories	**Power Supply Module**	
Office / Gaming / Laptop peripherals	PC & Networking Power	
Office Solutions Gaming Solutions Laptop Keyboard Module Touchpad Laptop Power Adapters USB Type C/PD Charger	PC Power Supply	Networking Adapter

Source: Company data, Morgan Stanley Research

4. Networking and surveillance

The product portfolio covers networking solutions, video surveillance, AIoT solutions, and server-related offerings. The solutions are designed to be user-friendly and adaptable to emerging trends in digital intelligence.

Exhibit 147: Lite-on: Offerings for networking and surveillance

Networking and Surveillance		
Video Surveillance	**Access Control & Alarms**	**Networking Solutions**
Home / Building / Module	Smart Hub / Access / Alarm	5G / WLAN / Switch / Module
Home Security Camera Building Surveillance Camera Camera Module	Smart Hub Access Control Alarm Kits	5G Total Solution Wireless LAN Wired Switch Wireless Module

Source: Company data, Morgan Stanley Research

5. Automotive electronics

Lite-On is able to leverage its core competencies in optoelectronics and power management, along with more than four decades of expertise in automotive electronics. This segment is dedicated to advancing vehicle-charging technology, ADAS (Advanced Driver Assistance Systems) solutions, and automotive lighting modules through high-end engineering. AC charging solutions are offered across multiple current ratings (32/40A, 48A, 80A, and an EU-specific variant), while DC charging solutions include a 30kW fast-charging offering.

Exhibit 148: Lite-on: Offerings for automotive electronics

Automotive Electronics		
ADAS Solutions	**AC Charging Solutions**	**DC Charging Solutions**
Vision & Lighting	32/40A - 48A - 80A - EU	Fast Charging
Automotive Vision Solutions Vehicle Lighting Solutions	IC 32/40A IC 48A IC 80A IC EU	DC 30kW

Source: Company data, Morgan Stanley Research

6. Roadway Intellimation System

This segment is mainly operated under the LEOTEK brand, a subsidiary of the LITEON Technology Group with dual headquarters in Silicon Valley and Taipei. It has served road infrastructure across more than 30 countries—including North America, Europe, the Middle East, and the Asia-Pacific region—since 1992. The segment comprises intelligent roadway lighting, traffic signal systems, and AI-driven predictive maintenance solutions, all developed to support smarter and more efficient road infrastructure.

Exhibit 149: Lite-on: Offerings for Roadway Intellimation System

Roadway Intellimation System		
Roadway Lighting Intellimation	**Intelligent Traffic Signal System**	**AI Predictive Maintenance Solution**
Smart LED Streetlights	AI-driven Signal Control	Condition Monitoring
		Includes AI warning and maintenance management, traffic flow data analysis and computation, traffic-adaptive signal control, and priority passage for emergency vehicles, etc
Roadway Lighting Intellimation	Intelligent Traffic Signal System	AI Predictive Maintenance Solution

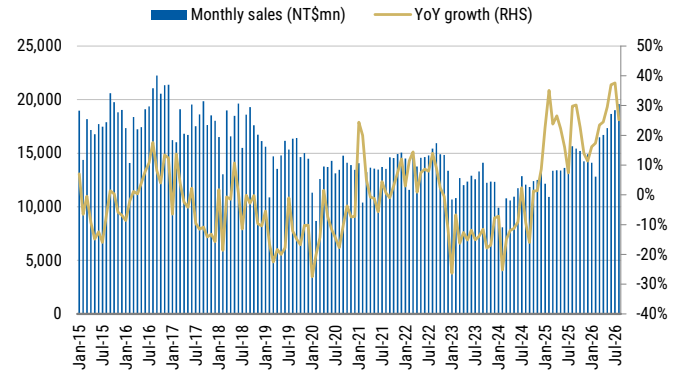
Source: Company data, Morgan Stanley Research

Monthly sales vs. share price

Lite-On's monthly sales have hovered around the NT$10-20bn level since 2013. Notably, we see a high correlation of ~76% between the company's monthly sales and its share price movements since 2023.

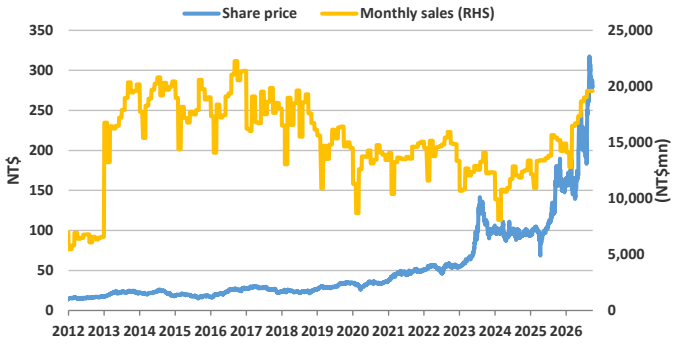
The company reported August monthly sales of NT$19,592mn (+3% MoM/+25% YoY).

Exhibit 150: Lite-on: Monthly sales vs. sales YoY growth



Source: TEJ, Morgan Stanley Research

Exhibit 151: Lite-on: Monthly sales vs. share price



Source: TEJ, Morgan Stanley Research

Key management & shareholding structure

Lite-On's board currently consists of five directors and four independent non-executive directors.

Its founder, Raymond Soong, and current Chairman, Tom Soong, together owned a stake of ~5% as of June 2026. Excluding the founder and Chairman, the management team owned a stake of ~0.1% as of June 2026.

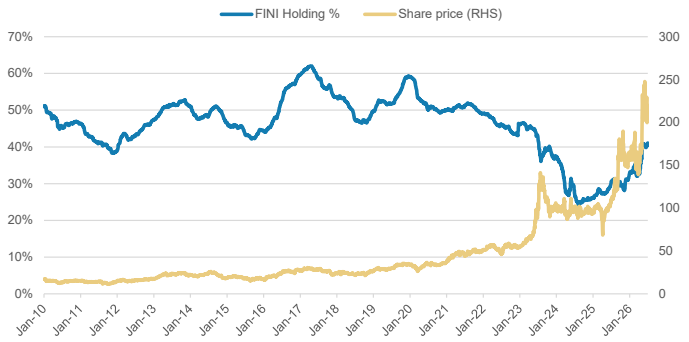
The FINI holding percentage as of July 7, 2026, was 41%, versus a previous peak of 62% in April 2017 and a historical average (since 2010) of 46%.

Exhibit 152: Lite-on: Shareholding structure

Shareholders	Ownership
Cathay MSCI Taiwan ESG Sustainability High Dividend Yield ETF	5.6%
Yuanta/P-shares Taiwan Dividend Plus ETF	5.5%
Ta-Rong Investment Co., Ltd	3.7%
Raymond Soong*	3.4%
Fuh Hwa Taiwan Technology Dividend Highlight ETF	3.3%
Labor Pension fund	2.6%
Ming-Hsing Investment Co., Ltd	2.6%
Ta-Sung Investment Co., Ltd*	2.1%
Chunghwa Post Co., Ltd	2.0%
Lite-On Technology Corporation (Treasury shares)*	1.7%
Total	**32.6%**

Source: Company data, Morgan Stanley Research. Note: Data as of August 2025. (*) are shown as insiders.

Exhibit 153: Lite-on: FINI holdings vs. share price since 2010



Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 154: Lite-on: FINI holdings vs. share price since 2023

Source: TEJ, Morgan Stanley Research. Note: Data as of Oct 5, 2026

Exhibit 155:

Lite-on: Management profiles

Name	Title	Education	Work Experience
Tom Soong	Chairman and Chief Sustainability Officer, LITEON Technology	Electrical Engineering, University of Southern California, USA; International Business Program Attendance, NTU-FUDAN EMBA	- 20+ years at LITEON - Special Assistant to Chairman & Vice Chairman, LITEON - CEO of LITEON Smart Life and Applications Business Group - General Manager, LITEON Shanghai Operational Center
Anson Chiu	President and Director, LITEON Technology	Bachelor of Industrial Management, Lunghwa University of Science and Technology	- Joined LITEON Technology in 1987 - 35+ years in global supply chain and business development' - CEO, LITEON Power Conversion Business Group - General Manager, LITEON Power Conversion Business Group - Director, NB Business Unit, Power Conversion Business Group
Raymond Soong	Founder and Director, LITEON Technology and LITEON Cultural Foundation	Fellow, Industrial Technology Research Institute; Honorary Doctor of Management, National Chiao Tung University; Honorary Doctor of Management, Taipei University of Technology	- Chairman, LITEON Group - Chairman & Founder, LITEON Cultural Foundation - Chief Engineer, Texas Instruments Taiwan Ltd. - Board of Councilors member, Doctorate College of Technology, USC
Harvey Chang	Director Representative, Lite-On Technology Corp.	MBA, The Wharton School, University of Pennsylvania	- Independent Director, LITEON Technology - Chairman, IC Broadcasting - Chairman, TVBS and Catchplay Media Holdings Ltd. - President & CEO, Taiwan Mobile - Senior VP & CFO, TSMC
Sarina Lin	Director Representative, Lite-On; Advisor to Chairman's Office, Lite-On	Doctor of Laws, National Taiwan University	- Chief Legal Officer, Lite-On - Special Assistant to Chairman and Attorney, MediaTek Inc. - Director, MediaTek Foundation - SVP, CHRO & Chief Legal Officer, MStar Semiconductor - Director & CEO, MStar Semiconductor Charity Foundation - Director and Supervisor, MStar Semiconductor
Albert Hsueh	Independent Director, LITEON Technology;	MBA, Bloomsburg University, Pennsylvania, U.S.A.	- Chairman, PricewaterhouseCoopers Taiwan - Director, Corporate Governance Association in Taiwan - Professor, National Taiwan University and Tsinghua University, College of Technology Management
Mike Yang	Independent Director, Lite-On Technology Corp.	Master of Science, Arizona State University	- Associate Vice President, ADI Corp. Procurement Dept. - Director, LIONS Taiwan Technology Inc. - Director, Quanta Culture & Education Foundation - Cloud computing and global technology leadership
MK Lu	Independent Director, Lite-On Technology Corp.	Honorary Doctor, Tatung University; Honorary Doctor of Engineering, National Chiao Tung University; Department of Electrical Engineering, Tatung University	- Chairman, Tatung Company - Chairman, Sino-American Silicon Products Inc. - Chairman, Actron Technology Corporation - President, Lite-On Semiconductor Corp. - President, Vishay Lite-On Power Semiconductor Corp. - Vice President, Silitek Corp.
Jesse Ding	Independent Director, Lite-On Technology Corp.	PhD program, The Wharton School, University of Pennsylvania; MBA, University of Detroit, USA	- Chairman, EnTie Commercial Bank - General Manager, EnTie Commercial Bank - General Manager, TAIPEI FUBON COMMERCIAL BANK CO., LTD - Financial accounting and leadership decision-making

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Panasonic Holdings (6752.T)

Our target price is based on P/E 19.5x our FY3/28 EPS forecast.

We apply: 36x to EPS generated by AI infrastructure-related businesses, 5x to EPS attributable to IRA subsidies in the automotive battery business, and 13.5x to EPS from all other businesses.

Risks to Upside

- Acceleration of restructuring and exit/scale-down of unprofitable businesses
- Stronger than expected growth of datacenter BBU, automotive battery business

Risks to Downside

- Deteriorating visibility for energy business
- Expectations for business portfolio revisions sink

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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(as of September 30, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
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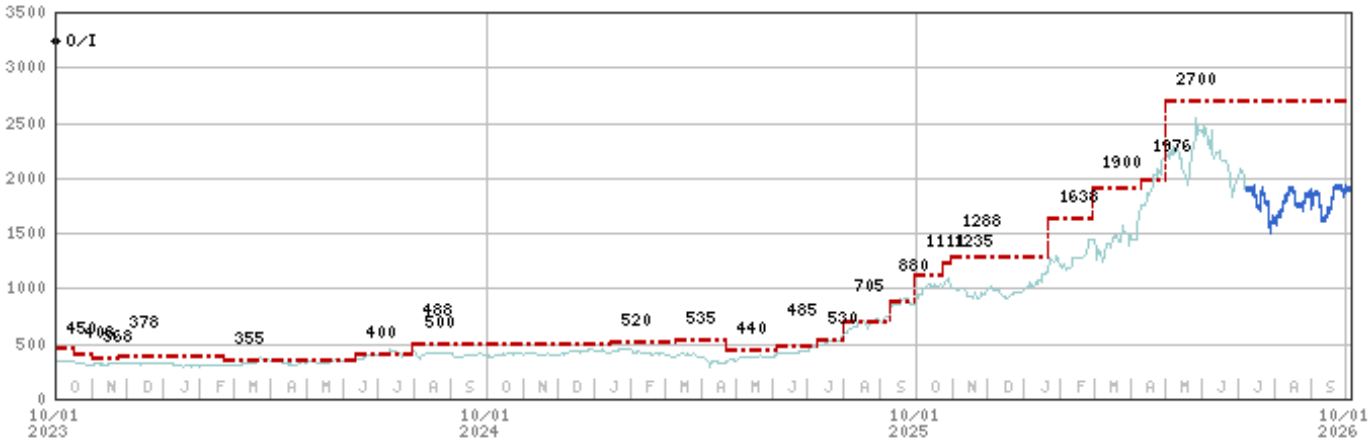
Stock Price, Price Target and Rating History (See Rating Definitions)

Delta Electronics Inc. (2308.TW) - As of 10/05/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : O/I

Price Target History: 9/9/21 : 345; 10/29/21 : 325; 2/25/22 : 303; 7/29/22 : 312; 9/21/22 : 326; 10/6/22 : 350; 11/23/22 : 368; 8/2/23 : 450; 10/17/23 : 408; 11/1/23 : 368; 11/23/23 : 378; 2/21/24 : 355; 6/12/24 : 400; 7/30/24 : 488; 8/1/24 : 500; 1/15/25 : 520; 3/11/25 : 535; 4/23/25 : 440; 6/4/25 : 485; 7/9/25 : 530; 8/1/25 : 705; 9/8/25 : 880; 9/30/25 : 1111; 10/24/25 : 1235; 10/30/25 : 1288; 1/21/26 : 1638; 2/27/26 : 1900; 4/10/26 : 1976; 5/1/26 : 2700

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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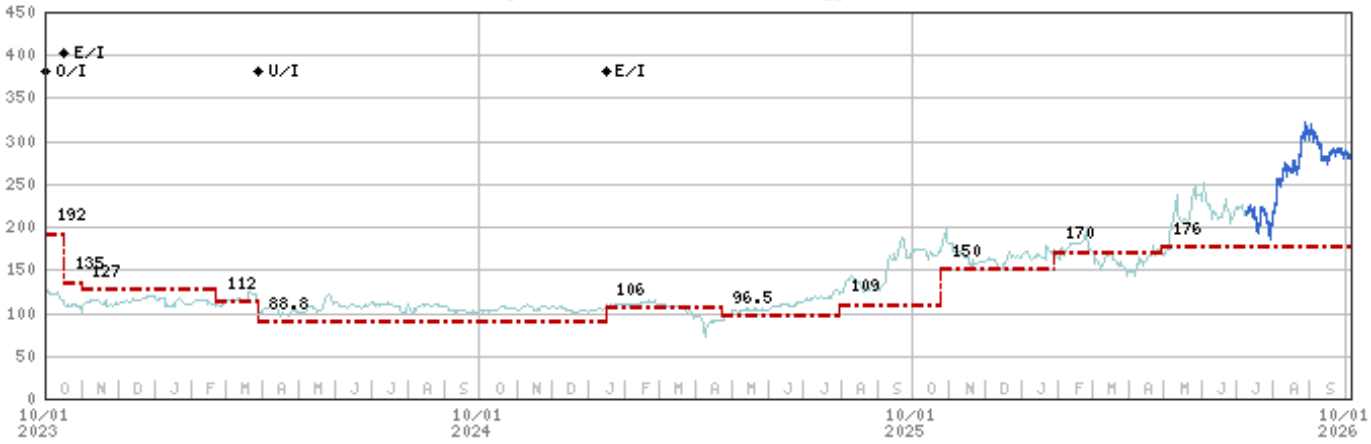
Lite-On Technology (2301.TW) - As of 10/05/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : E/I; 7/31/23 : O/I; 10/17/23 : E/I; 3/28/24 : U/I; 1/15/25 : E/I

Price Target History: 4/30/21 : 62; 7/19/22 : 65; 2/21/23 : 61.2; 7/31/23 : 192; 10/17/23 : 135; 10/31/23 : 127; 2/21/24 : 112; 3/28/24 : 88.8; 1/15/25 : 106; 4/23/25 : 96.5; 7/31/25 : 109; 10/24/25 : 150; 1/28/26 : 170; 4/29/26 : 176

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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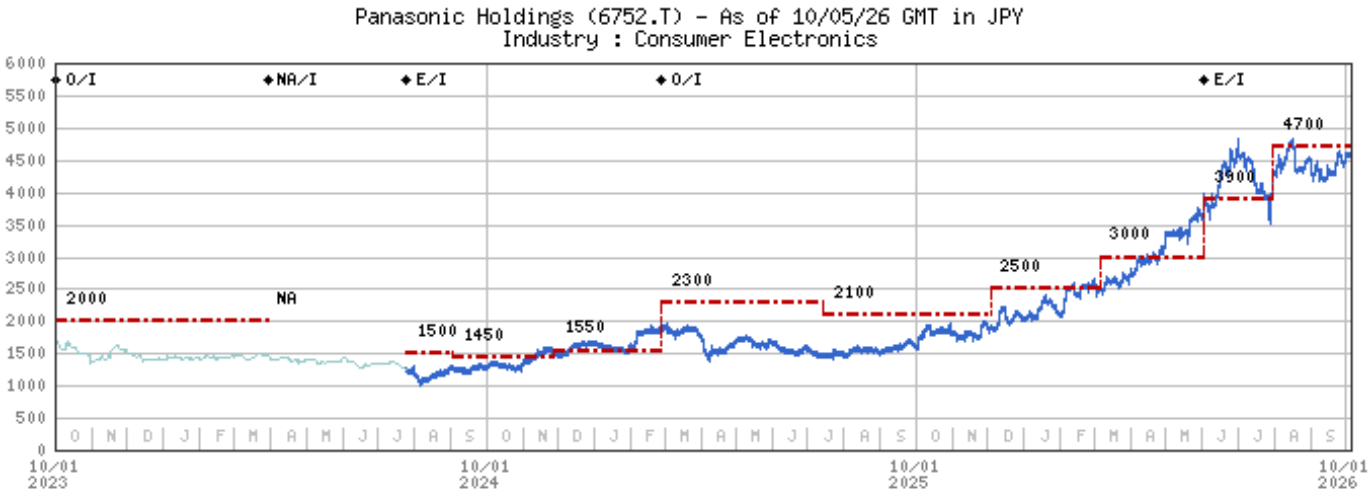

Stock Rating History: 10/1/21 : O/I; 3/29/24 : NA/I; 7/25/24 : E/I; 2/26/25 : O/I; 6/3/26 : E/I

Price Target History: 9/22/21 : 1900; 3/1/22 : 1700; 6/13/22 : 1500; 12/1/22 : 1600; 2/20/23 : 1500; 6/7/23 : 2000; 3/29/24 : NA; 7/25/24 : 1500; 9/3/24 : 1450; 11/28/24 : 1550; 2/26/25 : 2300; 7/14/25 : 2100; 12/3/25 : 2500; 3/6/26 : 3000; 6/3/26 : 3900; 7/31/26 : 4700

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/05/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$38.64
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.76
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,055.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,730.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.46
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$53.35
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.91
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$37.16
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$61.95
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43

Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$23.88
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$179.10
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,111.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.62
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,075.00
Advantech (2395.TW)	O (01/20/2021)	NT$740.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,180.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,585.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$39.05
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,550.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,485.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$890.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,325.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,005.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$146.50
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$68.20
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,450.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$345.50
Innolux (3481.TW)	E (04/07/2025)	NT$52.30
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,355.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$298.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.40
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$995.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$207.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.80
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,700.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.86
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$388.50
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,215.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$254.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.30
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$3,000.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.92
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,875.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,495.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$88.70
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$331.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,325.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$606.00

Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,150.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$626.00
Zhen Ding (4958.TW)	++	NT$576.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$570.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,780.00

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October 6, 2026 11:56 AM GMT

King Slide Works Co. Ltd. | Asia Pacific

Preliminary August Profit and September Monthly Sales (-31% MoM/+196% YoY)

Sentiment to stay soft in near term but 2027 outlook is promising.

King Slide announced its preliminary profit for August, per TWSE regulatory requirements:

- Revenue was NT$6,466mn (+1% MoM/+344% YoY), 33% of our estimate for 3Q26 of NT$19,555mn (+81% QoQ/+348% YoY) and 39% of consensus' NT $16,755mn (+55% QoQ/+284% YoY).
- Pretax income was NT$4,826mn (+216% YoY; pretax margin at 74.6%), 29% of our 3Q26 pretax income estimate of NT$16,377mn (margin at 83.7%) and 34% of consensus' NT$14,094mn (margin at 84.1%).
- Net income was NT$3,819mn (+212% YoY; EPS NT$40.07; NPM 59.1%), 30% of our 3Q26 net income estimate of NT$12,938mn (EPS NT$135.76; NPM 66.2%) and 34% consensus' NT$11,118mn (EPS NT$114.38; NPM 66.4%).

September monthly sales

- September sales of NT$4,464mn (-31% MoM/+196% YoY), driving 3Q26 revenue to NT$17,338mn (+60% QoQ/+297% YoY), were short of our estimate of NT$19,555mn (+81% QoQ/+348% YoY) and but ahead of the consensus forecast of NT$16,755mn (+55% QoQ/+284% YoY).

Our view:

- We think the sequential decline in September revenue was driven by slower momentum for some projects, which entered the later part of their product cycles. While sentiment should stay soft in the near term, we think the outlook into 2027 is still promising as customers start new product cycles.
- We maintain our view that King Slide's competitiveness in the comprehensive patent portfolio and strong execution are entrenched and accentuated amid fast generation migration in AI applications.
- Stay OW with PT of NT$15,000 (30x 2027e P/E).

MORGAN STANLEY TAIWAN LIMITED+

Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

King Slide Works Co. Ltd. (2059.TW, 2059 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$15,000.00
Up/downside to price target (%)	15
Shr price, close (Oct 6, 2026)	NT$13,100.00
52-Week Range	NT$15,085.00-2,875.00
Sh out, dil, curr (mn)	95
Mkt cap, curr (mn)	NT$1,248,392
EV, curr (mn)	NT$1,224,690
Avg daily trading value (mn)	NT$4,468

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	103.23	382.77	502.77	559.77
Prior EPS (NT$)**	-	-	-	-
EPS (NT$)§	100.25	343.67	499.35	574.20
Revenue, net (NT$ mn)	17,501	55,414	74,250	82,820
EBITDA (NT$ mn)	12,449	45,194	60,021	67,047
ModelWare net inc (NT $ mn)	9,837	36,477	47,913	53,345
P/E	36.3	34.2	26.1	23.4
RNOA (%)	216.6	818.3	534.3	549.4
ROE (%)	46.2	130.2	90.9	62.3
EV/EBITDA	26.8	26.7	19.5	17.0
Div yld (%)	0.9	0.9	1.2	1.3
FCF yld ratio (%)**	2.9	2.6	3.8	4.2
Leverage (EOP) (%)	(84.6)	(83.2)	(88.8)	(91.5)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

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AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
King Slide Works Co. Ltd. 2059.TW			
September 2026 sales	06 Oct 2026	Unchanged	↓ Modest downside surprise
Preliminary August Profit Announcement	06 Oct 2026	Unchanged	↓ Modest downside surprise

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

King Slide Works Co. Ltd. (2059.TW)

Base case, 30x 2027e P/E. We use price/earnings as our valuation methodology because we believe it better captures King Slide's earnings growth potential. We believe our target multiple is justified by an operating profit CAGR of 77% seen for 2025-28, implying a PEG of 0.4 vs. the average of 1.2 over 2010-24, when the stock was trading at 17-18x P/E with an operating profit CAGR of 14%.

Risks to Upside

- Stronger-than-expected AI capex
- Stronger server demand
- Faster market share gains

Risks to Downside

- Softer AI capex
- Weaker server demand
- Sluggish kitchen furniture demand
- Competition leading to slower share gains.

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(as of September 30, 2026)
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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
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Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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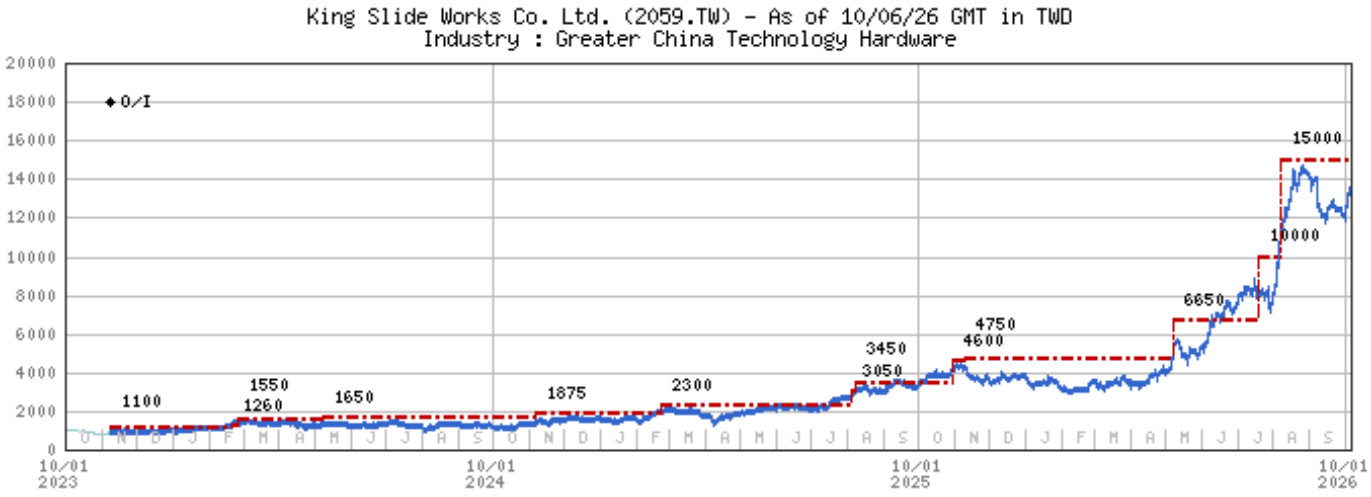
Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 11/8/23 : O/I

Price Target History: 11/8/23 : 1100; 2/20/24 : 1260; 2/26/24 : 1550; 5/8/24 : 1650; 11/7/24 : 1875; 2/22/25 : 2300; 8/4/25 : 3050; 8/8/25 : 3450; 10/30/25 : 4600; 11/10/25 : 4750; 5/8/26 : 6650; 7/20/26 : 10000; 8/7/26 : 15000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -·- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) ▬
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
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COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/06/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$39.36
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb164.88
BYD Electronics (0285.HK)	O (04/28/2023)	HK$23.90
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.35
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.32
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb389.00
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,045.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb22.79
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.46
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,855.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.97
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.43
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$52.30
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb49.43
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.49
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$5.00
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb71.35
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$38.14
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.80
Sunny Optical (2382.HK)	E (03/23/2026)	HK$64.10
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb260.77
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb16.43
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.26

Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb411.16
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$180.00
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb106.92
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb24.40
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb808.44
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$1,079.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$21.80
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb30.58
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,050.00
Advantech (2395.TW)	O (01/20/2021)	NT$738.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,195.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,685.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$37.45
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,530.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,325.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.72
Chenbro (8210.TW)	O (07/23/2025)	NT$885.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,350.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$2,050.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,415.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$339.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.10
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$13,100.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb32.31
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$299.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.79
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.31
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb184.03
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.05
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$1,000.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$206.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.90
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$5,995.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$5.02
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb58.20
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$380.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,260.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$256.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.93
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$55.10
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,920.00
Lenovo (0992.HK)	O (07/10/2026)	HK$35.40
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,865.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,505.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$90.00
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$333.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb129.96
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb373.01
Unimicron (3037.TW)	O (02/23/2026)	NT$1,310.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$567.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50

Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,290.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$625.00
Zhen Ding (4958.TW)	++	NT$574.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$551.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,755.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

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Kaori

NT$1,600.00 - HIGH CONVICTION OUTPERFORM

Research provided by SinoPac

6 October 2026

Taiwan

Technology

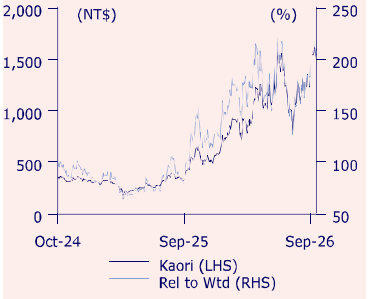
Reuters	8996.TW
Bloomberg	8996 TT
Priced on 6 October 2026	Taiwan Wtd @ 49,712.0
12M hi/lo	NT$1,655.00/401.00
12M price target	NT$2,710.00
±% potential	+69%
Shares in issue	92.5m
Free float (est.)	81.0%
Market cap	US$4.7bn
3M ADV	US$129.9m
Foreign s'holding	35.1%

Major shareholders
TU Shui-Cheng 3.6%
Tsai He-Ling 2.2%

SinoPac analysts:
Kevin SW Chen, SinoPac
Vanessa Huang, SinoPac

Stock performance (%)

	1M	3M	12M
Absolute	23.1	12.3	290.2
Relative	15.3	5.7	110.1
Abs (US$)	22.7	13.3	275.1



Source: Bloomberg

Growth in multiples

We forecast revenue to more than double next year

Bloom Energy's planned capacity expansion of up to 10GW strengthens Kaori's long-term SOFC growth outlook, while higher liquid cooling content and shipments should sustain liquid cooling momentum into 2027. CSP CDU heat exchangers and semiconductor equipment cooling could provide additional growth drivers from 4Q26. We raise our 2027/28 EPS forecasts by 14%/12% and increase our target price to NT$2,710 from NT$2,300. Reiterate HC O-PF rating.

Bloom Energy to double its capacity

Bloom Energy (BE) has announced plans to acquire a 158k sqft facility in Fremont, US, comparable in size to its existing Fremont manufacturing plant, which can produce up to 5GW. We believe the acquisition could support an eventual doubling of its fuel cell stack manufacturing capacity to 10GW. We expect its supply chain to support shipments of 2/4/6GW in 2026/27/28 to meet strong solid oxide fuel cell (SOFC). demand. To support this ramp, Kaori is expanding capacity to 6-7x its current level by end-2027. We now forecast Kaori's sales from BE to grow by 185%/66% YoY in 2027/28, versus our previous estimate of 101%/77%.

Strong liquid cooling momentum into 2027

With SpaceX planning an additional 6-8GW of datacentre capacity in 2027, we believe GPU server suppliers Supermicro and Dell will be key beneficiaries, supporting continued growth across their supply chains from 2H26 into 2027. Higher content value for Rubin manifolds, together with stronger CDU and rear-door heat exchanger (RDHx) shipments, should support Kaori's liquid cooling momentum. We forecast its liquid cooling revenue to grow by 60%/25% YoY in 2027/28.

Diversifying the growth engines

While SOFC and Supermicro-related liquid cooling remain Kaori's main growth drivers, we believe plate heat exchangers for CSP CDUs and liquid cooling applications in semiconductor equipment could emerge as additional growth drivers from 4Q26. These opportunities should broaden Kaori's customer and application exposure, supporting its longer-term growth outlook.

Earnings revisions and valuation

We raise our 2027/28SP EPS forecasts by 14%/12% to reflect higher expected demand and shipments for BE's fuel cells, alongside a stronger liquid cooling outlook. We increase our TP to NT$2,710 from NT$2,300, based on an unchanged 40x 2H27-1H28SP PE. Reiterate High Conviction Outperform rating.

Financials

Year to 31 December	**24A**	**25A**	**26SP**	**27SP**	**28SP**
Revenue (NT$m)	4,003	6,581	14,390	30,154	46,912
Rev forecast change (%)	-	-	*0.5*	*15.6*	*13.8*
Net profit (NT$m)	593	830	2,028	4,665	7,643
NP forecast change (%)	-	-	*(0.7)*	*15.7*	*13.4*
EPS (NT$)	6.56	9.07	21.47	48.35	79.21
SP/consensus (5) (EPS%)	-	-	97	114	107
EPS growth (% YoY)	1.7	38.2	136.8	125.2	63.8
PE (x)	244.0	176.5	74.5	33.1	20.2
Dividend yield (%)	0.2	0.2	0.5	0.9	1.5
ROE (%)	19.4	21.9	40.4	59.8	59.5
Net debt/equity (%)	(5.2)	(6.7)	22.1	10.9	3.2

Source: SinoPac

Change in target price

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Financials at a glance

Year to 31 December	2024A	2025A	2026SP	(% YoY)	2027SP	2028SP
Profit & Loss (NT$m)						
Revenue	**4,003**	**6,581**	**14,390**	***118.7***	**30,154**	**46,912**
Cogs (ex-D&A)	(2,650)	(4,587)	(10,010)		(20,537)	(31,642)
Gross Profit (ex-D&A)	**1,354**	**1,994**	**4,380**	***119.7***	**9,616**	**15,270**
SG&A and other expenses	(548)	(775)	(1,468)		(2,756)	(4,037)
Op Ebitda	**806**	**1,218**	**2,912**	***139***	**6,861**	**11,233**
Depreciation/amortisation	(159)	(195)	(367)		(997)	(1,631)
Op Ebit	**647**	**1,024**	**2,546**	***148.7***	**5,864**	**9,602**
Net interest inc/(exp)	(14)	(10)	(8)		(7)	(7)
Other non-Op items	117	39	49	*26.1*	(2)	(2)
Profit before tax	**750**	**1,052**	**2,586**	***145.9***	**5,855**	**9,593**
Taxation	(157)	(222)	(558)		(1,190)	(1,950)
Profit after tax	**593**	**830**	**2,028**	***144.5***	**4,665**	**7,643**
Minority interest	0	0	0		0	0
Net profit	**593**	**830**	**2,028**	***144.5***	**4,665**	**7,643**
Adjusted profit	**593**	**830**	**2,028**	***144.5***	**4,665**	**7,643**
Cashflow (NT$m)	**2024A**	**2025A**	**2026SP**	**(% YoY)**	**2027SP**	**2028SP**
Operating profit	**647**	**1,024**	**2,546**	***148.7***	**5,864**	**9,602**
Depreciation/amortisation	159	195	367	*88.4*	997	1,631
Working capital changes	331	(416)	(3)		(1,625)	(4,867)
Other items	(96)	(205)	(518)		(1,199)	(1,959)
Net operating cashflow	**1,041**	**598**	**2,392**	***300.1***	**4,037**	**4,406**
Capital expenditure	(190)	(313)	(3,617)		(3,000)	(2,460)
Free cashflow	**851**	**285**	**(1,225)**		**1,037**	**1,946**
M&A/Others	(4)	94	0		0	0
Net investing cashflow	**(195)**	**(219)**	**(3,617)**		**(3,000)**	**(2,460)**
Increase in loans	787	(81)	200		200	200
Dividends	(358)	(366)	(365)		(811)	(1,400)
Net equity raised/other	(4)	(132)	5		5	5
Net financing cashflow	**426**	**(579)**	**(160)**		**(606)**	**(1,195)**
Incr/(decr) in net cash	1,273	(200)	(1,385)		431	752
Exch rate movements	8	5	5	*0*	5	5
Balance sheet (NT$m)	**2024A**	**2025A**	**2026SP**	**(% YoY)**	**2027SP**	**2028SP**
Cash & equivalents	1,871	1,677	298	*(82.2)*	734	1,491
Accounts receivable	822	1,817	1,799	*(1)*	3,769	5,864
Other current assets	1,163	2,086	2,308	*10.6*	3,822	8,551
Fixed assets	1,966	2,075	5,332	*156.9*	7,335	8,164
Investments	145	252	298	*18*	298	298
Intangible assets	9	15	15	*0*	15	15
Other non-current assets	127	204	127	*(37.8)*	127	127
Total assets	**6,103**	**8,127**	**10,176**	***25.2***	**16,100**	**24,510**
Short-term debt	958	1,174	1,174	*0*	1,174	1,174
Accounts payable	362	1,529	1,729	*13.1*	3,589	5,545
Other current liabs	557	872	872	*0*	872	872
Long-term debt/CBs	735	224	424	*89.4*	624	824
Provisions/other LT liabs	53	152	97	*(36.1)*	107	118
Shareholder funds	3,395	4,171	5,874	*40.8*	9,728	15,971
Minorities/other equity	42	6	6	*0*	6	6
Total liabs & equity	**6,103**	**8,127**	**10,176**	***25.2***	**16,100**	**24,510**
Ratio analysis	**2024A**	**2025A**	**2026SP**	**(% YoY)**	**2027SP**	**2028SP**
Revenue growth (% YoY)	(7.4)	64.4	118.7		109.5	55.6
Ebitda margin (%)	20.1	18.5	20.2		22.8	23.9
Ebit margin (%)	16.2	15.6	17.7		19.4	20.5
Net profit growth (%)	2.9	39.9	144.5		130.0	63.8
Op cashflow growth (% YoY)	53.3	(42.6)	300.1		68.8	9.1
Capex/sales (%)	4.8	4.8	25.1		9.9	5.2
Net debt/equity (%)	(5.2)	(6.7)	22.1		10.9	3.2
Net debt/Ebitda (x)	-	-	0.4		0.2	0.0
ROE (%)	19.4	21.9	40.4		59.8	59.5
ROIC (%)	15.7	23.2	37.1		53.1	56.9

Source: SinoPac

Figure 1

Forecast revisions

	2026SP			2027SP			2028SP		
(NT$m)	old	new	% ch	old	new	% ch	old	new	% ch
Revenue	14,318	14,390	0%	26,078	30,154	16%	41,233	46,912	14%
Gross profit	4,027	4,014	0%	7,463	8,620	16%	12,024	13,639	13%
Opex	1,460	1,468	0%	2,387	2,756	15%	3,549	4,037	14%
Operating profit	2,566	2,546	0%	5,076	5,864	16%	8,475	9,602	13%
Pretax profit	2,605	2,586	0%	5,060	5,855	16%	8,460	9,593	13%
Net profit	2,043	2,028	0%	4,032	4,665	16%	6,740	7,643	13%
EPS (NT$)	21.40	21.02	(1%)	42.23	48.35	14%	70.60	79.21	12%
Margin									
Gross margin	28.1%	27.9%		28.6%	28.6%		29.2%	29.1%	
Operating margin	17.9%	17.7%		19.5%	19.4%		20.6%	20.5%	
Net margin	14.3%	14.1%		15.5%	15.5%		16.3%	16.3%	

Source: SinoPac

Figure 2

Quarterly P&L

(NT$m)	1Q26	2Q26	3Q26SP	4Q26SP	1Q27SP	2Q27SP	3Q27SP	4Q27SP	2025	2026SP	2027SP	2028SP
Sales	3,423	2,305	3,743	4,919	5,395	6,637	8,334	9,788	6,581	14,390	30,154	46,912
Gross profit	993	599	1,030	1,391	1,529	1,890	2,383	2,818	1,799	4,014	8,620	13,639
Opex	285	301	385	497	513	611	742	891	775	1,468	2,756	4,037
Operating profit	709	298	644	895	1,016	1,280	1,642	1,927	1,024	2,546	5,864	9,602
Non-OP	51	(6)	(3)	(2)	(2)	(2)	(2)	(2)	28	41	(9)	(9)
Pretax profit	760	293	641	893	1,014	1,278	1,639	1,925	1,052	2,586	5,855	9,593
Net profit	606	201	511	711	808	1,018	1,306	1,533	830	2,028	4,665	7,643
EPS (NT$)	6.54	2.16	5.40	7.53	8.37	10.55	13.54	15.89	8.97	21.02	48.35	79.21
Growth												
Sales QoQ	49%	(33%)	62%	34%	10%	23%	26%	17%				
Sales YoY	238%	55%	110%	115%	58%	188%	127%	99%	64%	119%	111%	56%
EPS QoQ	90%	(67%)	150%	35%	11%	26%	28%	17%				
EPS YoY	347%	62%	91%	119%	28%	388%	143%	111%	37%	134%	128%	64%
Margin												
Gross margin	29.0%	26.0%	27.5%	28.3%	28.3%	28.5%	28.6%	28.8%	27.3%	27.9%	28.6%	29.1%
Opex	8.3%	13.1%	10.3%	10.1%	9.5%	9.2%	8.9%	9.1%	11.8%	10.2%	9.1%	8.6%
OP margin	20.7%	12.9%	17.2%	18.2%	18.8%	19.3%	19.7%	19.7%	15.6%	17.7%	19.4%	20.5%
Non-OP	1.5%	(0.2%)	-0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.4%	0.3%	0.0%	0.0%
Net margin	17.7%	8.7%	13.6%	14.5%	15.0%	15.3%	15.7%	15.7%	12.6%	14.1%	15.5%	16.3%

Source: SinoPac

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Figure 3

Sales mix and GM assumptions

Sales (NT$m)	2025	2026SP	2027SP	2028SP
Heat Energy Product	4,878	12,073	25,439	37,548
Plate Heat Exchanger	1,702	2,317	4,715	9,364
Total	6,581	14,390	30,154	46,912
YoY (%)	2025	2026SP	2027SP	2028SP
Heat Energy Product	104%	147%	112%	48%
Plate Heat Exchanger	5%	36%	101%	99%
Total	64%	119%	111%	56%
Sales mix (%)	2025	2026SP	2027SP	2028SP
Heat Energy Product	74%	84%	84%	80%
Plate Heat Exchanger	26%	16%	16%	20%
Total	100%	100%	100%	100%
Gross margin (%)	2025	2026SP	2027SP	2028SP
Heat Energy Product	27%	28%	28%	28%
Plate Heat Exchanger	29%	29%	30%	32%
Total	27%	28%	29%	29%

Source: SinoPac

Figure 4

Forward PE

Source: SinoPac

Figure 5

Forward PB

Source: SinoPac

Figure 6

Target price change

	Target PE (x)	Valuation period	12-month rolling EPS (NT$)	Target Price (NT$)
Old	40	2H27-1H28	57.6	2,300
New	40	2H27-1H28	67.1	2,710

Source: SinoPac

Investment thesis

Kaori is well positioned to capitalise on rising demand for liquid cooling, and it expects related product shipments to ramp up. The fuel-cell business should be a huge growth driver in 2026-27 to meet AI power infrastructure demand in the US. In the longer term, we believe the company will also benefit from heat-pump demand as Europe is phasing out gas boilers, despite some noise about near-term demand. Kaori has been planning capacity expansion earlier than peers to fully capture demand, with rising exposure to large clients. We thus expect the company to enjoy robust earnings in coming years.

Catalysts

The major growth catalysts largely depend on fuel-cell projects for AI datacentres taking off, increasing content value in liquid cooling offerings, and a clearer outlook for hydrogen electrolyzer development.

Valuation details

We adopt a PE-based valuation methodology as we believe the market will mainly focus on the earnings growth outlook, which is better reflected with PE metrics. We base our valuation on 40x PE, which is around +1sd of its five-year average and supported by an 93% EPS Cagr in 2026-28SP, to account for the surging demand for AI server liquid cooling and power infrastructure. We are positive on the company's long-term outlook given its growing stationary fuel cell demand from AI datacentres, as well as its efforts to diversify customer exposure for liquid cooling business. We believe the company can maintain its competitive edge over other cooling players given its expertise in vacuum welding technology to manufacture high-quality products.

Investment risks

Downside risks: 1) increased competition for Bloom Energy's fuel cell business or new technology solutions replacing fuel cells; 2) slower kick-off of new fuel cell projects for AI datacentres; 3) weaker-than-expected liquid cooling adoption; 4) slower rollout of Europe/US government subsidies for heat pump installations.

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Detailed financials

Profit & Loss (NT$m)

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Revenue	**2,844**	**4,326**	**4,003**	**6,581**	**14,390**	**30,154**	**46,912**
Cogs (ex-D&A)	(1,944)	(2,973)	(2,650)	(4,587)	(10,010)	(20,537)	(31,642)
Gross Profit (ex-D&A)	**899**	**1,353**	**1,354**	**1,994**	**4,380**	**9,616**	**15,270**
Research & development costs	(81)	(84)	(113)	(218)	(364)	(551)	(807)
Selling & marketing expenses	(132)	(139)	(139)	(159)	(335)	(689)	(1,009)
Other SG&A	(227)	(289)	(296)	(398)	(769)	(1,516)	(2,221)
Other Op Expenses ex-D&A	-	-	-	-	-	-	-
Op Ebitda	**460**	**841**	**806**	**1,218**	**2,912**	**6,861**	**11,233**
Depreciation/amortisation	(114)	(129)	(159)	(195)	(367)	(997)	(1,631)
Op Ebit	**346**	**711**	**647**	**1,024**	**2,546**	**5,864**	**9,602**
Interest income	2	8	9	20	24	26	26
Interest expense	(13)	(33)	(22)	(31)	(32)	(33)	(32)
Net interest inc/(exp)	(12)	(25)	(14)	(10)	(8)	(7)	(7)
Associates/investments	(6)	(2)	12	3	0	0	0
Forex/other income	57	31	50	16	40	(2)	(2)
Asset sales/other cash items	-	-	-	-	-	-	-
Provisions/other non-cash items	5	14	55	20	9	0	0
Asset revaluation/Exceptional items	-	-	-	-	-	-	-
Profit before tax	**390**	**730**	**750**	**1,052**	**2,586**	**5,855**	**9,593**
Taxation	(89)	(153)	(157)	(222)	(558)	(1,190)	(1,950)
Profit after tax	**301**	**577**	**593**	**830**	**2,028**	**4,665**	**7,643**
Preference dividends	-	-	-	-	-	-	-
Profit for period	**301**	**577**	**593**	**830**	**2,028**	**4,665**	**7,643**
Minority interest	0	0	0	0	0	0	0
Net profit	**301**	**577**	**593**	**830**	**2,028**	**4,665**	**7,643**
Extraordinaries/others	0	0	0	0	0	0	0
Profit avail to ordinary shares	301	577	593	830	2,028	4,665	7,643
Dividends	(134)	(134)	(358)	(366)	(365)	(811)	(1,400)
Retained profit	167	442	236	464	1,663	3,854	6,243
Adjusted profit	**301**	**577**	**593**	**830**	**2,028**	**4,665**	**7,643**
EPS (NT$)	3.4	6.4	6.6	9.1	21.5	48.4	79.2
Adj EPS [pre excep] (NT$)	3.4	6.4	6.6	9.1	21.5	48.4	79.2
Core EPS (NT$)	3.4	6.4	6.6	9.1	21.5	48.4	79.2
DPS (NT$)	1.5	4.0	2.9	3.9	8.4	14.5	23.8

Profit & loss ratios

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Growth (%)							
Revenue growth (% YoY)	27.4	52.1	(7.4)	64.4	118.7	109.5	55.6
Ebitda growth (% YoY)	34.1	82.7	(4.2)	51.3	139.0	135.6	63.7
Ebit growth (% YoY)	50.3	105.5	(9.1)	58.3	148.7	130.3	63.7
Net profit growth (%)	101.8	91.5	2.9	39.9	144.5	130.0	63.8
EPS growth (% YoY)	101.8	91.5	1.7	38.2	136.8	125.2	63.8
Adj EPS growth (% YoY)	101.8	91.5	1.7	38.2	136.8	125.2	63.8
DPS growth (% YoY)	(0.3)	166.3	(27.5)	36.8	113.1	72.5	63.8
Core EPS growth (% YoY)	101.8	91.5	1.7	38.2	136.8	125.2	63.8
Margins (%)							
Gross margin (%)	31.6	31.3	33.8	30.3	30.4	31.9	32.6
Ebitda margin (%)	16.2	19.4	20.1	18.5	20.2	22.8	23.9
Ebit margin (%)	12.2	16.4	16.2	15.6	17.7	19.4	20.5
Net profit margin (%)	10.6	13.3	14.8	12.6	14.1	15.5	16.3
Core profit margin	10.6	13.3	14.8	12.6	14.1	15.5	16.3
Op cashflow margin	(10.1)	15.7	26.0	9.1	16.6	13.4	9.4
Returns (%)							
ROE (%)	15.4	24.3	19.4	21.9	40.4	59.8	59.5
ROA (%)	7.1	12.5	9.4	11.3	21.8	35.6	37.7
ROIC (%)	9.8	17.4	15.7	23.2	37.1	53.1	56.9
ROCE (%)	12.1	21.2	19.2	28.6	46.0	65.2	70.4
Other key ratios (%)							
Effective tax rate (%)	22.8	21.0	20.9	21.1	21.6	20.3	20.3
Ebitda/net int exp (x)	39.6	33.5	57.9	116.4	365.4	1,036.5	1,723.6
Exceptional or extraord. inc/PBT (%)	-	-	-	-	-	-	-
Dividend payout (%)	44.4	61.7	44.0	43.5	39.2	30.0	30.0

Source: SinoPac

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The Global Equities Database

Balance sheet (NT$m)

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Cash & equivalents	249	591	1,871	1,677	298	734	1,491
Accounts receivable	580	798	822	1,817	1,799	3,769	5,864
Inventories	1,180	1,235	1,052	1,853	2,075	3,589	8,318
Other current assets	236	78	110	232	232	232	232
Current assets	**2,245**	**2,702**	**3,856**	**5,580**	**4,404**	**8,325**	**15,906**
Fixed assets	1,625	1,839	1,966	2,075	5,332	7,335	8,164
Investments	159	132	145	252	298	298	298
Goodwill	0	0	0	0	0	0	0
Other intangible assets	0	3	9	15	15	15	15
Other non-current assets	128	154	127	204	127	127	127
Total assets	**4,156**	**4,829**	**6,103**	**8,127**	**10,176**	**16,100**	**24,510**
Short term loans/OD	838	255	958	1,174	1,174	1,174	1,174
Accounts payable	255	206	362	1,529	1,729	3,589	5,545
Accrued expenses	-	-	-	-	-	-	-
Taxes payable	0	0	0	0	0	0	0
Other current liabs	383	535	557	872	872	872	872
Current liabilities	**1,476**	**996**	**1,877**	**3,575**	**3,775**	**5,635**	**7,591**
Long-term debt/leases/other	595	1,092	735	224	424	624	824
Convertible bonds	-	-	-	-	-	-	-
Provisions/other LT liabs	26	38	53	152	97	107	118
Total liabilities	**2,097**	**2,125**	**2,666**	**3,950**	**4,296**	**6,366**	**8,533**
Share capital	894	894	915	925	965	965	965
Retained earnings	368	778	956	1,451	2,911	6,298	11,777
Reserves/others	784	1,037	1,524	1,795	1,998	2,465	3,229
Shareholder funds	**2,045**	**2,709**	**3,395**	**4,171**	**5,874**	**9,728**	**15,971**
Minorities/other equity	14	(5)	42	6	6	6	6
Total equity	**2,059**	**2,704**	**3,437**	**4,177**	**5,880**	**9,734**	**15,977**
Total liabs & equity	**4,156**	**4,829**	**6,103**	**8,127**	**10,176**	**16,100**	**24,510**
Total debt	1,433	1,346	1,694	1,397	1,597	1,797	1,997
Net debt	1,184	755	(178)	(280)	1,300	1,063	506
Adjusted EV	144,017	143,628	146,063	147,447	155,381	155,145	154,587
BVPS (NT$)	22.9	30.3	37.1	45.1	60.9	100.8	165.5

Balance sheet ratios

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Key ratios							
Current ratio (x)	1.5	2.7	2.1	1.6	1.2	1.5	2.1
Growth in total assets (% YoY)	21.8	16.2	26.4	33.2	25.2	58.2	52.2
Growth in capital employed (% YoY)	30.8	6.7	(5.8)	19.6	84.2	50.4	52.7
Net debt to operating cashflow (x)	(4.1)	1.1	-	-	0.5	0.3	0.1
Gross debt to operating cashflow (x)	(5.0)	2.0	1.6	2.3	0.7	0.4	0.5
Gross debt to Ebitda (x)	3.1	1.6	2.1	1.1	0.5	0.3	0.2
Net debt/Ebitda (x)	2.6	0.9	-	-	0.4	0.2	0.0
Gearing							
Net debt/equity (%)	57.5	27.9	(5.2)	(6.7)	22.1	10.9	3.2
Gross debt/equity (%)	69.6	49.8	49.3	33.5	27.2	18.5	12.5
Interest cover (x)	26.1	21.6	29.2	33.8	79.9	181.2	296.5
Debt cover (x)	(0.2)	0.5	0.6	0.4	1.5	2.2	2.2
Net cash per share (NT$)	(13.2)	(8.4)	2.0	3.1	(13.8)	(11.0)	(5.2)
Working capital analysis							
Inventory days	155.7	142.1	148.6	110.9	69.1	48.0	65.3
Debtor days	60.4	58.1	73.9	73.2	45.9	33.7	37.5
Creditor days	39.6	27.1	36.9	72.2	57.3	45.1	50.1
Working capital/Sales (%)	47.7	31.7	26.6	22.8	10.5	10.4	17.0
Capital employed analysis							
Sales/Capital employed (%)	87.7	125.1	122.8	168.8	200.4	279.3	284.6
EV/Capital employed (%)	4,441.3	4,152.4	4,481.3	3,783.3	2,164.1	1,436.9	937.8
Working capital/Capital employed (%)	41.9	39.6	32.7	38.5	21.0	29.0	48.5
Fixed capital/Capital employed (%)	50.1	53.2	60.3	53.3	74.3	67.9	49.5
Other ratios (%)							
PB (x)	69.9	52.8	43.1	35.5	26.3	15.9	9.7
EV/Ebitda (x)	313.1	170.9	181.3	121.0	53.4	22.6	13.8
EV/OCF (x)	(500.3)	211.5	140.3	246.6	65.0	38.4	35.1
EV/FCF (x)	(372.1)	435.0	171.6	517.6	(126.9)	149.6	79.4
EV/Sales (x)	50.6	33.2	36.5	22.4	10.8	5.1	3.3
Capex/depreciation (%)	91.9	284.1	126.0	169.7	1,004.2	301.0	150.8

Source: SinoPac

evalu@tor
The Global Equities Database

Cashflow (NT$m)

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Operating profit	346	711	647	1,024	2,546	5,864	9,602
Operating adjustments	(45)	(135)	(54)	(194)	(518)	(1,199)	(1,959)
Depreciation/amortisation	114	129	159	195	367	997	1,631
Working capital changes	(711)	(55)	331	(416)	(3)	(1,625)	(4,867)
Interest paid / other financial expenses	0	0	0	0	0	0	0
Tax paid	-	-	-	-	-	-	-
Other non-cash operating items	8	28	(42)	(11)	0	0	0
Net operating cashflow	(288)	679	1,041	598	2,392	4,037	4,406
Capital expenditure	(99)	(349)	(190)	(313)	(3,617)	(3,000)	(2,460)
Free cashflow	(387)	330	851	285	(1,225)	1,037	1,946
Acq/inv/disposals	26	-	-	92	-	-	-
Int, invt & associate div	(93)	3	(4)	2	-	-	-
Net investing cashflow	(167)	(346)	(195)	(219)	(3,617)	(3,000)	(2,460)
Increase in loans	341	139	787	(81)	200	200	200
Dividends	(134)	(134)	(358)	(366)	(365)	(811)	(1,400)
Net equity raised/others	(3)	(3)	(4)	(132)	5	5	5
Net financing cashflow	204	2	426	(579)	(160)	(606)	(1,195)
Incr/(decr) in net cash	(251)	335	1,273	(200)	(1,385)	431	752
Exch rate movements	2	(4)	8	5	5	5	5
Opening cash	499	259	591	1,871	1,677	298	734
Closing cash	249	591	1,871	1,677	298	734	1,491
OCF PS (NT$)	(3.2)	7.6	11.5	6.5	25.3	41.8	45.7
FCF PS (NT$)	(4.3)	3.7	9.4	3.1	(13.0)	10.7	20.2

Cashflow ratio analysis

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Growth (%)							
Op cashflow growth (% YoY)	(197.5)	nm	53.3	(42.6)	300.1	68.8	9.1
FCF growth (% YoY)	(298.8)	-	157.7	(66.5)	(529.9)	-	87.7
Capex growth (%)	(1.4)	251.5	(45.5)	64.5	1,055.3	(17.1)	(18.0)
Other key ratios (%)							
Capex/sales (%)	3.5	8.1	4.8	4.8	25.1	9.9	5.2
Capex/op cashflow (%)	(34.5)	51.4	18.3	52.4	151.2	74.3	55.8
Operating cashflow payout ratio (%)	-	52.4	25.1	60.4	33.2	34.7	52.0
Cashflow payout ratio (%)	-	19.7	34.3	61.2	15.3	20.1	31.8
Free cashflow payout ratio (%)	-	40.6	42.0	128.4	-	78.2	71.9

DuPont analysis

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Ebit margin (%)	12.2	16.4	16.2	15.6	17.7	19.4	20.5
Asset turnover (x)	0.8	1.0	0.7	0.9	1.6	2.3	2.3
Interest burden (x)	1.1	1.0	1.2	1.0	1.0	1.0	1.0
Tax burden (x)	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Return on assets (%)	7.1	12.5	9.4	11.3	21.8	35.6	37.7
Leverage (x)	1.9	1.9	1.8	1.9	1.8	1.7	1.6
ROE (%)	15.4	24.3	19.4	21.9	40.4	59.8	59.5

EVA® analysis

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Ebit adj for tax	267	562	511	807	1,996	4,672	7,650
Average invested capital	2,721	3,237	3,266	3,482	5,388	8,793	13,455
ROIC (%)	9.8	17.4	15.7	23.2	37.1	53.1	56.9
Cost of equity (%)	9.7	9.7	9.7	9.7	9.7	9.7	9.7
Cost of debt (adj for tax)	2.0	2.1	2.1	2.1	2.0	2.1	2.1
Weighted average cost of capital (%)	7.5	7.5	7.5	7.5	7.5	7.5	7.5
EVA/IC (%)	2.3	9.8	8.1	15.6	29.5	45.6	49.3
EVA (NT$m)	62	318	265	545	1,590	4,009	6,635

Source: SinoPac

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Companies mentioned

Kaori (8996 TT - NT$1,600.00 - HIGH CONVICTION OUTPERFORM)
Bloom Energy (N-R)
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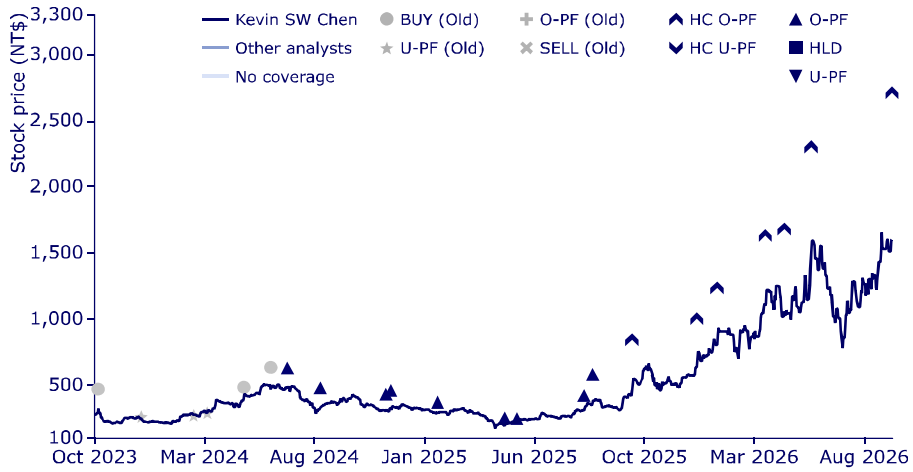
Recommendation history of Kaori Heat Treatment Co Ltd 8996 TT



Date	Rec	Target	Date	Rec	Target
LATEST	HC O-PF	2,710.00	20 Jan 2025	O-PF	370.00
17 Jun 2026	HC O-PF	2,300.00	17 Nov 2024	O-PF	460.00
11 May 2026	HC O-PF	1,680.00	10 Nov 2024	O-PF	430.00
15 Apr 2026	HC O-PF	1,630.00	12 Aug 2024	O-PF	480.00
08 Feb 2026	HC O-PF	1,234.00	28 Jun 2024	O-PF	630.00
11 Jan 2026	HC O-PF	1,000.00	05 Jun 2024	BUY	630.00
14 Oct 2025	HC O-PF	840.00	29 Apr 2024	BUY	480.00
21 Aug 2025	O-PF	580.00	10 Mar 2024	U-PF	290.00
09 Aug 2025	O-PF	420.00	20 Feb 2024	U-PF	271.00
09 May 2025	O-PF	245.00	10 Dec 2023	U-PF	260.00
22 Apr 2025	O-PF	250.00	12 Oct 2023	BUY	466.00

Note: At 6pm (HKT) on 28 June 2024, we changed our ratings to Outperform (O-PF), Hold (HLD) and Underperform (U-PF); and on 16 December 2024, we added the ratings: High-Conviction Outperform (HC O-PF) and High-Conviction Underperform (HC U-PF). Please see Key to CLSA/SinoPac investment rankings below for details. Source: CLSA, SinoPac

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ACES

NT$172.00 - OUTPERFORM

Research provided by SinoPac

6 October 2026

Taiwan

Technology

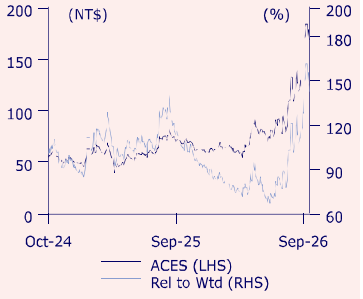
Reuters	3605.TW
Bloomberg	3605 TT
Priced on 6 October 2026	Taiwan Wtd @ 49,712.0
12M hi/lo	NT$183.50/53.94
12M price target	NT$270.00
±% potential	+57%
Shares in issue	163.8m
Free float (est.)	76.5%
Market cap	US$928m
3M ADV	US$58.2m
Foreign s'holding	21.6%

Major shareholders
Yuan Wan-Ting 5.3%
Keyan Investment Co., Ltd. 4.0%

SinoPac analysts:
Vanessa Huang, SinoPac
Kevin SW Chen, SinoPac

Stock performance (%)

	1M	3M	12M
Absolute	45.8	96.1	142.0
Relative	36.5	84.6	30.3
Abs (US$)	45.3	97.9	132.6



Source: Bloomberg

Wired for growth

Initiate coverage with an Outperform rating and target price of NT$270

ACES is entering a two-engine growth phase. It is a key power cable supplier to Bloom Energy as Bloom doubles its fuel-cell capacity, and it is expanding its high-speed connector content from AI server daughterboards to ASIC motherboards, as a high-speed cable leg comes on stream. We forecast a 72% EPS Cagr over 2026-28, with OPM expanding from 7.4% in 2026 to 11.0% in 2027. Our TP of NT$270 is based on 20x 2H27-1H28SP EPS, and we initiate coverage at Outperform.

Bloom Energy the key earnings driver in 2027

ACES's multiyear partnership with Bloom locks in its share as Bloom plans to lift capacity from 2GW in 2026 to 4GW/6GW in 2027/28, based on our checks. ACES has shipped fuel-cell power cables to Bloom since 1Q26 and will add external DLO high-voltage cables, with ASP of US$80-100k per set. We expect Bloom power cables to be NT$5.8bn of sales and account for 28% of group revenue in 2027. Power products need little incremental R&D and benefit from strong operating leverage. We expect overall OPM to expand from 7.4% in 2026 to 11.0% in 2027.

High-speed products a new driver

Previously, ACES supplied only daughterboard connectors, at 10-20 units per rack. On ASIC platforms, where ODMs have more say in supplier selection, ACES has secured the opportunity to ship both motherboard and daughterboard connectors (about 30 units per rack) and broadens its connector product offerings. With Gen6 connectors carrying around a 15% ASP premium over Gen5, and ACES as one of only four vendors qualified by CSPs at Gen6, we forecast server revenue growth of 107%/74% YoY in 2026/27. The newly added Gen6 PCIe cable product, qualified by two CSPs, began shipping in April 2026 at 20% of GM, with room to rise towards 30% as yield improves. Hence, we expect GM to recover to 26% in 2027.

Capacity in place ahead of demand

ACES has committed firm capacity ahead of demand. For power cables, it has leased a new facility next to its Philippine plant, with new capacity planned to come online in 4Q26. For server products, it plans to raise connector capacity from 4m to 5m units/mo and will expand high-speed cable production lines from two to eight.

Initiate coverage with an Outperform rating

We forecast 2027/28SP EPS of NT$11.19/NT$14.93, with our 2027 forecast 48% ahead of consensus. Our TP of NT$270 is based on 20x 2H27-1H28SP EPS, backed by a 72% EPS Cagr in 2026-28SP driven by power cable ramp-up and high-speed connector shipment growth. We initiate coverage with an Outperform rating.

Financials

Year to 31 December	24A	25A	26SP	27SP	28SP
Revenue (NT$m)	9,771	10,860	13,199	20,531	26,733
Net profit (NT$m)	344	661	830	1,961	2,617
EPS (NT$)	2.40	4.35	5.07	11.19	14.93
SP/consensus (4) (EPS%)	-	-	104	148	-
EPS growth (% YoY)	nm	81.5	16.6	120.5	33.4
PE (x)	71.8	39.5	33.9	15.4	11.5
Dividend yield (%)	0.4	0.9	1.0	2.3	3.0
FCF yield (%)	(1.1)	(0.4)	0.4	0.9	0.2
PB (x)	4.1	3.6	3.6	3.0	2.5
ROE (%)	6.0	9.5	10.3	21.1	23.6
Net debt/equity (%)	21.9	7.9	(0.1)	(1.1)	4.1

Source: SinoPac

Initiating coverage

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Great Solid Meh

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Financials at a glance

Year to 31 December	2024A	2025A	2026SP	(% YoY)	2027SP	2028SP
Profit & Loss (NT$m)						
Revenue	**9,771**	**10,860**	**13,199**	***21.5***	**20,531**	**26,733**
Cogs (ex-D&A)	(6,710)	(7,317)	(9,157)		(14,566)	(19,258)
Gross Profit (ex-D&A)	**3,061**	**3,543**	**4,042**	***14.1***	**5,965**	**7,475**
SG&A and other expenses	(2,043)	(2,227)	(2,313)		(3,085)	(3,923)
Op Ebitda	**1,018**	**1,316**	**1,729**	***31.4***	**2,881**	**3,551**
Depreciation/amortisation	(740)	(719)	(758)		(618)	(491)
Op Ebit	**278**	**597**	**970**	***62.6***	**2,263**	**3,060**
Net interest inc/(exp)	(62)	(48)	(38)		(38)	(38)
Other non-Op items	170	267	114	*(57.4)*	196	209
Profit before tax	**387**	**815**	**1,046**	***28.4***	**2,421**	**3,231**
Taxation	(43)	(154)	(216)		(460)	(614)
Profit after tax	**344**	**661**	**830**	***25.6***	**1,961**	**2,617**
Minority interest	0	0	0		0	0
Net profit	**344**	**661**	**830**	***25.6***	**1,961**	**2,617**
Adjusted profit	**344**	**661**	**830**	***25.6***	**1,961**	**2,617**
Cashflow (NT$m)	**2024A**	**2025A**	**2026SP**	**(% YoY)**	**2027SP**	**2028SP**
Operating profit	**278**	**597**	**970**	***62.6***	**2,263**	**3,060**
Depreciation/amortisation	740	719	758	*5.4*	618	491
Working capital changes	(184)	29	(379)		(1,451)	(2,140)
Other items	137	(134)	(150)		(311)	(451)
Net operating cashflow	**971**	**1,210**	**1,199**	***(0.9)***	**1,119**	**960**
Capital expenditure	(1,253)	(1,308)	(1,075)		(855)	(887)
Free cashflow	**(283)**	**(98)**	**124**		**264**	**73**
M&A/Others	137	324	0		0	0
Net investing cashflow	**(1,117)**	**(984)**	**(1,075)**		**(855)**	**(887)**
Increase in loans	62	508	1,000	*96.9*	-	-
Dividends	0	(112)	(239)		(292)	(689)
Net equity raised/other	(43)	(252)	0		0	0
Net financing cashflow	**19**	**144**	**761**	***427.5***	**(292)**	**(689)**
Incr/(decr) in net cash	(127)	370	885	*139.1*	(28)	(616)
Exch rate movements	70	112	0		0	1
Balance sheet (NT$m)	**2024A**	**2025A**	**2026SP**	**(% YoY)**	**2027SP**	**2028SP**
Cash & equivalents	2,001	2,483	3,368	*35.6*	3,341	2,725
Accounts receivable	3,352	3,331	3,473	*4.3*	5,133	7,426
Other current assets	1,537	1,597	2,640	*65.3*	3,957	5,098
Fixed assets	4,415	4,871	5,136	*5.4*	5,526	6,104
Investments	1,023	1,267	888	*(29.9)*	888	888
Intangible assets	154	140	140	*0*	140	140
Other non-current assets	1,115	1,161	1,161	*0*	1,161	1,161
Total assets	**13,596**	**14,849**	**16,808**	***13.2***	**20,146**	**23,544**
Short-term debt	428	913	1,169	*28*	1,037	1,040
Accounts payable	1,961	2,016	2,833	*40.5*	4,338	5,643
Other current liabs	1,127	1,086	1,076	*(1)*	1,096	1,086
Long-term debt/CBs	2,991	2,194	2,194	*0*	2,194	2,194
Provisions/other LT liabs	628	706	880	*24.7*	1,156	1,328
Shareholder funds	6,260	7,726	8,447	*9.3*	10,117	12,045
Minorities/other equity	202	208	208	*0*	208	208
Total liabs & equity	**13,596**	**14,849**	**16,808**	***13.2***	**20,146**	**23,544**
Ratio analysis	**2024A**	**2025A**	**2026SP**	**(% YoY)**	**2027SP**	**2028SP**
Revenue growth (% YoY)	15.1	11.1	21.5		55.5	30.2
Ebitda margin (%)	10.4	12.1	13.1		14.0	13.3
Ebit margin (%)	2.9	5.5	7.4		11.0	11.4
Net profit growth (%)	nm	92.2	25.6		136.3	33.4
Op cashflow growth (% YoY)	46.3	24.7	(0.9)		(6.6)	(14.2)
Capex/sales (%)	12.8	12.0	8.1		4.2	3.3
Net debt/equity (%)	21.9	7.9	(0.1)		(1.1)	4.1
Net debt/Ebitda (x)	1.4	0.5	-		-	0.1
ROE (%)	6.0	9.5	10.3		21.1	23.6
ROIC (%)	3.5	6.3	9.3		19.2	20.9

Source: SinoPac

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Bloom Energy the key earnings driver in 2027

Each set of power cables for Bloom carries dollar content of US$80-100k

We see tremendous advantages for ACES from Bloom Energy's strong demand backed by its aggressive expansion plan. ACES supplies inner power cables used inside each fuel cell in 2026 and will begin supplying external diesel locomotive (DLO) cables that connect fuel cell stacks into a fuel cell module (energy server). Each set (including inner cable and external cable) carries dollar content of US$80-100k.

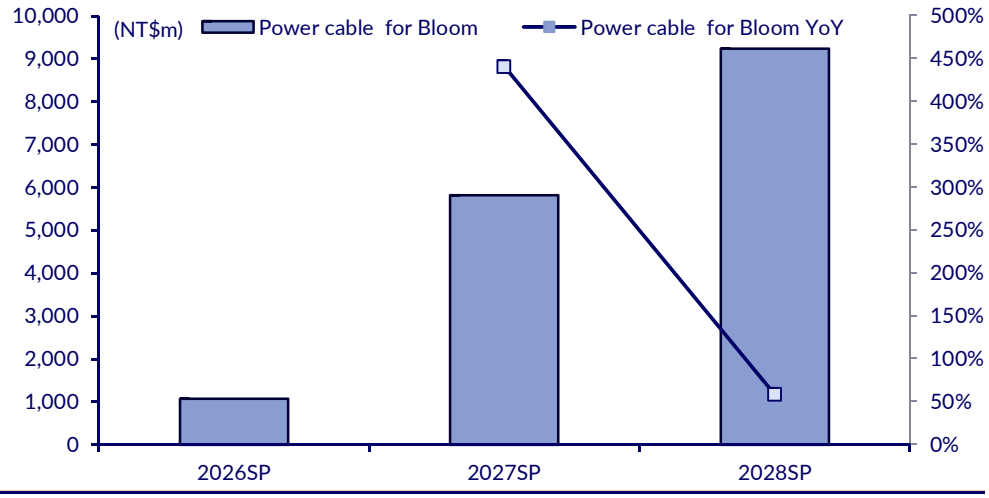
ACES holds at least a 30% share

Furthermore, ACES has worked with Bloom on high-voltage power cables for SOFC for around six years, even though the real contribution did not transpire until 2026. On the back of this long-standing relationship, ACES is the key supplier during the development phase and holds at least a 30% share in the mass production stage. Together with these considerations, we expect inner cables to contribute revenue of NT$5,113m/NT$8,074m in 2027/28, while external cables will add revenue of NT$707m/NT$1,154m.

We expect power cables for Bloom to grow 441%/59% in 2027/28

Figure 1

Power cable revenue from Bloom Energy and YoY change



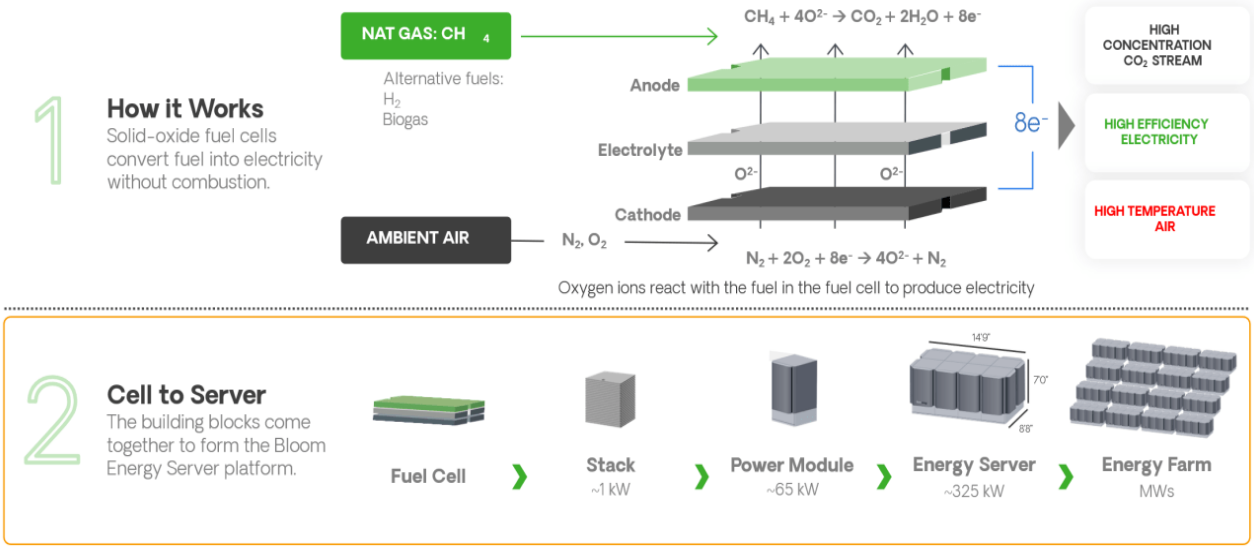
Source: SinoPac

Bloom delivers compelling time-to-power

As power scarcity remains the key bottleneck for AI datacentre construction, Bloom serves as a suitable solution and offers fuel cell systems that generate electricity from natural gas. Its modular, on-site, 24/7 power platform stands apart in the power sector. Against a conventional grid built on centralised generation, Bloom delivers compelling time-to-power, cleaner air and water emissions profile, native-DC compatibility with emerging 800V AI datacentre architectures, and a solution that communities and nearby residents accept more readily.

Figure 2

Bloom Energy overview



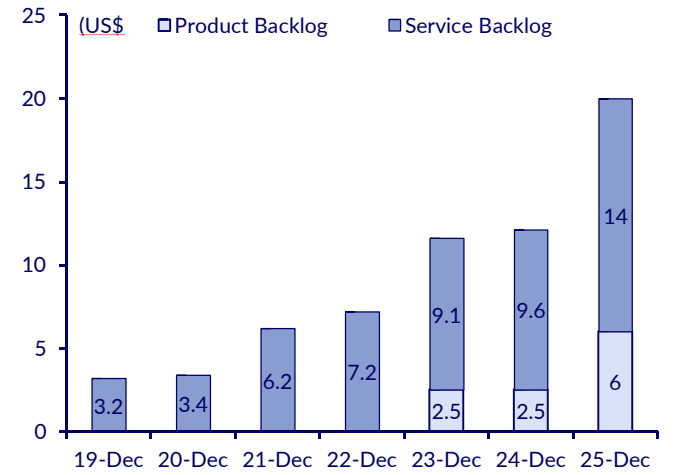
Source: Bloom Energy

We expect Bloom to increase capacity from 2GW to 4GW/6GW in 2027/28

As of year-end 2025, Bloom's product backlog grew about 140% YoY to roughly US$6bn. Amid the surging demand, Bloom is improving backlog conversion and stepping up capacity expansion to capture the strong growth opportunity. In 2Q26, Bloom raised its full-year revenue guidance to US$3.9-4.2bn, up from US$3.4-3.8bn, and plans to double annual capacity from about 1GW in 2025, split across its California and Delaware sites, to 2GW by the end of 2026; and will further expand the capacity to 4GW/6GW in 2027/28, based on our channel checks.

Figure 3

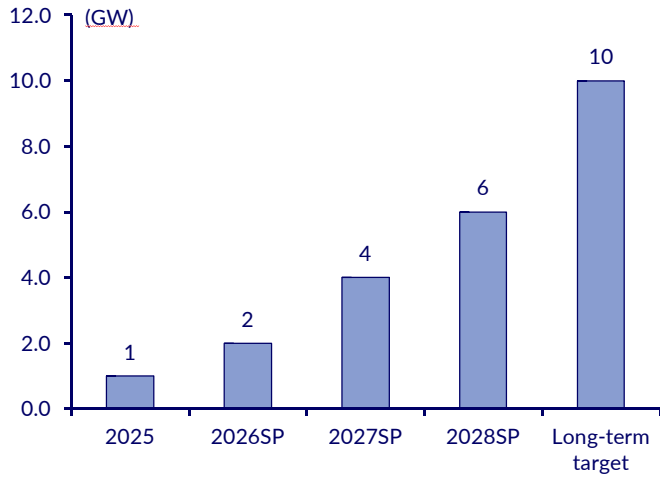
Bloom product backlog grew about 140% YoY in 2025



Source: SinoPac

Figure 4

Bloom Energy annual capacity forecast



Source: Sinopac

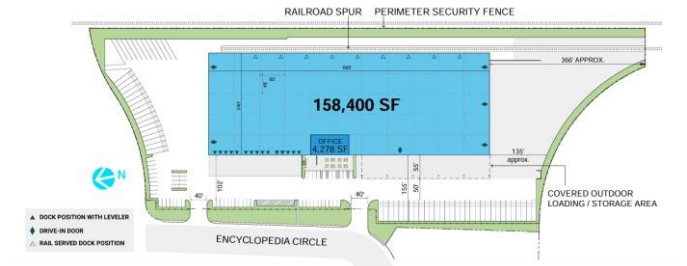
We believe Bloom has great potential. On September 2026, Bloom announced that it leased a 158,000-square-foot facility at Encyclopedia Circle to expand its manufacturing operations. The size of the new facility is notable since it is nearly

The newly leased facility could add another 5 GW, implying a long-term capacity potential of up to 10 GW

as large as Bloom's existing 164,000-square-foot manufacturing plant in Fremont. The existing plant can ultimately accommodate approximately 5GW of annual production capacity, which suggests the newly leased facility could add another 5GW for fuel cell stacks, implying a long-term capacity potential of up to 10GW. Given the multiyear partnership that secures ACES's share, we have strong conviction that ACES is well positioned to benefit from Bloom's capacity expansion.

Figure 5

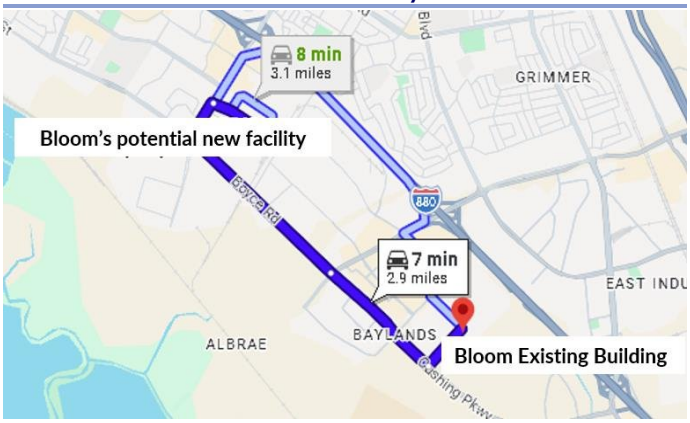
The newly leased facility has similar floor-area to existing plant



Source: LoopNet

Figure 6

Potential location of Bloom's new facility



Source: SinoPac, Google Maps

Figure 7

Bloom Energy potential new facility site

Source: LoopNet, Sinopac

We view Bloom's growth as sustainable, supported by favourable industry tailwinds. On financing, Bloom and Brookfield are partnering to develop AI datacentre power

infrastructure, with Bloom providing its fuel cell-based power generation systems and Brookfield providing capital and infrastructure development capabilities. The strategic financing framework expanded on June 30, 2026, from the original $5bn (October 2025) to $25bn, a fivefold increase. On customer wins, Nebius dropped its previously planned gas and combustion turbines in May 2026 and switched to Bloom's SOFC solution, signing a 10-year master agreement worth up to $2.6bn. The first phase covers 328MW of installed capacity, or roughly 250MW of guaranteed capacity, and is expected to come online within 2026. Bloom also expanded its Oracle contract from an initial 1.2GW to up to 2.8GW and signed deals including AEP (up to 1 GW) and Equinix (over 100 MW across 19 datacentres), broadening its customer base materially.

Wide range of power cable products

DLO high-voltage cables launch might add another NT$707m in revenue

ACES supplies Bloom with a variety of power cables across different specifications, including Power Conversion System (PCS) cables, hot box cables and Diesel Locomotive (DLO) high-voltage cables, rather than a single product type. In 2026, ACES only supplies PCS cables and hot box cables, while the company will expand its product offerings to DLO high-voltage cables from 2027, which we expect to contribute another NT$707m of sales for the whole year.

The power cable portfolio includes inner cable and external cable

The PCS cables and hot box cables are the inner power cables. Hot box cables interconnect individual fuel cells within the hot box, where power generation creates significant heat and therefore requires materials with higher durability and heat resistance. Kaori and MTAR supply hot boxes for fuel cell power generation systems, which require ACES's hot box cables as one of their key components.

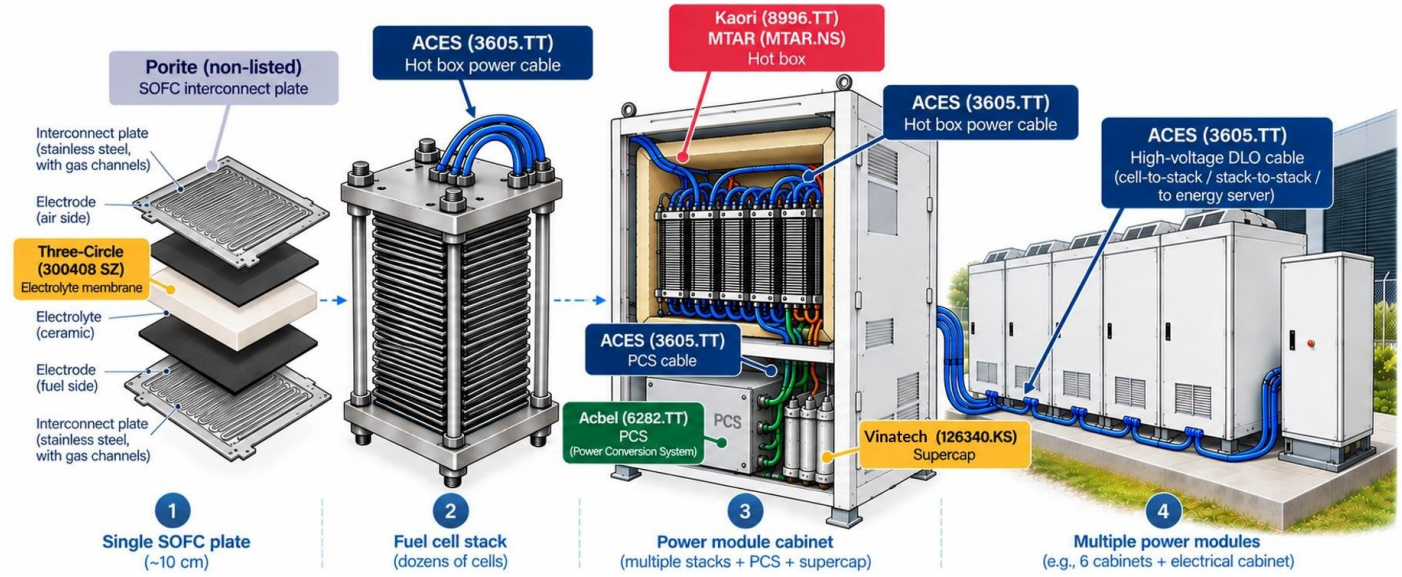
On the other hand, PCS cables are installed inside the Power Conversion System (PCS) and are supplied to AcBel, a PCS solution provider. The PCS converts DC output into usable power and manages load changes. As SOFCs generate DC power with voltage varying according to load, the output cannot be directly connected to a datacentre.

The external power cable portfolio is mainly DLO high-voltage cables, which transmit high-voltage DC power between fuel cell platforms, power modules and microgrid systems. As external cables need to withstand higher voltage levels, their dollar content is expected to be slightly higher than that of inner power cables.

Lastly, ACES also ships cables directly to Bloom Energy for development purposes.

Figure 8

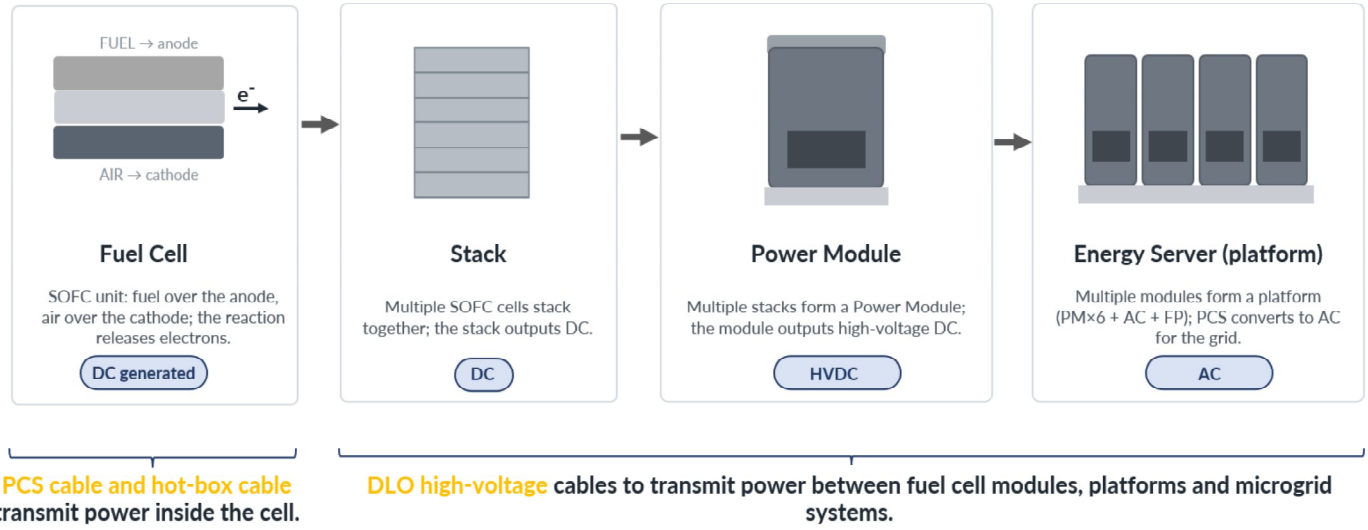
ACES power cable product and Bloom energy suppliers



Source: SinoPac, Bloom Energy

Figure 9

ACES supplies Bloom with a variety of power cables across different specifications



Source: SinoPac

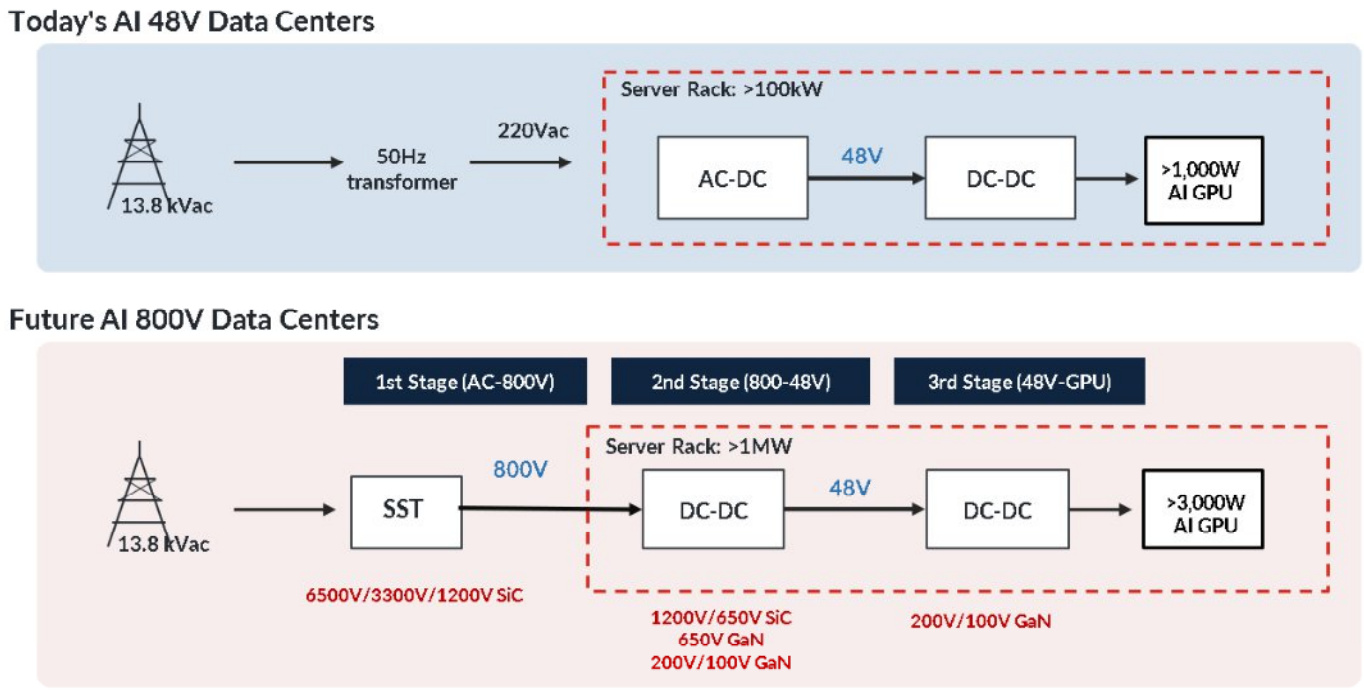
DLO high-voltage cable opportunity heading into 800V DC era

We also see DLO cables, which are currently used outside the fuel cell, as having significant potential after introducing 800V architecture. Under conventional AC architecture, grid power, onsite turbines and reciprocating engines supply AC power, and the downstream transformer, switchgear and UPS cables belong to the utility/EPC supply chain, which ACES does not address. Bloom can deliver native DC power directly to the rack, so the DC output cables might become part of Bloom's system. We see this opening as a new content opportunity for ACES as a qualified high-voltage power cable supplier.

Within Bloom's energy server under 800V architecture, the longer series stack of fuel cells and the replacement of the inverter with DC-DC conversion require a redesign of the internal harness to high-voltage specifications, which we believe positions ACES to ship a growing volume of high-voltage DLO cables over time.

Figure 10

800V DC architecture



Source: SinoPac, TrendForce

High-speed products a new driver

We expect ACES's server business to grow 107% in 2026

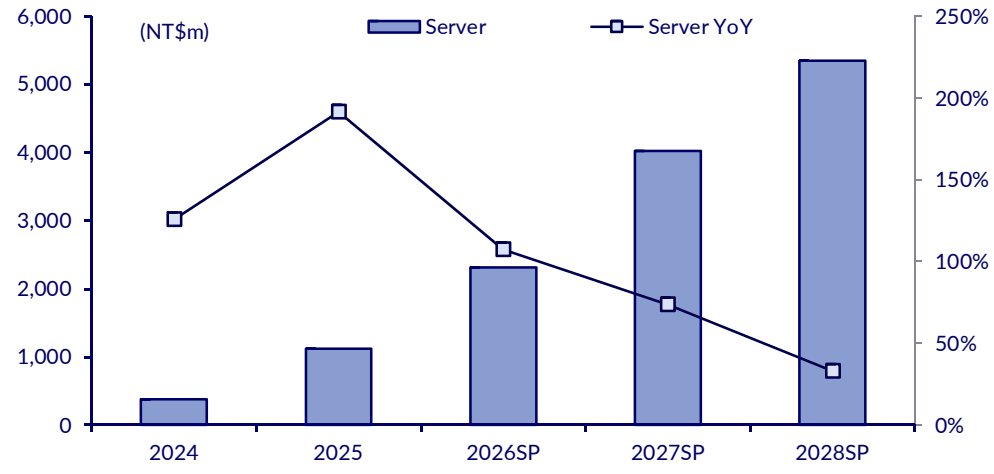
In 3Q24, ACES passed product qualification with ODMs for the high-speed connectors used on Nvidia GB200 networking cards (NICs) and accelerator cards, marking its move from traditional PC peripherals into the server space and Nvidia supply chain. In 2025, server revenue accounted for 10% of ACES's total revenue.

Looking ahead, we expect ACES's server business to grow 107% in 2026, thanks to its new product offerings, higher content per rack of high-speed products, and strong contribution from two major ASIC customers. Looking ahead, with the new connector capacity coming online and increasing ASP driven by connector spec upgrade, we expect server revenue to further grow 74%/33% in 2027/28SP.

Expect server revenue to grow a further 74%/33% in 2027/28SP

Figure 11

Server sales and YoY

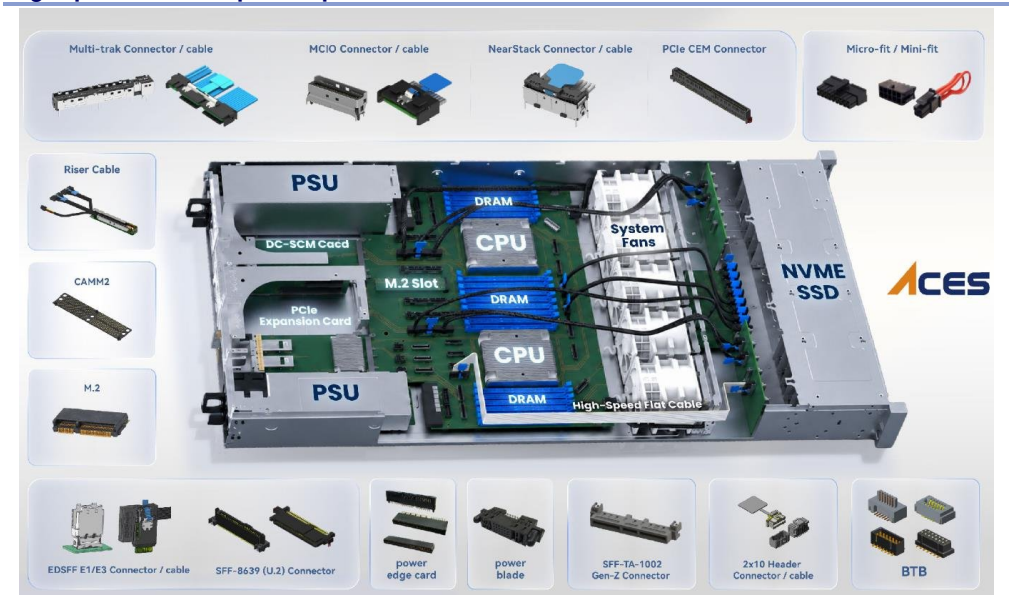


Source: SinoPac

Figure 12

High-speed connector product portfolio

High speed connector products include multi-track, board-to-board and E1/E3 connectors



Source: ACES

Connectors for ASIC carry up to 3x usage

Broadening connector footprint in ASICs

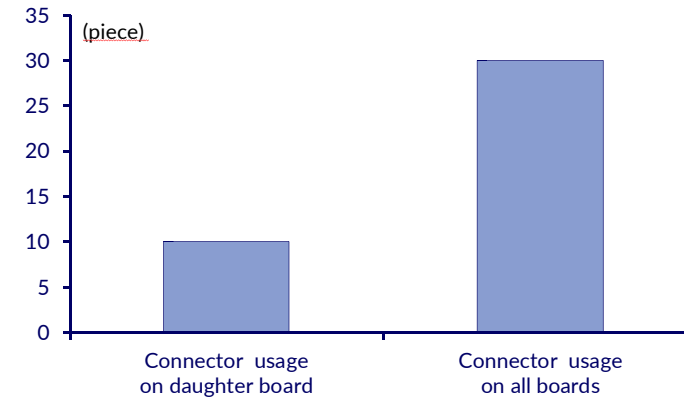
Since Nvidia has strict control over its supply chain, ACES previously supplied only the daughterboard high-speed connectors for GB200/300 servers, at roughly 10 to 20 units per rack. On ASIC platforms, since ODMs have greater influence over supplier selection, ACES secures share on both the motherboard and the daughterboard, lifting content to around 30 units per rack.

ACES broadening its connector product offerings in 2026

Besides, ACES historically supplied only MCIO connectors to these ASIC customers, but as further qualifications clear, the company will broaden its connector product offerings in 2026 to include Multi-Track, board-to-board, and E1/E3 connectors. We expect the contribution from ASIC customers to scale further in 2H26, driving server revenue to grow 64% HoH.

Figure 13

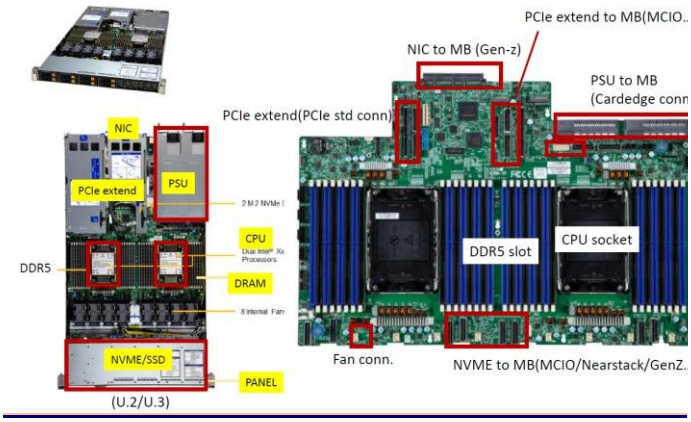
ACES high-speed connector content per rack increase



Source: SinoPac

Figure 14

ACES provides MCIO and a wide range of high-speed connectors



Source: ACES

Migration from Gen5 to Gen6 is the first major shift since 2022

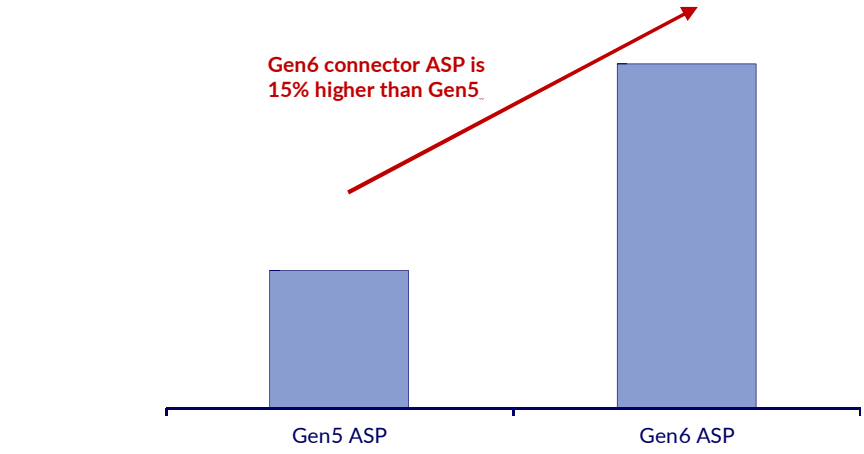
Riding the Gen6 ASP curve with a compatibility advantage

Aside from the spectacular growth in the connector shipments, we expect ACES to benefit from the ongoing Gen6 connector upgrade cycle, with ASPs expected to increase by 15%. The AI server trays currently adopt a mix of Gen5 and Gen6 connectors, but we expect the penetration of Gen6 connectors to gradually increase going forward as the migration from Gen5 to Gen6 is the first major generational shift since Genoa introduced Gen5 in 2022.

Gen6 connector ASP is 15% higher than Gen5

Figure 15

Gen6 connector ASP is 15% higher than Gen5



Source: SinoPac

Only 3-4 vendors are qualified by CSPs at Gen6

Only three to four connector vendors are qualified by CSPs at Gen6, and ACES is one of them. Furthermore, ACES can supply both Gen5 and Gen6 platforms under its existing qualification, with no need to re-sample or re-qualify for Gen5. This is because Gen6 connectors keep footprint backward compatibility with Gen5. By contrast, other peers must re-sample and run through the full qualification process again to move up to Gen6.

About the PCIe migration and Gen6 spec upgrade

PCIe has been used in on-board and in-chassis interconnects

PCIe has long been used for on-board and in-chassis interconnects. In the Vera Rubin platform, PCIe Gen6 serves as the in-tray bridge to front-of-tray modules such as CX-9 SuperNICs, BlueField-4 DPUs and NVMe storage. GPU-to-GPU scale-up runs on NVLink, with UALink as the open alternative; these fabrics work alongside PCIe.

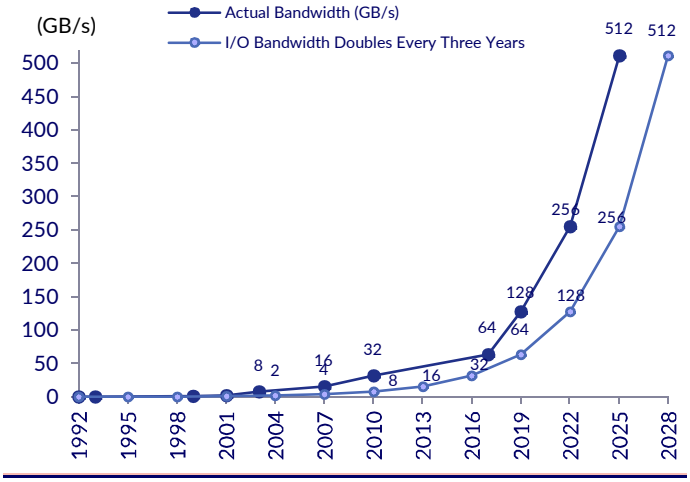
In order to address ever-larger training models, the PCIe must scale with the fast-growing size of model parameters and datasets. The move from Gen5 to Gen6 is not a simple speed doubling, even though per-lane bandwidth rises from 32 GT/s to 64 GT/s. It reworks the link from the coding scheme up. Gen1 through Gen5 doubled I/O speed using NRZ signaling. Gen6 switches to PAM4 four-level signaling and, for the first time, adds FEC (Forward Error Correction), which lets the receiver detect and correct transmission errors on its own.

Each generation upgrade raises barriers to entry

Each generation doubles bandwidth while tightening signal-integrity specs, which demands higher-precision connector manufacturing. This creates a real barrier to entry for high-speed connector suppliers.

Figure 16

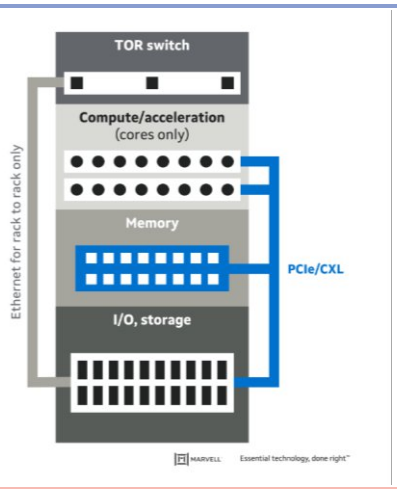
PCIe doubles its bandwidth with each generation



Source: Marvell

Figure 17

PCIe is used for on-board and in-chassis interconnects



Source: Marvell

Figure 18

Migration comparison: PCIe 4/5/6

Feature	PCIe 4.0 (Gen 4)	PCIe 5.0 (Gen 5)	PCIe 6.0 (Gen 6)
Data Rate per Lane	16 GT/s	32 GT/s	64 GT/s (2x Gen 5)
Signalling Technology	NRZ	NRZ	PAM4 (Pulse Amplitude Modulation, 4 levels)
Encoding	128b/130b	128b/130b	FLIT-based (256-byte fixed-size packets)
Error Correction	Basic CRC	Basic CRC	Forward Error Correction (FEC) + CRC
Bandwidth per Lane (uni-directional)	~2 GB/s	~4 GB/s	~8 GB/s
Total Bandwidth x16 (bi-directional)	64 GB/s	128 GB/s	256 GB/s
Power Efficiency	Basic	Basic	Introduced L0p low-power state for scaling
Security Enhancements	None	CMA, IDE (added via PCIe 5.0 ECN)	Enhanced IDE (FLIT-mode, partial header encryption)
Target Applications	High-end gaming, SSDs, etc.	AI/ML, HPC, high-performance storage; content creation and demanding workloads	Advanced AI, 800G Ethernet, NVMe, CXL
Adoption	Mature, widely adopted	Mainstream, datacentre standard	Emerging, ramping in AI/datacentre
Spec Release	2017	2019	2022
Backward Compatibility	Backward compatible with previous versions	Backward compatible, optimised on newer hardware	Full compatibility with earlier gens

Source: SinoPac

Figure 19

Core highlights of PCIe Gen6

Core upgrade	Before/challenge	Why it matters
PAM4 encoding (2x bandwidth)	NRZ: two levels (0/1), 1 bit per cycle	Doubles the rate to 64 GT/s (256 GB/s x16 bi-dir) with no frequency increase – same Nyquist as Gen 5 – avoiding severe high-frequency signal loss
FEC error correction	PAM4's fine voltage steps are noise-sensitive, pushing BER far above NRZ	Lightweight FEC + CRC-retry design keeps added latency near zero, unlike typical FEC
FLIT packaging	Variable packet sizes	Simpler decoding and near-0% encoding overhead, boosting real throughput
L0p low-power state	Saving power meant restarting the whole link	Idle lanes sleep on demand while active lanes keep running – saves power without dropping the connection

Source: SinoPac

High-speed cables emerging as a potential driver

ACES previously supplied connectors only. After several years of effort, its Gen6 PCIe cable has now been qualified by two CSP clients, and the company began shipping the in-rack cable product in April 2026. We expect high-speed cable shipments to enlarge after the new capacity comes online and more CSP orders in 2H26. Hence, we expect high-speed cable revenue to grow from NT$268m in 2026 to NT$793/NT$982m in 2027/28.

Currently, its high-speed cable product carries gross margin of merely 20%, but it is likely to edge up to around 30% as the scale enlarges, and yield improves.

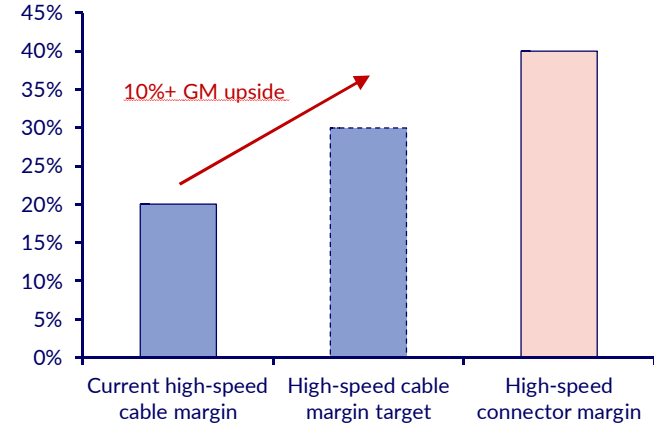
Figure 20

ACES high-speed cable product offerings

Source: ACES

Figure 21

Gross margin of ACES high-speed products



Source: SinoPac

Capacity in place ahead of demand

Connector capacity is set to rise from 4m units/month to 5m units/month

ACES is rapidly expanding its high-speed cable and connector capacity, with high-speed cable production lines expected to increase from two in 2025 to eight in 2H26, while it will proactively ramp up connector monthly capacity from 4m units in July 2026 to 5m units by the end of 2026 to address customer demand ahead of order placement.

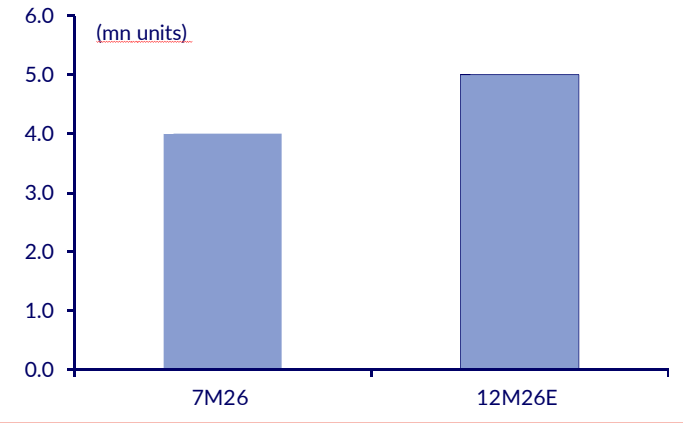
ACES expects connector shipments to reach 50m units in 2027

Supported by strong customer demand and sufficient capacity, ACES expects total connector shipments to reach 25m units in 2H26 and 50m units in 2027, representing a significant increase from 15m units in 1H26.

As for power cables for Bloom Energy, ACES has leased a new facility next to its Philippine plant, with the new power cable capacity planned to come online in 4Q26.

Figure 22

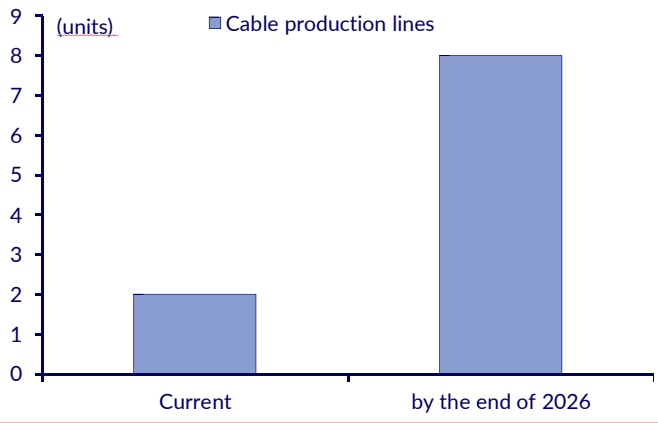
ACES connector monthly capacity expansion plan



Source: ACES, SinoPac

Figure 23

ACES high-speed cable production lines will increase 4x by 2026

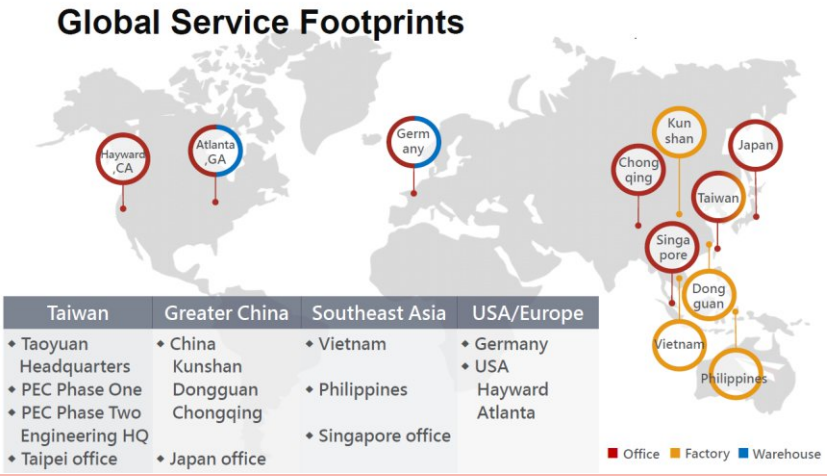


Source: ACES, SinoPac

ACES has manufacturing facilities across key regions worldwide

Figure 24

ACES global production footprint



Source: ACES

Networking: stable growth

Diversified growth engines

ACES's networking product portfolio includes set-top boxes, WiFi devices, routers, and other networking equipment. We expect the business to benefit from technology upgrades and equipment replacement driven by advances in wireless communications, IoT, cloud computing, and cybersecurity. However, tariff and inflationary pressures, together with component shortages among Asian customers, are likely to limit growth. We expect networking product revenue to grow by 34% and 14% in 2026 and 2027, respectively.

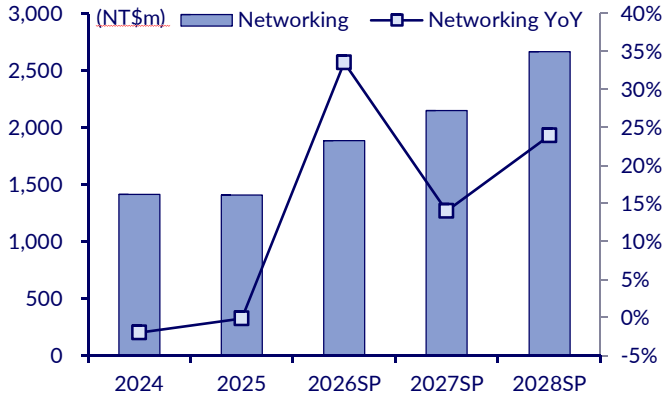
Figure 25

ACES Networking product offerings

Source: ACES

Figure 26

Networking sales and YoY



Source: SinoPac

Automotive: rising electronic content

ACES's product range spans ADAS, reversing radar and camera, body control, smart cockpit, smart antenna, and high-voltage connectors and harnesses, extending into applications such as EV charging guns. It has entered a US EV major as well as European and Chinese Tier-1 supply chains. Going forward, it will continue to focus on high-voltage, high-frequency, and high-speed connection technologies. The company expects the automotive sector to stay flat in 2027 to reflect muted demand and limited internal resource allocation, as it focuses on AI-related business at this stage. In the long-term, growth from the automotive sector will be driven by ADAS upgrades, which lifts demand for sensors and compute, alongside rising EV penetration, which adds high-value electronic systems and power semiconductor content.

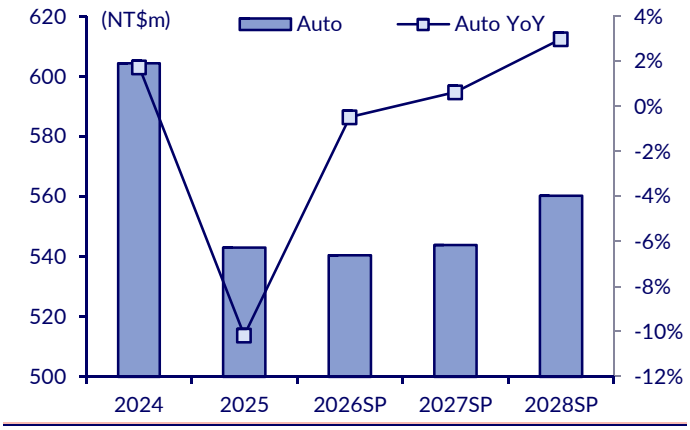
Figure 27

ACES Automotive product portfolio

Source: ACES

Figure 28

Automotive sales and YoY change



Source: SinoPac

Industrial: a steady UPS-driven base

ACES supplies industrial power cords, used mainly in uninterruptible power systems (UPS). A UPS is an electrical device that acts as a reliable buffer when the main power supply behaves abnormally, such as during a fault, outage, or unstable voltage. It keeps automation controllers, sensors, and actuators continuously powered so equipment keeps running. When a power problem occurs, the UPS supplies stable backup power to prevent equipment failure, data corruption, or system crashes. We expect ACES's industrial business to grow steadily, with growth of 9% and 14% in 2026 and 2027, respectively.

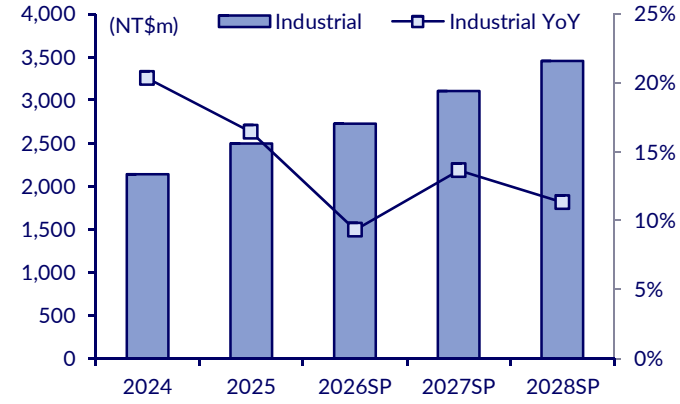
Figure 29

ACES Industrial product portfolio

Source: ACES

Figure 30

Industrial sales and YoY



Source: SinoPac

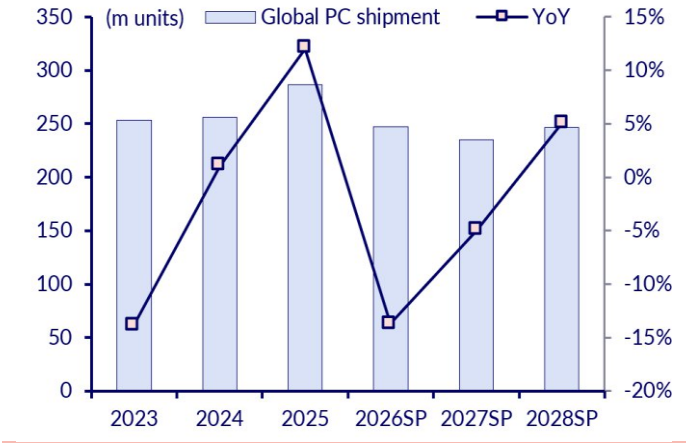
Muted 2026, with Windows 12 driving a replacement cycle in late 2027

Notebook and Consumer

Under the twin pressures of softening end-consumer demand and continued supply-chain cost increases, we expect ACES's full-year NB revenue to decrease 12% in 2026 and return to a growth trajectory in 2027. However, we remain cautiously optimistic about 2027 with the launch of Windows 12, combined with easing component shortages and broader AI adoption, which should drive consumer replacement demand, lifting global NB shipments -5%/5% year over year in 2027/28, which is a gradual recovery following a weaker 2026.

Figure 31

Global NB shipment forecast



Source: SinoPac

Figure 32

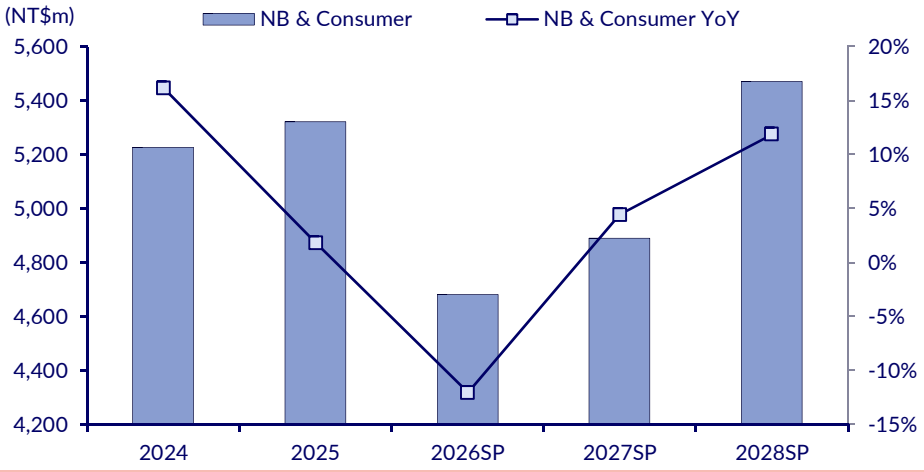
ACES Consumer product portfolio

Source: ACES

We expect NB and Consumer to return to a growth trajectory from 2027

Figure 33

NB and Consumer sales and YoY



Source: SinoPac

Diversifying into new growth opportunities

We found that ACES is diversifying its products' end-application by developing new specification and engaging with new clients. Hence, we believe in its long-term balanced growth.

Power rack: leveraging its long-standing relationship with Schneider

ACES has long-standing relationship with Schneider Electric

ACES has a long-standing relationship with Schneider Electric at its Philippine plant. The partnership dates back more than 20 years, with ACES's Philippine plant initially serving Schneider almost exclusively. Over the years, the product portfolio has expanded from conventional power cables to uninterruptible power systems (UPS) and, more recently, power racks for AI datacentres.

ACES ramped-up high-voltage power cables for power racks in 2Q26

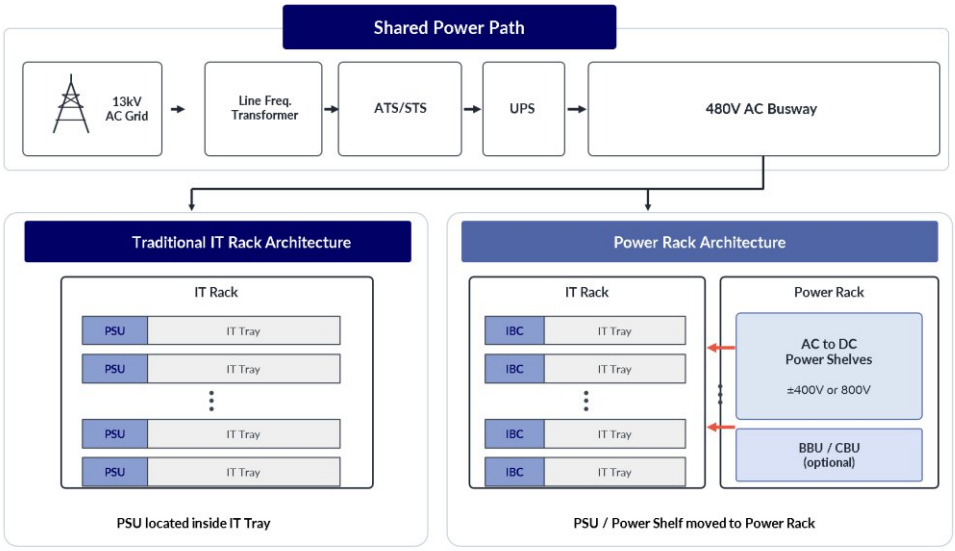
ACES began shipping high-voltage power cables for power racks in 1Q26 and entered the volume ramp-up phase in 2Q26. We remain positive on power rack adoption as the PSU in every IT tray hits a physical limit as rack power reaches 1MW, occupies rack space and caps compute density. Moving power shelves and BBU/CBU into a dedicated power rack decouples power from computing and frees

the IT rack for near-full compute use. It also enables 800V HVDC high-voltage delivery cuts current and I^2R losses, lifting efficiency from the 48V era's 80-90% towards >90%, while scaling per-rack power module and high-voltage cable content. Given the industry trend of increasing in-rack space density and surging single-rack voltage, we expect ACES's high-voltage power cable business to continue growing.

PSU/Power Shelf moved to Power Rack

Figure 34

Traditional rack architecture versus the new architecture



Source: SinoPac, TI, Tech news

Schneider showcased power rack product at Computex 2026

Figure 35

Schneider power rack product

Source: SinoPac

TP of NT$270

Initiate with an Outperform rating

We initiate coverage of ACES with an Outperform rating given its dual growth engine from Bloom's strong expansion and high-speed connector customer wins and generation upgrades. We forecast 2027/28 EPS of NT$11.19/NT$14.93, with our 2027 forecast 48% ahead of consensus. Our TP of NT$270 is based on 20x 2H27-1H28SP PE, backed by a 72% EPS Cagr in 2026-28SP driven by Bloom's significant ramp and larger content in server products. Initiate coverage with an Outperform rating.

Company profile

ACES was founded in 1996, and it mainly produces connectors, micro-coaxial cable, high-speed transmission cables, and RF connectors. Connectors and cable assemblies link two separate subsystems or operating units and carry the signals between them, allowing the overall system to function properly. They are used across a wide range of electronic products, with main applications spanning notebooks and PC peripherals, consumer electronics, automotive electronics, cloud servers, networking equipment, and industrial equipment.

To keep pace with the push towards lighter, thinner, and smaller notebooks and consumer devices, ACES developed high-temperature-resistant, fine-pitch, and low-profile products at an early stage. At the same time, it is moving actively into the high-frequency, high-speed connectors, high-speed cables, and power supplies required for cloud servers, as well as the connector and cable assemblies used in automotive electronics.

Figure 36

ACES product portfolio

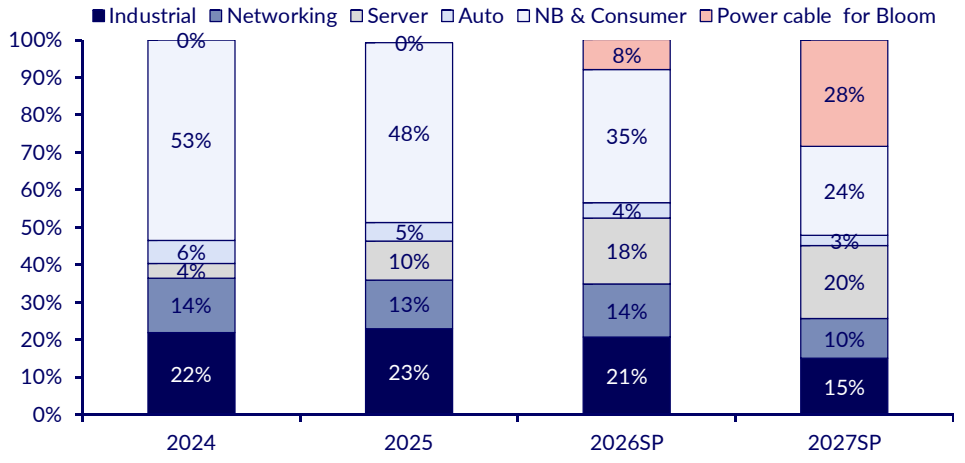
Product category	2Q26 revenue breakdown	Average GM	Product application
Industry	21%	15%	UPS and IPC
Networking	15%	30-35%	Signal set-top box, WiFi router and devices
Server	16%	35-40%	MCIO, Multi-Track, board-to-board, and E1/E3 connectors
High-voltage cable for Bloom	4%	20%	PCS cables, hot-box cables and DLO high-voltage cables
Automotive	4%	15%	ADAS, Intelligent Cockpit and Vehicle-to-Everything (V2X)
Consumer and NB	40%	20-25%	Smartphone and Notebook

Source: SinoPac

AI-related product contribution increasing

Figure 37

Sales mix by product



Source: SinoPac

ACES has leveraged M&A to gain new capabilities, plants and technology

Company development

ACES has used M&A well to gain new capabilities, plants and technology, which in turn opened doors to new customers and markets. In 2015, it bought MEC IMEX to add cable assembly capability and a plant in the Philippines, which took it into the industrial market. In 2017, the CPC deal gave it access to the European auto connector market and a plant in Vietnam. In 2019, it bought Kuang Ying to add metal stamping capacity and reach Taiwan telecom customers. The key deal came in 2021, when ACES bought 100% of Genesis, a US high-speed interconnect maker based in Atlanta. The main goal was to gain Genesis's long-standing ties with US CSP and cloud customers. In 2025, ACES opened its Phase Two R&D headquarters in Taoyuan, which focuses on cloud connectors and high-speed cables. We believe 10 years of M&A have turned ACES from a consumer connector maker into a supplier of high-speed interconnect solutions for AI servers and the cloud.

Figure 38

ACES M&A activity

Year	Type/entity	Capability gained	Market entered	Representative products
2011	Establishment	Precision connector manufacturing capability	Core connectors	Board connectors
2015	M&A – MEC	Cable assembly product-line manufacturing capability	Industrial	Wire harness/cable assembly
2017	M&A – CPC	European automotive connector capacity	European Automotive	FAKRA automotive RF connectors
2019	M&A – KY	Metal stamping-part manufacturing capability	Taiwan Telecom	Stamping parts
2021	M&A – Genesis	High-speed cable technology	US Telecom/ Server	High-speed cable/ SlimSAS
2025	R&D expansion	2 RD HQ for Cloud-computing connector/high-speed cable	Cloud/ Datacentre	High-speed backplane connector

Source: SinoPac, ACES

Shareholding structure

ACES's shareholding structure is relatively stable, with the top 10 owning about 30%, including Directors and Supervisors' total share of 9%. Foreign investor holdings have increased since the company entered the high-speed connectors industry and were around 22% in August 2026.

Figure 39

Top-10 shareholders

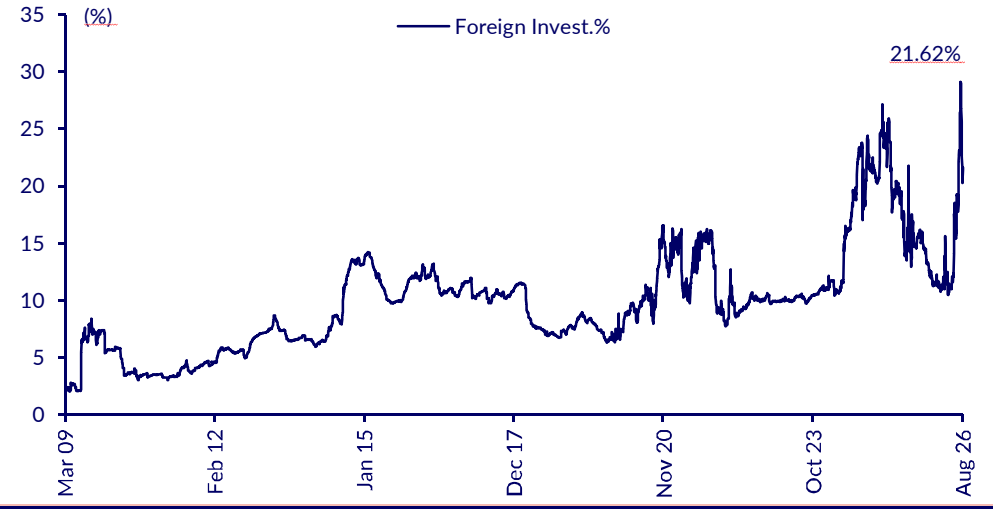
	Holding %
Yuan Wan-Ting (Chairman)	5.28
Keyan Investments Co., Ltd. (owned by Chairman's family)	3.98
Hsu Chang Fei (Chairman's spouse)	3.76
Wei Ji Investments Co., Ltd.	3.55
Deutsche Bank AG & Albula Investment Fund	3.54
CITI & HK First Securities Agent Invest	2.97
Yuan Hsiang Feng (Chairman's family)	2.89
Taishin International & Aces Electronics	1.92
Liao Ming Shan	1.25
Yuan Chen Ting (Chairman's family)	1.05
Top-10 shareholders	**30.19**
Director and supervisors	**9.05**

Source: SinoPac

Foreign investors hold about 22% of shares

Figure 40

Foreign investor holdings



Source: TEJ, Sinopac

Financial forecasts

We forecast sales to rise by 22%/56%/30% in 2026/27/28, fueled by robust high-speed connector shipments from 2H26, rising Gen6 penetration that drives high-speed connector ASP expansion, and Bloom's sustained, large-scale capacity expansion, which should provide a durable growth engine.

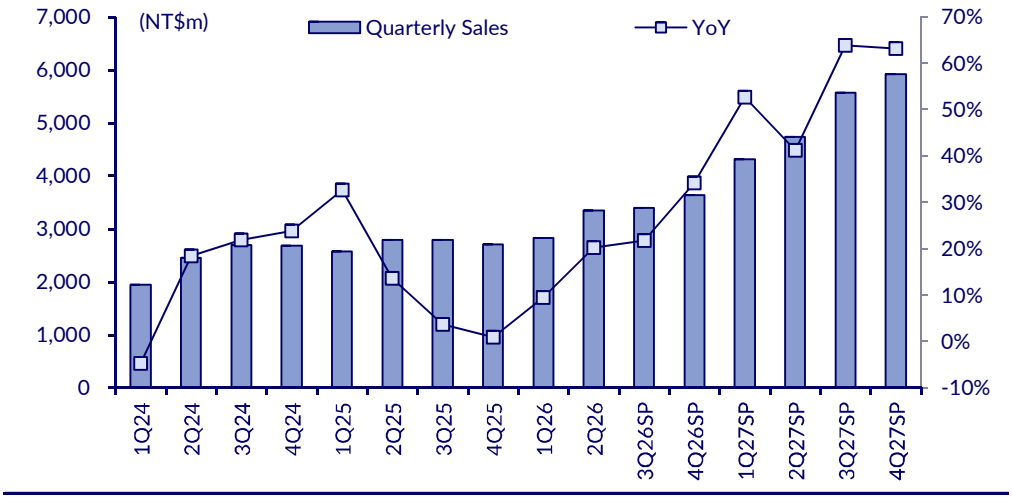
ASIC customer wins to help server revenue grow 74% YoY in 2027SP

AI product sales, both connectors and cables, will make a more meaningful revenue contribution in 2H26SP given ACES's broadening ASIC customer base. We expect shipment contribution from high speed connectors and cables to help its server revenue grow a further 107%/74% YoY into 2026/2027.

Growth rate of sales to accelerate from 2026SP

Figure 41

Quarterly sales and YoY change



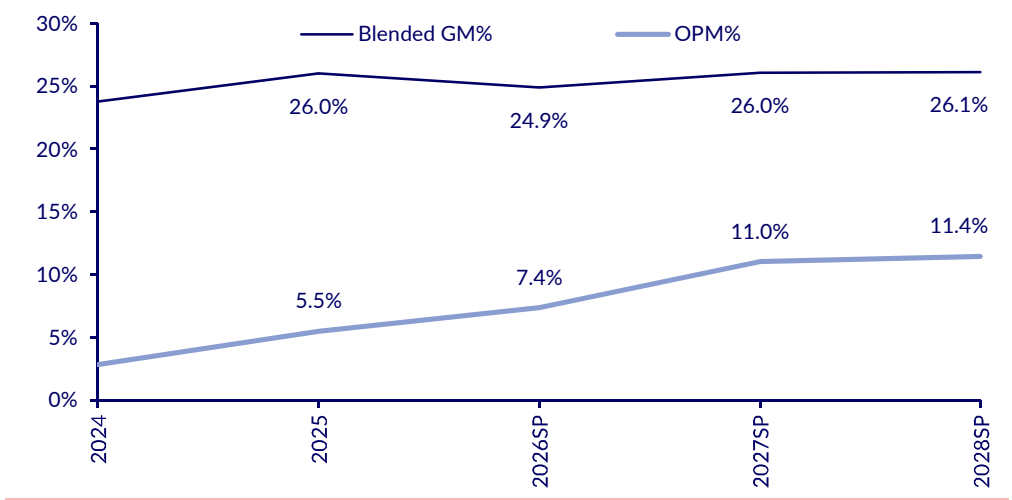
Source: SinoPac

Given the high-speed connectors carry higher gross margin of 40% and cable's GM keeps improving with the learning curve, we forecast ACES's GM to expand from 24.9% in 2026 to 26.0%/26.1% in 2027/28, partially offset by the strong power cables ramp which may dilute the overall growth margin.

Both GM and OPM steadily improving

Figure 42

GM and OPM



Source: SinoPac

Figure 43

Quarterly P&L

(NT$m)	1Q26	2Q26	3Q26SP	4Q26SP	1Q27SP	2Q27SP	3Q27SP	4Q27SP	2025	2026SP	2027SP	2028SP
Sales	2,822	3,350	3,396	3,631	4,307	4,733	5,567	5,924	10,860	13,199	20,531	26,733
Gross profit	694	859	839	892	1,116	1,240	1,468	1,524	2,824	3,284	5,348	6,983
Opex	546	613	562	592	664	730	827	863	2,227	2,313	3,085	3,923
Operating profit	147	246	277	300	451	510	641	660	597	970	2,263	3,060
Non-OP	(4)	13	32	35	40	42	37	39	218	76	159	171
Pretax profit	143	259	309	335	492	552	679	699	815	1,046	2,421	3,231
Net profit	108	201	250	271	398	447	550	566	661	830	1,961	2,617
EPS (NT$)	0.67	1.22	1.53	1.66	2.27	2.55	3.13	3.23	4.35	5.07	11.19	14.93
Growth												
Sales QoQ	4%	19%	1%	7%	19%	10%	18%	6%				
Sales YoY	9%	20%	22%	34%	53%	41%	64%	63%	11%	22%	56%	30%
EPS QoQ	(19%)	82%	26%	8%	37%	12%	23%	3%				
EPS YoY	(51%)	81%	4%	103%	241%	110%	105%	95%	73%	17%	121%	33%
Margin												
Gross margin	24.6%	25.6%	24.7%	24.6%	25.9%	26.2%	26.4%	25.7%	26.0%	24.9%	26.0%	26.1%
Opex	19.4%	18.3%	16.6%	16.3%	15.4%	15.4%	14.9%	14.6%	20.5%	17.5%	15.0%	14.7%
OP margin	5.2%	7.3%	8.2%	8.3%	10.5%	10.8%	11.5%	11.1%	5.5%	7.4%	11.0%	11.4%
Non-OP	(0.2%)	0.4%	0.9%	1.0%	0.9%	0.9%	0.7%	0.7%	2.0%	0.6%	0.8%	0.6%
Net margin	3.8%	6.0%	7.4%	7.5%	9.3%	9.4%	9.9%	9.6%	6.1%	6.3%	9.6%	9.8%

Source: SinoPac

Figure 44

Sales mix and GM assumptions

Sales (NT$m)	2025	2026SP	2027SP	2028SP
Industrial	2,498	2,732	3,104	3,456
Networking	1,412	1,885	2,150	2,665
Server	1,118	2,319	4,025	5,353
Power cable for Bloom	-	1,076	5,819	9,229
Auto	543	540	544	560
NB & Consumer	5,321	4,680	4,888	5,470
Total	10,860	13,199	20,531	26,733

YoY (%)	2025	2026SP	2027SP	2028SP
Industrial	16%	9%	14%	11%
Networking	0%	34%	14%	24%
Server	191%	107%	74%	33%
Power cable for Bloom	0%	0%	441%	59%
Auto	(10%)	0%	1%	3%
NB & Consumer	2%	(12%)	4%	12%
Total	11%	22%	56%	30%

Sales mix (%)	2025	2026SP	2027SP	2028SP
Industrial	23%	21%	15%	13%
Networking	13%	14%	10%	10%
Server	10%	18%	20%	20%
Power cable for Bloom	0%	8%	28%	35%
Auto	5%	4%	3%	2%
NB & Consumer	49%	35%	24%	20%
Total	100%	100%	100%	100%

Gross margin (%)	2025	2026SP	2027SP	2028SP
Industrial	15.0%	14.7%	15.0%	16.0%
Networking	34.0%	34.2%	35.0%	35.0%
Server	35.0%	37.9%	42.2%	42.1%
Power cable for Bloom	0.0%	20.0%	20.3%	20.5%
Auto	18.0%	14.8%	15.0%	15.0%
NB & Consumer	27.8%	23.5%	23.9%	23.3%
Total	26.0%	24.9%	26.0%	26.1%

Source: SinoPac

Cashflow and balance sheet

Capex-to-sales ratio of 12%/8% in 2025/26

ACES's balance sheet shows healthy debt exposure with an improving net cash level. Given its plan to massively expand capacity at its Taiwan plants, we expect Capex to remain at a high level of NT$1,075m in 2026 and then slightly decline to NT$855m in 2027. We forecast its depreciation-to-sales ratio to be around 3% in 2027, which is at the low end of its historical level.

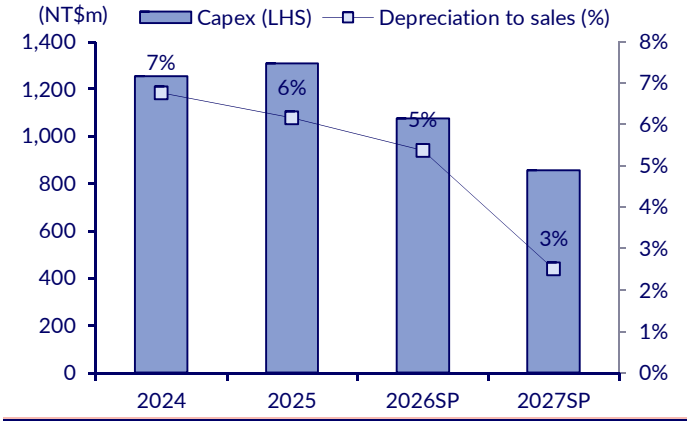
We expect ACES's ROE to expand from 10% in 2026 to 21% in 2027 and 23% in 2028, as the volume ramp of AI high-speed connectors and Bloom Energy power cables drives a richer product mix and gross margin expansion, delivering 121% EPS growth in 2027.

Free cashflow per share of NT$-0.6/0.8 in 2025/26

We expect ACES's FCF to turn positive in 2026 as operating cashflow scales with the AI high-speed connector and Bloom power cable ramp while Capex normalises after the 2026 capacity buildout. Looking ahead, we expect ACES's free cashflow per share to be NT$0.8-1.5 in coming years.

Figure 45

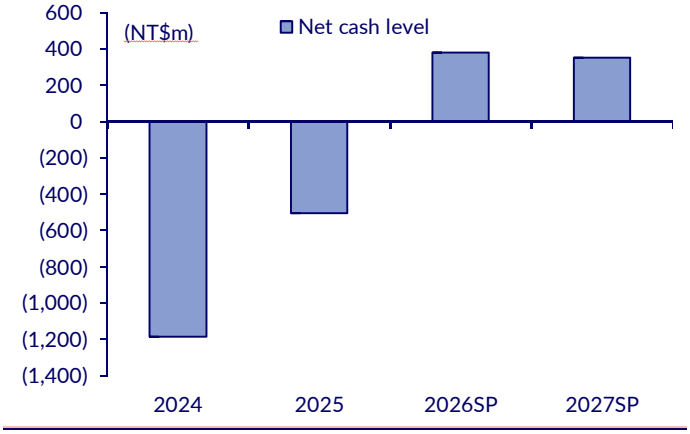
Capex and depreciation to sales



Source: SinoPac

Figure 46

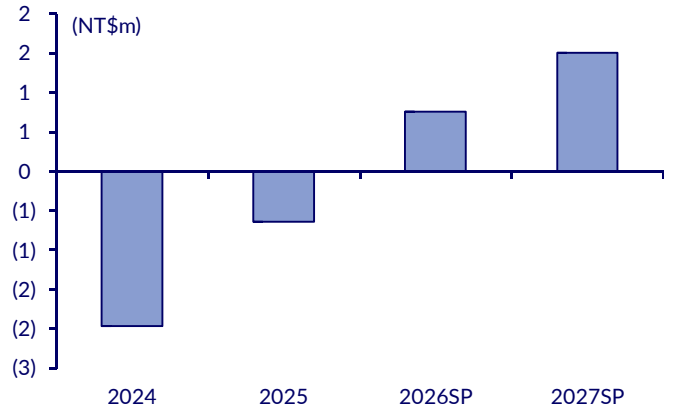
Net cash position



Source: SinoPac

Figure 47

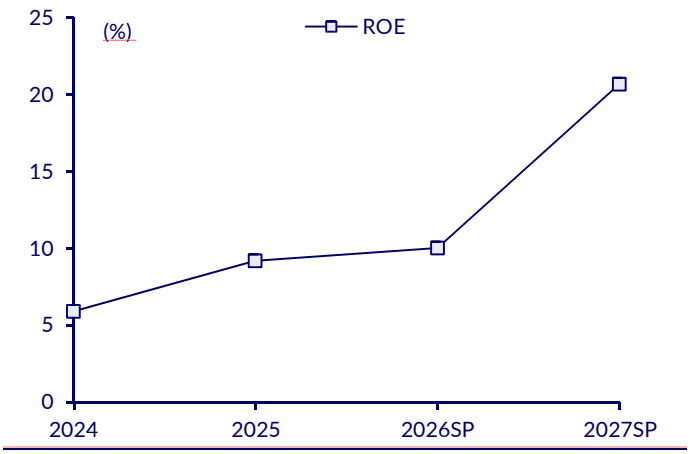
Free cashflow per share



Source: SinoPac

Figure 48

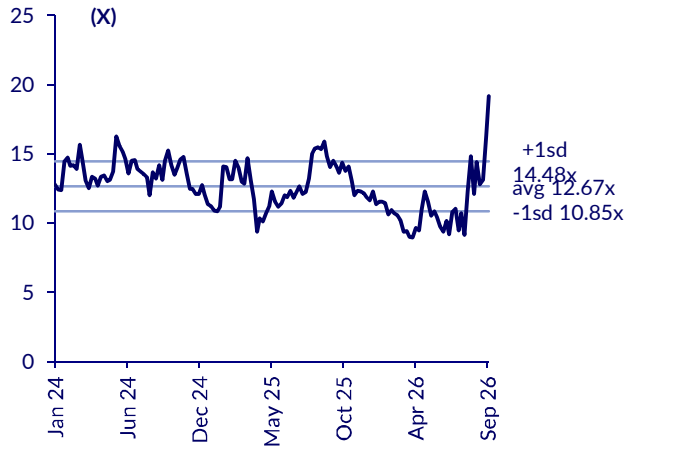
ROE



Source: SinoPac

Figure 49

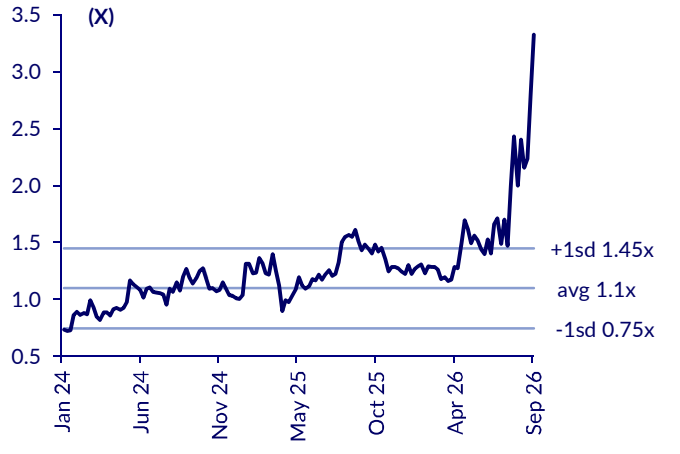
Forward PE



Source: SinoPac

Figure 50

Forward PB



Source: SinoPac

Figure 49

Balance sheet

(NT$m)	2021	2022	2023	2024	2025	2026SP	2027SP
Cash & bank deposits	2568	2459	2058	2001	2483	3368	3341
ST marketable securities	100	0	0	2	70	70	70
Accounts receivable (trade debtors)	2991	2779	2743	3352	3331	3473	5133
Inventories	1663	1454	1189	1394	1436	2479	3796
Prepaid, deferred tax and others	106	141	87	79	95	95	95
Other current assets	166	165	98	141	91	91	91
Total Current Assets	7595	6998	6175	6969	7506	9577	12525
LT Investment	1136	1130	901	954	1196	818	818
Gross Fixed	5869	6393	6507	8040	8692	9767	10622
Unfinished projects	186	706	1229	1142	1575	1475	1525
Accum. Deprec.	(3227)	(3671)	(3996)	(4768)	(5396)	(6105)	(6621)
Fixed assets	2829	3428	3741	4415	4871	5136	5526
Deferred assets	78	81	81	69	70	70	70
Intangible assets	156	154	192	154	140	140	140
Other LT assets	1079	1049	1161	1036	1066	1066	1066
Total Assets	12873	12840	12251	13596	14849	16808	20146
Short-term borrowings	1044	1429	1467	195	794	794	794
Bills Issued	0	0	0	0	0	0	0
Accounts payable	1922	1624	1409	1961	2016	2833	4338
Curr. Of L-T Debt	240	1119	774	233	119	375	243
Others current liabilities	1337	1220	1014	1127	1086	1076	1096
Total Current liabilities	4544	5392	4664	3516	4016	5078	6471
Long-term debt	2059	1216	1737	2991	2194	2194	2194
Deferred Taxations	302	317	308	356	419	419	419
Other Liabilities	436	260	285	272	287	462	737
Total Liabilities	7340	7185	6994	7134	6915	8152	9821
Common Stocks	1344	1344	1344	1419	1624	1753	1753
Reserves	1654	1691	1719	2312	2962	3045	3241
Retained earnings	2555	2492	2236	2529	3141	3649	5123
Others	(21)	128	(43)	202	208	208	208
Shareholder funds	5532	5655	5257	6462	7934	8655	10325

Source: SinoPac

ACES has had a dividend payout ratio of around 29-36% in the past five years, except for 2023, when the company reported a loss. We expect to maintain a payout ratio around 35% as the ACES's earnings significantly improved, but the actual dividend payout depends on future investments and business plans.

Figure 50

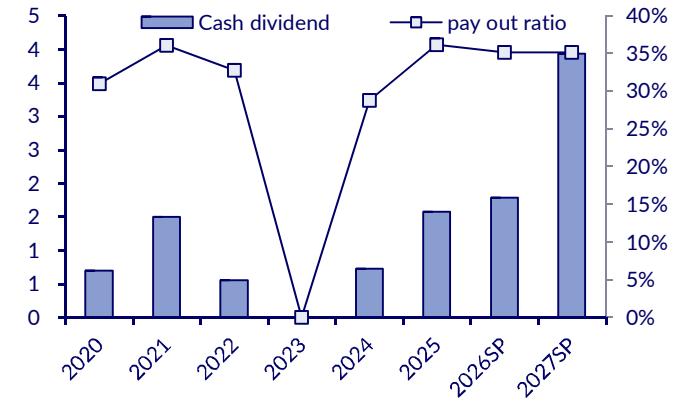
Cash dividend payout

(NT$m)	2020	2021	2022	2023	2024	2025
Net profit	251	511	225	(267)	344	661
Payout ratio	31%	36%	33%	0%	29%	36%
Cash dividend (NT$)	0.7	1.5	0.6	0.0	0.7	1.6
Dividend yield	0.4%	0.9%	0.3%	0.0%	0.4%	0.9%

Source: SinoPac

Figure 51

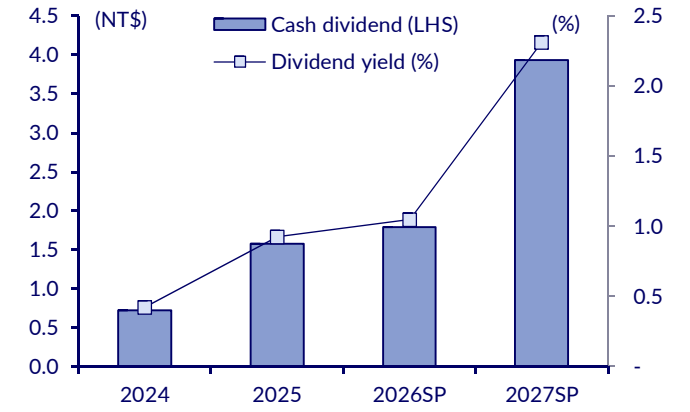
DPS and payout ratio



Source: SinoPac

Figure 52

Cash dividend and yield



Source: SinoPac

Figure 53

Cashflow statement

(NT$m)	2021	2022	2023	2024	2025	2026SP	2027SP
Net income	508	223	(268)	344	661	830	1,961
Depreciation	564	651	651	661	669	709	516
Amortization	29	48	55	78	50	49	102
Change in working cap.	(520)	(27)	0	(184)	29	(379)	(1,451)
Investment income	(31)	(10)	11	(3)	37	(10)	(9)
Others non-cash op	219	298	214	74	(236)	-	-
Cash flow from operation	769	1,183	663	971	1,210	1,199	1,119
Capex	(1,020)	(1,116)	(887)	(1,253)	(1,308)	(1,075)	(855)
LT investment decrease (increase)	(217)	(17)	21	-	378	-	-
ST investment decrease (increase)	243	74	22	(36)	(58)	-	-
Others CFI	111	(76)	(206)	173	3	-	-
Cash flow from investment	(883)	(1,136)	(1,050)	(1,117)	(984)	(1,075)	(855)
ST borrowing	(159)	354	52	(1,272)	599	-	-
LT borrowing	314	16	150	336	(91)	-	-
Bonds Issued	603	-	-	999	-	1,000	-
Bonds Redeemed	-	-	-	(1)	-	-	-
Dividend paid	(86)	(202)	(74)	-	(112)	(239)	(292)
Dir.&Emp.Bonus	-	-	-	-	-	-	-
Others CFF	328	(207)	(85)	(43)	(252)	-	-
Cash flow from financing	1,000	(39)	43	19	144	761	(292)
Net change in cash	887	8	(343)	(127)	370	885	(28)
Exchange Influence	(59)	(117)	(57)	70	112	-	-
Beginning cash	1,740	2,568	2,459	2,058	2,001	2,483	3,368
Ending cash	2,568	2,459	2,058	2,001	2,483	3,368	3,341

Source: SinoPac

ACES Enters from Gen6 with Vertical Integration and flexible lead time

CSPs are searching for a supplier able to provide both connectors and cables

Competitive positioning

In high-speed connectors, ACES competes primarily with overseas peers including Amphenol (APH), TE, and Molex. APH's edge lies in a development-technology lead of one to two generations, while ACES's relative strength rests on customers' recognition on delivering quality comparable to the overseas majors, alongside more flexible lead times, strong capacity-expansion support, and fast ramp-up.

The structural opportunity comes from the supply chain integration trend. As transmission speeds rise, CSPs increasingly prefer suppliers which can integrate high-speed connectors with high-speed cables and ship assembled, finished goods.

ACES is one of the few Taiwanese players with this vertical-integration footprint

ACES is one of the few Taiwanese players with this vertical-integration footprint: it holds wire-drawing technology transferred in-house from ITRI, produces everything from wire drawing through assembly on its own, and its commitment to moving into high-speed cables has been noted by AWS and Meta, backed by roughly two years of qualification work already completed.

By contrast, other Taiwanese peers have yet to build comparable vertical integration. Customers have historically split connectors and cables across separate suppliers, with cable suppliers represented by BizLink (3665 TT) and Longwell (6290 TT), while peers such as Lotes (3533 TT), Alltop (3526 TT), Argosy (3217 TT) and JPC Connectivity (6197 TT) are likewise strong in connectors but lack in-house cable capability.

Gen6 suppliers include ACES, APH, TE, and Lotes

Since the Gen5 qualified-supplier base was already fragmented, the Gen6 bar of combined connector-plus-cable capability becomes ACES's entry point, bringing ACES into design-in from Gen6. The qualified connector supplier base is small, with suppliers including APH, TE, and ACES, with Lotes added subsequently.

Peer comparison

ACES's primary local competitors in connectors include Lotes, Alltop, Argosy and JPC Connectivity. In the cable industry, it competes with Bizlink and Longwell. ACES's foreign peers include Amphenol, TE and FIT.

Figure 54

Peer comparison

Company	Code	Mkt Cap (US$bn)	Rec	Earnings growth (%)			PE (x)			PB (x)			ROE (%)		
				26SP	27SP	28SP	26SP	27SP	28SP	26SP	27SP	28SP	26SP	27SP	28SP
Taiwan															
ACES	3605 TT	0.93	O-PF	17	121	33	33.1	15.0	11.3	3.5	2.9	2.4	10	21	23
Bizlink	3665 TT	18.2	O-PF	50	94	20	37.7	19.9	16.5	9.5	6.8	5.2	27	40	36
Longwell	6290 TT	1.5	N-R	23	11	na	28.6	25.8	na	5.0	4.9	na	18	20	na
Lotes	3533 TT	6.5	HLD	11	19	29	24.0	20.1	15.6	4.7	4.2	3.6	21	22	25
Alltop	3526 TT	0.6	N-R	25	33	na	16.0	12.8	na	4.5	4.1	na	32	42	na
Argosy	3217 TT	0.4	N-R	2	9	na	10.3	10.0	na	2.4	2.4	na	29	31	na
JPC	6197 TT	1.3	N-R	83	42	na	30.0	21.1	na	8.3	5.8	na	29	52	na
Foreign															
Amphenol	APH US	211.3	N-R	60	23	na	51.5	32.1	23.1	16.0	11.7	8.5	41	38	32
TE Connectivity	TEL US	63.0	N-R	31	14	na	25.3	19.0	14.9	5.1	4.7	4.1	25	28	25
FIT	6088 HK	4.6	N-R	68	46	na	25.2	17.0	9.5	1.7	1.6	1.4	9	13	13

Source: Sinopac, Bloomberg

Target price calculation

We initiate coverage of ACES with an O-PF rating and a target price of NT$270, based on 20x 2H27-1H28SP EPS, supported by a strong 72% EPS Cagr in 2026-28SP. We use a PE-based methodology for ACES, in line with other connector companies. We initiate coverage with an Outperform rating.

Our target multiple of 20x is in line with the 2027SP average PE of 20.4x for Taiwan-listed connector and cable peers with direct AI datacentre exposure, namely BizLink, Lotes and JPC Connectivity. We view this as a reasonable anchor. ACES offers faster earnings growth than the group, with 2027SP EPS growth of 121% versus the peer average of 52%, while its 2027-28SP ROE of 21-24% sits at the low end of the peer range of 22-52%.

We also believe the target PE premium is justified as the AI revenue contribution, including AI data transmission and power, rises from 4% in 2024 to 48% in 2027SP. The company maintains its key supplier position at Bloom as the company ramps up its capacity expansion, while ASIC customer wins and higher connector content from the Gen6 upgrade cycle should provide additional growth drivers. As the company gradually shifts away from its legacy consumer electronics base and its exposure to cyclical downturns, we see room for a structural rerating.

Figure 55

Target price calculation

	Target PE (x)	Valuation period	12-month rolling EPS (NT$)	Target Price (NT$)
New	20	2H27-1H28SP	13.5	270

Source: SinoPac

Key risks

We see four key operational risks for ACES:

Raw material cost volatility: Copper prices drive a large share of Cogs for ACES's cable products, while connectors carry far less exposure. ACES currently reprices its cable products with customers on a quarterly basis, which has allowed it to sustain a long-term GM of around 20% on power cables. However, the pass-through might be lagging, so sharp copper price swings could cause quarterly GM to fluctuate before repricing impact catches up.

Delays in ODM pull-in: ACES ships its high-speed connectors directly to ODM customers. Because rack assembly sits at the final stage of the AI server supply chain, any upstream disruption, such as components shortages and assembly yield issue, could push out rack shipment schedules or reduce volumes. This could in turn lead ODMs to cut or defer orders for high-speed connectors from ACES, creating quarter-to-quarter volatility in server revenue.

Slower-than-expected ramp-up at Bloom Energy: Power cables for Bloom will account for 28%/35% of ACES's revenue in 2027/28SP, which makes Bloom's ramp-up the single largest swing factor in our forecasts. Delays could arise on both the supply and demand sides. On the supply side, Bloom could fall behind its capacity plan if new facility fit-out, equipment installation or hiring takes longer than expected. On the demand side, its datacentre customers could defer deployments due to grid interconnection, gas supply or financing hurdles, or due to regulatory headwinds such as tighter permitting and emissions rules for on-site gas-fired generation, utility restrictions on behind-the-meter power, or changes to fuel cell tax incentives. Either scenario would directly reduce ACES's power cable shipments. As volumes scale, Bloom may also qualify a second cable supplier, which would add pressure to ACES's share and ASP.

Prolonged NB weakness: NB and Consumer remains ACES's largest segment at 35% of 2026SP revenue, and our forecasts assume it returns to 4%/12% growth in 2027/28SP on a Windows 12-driven replacement cycle. If end-demand stays soft or the replacement cycle slips, a further NB decline could offset part of the growth from Bloom and AI servers and dilute group GM.

Investment thesis

ACES is evolving from a notebook connector maker into a supplier of AI interconnects and power cables, with two growth engines ramping at the same time. The first is Bloom Energy. ACES has co-developed SOFC power cables with Bloom for about six years, and it is currently one of three suppliers of Bloom's power cable assemblies, with around one-third of the total. It ships inner PCS and hot-box cables in 2026 and adds external DLO high-voltage cables from 2027, with order visibility extending to 2028 and beyond. The second engine is AI/ASIC servers. On ASIC platforms, ODMs have more say in supplier selection, so ACES wins both motherboard and daughterboard sockets, which lifts content from 10-20 to ~30 units per rack. Gen6 raises connector ASP by ~15%, and a new in-chassis PCIe Gen6 cable line, qualified by two CSPs, adds a higher-content stream. ACES integrates vertically from wire drawing to final assembly, which makes it one of the few Taiwanese suppliers able to ship connector-plus-cable solutions as CSPs consolidate their vendor base.

Catalysts

(1) Bloom's capacity ramp from 2GW to 4GW/6GW in 2026-28 and new data-center wins (Oracle, Nebius, AEP, Equinix) converting into cable orders; (2) DLO high-voltage cable shipments starting in 2027; (3) ASIC customer ramps in 2H26 and a broader product range beyond MCIO into Multi-Track, board-to-board and E1/E3 connectors; (4) rising Gen6 penetration on AI server trays; (5) high-speed cable lines expanding from two to eight by end-2026, with GM improving from ~20% toward ~30%.

Valuation details

We adopt a PE-based valuation methodology as we believe the market will mainly focus on the earnings growth outlooks of high-speed connector makers, which is better reflected with PE metrics. We assign a target multiple of 20x PE, more than +1sd of the three-year average, to reflect a strong outlook with margin expansion and an EPS Cagr of 72% in 2026-28 on solid Bloom Energy demand and increasing high speed connector content.

Investment risks

There could be downside risk to sales growth and margins if: (1) Bloom's deployments or backlog conversion slow on financing, gas supply or permitting delays, or ACES loses share among Bloom's cable suppliers; (2) ASIC programs ramp more slowly or CSP Capex is cut; (3) Gen6 adoption is slower than expected, or competition from Amphenol, TE, Molex and Lotes pressures ASP; (4) high-speed cable yield and GM improve more slowly than we assume; (5) NB and consumer demand weakens further on component shortages, tariffs and inflation; (6) TWD appreciation or higher copper prices squeeze margins.

ACES scores high in balance-sheet-quality

Figure 56

Earnings and balance-sheet risk scores (lower the better)

	Score	Comments
Earnings-quality flags		
Capex indiscipline	0	
Cash burn	1	Capacity expansion for high-speed connectors, high-speed cable and power cable products
Rising non-core or intangibles	0	
Rising working capital	0	
Poor cash conversion	0	
Earnings-quality risk score (EQRS)	**1/5**	
Balance-sheet-quality flags		
Cash burn	1	Capacity expansion for high-speed connectors, high-speed cable and power cable products
Excessive leverage	0	
Frequent fundraising	0	
Liquidity concerns	0	
Operational stress	0	
Balance-sheet-quality risk score (BQRS)	**1/5**	

Source: SinoPac

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Click to rate this research
Great
Solid
Meh

Detailed financials

Profit & Loss (NT$m)

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Revenue	**10,393**	**8,486**	**9,771**	**10,860**	**13,199**	**20,531**	**26,733**
Cogs (ex-D&A)	(7,461)	(6,028)	(6,710)	(7,317)	(9,157)	(14,566)	(19,258)
Gross Profit (ex-D&A)	**2,932**	**2,458**	**3,061**	**3,543**	**4,042**	**5,965**	**7,475**
Research & development costs	(574)	(621)	(593)	(615)	(861)	(1,489)	(1,844)
Selling & marketing expenses	(686)	(585)	(593)	(621)	(666)	(847)	(1,153)
Other SG&A	(916)	(876)	(858)	(991)	(787)	(749)	(926)
Other Op Expenses ex-D&A	-	0	-	0	-	-	-
Op Ebitda	**755**	**376**	**1,018**	**1,316**	**1,729**	**2,881**	**3,551**
Depreciation/amortisation	(699)	(706)	(740)	(719)	(758)	(618)	(491)
Op Ebit	**57**	**(331)**	**278**	**597**	**970**	**2,263**	**3,060**
Interest income	27	50	42	31	34	35	35
Interest expense	(85)	(109)	(104)	(79)	(72)	(73)	(73)
Net interest inc/(exp)	(58)	(59)	(62)	(48)	(38)	(38)	(38)
Associates/investments	(23)	(7)	13	295	124	181	195
Forex/other income	212	23	108	(30)	(85)	(56)	(57)
Asset sales/other cash items	-	-	-	-	-	-	-
Provisions/other non-cash items	130	105	49	2	75	72	71
Asset revaluation/Exceptional items	(3)	-	-	-	-	-	-
Profit before tax	**315**	**(270)**	**387**	**815**	**1,046**	**2,421**	**3,231**
Taxation	(92)	2	(43)	(154)	(216)	(460)	(614)
Profit after tax	**223**	**(268)**	**344**	**661**	**830**	**1,961**	**2,617**
Preference dividends	-	-	-	-	-	-	-
Profit for period	**223**	**(268)**	**344**	**661**	**830**	**1,961**	**2,617**
Minority interest	2	2	0	0	0	0	0
Net profit	**225**	**(267)**	**344**	**661**	**830**	**1,961**	**2,617**
Extraordinaries/others	0	0	0	0	0	0	0
Profit avail to ordinary shares	225	(267)	344	661	830	1,961	2,617
Dividends	(202)	(74)	0	(112)	(239)	(292)	0
Retained profit	24	(340)	344	550	591	1,670	2,617
Adjusted profit	**225**	**(267)**	**344**	**661**	**830**	**1,961**	**2,617**
EPS (NT$)	1.7	(2.0)	2.4	4.4	5.1	11.2	14.9
Adj EPS [pre excep] (NT$)	1.7	(2.0)	2.4	4.3	5.1	11.2	14.9
Core EPS (NT$)	1.7	(2.0)	2.4	4.4	5.1	11.2	14.9
DPS (NT$)	0.6	-	0.7	1.6	1.8	3.9	5.2

Profit & loss ratios

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Growth (%)							
Revenue growth (% YoY)	(1.7)	(18.3)	15.1	11.1	21.5	55.5	30.2
Ebitda growth (% YoY)	(31.8)	(50.3)	171.0	29.2	31.4	66.7	23.3
Ebit growth (% YoY)	(89.0)	(682.8)	nm	114.2	62.6	133.2	35.2
Net profit growth (%)	(55.9)	(218.3)	nm	92.2	25.6	136.3	33.4
EPS growth (% YoY)	(59.7)	(216.7)	nm	81.5	16.6	120.5	33.4
Adj EPS growth (% YoY)	(59.7)	(216.7)	nm	81.5	16.6	120.5	33.4
DPS growth (% YoY)	(63.3)	(100.0)	72,087,981,900.0	118.1	13.4	120.5	33.4
Core EPS growth (% YoY)	(59.7)	(216.7)	nm	81.5	16.6	120.5	33.4
Margins (%)							
Gross margin (%)	28.2	29.0	31.3	32.6	30.6	29.1	28.0
Ebitda margin (%)	7.3	4.4	10.4	12.1	13.1	14.0	13.3
Ebit margin (%)	0.5	(3.9)	2.9	5.5	7.4	11.0	11.4
Net profit margin (%)	2.2	(3.1)	3.5	6.1	6.3	9.6	9.8
Core profit margin	2.2	(3.1)	3.5	6.1	6.3	9.6	9.8
Op cashflow margin	11.4	7.8	9.9	11.1	9.1	5.5	3.6
Returns (%)							
ROE (%)	4.1	(4.9)	6.0	9.5	10.3	21.1	23.6
ROA (%)	0.3	(2.6)	1.9	3.4	4.9	9.9	11.3
ROIC (%)	0.7	(5.0)	3.5	6.3	9.3	19.2	20.9
ROCE (%)	0.9	(4.8)	3.7	7.3	11.4	24.2	26.9
Other key ratios (%)							
Effective tax rate (%)	29.1	0.6	11.1	18.8	20.6	19.0	19.0
Ebitda/net int exp (x)	13.1	6.4	16.5	27.2	45.3	76.1	93.7
Exceptional or extraord. inc/PBT (%)	-	-	-	-	-	-	-
Dividend payout (%)	32.8	-	30.1	36.1	35.1	35.1	35.1

Source: SinoPac

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Balance sheet (NT$m)

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Cash & equivalents	2,459	2,058	2,001	2,483	3,368	3,341	2,725
Accounts receivable	2,779	2,743	3,352	3,331	3,473	5,133	7,426
Inventories	1,454	1,189	1,394	1,436	2,479	3,796	4,937
Other current assets	165	98	142	161	161	161	161
Current assets	**6,857**	**6,088**	**6,889**	**7,411**	**9,482**	**12,430**	**15,250**
Fixed assets	3,428	3,741	4,415	4,871	5,136	5,526	6,104
Investments	1,211	981	1,023	1,267	888	888	888
Goodwill	0	0	0	0	0	0	0
Other intangible assets	154	192	154	140	140	140	140
Other non-current assets	1,191	1,248	1,115	1,161	1,161	1,161	1,161
Total assets	**12,840**	**12,251**	**13,596**	**14,849**	**16,808**	**20,146**	**23,544**
Short term loans/OD	2,548	2,241	428	913	1,169	1,037	1,040
Accounts payable	1,624	1,409	1,961	2,016	2,833	4,338	5,643
Accrued expenses	-	-	-	-	-	-	-
Taxes payable	0	0	0	0	0	0	0
Other current liabs	1,220	1,014	1,127	1,086	1,076	1,096	1,086
Current liabilities	**5,392**	**4,664**	**3,516**	**4,016**	**5,078**	**6,471**	**7,769**
Long-term debt/leases/other	1,216	1,737	2,991	2,194	2,194	2,194	2,194
Convertible bonds	-	-	-	-	-	-	-
Provisions/other LT liabs	577	593	628	706	880	1,156	1,328
Total liabilities	**7,185**	**6,994**	**7,134**	**6,915**	**8,152**	**9,821**	**11,291**
Share capital	1,344	1,344	1,419	1,624	1,753	1,753	1,753
Retained earnings	2,492	2,236	2,529	3,141	3,649	5,123	6,789
Reserves/others	1,691	1,719	2,312	2,962	3,045	3,241	3,502
Shareholder funds	**5,528**	**5,300**	**6,260**	**7,726**	**8,447**	**10,117**	**12,045**
Minorities/other equity	128	(43)	202	208	208	208	208
Total equity	**5,655**	**5,257**	**6,462**	**7,934**	**8,655**	**10,325**	**12,253**
Total liabs & equity	**12,840**	**12,251**	**13,596**	**14,849**	**16,808**	**20,146**	**23,544**
Total debt	3,764	3,978	3,419	3,107	3,363	3,230	3,234
Net debt	1,305	1,920	1,418	624	(5)	(110)	508
Adjusted EV	22,636	22,786	25,037	26,162	27,999	27,844	28,413
BVPS (NT$)	41.1	39.4	42.1	47.2	48.2	57.7	68.7

Balance sheet ratios

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Key ratios							
Current ratio (x)	1.3	1.3	2.0	1.8	1.9	1.9	2.0
Growth in total assets (% YoY)	(0.3)	(4.6)	11.0	9.2	13.2	19.9	16.9
Growth in capital employed (% YoY)	10.0	4.0	10.0	8.5	1.1	18.3	25.2
Net debt to operating cashflow (x)	1.1	2.9	1.5	0.5	-	-	0.5
Gross debt to operating cashflow (x)	3.2	6.0	3.5	2.6	2.8	2.9	3.4
Gross debt to Ebitda (x)	5.0	10.6	3.4	2.4	1.9	1.1	0.9
Net debt/Ebitda (x)	1.7	5.1	1.4	0.5	-	-	0.1
Gearing							
Net debt/equity (%)	23.1	36.5	21.9	7.9	(0.1)	(1.1)	4.1
Gross debt/equity (%)	66.6	75.7	52.9	39.2	38.9	31.3	26.4
Interest cover (x)	1.0	(2.6)	3.1	7.9	13.9	31.5	42.3
Debt cover (x)	0.3	0.2	0.3	0.4	0.4	0.3	0.3
Net cash per share (NT$)	(9.7)	(14.1)	(9.9)	(4.1)	0.0	0.6	(2.9)
Working capital analysis							
Inventory days	69.7	71.6	63.3	64.3	72.1	75.4	80.7
Debtor days	101.3	118.8	113.8	112.3	94.1	76.5	85.7
Creditor days	79.3	82.2	82.6	90.3	89.2	86.2	92.2
Working capital/Sales (%)	14.9	18.9	18.4	16.8	16.7	17.8	21.7
Capital employed analysis							
Sales/Capital employed (%)	152.4	119.7	125.3	128.3	154.3	202.9	211.1
EV/Capital employed (%)	331.9	321.4	321.0	309.1	327.3	275.1	224.3
Working capital/Capital employed (%)	22.8	22.7	23.1	21.6	25.8	36.1	45.8
Fixed capital/Capital employed (%)	50.3	52.8	56.6	57.6	60.0	54.6	48.2
Other ratios (%)							
PB (x)	4.2	4.4	4.1	3.6	3.6	3.0	2.5
EV/Ebitda (x)	30.0	60.6	24.6	19.9	16.2	9.7	8.0
EV/OCF (x)	19.1	34.4	25.8	21.6	23.4	24.9	29.6
EV/FCF (x)	339.1	(101.7)	(88.5)	(266.4)	225.8	105.4	390.2
EV/Sales (x)	2.2	2.7	2.6	2.4	2.1	1.4	1.1
Capex/depreciation (%)	171.4	136.2	189.6	195.6	151.5	165.8	247.6

Source: SinoPac

evaluator
The Global Equities Database

Cashflow (NT$m)

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Operating profit	57	(331)	278	597	970	2,263	3,060
Operating adjustments	167	63	66	65	(140)	(302)	(443)
Depreciation/amortisation	699	706	740	719	758	618	491
Working capital changes	(27)	0	(184)	29	(379)	(1,451)	(2,140)
Interest paid / other financial expenses	(10)	11	(3)	37	(10)	(9)	(9)
Tax paid	-	-	-	-	-	-	-
Other non-cash operating items	298	214	74	(236)	0	0	1
Net operating cashflow	1,183	663	971	1,210	1,199	1,119	960
Capital expenditure	(1,116)	(887)	(1,253)	(1,308)	(1,075)	(855)	(887)
Free cashflow	67	(224)	(283)	(98)	124	264	73
Acq/inv/disposals	57	43	(36)	321	-	-	-
Int, invt & associate div	(76)	(206)	173	3	-	-	-
Net investing cashflow	(1,136)	(1,050)	(1,117)	(984)	(1,075)	(855)	(887)
Increase in loans	369	202	62	508	1,000	-	-
Dividends	(202)	(74)	0	(112)	(239)	(292)	(689)
Net equity raised/others	(207)	(85)	(43)	(252)	0	0	0
Net financing cashflow	(39)	43	19	144	761	(292)	(689)
Incr/(decr) in net cash	8	(343)	(127)	370	885	(28)	(616)
Exch rate movements	(117)	(57)	70	112	0	0	1
Opening cash	2,568	2,459	2,058	2,001	2,483	3,368	3,341
Closing cash	2,459	2,058	2,001	2,483	3,368	3,341	2,725
OCF PS (NT$)	8.8	4.9	6.8	8.0	7.3	6.4	5.5
FCF PS (NT$)	0.5	(1.6)	(2.0)	(0.6)	0.8	1.5	0.4

Cashflow ratio analysis

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Growth (%)							
Op cashflow growth (% YoY)	53.8	(43.9)	46.3	24.7	(0.9)	(6.6)	(14.2)
FCF growth (% YoY)	-	(435.5)	-	-	-	113.0	(72.4)
Capex growth (%)	9.4	(20.5)	41.3	4.4	(17.8)	(20.4)	3.7
Other key ratios (%)							
Capex/sales (%)	10.7	10.5	12.8	12.0	8.1	4.2	3.3
Capex/op cashflow (%)	94.4	133.8	129.1	108.1	89.7	76.4	92.4
Operating cashflow payout ratio (%)	6.3	0.0	10.7	19.7	24.3	61.6	95.8
Cashflow payout ratio (%)	17.0	11.1	0.0	9.2	19.9	26.1	0.0
Free cashflow payout ratio (%)	302.0	-	-	-	192.7	110.5	0.0

DuPont analysis

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Ebit margin (%)	0.5	(3.9)	2.9	5.5	7.4	11.0	11.4
Asset turnover (x)	0.8	0.7	0.8	0.8	0.8	1.1	1.2
Interest burden (x)	5.6	0.8	1.4	1.4	1.1	1.1	1.1
Tax burden (x)	0.7	1.0	0.9	0.8	0.8	0.8	0.8
Return on assets (%)	0.3	(2.6)	1.9	3.4	4.9	9.9	11.3
Leverage (x)	2.3	2.3	2.2	2.0	1.9	1.9	1.9
ROE (%)	4.1	(4.9)	6.0	9.5	10.3	21.1	23.6

EVA® analysis

Year to 31 December	2022A	2023A	2024A	2025A	2026SP	2027SP	2028SP
Ebit adj for tax	40	(329)	248	484	770	1,833	2,479
Average invested capital	6,079	6,558	7,137	7,741	8,320	9,562	11,842
ROIC (%)	0.7	(5.0)	3.5	6.3	9.3	19.2	20.9
Cost of equity (%)	9.7	9.7	9.7	9.7	9.7	9.7	9.7
Cost of debt (adj for tax)	1.8	2.5	2.2	2.0	2.0	2.0	2.0
Weighted average cost of capital (%)	8.1	8.2	8.2	8.1	8.1	8.1	8.1
EVA/IC (%)	(7.4)	(13.2)	(4.7)	(1.9)	1.1	11.0	12.8
EVA (NT$m)	(451)	(868)	(335)	(145)	95	1,056	1,517

Source: SinoPac

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For purposes of regulatory disclosure only, our High-Conviction Outperform and Outperform ratings fall into a buy rating category; our Hold rating falls into a hold rating category; and our High-Conviction Underperform and Underperform ratings fall into a sell rating category.

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Macquarie Equity Research
06 October 2026

MarqueeIdeas

TSMC

Outperform

3Q26 preview: The core AI holding

Key Points

- N2 ramp-up appears to be progressing faster than our expectations.
- We expect strong N2 pricing to more than offset depreciation dilution, leaving room for GPM upside to guidance.
- GPM dilution risk from any potential Texas investment may anyway be further out than investors fear. We retain Outperform (Marquee buy).

- **N2 ramp running ahead of plan.** Jul-Aug 2026 rev reached NT$982bn, with a record NT$515bn (+53% YoY) in Aug. Even a flat MoM rev in Sep would put 3Q26E rev ~2% above the top end of guidance (NT $1.43tn-1.47tn). We believe N2 is the main driver. Mgmt had guided to a steep N2 ramp in 2H26, and news reports indicate TSMC is targeting first-year N2 output ~45% above N3's starting level.
- **Pricing to more than offset depreciation dilution.** 3Q26 GPM guide of 65-67% already absorbs a 3-4ppt dilution from N2, mainly higher depreciation in COGS. We think the guide is conservative. N2 carries a meaningful price premium over N3, and faster loading spreads fixed costs sooner. We expect 3Q26E GPM at high end of guidance and N2 drag to narrow from 4Q26E.
- **Overseas expansion's GPM impact to come later than feared.** According to news reports, TSMC is reportedly evaluating a Texas investment on top of its US$265bn Arizona commitment, with each fab costing at least US$20bn. The plan apparently hinges on a US extension of advanced manufacturing tax credit expiring at end-2026. We think GPM impact would likely come later than investors fear. An advanced fab takes 5+ years from planning to volume output, so we see no Texas dilution before 2030. Near term, overseas dilution stays with Arizona, guided at 2-3ppt in early stages.
- **What to watch on 15 Oct.** 4Q26 GPM guidance, N2 rev share, 2027 pricing, and the reported Texas timeline. N2 pricing power is the key source of margin upside into 2027.

Earnings changes: We raise 2026E/27E/28E net profit by 2%/5%/5% to reflect better-than-expected N2 ramp and ASP upside driven by stronger AI demand.

Valuation: We lift our TP 5% to NT$4,410, from NT$4,200, based on an unchanged 21x 2028E PE; we project 47% EPS CAGR over 2025-28E.

Catalysts: New capacity increases for N2/A16/A14; smart device replacement demand; new AI product launches.

Investment Thesis and Recommendation

Our three-year outlook remains very positive, underpinned by strong HPC/on-device AI demand growth, pricing power for future products, and continued market-share gains. More TSMC research here.

Semicons & Semicon Equip
Taiwan

Arthur Lai

Katherine Zheng

2330 TT	**Outperform**
Price (at 6 Oct 2026)	**TWD2,585**
12-month target	**TWD4,410.00**
12-month TSR (%)	**71.6**
Volatility Index	**Medium**
Market Cap (Local) (m)	67,035,176
Market Cap (USD) (m)	2,110,779
Free Float (%)	93
30-day avg turnover (USD) (m)	1,542.3

Investment Fundamentals

Year end 31 Dec	2025A	2026E	2027E	2028E
Revenue (bn)	3,809	5,612	8,534	11,512
Revenue growth (%)	31.6	47.3	52.1	34.9
EBIT (bn)	1,936	3,345	5,077	6,383
EBIT growth (%)	46.4	72.8	51.8	25.7
Reported profit (bn)	1,718	2,936	4,334	5,440
Adjusted profit (bn)	1,718	2,936	4,334	5,440
EPS rep [TWD]	66.2	113.2	167.1	209.8
EPS rep growth (%)	46.4	70.9	47.6	25.5
Net debt/equity (%)	(31.8)	(40.4)	(51.6)	(60.7)
ROA (%)	26.5	34.4	36.5	33.0
ROE (%)	35.4	44.5	45.2	39.6
PER rep (x)	39.0	22.8	15.5	12.3
EV/EBITDA (x)	24.6	15.3	10.1	7.8
P/BV (x)	12.4	8.6	5.9	4.2
Total div yield (%)	0.7	0.9	1.0	1.2

Quant (rank vs. global sector)	2 / 705

2330 TT rel TAIEX performance, & rec history

Source: FactSet, Macquarie Research, Oct 2026 (all figures in TWD unless noted, TP in TWD)

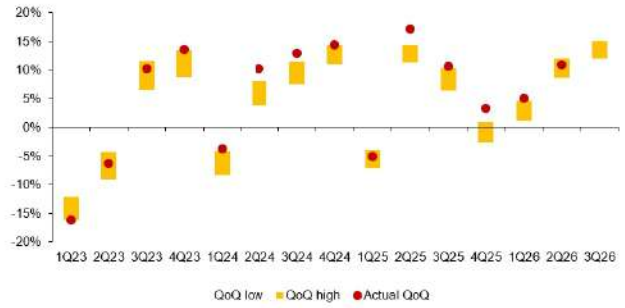
Key Risks to Investment Thesis

- A larger/longer inventory correction.
- End-demand weakening more than we expect.
- Higher-than-expected material costs/inflation.
- Unfavourable FX movements.

Company Description

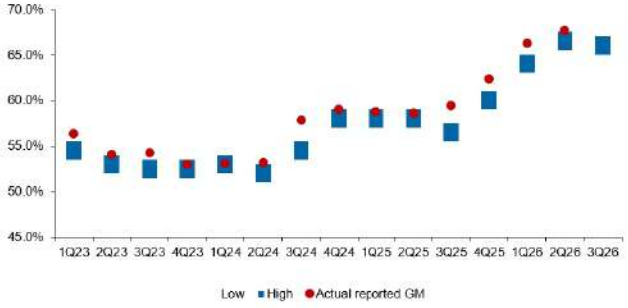
TSMC is the world's largest semiconductor foundry with over 50% market share. It has a diversified exposure to Mobile processors (Apple, Mediatek, Qualcomm), High Performance Computing (AMD, AWS, Broadcom, Intel, Nvidia), Automotives (MobilEye, STMicro, Tesla) and Industrial and Consumer electronics. TSMC has over 90% market share in leading-edge EUV nodes. The majority of its capacity is located in Taiwan, but some is also in China, Japan and the USA.

Figure 1 - TSMC revenue QoQ guidance vs actual



Source: TSMC, Macquarie Research, Oct 2026

Figure 2 - TSMC GPM guidance vs actual



Source: TSMC, Macquarie Research, Oct 2026

TSMC (2330 TT) TWD/(bn) unless otherwise noted

Income Statement Dec FY	2025A	2026E	2027E	2028E	Q2/26A	Q3/26E
Revenue	**3,809**	**5,612**	**8,534**	**11,512**	**1,270**	**1,504**
Cost of Goods Sold	1,528	1,845	2,834	4,282	410.1	490.3
Gross Profit	**2,281**	**3,767**	**5,700**	**7,230**	**860.3**	**1,014**
EBITDA	**2,624**	**4,227**	**6,390**	**8,293**	**965.1**	**1,151**
Depreciation	679.7	872.7	1,303	1,899	196.2	245.2
Amortisation	8.4	9.1	9.4	10.2	2.3	2.4
EBIT	**1,936**	**3,345**	**5,077**	**6,383**	**766.6**	**903.0**
Net Interest Income	93.4	102.1	104.8	169.8	27.0	22.6
Associates	5.5	5.7	5.2	4.0	1.4	1.3
Exceptionals	0.0	0.0	0.0	0.0	0.0	0.0
Forex Gains / Losses	13.8	6.6	0.0	0.0	0.4	0.0
Other Pre-Tax Income	(7.1)	61.8	0.0	0.0	67.0	0.0
Pre-Tax Profit	**2,042**	**3,521**	**5,187**	**6,557**	**862.4**	**926.8**
Tax Expense	(326.3)	(586.0)	(855.9)	(1,115)	(155.6)	(148.3)
Net Profit	**1,715**	**2,935**	**4,331**	**5,442**	**706.8**	**778.5**
Minority Interests	2.5	0.4	2.6	(2.0)	(0.2)	0.5
Reported Earnings	**1,718**	**2,936**	**4,334**	**5,440**	**706.6**	**779.0**
Adjusted Earnings	**1,718**	**2,936**	**4,334**	**5,440**	**706.6**	**779.0**
Basic Shares Outstanding	25.9	25.9	25.9	25.9	25.9	25.9
Diluted Shares Outstanding	25.9	25.9	25.9	25.9	25.9	25.9
EPS (rep) [TWD]	66.2	113.2	167.1	209.8	27.2	30.0
Total DPS [TWD]	18.0	23.0	27.0	30.0	6.0	6.0
Ratio	**2025A**	**2026E**	**2027E**	**2028E**		
Revenue Growth (%)	31.6	47.3	52.1	34.9	-	-
EBITDA Growth (%)	32.2	61.1	51.2	29.8	-	-
EBIT Growth (%)	46.4	72.8	51.8	25.7	-	-
EPS Growth (adj) (%)	46.4	70.9	47.6	25.5	-	-
Gross Profit Margin (%)	59.9	67.1	66.8	62.8	-	-
EBITDA Margin (%)	68.9	75.3	74.9	72.0	-	-
EBIT Margin (%)	50.8	59.6	59.5	55.4	-	-
Net Profit Margin (%)	45.1	52.3	50.8	47.3	-	-
Payout Ratio (%)	27.2	20.3	16.2	14.3	-	-
PE (rep) (x)	39.0	22.8	15.5	12.3	-	-
EV/EBITDA (x)	24.6	15.3	10.1	7.8	-	-
EV/EBIT (x)	33.3	19.3	12.7	10.1	-	-
Price/Book (x)	12.4	8.6	5.9	4.2	-	-
Total Div Yield (%)	0.7	0.9	1.0	1.2	-	-

Balance Sheet	2025A	2026E	2027E	2028E
Cash	2,768	4,260	7,237	11,311
Receivables	282.1	896.0	1,231	1,692
Inventories	288.1	466.7	641.3	881.0
Investments	473.1	611.7	611.7	611.7
Fixed Assets	3,692	4,916	6,163	7,454
Intangibles	49.9	48.1	48.1	48.1
Other Assets	380.1	324.3	359.2	407.2
Total Assets	**7,933**	**11,523**	**16,292**	**22,404**
Payables	767.6	1,773	2,437	3,348
Short Term Debt	136.9	170.9	206.3	238.0
Long Term Debt	896.1	926.0	1,118	1,289
Other Liabilities	671.7	832.5	1,077	1,413
Total Liabilities	2,472	3,703	4,838	6,288
Shareholders' Funds	5,363	7,656	11,290	15,952
Minority Interests	41.2	42.0	42.0	42.0
Other	56.8	122.1	122.1	122.1
Total S/H Equity	**5,461**	**7,820**	**11,454**	**16,116**
Total Liab & S/H Funds	7,933	11,523	16,292	22,404
Net Debt / Equity (%)	(31.8)	(40.4)	(51.6)	(60.7)
ROE (%)	35.4	44.5	45.2	39.6
ROA (%)	26.5	34.4	36.5	33.0
ROIC (%)	50.6	74.8	91.0	95.6
Cash Flow Analysis	**2025A**	**2026E**	**2027E**	**2028E**
EBITDA	**2,624**	**4,227**	**6,390**	**8,293**
Tax Paid	(326.3)	(586.0)	(855.9)	(1,115)
Chgs in Working Cap	34.2	398.2	363.1	498.7
Net Interest Paid	93.4	102.1	104.8	169.8
Other	(150.5)	(156.2)	7.8	2.0
Operating Cashflow	**2,275**	**3,985**	**6,010**	**7,849**
Acquisitions	(29.7)	(138.6)	0.0	0.0
Capex	(1,272)	(1,978)	(2,560)	(3,200)
Other	157.7	135.7	0.0	0.0
Investing Cashflow	**(1,144)**	**(1,981)**	**(2,560)**	**(3,200)**
Dividend (Ordinary)	(466.8)	(596.4)	(700.2)	(778.0)
Equity Raised	0.0	0.0	0.0	0.0
Debt Movements	14.7	64.0	227.0	203.3
Other	11.7	(18.0)	0.0	0.0
Financing Cashflow	**(440.3)**	**(550.5)**	**(473.2)**	**(574.7)**
Net Chg in Cash/Debt	**640.2**	**1,492**	**2,977**	**4,074**
Free Cashflow	**1,003**	**2,007**	**3,450**	**4,649**

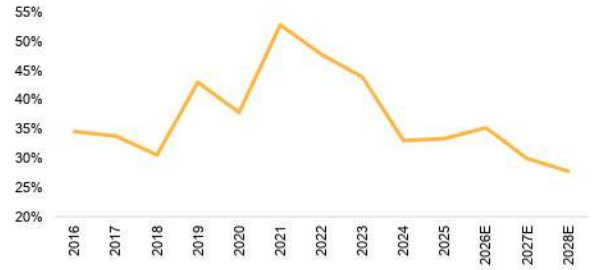
Source: Company data, Macquarie Research Oct 2026

Figure 3 - TSMC revision snapshot

(NT$m)	FY26			FY27			FY28		
	New	old	change	New	old	change	New	old	change
Revenue	5,611,930	5,498,290	2%	8,534,416	8,211,676	4%	11,512,006	11,076,210	4%
Gross profit	3,766,862	3,687,754	2%	5,700,266	5,483,644	4%	7,229,540	6,952,394	4%
Op profit	3,345,183	3,263,493	3%	5,077,360	4,849,395	5%	6,383,299	6,086,238	5%
Pretax profit	3,521,391	3,441,259	2%	5,187,342	4,955,782	5%	6,557,104	6,249,566	5%
Net profit	2,935,803	2,868,047	2%	4,334,024	4,138,078	5%	5,440,429	5,187,139	5%
Margins									
Gross margin	67.1%	67.1%	6 bps	66.8%	66.8%	2 bps	62.8%	62.8%	4 bps
Op margin	59.6%	59.4%	26 bps	59.5%	59.1%	44 bps	55.4%	54.9%	51 bps
Net margin	52.3%	52.2%	16 bps	50.8%	50.4%	40 bps	47.3%	46.8%	43 bps

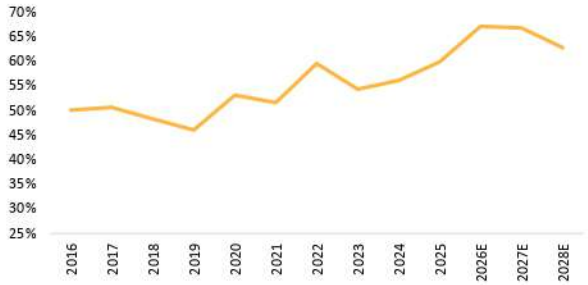
Source: TSMC, Macquarie Research, Oct 2026

Figure 4 - TSMC: Capex to revenue ratio



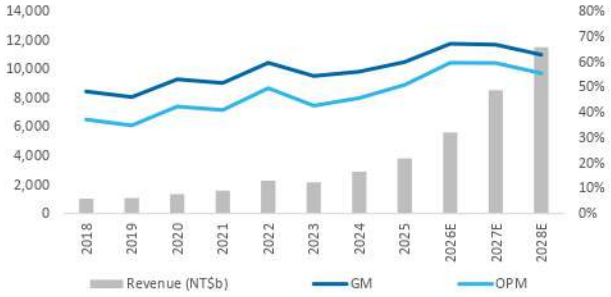
Source: TSMC, Macquarie Research, Oct 2026

Figure 5 - TSMC: GPM trend



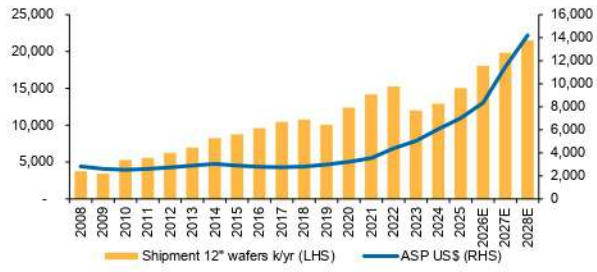
Source: TSMC, Macquarie Research, Oct 2026

Figure 6 - TSMC: Revenue in NT$bn and margins



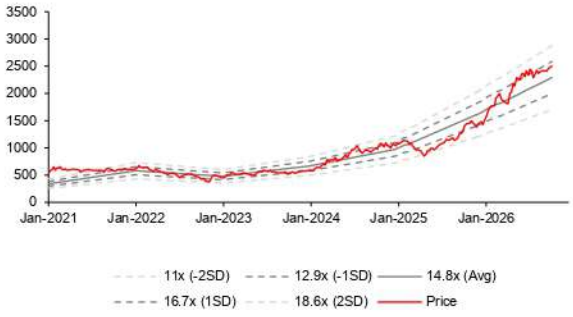
Source: TSMC, Macquarie Research, Oct 2026

Figure 7 - TSMC: Wafer shipment, ASP



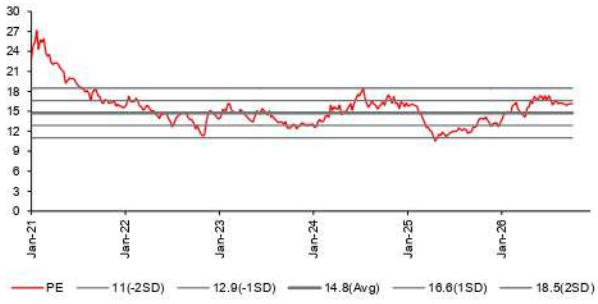
Source: TSMC, Macquarie Research, Oct 2026

Figure 8 - TSMC: 12-month forward PEx



Source: Factset, Macquarie Research, Oct 2026

Figure 9 - TSMC: 12-month Forward PE band



Source: Factset, Macquarie Research, Oct 2026

Key Quant Findings
(top 5% of the Macquarie Quant Alpha model)
The quant model currently holds a strong positive view on TSMC. The strongest style exposure is Profitability, indicating this stock is efficiently converting investments to earnings; proxied by ratios like ROE or ROA. The weakest style exposure is Valuations, indicating this stock is over-priced in the market relative to its peers.

Macquarie Alpha Model: Key rankings

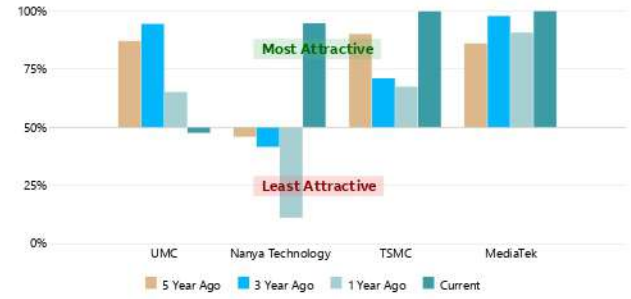
The Macquarie Quant's flagship Alpha model is a dynamic multi-factor model based on a staple of quant factors such as value, momentum, revisions, quality, and risk.

	Global	Market (Country)	Sector
	Whole Universe	Taiwan	Semiconductors & Semiconductor Equip.
Macquarie Alpha Model	**2/18805**	**2/987**	**2/705**
Fundamental (Consensus) *	**2828/18805**	**78/987**	**140/705**

** based on Total Shareholder Return = Consensus Price target / Current Price*

Current and Historical Alpha Model Rank

The chart shows the Macquarie Alpha model market ranking against the company's peers and over recent history.



Drivers of Stock Return

Breakdown of 1-year total return (local currency) into returns from dividends, changes in forward earnings estimates and the resulting change in earnings multiple.

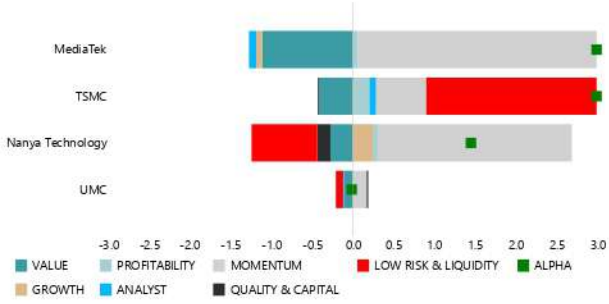
Alpha Model Decomposition

The Macquarie Alpha is decomposed into its sector and market relative factor & styles exposures (a higher/better percentile is coded in green, whilst lower in red).

Factors / Styles	Percentile relative to sectors (/705)	Percentile relative to market (/987)	Core factors in definition
ALPHA	100%	100%	Built from the styles below
VALUE	70%	20%	Book, CF, Yield, Earnings Multiples
ANALYST	83%	97%	Revisions (Earnings, Recommendations)
MOMENTUM	60%	88%	Price Momentum
GROWTH	55%	75%	EPS, Sales (Forecast, Historic)
PROFITABILITY	96%	99%	ROE, Margin, Asset Turnover
QUALITY	81%	91%	Accruals, Earn Stability, Cash Conversion
CAPITAL	60%	22%	Investment/Capex, Net share issuance
LIQUIDITY	100%	100%	Size, Turnover, Analyst Coverage
LOW RISK	88%	75%	Beta, Volatility, Earn.Cert, Leverage
TECHNICAL	60%	41%	MACD, RSI, Bollinger, Williams R, etc

Factors driving the Alpha Model vs peers

For the comparable firms this chart shows the key underlying styles and their contribution to the current overall raw Alpha score.



Macquarie Style Returns over last year

Recent performance to Macquarie style factors

	Monthly Factor Long-Short Returns for												Last 5 Years (ann)	Last 10 Years (ann)
Asia Ex JP	Sep-26	Aug-26	Jul-26	Jun-26	May-26	Apr-26	Mar-26	Feb-26	Jan-26	Dec-25	Nov-25	Oct-25		
ALPHA	5%	6%	10%	1%	3%	3%	4%	0%	0%	3%	3%	2%	11%	7%
VALUE	2%	1%	19%	-5%	1%	-7%	6%	0%	-2%	3%	5%	4%	13%	5%
ANALYST	1%	14%	-12%	7%	4%	12%	-2%	0%	6%	5%	-2%	3%	16%	14%
MOMENTUM	1%	14%	-35%	15%	17%	19%	-4%	5%	8%	6%	-1%	-2%	-1%	4%
GROWTH	-3%	8%	-19%	9%	7%	7%	-4%	-1%	4%	2%	-2%	0%	4%	2%
PROFITABILITY	1%	1%	3%	6%	1%	-4%	3%	0%	-4%	1%	3%	0%	-2%	-2%
QUALITY	2%	0%	7%	-3%	-2%	-4%	2%	-1%	-4%	-1%	1%	-4%	-4%	-2%
CAPITAL	1%	-7%	14%	-7%	-1%	-12%	2%	-2%	2%	-1%	3%	-3%	2%	-4%
LIQUIDITY	2%	-1%	-11%	8%	10%	7%	-1%	3%	1%	2%	-1%	-1%	-2%	3%
LOW RISK	5%	-13%	29%	-6%	-8%	-13%	6%	-2%	-8%	-2%	5%	1%	-7%	-7%

Source (all charts): FactSet, Refinitiv, and Macquarie Quant. For more details on the Macquarie Alpha model or for more customised analysis and screens, please contact the Macquarie Global Quantitative Team: maccapequitiesresearchquantglobal@macquarie.com. Explanation for items on this page can be found at https://www.macquarieinsights.com/rp/d/r/p/OTUyMzg1

Analysts

Arthur Lai
852 3922 4216
art.lai@macquarie.com
Macquarie Capital Limited

Katherine Zheng
+852 6886 4436
kat.zheng@macquarie.com
Macquarie Capital Limited

Important Disclosures

Recommendation definitions	Volatility index definition	Financial definitions
Macquarie – Asia and USA Outperform – expected return >10% Neutral – expected return from -10% to +10% Underperform – expected return <-10% **Macquarie – Australia/New Zealand** Outperform – expected return >10% Neutral – expected return from 0% to 10% Underperform – expected return <0% During periods of share price volatility, recommendations and target prices may occasionally and temporarily be inconsistent with the above definitions. **Recommendations** – 12 months **12-month target** – Expected share price in 12 months **Valuation** – The company's estimated fair value share price based on the disclosed valuation methodology **Note**: Quant recommendations may differ from Fundamental Analyst recommendations	This is calculated from the volatility of historical price movements. **Very high** – highest risk – Stock should be expected to move up or down 60-100% in a year – investors should be aware this stock is highly speculative. **High** – stock should be expected to move up or down at least 40-60% in a year – investors should be aware this stock could be speculative. **Medium** – stock should be expected to move up or down at least 25-40% in a year. **Low** – stock should be expected to move up or down at least 15-25% in a year. * Applicable to select stocks in Asia/Australia/NZ **Note:** expected return is reflective of a Medium Volatility stock and should be assumed to adjust proportionately with volatility risk	All "Adjusted" data items have had the following adjustments made: **Added back:** goodwill amortisation, provision for catastrophe reserves, IFRS derivatives & hedging, IFRS impairments & IFRS interest expense Excluded: non recurring items, asset revals, property revals, appraisal value uplift, preference dividends & minority interests **EPS** = adjusted net profit / efpowa* **ROA** = adjusted ebit / average total assets **ROA Banks/Insurance** = adjusted net profit /average total assets **ROE** = adjusted net profit / average shareholders funds **Gross cashflow** = adjusted net profit + depreciation *equivalent fully paid ordinary weighted average number of shares All Reported numbers for Australian/NZ listed stocks are modelled under IFRS (International Financial Reporting Standards).

Recommendation proportions for quarter ending 30 Sept 2026

	AU/NZ	Asia	USA	
Outperform	62.24%	73.44%	63.24%	(for global coverage by Macquarie, 1.61% of stocks followed are investment banking clients)
Neutral	30.77%	14.98%	36.76%	(for global coverage by Macquarie, 0.41% of stocks followed are investment banking clients)
Underperform	6.99%	11.58%	0.00%	(for global coverage by Macquarie, 0.82% of stocks followed are investment banking clients)

Company-Specific Disclosures

Company Name	Disclosure
TSMC (2330 TT) **Outperform** 12-month target: TWD4,410.00 - PER Valuation: TWD 4,410.00 - PER Price: TWD2,585	None

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Recommendation history

Company name	Date	Recommendation	Target price
TSMC (2330 TT)	6-Oct-2026	Outperform	TWD 4410.00
	16-Jul-2026	Outperform	TWD 4200.00
	18-Jun-2026	Outperform	TWD 3380.00
	16-Apr-2026	Outperform	TWD 2550.00
	16-Jan-2026	Outperform	TWD 2400.00
	6-Jan-2026	Outperform	TWD 2150.00
	16-Oct-2025	Outperform	TWD 1710.00
	3-Oct-2025	Outperform	TWD 1620.00
	17-Jul-2025	Outperform	TWD 1310.00

7-Jul-2025	Outperform	TWD 1282.00
18-Apr-2025	Outperform	TWD 1125.00
10-Apr-2025	Outperform	TWD 990.00
7-Jan-2025	Outperform	TWD 1450.00
18-Oct-2024	Outperform	TWD 1500.00
10-Oct-2024	Outperform	TWD 1350.00
19-Jul-2024	Outperform	TWD 1295.00
5-Jul-2024	Outperform	TWD 1280.00
19-Apr-2024	Outperform	TWD 1000.00
18-Jan-2024	Outperform	TWD 710.31
19-Oct-2023	Outperform	TWD 725.43

Sensitivity analysis:
Clients receiving this report can request access to a model which allows for further in-depth analysis of the assumptions used, and recommendations made, by the author relating to the subject companies covered. Contact https://www.macquarieinsights.com/contacts for access requests.

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上銀

(2049.TW/2049 TT)

FTSE4Good TIP Taiwan ESG Index stock

增加持股・維持

收盤價 October 6 (NT$)	339.5
12 個月目標價 (NT$)	482.0
前次目標價 (NT$)	478.0
調升 (%)	0.8
上漲空間 (%)	42.0

焦點內容

1. 3Q26 營收較我們預估高出 6%，顯示利潤率與獲利上修可期
2. 客戶資本支出持續增加，加上製造業活動復甦範圍擴大，可望推動本波上升週期延續至 2027 年
3. 我們預期市場共識進一步上修獲利預估，加上能見度改善，將帶動評價提升

交易資料表

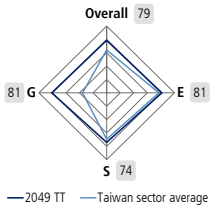
市值：(NT$bn/US$mn)	120.1 / 3,771
流通在外股數 (百萬股)：	353.8
外資持有股數 (百萬股)：	101
3M 平均成交量 (百萬股)：	2.89
52 週股價 (低 \ 高) (NT$):	181.0 –409.5

股價表現	3M	6M	12M
絕對表現 (%)	-6.7	48.9	56.5
相對表現 (%)	-13.7	-4.1	-29.7

每股盈餘

NT$	1Q	2Q	3Q	4Q
2025	1.36A	0.38A	1.26A	1.31A
2026	1.64A	2.40A	2.89F	2.46F
2027	2.47F	3.47F	3.60F	3.14F

ESG 分數評等



資料來源：Refinitiv、凱基

凱基投顧

莊政翰
886.2.2181.8719
angus.chuang@kgi.com

張育程
886.2.2181.8735
eden.chang@kgi.com

重要免責聲明，詳見最終頁

上升週期延續，獲利調升可期

重要訊息

上銀 3Q26 營收年增率加速至 35%，分別優於市場共識及我們預估 3%與 6%。儘管人力短缺可能限制短期營收動能，我們認為客戶資本支出持續上修，加上出口成長加速帶動製造業活動升溫，將支撐本波上升週期延續至 2027 年。

評論及分析

3Q26 營收表現超前。9 月營收達 27.2 億元，年增 30.1%、月增 0.4%，帶動 3Q26 營收年增 35%、季增 8%。受惠 AI 與半導體相關應用訂單強勁，以及產品漲價，3Q26 營收表現優於季節性，並分別較市場共識與我們預估高出 3%與 6%。管理層表示亞洲及美國市場需求維持強勁，歐洲訂單亦開始出現復甦跡象。考量產能持續滿載及產品組合有利，我們預估毛利率將季增 0.8 個百分點至 35.0%。加上營運槓桿效益，我們預期營益率進一步走升將帶動每股盈餘達 2.89 元，較凱基原估與市場共識高出 8%、7%。

自動化上升週期可望延續至 2027 年。上銀指出目前主要瓶頸仍為人力短缺，因接單持續超過產能供給，滾珠螺桿與線性滑軌交期分別維持於六個月及五個月的高檔水準。雖然公司已積極增聘移工並提高加班時數，但我們認為相關措施轉化為明顯營收貢獻仍需時間。受惠有利的產業動態，我們目前預估 4Q26 營收將穩健年增 21%，季減幅度收斂至 3%。展望後市，我們預期客戶資本支出持續上修，加上出口成長加速及 PMI 指標回升所反映的製造業活動增溫，可望支撐本波自動化上升週期延續至 2027 年。

獲利進一步上修可望推升評價。考量 3Q26 營收優於預期，以及 AI 驅動的半導體需求持續擴大至更多應用領域，我們預期市場將在 11 月 10 日財報公布前進一步調升上銀獲利預估。屆時，我們預期管理層將釋出樂觀的 2027 年展望，反映客戶持續擴大資本支出與產能，進一步強化公司成長動能。我們分別調升 2026-27 年獲利預估 4%/1%，且隨著投資人對本波上升週期延續性的信心增強，我們預期市場共識獲利預估仍有進一步上修空間，進而帶動評價提升。

投資建議

我們持續看好上銀，預期自動化上升週期延續加上共識獲利上修將推升股價上檔。目標價調整為 482 元，根據 35 倍本益比。重申「增加持股」評等。

投資風險

下檔風險包括資本支出週期延宕、新台幣升值。

主要財務數據及估值

	Dec-24A	Dec-25A	Dec-26F	Dec-27F	Dec-28F
營業收入 (NT$百萬)	24,392	24,263	29,843	33,202	36,172
營業毛利 (NT$百萬)	7,218	6,978	10,050	11,450	12,620
營業利益 (NT$百萬)	2,059	1,643	3,821	5,143	6,112
稅後淨利 (NT$百萬)	1,972	1,526	3,319	4,485	5,268
每股盈餘 (NT$)	5.57	4.31	9.38	12.68	14.89
每股現金股利 (NT$)	2.40	2.00	4.35	5.88	6.90
每股盈餘成長率 (%)	(3.1)	(22.6)	117.5	35.1	17.5
本益比 (x)	60.9	78.7	36.2	26.8	22.8
股價淨值比 (x)	3.3	3.2	3.1	2.9	2.7
EV/EBITDA (x)	26.6	29.4	15.7	12.7	10.8
淨負債比率 (%)	7.2	5.6	1.1	Net cash	Net cash
殖利率 (%)	0.7	0.6	1.3	1.7	2.0
股東權益報酬率 (%)	5.5	4.1	8.7	11.1	12.3

資料來源：公司資料，凱基

圖 1：3Q26 與 4Q26 財務預估修正暨市場共識比較

	3Q26F							4Q26F						
百萬元	修正後	修正前	調幅 (%)	QoQ (%)	YoY (%)	市場共識	差異 (%)	修正後	修正前	調幅 (%)	QoQ (%)	YoY (%)	市場共識	差異 (%)
營收	8,110	7,682	5.6	8.0	35.3	7,885	2.9	7,845	7,398	6.0	(3.3)	20.6	7,998	(1.9)
毛利	2,839	2,673	6.2	10.7	65.5	2,735	3.8	2,608	2,445	6.7	(8.1)	48.3	2,766	(5.7)
營業利益	1,196	1,118	7.0	15.7	191.3	1,154	3.6	961	891	7.8	(19.6)	204.7	1,096	(12.3)
稅前淨利	1,210	1,128	7.3	12.1	111.1	1,182	2.3	991	925	7.1	(18.1)	89.6	1,164	(14.9)
稅後淨利	1,022	950	7.6	20.6	129.2	954	7.1	871	815	6.8	(14.8)	88.4	943	(7.7)
每股盈餘 (元)	2.89	2.68	7.6	20.6	129.2	2.70	7.1	2.46	2.30	6.8	(14.8)	88.4	2.67	(7.7)
毛利率 (%)	35.0	34.8	0.2 ppts	0.8 ppts	6.4 ppts	34.7	0.3 ppts	33.2	33.0	0.2 ppts	(1.8) ppts	6.2 ppts	34.6	(1.3) ppts
營業利益率 (%)	14.7	14.5	0.2 ppts	1.0 ppts	7.9 ppts	14.6	0.1 ppts	12.2	12.0	0.2 ppts	(2.5) ppts	7.4 ppts	13.7	(1.5) ppts
淨利率 (%)	12.6	12.4	0.2 ppts	1.3 ppts	5.2 ppts	12.1	0.5 ppts	11.1	11.0	0.1 ppts	(1.5) ppts	4.0 ppts	11.8	(0.7) ppts

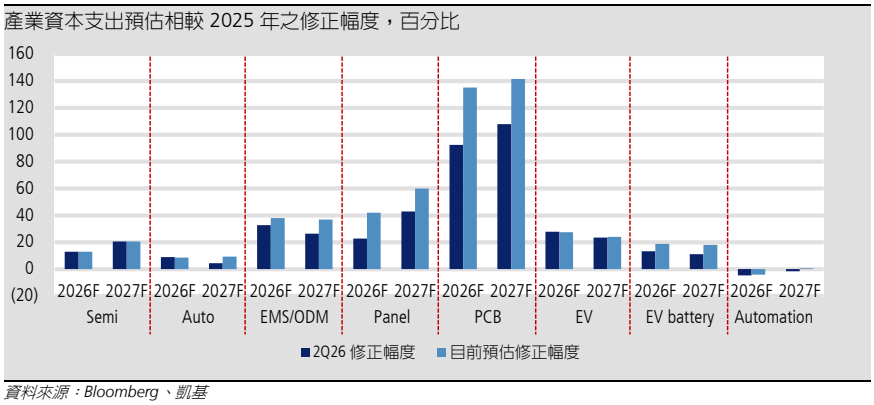
資料來源：Bloomberg；凱基預估

圖 2：2026-28 年財務預估修正暨市場共識比較

	2026F						2027F						2028F					
百萬元	修正後	修正前	調幅 (%)	YoY (%)	市場共識	差異 (%)	修正後	修正前	調幅 (%)	YoY (%)	市場共識	差異 (%)	修正後	修正前	調幅 (%)	YoY (%)	市場共識	差異 (%)
營收	29,843	28,967	3.0	23.0	29,548	1.0	33,202	32,383	2.5	11.3	33,959	(2.2)	36,172	35,298	2.5	8.9	33,617	7.6
毛利	10,050	9,721	3.4	44.0	9,940	1.1	11,450	11,131	2.9	13.9	11,753	(2.6)	12,620	12,310	2.5	10.2	11,636	8.5
營業利益	3,821	3,673	4.0	132.6	3,828	(0.2)	5,143	4,981	3.3	34.6	4,973	3.4	6,112	5,962	2.5	18.8	5,086	20.2
稅前獲利	3,999	3,851	3.9	126.7	3,994	0.1	5,298	5,227	1.4	32.5	5,087	4.1	6,314	6,262	0.8	19.2	5,094	24.0
稅後淨利	3,319	3,204	3.6	117.5	3,226	2.9	4,485	4,429	1.3	35.1	4,098	9.4	5,268	5,227	0.8	17.5	4,052	30.0
每股盈餘 (元)	9.38	9.06	3.6	117.5	9.12	2.9	12.68	12.52	1.3	35.1	11.58	9.4	14.89	14.77	0.8	17.5	11.45	30.0
毛利率 (%)	33.7	33.6	0.1 ppts	4.9 ppts	33.6	0.0 ppts	34.5	34.4	0.1 ppts	0.8 ppts	34.6	(0.1) ppts	34.9	34.9	0.0 ppts	0.4 ppts	34.6	0.3 ppts
營利率 (%)	12.8	12.7	0.1 ppts	6.0 ppts	13.0	(0.1)ppts	15.5	15.4	0.1 ppts	2.7 ppts	14.6	0.8 ppts	16.9	16.9	0.0 ppts	1.4 ppts	15.1	1.8 ppts
淨利率 (%)	11.1	11.1	0.1 ppts	4.8 ppts	10.9	0.2 ppts	13.5	13.7	(0.2) ppts	2.4 ppts	12.1	1.4 ppts	14.6	14.8	(0.2) ppts	1.1 ppts	12.1	2.5 ppts

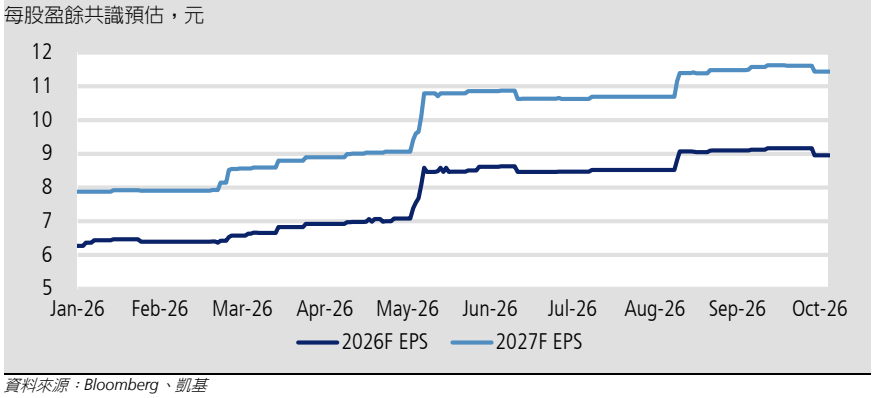
資料來源：Bloomberg；凱基預估

圖 3：產業資本支出預估持續上修



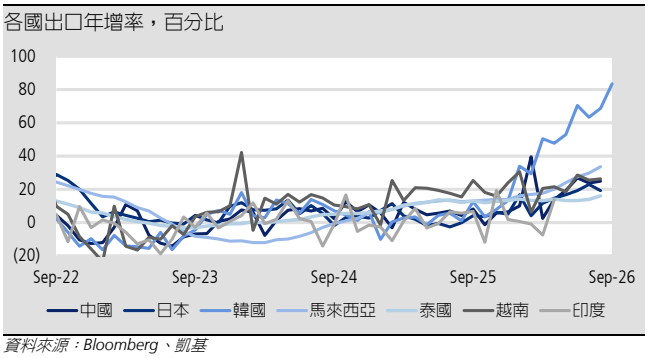
資料來源：Bloomberg、凱基

圖 4：市場共識對上銀獲利將進一步上修



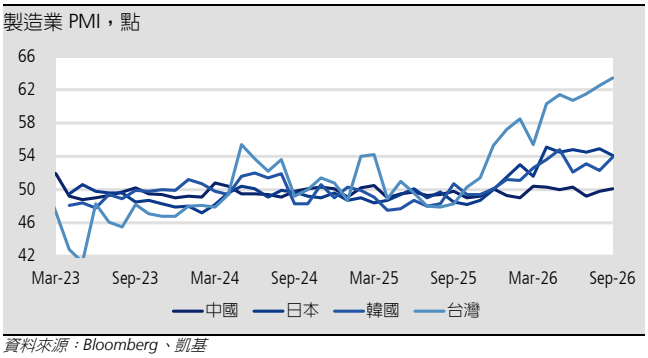
資料來源：Bloomberg、凱基

圖 5：亞洲出口成長加速



資料來源：Bloomberg、凱基

圖 6：亞洲製造業 PMI 持續走高



資料來源：Bloomberg、凱基

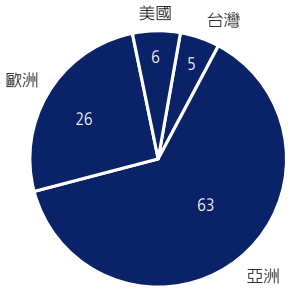
圖 7: 公司概況

上銀為全球第二大線性滑軌產品製造商，僅次於 THK (6481 JP)，公司研發並製造線性移動零組件，包括線性滑軌與滾珠螺桿。線性滑軌為公司收入大宗，佔整體營收的 61%，其次為滾珠螺桿 (20%)、工業機器人 (9%) 與其他 (10%)。目前亞洲 (不含台灣) 為其最大市場，營收佔比達 63%，其次為歐洲 (26%)、美洲 (6%) 與台灣 (5%)。

資料來源：凱基

圖 8: 亞洲為上銀主要銷售市場

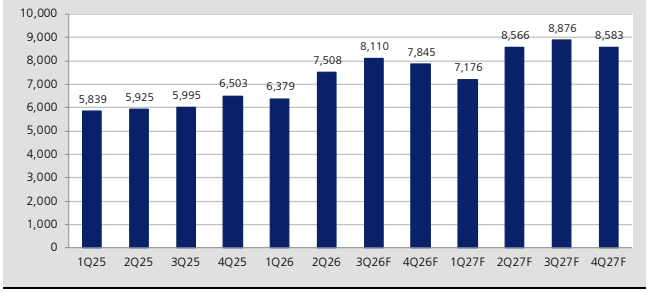
營收地區占比，百分比



資料來源：凱基

圖 9: 季營業收入

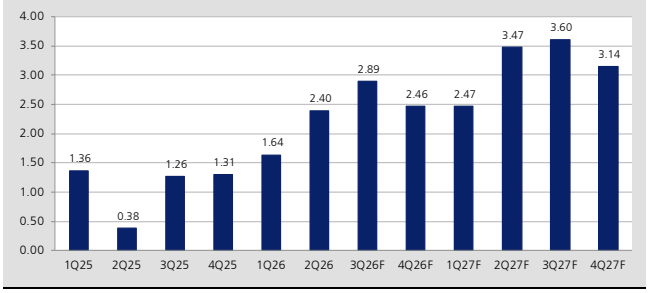
季營業收入，百萬元



資料來源：凱基

圖 10: 每股盈利

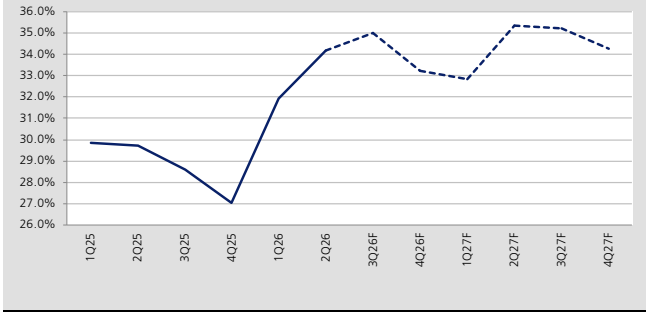
每股盈利，元



資料來源：凱基

圖 11: 毛利率

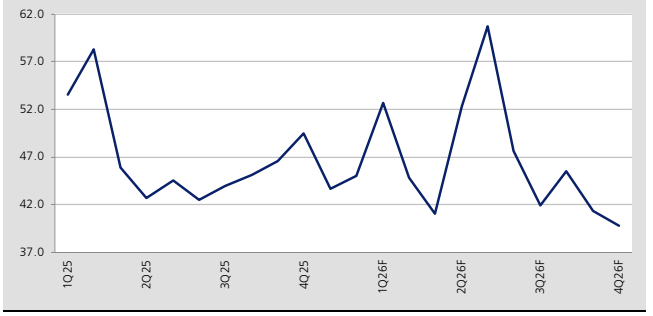
毛利率，百分比



資料來源：凱基

圖 12: 本益比

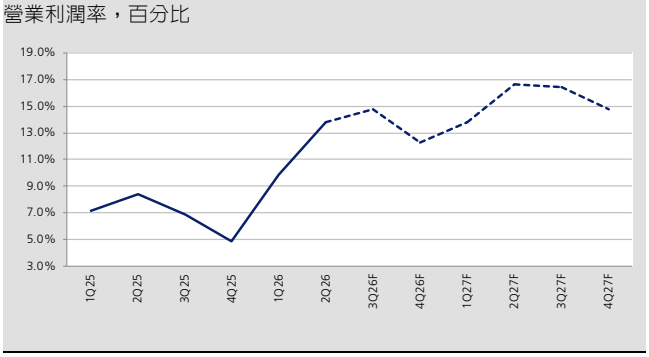
本益比，倍



資料來源：凱基

圖 13: 營業利潤率

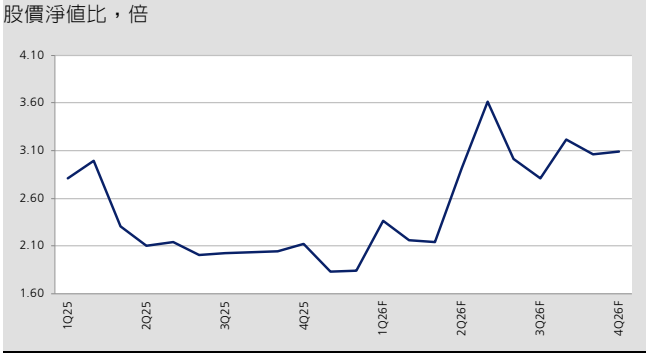
營業利潤率，百分比



資料來源：凱基

圖 14: 股價淨值比

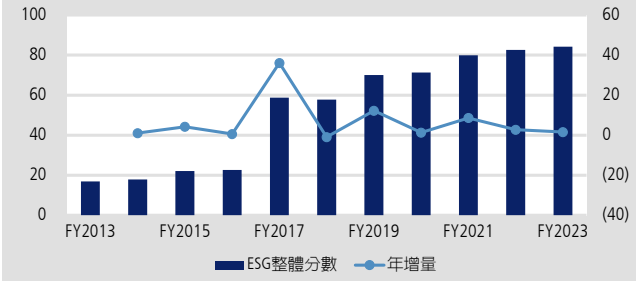
股價淨值比，倍



資料來源：凱基

圖 15: ESG 整體分數

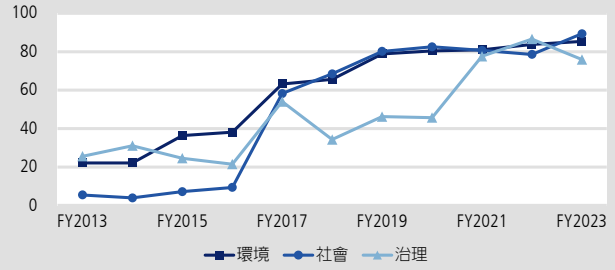
ESG 整體分數，分 (左軸)；年增量，分 (右軸)



資料來源：Refinitiv；凱基；公司資料

圖 16: ESG 各項分數

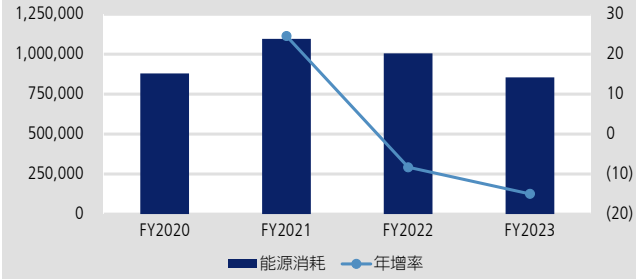
ESG 分數，分



資料來源：Refinitiv；凱基；公司資料

圖 17: 能源消耗

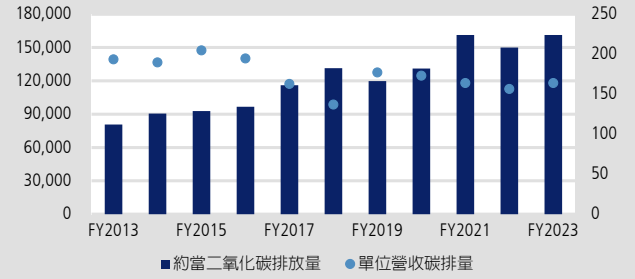
能源消耗，十億焦耳 (左軸)；年增率，百分比 (右軸)



資料來源：Refinitiv；凱基；公司資料

圖 18: 碳排量

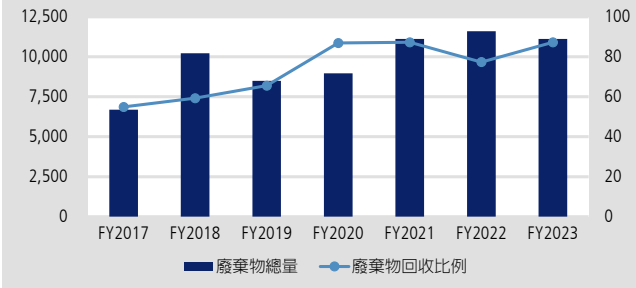
約當二氧化碳排放量，噸 (左軸)；單位營收碳排量，噸/百萬美元 (右軸)



資料來源：Refinitiv；凱基；公司資料

圖 19: 廢棄物總量

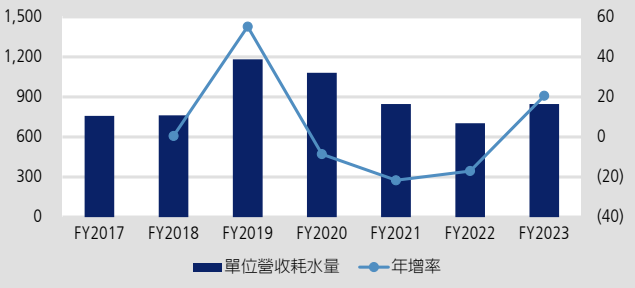
廢棄物總量，噸 (左軸)；廢棄物回收比例，百分比 (右軸)



資料來源：Refinitiv；凱基；公司資料

圖 20: 耗水量

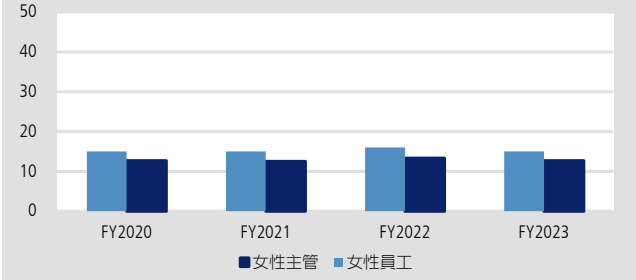
單位營收耗水量，立方公尺/百萬美元 (左軸)；年增率，百分比 (右軸)



資料來源：Refinitiv；凱基；公司資料

圖 21: 性別多樣性

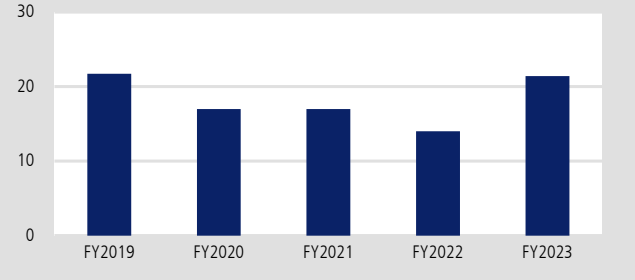
女性主管及女性員工比例，百分比



資料來源：Refinitiv；凱基；公司資料

圖 22: 員工流動率

員工流動率，百分比



資料來源：Refinitiv；凱基；公司資料

ESG 圖表定義

項目	定義	說明
能源消耗	直接與間接能源消耗總量(十億焦耳) - 公司運營範圍內消耗之能源總量 - 能源使用總量 = 直接能源消耗總量 + 間接能源消耗量 - 總能源使用量包括購買的能源、生產的能源 - 就公用事業而言，輸電/電網損耗為其業務活動之一部分，被視為總能源消耗，數據不包括為滿足能源使用而生產的電力（公用事業為出售而生產） - 就公用事業而言，用於能源生產的煤、天然氣或核能等原料不屬於「總能源使用」項下	
購買之再生能源	再生能源購買總量(十億焦耳) - 公司各種來源之能源消耗量與購買的能源中屬於自然界可再生者（太陽能、風能、水能、生物質能、地熱能）之量 - 如無證據顯示再生能源由公司生產，則所報告的能源數據視為購買的再生能源	
再生能源使用率	再生能源占總能源消耗量之比例	
約當二氧化碳排放量	直接 CO2 與 CO2 約當排放量(公噸) - 公司擁有或控制的來源的直接排放量（範圍 1 排放量） - 相關氣體：二氧化碳 (CO2)、甲烷 (CH4)、一氧化二氮 (N2O)、氫氟碳化物 (HFCS)、全氟化合物 (PFCS)、六氟化硫 (SF6)、三氟化氮 (NF3)	
單位營收碳排量	直接 CO2 與 CO2 約當排放量（公噸）銷售（百萬元）比 - 公司擁有或控制的來源的直接排放量（範圍 1 排放量） - 相關氣體：二氧化碳 (CO2)、甲烷 (CH4)、一氧化二氮 (N2O)、氫氟碳化物 (HFCS)、全氟化合物 (PFCS)、六氟化硫 (SF6)、三氟化氮 (NF3)	
廢棄物總量	廢棄物總量(公噸) - 總廢棄物 = 無害廢棄物 + 有害廢棄物 - 僅考慮固體廢棄物，但如液體廢棄物以公噸為單位呈報，則會將其加入求得包含液體廢棄物之總量 - 對於採礦、石油與天然氣等行業，tailings、廢石、煤、飛灰等廢棄物亦考慮在內	
廢棄物回收比例	公司呈報的廢棄物回收比例 - 廢棄物回收比例 = 廢棄物回收量/總廢棄物*100 - 廢棄物轉化為能源或經由廢棄物焚燒產生能源視為廢棄物回收 - 經由堆肥回收的廢棄物視為回收之廢棄物	
耗水量	耗水量(立方公尺) - 由呈報組織直接或經水公司等中介機構從任何水源抽取的總水量 - 水井、城鎮/公用事業/市政用水、河水、地表水等不同之水源均予以考慮	
環保支出	環保支出總金額 - 所有用於環境保護，或防止、減少、控制環境因素、影響、危害的投資與支出，亦包括處置、處理、衛生、清理支出	
員工流動率	員工流動率 - 包括任何原因（自願或非自願）離開公司的員工，如辭職、退休、自然離職/死亡、醫療失能、冗員、裁員、重組、解僱、裁減或定期合約到期 - 員工流動率 =（離開之員工/平均員工人數）*100 - 平均員工人數 =（本年末員工人數 + 去年末員工人數）/2 - 本財年末員工數 =上個財年末員工數 + 新員工數 - 離開之員工數	
女性主管	女性主管百分比 - 女性主管占公司主管的百分比 - 如有不同階層之百分比，如最高階、高階、中階、初階，則會考慮中階女性主管之百分比 - 女性主管百分比 = 女性主管人數/主管總數*100	
女性員工	女性員工百分比 - 女性員工占公司員工總數的百分比 - 女性員工百分比 = 女性人數/員工總數*100	
教育訓練總時數	所有員工教育訓練總時數 - 僅考慮員工教育訓練時數 - 包括一般員工所有類型的教育訓練（如健康與安全、環境、急難事件救援、技能與職業發展） - 如果數據以天為單位，則乘以 8，係假設 1 天 = 8 小時工作	
每位員工教育訓練時數	平均每年每位員工總教育訓練時數	

資料來源：Refinitiv；凱基

損益表

	季度								年度		
	Mar-26A	Jun-26A	Sep-26F	Dec-26F	Mar-27F	Jun-27F	Sep-27F	Dec-27F	Dec-26F	Dec-27F	Dec-28F
損益表（NT$百萬）											
營業收入	**6,379**	**7,508**	**8,110**	**7,845**	**7,176**	**8,566**	**8,876**	**8,583**	**29,843**	**33,202**	**36,172**
營業成本	(4,342)	(4,943)	(5,271)	(5,237)	(4,819)	(5,537)	(5,752)	(5,644)	(19,793)	(21,751)	(23,552)
營業毛利	**2,038**	**2,565**	**2,839**	**2,608**	**2,357**	**3,030**	**3,125**	**2,939**	**10,050**	**11,450**	**12,620**
營業費用	(1,407)	(1,531)	(1,643)	(1,647)	(1,368)	(1,602)	(1,665)	(1,673)	(6,228)	(6,307)	(6,508)
營業利益	**631**	**1,034**	**1,196**	**961**	**989**	**1,428**	**1,460**	**1,266**	**3,821**	**5,143**	**6,112**
折舊	(612)	(615)	(869)	(1,382)	(918)	(918)	(918)	(918)	(3,478)	(3,671)	(3,858)
攤提	(17)	(19)	(13)	(3)	(13)	(13)	(13)	(13)	(51)	(51)	(51)
EBITDA	**1,260**	**1,667**	**2,078**	**2,345**	**1,920**	**2,359**	**2,391**	**2,197**	**7,351**	**8,866**	**10,022**
利息收入	19	29	28	26	32	34	34	35	101	134	177
投資利益淨額	16	10	9	8	12	10	11	11	42	44	53
其他營業外收入	26	38	40	42	40	39	40	41	145	160	176
總營業外收入	**60**	**76**	**77**	**76**	**84**	**83**	**84**	**87**	**289**	**338**	**406**
利息費用	(54)	(57)	(48)	(52)	(55)	(54)	(56)	(59)	(211)	(223)	(236)
投資損失	-	-	-	-	-	-	-	-	-	-	-
其他營業外費用	83	26	(15)	5	10	8	10	12	100	40	32
總營業外費用	**29**	**(30)**	**(63)**	**(46)**	**(45)**	**(46)**	**(46)**	**(47)**	**(111)**	**(183)**	**(204)**
稅前純益	**719**	**1,080**	**1,210**	**991**	**1,028**	**1,465**	**1,499**	**1,306**	**3,999**	**5,298**	**6,314**
所得稅費用[利益]	(170)	(263)	(248)	(199)	(216)	(322)	(299)	(276)	(880)	(1,113)	(1,326)
少數股東損益	30	31	60	79	60	86	74	80	200	300	280
非常項目前稅後純益	**579**	**848**	**1,022**	**871**	**872**	**1,229**	**1,274**	**1,111**	**3,319**	**4,485**	**5,268**
非常項目	0	(0)	-	(0)	-	-	-	-	-	-	-
稅後淨利	**579**	**848**	**1,022**	**871**	**872**	**1,229**	**1,274**	**1,111**	**3,319**	**4,485**	**5,268**
每股盈餘 (NT$)	1.64	2.40	2.89	2.46	2.47	3.47	3.60	3.14	9.38	12.68	14.89
獲利率 (%)											
營業毛利率	31.9	34.2	35.0	33.2	32.8	35.4	35.2	34.2	33.7	34.5	34.9
營業利益率	9.9	13.8	14.7	12.2	13.8	16.7	16.4	14.7	12.8	15.5	16.9
EBITDA Margin	19.8	22.2	25.6	29.9	26.8	27.5	26.9	25.6	24.6	26.7	27.7
稅前純益率	11.3	14.4	14.9	12.6	14.3	17.1	16.9	15.2	13.4	16.0	17.5
稅後純益率	9.1	11.3	12.6	11.1	12.2	14.3	14.3	12.9	11.1	13.5	14.6
季成長率 (%)											
營業收入	(1.9)	17.7	8.0	(3.3)	(8.5)	19.4	3.6	(3.3)			
營業毛利	15.9	25.9	10.7	(8.1)	(9.6)	28.5	3.1	(5.9)			
營業收益增長	99.9	64.0	15.7	(19.6)	2.9	44.4	2.2	(13.3)			
EBITDA	36.7	32.3	24.7	12.8	(18.1)	22.9	1.4	(8.1)			
稅前純益	37.6	50.1	12.1	(18.1)	3.8	42.4	2.3	(12.8)			
稅後純益	25.4	46.3	20.6	(14.8)	0.2	40.9	3.7	(12.8)			
年成長率 (%)											
營業收入	9.3	26.7	35.3	20.6	12.5	14.1	9.5	9.4	23.0	11.3	8.9
營業毛利	17.0	45.6	65.5	48.3	15.7	18.1	10.1	12.7	44.0	13.9	10.2
營業收益	50.6	107.4	191.3	204.7	56.9	38.1	22.1	31.7	132.6	34.6	18.8
EBITDA	25.4	54.0	109.5	154.5	52.4	41.5	15.0	(6.3)	83.7	20.6	13.0
稅前純益	19.9	1475.5	111.1	89.6	43.0	35.7	23.8	31.9	126.7	32.5	19.2
稅後純益	20.0	526.8	129.2	88.4	50.5	44.9	24.6	27.6	117.5	35.1	17.5

資料來源：公司資料，凱基

資產負債表

NT$百萬	Dec-24A	Dec-25A	Dec-26F	Dec-27F	Dec-28F
資產總額	**54,418**	**54,176**	**57,660**	**61,324**	**65,158**
流動資產	**19,983**	**19,864**	**24,163**	**28,955**	**34,145**
現金及短期投資	7,354	7,625	9,879	13,255	17,209
存貨	7,407	7,094	8,123	8,927	9,639
應收帳款及票據	4,467	4,418	5,434	6,046	6,569
其他流動資產	755	727	727	727	727
非流動資產	**34,434**	**34,312**	**33,496**	**32,369**	**31,014**
長期投資	1,907	1,490	1,532	1,576	1,629
固定資產	29,367	29,836	28,978	27,806	26,398
什項資產	3,160	2,986	2,986	2,986	2,986
負債總額	**17,343**	**16,702**	**18,606**	**20,165**	**21,454**
流動負債	**8,322**	**8,530**	**11,093**	**11,782**	**12,761**
應付帳款及票據	3,318	3,251	3,722	4,091	4,417
短期借款	2,465	3,011	4,271	4,051	4,341
什項負債	2,540	2,268	3,100	3,641	4,004
長期負債	**9,021**	**8,173**	**7,513**	**8,383**	**8,693**
長期借款	7,563	6,712	6,052	6,922	7,232
其他負債及準備	988	1,041	1,041	1,041	1,041
股東權益總額	**37,074**	**37,474**	**39,054**	**41,159**	**43,704**
普通股本	3,538	3,538	3,538	3,538	3,538
保留盈餘	20,520	20,949	22,730	25,135	27,960
少數股東權益	145	128	(72)	(372)	(652)
優先股股東資金	-	-	-	-	-

主要財務比率

	Dec-24A	Dec-25A	Dec-26F	Dec-27F	Dec-28F
年成長率					
營業收入	(1.0%)	(0.5%)	23.0%	11.3%	8.9%
營業收益增長	(22.3%)	(20.2%)	132.6%	34.6%	18.8%
EBITDA	(11.7%)	(8.9%)	83.7%	20.6%	13.0%
稅後純益	(3.1%)	(22.6%)	117.5%	35.1%	17.5%
每股盈餘成長率	(3.1%)	(22.6%)	117.5%	35.1%	17.5%
獲利能力分析					
營業毛利率	29.6%	28.8%	33.7%	34.5%	34.9%
營業利益率	8.4%	6.8%	12.8%	15.5%	16.9%
EBITDA Margin	18.0%	16.5%	24.6%	26.7%	27.7%
稅後純益率	8.1%	6.3%	11.1%	13.5%	14.6%
平均資產報酬率	3.7%	2.8%	5.9%	7.5%	8.3%
股東權益報酬率	5.5%	4.1%	8.7%	11.1%	12.3%
穩定\償債能力分析					
毛負債比率 (%)	27.0%	25.9%	26.4%	26.7%	26.5%
淨負債比率	7.2%	5.6%	1.1%	Net cash	Net cash
利息保障倍數 (x)	13.6	9.2	20.0	24.7	27.7
利息及短期債保障倍數 (x)	0.4	0.3	0.5	0.5	0.6
Cash Flow Int. Coverage (x)	20.3	19.9	24.8	31.9	34.7
Cash Flow/Int. & ST Debt (x)	1.4	1.3	1.2	1.7	1.8
流動比率 (x)	2.4	2.3	2.2	2.5	2.7
速動比率 (x)	1.5	1.5	1.4	1.7	1.9
淨負債 (NT$百萬)	2,674	2,098	444	(2,282)	(5,636)
每股資料分析					
每股盈餘 (NT$)	5.57	4.31	9.38	12.68	14.89
每股現金盈餘 (NT$)	10.23	12.09	14.79	20.12	23.22
每股淨值 (NT$)	104.38	105.56	110.59	117.39	125.37
調整後每股淨值 (NT$)	104.38	105.56	110.59	117.39	125.37
每股營收 (NT$)	68.94	68.58	84.35	93.84	102.24
EBITDA/Share (NT$)	12.42	11.31	20.78	25.06	28.33
每股現金股利 (NT$)	2.40	2.00	4.35	5.88	6.90
資產運用狀況					
資產周轉率 (x)	0.46	0.45	0.53	0.56	0.57
應收帳款周轉天數	67.0	66.5	66.5	66.5	66.5
存貨周轉天數	157.9	149.8	149.8	149.8	149.8
應付帳款周轉天數	70.7	68.6	68.6	68.6	68.6
現金轉換周轉天數	154.2	147.6	147.6	147.6	147.6

資料來源：公司資料，凱基

損益表

NT$百萬	Dec-24A	Dec-25A	Dec-26F	Dec-27F	Dec-28F
營業收入	**24,392**	**24,263**	**29,843**	**33,202**	**36,172**
營業成本	(17,174)	(17,285)	(19,793)	(21,751)	(23,552)
營業毛利	**7,218**	**6,978**	**10,050**	**11,450**	**12,620**
營業費用	(5,159)	(5,335)	(6,228)	(6,307)	(6,508)
營業利益	**2,059**	**1,643**	**3,821**	**5,143**	**6,112**
總營業外收入	**277**	**251**	**289**	**338**	**406**
利息收入	95	87	101	134	177
投資利益淨額	35	22	42	44	53
其他營業外收入	146	142	145	160	176
總營業外費用	**(86)**	**(131)**	**(111)**	**(183)**	**(204)**
利息費用	(178)	(215)	(211)	(223)	(236)
投資損失	-	-	-	-	-
其他營業外費用	92	85	100	40	32
稅前純益	**2,250**	**1,764**	**3,999**	**5,298**	**6,314**
所得稅費用[利益]	(387)	(354)	(880)	(1,113)	(1,326)
少數股東損益	109	116	200	300	280
非常項目	-	0	-	-	-
稅後淨利	**1,972**	**1,526**	**3,319**	**4,485**	**5,268**
EBITDA	**4,393**	**4,001**	**7,351**	**8,866**	**10,022**
每股盈餘 (NT$)	5.57	4.31	9.38	12.68	14.89

現金流量

NT$百萬	Dec-24A	Dec-25A	Dec-26F	Dec-27F	Dec-28F
營運活動之現金流量	**3,619**	**4,277**	**5,233**	**7,117**	**8,216**
本期純益	1,972	1,526	3,319	4,485	5,268
折舊及攤提	2,334	2,358	3,529	3,723	3,910
本期運用資金變動	(168)	842	(1,574)	(1,047)	(909)
其他營業資產及負債變動	(520)	(449)	(42)	(44)	(53)
投資活動之現金流量	**(3,883)**	**(2,818)**	**(2,671)**	**(2,551)**	**(2,501)**
投資用短期投資出售[新購]	13	2	-	-	-
本期長期投資變動	-	-	-	-	-
資本支出淨額	(3,714)	(2,766)	(2,620)	(2,500)	(2,450)
其他資產變動	(182)	(54)	(51)	(51)	(51)
自由現金流	**95**	**1,693**	**2,265**	**4,187**	**5,328**
融資活動之現金流量	**(19)**	**(1,278)**	**(308)**	**(1,189)**	**(1,760)**
短期借款變動	(98)	618			
長期借款變動	1,101	(935)	600	650	600
現金增資	-	-			
已支付普通股股息	(884)	(849)	(708)	(1,539)	(2,080)
其他融資現金流	(138)	(112)	(200)	(300)	(280)
匯率影響數	**73**	**89**			
本期產生現金流量	**(210)**	**270**	**2,254**	**3,376**	**3,954**

投資回報率

	Dec-24A	Dec-25A	Dec-26F	Dec-27F	Dec-28F
1－營業成本/營業收入					
－銷管費用/營業收入	21.2%	22.0%	20.9%	19.0%	18.0%
＝營業利益率	**8.4%**	**6.8%**	**12.8%**	**15.5%**	**16.9%**
1／(營業運用資金/營業收入	0.3	0.3	0.3	0.2	0.2
＋淨固定資產/營業收入	1.2	1.2	1.0	0.8	0.7
＋什項資產/營業收入)	0.1	0.1	0.1	0.0	0.0
＝資本周轉率	**0.6**	**0.6**	**0.8**	**0.9**	**1.0**
營業利益率	8.4%	6.8%	12.8%	15.5%	16.9%
x 資本周轉率	0.6	0.6	0.8	0.9	1.0
x (1－有效現金稅率)	82.8%	79.9%	78.0%	79.0%	79.0%
＝稅後 ROIC	**4.5%**	**3.4%**	**7.8%**	**10.9%**	**13.2%**

資料來源：公司資料，凱基

上銀 – 以往評級及目標價

日期	評級	目標價	收盤價
2026-08-12	增加持股	478.0	369.0
2026-05-28	增加持股	478.0	388.0
2026-05-08	增加持股	370.0	319.5
2026-04-21	增加持股	350.0	301.0
2026-03-06	增加持股	316.0	230.0
2026-02-26	增加持股	308.0	230.0
2026-01-29	增加持股	308.0	257.5
2025-11-11	持有	195.0	209.5
2025-08-12	持有	195.0	213.5
2025-07-24	持有	195.0	217.0

資料來源：TEJ，凱基

凱基證券集團據點

中國	上海	上海市靜安區南京西路 1601 號越洋國際廣場 1507 室 郵政編號：200040
台灣	台北	104 台北市明水路 700 號 電話 886.2.2181.8888，傳真 886.2.8501.1691
香港		香港灣仔港灣道十八號中環廣場四十一樓 電話 852.2878.6888，傳真 852.2878.6800
泰國	曼谷	8th - 11th floors, Asia Centre Building 173 South Sathorn Road, Bangkok 10120, Thailand 電話 66.2658.8888，傳真 66.2658.8014
新加坡		珊頓大道 4 號#13-01 新交所第二大廈 郵政編號：068807 電話 65.6202.1188，傳真 65.6534.4826
印尼		Sona Topas Tower Fl.11 Jl. Jend. Sudirman kav.26 JakartaSelatan 12920 Indonesia 電話 62 21 250 6337

股價說明

等級	定義
增加持股 (OP)	對個股持正面看法，預期個股未來十二個月的表現超越凱基證券集團所追蹤的相關市場的總報酬。
持有 (N)	對個股持中性看法，預期個股未來十二個月的表現符合凱基證券集團所追蹤的相關市場的總報酬。
降低持股 (U)	對個股持負面看法，預期個股未來十二個月的表現低於凱基證券集團所追蹤的相關市場的總報酬。
未評等 (NR)	凱基證券未對該個股加以評等。
受法規限制未評等 (R)	受凱基證券集團內部政策和/或相關法令限制使凱基證券集團無法進行某些形式的資訊交流，其中包括提供評等給投資人參考。
	*總報酬 = (十二個月目標價-現價)/現價

免責聲明

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法說重點摘要

矽格&台星科

(6257 TT, NT$286, 未評等) & (3265 TT, NT$197.5, 未評等)

重點摘要

1. 矽格以 IC 測試為主，聚焦 AI ASIC、CPU/GPU 及矽光子等高階測試。台星科主要提供晶圓凸塊、覆晶封裝與晶圓測試。矽格目前持有台星科股權約 51.9%。
2. 台星科和矽格連袂調高 2026 年資本支出，其中台星科調高至 134 億元(年增 705%)，矽格追加到 88 億元(年增 66%)，預示了整個產業產能吃緊，並且 AI 外溢需求已經大幅排擠了原本封測產能。AI 相關的需求特別是伺服器 CPU 以及矽光子的需求，將大幅帶動公司明年的業績成長。矽格目前的 2027 年營收展望為年增 30%以上。
3. 不單是產品組合優化，矽格對於產業的價格趨勢也是正面看法，而資本支出的選擇則是不大幅增加折舊的前提下謹慎規劃，並且有部分大客戶有預定產能，拉高了整體的訂單能見度。我們對於整體封測產業在 AI 的帶動之下維持正面且樂觀的看法。矽格目前交易在 2027 年 EPS 市場共識值的 24 倍。

矽格營運概況

- 2Q26 營收 58.8 億元，年增 23.1%，營業利益 15.1 億元，年增 34.4%，歸屬母公司淨利 12.2 億元，年增 264%，EPS 2.55 元。
- 2026 年 1-8 月營收占比為 AI CPU/GPU、ASIC 及矽光子 24%、智慧型手機 30%、車用/醫療 6%、網通/IoT 21%、消費電子/智慧家庭 19%。此外，記憶體占約 10%，營收年增近 30%。
- 2026 年 CapEx 提高至 88 億元，較原始規畫增加 48%，7 月子公司矽格聯測追加 21 億元，主要擴充美系 CPU/GPU 客戶測試產能。

矽格未來展望

- 4Q26 動能來自 AI 伺服器、AI Connectivity、矽光子、2 奈米高階手機晶片、低軌衛星及高速網通；目標 2027 年 AI CPU/GPU、ASIC 及矽光子占比超過 30%。
- 湖口二廠目前產能開出 40%，預計 2027 年逐步開滿；中興三廠預計 4Q26 完成廠房工程、1Q27 量產，主要支援 AI ASIC、高速網通及高階晶片需求。

台星科營運概況

- 2Q26 營收 14.1 億元，年增 20.2%，營業利益 3.38 億元，年增 35%，歸屬母公司淨利 2.85 億，年增 217%，EPS 2.09 元
- 2026 年 1-8 月營收占比為 HPC 49%、Memory 8%、CE/3C 34%、IoT 9%，預期 2027-2028 年 HPC 占比持續提升。

凱基投顧

潘俊宏
886.2.2181.8706
Felix.Pan@kgi.com

重要免責聲明，詳見最終頁

台星科未來展望

- N2 產品認證完成並陸續量產，55×55mm FCBGA 已進入量產，預計 2027 年放量。
- CP 為 2027 年後主要擴充產品線，聚焦 AI Data Center 及 N2 CPU 晶圓測試。FT 主要對應 AI Server/CoWoS，涵蓋 Burn-In 及 SLT，亦為 2027 年成長動能。

矽格 Q&A

- 光通訊相關布局及營收、獲利貢獻展望？
  - 矽格布局全方位 AI Connectivity 及 One-stop 光通訊測試服務，涵蓋測試技術、設備、工程至量產，預期 2027 年光通訊營收倍數成長，光通訊獲利優於傳統測試。
- 頎邦進入光測試對矽格的影響?
  - 頎邦主要仍以 Bumping 為主，部分客戶於頎邦完成 Bumping 後，後續測試仍交由矽格。
- 目前訂單能見度？
  - 能見度高，多數主要客戶已簽訂一年以上的產能保障協議。
- AI ASIC、矽光子及高速運算 2027 年營收占比？
  - 2026 年目前占比約 24%，2027 年目標超過 30%。
- 因應 AI 晶片測試需求提升，2027 年如何擴充產能？
  - 會先與主要客戶確認未來產能需求，並簽訂產能保障協議後再進行投資，湖口二廠即因客戶需求強勁、自建廠房時程過長，故提前購入以銜接產能。
- 2026 年 CapEx 是否還會上修？
  - 目前仍有數個重要客戶洽談合作，若客戶需求非常緊迫，不排除未來數月再上修 CapEx 以滿足需求。
- 如何因應客戶要求分散生產據點？
  - 目前矽格在台灣有 12 座工廠、中國 2 座、日本新增 1 座工廠；暫無赴美設廠計劃，但公司以服務客戶為主，只要有需求，皆可評估於不同地區布局。
- 湖口二廠滿載後產值多少？
  - 約 80%即可視為接近滿載，滿載後單月營收估約 2-3 億元。
- 中興三廠量產後營收規模？
  - 中興三廠共七層，其中五層規劃為 clean room，每層約可容納 100 台設備，產能會逐年爬坡，不會在一年內達滿載。
- 湖口二廠是否專門測試 EIC/PIC？不同客戶設備能否共用？
  - 湖口二廠不限特定產品或客戶；EIC/PIC 測試設備主要為公司自製，僅需依不同客戶需求微調，因此設備共用性高。

- 如何看 2027 年測試價格？
  - AI 及光通訊晶片複雜度提升後，測試與良率的重要性增加，矽格表示目前測試價格仍在漲價。

台星科 Q&A

- 3Q26 營收受到客戶 wafer 供應不及影響，是否會遞延至 4Q26？
  - 影響主要來自 N2 及 N3 產品，3Q26 未到位部分預計於 4Q26 回補。
- 目前 CapEx 大幅增加，未來轉換為營收的效率及主要應用為何？
  - 新增 CapEx 以測試設備占比較高，主要對應 Data Center 產品，包括 CPU、ASIC 及矽光相關應用
  - 設備交期最長可達約一年，因此 capex 下單與營收貢獻通常存在至少半年至一年時間差，主要營收貢獻將落在 2027 年。
- 是否可以用 1 元 CapEx 對應約 0.5 元營收估算？
  - 針對公司 2027 年以晶圓測試為主的投資，管理層確認上述估算合理。

凱基證券集團據點

中國	上海	上海市靜安區南京西路 1601 號越洋國際廣場 1507 室 郵政編號：200040
台灣	台北	104 台北市明水路 700 號 電話 886.2.2181.8888 • 傳真 886.2.8501.1691
香港		香港灣仔港灣道十八號中環廣場四十一樓 電話 852.2878.6888 • 傳真 852.2878.6800
泰國	曼谷	8th - 11th floors, Asia Centre Building 173 South Sathorn Road, Bangkok 10120, Thailand 電話 66.2658.8888 • 傳真 66.2658.8014
新加坡		珊頓大道 4 號#13-01 新交所第二大廈 郵政編號：068807 電話 65.6202.1188 • 傳真 65.6534.4826
印尼		Sona Topas Tower Fl.11 Jl. Jend. Sudirman kav.26 JakartaSelatan 12920 Indonesia 電話 62 21 250 6337

股價說明

等級	**定義**
增加持股 (OP)	對個股持正面看法，預期個股未來十二個月的表現超越凱基證券集團所追蹤的相關市場的總報酬。
持有 (N)	對個股持中性看法，預期個股未來十二個月的表現符合凱基證券集團所追蹤的相關市場的總報酬。
降低持股 (U)	對個股持負面看法，預期個股未來十二個月的表現低於凱基證券集團所追蹤的相關市場的總報酬。
未評等 (NR)	凱基證券未對該個股加以評等。
受法規限制未評等 (R)	受凱基證券集團內部政策和/或相關法令限制使凱基證券集團無法進行某些形式的資訊交流，其中包括提供評等給投資人參考。
	*總報酬 = (十二個月目標價-現價)/現價

免責聲明

凱基證券投資顧問股份有限公司（以下簡稱本公司）為凱基金控集團之成員。本報告之內容皆來自本公司認可之資料來源，但不保證其完整性及精確性。報告內容所提及之各項業務、財務等相關檔案資料及所有的意見及預估皆基於本公司於特定日期所做之判斷，故有其時效性限制，邇後若有變更時，本公司將不做預告或更新。本報告內容僅供參考，並不提供或遊說客戶為買賣股票之投資依據。投資人應審慎考量本身之投資風險，並就投資結果自行負責。本公司及所屬集團成員，暨其主管或員工皆有可能持有報告中所提及的證券。本公司及所屬集團成員並可能經常提供投資銀行或其他服務給報告中提及之公司或向其爭取相關業務。本報告之著作權為本公司所有，非經本公司同意，本報告全文或部份內容，不得以任何形式或方式引用、轉載或轉寄。

2026/10/06

環保　　重新覆蓋

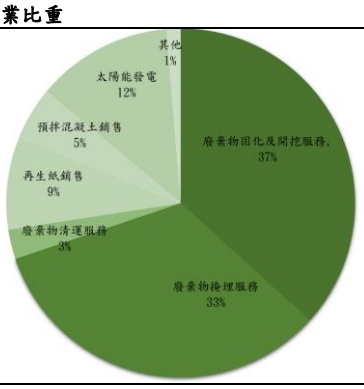
可寧衛 (8422 TT)　　買進(Buy)

潛在漲幅	25.7%
前日收盤價	25.30 元
目標價	31.80 元
前次目標價	-
前次評等	-

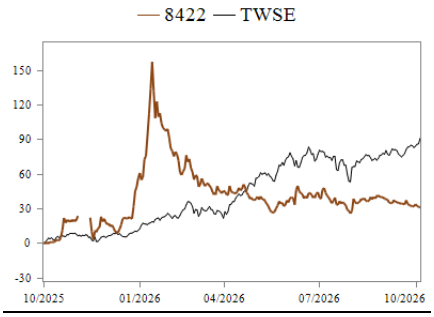
公司基本資訊

目前股本(百萬元)	1,359
市值(億元)	344
目前每股淨值(元)	7.57
外資持股比(%)	5.47
董監持股比(%)	64.52

營業比重



個股與大盤走勢(%)



王淑君
Jenny.wang@ctbcsis.com

◎研究分析標的公司為本公司關係企業之客戶。

既有處理業務穩固，新廠與再利用事業接續貢獻

- **營收規模擴大，後續獲利動能逐步增強**
- **整治專案支撐營運，再生粒料與綠電增添成長動能**
- **與台灣人壽合作的南區新焚化廠，預計 2029 年起成為長期動能**

營收規模擴大，後續獲利動能逐步增強：可寧衛 1H26 合併營收 33.82 億元，毛利率 37%、營益率 27%，EPS 0.55 元。南區 BOT 建造收入自 5 月起依工程進度認列，上半年約 10.3 億元，帶動合併營收擴大；7、8 月分別認列 0.8 億元及 2.6 億元，後續金額將隨施工進度變動。展望下半年，205 兵工廠整治案認列可望高於上半年，南區焚化舊廠的處理量與發電效率持續改善，虧損可望收斂。明年則隨大承再生粒料業務逐步投入營運，為營收及獲利增添新動能。

整治專案支撐營運，再生粒料與綠電增添成長動能：205 兵工廠既有合約 24 億元，因實際挖出的有害廢棄物比率高於原預期，固化與掩埋需求增加；既有工作預計於 1H27 完成，預期另有數億元潛在追加預算。新業務方面，大承再生粒料廠規劃月處理量約 9,000 噸，待試運轉及審查完成後，可望成為後續的成長動能；其產品可供吉鼎製成道路回填用低強度混凝土，而吉鼎現有一般混凝土業務已開始獲利。此外，漁電共生一期 69MW 已併聯，二期規劃約 70MW，目前仍在申設。

與台灣人壽合作的南區新焚化廠，預計 2029 年起成為長期動能：可寧衛與台壽共同投資吉信，持股分別為 55%及 45%；BOT 案包含約 3.5 年興建期及後續 20 年營運期。興建期間同步操作的舊廠，經設備調整後每日處理量已回升至 1,000~1,200 噸、每噸發電效率恢復至約 300 度，2H26 舊廠虧損可望收斂。新廠規劃每日處理量 1,350 噸、年發電量約 3.5 億度，預期於 2029 年開始貢獻處理費與售電收入；穩定營運後年營收規模約 15~18 億元。我們預估可寧衛 2026/2027 年 EPS 1.31/1.67 元，給予「買進」評等，目標價 31.8 元(2027 年 EPSx19 倍)。

風險因子：專案認列與追加預算不如預期、投產及工程遞延、設備運轉異常。

(百萬元)	2022	2023	2024	2025	2026F	2027F
營業收入淨額	5,244	4,265	5,169	4,780	6,594	7,165
營業毛利淨額	2,207	1,809	2,028	2,540	2,847	3,545
營業利益	1,670	1,261	1,439	1,940	2,216	2,835
稅後純益	1,343	994	1,172	1,459	1,775	2,275
公告基本 EPS(元)	12.33	9.12	1.07	1.29	--	--
調整後 EPS(元)	12.33	9.10	10.48	1.22	1.31	1.67
營業收入 YoY(%)	73.48%	-18.66%	21.19%	-7.53%	37.94%	8.67%
稅後純益 YoY(%)	16.46%	-25.99%	17.92%	24.46%	21.70%	28.14%
本益比	1.20	1.70	15.27	23.46	19.37	15.11
本淨比	0.26	0.27	0.27	0.43	3.52	2.91
股息(元)	9.99	10.00	9.94	1.20	1.22	1.56
殖利率(%)	4.79%	5.24%	5.21%	4.24%	4.82%	6.15%
ROE(%)	22.26%	15.85%	17.91%	19.10%	19.51%	21.08%

可寧衛以事業廢棄物處理起家，業務涵蓋有害廢棄物固化、最終掩埋及污染場址整治，並逐步拓展至再生粒料、混凝土、焚化廠營運及再生能源。公司於高雄岡山的兩座掩埋場持續營運，分別服務有害及一般事業廢棄物。掩埋場許可與可用容量不易取得，集團提供從開挖、檢測、中間處理到最終處置的服務。

圖表一、可寧衛集團事業群

資料來源：可寧衛，中信投顧整理

205 案包括污染土壤開挖、分類檢測及後續處理。現場有害物質比率較原估高，帶動固化與掩埋需求，也使既有預算消耗較快。工程按實際施作進度認列，雨天、現場檢測及業主預算程序均會影響單季金額。預期將有數億元潛在追加預算，可提高剩餘工期的收入貢獻。

南區 BOT 案於新廠興建期間同步操作舊廠。舊廠設備使用年限長，接手後需逐步檢修與調整；近期每日處理量回升至 1,000~1,200 噸，發電效率亦有改善。處理量提升可增加處理費收入，發電效率改善則有助於售電及降低外購電需求。可寧衛與台灣人壽共同投資吉信，持股分別為 55%及 45%，承作南區焚化廠 BOT 案。合約期間共 23.5 年，包含 3.5 年興建期及 20 年操作期，總投資金額約 150 億元。新廠規劃三爐兩機，每日處理量 1,350 噸、年發電量約 3.5 億度；完工後將透過廢棄物處理費與售電提供長期營運貢獻，穩定運作後年營收規模估約 15～18 億元。

圖表二、可寧衛取得高雄南區 BOT 案

資料來源：可寧衛，中信投顧整理

大承再生粒料廠透過固化及乾燥製程處理污泥、底渣等廢棄物，設計處理量約每月 9,000 噸。產出的再生粒料可供吉鼎製成控制性低強度回填材料(CLSM)，應用於道路等工程。大承目前仍在試運轉與審查階段，待正式投入營運、收料量逐步提高後，有望成為後續的獲利成長動能。吉鼎現有一般混凝土業務已開始貢獻營收與獲利，後續 CLSM 業務則仰賴大承穩定供料及客戶導入。月設計處理量 9,000 噸，預估大承滿載年化處理收入約 4~5.5 億元。

圖表三、可寧衛集團展現循環經濟

資料來源：可寧衛，中信投顧整理

圖表四、季度財報與預估差異

損益表（單位：百萬元）	2Q26					3Q26(F)				
	實際數	中信預估	差異	市場共識	差異	調整後	調整前	差異	市場共識	差異
營業收入	2,268	-	-	-	-	1,574	-	-	-	-
營業毛利	720	-	-	-	-	763	-	-	-	-
營業利益	553	-	-	-	-	609	-	-	-	-
稅前純益	547	-	-	-	-	583	-	-	-	-
稅後淨利	459	-	-	-	-	479	-	-	-	-
EPS(元)	0.34	-	-	-	-	0.35	-	-	-	-
財務比率(%/百分點)										
毛利率	31.7%	-	-	-	-	48.5%	-	-	-	-
營益率	24.4%	-	-	-	-	38.7%	-	-	-	-
稅後淨利率	20.2%	-	-	-	-	30.4%	-	-	-	-

資料來源：Cmoney、Bloomberg，中信投顧整理

圖表五、年度財報與預估差異

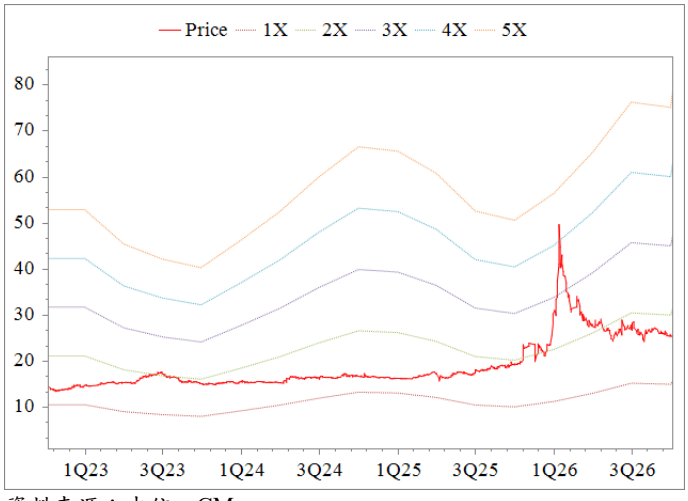
損益表（單位：百萬元）	2026(F)					2027(F)				
	調整後	調整前	差異	市場共識	差異	調整後	調整前	差異	市場共識	差異
營業收入	6,594	-	-	-	-	7,165	-	-	-	-
營業毛利	2,847	-	-	-	-	3,545	-	-	-	-
營業利益	2,216	-	-	-	-	2,835	-	-	-	-
稅前純益	2,153	-	-	-	-	2,772	-	-	-	-
稅後淨利	1,775	-	-	-	-	2,275	-	-	-	-
EPS(元)	1.31	-	-	-	-	1.67	-	-	-	-
財務比率(%/百分點)										
毛利率	43.2%	-	-	-	-	49.5%	-	-	-	-
營益率	33.6%	-	-	-	-	39.6%	-	-	-	-
稅後淨利率	26.9%	-	-	-	-	31.8%	-	-	-	-

資料來源：Cmoney、Bloomberg，中信投顧整理

2026/10/06

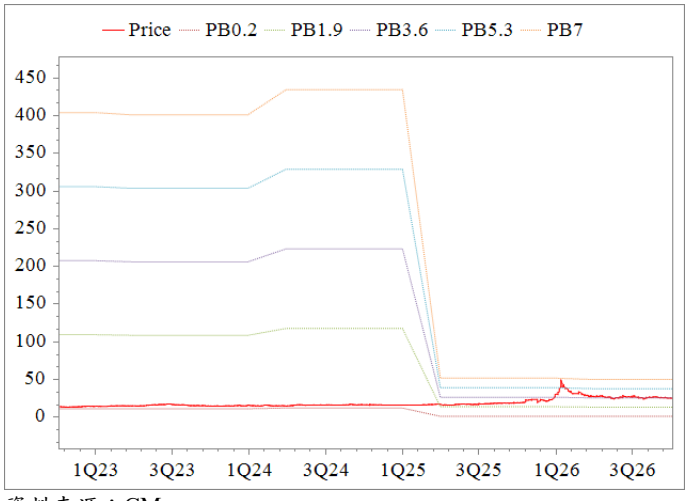
(百萬元)	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F
營業收入	1,114	2,268	1,574	1,638	1,778	1,455	1,868	2,065
營業成本	577	1,548	811	811	880	735	953	1,053
營業毛利	537	720	763	827	898	720	915	1,012
營業費用	154	167	154	156	172	175	179	184
營業利益	383	553	609	672	725	545	736	828
營業外收支淨額	-38	-6	-26	6	-21	-24	-26	8
稅前純益	345	547	583	678	705	521	710	836
所得稅	89	123	140	163	169	125	170	201
稅後純益	281	459	479	557	578	428	583	686
調整後 EPS(元)	2.07	3.38	3.52	4.10	4.26	3.15	4.29	5.05
毛利率(%)	48.21%	31.74%	48.50%	50.50%	50.50%	49.50%	49.00%	49.00%
營業利益率(%)	34.38%	24.37%	38.70%	41.00%	40.80%	37.50%	39.40%	40.10%
稅後純益率(%)	25.23%	20.24%	30.43%	33.98%	32.54%	29.41%	31.18%	33.24%
營業收入								
QoQ(%)	11.99%	103.67%	-30.61%	4.08%	8.54%	-18.18%	28.41%	10.53%
YoY(%)	-23.79%	78.77%	49.08%	64.74%	59.66%	-35.86%	18.69%	26.04%
營業利益								
QoQ(%)	9.76%	44.38%	10.17%	10.27%	8.01%	-24.80%	34.92%	12.49%
YoY(%)	-37.57%	-9.86%	66.90%	92.52%	89.45%	-1.33%	20.84%	23.28%
稅後純益								
QoQ(%)	0.50%	63.39%	4.34%	16.22%	3.94%	-26.06%	36.18%	17.80%
YoY(%)	-36.83%	1.35%	69.99%	99.12%	105.93%	-6.80%	21.63%	23.29%

PER Band



資料來源：中信、CMoney

PBR Band



資料來源：CMoney

2026/10/06

資產負債表

(百萬元)	2022	2023	2024	2025	2Q26
資產總計	12,570	15,306	20,041	22,291	26,499
流動資產	4,916	3,526	4,601	5,088	7,725
現金及約當現金	3,389	1,476	2,309	3,223	4,116
應收帳款與票據	517	559	1,150	776	1,128
存貨	56	62	63	49	30
採權益法之投資	1,104	1,221	1,408	1,399	1,447
不動產、廠房設備	5,936	9,575	12,218	14,152	14,665
負債總計	5,969	8,501	12,655	12,774	14,762
流動負債	1,569	2,762	6,128	2,402	7,368
應付帳款及票據	61	71	57	48	35
非流動負債	4,399	5,738	6,527	10,372	7,394
權益總計	6,601	6,805	7,386	9,518	11,737
普通股股本	1,089	1,092	1,101	1,137	1,357
保留盈餘	3,246	3,695	4,104	4,780	4,163
母公司業主權益	6,287	6,255	6,833	8,443	10,274
負債及權益總計	12,570	15,306	20,041	22,291	26,499

損益表

(百萬元)	2023	2024	2025	2026F	2027F
營業收入淨額	4,265	5,169	4,780	6,594	7,165
營業成本	2,456	3,141	2,240	3,746	3,620
營業毛利淨額	1,809	2,028	2,540	2,847	3,545
營業費用	549	589	599	631	710
營業利益	1,261	1,439	1,940	2,216	2,835
EBITDA	--	--	--	2,732	3,289
業外收入及支出	8	49	-98	-63	-63
稅前純益	1,269	1,488	1,842	2,153	2,772
所得稅	289	313	414	515	665
稅後純益	994	1,172	1,459	1,775	2,275
公告基本 EPS(元)	9.12	1.07	1.29	--	--
調整後 EPS(元)	9.10	10.48	1.22	1.31	1.67

註：稅後純益係指本期淨利歸屬於母公司業主。

現金流量表

(百萬元)	2022	2023	2024	2025	2Q26
營業活動現金	1,289	473	1,216	2,125	-396
稅前純益	1,702	1,269	1,488	1,842	547
折舊及攤銷	189	168	364	462	149
營運資金變動	43	-36	-596	387	56
其他營運現金	-645	-927	-41	-566	-1,148
投資活動現金	-2,340	-3,782	-2,857	-2,229	-521
資本支出淨額	-2,342	-3,745	-2,828	-2,221	-438
長期投資變動	-25	-96	-66	-76	-86
其他投資現金	27	59	36	68	3
籌資活動現金	3,021	1,397	2,472	1,018	1,979
長借/公司債變動	3,984	1,360	2,577	3,191	20
現金增資	0	0	0	0	0
發放現金股利	-1,089	-1,089	-1,092	-1,119	0
其他籌資現金	126	1,126	988	-1,055	1,959
淨現金流量	1,971	-1,912	832	914	1,063
期初現金	1,418	3,389	1,476	2,309	3,053
期末現金	3,389	1,476	2,309	3,223	4,116

資料來源：中信、CMoney

比率分析

(百萬元)	2023	2024	2025	2026F	2027F
成長力分析(%)					
營業收入淨額	-18.66%	21.19%	-7.53%	37.94%	8.67%
營業毛利淨額	-18.03%	12.06%	25.26%	12.11%	24.50%
營業利益	-24.51%	14.13%	34.85%	14.22%	27.90%
稅後純益	-26.86%	19.99%	21.49%	21.70%	28.14%
獲利能力分析(%)					
毛利率	42.42%	39.23%	53.13%	43.18%	49.47%
EBITDA(%)	--	--	--	41.44%	45.91%
營益率	29.56%	27.84%	40.60%	33.61%	39.56%
稅後純益率	22.96%	22.73%	29.87%	26.93%	31.75%
ROE(%)	15.85%	17.91%	19.10%	19.51%	21.08%

2026/10/06

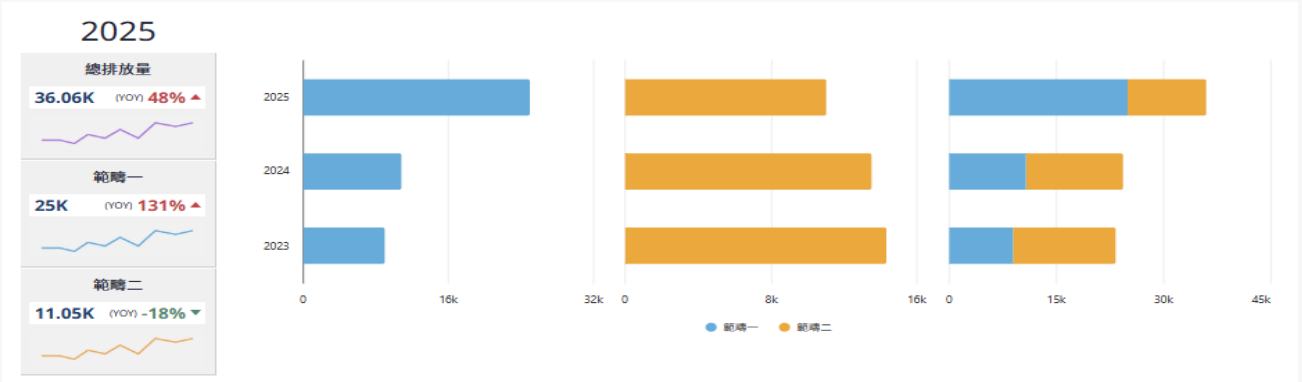
ESG 資訊-環境保護(Environmental)

E

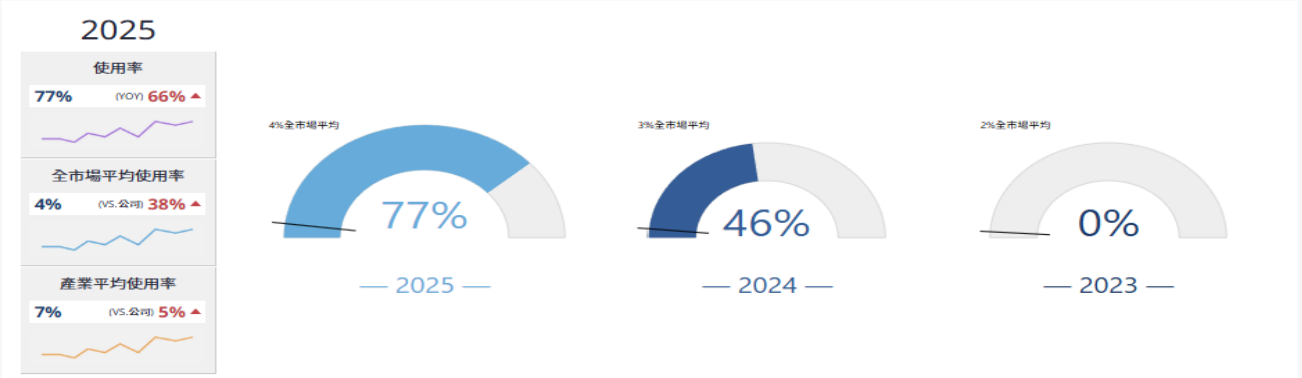
2025年 可寧衛*（全市場平均）

變動率(YOY)可能受盤查邊界異動影響，詳情請參閱ESG數位平台原始資料

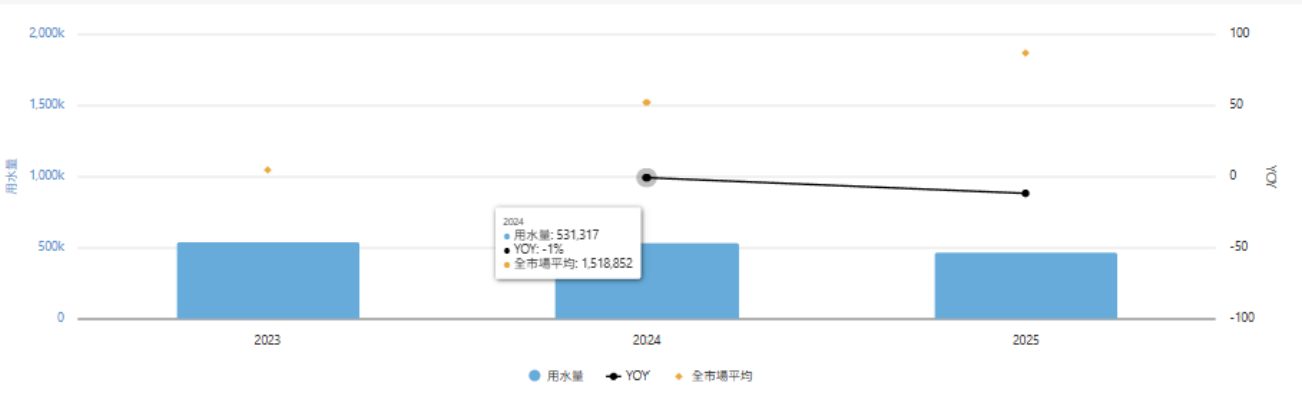
溫室氣體(範疇一/二)排放量



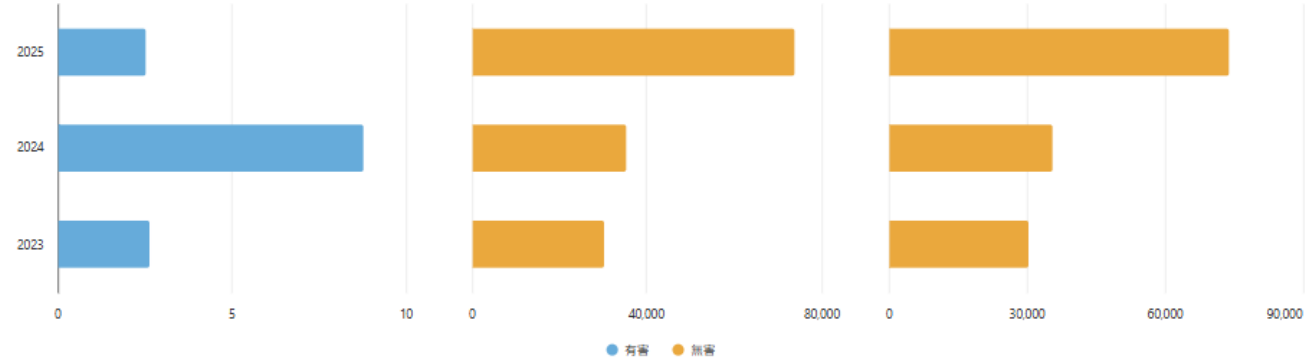
再生能源使用率



用水量



廢棄物量



2025

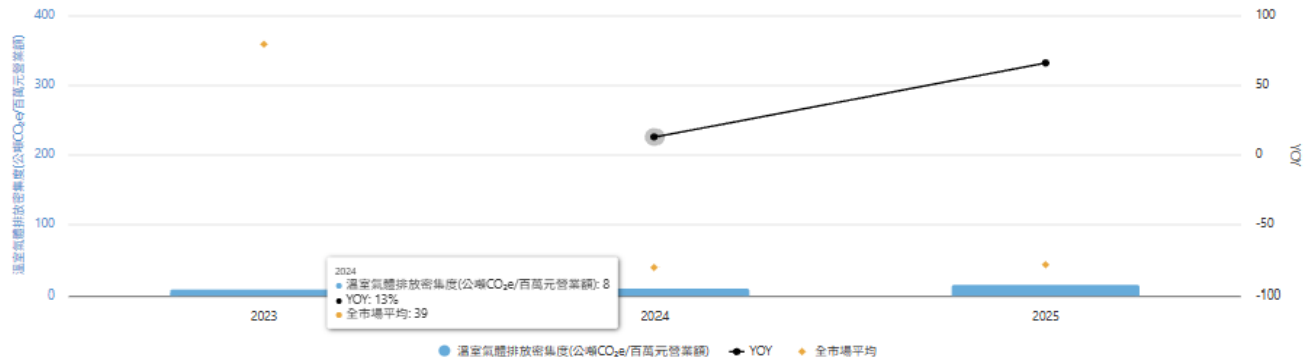
有害	無害	總量
0K (YOY) -71% ▼	73.79K (YOY) 108% ▲	73.79K (YOY) 108% ▲

密集度

2025

溫室氣體排放密集度	用水密集度	廢棄物密集度
13.65	97.544	15.438

溫室氣體排放密集度(公噸CO_2e/百萬元營業額)



2025

溫室氣體排放密集度(公噸CO_2e/百萬元營業額)	全市場平均溫室氣體排放密集度(公噸CO_2e/百萬元營業額)	產業平均溫室氣體排放密集度(公噸CO_2e/百萬元營業額)
0.01K (YOY) 66% ▲	0.04K (YOY) 10% ▲	0.01K (YOY) -19% ▼

ESG 資訊-社會責任(Social)

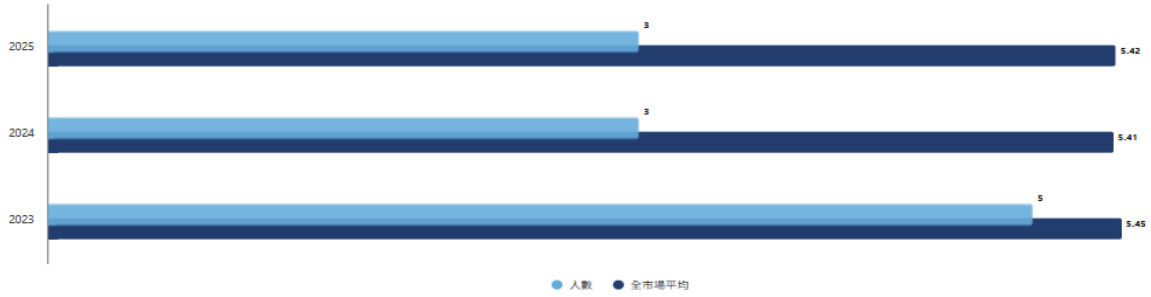
S 2025年 可寧衛*（全市場平均）

管理職女性主管占比

2025

占比	全市場平均占比	產業平均占比
25% (YOY) 200% ▲	28% (YOY) 2% ▲	21% (YOY) 7% ▲

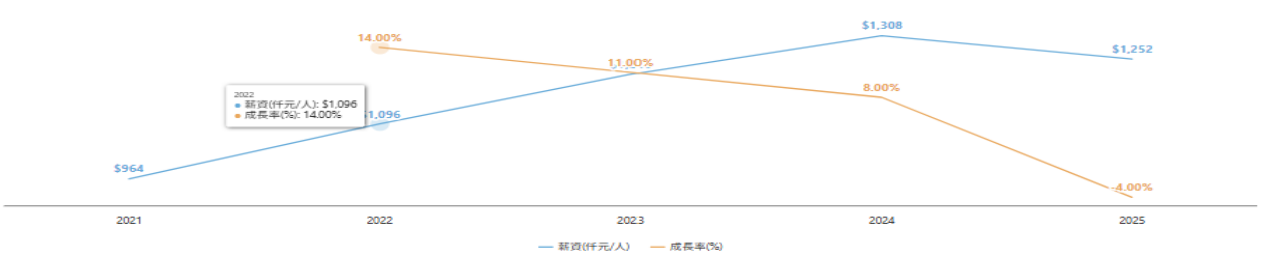
職業災害人數



2025

人數	全市場平均人數	產業平均人數
3 (YOY) 0% —	5 (VS.公司) 0% —	6 (VS.公司) 234% ▲

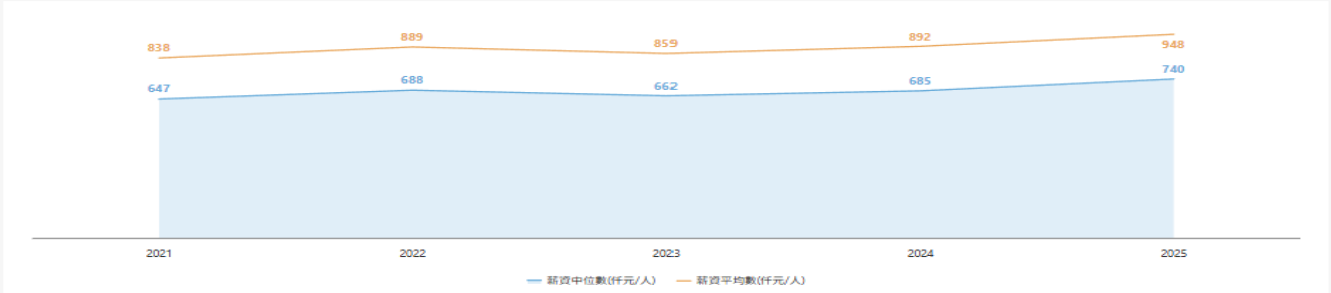
員工薪資年度趨勢



2025

員工薪資平均數	與去年同期比	與同產業薪資比
1,252	-4% ▼	+10.88%

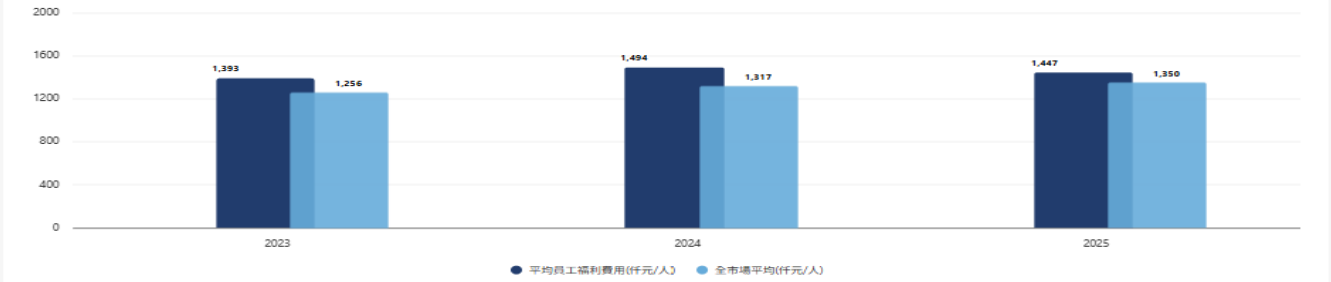
非擔任主管職務之全時員工薪資



2025

平均數	中位數	差異%
948 (YOY) 6% ▲	740 (YOY) 8% ▲	0% (YOY) -7% ▼

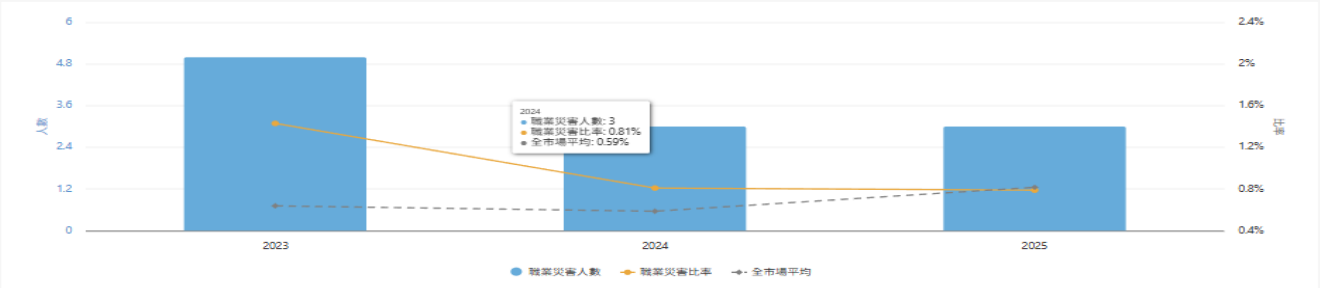
平均員工福利費用



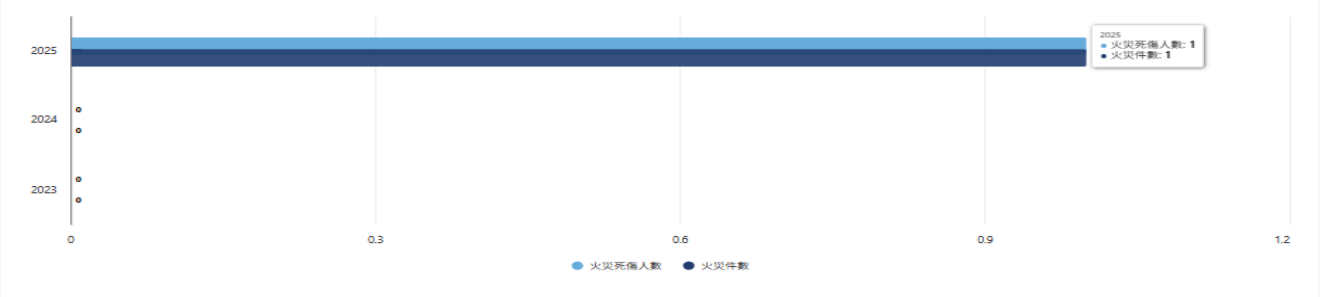
2025

平均福利	全市場平均福利	產業平均福利
1,447 (YOY) -3% ▼	1,350 (YOY) 3% ▲	1,316 (YOY) 12% ▲

職業災害人數與比率

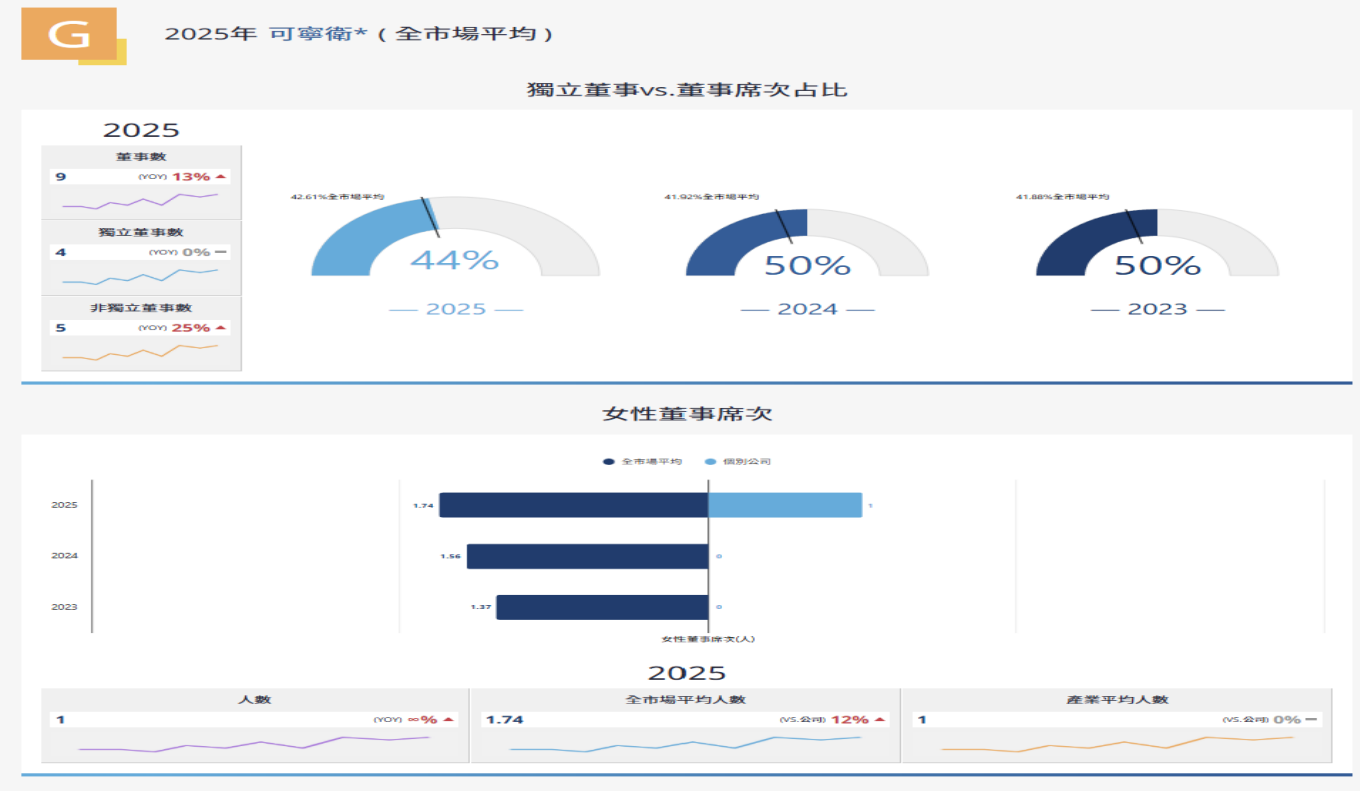


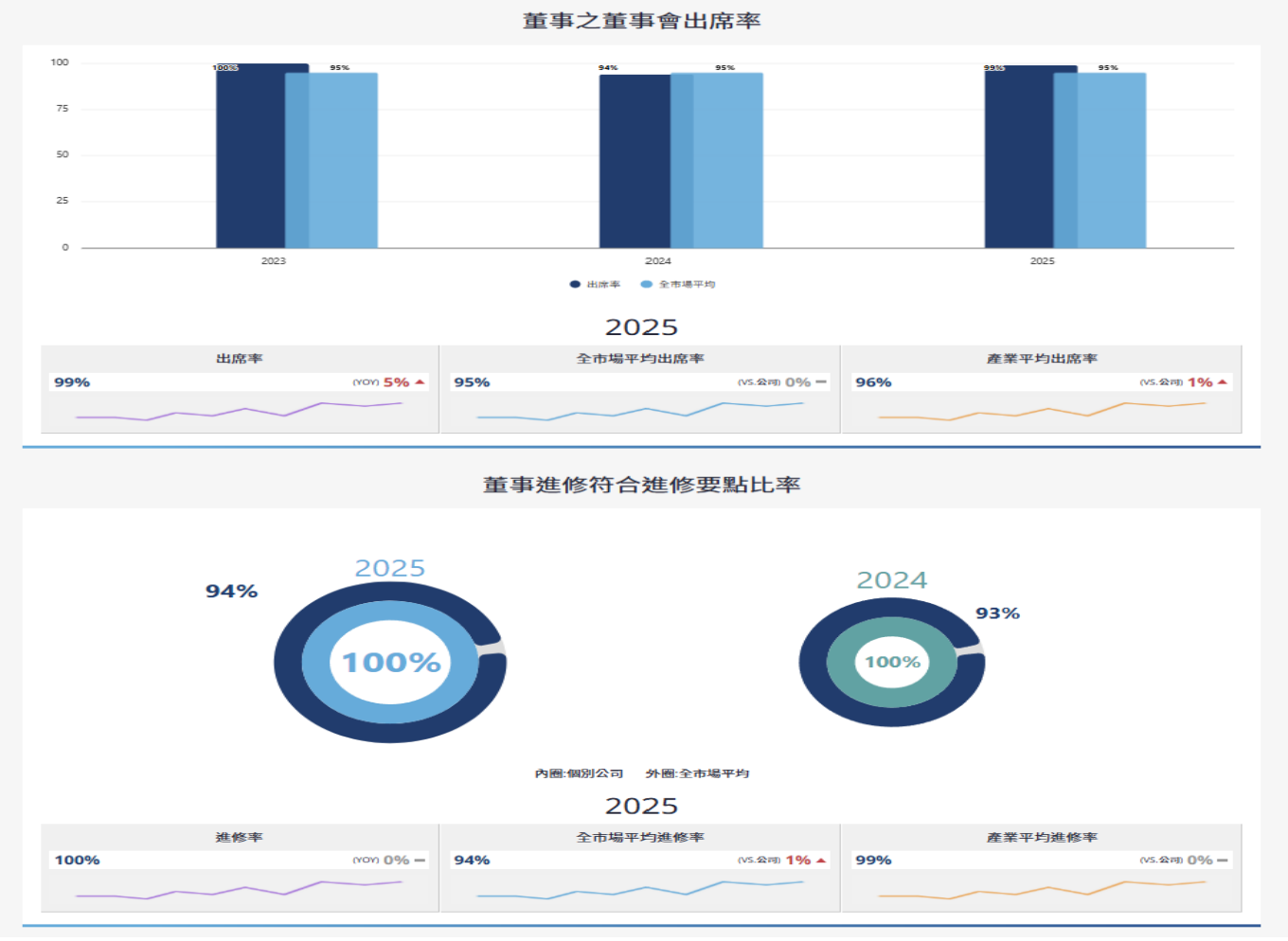
火災件數與死傷人數



2026/10/06

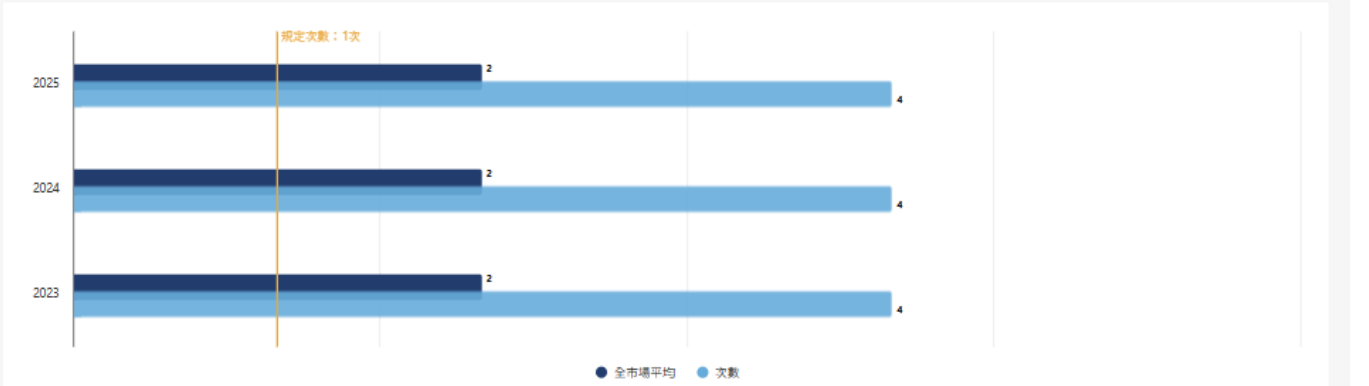
ESG 資訊-公司治理(Governance)





2026/10/06

法說會次數



2025

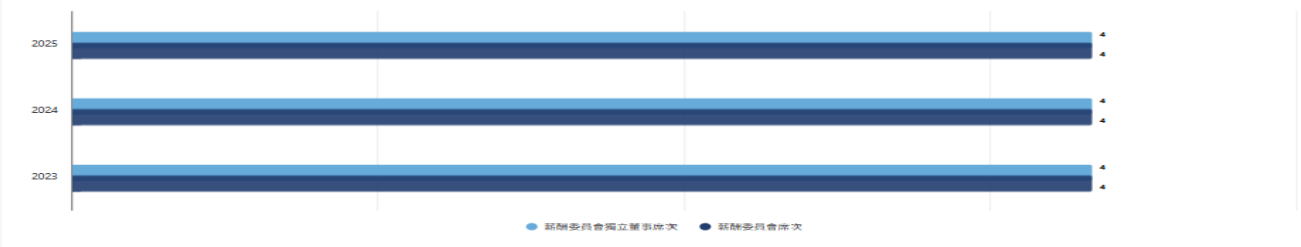
次數	全市場平均次數	產業平均次數
4 (YOY) 0%	2 (VS.公司) 0%	3 (VS.公司) 0%

薪酬委員會出席率

2025

進修率	全市場平均進修率	產業平均進修率
100% (YOY) 0%	97% (VS.公司) 1%	97% (VS.公司) 0%

薪酬委員會席次



資料來源:證交所

2026/10/06

投資評等說明

個股評等	Rating	定義
買進	Buy (B)	相較於市場指數之回報潛力極具吸引力
增加持股	Overweight (OW)	相較於市場指數之回報潛力略有吸引力
中立	Neutral (N)	相較於市場指數之回報潛力相似
降低持股	Underweight (UW)	相較於市場指數之回報潛力稍低
賣出	Sell (S)	相較於市場指數之回報潛力較低
未評等	Not Rated (NR)	

免責聲明

【中國信託證券投資顧問股份有限公司獨立經營管理】

本報告係本公司於特定日期就現有資訊所提出，雖已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，本公司將不做預告或更新；而且本報告並非依特定投資目的或特定對象之財務狀況所完成，因此，本報告之目的並非針對客戶於買賣證券、選擇權、期貨或其他證券相關商品提供詢價或推介服務，亦非進行交易之要約。本公司出具之短、中、長期策略性報告所做出之投資分析與建議，亦可能會隨市場波動隨時調整，而與個股研究報告之意見未必一致。投資人應自行考量本身之需求、投資風險及承受度，並就投資結果自行負責，本公司不做任何獲利保證，亦不就投資盈虧負擔任何法律責任。非經本公司同意，任何人不得將本報告加以引用或轉載。

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中國信託證券投資顧問股份有限公司
台北市南港區經貿二路 188 號 5 樓及 18 樓 / (02) 5569-1988
營業執照字號：114 金管投顧新字第 014 號

6670 復盛應用(Fusheng)

報告日期：2026/10/06　　研 究 員：洪 國 峰 (Titan Horng)

投資評等：買進
目標價位：346.0 元
12.0X 2027 年 EPS
參考價位：270.0 元
9.4X 2027 年 EPS
2026/10/05 收盤

壹、訪 談／法 說 內 容

重點摘要：

- **3Q26 毛利率受到傳統淡季經濟規模尚能顯現、原料成本增加、人民幣升值等不利因素，預估毛利率落在 23%水準(原毛利率預估值 24%)。**
- **4Q26 進入傳統旺季，加上部分客戶訂單由原本 1Q27 提早至 4Q26 下單，宏遠預估 4Q26 營收金額 97.16 億元(原 4Q26 營收預估值 94.9 億元)，YoY+9.87%，QoQ+25.51%。**
- **復盛應用憑藉多年製程實力包括金屬成型、鍛造與鑄造能力延伸到 AI 水冷散熱零組件，3Q26 已開始小量出貨，營收占比仍低，預期 2027 年出貨量將持續增加，後續可持續觀察。**

一. 預估 3Q26 稅後 EPS 5.27 元

復盛應用 07~08/2026 營收金額 23.66 億元與 26.76 億元，YoY+25.09%與 45.02%，MoM-2.49%與 13.13%，預估復盛應用 3Q26 營收金額 77.41 億元，YoY+31.32%，QoQ-4.13%，預估營業利益 10.22 億元，YoY+46.1%，QoQ-18.67%，預估稅後盈餘 7.34 億元，YoY+51.07%，QoQ-14.27%，預估稅後 EPS 5.27 元。

二. 預估 2026 年稅後 EPS 26.30 元

復盛 2026 年在高爾夫球業務受惠主力客戶訂單推升營運穩定，加上伯鑫貢獻全年度營運，另外業外無匯兌損失衝擊，整體獲利恢復成長態勢。宏遠預估復盛 2026 年營收金額 340.00 億元，YoY+14.13%，預估營業利益 50.68 億元，YoY+9.4%，預估稅後盈餘 36.65 億元，YoY+17.43%，預估稅後 EPS 26.30 元。

三. 預估 2027 年稅後 EPS 28.80 元

復盛 2027 年在高爾夫球業務穩定小幅成長，加上民盛營運恢復成長態勢，另外新業務事業開始小量出貨，有助復盛應用 2027 年營運持續成長。宏遠預估復盛 2026 年營收金額 355.00 億元，YoY+4.41%，預估營業利益 56.03 億元，YoY+10.57%，預估稅後盈餘 40.14 億元，YoY+9.51%，預估稅後 EPS 28.80 元。

四. 投資建議買進，目標價 346 元

觀察復盛應用歷史 PER 8~15X， 2026 年 ROE 21.4%，PER 10.3X，2027 年 ROE 19.4%，PER 9.4X，評價處在歷史低檔水準，加上復盛應用已切入水冷散熱零組件供應鏈，有助評價提升，故投資建議買進，目標價 346 元。(12X 2027 年稅後 EPS)

本報告僅供宏遠投顧內部及客戶參考，雖已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之實際狀況與風險，本公司恕不負擔任何法律責任及做任何保證。非經本公司同意，不得將本報告加以引用或轉載

貳、產 業 及 營 運 概 況

一.公司簡介

復盛(6670.TT)成立於 1964 年，並於 1997 年掛牌上市，復盛原有三大事業部，分別為機械(空壓機)、運動器材(高爾夫球)與電子事業部(導線架)。復盛 2008 年經勇德國際私募基金收購下市，將復盛公司一分為三，將運動器材部門成立為復盛應用材料公司，機械部門沿用復盛公司，電子事業部成立為復盛精密。復盛應用材料資本額 13.94 億元，並於 2018 年掛牌上市。旗下生產據點有八處，遍佈中國廣東、越南與台灣為主。客戶包括高爾夫球品牌廠，如 Callaway、TITLEIST、DUNLOP、HONMA、TaylorMade、PING、PUMA、MIZUNO、Honeywell、YAMAHA、SAFRAN、SRIXON、Cleveland 等。

圖一、復盛公司沿革

資料來源：公司、宏遠投顧整理

二. 生產基地與產能概況

復盛生產基地有八處，遍佈台灣、中國大陸、越南與泰國。

在台灣方面，有高雄與桃園三座工廠，其中高雄廠主要生產球棒，桃園廠主要以高球頭的鈦合金前段製程、鈦合金鍛造製程為主，嘉華盛位於高雄岡山，主要以航太引擎、油路系統產品為主。

中國大陸主要位於廣東中山，廣盛一廠主要生產碳纖產品與球棒為主，廣盛二廠以高球頭鈦合金後段製程、鋼類球頭前後段製程與鐵頭前後段製程為主，奧博主要以高爾夫球桿頭零配件為主。越盛廠，主要以鋼類球頭前後段製程與鐵頭前後段製程為主。

美國 FSPT 廠，主要生產車用滑輪葉片與航太產品為主。民盛應用生產基地位於台灣與泰國，主要生產滑冰靴、越野摩托車靴與護具為主。

本報告僅供宏遠投顧內部及客戶參考，雖已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之實際狀況與風險，本公司恕不負擔任何法律責任及做任何保證。非經本公司同意，不得將本報告加以引用或轉載

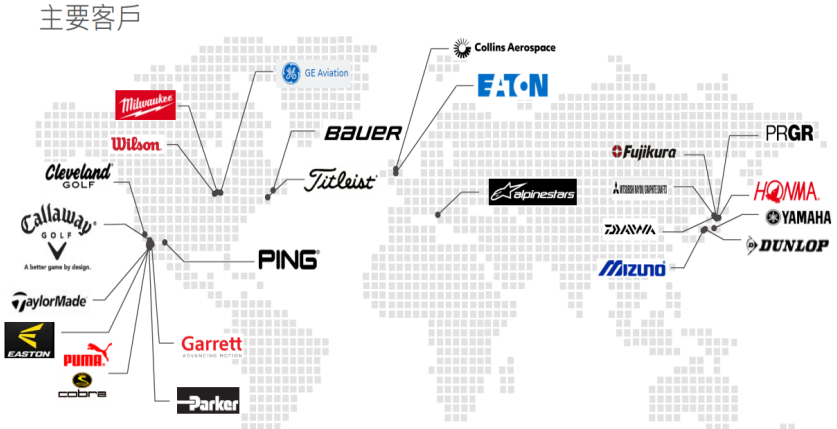
圖二、復盛生產基地

資料來源：公司、宏遠投顧整理

三. 2026 年營運概況及展望

觀察高爾夫球桿頭全年市場量在 4,000 萬顆(疫情前)，復盛年產能在 1,600~1,700 萬顆計算，復盛市占率約 40%左右，若以中高階市場計算，市占率將超過 50%。客戶包括高爾夫球品牌廠，如 Callaway、Cleveland、DUNLOP、HONMA、Honeywell、MIZUNO、PING、PUMA、SAFRAN、SRIXON、TaylorMade、TITLEIST、YAMAHA 等，前三大客戶營收比重在 50~55%間。

圖三、復盛客戶群



資料來源：公司、宏遠投顧整理

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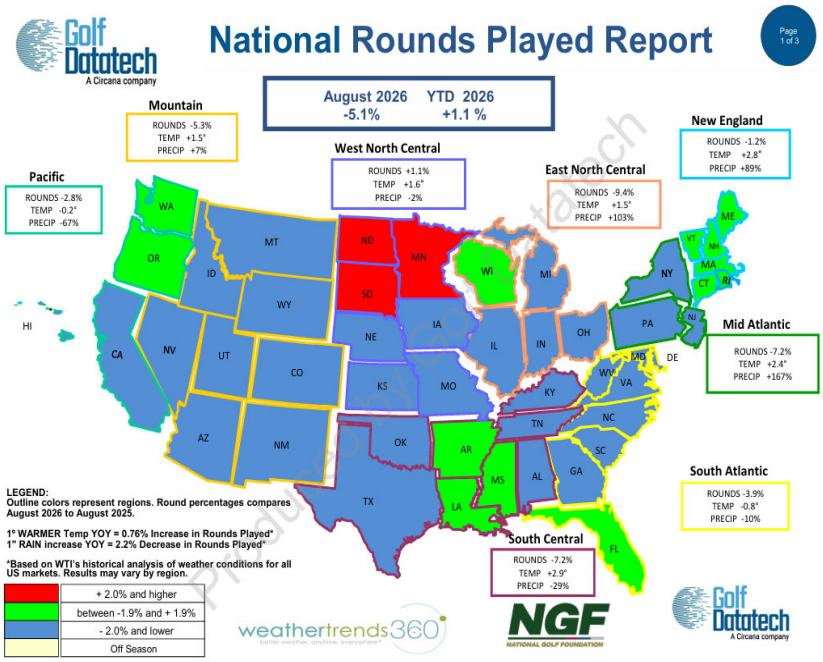
復盛 01~08/2026 營收金額 215.85 億元，YoY+14.88%。觀察美國 2026 年前八月累計打球人次成長 1.1%。上述顯示，美國高爾夫球打球人次並未因為疫情結束而呈現衰退情況，高爾夫球業務受惠新客戶在越南生產基地訂單開始放量，加上原有客戶新產品持續貢獻，預估 2026 年高爾夫球產業營收金額 273.00 億元，YoY+17.49%(包括高球頭、組桿與高爾夫配件等)；在非高爾夫球產業在伯鑫 2026 年貢獻全年度營運，加上力野持續成長，預估 2026 年非高爾夫球產業營收金額 37.2 億元，YoY+6.93%(航太、球棒與其他)；Minson 2026 年受到 EASTON 訂單減少拖累，預估民盛 2026 年營收金額 29.8 億元，YoY-3.09%。綜合上述，宏遠預估復盛 2026 年合併營收金額 340 億元，YoY+14.13%。

圖四、美國 2026 年前八月打球人次概況



資料來源：公司、宏遠投顧整理

預期 2027 年，高爾夫球業務在每年呈現微幅成長情況下，預估 2027 年高爾夫球產業營收金額 283.00 億元，YoY+3.66%(包括高球頭、組桿與高爾夫配件等)；在非高爾夫球產業在航太、伯鑫與新業務拓展帶動，預估 2027 年非高爾夫球產業營收金額 40 億元，YoY+7.53%(航太、球棒與其他)；Minson 2027 年受惠 Alpinestar 與 EASTON 兩大客戶成長帶動，預估民盛 2027 年營收金額 32.00 億元，YoY+7.38%。綜合上述，宏遠預估復盛 2027 年合併營收金額 355 億元，YoY+4.44%。

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參、業 績 及 獲 利 預 估

預估 2026 年稅後盈餘 36.65 億元，YoY+17.43%，稅後 EPS 26.30 元

復盛 2026 年在高爾夫球業務受惠主力客戶訂單推升營運穩定，加上伯鑫貢獻全年度營運，另外業外無匯兌損失衝擊，整體獲利恢復成長態勢。宏遠預估復盛 2026 年營收金額 340.00 億元，YoY+14.13%，預估營業利益 50.68 億元，YoY+9.4%，預估稅後盈餘 36.65 億元，YoY+17.43%，預估稅後 EPS 26.30 元。

預估 2027 年稅後盈餘 40.14 億元，YoY+9.5%，稅後 EPS 28.80 元

復盛 2027 年在高爾夫球業務穩定小幅成長，加上民盛營運恢復成長態勢，另外新業務事業開始小量出貨，有助復盛應用 2027 年營運持續成長。宏遠預估復盛 2026 年營收金額 355.00 億元，YoY+4.41%，預估營業利益 56.03 億元，YoY+10.57%，預估稅後盈餘 40.14 億元，YoY+9.51%，預估稅後 EPS 28.80 元。

推估 2026 年股利政策將配發現金股利 17.5 元，現金殖利率 6.5%

復盛配發現金股利為主，股利發放率 65~77%。復盛 2025 年股利政策，配發現金股利 16.00 元，股利發放率 71%。宏遠推估復盛 2026 年股利政策，將配發現金股利 17.50 元，現金殖利率 6.5%。

表一、歷年股利政策

	2021年	2022年	2023年	2024年	2025年	2026年(F)
每股盈餘(元)	18.03	36.64	18.18	28.17	22.39	26.30
現金股利(元)	12.00	22.00	12.31	19.70	16.00	17.50
股票股利(元)	0.00	0.00	0.00	0.00	0.00	0.00
股利發放率(%)	67%	60%	68%	70%	71%	67%

資料來源：公司、宏遠投顧整理

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圖五、PE Band

資料來源: CMoney、宏遠投顧整理

肆、財 務 簡 表

單位：佰萬元

	2025(A)	26Q1(A)	26Q2(A)	26Q3(F)	26Q4(F)	2026(F)	27Q1(F)	27Q2(F)	27Q3(F)	27Q4(F)	2027(F)
營業收入	29,790	8,469	8,074	7,741	9,716	34,000	8,963	8,397	8,210	9,930	35,500
營業毛利	7,580	2,033	2,072	1,785	2,430	8,320	2,180	2,066	1,990	2,586	8,822
營業費用	2,948	768	815	763	907	3,252	809	732	768	909	3,219
營業利益	4,632	1,265	1,257	1,022	1,523	5,068	1,371	1,334	1,222	1,677	5,603
營業外淨收入（支出）	-188	67	-38	26	12	67	15	33	25	-43	30
稅前純益	4,444	1,333	1,219	1,048	1,535	5,135	1,386	1,367	1,247	1,634	5,633
稅後純益	3,121	970	856	734	1,105	3,665	1,012	929	880	1,193	4,014
稅前 EPS（元）	32.23	9.56	8.75	7.52	11.01	36.84	9.94	9.81	8.94	11.72	40.41
稅後 EPS（元）	22.39	6.96	6.14	5.27	7.93	26.30	7.26	6.67	6.31	8.56	28.80
股本	1,379	1,379	1,394	1,394	1,394	1,394	1,394	1,394	1,394	1,394	1,394
稅後股東權益報酬率 %	21.99	6.24	5.08	4.14	5.99	21.45	5.17	4.52	4.09	5.33	19.11
每股淨值（元）	111.50	120.98	127.12	132.39	140.31	140.31	147.57	154.24	160.56	169.11	169.11
毛利率 %	25.45	24.00	25.67	23.06	25.01	24.47	24.32	24.60	24.24	26.04	24.85
營利率 %	15.55	14.94	15.57	13.21	15.67	14.90	15.29	15.88	14.88	16.89	15.78
稅前純益與前期比較 %	-13.59	-1.53	-0.64	-1.55	2.25	15.55	-0.34	0.82	-1.09	1.27	9.70
稅前純益率 %	14.92	15.73	15.10	13.54	15.80	15.10	15.46	16.28	15.19	16.45	15.87
稅後純益率 %	10.48	11.46	10.60	9.48	11.37	10.78	11.29	11.07	10.72	12.01	11.31

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永續發展報告

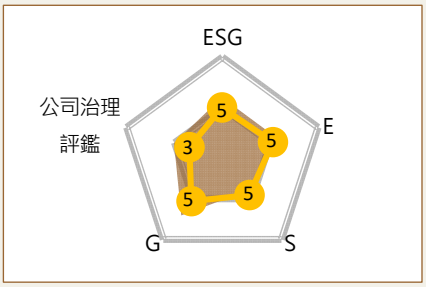
6670 復盛應用 永續發展概況

CMoney ESG Rating　　評鑑年季：2026Q3

CMoney ESG Rating		交易所公司治理評鑑	產業排名
6		21%至35%	7/27
E	6		
S	4		
G	7		

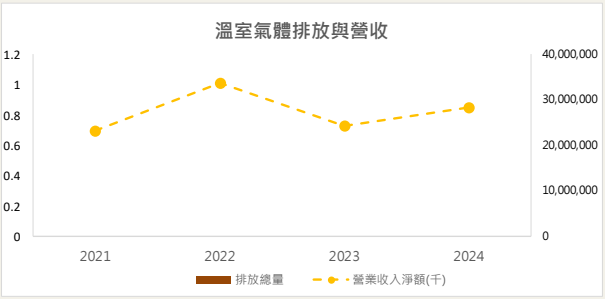
※評等1-10分，10分最高

永續報告書連結	
enplus.twse.com.tw/api/api/MopsSustainReport/data/FileStream?id=d51ebe0b-f8f8-481f-a76f-26	
報告書保證確信	驗證單位：會計師確信認證單位：安永聯合會計師事務所驗證方法：ISAE 3000



環境保護 Environment

溫室氣體

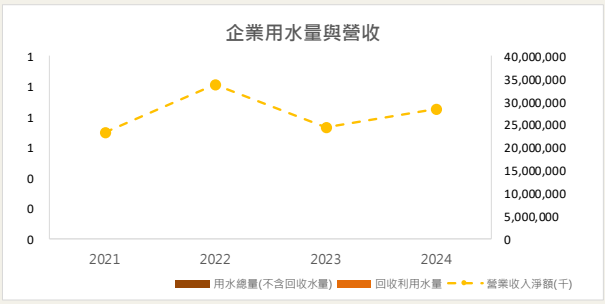


摘要	公司未揭露溫室氣體盤查數據		
溫室氣體盤查數據			
年度	排放總量(公噸)	範疇一	範疇二
2024	未揭露	-	-
2023	未揭露	-	-
2022	未揭露	-	-
2021	未揭露	-	-

*排放總量(公噸)：範疇一+範疇二溫室氣體排放數據
*範疇一：組織擁有或控制的營運據點的溫室氣體排放
*範疇二：組織所購買或取得之電力、用之於供熱、製冷或蒸汽而產生的溫室氣體排放

資料來源：CMoney、企業揭露資訊

水資源



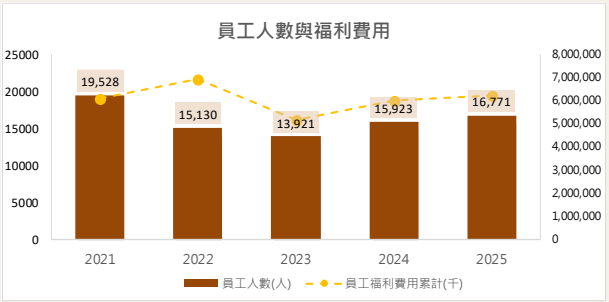
摘要	公司未揭露溫室氣體盤查數據		
企業用水量盤查數據			
年度	用水總量(公噸)	回收水量(公噸)	排水量(公噸)
2024	未揭露	-	-
2023	未揭露	-	-
2022	未揭露	-	-
2021	未揭露	-	-

*用水總量：依取水來源統計之總使用水量
*回收水量：將已用水和廢水透過循環處理成可再利用的水資源(包含製成回收水)
*排水量：排放的汙水總量

資料來源：CMoney、企業揭露資訊

社會責任 Social Responsibility

員工人數



摘要	2025年員工人數增加848人，營收增加5.01%，人均福利費用減少0.01%		
員工人數統計			
年度	員工人數(人)	員工人數增減(人)	福利費用(千)
2025	16,771	848	6,253,931
2024	15,923	2,002	6,012,978
2023	13,921	-1,209	5,184,748
2022	15,130	-4,398	6,926,229
2021	19,528	1,418	6,115,909

資料來源：CMoney、企業揭露資訊

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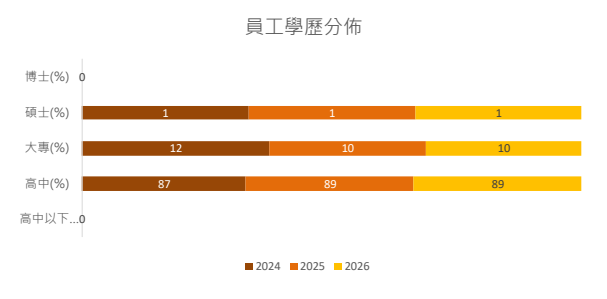
社會責任 Social Responsibility



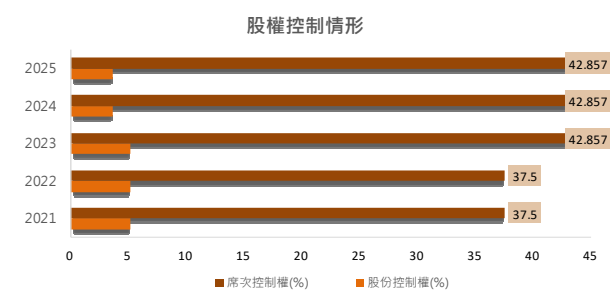
摘要 2026年員工學歷分佈，碩士以上學歷佔1(%)

員工學歷分布			
學歷/年度	2024	2025	2026
博士(%)	-	-	-
碩士(%)	1.0	1.0	1.0
大專(%)	12.0	10.0	10.0
高中(%)	87.0	89.0	89.0
高中以下(%)	-	-	-

資料來源：CMoney、企業揭露資訊

公司治理 Corporate Governance

股權控制



摘要 2025，最終控制者(董事長、副董事長、總經理、副總經理)掌握董監席次控制權增加0%，股份控制權減少0.04%

最終控制者股權控制情形			
年度	席次控制權(%)	股份控制權(%)	股權集中度
2025	42.86	3.61	0.0013
2024	42.86	3.66	0.0014
2023	42.86	5.15	0.0023
2022	37.50	5.15	0.0025
2021	37.50	5.21	0.0024

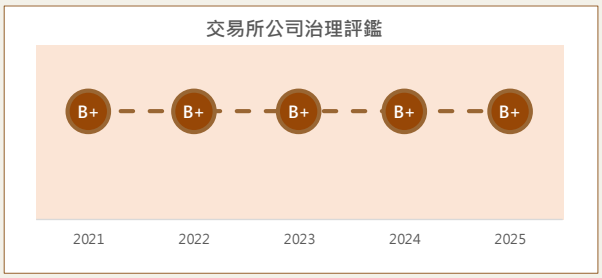
資料來源：CMoney、企業揭露資訊

公司治理評鑑

評鑑年度	評鑑結果	等級	加權項目比重			
			維護股東權益及平等對待股東	強化董事會結構與運作	資訊透明度	落實企業社會責任
2025	21%至35%	B+	0.16	0.25	0.1	0.49
2024	21%至35%	B+	0.19	0.29	0.17	0.35
2023	21%至35%	B+	0.22	0.31	0.19	0.28
2022	21%至35%	B+	0.2	0.33	0.23	0.24
2021	21%至35%	B+	0.2	0.33	0.26	0.21

資料來源：CMoney、交易所

[註]等級轉換：前5%：「A+」；6%至20%：「A」；21%至35%：「B+」；36%至50%：「B」；51%至65%：「C+」；66%至80%：「C」；81%至100%：「D」



企業功能委員會

功能委員會			
審計委員會(非強制設置)		審計委員會(強制設置)	★
提名委員會		資訊安全委員會	
公司治理委員會		永續發展委員會	
薪資報酬委員會	★	誠信經營委員會	
風險管理委員會		策略委員會	

企業裁罰資訊

裁罰資訊

今年以來沒有裁罰資訊

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Fubon Research
2026 年 10 月 6 日

南亞科 (2408 TT)

DRAM 明年仍緊缺

買進	
股價	目標價
NT$530.00	NT$660
○訪談摘要	○初次評論
●公司動態	●獲利調整
●獲利預覽	○評等調整
流通在外股數 (百萬)	3,450
市值 (百萬台幣)	1,828,610
市值 (百萬美元)	57,544
外資持股率 (%)	16.97
投信持股率 (%)	5.23
董監持股率 (%)	26.33
52 週高價 (元)	543.00
52 週低價 (元)	70.77
當沖比率	46%

ESG 評等	8	產業排名	5/207

資料來源：CMoney

盈餘預估及評等修正

	新	舊	%
評 等	買進	買進	
目標價	660	548	20
EPS 預估	73.43	59.16	24
目標 PE	9	9.3	

主要財務報表分析 至 12 月 31 日

百萬元	FY26F	FY27F	FY28F
營收淨額	393,222	614,311	664,235
年成長率(%)	490.5	56.2	8.1
營業利益	306,366	504,982	436,185
年成長率(%)	5,743.0	64.8	-13.6
稅後淨利	253,351	416,316	360,923
年成長率(%)	3,730.7	64.3	-13.3
每股淨利(NT$)	73.43	120.66	104.61
每股現金股利	40.0	60.0	60.0
本益比(x)	7.2	4.4	5.1
本淨比(x)	3.3	2.1	1.7
現金股利殖利率	7.5	11.3	11.3
淨值報酬率(%)	75.7	64.6	41.1

資料來源：CMoney，富邦投顧

夏武正
886-2-27815995#37011
richard.hsia@fubon.com

目標價上調至 660 元，潛在上漲空間+25%

1. 3Q26 營收 825.49 億元，QoQ+68%；其中價漲料逾 6 成，優於富邦預期。
2. 產業供不應求將持續，2027 年毛利率料逾 8 成。
3. 市場正逐步拉高 DRAM 明年仍供不應求之預期，預料南亞科評價可望上調。

3Q26 價漲料逾 6 成，單季 EPS 估 26.22 元

南亞科累積 7-9 月合併營收 1336.49 億元，QoQ+61.9%，YoY+611.7%，數字優於富邦預期。由於公司銷售量大致 QoQ 持平，絕大部分季成長來自售價上揚；因此我們預估本季毛利率約 87.0%，較前季上揚 7.5 個百分點。簡言之，我們預估南亞科 3Q26 營業利益估約 1,102 億元，稅前淨利估 1,117 億元，歸屬母公司稅後淨利估約 905 億元，EPS 估約 26.22 元。

DRAM 明年仍緊缺，價格修正料將延後

市調機構及美光法說不約而同指出，DRAM 產業 2027 年供不應求將較今年更加嚴峻，可見未來尚無法看到供需平衡的時間點。其次，南亞科 7 月法說已指出當時已簽定之長短期 LTA 供貨合約佔總產能 50%，且下游客戶已依照 LTA 基礎價格低標提供一定的預付款。無獨有偶，美光 10 月 1 日(台北時間)指出已簽署 26 份 SCA(策略客戶協議)，目前協議涵蓋了約 2030 年以前約 35%的營收。簡言之，我們目前已將 DRAM 價格修正的時點延後到 4Q27 或更晚。

產業供不應求將持續，27 年毛利率料逾 8 成

展望 4Q26，我們預期 Consumer DRAM 漲幅約+3~8%、其他傳統 DRAM 漲幅約 10~15%；4Q26EPS 估約 25.11 元，全年 EPS 約 73.43 元。至於 2027 年，Fab 5A(泰林新廠)預計 2H 開始有意義挹注產出；由於 DDR4 升級到 DDR5 的情況擴大，銷售組合預料逐漸往 DDR5 放大，因此每 GB 平均售價可能因此出現 QoQ 下滑。我們預估南亞科合併營收約 6,143 億元，YoY+56.2%；其中位元銷售量年增估約二成、ASP 年增約估三成。簡言之，富邦預估南亞科 2027 年毛利率約達 86.3%，營業利益估約 5,050 億元，歸屬母公司稅後淨利估約 4,163 億元，EPS 估約 120.66 元。

富邦不同之處

富邦修正南亞科最新之 2026-27 年 EPS 預估值分別為 73.43、120.66 元，相較目前 CMoney 同業平均預估值分別+10%、+12%，數字相對樂觀。

評價與風險

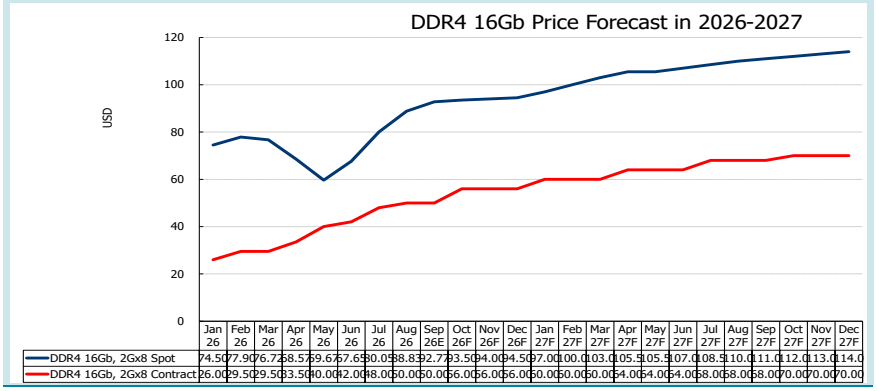
我們預料美光 10 月 1 日法說後，一部分投資人將逐步認同 DRAM 明年仍將供不應求，售價也不會出現修正，南亞科股價預料將逐步墊高。簡言之，我們維持南亞科買進的投資評等，目標價上調至 660 元(約 FY26 的 9.0PER 或 FY26 4.6XPBR)。南亞科的股價風險來自(1)高階伺服器產品或私募股東合作進度不如預期、(2)AI 產業展望出現意外的結構修正。

圖表 1：獲利預估修正 單位: 新台幣百萬元

	FY25	FY26F			FY27F		
		原估	新估	差異	原估	新估	差異
營業收入	66,587	330,184	393,222	19.1%	416,219	614,311	47.6%
銷貨毛利	14,986	264,796	326,407	23.3%	328,045	530,128	61.6%
營業利益	5,243	246,608	306,366	24.2%	308,131	504,982	63.9%
稅後淨利	6,614	204,108	253,351	24.1%	253,701	416,316	64.1%
EPS (元)	2.13	59.16	73.43	24.1%	73.53	120.66	64.1%
重要比率							
毛利率	22.5%	80.2%	83.0%		78.8%	86.3%	
營益率	7.9%	74.7%	77.9%		74.0%	82.2%	
淨利率	9.9%	61.8%	64.4%		61.0%	67.8%	

資料來源：公司，CMoney、富邦投顧

圖表 2：標準型 DDR4 16Gb 合約與現貨價預估(2026/1~2027/12)



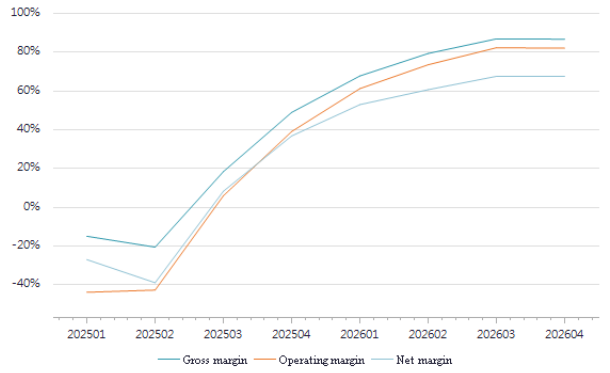
資料來源：DRAMExchange 2026/9/17、富邦投顧

圖表 3：單季預測

單位: 百萬元	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F	2028F
損益模型											
營業收入	49,087	82,549	133,649	127,937	134,210	148,814	172,187	159,100	393,222	614,311	664,235
銷貨成本	15,761	16,932	17,337	16,786	16,349	18,653	21,678	27,502	66,816	84,183	202,842
銷貨毛利	33,325	65,617	116,312	111,152	117,861	130,160	150,508	131,598	326,407	530,128	461,393
營業費用	3,205	4,793	6,086	5,956	6,157	6,232	6,502	6,255	20,041	25,146	25,208
營業利益	30,111	60,826	110,226	105,195	111,704	123,928	144,007	125,343	306,366	504,982	436,185
業外收支淨額	1,607	1,503	1,446	1,774	1,965	2,136	1,771	2,056	6,330	7,927	9,101
稅前淨利	31,718	62,329	111,672	106,969	113,668	126,064	145,778	127,399	312,696	512,909	445,286
稅後淨利	26,059	50,192	90,455	86,645	93,208	101,833	118,081	103,194	253,351	416,316	360,923
EPS	8.41	14.66	26.22	25.11	27.02	29.52	34.22	29.91	73.43	120.66	104.61
獲利能力(%)											
毛利率	67.9%	79.5%	87.0%	86.9%	87.8%	87.5%	87.4%	82.7%	83.0%	86.3%	69.5%
營益率	61.3%	73.7%	82.5%	82.2%	83.2%	83.3%	83.6%	78.8%	77.9%	82.2%	65.7%
淨利率	53.1%	60.8%	67.7%	67.7%	69.4%	68.4%	68.6%	64.9%	64.4%	67.8%	54.3%
年成長率(%)											
營業收入	582.9%	684.2%	611.7%	325.1%	173.4%	80.3%	28.8%	24.4%	490.5%	56.2%	8.1%
銷貨毛利	--	--	3,257.9%	653.0%	253.7%	98.4%	29.4%	18.4%	2,078.1%	62.4%	-13.0%
營業利益	--	--	9,749.0%	792.9%	271.0%	103.7%	30.6%	19.2%	5,743.0%	64.8%	-13.6%
稅後淨利	--	--	5,686.0%	681.1%	257.7%	102.9%	30.5%	19.1%	3,730.7%	64.3%	-13.3%
EPS	--	--	5,143.4%	601.5%	221.2%	101.3%	30.5%	19.1%	3,347.5%	64.3%	-13.3%
季成長率(%)											
營業收入	63.1%	68.2%	61.9%	-4.3%	4.9%	10.9%	15.7%	-7.6%			
銷貨毛利	125.8%	96.9%	77.3%	-4.4%	6.0%	10.4%	15.6%	-12.6%			
營業利益	155.6%	102.0%	81.2%	-4.6%	6.2%	10.9%	16.2%	-13.0%			
稅後淨利	134.9%	92.6%	80.2%	-4.2%	7.6%	9.3%	16.0%	-12.6%			
EPS	134.9%	74.3%	78.8%	-4.2%	7.6%	9.3%	16.0%	-12.6%			

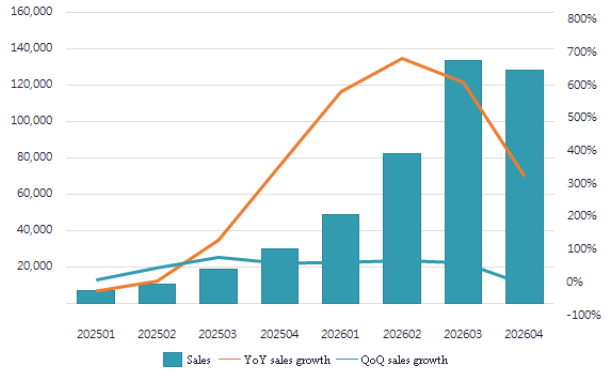
資料來源：公司、CMoney、富邦投顧

圖表 4：毛利率、營益率、稅後營益率 單位：%



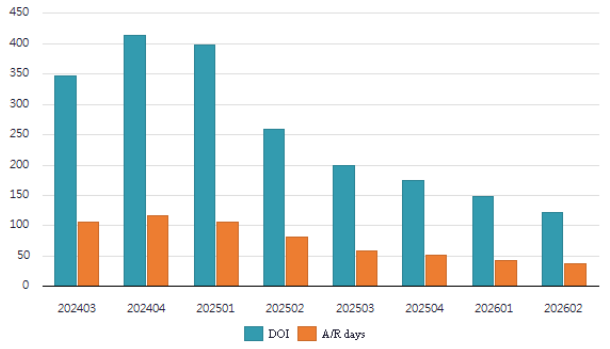
資料來源：CMoney、富邦投顧

圖表 5： 季營收變化 單位：百萬元/%



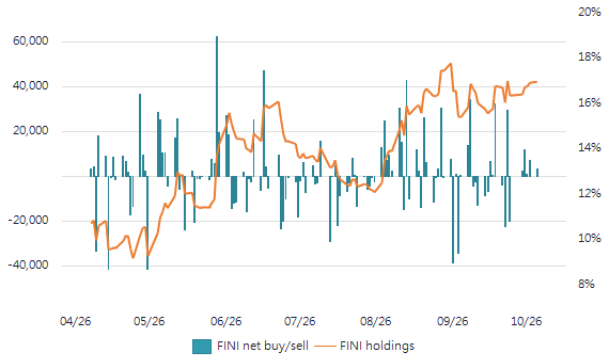
資料來源：CMoney、富邦投顧

圖表 6：存貨及 A/R 週轉天數 單位：天



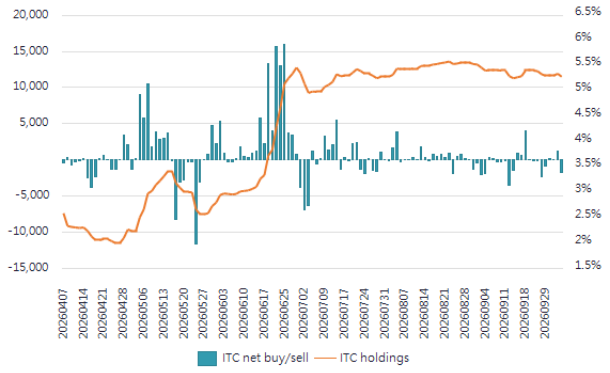
資料來源：CMoney、富邦投顧

圖表 7：外資買賣超張數及持股比率 單位：張/%



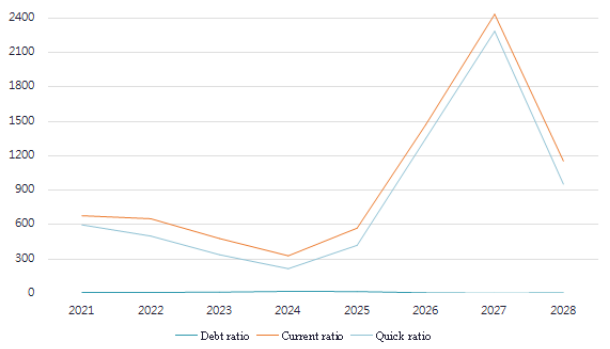
資料來源：CMoney、富邦投顧

圖表 8：投信買賣超張數及持股比率 單位：張/%



資料來源：CMoney、富邦投顧

圖表 9：負債比率與財務比率 單位：%



資料來源：CMoney、富邦投顧

ESG永續發展專區

CMoney Taiwan ESG Rating

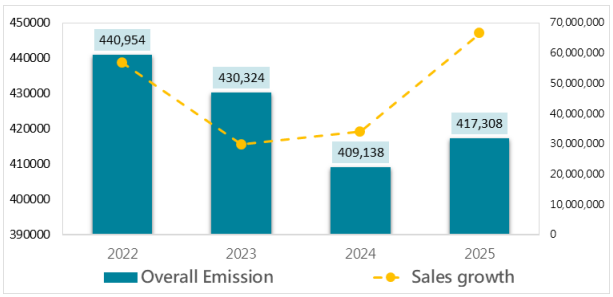
CMoney ESG Rating	E(環境保護)	S(社會責任)	G(公司治理)
8/10	7 /10	6 /10	8 /10

交易所產業類別	產業排名	交易所公司治理評鑑
電子–半導體	5 /207	6%至 20%

環境保護 Environment

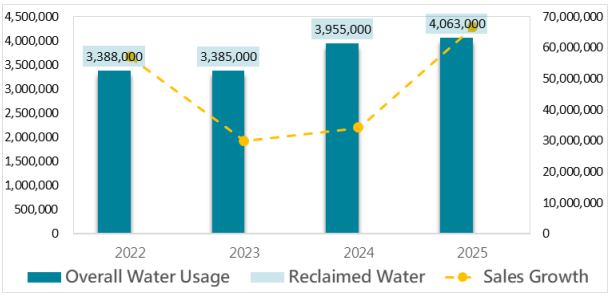
溫室氣體排放與營收

摘要			
溫室氣體盤查數據			
年度	排放總量(公噸)	直接排放(公噸)	間接排放(公噸)
2025	417,308	51,088	366,220
2024	409,138	39,063	370,075
2023	430,324	56,654	373,670
2022	440,954	59,788	381,166



企業用水量與營收

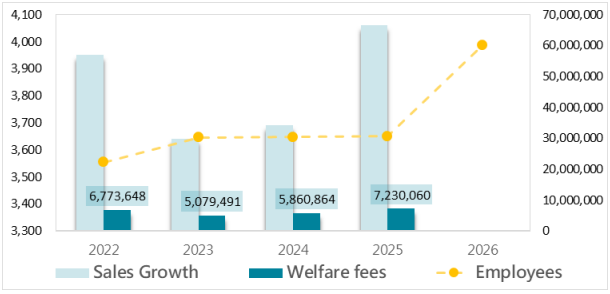
摘要			
企業用水量盤查數據			
年度	用水總量(公噸)	回收水量(公噸)	排水量(公噸)
2025	4,063,000	--	3,017,000
2024	3,955,000	--	2,864,000
2023	3,385,000	--	2,446,000
2022	3,388,000	--	2,695,000



社會責任 Social Responsibility

員工人數與福利費用

摘要			
員工人數與福利費用統計			
年度	員工人數(人)	員工人數增減(人)	福利費用
2026	3,986	335	--
2025	3,651	3	7,230,060
2024	3,648	2	5,860,864
2023	3,646	92	5,079,491
2022	3,554	0	6,773,648



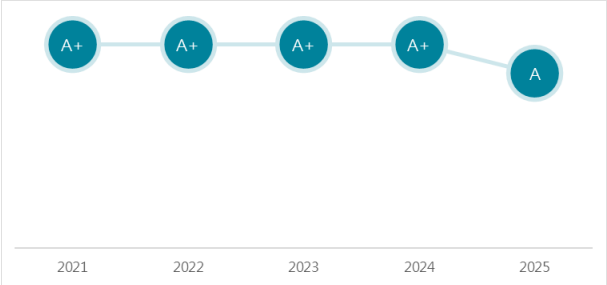
公司治理 Corporate Governance

公司治理評鑑

評鑑年度	評鑑結果	等級	加權項目比重			
			構面一	構面二	構面三	構面四
2025	6%至 20%	A	0.16	0.25	0.10	0.49
2024	前 5%	A+	0.19	0.29	0.17	0.35
2023	前 5%	A+	0.22	0.31	0.19	0.28
2022	前 5%	A+	0.20	0.33	0.23	0.24

[註]等級轉換：前 5%：「A+」; 6%至 20%：「A」; 21%至 35%：「B+」; 36%至 50%：「B」; 51%至 65%：「C+」; 66%至 80%：「C」; 81%至 100%：「D」

資料來源：交易所、公司、CMoney、富邦投顧



損益表					百萬台幣
	FY24	FY25	FY26F	FY27F	FY28F
營收淨額	34,132	66,587	393,222	614,311	664,235
銷貨成本	34,552	51,600	66,816	84,183	202,842
銷貨毛利	(421)	14,986	326,407	530,128	461,393
營業費用	10,134	9,741	20,041	25,146	25,208
營業利益	(10,555)	5,243	306,366	504,982	436,185
EBITDA	9,973	22,735	320,028	524,110	463,526
利息淨收支	2,974	1,907	4,329	6,677	7,538
投資淨損益	(21)	(7)	1,000	1,250	1,562
匯兌淨損益	697	(214)	946	0	0
其他收支	381	1,041	54	0	0
業外收支淨額	3,998	2,639	6,330	7,927	9,101
稅前淨利	(6,557)	7,882	312,696	512,909	445,286
所得稅	(1,474)	1,269	59,339	96,595	84,365
少數股權	0	0	6	(2)	(2)
稅後淨利	(5,083)	6,614	253,351	416,316	360,923
基本 EPS(NT$)	(1.64)	2.13	73.43	120.66	104.61
完全稀釋 EPS(NT$)	0.00	2.10	73.43	120.66	104.61
股票股利(NT$)	0.00	0.00	0.00	0.00	0.00
現金股利(NT$)	0.00	1.35	40.00	60.00	60.00
年成長率(%)					
營收淨額	14.2	95.1	490.5	56.2	8.1
銷貨毛利	--	--	2,078.1	62.4	(13.0)
營業利益	--	--	5,743.0	64.8	(13.6)
稅後淨利	--	--	3,730.7	64.3	(13.3)
基本 EPS	--	--	3,347.5	64.3	(13.3)
獲利能力(%)					
毛利率	(1.2)	22.5	83.0	86.3	69.5
營益率	(30.9)	7.9	77.9	82.2	65.7
稅後淨利率	(14.9)	9.9	64.4	67.8	54.3
EBITDA 率	29.2	34.1	81.4	85.3	69.8

財務比率					
	FY24	FY25	FY26F	FY27F	FY28F
流動比率(%)	329	571	1,473	2,433	1,155
速動比率(%)	217	421	1,353	2,285	953
淨負債股權比率(%)	(25)	(25)	(48)	(58)	(51)
稅後淨利率(%)	(14.9)	9.9	64.4	67.8	54.3
資產周轉率(x)	0.2	0.3	1.1	0.9	0.7
財務槓桿(x)	1.3	1.2	1.1	1.0	1.1
已動用資本/淨值(x)	0.7	0.7	0.5	0.4	0.5
總資產報酬率(%)	(2.5)	3.2	47.1	51.0	35.0
淨值報酬率(%)	(3.1)	3.9	75.7	64.6	41.1
資本報酬率(%)	(5.7)	3.5	89.7	78.0	49.8

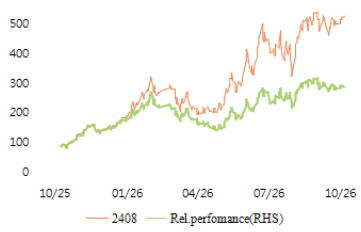
財務比率					
	FY24	FY25	FY26F	FY27F	FY28F
股息保障倍數(x)	--	--	29.5	2.7	1.2
所得稅保障倍數(x)	9.6	4.0	2.1	3.5	2.6
資本支出保障倍數(x)	(0.9)	0.4	2.3	3.1	2.7
應收帳款週轉率(x)	8.3	4.0	3.7	7.3	5.5
存貨周轉率(x)	1.0	1.9	2.1	2.4	1.6
應付帳款周轉率(x)	6.7	8.0	3.8	5.3	3.5
應收帳款周轉天數	44	91	99	50	66
存貨週轉天數	373	193	176	154	235
應付帳款週轉天數	55	46	96	69	106
現金循環週期(天)	363	238	179	135	195

資產負債表					百萬台幣
	FY24	FY25	FY26F	FY27F	FY28F
現金及約當現金	61,903	58,074	250,397	458,155	489,330
應收款項	4,132	16,558	106,352	83,738	119,759
存貨	35,318	27,288	32,219	35,456	130,424
其他流動資產	6,593	6,621	6,621	6,621	6,621
流動資產	107,946	108,541	395,589	583,970	746,134
長期投資	5,396	6,659	7,038	5,951	4,768
固定資產	84,327	85,031	126,631	218,054	272,814
其他長期資產	9,037	8,221	8,222	8,221	8,221
總資產	206,706	208,453	537,480	816,196	1,031,937
短期借款	14,536	5,060	1,547	808	808
商業本票	6,734	0	0	0	0
應付款項	5,180	6,481	17,505	15,842	58,717
其他流動負債	6,330	7,480	7,806	7,355	5,094
流動負債	32,780	19,020	26,858	24,005	64,619
長期借款	0	10,200	8,638	0	0
其他長期負債	8,873	8,694	3,551	1,387	1,508
總負債	41,653	37,914	39,046	25,392	66,127
普通股股本	30,986	30,986	30,986	30,986	30,986
公積	32,834	33,081	111,799	111,799	111,799
保留盈餘	97,478	104,102	353,279	645,650	820,656
其他	3,755	2,369	2,369	2,370	2,369
股東權益	165,053	170,538	498,433	790,805	965,810
負債加股東權益	206,706	208,453	537,480	816,196	1,031,937

現金流量表					
	FY24	FY25	FY26F	FY27F	FY28F
稅後淨利	(5,083)	6,614	253,351	416,316	360,923
折舊與攤提	16,144	14,281	11,661	17,878	25,778
投資收入	21	7	(1,000)	(1,250)	(1,562)
營運資金增減	(4,859)	(3,095)	(83,701)	17,714	(88,114)
其他	(4,270)	762	(4,082)	(1,272)	730
來自營運之現金流量	1,952	18,568	176,229	449,386	297,755
資本支出	(16,141)	(13,441)	(53,000)	(109,000)	(80,237)
長投增減	(613)	1,263	(1,127)	(7,475)	(325)
其他	(15,576)	(15,378)	0	(1)	0
來自投資之現金流量	(16,188)	(14,110)	(54,127)	(116,476)	(80,562)
自由現金流量	(14,236)	4,459	122,102	332,910	217,193
借款增（減）	0	17,200	(3,512)	(739)	0
現金股利支出	--	--	(4,174)	(123,945)	(185,918)
發行公司債	4,000	0	0	0	0
償還公司債	--	--	0	0	0
現金增資	--	--	78,718	0	0
其他	9,707	(23,517)	77,906	(468)	(100)
來自融資之現金流量	13,707	(6,317)	70,220	(125,152)	(186,018)
匯率調整	3,619	(1,970)	0	0	0
本期產生之現金流量	3,091	(3,829)	192,323	207,759	31,174

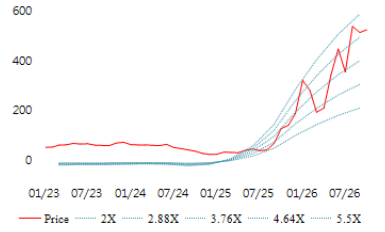
比率分析					
	FY24	FY25	FY26F	FY27F	FY28F
本益比(x)	(323.2)	248.8	7.2	4.4	5.1
本營比(x)	48.1	24.7	4.2	2.7	2.5
本淨比(x)	9.9	9.6	3.3	2.1	1.7
P/CFPS(x)	841.4	88.4	9.3	3.7	5.5
PE/growth (x)	--	--	0.0	0.1	(0.4)
P/FCFPS(x)	(115.7)	320.3	13.3	4.8	7.5
EV/EBITDA (x)	178.6	78.5	5.0	2.6	2.9
現金股利殖利率(%)	0.0	0.3	7.5	11.3	11.3

股價及相對大盤績效圖



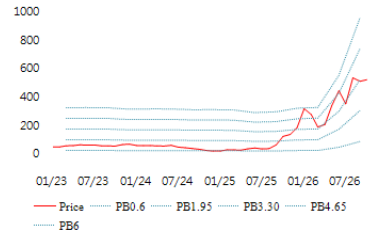
資料來源：CMoney、富邦投顧

未來 12 個月本益比區間圖



資料來源：CMoney、富邦投顧

未來 12 個月本淨比區間圖



資料來源：CMoney、富邦投顧

免責宣言

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富邦的股票評等標準

評等	定義
買進	預估未來6個月內的絕對報酬超過15%
中立	預估未來6個月內有絕對報酬介於15%與負15%之間
賣出	預估未來6個月內的絕對報酬高於負15%
未評等	由於富邦目前與該公司有特定交易或沒有足夠的基本資料判斷該公司評等
評估中	目前正在研議個股的投資評等，將於3到6個月內提供投資評等

產業評等	定義
優於大盤	預估該產業在未來6個月內會比大盤指數表現突出
持平	預估該產業在未來6個月內與大盤指數表現相較持平
劣於大盤	預估該產業在未來6個月內會比大盤指數表現較差

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Fubon's Rating System

Rating	Definition
Buy	Expected absolute return to be over 15% within the next 6 months
Neutral	Expected absolute return to be level within the next 6 months
Sell	Expected absolute return to be over negative 15% within the next 6 months
Not Rated (NR)	Pursuant to Fubon acting in deals involving the company or there is not a sufficient fundamental basis for determining an investment rating
Under Review	Fubon is in the process of determining an investment rating and will assign a rating within the 3- to 6-month horizon

Sector Rating	Definition
Overweight	Sector expected to outperform the broader market indexes within the next 6-12 months
Market Weight	Sector expected to perform in line with the broader market averages within the next 6-12 months
Underweight	Sector expected to underperform the broader market indexes within the next 6-12 months.

Fubon Research
2026 年 10 月 6 日

邁科 (6831 TT)

AISC 散熱指標股

買進

股價	目標價
NT$489.00	NT$600
●訪談摘要	○初次評論
●公司動態	●獲利調整
○獲利預覽	○評等調整

流通在外股數 (百萬)	68
市值 (百萬台幣)	33,010
市值 (百萬美元)	1,039
外資持股率 (%)	13.94
投信持股率 (%)	0.77
董監持股率 (%)	11.69
52 週高價 (元)	956.00
52 週低價 (元)	177.18
當沖比率	28%

ESG 評等	4	產業排名	95/109

資料來源：CMoney

2027 年盈餘預估及評等修正

	新	舊	%
評 等	買進	買進	
目標價	600	430	40
EPS 預估	25.0	20.5	22
目標 PE	24	21	

主要財務報表分析 至 12 月 31 日

百萬元	FY25	FY26F	FY27F
營收淨額	3,703	5,492	10,734
年成長率(%)	84.7	48.3	95.4
營業利益	426	847	2,216
年成長率(%)	234.4	98.8	161.6
稅後淨利	301	624	1,688
年成長率(%)	183.8	107.5	170.5
每股淨利(NT$)	4.94	9.25	25.00
每股現金股利	3.0	6.0	16.0
本益比(x)	99.0	52.9	19.6
本淨比(x)	11.7	10.2	7.3
現金股利殖利率	0.6	1.2	3.3
淨值報酬率(%)	13.7	20.6	43.5

資料來源：CMoney，富邦投顧

楊惟婷
886-2-27815995#37015
weiting.yang@fubon.com

2026~2027 年 AI 伺服器散熱需求強勁增長，維持買進建議，目標價 600 元，
1. 4Q26 客戶拉貨動能放大，營收獲利達全年高點。
2. 2026 年 ASIC、NIC 出貨動能佳，伺服器散熱營收倍增，成長集中下半年。
3. 2027 年伺服器散熱出貨版圖再擴大，拓展其他 CSP 客戶有成。

4Q26 客戶拉貨動能放大，營收獲利達全年高點

Trainium 3 氣冷散熱的 2.5D VC 5 月底至 6 月進入量產，7~8 月份出貨爬升速度低於預期，9 月出貨量持穩，下修邁科 3Q26 營收預估至 16.3 億，QoQ+55.8%，YoY+41.2%，營收規模擴張，伺服器散熱佔比回升，估推高毛利率回升至 26.7%，EPS 估 3.35 元。4Q26 預期 AWS Trainium 3 氣冷散熱拉貨動能可較前一季放大，加上新增 AWS Graviton 5 氣冷散熱模組的營收貢獻，估計 4Q26 邁科營收季增 20.3%至 19.6 億，YoY+108.5%，達全年高點，營益率續增，EPS 估 4.24 元。

2026 年 ASIC、NIC 出貨動能佳，伺服器散熱營收倍增

邁科伺服器散熱營收貢獻主要來自 AWS AI 伺服器，目前已出貨包含 ASIC、AI 加速卡、Switch 的散熱 VC。雖 3Q26 伺服器散熱拉貨低於預期，但受惠 AWS Trainium 3 及 AI 加速卡散熱終端需求量較去年攀升及邁科市佔率增長，富邦仍預估 2026 年邁科的伺服器散熱營收年增超過 100%，營收佔比由 56%拉高到 78%。其他 NB 及顯示卡等消費性散熱產品營收則呈現年減，合計預估 2026 年邁科營收 54.9 億，YoY+48.3%，營益率在經濟規模及產品組合正向發展下，擴大到 15.4%，EPS 估 9.25 元。

2027 年伺服器散熱出貨版圖再擴大，拓展其他 CSP 客戶有成

4Q26 邁科出貨 AWS Graviton 5 氣冷散熱模組給另一個組裝客戶，首度跨入 AWS CPU 散熱，2027 年 AWS CPU 散熱市佔率估計逐步拉升至 30%；至於 AWS Trainium 3 液冷散熱，邁科為客戶的新進水冷板供應商，估計初期供應比重低於同業，將視後續液冷產品良率攀升拉高供應比重；另外邁科也積極接洽其他 CSP 客戶，預計今年取得相關認證，有望明年開始出貨 AWS 之外其他 CSP 客戶的 SWITCH 散熱產品。2027 年邁科仍是 AWS ASIC 氣冷散熱最大供應商，隨 AWS ASIC 出貨量增，及新增 CPU 氣冷、ASIC 液冷及其他 CSP 客戶散熱產品出貨，估計邁科 2027 年伺服器散熱營收可較今年倍增。初估 2027 年邁科營收 107.3 億，YoY+95.4%，獲利能力持續優化，EPS 估 25.0 元。

富邦不同之處

邁科 2026、2027 年來自 AI 伺服器散熱營收倍增，強勁的營收、獲利成長，優於其他散熱同業，給予以接近龍頭的的 24 倍本益比評價。

評價與風險

AI 伺服器散熱同業本益比區間 20~27 倍。AI 伺服器散熱驅動 2026~2027 年邁科營收獲利強勁增長>CSP 業者加速 ASIC 佈局及 AI 伺服器晶片功耗攀升，正面看待邁科後續年度的營運動能延續，富邦以 2027 年獲利的 24 倍本益比評價，目標價上調至 600 元，維持買進建議。投資風險有 AI 伺服器需求不如預期、大客戶供應比重走低。

圖表 1: 獲利預估修正 **單位: 新台幣百萬元**

	FY25	FY26F			FY27F		
		原估	新估	差異	原估	新估	差異
營業收入	3,703	7,452	5,492	-26.3%	9,610	10,734	11.7%
銷貨毛利	868	2,050	1,363	-33.5%	2,693	2,993	11.1%
營業利益	426	1,395	847	-39.3%	1,877	2,216	18.1%
稅後淨利	301	1,022	624	-38.9%	1,384	1,688	22.0%
EPS (元)	4.94	15.15	9.25	-38.9%	20.50	25.00	21.9%
重要比率							
毛利率	23.5%	27.5%	24.8%		28.0%	27.9%	
營益率	11.5%	18.7%	15.4%		19.5%	20.6%	
淨利率	8.1%	13.7%	11.4%		14.4%	15.7%	

資料來源：公司，CMoney、富邦投顧

圖表 2: 單季預測

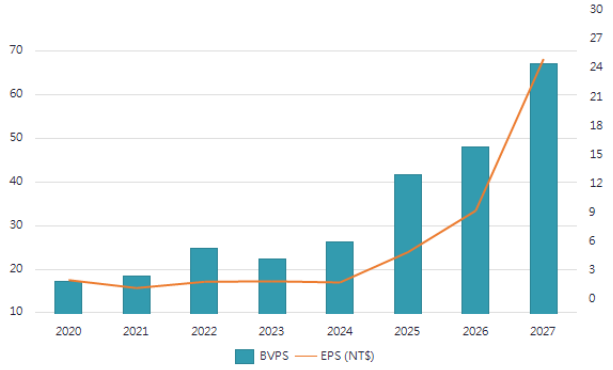
單位: 百萬元	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2025	2026F	2027F
損益模型											
營業收入	854	1,047	1,630	1,961	2,121	2,345	3,011	3,257	3,703	5,492	10,734
銷貨成本	699	822	1,195	1,414	1,541	1,698	2,164	2,338	2,835	4,130	7,741
銷貨毛利	156	225	435	547	580	647	847	919	868	1,363	2,993
營業費用	100	112	135	169	159	179	210	230	442	515	777
營業利益	56	112	301	378	421	468	638	689	426	847	2,216
業外收支淨額	0	-22	-7	-7	-6	-6	-6	-6	-78	-36	-24
稅前淨利	56	91	294	371	415	462	632	683	348	812	2,192
稅後淨利	45	68	226	286	320	356	486	526	301	624	1,688
EPS	0.66	1.00	3.35	4.24	4.73	5.27	7.21	7.79	4.94	9.25	25.00
獲利能力(%)											
毛利率	18.2%	21.5%	26.7%	27.9%	27.3%	27.6%	28.1%	28.2%	23.5%	24.8%	27.9%
營益率	6.6%	10.7%	18.5%	19.3%	19.8%	20.0%	21.2%	21.2%	11.5%	15.4%	20.6%
淨利率	5.2%	6.5%	13.9%	14.6%	15.1%	15.2%	16.1%	16.1%	8.1%	11.4%	15.7%
年成長率(%)											
營業收入	50.7%	0.6%	41.2%	108.5%	148.2%	124.0%	84.7%	66.1%	84.7%	48.3%	95.4%
銷貨毛利	30.7%	7.3%	35.6%	149.4%	273.0%	188.0%	94.7%	68.0%	87.6%	57.0%	119.6%
營業利益	150.6%	1.9%	41.6%	367.1%	652.4%	316.2%	112.0%	82.3%	234.4%	98.8%	161.6%
稅後淨利	232.5%	166.2%	51.4%	154.1%	617.7%	426.9%	115.0%	83.9%	183.8%	107.5%	170.5%
EPS	200.0%	138.1%	34.5%	135.4%	617.3%	426.7%	115.1%	84.0%	174.4%	87.2%	170.3%
季成長率(%)											
營業收入	-9.2%	22.5%	55.7%	20.3%	8.2%	10.6%	28.4%	8.2%			
銷貨毛利	-29.1%	44.5%	93.6%	25.7%	6.0%	11.6%	30.9%	8.5%			
營業利益	-30.9%	100.9%	167.7%	25.6%	11.4%	11.2%	36.3%	8.0%			
稅後淨利	-60.4%	51.5%	234.5%	26.5%	11.9%	11.3%	36.5%	8.2%			
EPS	-63.3%	51.5%	235.0%	26.5%	11.7%	11.3%	36.8%	8.1%			

資料來源：公司、CMoney、富邦投顧

圖表 3 ： 毛利率、營益率、稅後營益率 單位： %

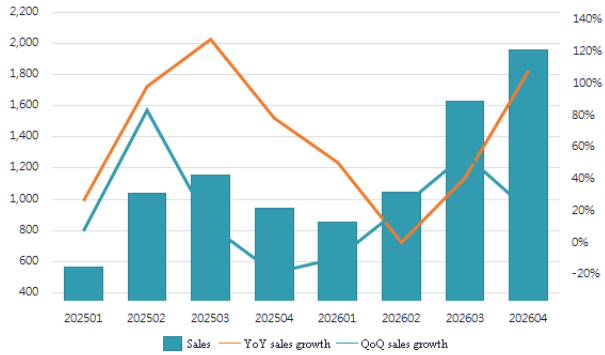
資料來源：CMoney、富邦投顧

圖表 4 : EPS 與 BVPS 趨勢 單位: 元/%



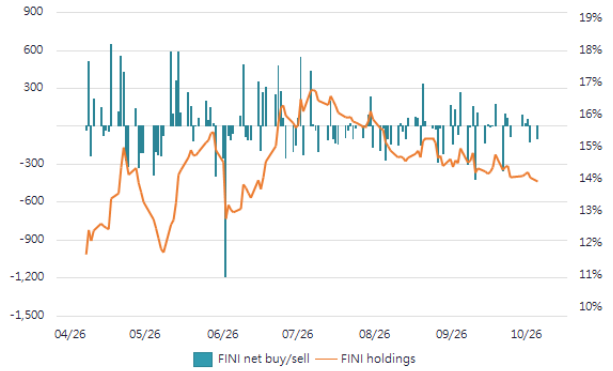
資料來源：CMoney、富邦投顧

圖表 5： 季營收變化 單位: 百萬元/%



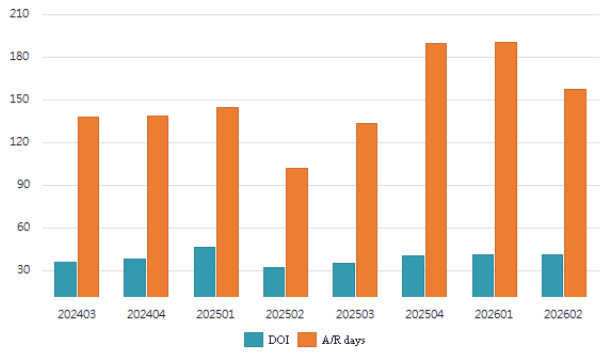
資料來源：CMoney、富邦投顧

圖表 6： 外資買賣超張數及持股比率 單位: 張/%



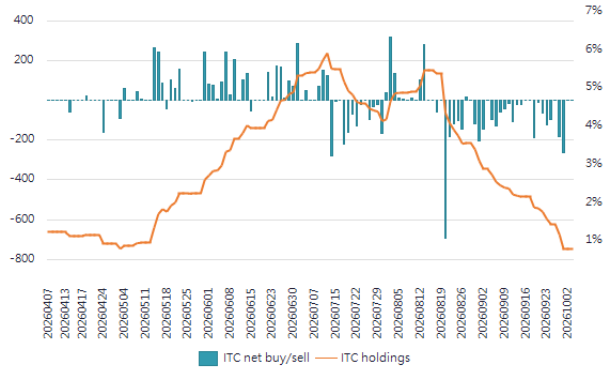
資料來源：CMoney、富邦投顧

圖表 7： 存貨及 A/R 週轉天數 單位: 天



資料來源：CMoney、富邦投顧

圖表 8: 投信買賣超張數及持股比率 單位: 張/%



資料來源：CMoney、富邦投顧

ESG永續發展專區

CMoney Taiwan ESG Rating

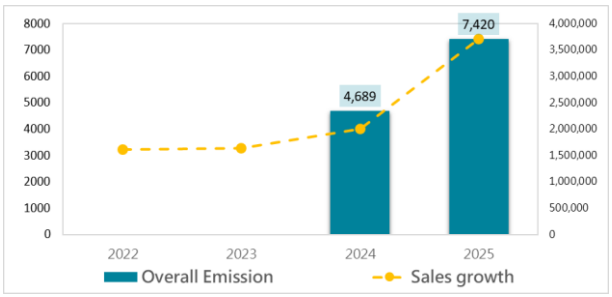
CMoney ESG Rating	E(環境保護)	S(社會責任)	G(公司治理)
4/10	4 /10	3 /10	4 /10

交易所產業類別	產業排名	交易所公司治理評鑑
電子–電腦及週邊設備	95 /109	不列入受評

環境保護 Environment

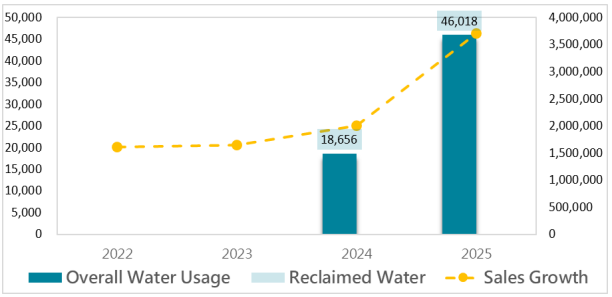
溫室氣體排放與營收

摘要			
溫室氣體盤查數據			
年度	排放總量(公噸)	直接排放(公噸)	間接排放(公噸)
2025	7,420	85	7,335
2024	4,689	37	4,652
2023	--	--	--
2022	--	--	--



企業用水量與營收

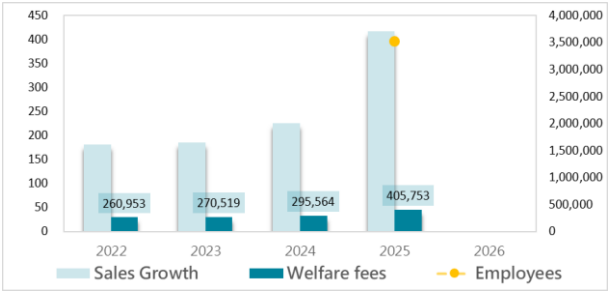
摘要			
企業用水量盤查數據			
年度	用水總量(公噸)	回收水量(公噸)	排水量(公噸)
2025	46,018	--	--
2024	18,656	--	--
2023	--	--	--
2022	--	--	--



社會責任 Social Responsibility

員工人數與福利費用

摘要			
員工人數與福利費用統計			
年度	員工人數(人)	員工人數增減(人)	福利費用
2026	--	--	--
2025	396	--	405,753
2024	--	--	295,564
2023	--	--	270,519
2022	--	--	260,953



公司治理 Corporate Governance

公司治理評鑑

評鑑年度	評鑑結果	等級	加權項目比重			
			構面一	構面二	構面三	構面四
2025	不列入受評	--	0.16	0.25	0.10	0.49
2024	--	--	--	--	--	--
2023	--	--	--	--	--	--
2022	--	--	--	--	--	--

[註]等級轉換：前 5%：「A+」; 6%至 20%：「A」; 21%至 35%：「B+」; 36%至 50%：「B」; 51%至 65%：「C+」; 66%至 80%：「C」; 81%至 100%：「D」

資料來源：交易所、公司，CMoney、富邦投顧

損益表					百萬台幣
	FY23	FY24	FY25	FY26F	FY27F
營收淨額	1,642	2,005	3,703	5,492	10,734
銷貨成本	1,258	1,542	2,835	4,130	7,741
銷貨毛利	383	463	868	1,363	2,993
營業費用	274	335	442	515	777
營業利益	109	127	426	847	2,216
EBITDA	217	255	500	982	2,364
利息淨收支	8	8	(9)	(4)	(4)
投資淨損益	21	(1)	(15)	0	0
匯兌淨損益	7	50	(28)	(4)	0
其他收支	(15)	(45)	(41)	(27)	(20)
業外收支淨額	25	14	(78)	(36)	(24)
稅前淨利	134	141	348	812	2,192
所得稅	31	35	48	187	504
少數股權	0	0	0	1	0
稅後淨利	103	106	301	624	1,688
基本 EPS(NT$)	1.92	1.80	4.94	9.25	25.00
完全稀釋 EPS(NT$)	1.91	1.79	4.94	9.25	25.00
股票股利(NT$)	0.55	0.00	0.00	0.00	0.00
現金股利(NT$)	0.78	1.25	3.00	6.00	16.00
年成長率(%)					
營收淨額	2.1	22.1	84.7	48.3	95.4
銷貨毛利	5.7	20.7	87.6	57.0	119.6
營業利益	0.9	16.9	234.4	98.8	161.6
稅後淨利	18.2	2.4	183.8	107.5	170.5
基本 EPS	2.7	(6.3)	174.4	87.2	170.3
獲利能力(%)					
毛利率	23.4	23.1	23.5	24.8	27.9
營益率	6.6	6.4	11.5	15.4	20.6
稅後淨利率	6.3	5.3	8.1	11.4	15.7
EBITDA 率	13.2	12.7	13.5	17.9	22.0

財務比率					
	FY23	FY24	FY25	FY26F	FY27F
流動比率(%)	194	167	185	159	153
速動比率(%)	170	147	168	128	123
淨負債股權比率(%)	(54)	(32)	(26)	15	27
稅後淨利率(%)	6.3	5.3	8.1	11.4	15.7
資產周轉率(x)	0.9	0.8	0.9	1.0	1.4
財務槓桿(x)	1.7	1.8	1.8	2.0	2.0
已動用資本/淨值(x)	0.6	0.9	0.8	1.1	1.2
總資產報酬率(%)	5.4	3.7	6.0	9.8	18.4
淨值報酬率(%)	9.3	7.8	13.7	20.6	43.5
資本報酬率(%)	10.4	9.8	16.3	26.9	56.6

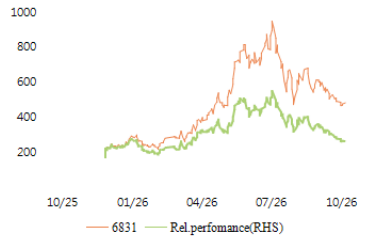
財務比率					
	FY23	FY24	FY25	FY26F	FY27F
股息保障倍數(x)	15.0	(8.4)	(7.5)	(3.5)	(0.4)
所得稅保障倍數(x)	11.3	(10.5)	(11.8)	(3.8)	(0.3)
資本支出保障倍數(x)	(2.1)	(0.7)	(1.7)	(1.3)	(0.2)
應收帳款週轉率(x)	2.6	2.4	2.0	1.7	1.9
存貨周轉率(x)	6.8	7.6	8.6	4.5	5.9
應付帳款周轉率(x)	3.0	3.1	3.3	2.0	2.1
應收帳款周轉天數	142	152	183	218	188
存貨週轉天數	53	48	43	81	62
應付帳款週轉天數	120	119	112	179	170
現金循環週期(天)	75	81	114	120	80

資產負債表					百萬台幣
	FY23	FY24	FY25	FY26F	FY27F
現金及約當現金	609	528	1,106	(92)	(818)
應收款項	637	836	1,859	3,274	5,524
存貨	184	204	331	919	1,322
其他流動資產	56	106	369	563	764
流動資產	1,485	1,673	3,665	4,664	6,792
長期投資	--	--	--	0	0
固定資產	329	809	1,017	1,393	2,066
其他長期資產	103	348	336	336	336
總資產	1,917	2,830	5,018	6,393	9,194
短期借款	0	20	383	383	383
商業本票	--	--	--	0	0
應付款項	414	504	871	2,027	3,603
其他流動負債	350	475	725	523	466
流動負債	764	999	1,980	2,933	4,452
長期借款	--	--	--	0	0
其他長期負債	18	257	220	220	220
總負債	781	1,257	2,200	3,153	4,672
普通股股本	509	600	675	675	675
公積	401	701	1,632	1,632	1,632
保留盈餘	246	276	502	923	2,206
其他	(20)	(4)	10	10	10
股東權益	1,135	1,573	2,818	3,240	4,523
負債加股東權益	1,917	2,830	5,018	6,393	9,194

現金流量表					
	FY23	FY24	FY25	FY26F	FY27F
稅後淨利	103	106	301	624	1,688
折舊與攤提	78	113	136	166	168
投資收入	(21)	1	15	0	0
營運資金增減	(130)	(129)	(783)	(847)	(1,077)
其他	149	46	91	(121)	(121)
來自營運之現金流量	180	137	(240)	(178)	658
資本支出	166	(508)	(325)	(525)	(825)
長投增減	(15)	0	0	(117)	(117)
其他	(178)	(521)	(442)	0	0
來自投資之現金流量	140	(518)	(442)	(642)	(942)
自由現金流量	320	(381)	(682)	(820)	(284)
借款增（減）	--	--	--	0	0
現金股利支出	(23)	(44)	(75)	(202)	(405)
發行公司債	--	--	--	0	0
償還公司債	--	--	--	0	0
現金增資	0	360	994	0	0
其他	(290)	340	1,320	(38)	(37)
來自融資之現金流量	(313)	296	1,245	(240)	(442)
匯率調整	(1)	4	16	0	0
本期產生之現金流量	6	(81)	578	(1,060)	(726)

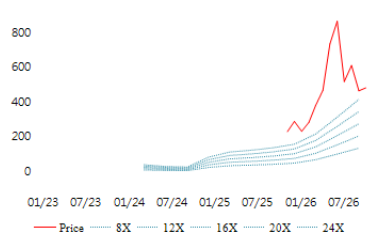
比率分析					
	FY23	FY24	FY25	FY26F	FY27F
本益比(x)	254.7	271.7	99.0	52.9	19.6
本營比(x)	15.2	14.6	8.9	6.0	3.1
本淨比(x)	21.9	18.7	11.7	10.2	7.3
P/CFPS(x)	138.6	214.3	(137.3)	(185.4)	50.2
PE/growth (x)	95.3	(43.5)	0.6	0.6	0.1
P/FCFPS(x)	71.9	(79.1)	(58.4)	(47.0)	(197.6)
EV/EBITDA (x)	149.5	127.3	64.5	34.1	14.5
現金股利殖利率(%)	0.2	0.3	0.6	1.2	3.3

股價及相對大盤績效圖



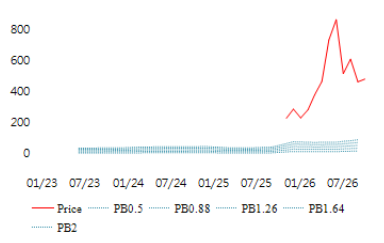
資料來源：CMoney、富邦投顧

未來 12 個月本益比區間圖



資料來源：CMoney、富邦投顧

未來 12 個月本淨比區間圖



資料來源：CMoney、富邦投顧

免責宣言

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2. 分析師的報酬與本研究報告內容表述的個別建議或觀點無關。

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其他揭露事項

本公司關係企業富邦證券過去 12 個月間曾擔任報告標的公司證券之主(協)辦承銷商。

富邦的股票評等標準

評等	定義
買進	預估未來6個月內的絕對報酬超過 15%
中立	預估未來6個月內有絕對報酬介於15%與負15%之間
賣出	預估未來6個月內的絕對報酬高於負15%
未評等	由於富邦目前與該公司有特定交易或沒有足夠的基本資料判斷該公司評等
評估中	目前正在研議個股的投資評等，將於3到6個月內提供投資評等

產業評等	定義
優於大盤	預估該產業在未來6個月內會比大盤指數表現突出
持平	預估該產業在未來6個月內與大盤指數表現相較持平
劣於大盤	預估該產業在未來6個月內會比大盤指數表現較差

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Fubon's Rating System

Rating	Definition
Buy	Expected absolute return to be over 15% within the next 6 months
Neutral	Expected absolute return to be level within the next 6 months
Sell	Expected absolute return to be over negative 15% within the next 6 months
Not Rated (NR)	Pursuant to Fubon acting in deals involving the company or there is not a sufficient fundamental basis for determining an investment rating
Under Review	Fubon is in the process of determining an investment rating and will assign a rating within the 3- to 6-month horizon

Sector Rating	Definition
Overweight	Sector expected to outperform the broader market indexes within the next 6-12 months
Market Weight	Sector expected to perform in line with the broader market averages within the next 6-12 months
Underweight	Sector expected to underperform the broader market indexes within the next 6-12 months.

台灣股市｜電子上游

投 資 速 報

尖點(8021 TT)

買進 目標價 NT$730 (潛在報酬+30.3%)

2026-10-06

AI 板推升高長徑比鑽針需求

■ 結論與建議

尖點 2Q26 受惠新產能逐步開出、1H26 產品報價調整效益反映，以及高階鍍膜鑽針比重持續提升，單季營收 18.26 億元(QoQ +36.3%、YoY +80.3%)，毛利率 41.8%、營益率 28.1%同創歷史新高，稅後 EPS 2.05 元。7、8 月營收分別為 7.30、7.71 億元，連續改寫單月新高。展望後續，AI 伺服器主板朝高層數、厚板化與高階板材演進，帶動鑽針長徑比由 20 倍往 30~50 倍以上發展，並使單針壽命大幅縮短，鑽針呈「量價齊揚」格局。

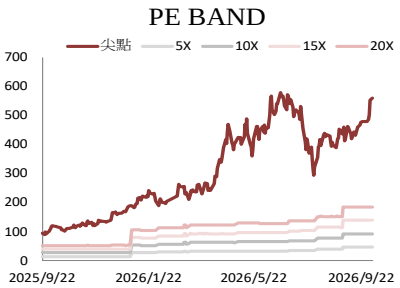
尖點規劃鑽針月產能由 2025 年底 3,200 萬支擴增至 2026 / 2027 / 2028 年底 4,500 / 7,000 / 9,000 萬支，新增產能全數投入高階產品。研究部預估尖點 2026 / 2027 年營收 77.90 / 154.75 億元(YoY+76.6% / +98.7%)，毛利率 42.5% / 51.3%，稅後淨利 13.87 / 38.46 億元，EPS 9.41 / 26.09 元。

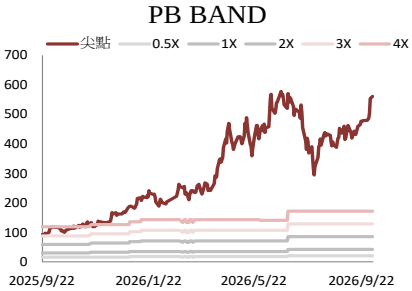
評價方面，尖點為台灣最大 PCB 鑽針廠，具備「鑽針製造＋鑽孔服務」雙軸佈局帶來的製程數據與客戶黏著優勢，且擴產進度領先產業。研究部以 2027 年 EPS 26.09 元給予 28 倍本益比評價，目標價 730 元，潛在漲幅 30.3%，評等為「買進」。

前次報告	
日期	2026/3/9
評等	買進
目標價(元)	260

股價資訊	
收盤價(元)	561.00
總市值(億元)	825.4
股本(百萬)	1,474
一年高/低價	581 / 89.7
月成交均量(張)	5,776
月成交均值(百萬)	2,734.2

股價表現 & PE/PB





研究員/吳建毅
115050@concords.com.tw

■ 重點摘要

公司簡介

尖點科技(8021)成立於 1996 年，總部位於新北市樹林區，為全球主要 PCB 精密微型鑽針供應商之一，並於 2008 年掛牌上市。公司核心業務以 PCB 鑽針及鑽孔加工服務為主，鑽針產品涵蓋微型鑽針、銑刀及槽刀等，主要應用於 PCB 機械鑽孔製程；2010 年起進一步拓展專業鑽孔加工，服務範圍涵蓋機械鑽孔與雷射鑽孔，形成由耗材供應延伸至加工服務的完整解決方案。

公司目前於台灣及中國設有鑽針生產基地，並在台灣、中國、日本、韓國及泰國等地建立生產與服務據點，以貼近全球 PCB 客戶需求。隨 AI 伺服器、高速 Switch 及光通訊快速發展，PCB 朝高層數、高密度與高縱橫比方向升級，鑽孔數量增加且孔徑持續縮小，使鑽針耗用量與技術門檻同步提高。尖點近年亦積極提升高階鍍膜鑽針及先進鑽孔能力，透過產品組合優化與產能擴充，掌握 AI 帶動高階 PCB 需求成長的產業趨勢。

2Q26 營運概況

尖點 2Q26 營收 18.26 億元，QoQ+36.3%，YoY+80.3%；毛利率 41.8%，

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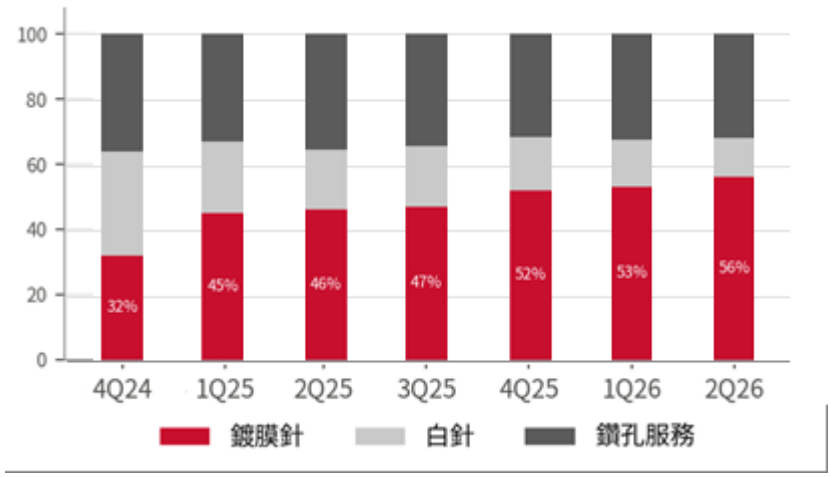
QoQ+5.2ppts，YoY+12.4ppts；營益率 28.1%，QoQ+8.0ppts，YoY+14.2ppts;稅後淨利 3.02 億元，QoQ+79.1%，YoY+285.5%;EPS 2.05 元，營收、毛利率與獲利同創單季新高。1H26 EPS 合計約 3.20 元，已超越 2025 全年(2.64 元)。1Q26 營業費用率一度上揚至 16%左右，主因營收成長下獎金與底薪提列增加，屬旺季提前反映之正常現象；2Q26 起費用率已降至 14.3%(QoQ-2.0ppts)，產能利用率方面，2Q26 鑽針製造稼動率約 88%，鑽孔服務稼動率約 75%。高階鍍膜鑽針銷量比重由 2024 年 32%、2025 年 48%，進一步提升至 1H26 的尊督 56%，產品結構轉向高單價、高技術密度產品，為推升 ASP 與獲利能力的主要引擎。

產品組合變化

2Q26 鑽針(含白針、鍍膜針)營收 12.42 億元，占比 68%，QoQ+38%、YoY+90%；鑽孔加工服務營收 5.84 億元，占比 32%，QoQ+34%、YoY+62%。其中高階鍍膜產品占鑽針銷售比重由 4Q25 的 52%，提升至 1Q26 的 53%、2Q26 約 55～56%，展望 2H26 可望上看 55～60%，為 ASP 與毛利率結構性上行的核心引擎。

原料方面，國際鎢粉及鎢鋼價格自 2H25 起持續上漲，尖點已於 1H26 完成一波成品價格調整，2H26 暫無調價計畫，將以製程成本優化及提高高階鍍膜產品比重稀釋材料成本壓力。尖點鎢鋼棒主要向日系供應商採購，其料源已轉向中國以外地區，因此中國境內 APT 報價自高點回落與尖點採購成本連動有限；在鎢料供給偏緊下，穩定料源亦成為客戶選擇供應商的重要考量，預期成長動能將由「漲價」轉為「規格升級」接棒，往高長徑比產品發展即為變相漲價

圖一、產品組合占比



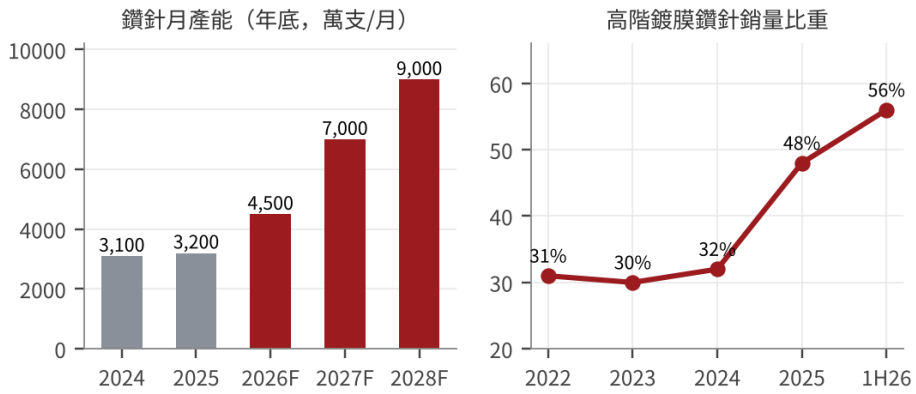
資料來源:公司法說、康和投顧整理

產能擴廠進度

目前鑽針月產能規劃為：2024 年底 3,100 萬支、2025 年底 3,200 萬支(實際基期)，2026 年底規劃達 4,500 萬支、2027 年底規劃達 7,000 萬支、2028 年底規劃達 9,000 萬支，新增產能將完全聚焦高階(鍍膜)產品。2027 年新

增產能上半年主要利用既有廠房空間擴充，下半年則涉及中壢、上海全新廠房投入，故產能雖全年各季陸續開出，仍以 3、4 季釋出效益最明顯。中壢新廠最快 10 月投產(先期開出 200 萬支，10 月底至 11 月即可出貨、貢獻 4Q26 營收)，後續至 2027 年可望再開出 600 萬支；上海新廠與中壢新廠均按原訂計畫順利推進，現階段無重大設備交期或產能瓶頸，主要設備均已提早議定交期。2Q26 鑽針製造稼動率達 88%(已計入新擴充產能)，鑽孔代工服務稼動率約 75%；台灣缺工問題持續存在，公司透過外籍勞工、僑生與產學合作等多元管道擴充人力，以支應產能爬坡所需。

圖二、月產能規劃與高階鍍膜鑽針銷量比重



資料來源:公司法說、康和投顧整理

2H26 營運逐季走高

尖點 7、8 月合併營收分別為 7.30 / 7.71 億元，YoY+102.6% / +97.8%，合計 15.01 億元，已達研究部 3Q26 預估營收 21.65 億元 69%。獲利表現更為亮眼，公司自結 8 月稅前淨利 2.24 億元、年增 251.6%，歸母公司淨利 1.42 億元、年增 300.4%，單月 EPS 0.97 元，加計 7 月 EPS 0.92 元，7～8 月 EPS 已達 1.89 元，前 8 月累計 EPS 約 5.14 元，顯示高階產品組合改善帶動獲利增速明顯高於營收。

主要動能來自 AI Server PCB 朝高層數、厚板及高速材料升級，高階鍍膜鑽針需求快速增加。隨 AI Server 主板由過去約 22～24 層持續往 40 層以上發展，部分 Midplane 已達 44 層，板厚亦由約 5mm 朝 7mm 以上增加，加上 M8、M9 高階 CCL 材料硬度提高，使鑽針磨耗增加、使用壽命下降，進一步推升單位 PCB 耗針量。

PCB 上游持續升級，鑽針供需持續緊俏

隨著 AI 伺服器與交換器傳輸速率由 800G 持續往 1.6T、3.2T 升級，高速 CCL 規格亦由 M7 / M8 朝 M9 演進，以滿足更低損耗與更高訊號穩定性的需求。隨著材料填料比例逐代提高，加上 M9 高階方案(M9Q)進一步導入硬度更高的石英布(Q 布)，PCB 鑽孔難度與鑽針磨耗明顯增加。同樣板厚下，主針壽命已由 M6～M7 的約 600～800 次，下降至 M8 的 400～500 次，到了 M9 更僅剩約 100～200 次，相較傳統 PCB 約 8,000～10,000 次的鑽孔壽命，耗用速度呈現大幅提升。

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從 CCL 材料來看，M9 材料已開始進入放量前期。台光電 M9 於 1H26 已小量試產出貨，預估營收占比可由 2H26 的中至高個位數，於 2027 年進一步提升至雙位數。隨 VR200 Midplane 於 2027 年放量，加上 AWS Trainium 4、Google TPU 等次世代 ASIC 平台亦評估導入 M9，高階 CCL 滲透率可望持續提升。

其中，Q 布為 M8 升級至 M9 後鑽針壽命呈現斷崖式下滑的主要原因之一。由於石英布硬度極高，對刀具磨耗明顯，單次鑽孔製程極限約僅 26 層，以 78 層高階背板為例，甚至必須拆分為 26 層×3 段製作。此外，隨 PCB 板厚增加，一套鑽針配置亦由過去約 3 根增加至 4～5 根，新增鑽針多為高階鍍膜針，進一步提升單板鑽針使用量與產品 ASP。

另一方面，HDI 與 mSAP 製程滲透率提升，也同步推升鑽針需求。隨線路密度提高，機械鑽孔孔徑由過去約 0.25～0.275mm，逐步縮小至 0.175mm，甚至進一步朝 0.1mm 發展，單板孔數亦持續增加。雖然部分微孔可改採雷射鑽孔，但 HDI 核心層仍須使用機械鑽孔，因此雷射製程對機械鑽針的替代程度仍相對有限。

圖三、CSP 業者 PCB/CCL 供應鏈

	平台 / 代號 (Code name)	時程 (Timeline)	PCB 規格 (層數 / 技術)	PCB 主要供應商	CCL 規格	CCL 主要供應商	應用示意
NVIDIA	GB200	1H25	22L HDI 24L (switch)	VGT, Unimicron VGT, TTM, Unimicron (switch)	M8/M4 M8/M2 (switch)	Doosan Shengyi, EMC (switch)	GB200 / Rubin AI Server
	GB300	1H26	22L HDI 24L (switch)	VGT, Unimicron VGT, TTM, Unimicron (switch)	M8/M4 M8/M2 (switch) M8/M4	Doosan Shengyi, EMC (switch)	
	Vera Rubin	2H26	28L HDI 32L (switch) 44L (mid-plane)	VGT, Unimicron, ZDT VGT, WUS, TTM, Unimicron (switch)	M8/M4 M8 (switch) M8 (mid-plane)	Doosan Shengyi, EMC, Nanya (switch)	
	Rubin Ultra	2H27	30L HDI 32L (switch) 44L (mid-plane)	VGT, Unimicron, ZDT VGT, WUS, TTM, Unimicron, SCC (switch)	M8/M6 M8/M8+ PTFE (switch) M9 (mid-plane)	Doosan Shengyi, EMC/Taiflex (switch)	
Google	TPU v7 Ironwood	1H26	26L	ISU, LCS, WUS, TTM	M7	Panasonic, EMC	Google TPU AI Server
	TPU v8i Sunfish	4Q26	32L M+M	ISU, LCS, WUS, TTM, Dynamic	M8/M6	Panasonic, EMC, Nanya	
	TPU v8t Zebrafish	1Q27	34L M+M	ISU, LCS, WUS, TTM, GCE, Dynamic	M8/M6	Panasonic, EMC, Doosan	
	TPU v9 Humulish	2028	HDI	ISU, LCS, WUS, TTM, Unimicron, Dynamic	M9/M8	Panasonic, EMC, Doosan	
amazon	Trainium 2e	1H26	26L	GCE, Shengyi, FHT	M8	EMC, TUC	Amazon Trainium AI Server
	Trainium 3	2H26	26L/40L 22L (switch)	GCE, Shengyi, FHT, WUS, ZDT	M8	EMC, TUC	
	Trainium 4	2H27	34L 30L (switch)	GCE, Shengyi, ZDT	M9?	EMC, TUC, Nanya	
AMD	MI450	4Q26	48L	SCC (major), Unimicron, Dynamic	M8	EMC, Doosan	AMD Instinct AI Server
	MI500	4Q27	48-52L	SCC (major), Unimicron, Dynamic	M9	EMC, Doosan	
OpenAI	Jalapeno	4Q26	40L	GCE and others	M8?	EMC, Doosan	OpenAI ASIC AI Server

資料來源:Aletheia、康和投顧整理

鑽孔服務提升客戶黏著度，27 年獲利翻倍

鑽孔服務除為鑽針提供穩定出海口，更使尖點直接取得高難度板材的第一線加工數據，回饋刀形設計與鍍膜配方迭代，並配合客戶開發 PTFE 等新一代配方，從服務取得數據、製造反覆優化的循環不易複製，有助提升客戶黏著度。

研究部預估尖點 2026 年營收 77.90 億元（YoY+76.6%），其中鑽針營收 54.76 億元（YoY +86.7%）、鑽孔服務 23.14 億元（YoY +56.7%）；毛利率 42.5%、營益率 28.1%，稅後淨利 13.87 億元（YoY +256.1%），EPS 9.41 元。2027 年在新增產能放量及高長徑比產品滲透下，營收 154.75 億元（YoY+98.7%），鑽針營收占比由 2025 年 67%提升至 80%，帶動毛利

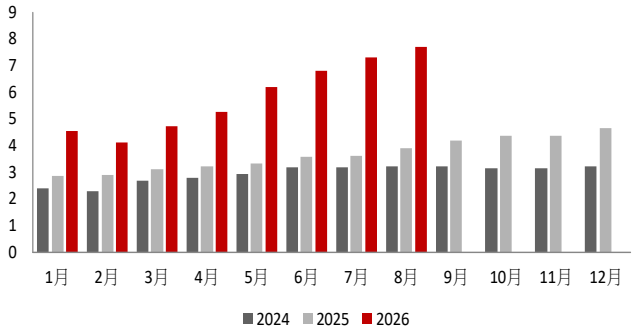
率升至 51.3%、營益率 38.2%，稅後淨利 38.46 億元（YoY+177.4%），EPS 26.09 元，以 2027 年 EPS 26.09 元給予 28 倍本益比評價，目標價 730 元，潛在漲幅 30.3%，評等為「買進」。

	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2025	2026F	2027F
營收(百萬元)	1,340	1,826	2,165	2,460	2,862	4,410	7,790	15,475
QOQ(%)	(0.1)	36.3	18.6	13.6	16.3	-	-	-
YOY(%)	51.0	80.3	85.2	83.4	113.6	24.5	76.6	98.7
毛利率(%)	36.6	41.8	43.8	45.0	47.4	30.9	42.5	51.3
營業利益率(%)	20.1	28.1	29.9	30.9	33.4	14.9	28.1	38.2
稅後淨利(百萬元)	169	302	422	494	623	389	1,387	3,846
QOQ(%)	12.7	79.1	39.6	17.2	26.0	-	-	-
YOY(%)	225.0	285.5	285.6	230.0	269.1	89.0	256.1	177.4
EPS(元)	1.14	2.05	2.86	3.35	4.22	2.64	9.41	26.09

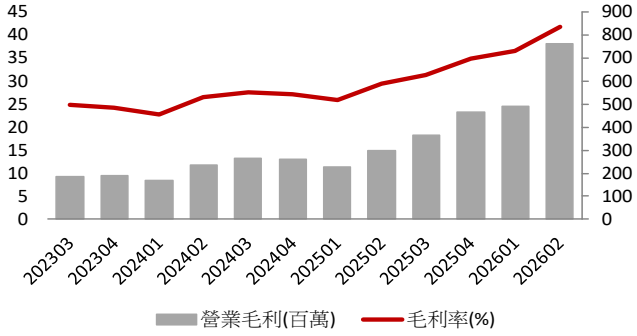
EPS以股本147.4百萬股計算

財務現況

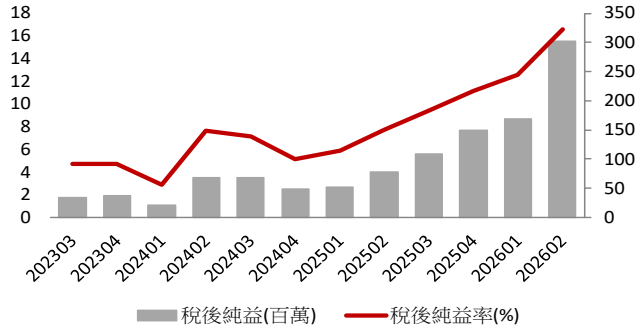
近三年營收狀況(億)



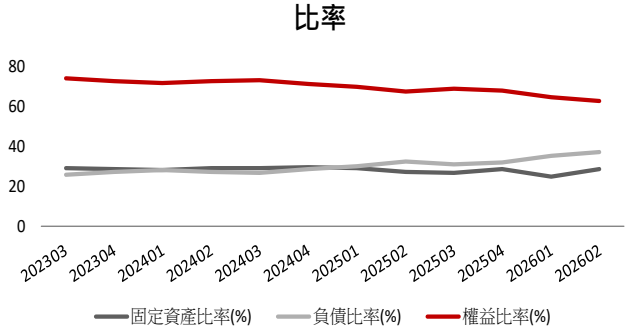
近12季_營業毛利 & 毛利率



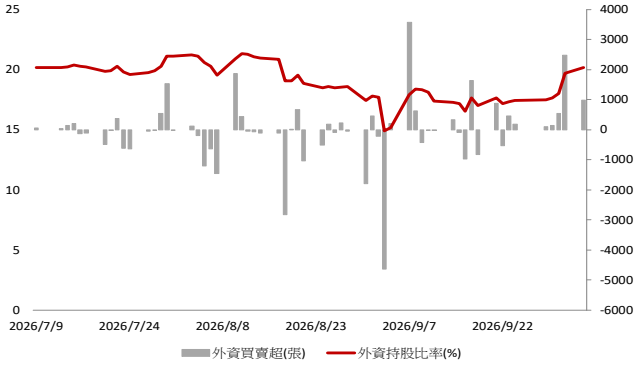
近12季_稅後純益 & 稅後純益率



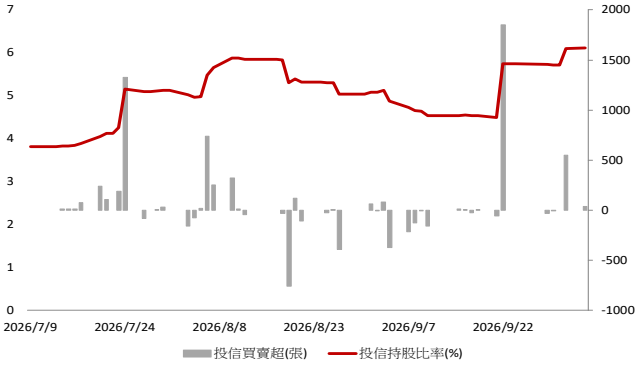
近12季_固定資產比率 & 負債比率 & 權益比率



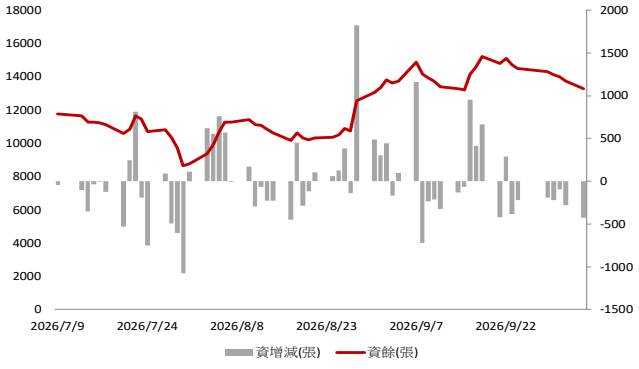
外資_持股比例 & 買賣超



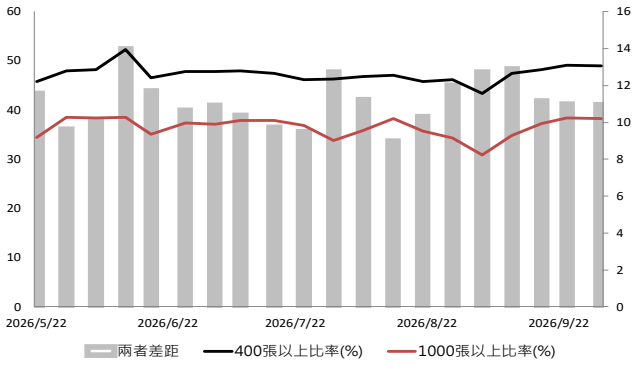
投信_持股比例 & 買賣超



融資餘額 & 資增減



近20周持股400張 & 1000張以上比例變化



評等說明

1) 強力買進/Strong Buy：我們對該評價個股未來展望正向，並預估於未來 12 個月投資期間，潛在上漲空間大於 50%
2) 買進/Buy：我們對該評價個股未來展望正向，並預估於未來 12 個月投資期間，潛在上漲空間大於 25%
3) 逢低買進/Trading Buy：我們對該評價個股未來展望正向，並預估於未來 12 個月投資期間，潛在上漲空間小於 25%
4) 未評等/Not Rated：至目前我們沒有足夠基本資料判斷該公司評等；或此期間康和與該公司有特定交易，基於規範或康和政策，限制予該公司評等

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4971 TT 英特磊

英特磊的 InP、GaAs 等 III-V 族磊晶晶圓受惠 AI DC 高速光傳輸需求，中長期具成長機會；惟 InP 營收受限於基板供應，GaAs VCSEL 出貨仍取決於客戶產品驗證與量產進度，營運能見度低。考量 III-V 族材料長期成長潛力，給予目標價 630 元(PER 70x 2027E)，惟目前上檔空間有限，投資評等「中立」，建議回檔再布局。

投資評等	中立
目標價位	630 元

上市/上櫃指數	432.48
股價	584

公司基本資料

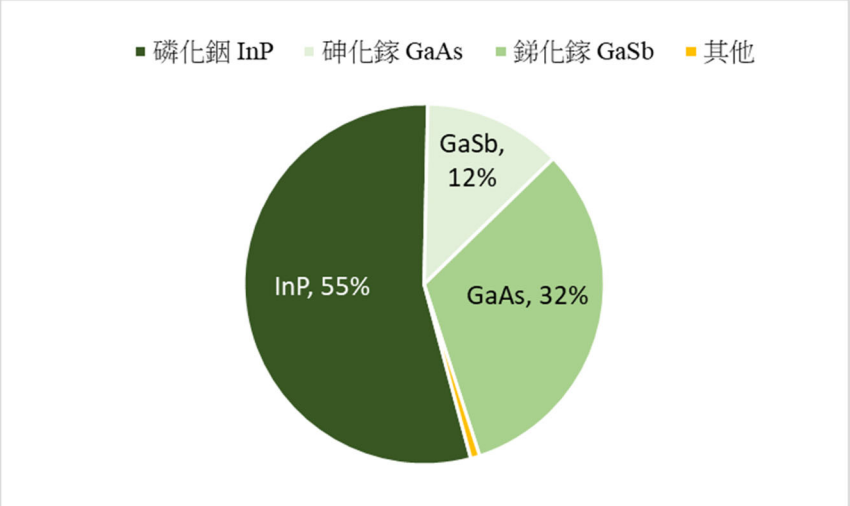
流通在外股數(百萬股)	47.2
市值(百萬元)	27,560
1M 平均成交量(張數)	1,720.74
外資持股率(%)	13.37
投信持股率(%)	0
董監持股率(%)	3.81
自營持股率(%)	1.47
融資使用率(%)	37.69
融券使用率(%)	0.97
券資比(%)	2.56
TCRI 信用評等	6
CSR 企業總分	0

重點摘要

一、英特磊以 MBE 布局高精度磊晶，InP 為現階段營收主力但受限基板短缺。

英特磊主要從事III-V族磊晶晶圓製造，並以MBE設備進行磊晶成長。相較量產導向的MOCVD，MBE在材料組成、層厚及磊晶結構控制上具較高精細度與製程調整彈性，部分設備亦可依需求生產GaAs或InP產品，有利客製化生產。目前英特磊產品包含砷化鎵(GaAs)、磷化銦(InP)、銻化鎵(GaSb)及氮化鎵(GaN)等材料晶圓，就2026年上半年營收而言，InP佔比最高，其次為GaSb、GaAs。

圖一、1H26英特磊各材料產品營收占比



資料來源：公司法說會資料；福邦投顧整理

InP材料特性有利高速、高頻元件設計，受惠AI資料中心高速傳輸需求成長，目前英特磊InP晶圓應用於100G/200G PD，相關營收占比約20%~30%，惟營收成長受限於InP基板供應不足。短期內，英特磊除仰賴客戶提供基板，或向日廠住友爭取配額外，亦透過既有基板研磨提高材料利用率；中長期則持續與其他日

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系InP基板供應商洽談，合作夥伴德商Freiberger亦已開始供應InP基板，惟供應穩定性不足，目前占比約5%，短期尚難放量。

GaSb主要用於紅外線偵測，以國防應用為主，GaAs既有產品以高頻通訊元件為主，應用涵蓋基地台與衛星通訊。就產能部份，目前英特磊配有16台MBE設備，其中8台用於生產InP晶圓。

二、3Q26 InP 基板缺貨未解，然因 GaAs 拉貨及 GaSb 國防訂單回補，研究員預估營收回溫至 3.17 億元，QoQ +23.7%、YoY +25.9%，EPS 1.32 元，QoQ +205.5%、YoY +75.1%。

3Q26營收回升主要來自既有GaAs高頻傳輸產品需求增加及GaSb訂單遞延出貨，帶動本季營收改善，相較之下，InP基板供應僅較2Q26小幅改善，因此公司預期3Q26營收仍略低於1Q26。

隨營收規模回升及產品組合改善，預估3Q26毛利率將回升至43.4%。較2Q26提升9.4個百分點，預估EPS亦回升至1.32元，QoQ +205.5%、YoY +75.1%。

三、4Q26 整合 MBE 設備銷售與服務的十年期專案開始貢獻，帶動營收及獲利續增，研究員預估營收 3.62 億元，QoQ +14.2%、YoY +13.7%，EPS 1.67 元，QoQ +27.5%、YoY +7.4%。

4Q26英特磊新增動能主要來自為期10年的機台專案開始認列營收，該專案涵蓋機台設計、於英特磊廠房內建置MBE設備，期間提供設備租賃、維護及磊晶服務，後續再進行設備出售，相關營收依履約進度分階段認列，惟公司仍需在客戶機台專案與VCSEL產能建置間進行資源配置。另外，預期InP基板短缺仍未明顯改善，100G/200G PD營收貢獻持平。

研究員預估4Q26營收3.62億元，QoQ +14.2%、YoY +13.7%。毛利率44.3%，較3Q26提升0.9個百分點；EPS估1.67元，QoQ +27.5%、YoY +7.4%。

四、新增 MBE 產能奠定英特磊 2027 成長基礎，惟 InP 與 VCSEL 實際貢獻仍視基板供應量及客戶導入進度而定。

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研究員預估 2027 年營收約 16.37 億元，YoY +29.2%，EPS 9 元，YoY +67.9%。

隨光模組及單通道傳輸速率提升，元件對磊晶結構精度要求更高，將凸顯MBE優勢。在美國晶片法支持下，英特磊預計2026年底至2027年初將新增1~2台設備，並以GaAs及InP可共用機型為主，再依需求調配產能，有望成為2027年營運成長基礎。

InP方面，100G/200G PD需求仍高於供給，公司認為2027年中左右基板供應狀況可望較為明朗，屆時InP出貨較具成長空間。GaAs方面，鎖定1.6T AOC應用的200G VCSEL已進入量產準備階段，惟實際出貨及營收貢獻仍取決於客戶驗證與導入進度。此外，長期機台專案營收陸續認列，亦為另一營收成長來源。研究預估2027年營收約16.37億元，YoY +29.2%，預估毛利率48.1%，年增5.2個百分點，EPS估約9元，YoY +67.9%。

新品布局方面，英特磊正開發QD Laser所需的GaAs磊晶，以及HBT驅動器所需的InP磊晶。QD Laser波長1310nm，可搭配單模光纖，相較200G VCSEL應用於短距離傳輸AOC，QD Laser可支援較長距離傳輸，也可做為搭配矽光PIC的CW雷射。QD Laser溫度穩定性較佳，可降低雷射光源的溫控需求，目前仍處於驗證階段，尚未進入量產。另一方面，InP HBT量產進度取決於客戶HBT元件製程進展，因此兩項新品短期內難有明顯營收貢獻。

五、評等為「中立」的理由

1. 英特磊同樣受惠 AI 資料中心高速光傳輸需求成長，InP、GaAs 等 III-V 族磊晶晶圓在中長期具成長機會。然而，公司現階段 InP 營收成長仍受限於基板供應，GaAs VCSEL 出貨則取決於客戶終端元件驗證及量產進度，加上長期機台專案對整體營收貢獻有限，使公司營運能見度較低、季度波動亦相對較大。此外，QD Laser 與 InP HBT 等新品仍處驗證或開發階段，預期 2027 年尚難形成明顯營收貢獻。

2. 考量 AI 高速傳輸需求帶動 III-V 族材料長期成長，市場通常願意給予較高評價；惟相較聯亞，英特磊營運規模較小，且受 InP 基板供應、客戶終端產品驗證進

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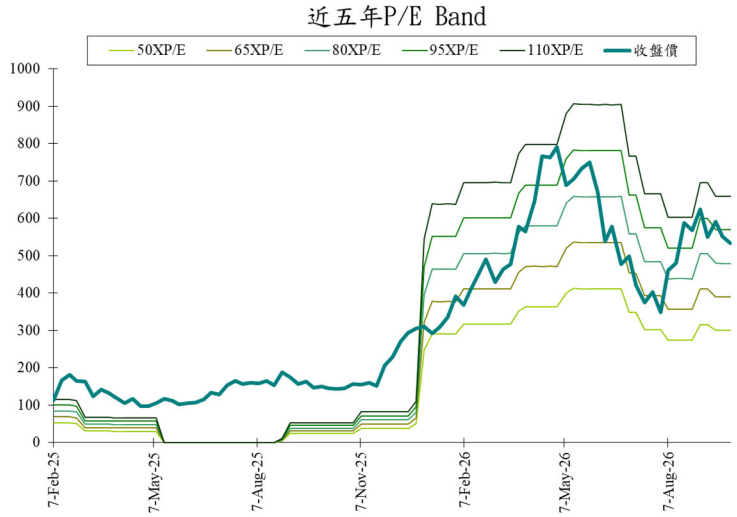
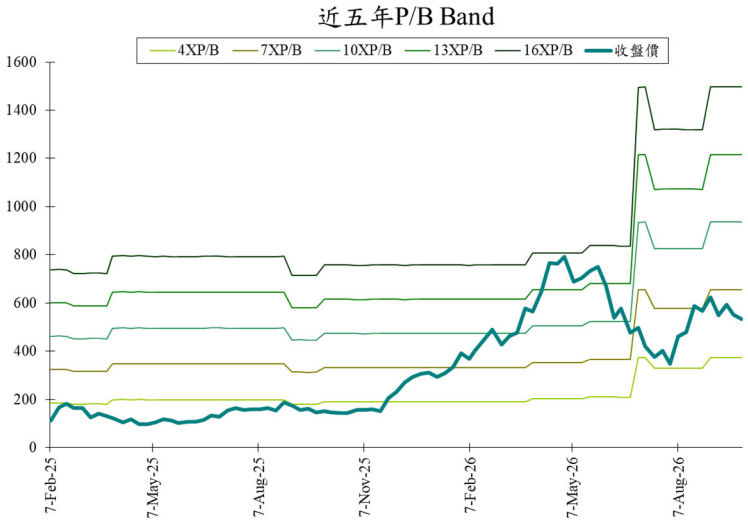
度影響，獲利波動較大，因此估值較聯亞約 80~100 倍的本益比區間有所折價。研究員給予英特磊 2027E PER 60~70x，若以 2027E EPS 約 9 元計算，合理股價區間約 540~630 元，目前上檔空間有限，因此給予投資評等「中立」。

圖二、英特磊近五年 PER 區間



資料來源：CMoney；福邦投顧整理

圖三、英特磊近五年PBR區間



資料來源：CMoney；福邦投顧整理

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分季損益表

單位：新台幣(百萬)

	2025	1Q26	2Q26	3Q26(f)	4Q26(f)	2026(f)	1Q27(f)	2Q27(f)	3Q27(f)	4Q27(f)	2027(f)
營業收入淨額	**1,079**	**332**	**256**	**317**	**362**	**1,267**	**385**	**403**	**419**	**429**	**1,637**
營業成本	695	173	169	179	202	723	202	211	217	221	850
營業毛利	**385**	**159**	**87**	**138**	**160**	**544**	**184**	**192**	**202**	**209**	**787**
營業費用											
推銷費用	5	1	2	2	2	6	2	2	2	2	7
管理費用	143	46	41	43	45	175	46	48	50	51	195
研究發展費用	51	14	22	20	21	76	21	22	22	23	87
預期信用減損損益	11	1	1	1	1	4	1	1	1	1	4
營業費用合計	**211**	**62**	**66**	**65**	**68**	**262**	**70**	**72**	**74**	**76**	**293**
非常項目淨額	-	0	0	0	0	0	0	0	0	0	0
營業淨利(損)	**174**	**97**	**22**	**72**	**92**	**282**	**114**	**120**	**128**	**132**	**494**
營業外收支											
利息收入	-	0	0	0	0	0	0	0	0	0	0
利息支出	2	0	1	0	0	0	0	0	0	0	0
投資損益淨額	(107)	0	0	0	0	0	0	0	0	0	0
其他收支淨額	4	1	1	1	1	4	1	1	1	1	4
營業外收支淨額合計	**(105)**	**1**	**0**	**1**	**1**	**3**	**1**	**1**	**1**	**1**	**4**
稅前淨利	**69**	**97**	**22**	**73**	**93**	**285**	**115**	**121**	**129**	**133**	**498**
所得稅費用(利益)	4	18	3	11	14	43	17	18	19	20	75
非控制權益	0	0	0	0	0	0	0	0	0	0	0
稅後淨利	**65**	**79**	**19**	**62**	**79**	**239**	**97**	**103**	**109**	**113**	**423**
發行股數（百萬股）	40	40	47	47	47	47	47	47	47	47	47
每股盈餘(元)	**1.61**	**1.97**	**0.40**	**1.32**	**1.67**	**5.08**	**2.06**	**2.18**	**2.32**	**2.40**	**8.96**
季成長分析 (QoQ,%)											
營收成長率		4.3	-22.9	23.7	14.2		6.4	4.6	4.1	2.3	
營業毛利成長率		23.1	-45.2	58.0	16.6		14.6	4.6	5.2	3.2	
營業利益成長率		30.9	-77.6	234.3	27.4		23.5	5.5	6.6	3.6	
稅前淨利成長率		30.5	-77.6	236.3	27.0		23.3	5.4	6.6	3.6	
稅後淨利成長率		27.8	-76.0	229.2	27.0		23.3	5.4	6.6	3.6	
年成長分析 (YoY,%)											
營收成長率	50.5	22.8	7.3	25.9	13.7	17.4	16.0	57.3	32.3	18.6	29.2
營業毛利成長率	239.1	69.4	5.9	72.6	24.3	41.4	15.7	120.7	46.9	30.1	44.6
營業利益成長率	-286.9	127.5	-9.3	113.8	24.7	62.5	17.7	455.0	77.0	44.0	74.8
稅前淨利成長率	-150.6	-251.0	-12.3	113.4	24.8	311.9	17.9	455.3	75.9	43.5	74.5
稅後淨利成長率	-149.6	-231.8	-43.5	107.6	28.3	265.7	23.8	443.6	75.9	43.5	77.1
獲利分析 (%)											
營業毛利率	35.7	47.8	34.0	43.4	44.3	42.9	47.7	47.7	48.2	48.6	48.1
營業利益率	16.1	29.1	8.4	22.8	25.4	22.3	29.5	29.7	30.5	30.8	30.2
稅前淨利率	6.4	29.3	8.5	23.1	25.7	22.5	29.7	30.0	30.7	31.1	30.4
稅後淨利率	6.0	23.7	7.4	19.6	21.8	18.8	25.3	25.5	26.1	26.4	25.8
實質所得稅率	5.7	19.0	13.2	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0

資料來源:Cmoney，福邦投顧整理

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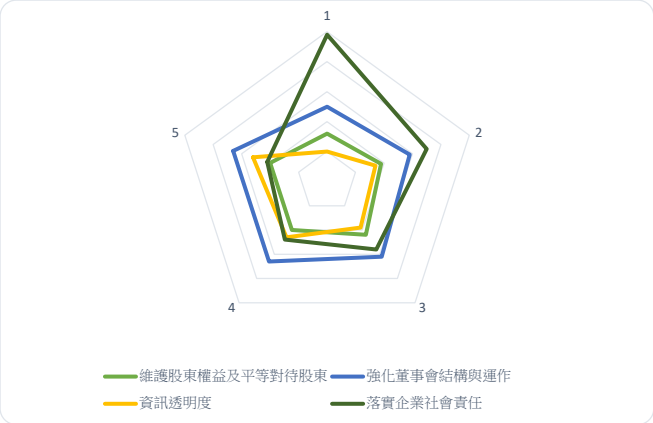
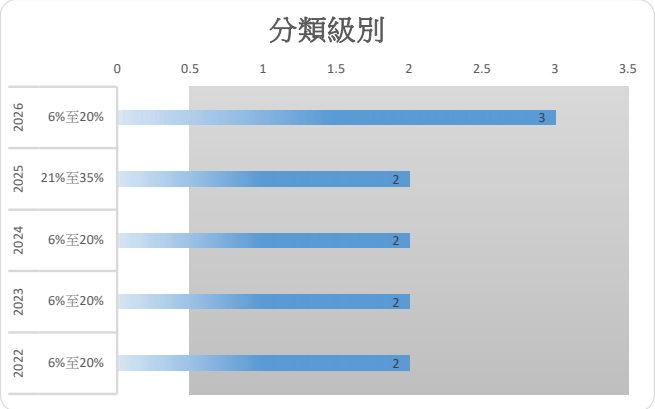
公司治理評鑑

分類級別 1	分類級別 2	分類級別 3	分類級別 4	分類級別 5	分類級別 6	分類級別 7
前 5%	6%至 20%	21%至 35%	36%至 50%	51%至 65%	66%至 80%	81%至 100%





資料來源：CMoney

說明：

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福邦投顧投資評等

強力買進：未來六個月預期報酬率+30%以上

買　　進：未來六個月預期報酬率+15~30%

中　　立：未來六個月預期報酬率+/-15%區間

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陳祐竹
(886-2)2563-1666

儒億(6604)

JUOKU TECHNOLOGY CO., LTD.

買進

前一日(10/05)收盤價：28.2 元
承銷價：24
目標價：45

儒億為台灣車燈 ODM/OEM 大廠，母公司堤維西持股 60.88%(釋股後降至 52.5%)，2025/10/28 登錄興櫃，2026/10/6 轉上櫃，承銷價 24 元，競拍最低得標價為 23.5 元，最高得標價為 26 元，加權平均得標價爲 24.09 元，前一日收盤價 28.2 元。雖然汽車市場規模相對成熟，總銷量變化不大，但結構正在進行深度轉型，儒億與日本斯坦雷(Stanley Electric)長期技術合作，具備車燈自主設計製造能力，隨車輛智慧化及電動化趨勢使產品單價持續提高，獲利結構可望逐步轉佳，目前台灣車市落底回穩，且台灣及印尼的自製率法規紅利使儒億迎來結構性利多，長期營運展望正向。綜合以上，研究部預估 2026 年 EPS 4.11 元，參考國內車燈同業如大億(1521)、麗清(3346)、華勝-KY (2248)等本益比區間為 8~20x，給予 11x 本益比評價，目標價 45 元，依承銷價 24 元計算，給予買進投資建議。

- 公司簡介

  儒億成立於 1993 年，為台灣車燈 ODM/OEM 大廠，2025/10/28 登錄興櫃，2026/10/6 轉上櫃，母公司堤維西原持股 60.88%、釋股後降至 52.5%，上櫃後資本額 5.07 億元，承銷價 24 元。2025 年營收比重：汽機車燈具 54%、模具 19%、車內開關 13%、電路板模組 6%、其他 8%。2025 年銷售區域別：台灣 58%、日本 16%、馬來西亞 14%、印尼 5%、美國 3%。主要客戶包含中華汽車、本田汽車、國瑞汽車、PROTON、PERODUA 等。儒億車燈廠、LED 模組廠、模具廠位於台南市，電裝廠位於新北市，另於 2012 年與印尼最大汽車零部件廠 PT Astra Otoparts 合資創立 PT Astra Juoku Indonesia(AJI)，儒億持股 50%並以權益法認列，在印尼當地製造與開發 OEM 車燈

與相關燈具，其主要客戶為 Toyota 及 Honda 集團，AJI 對儒億獲利貢獻度達 45~50%。

圖一、主要產品

資料來源：公司簡報、第一金投顧彙整

圖二、主要客戶

資料來源：公司簡報、第一金投顧彙整

- **<u>產業概況及競爭優勢</u>**

研調機構 Global Market Insights 預估全球車燈產值 2022~2032 年將以 CAGR+8.5%穩健擴張，TrendForce 亦指出配備自適應遠近光(AFS/ADB)的車款比重 2030 年將突破 20%，確立智慧車燈發展的必然趨勢。儒億的核心競爭利基包括(1)車廠 Tier 供應商進入門檻高：基於嚴格品質認證、長期的

產品開發驗證週期等原因，車廠尋找合格供應商至下單需 3~5 年時間，一旦進入不會輕易更換，因此長期訂單穩定；(2)日系技術合作與自主研發雙軌並進 : 儒億長期與日本斯坦雷(Stanley Electric)技術合作，兼具日規安全檢驗標竿與自主光學設計量能；(3)東南亞布局完整：攜手印尼零組件龍頭 PT Astra Otoparts 布局合資廠 AJI(儒億持股 50%)，掌握印尼半數市佔的 Astra 體系，具有成本競爭力的供應量能，可靈活因應全球供應鏈重組趨勢。

- **營運概況**

儒億 2025 年營收 20.71 億元，YoY-4.3%，毛利率 13.38%，稅後淨利 1.1 億元，YoY+8.9%，稅後 EPS 2.73 元，去年台美貿易談判之不確定性使消費者轉為觀望，整體車市疲弱，壓抑 OEM 訂單動能。雖 26Q1 台灣車市延續關稅政策影響，消費者持續觀望，但馬來西亞第二國產車 Perodua 首款電動車 QV-E 之車燈模具開始出貨，以及業外轉投資 AJI 受惠新車量產及模具開發收入、26H1 獲利成長 30%，26H1 儒億營收 10.83 億元，YoY+10.8%，毛利率 15.83%，稅後淨利 9755 萬元，稅後 EPS 2.23 元。隨著各品牌陸續推出新產品，台灣車市買氣自 26Q2 緩步回溫，8M26 台灣車市新車掛牌數 27 萬輛，YoY+ 2.4%，加上 Perodua 車燈持續放量，7~8 月營收分別 YoY+33%/23%，北美新能源車專案將於 26Q4 開始出貨。

- **營運展望**

台美關稅尚未完全定案，使台灣車市買氣並未顯著彈升，但目前已落底回穩，新款上市及財富效果刺激換購需求，預估 2026~2027 年台灣新車銷售達中低個位數增長；雖然汽車市場規模相對成熟，總銷量變化不大，但結構正在進行深度轉型，例如傳統燈具轉向 LED 及智慧燈具，車燈結合感測與通訊功

能，儒億目前具備 AFS(主動轉向式頭燈)及 ADB(自適應式遠光頭燈)的自主設計製造能力，整體智能產品比重已達 30%，隨車輛智慧化及電動化趨勢使產品單價持續提高，獲利結構可望逐步轉佳。此外，台灣經濟部自 2024/8/1 實施「國產車組裝在地化供應鏈比重」新制，2024 年前在台上市之國產車款，其零件國產化自製率須於 2026/8 後全面提升至 35%，新導入車款上市後三年自製率須自 15%逐年遞增至 25%及 35%，具備自主設計與高自製率的供應商迎來結構性利多。

轉投資 AJI 方面，印尼為東協第二大汽車市場，雖短期消費力受通膨壓抑，但中產階級擴張及都市化的長期趨勢使其具備龐大需求潛力，且印尼政府推行電動車與新能源車本地含量規定(TKDN)，本地生產之新能源車須於 2026 年達成 40%本地自製率，2027~2029 年提高至 60%，2030 年達 80%，AJI 在印尼設廠逾十年，具備先行者優勢，Toyota 集團透過 Astra 體系長期掌控印尼汽車約 50%市佔率，其成長性高於整體印尼車市，為儒億提供穩健之業外收益。綜合以上，研究部預估 2026 年營收 24.24 億元，YoY+17.1%，毛利率 15.98%，稅後淨利 2.08 億元，YoY+89.2%，以上櫃後股本計算 EPS 4.11 元。

- **結論與建議**

儒億為台灣車燈 ODM/OEM 大廠，母公司堤維西持股 60.88%(釋股後降至 52.5%)，2025/10/28 登錄興櫃，2026/10/6 轉上櫃，承銷價 24 元，競拍最低得標價為 23.5 元，最高得標價為 26 元，加權平均得標價爲 24.09 元，前一日收盤價 28.2 元。雖然汽車市場規模相對成熟，總銷量變化不大，但結構正在進行深度轉型，儒億與日本斯坦雷(Stanley Electric)長期技術合作，具備車燈自主設計製造能力，隨車輛智慧化及電動化趨

勢使產品單價持續提高，獲利結構可望逐步轉佳，目前台灣車市落底回穩，且台灣及印尼的自製率法規紅利使儒億迎來結構性利多，長期營運展望正向。綜合以上，研究部預估 2026 年 EPS 4.11 元，參考國內車燈同業如大億(1521)、麗清(3346)、華勝-KY(2248)等本益比區間為 8~20x，給予 11x 本益比評價，目標價 45 元，依承銷價 24 元計算，給予買進投資建議。

綜合損益表

6604綜合損益表

單位:Mn,%,元	2024	2025H1	2025H2	2025	2026H1	2026H2E	2026E
營業收入淨額	2,163.19	977.23	1,093.44	2,070.67	1,083.11	1,340.83	2,423.93
毛利率	13.15%	12.83%	13.88%	13.38%	15.83%	16.10%	15.98%
營業毛利	284.38	125.37	151.74	277.11	171.48	215.87	387.35
營業利益率	2.36%	2.59%	4.32%	3.50%	5.51%	6.60%	6.12%
營業利益	51.00	25.33	47.21	72.55	59.72	88.55	148.28
營業外收入及支出	82.66	19.40	51.71	71.11	55.84	65.34	121.18
稅前純益	133.66	44.73	98.92	143.65	115.56	153.89	269.45
母公司業主－稅後純益	101.13	39.13	71.00	110.13	97.55	110.80	208.35
普通股股本	387.31	387.31	437.31	437.31	437.31	507.31	507.31
未稀釋EPS	2.61	1.01	1.62	2.73	2.23	2.18	4.11
稀釋後EPS	2.61	1.01	1.62	2.73	2.23	2.18	4.11
QoQ(%)							
營收		-	-		-	-	
營業利益		-	-		-	-	
歸屬於母公司-稅後純益		-	-		-	-	
YoY(%)							
營收		-	-	-4.28%	10.83%	22.63%	17.06%
營業利益		-	-	42.25%	135.76%	87.56%	104.39%
歸屬於母公司-稅後純益		-	-	8.90%	149.32%	56.06%	89.19%

本資料內容僅供參考，投資人應審慎考量自身之投資需求與風險，本公司恕不負任何法律責任，亦不作任何保證

第一金證券投資顧問股份有限公司(First Capital Management Inc.)台北市長安東路一段 22 號 11F TEL(02) 2563-6262

個股報告

2026 年 10 月 6 日

研究員：薛雅月 rita.hsueh@capital.com.tw

前日收盤價	286.00 元
目標價	
3 個月	340.00 元
12 個月	340.00 元

近期報告日期、評等及前日股價

09/14/2026	Buy	234.50
08/27/2026	Buy	227.50
08/05/2026	Buy	170.50
05/28/2026	Buy	95.50

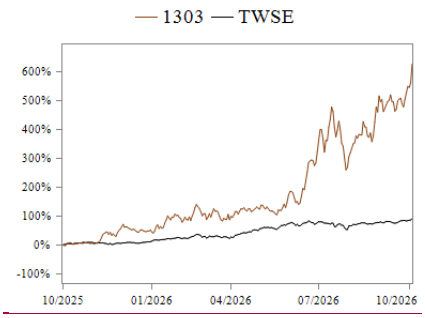
公司基本資訊

目前股本(百萬元)	79,308
市值(億元)	22,682
目前每股淨值(元)	60.24
外資持股比(%)	30.40
投信持股比(%)	6.95
董監持股比(%)	18.21
融資餘額(張)	36,950
現金股息配發率(%)	140.35

產品組合

- 電路板:15.46%
- 聚酯纖維:13.95%
- 銅箔基板:10.91%
- 其他:59.68%

股價相對大盤走勢



南亞(1303 TT) Buy

產品組合改善有利營運成長動能，投資建議為 Buy。

投資建議：電子材料高階產品比重持續提升，產品組合改善有利中長期營運成長動能，投資建議為 Buy，目標價 340 元。

電子材料產品組合改善，營收及營益率持續提升：電子材料展望：(1)銅箔:目前高階產品以 HVLP3 及 HVLP4 為主，高階銅箔因需求強勁加上加工產出速度緩慢，故產品售價及加工費都處於調漲階段，HVLP5 及以上的銅箔則在開發中；(2)銅箔基板:消費型產品漲價趨勢不變，M7 及 M8 有望於 4Q26 開始供應大筆訂單，高階產品比重將快速拉升；M9 及 M10 規格優先採用 Low-DK2 搭配自有的碳氫樹酯，此配方回饋數據可達理想狀態；隨產品結構逐步改善，未來銅箔基板有望為電子材料中獲利率最佳者。此外，公司目前透過整改既有及退役的產線來擴充產能，2026 年底產能將自 618 萬張/月增至 650 萬張/月，2027 年則有望達 700 萬張/月以上；(3)玻纖布:由日東紡提供 Low-Dk2 原紗再製成玻纖布的部分於 2H26 開始穩定放量，進而支援高階銅箔基板的開發。整體而言，公司電子材料營運策略在中階產品拓展銷售加上高階產品開發之下，未來營收及營益率皆有望獲得提升。

預估 2026 年稅後 EPS 13.92 元：南亞 1Q26 稅後 EPS 1.8 元。2Q26 因電子材料產品產能利用率及售價皆有提升，轉投資南亞科及台塑化獲利貢獻下，單季稅後 EPS 3.37 元。展望後市，電子材料產品組合結構改善，營收維持成長態勢；化工產品 EG 因中東戰事受損的產能短期內恢復機率不高，目前供應吃緊的狀態仍是持續，麥寮廠已啟動一套 EG 產線，4Q26 將全能運轉；聚酯產品方面，紡織及瓶用粒市況較為不佳，將積極擴大電子用途的比重；塑膠加工部門表現則是維持穩健。預估南亞 3Q26 及 4Q26 稅後 EPS 分別為 4.35 元及 4.39 元；預估 2026 年稅後獲利 1103.63 億元，稅後 EPS 13.92 元。2027 年電子材料高階產品比重提高、帶動營收及營益率持續增長，以及業外南亞科獲利貢獻增加之下，預估南亞稅後獲利 1550.8 億元，稅後 EPS 19.55 元。

(百萬元)	2026F	2027F	2028F	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F
營業收入淨額	342,693	449,040	570,874	64,177	64,467	68,596	83,647	92,752	97,698	99,746	107,665	115,337
營業毛利淨額	67,591	125,736	195,764	5,321	5,970	8,187	15,808	20,020	23,576	25,799	29,070	32,699
營業利益	48,226	103,211	170,427	1,036	1,329	3,742	11,357	14,826	18,301	20,612	23,687	26,932
稅後純益	110,363	155,080	171,996	3,256	4,925	14,255	26,753	34,538	34,816	35,033	37,703	40,672
稅後 EPS(元)	13.92	19.55	21.69	0.41	0.62	1.80	3.37	4.35	4.39	4.42	4.75	5.13
毛利率(%)	19.72%	28.00%	34.29%	8.29%	9.26%	11.93%	18.90%	21.58%	24.13%	25.86%	27.00%	28.35%
營業利益率(%)	14.07%	22.98%	29.85%	1.61%	2.06%	5.45%	13.58%	15.98%	18.73%	20.66%	22.00%	23.35%
稅後純益率(%)	32.20%	34.54%	30.13%	5.07%	7.64%	20.78%	31.98%	37.24%	35.64%	35.12%	35.02%	35.26%
營業收入 YoY/QoQ(%)	31.85%	31.03%	27.13%	-2.34%	0.45%	6.41%	21.94%	10.89%	5.33%	2.10%	7.94%	7.13%
稅後純益 YoY/QoQ(%)	2,342.41%	40.52%	10.91%	N.A	51.27%	189.41%	87.68%	29.10%	0.81%	0.62%	7.62%	7.87%

註：稅後純益係指本期淨利歸屬於母公司業主；稅後 EPS 以股本 793.08 億元計算。

公司 ESG 政策:

南亞主張「綠色生產」，積極管理其營運過程對環境的衝擊 — 包括能源消耗、資源利用、排放與廢棄物處理等。

公司進行溫室氣體盤查 (依據國際標準與地方法規) — 包含碳排查，並導入第三方認證與驗證機制。

為降低排放與污染，南亞在製程中導入技術改善，例如對含全氟化物 (PFCs) 的製程氣體進行削減，PFCs 的排放削減率達 93%。

公司也推動節能與資源回收／減量 —累積節能量 (節省能源消耗) 相當可觀。

能源管理也有具體政策：設置水回收系統、雨水回收設施、儲水池與滯洪池，並制定緊急應變計畫，以因應氣候變遷與水資源短缺問題。

同時，公司在產品與供應鏈層面也強調綠色採購與循環經濟 — 例如對回收塑膠廢棄物 (包括消費後廢棄的塑膠品) 進行再製加工，用於塑膠棧板等產品，減少對原生資源 (如木材) 的依賴。。

南亞投入教育與人才培育，尤其在科技／半導體產業鏈相關領域—例如協助培育新進人才，並透過與學校合作、產學計畫推動多元人才養成。

在其子公司 (或關係企業) 中，也有針對供應鏈的永續性管理與供應商評估，強化供應商在環境與社會責任上的表現，提醒企業不只是自己，而是整個價值鏈共同承擔責任。

在產品安全與客戶服務面，南亞亦重視品質管控與客戶滿意度，並設有制度 (包括內部稽核、品保程序、滿意度調查／改善機制) 以維護消費者權益與信任。

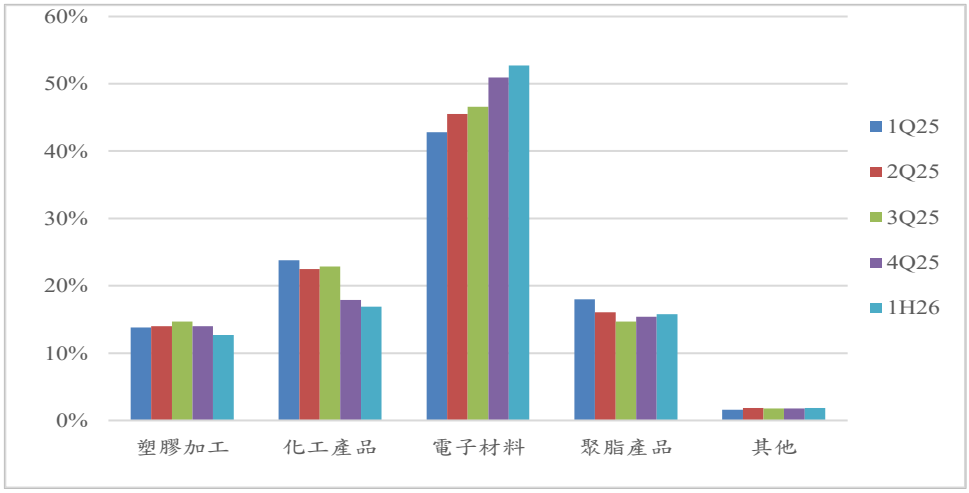
2026 年 10 月 6 日

公司簡介:

南亞業務包含:(1)塑膠加工類:軟質膠布、硬質膠布、PU 合成皮等;(2)化工產品類:乙二醇(EG)、丙二酚(BPA)、丁二醇(1,4BG)、可塑劑等;(3)聚酯產品類:聚酯棉、聚酯粒、聚酯加工絲及聚酯薄膜;(4)電子材料類:玻纖絲、玻纖布、環氧樹脂、銅箔、銅箔基板及印刷電路板。

南亞 1H26 營收比重中,電子材料占比 52.7%,化工產品占比 16.9%,聚酯產品占比 15.8%,塑膠產品占比 12.7%,其他項目則為 1.9%。受惠於 AI 伺服器、高階網通設備需求強勁,電子材料部門營收比重呈現逐季增加,化工產品部門則因石化景氣低迷,營收比重呈現下滑。

圖一、南亞各部門營收比重變化

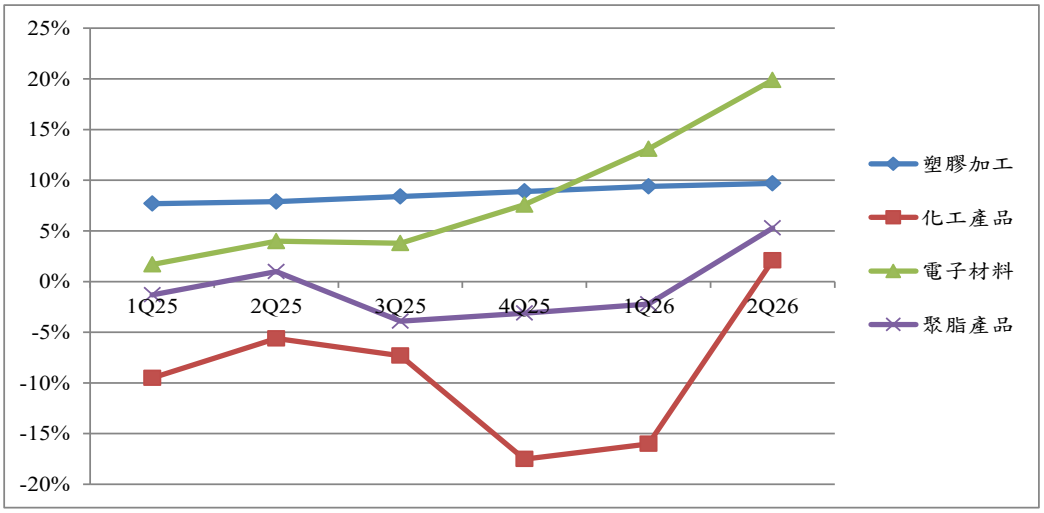


資料來源:公司簡報,群益投顧彙整

電子材料部門營益率持續攀高:

隨 AI 與高階網通投資持續擴張,中高階產品需求急速增長,加上銅價上漲進而推升產品售價,電子材料部門營益率顯著改善,2Q26 電子材料部門營益率已由 1Q26 的 13.1%攀升至 19.9%。

圖二、南亞各部門營益率變化



資料來源:公司簡報,群益投顧彙整

電子材料產品組合改善，2H26 營收及營益率皆有望獲得提升:

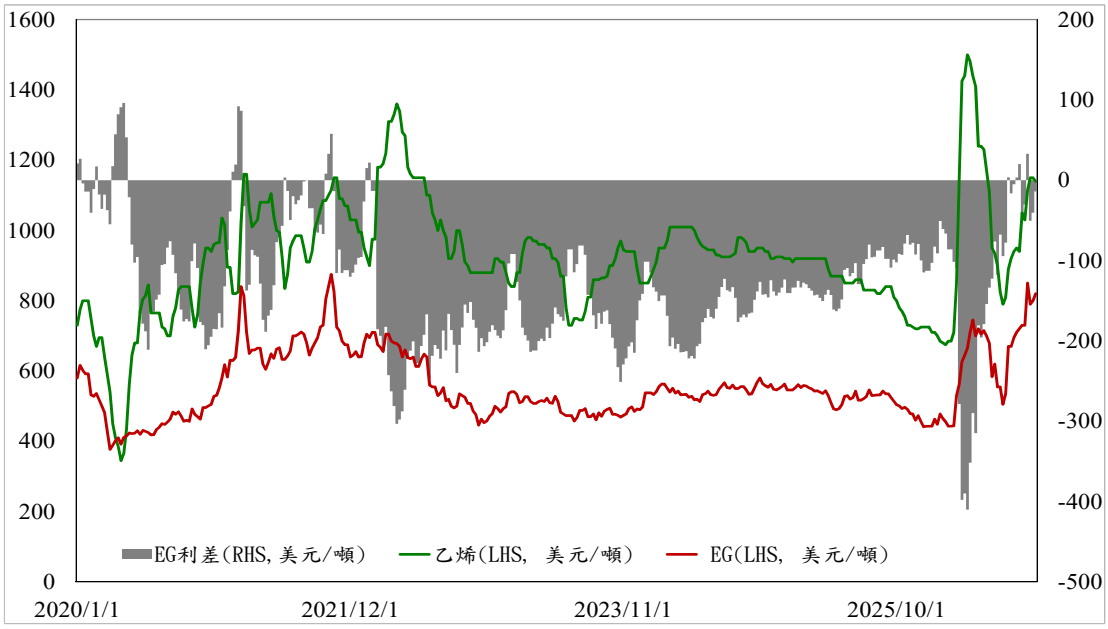
為因應 AI 相關應用需求持續成長，公司營運重心將持續將重心放在電子材料高階產品的開發及認證，08/2026 營收中電子材料比重已達 59.8%。

電子材料展望: (1)銅箔:目前高階產品以 HVLP3 及 HVLP4 為主，高階銅箔因需求強勁加上加工產出速度緩慢，故產品售價及加工費都處於調漲階段，HVLP5 及以上的銅箔則在開發中；(2)銅箔基板:消費型產品漲價趨勢不變，M7 及 M8 有望於 4Q26 開始供應大筆訂單，高階產品比重將快速拉升；M9 及 M10 規格優先採用 Low-DK2 搭配自有的碳氫樹酯，此配方回饋數據可達理想狀態；隨產品結構逐步改善，未來銅箔基板有望為電子材料中獲利率最佳者。此外，公司目前透過整改既有及退役的產線來擴充產能，2026 年底產能將自 618 萬張/月增至 650 萬張/月，2027 年則有望達 700 萬張/月以上；(3)玻纖布:由日東紡提供 Low-Dk2 原紗再製成玻纖布的部分於 2H26 開始穩定放量，進而支援高階銅箔基板的開發。整體而言，公司電子材料營運策略在中階產品拓展銷售加上高階產品開發之下，未來營收及營益率皆有望獲得提升。

EG 利差改善:

化工產品方面，中東衝突導致部分 EG 產能受損，供應減少之下，中國 EG 進口量減少、港口庫口下降，進而支撐報價走揚，利差同步獲得改善。

圖三、EG 報價及現貨利差走勢



資料來源：Bloomberg，群益投顧彙整

預估 2026 年稅後 EPS 13.92 元:

南亞 1Q26 營收 QoQ+6.41%，其中電子材料營收比重已達五成以上，主要受惠於需求持續強勁、價量俱揚所致；聚酯產品因美國市場增加境內採購取代進口，南卡廠營收有所增加；塑膠加工產品因亞洲及美國需求略為增溫，業績小幅成長；化工產品營收則持平。營業利益 QoQ 正增長，主因電子材料產品有顯著貢獻，塑膠加工及聚酯產品營運則略優於 4Q25。業外方面，轉

投資台塑化及南亞科技獲利貢獻皆較上季顯著增加，加上匯兌利益 4.9 億元，單季稅後獲利 142.55 元，稅後 EPS 1.8 元。

2Q26 營收 QoQ+21.9%，受惠於 AI 及周邊產業加速發展，電子材料產品價量俱揚，營收增長最為顯著；另受惠於戰爭局勢推升產品報價，以及美國市場增加當地採購，聚酯、化工及塑膠等產品營收皆有所增加。營業利益 QoQ 大幅增長，主因電子材料產品積極發展 AI 伺服器、高效運算、高階網通用材料，產能利用率及售價皆有提升；化工產品享有庫存利益，美國 EG 產線全能運轉下亦由虧轉盈。業外方面，轉投資南亞科技獲利貢獻較上季顯著增加，加上台塑化獲利貢獻及股利收入等，單季稅後 EPS 3.37 元。

展望後市，電子材料產品組合結構改善，營收維持成長態勢；化工產品 EG 因中東戰事受損的產能短期內恢復機率不高，目前供應吃緊的狀態仍是持續，麥寮廠已啟動一套 EG 產線，4Q26 將全能運轉；聚酯產品方面，紡織及瓶用粒市況較為不佳，將積極擴大電子用途的比重；塑膠加工部門表現則是維持穩健。

整體而言，未來本業營運的增長仍將來自於電子材料營收及營益率的提升，而業外在南亞科獲利貢獻持續增加之下，預估南亞 3Q26 及 4Q26 稅後 EPS 分別為 4.35 元及 4.39 元。

綜上所述，2026 年受惠於 AI 伺服器市規模持續擴大，南亞電子材料產品營收將維持增長態勢，產品組合的改善有望提升整體營運表現，加上業外南亞科獲利貢獻大幅改善，預估南亞 2026 年稅後獲利 1103.63 億元，稅後 EPS 13.92 元。

2027 年電子材料高階產品比重提高、帶動營收及營益率持續增長，以及業外南亞科獲利貢獻增加之下，預估南亞稅後獲利 1550.8 億元，稅後 EPS 19.55 元。

投資建議:

電子材料高階產品比重持續提高，產品組合改善有利中長期營運成長動能，投資建議為 Buy，目標價 340 元。

Forward PB Band

資料來源：CMoney，群益預估彙整

資產負債表

(百萬元)	2024	2025	2026F	2027F	2028F
資產總計	617,473	610,598	782,417	1,002,599	1,185,337
流動資產	185,502	177,836	315,248	524,078	758,784
現金及約當現金	66,445	46,150	97,693	152,884	226,532
應收帳款與票據	41,852	46,344	65,947	77,500	106,431
存貨	51,696	51,303	66,266	71,913	88,408
採權益法之投資	170,623	184,104	212,096	221,332	168,474
不動產、廠房設備	218,273	206,478	206,478	206,478	219,478
負債總計	**254,870**	**238,800**	**302,406**	**405,814**	**466,951**
流動負債	140,671	134,700	233,517	351,730	448,985
應付帳款及票據	15,053	13,226	19,934	18,685	25,712
非流動負債	114,199	104,099	62,181	48,625	5,481
權益總計	362,603	371,799	480,010	596,785	718,386
普通股股本	**79,308**	**79,308**	**79,308**	**79,308**	**79,308**
保留盈餘	229,118	230,145	338,356	455,131	576,732
母公司業主權益	346,583	355,666	459,683	563,214	663,832
負債及權益總計	**617,473**	**610,598**	**782,417**	**1,002,599**	**1,185,337**

現金流量表

(百萬元)	2024	2025	2026F	2027F	2028F
營業活動現金	18,786	8,649	172,917	197,281	201,122
稅前純益	4,523	6,459	137,258	210,406	241,223
折舊及攤銷	22,702	22,518	22,713	22,300	23,704
營運資金變動	-5,952	-5,926	-27,859	-18,448	-38,401
其他營運現金	-2,487	-14,401	40,806	-16,977	-25,403
投資活動現金	-17,900	-11,332	-18,613	-15,466	-1,240
資本支出淨額	-12,447	-11,168	-12,000	-13,000	-13,000
長期投資變動	-2,562	955	-200	-350	-350
其他投資現金	-2,891	-1,120	-6,413	-2,116	12,110
籌資活動現金	-20,112	-16,751	-102,761	-126,623	-126,235
長借/公司債變動	-9,542	4,183	-41,918	-13,556	-43,143
現金增資	0	0	0	0	0
發放現金股利	-5,615	-5,562	-6,345	-51,550	-71,377
其他籌資現金	-4,955	-15,372	-54,498	-61,516	-11,714
淨現金流量	-13,856	-20,295	51,543	55,192	73,647
期初現金	80,301	66,445	46,150	97,693	152,884
期末現金	66,445	46,150	97,693	152,884	226,532

損益表

(百萬元)	2024	2025	2026F	2027F	2028F
營業收入淨額	259,608	259,912	342,693	449,040	570,874
營業成本	241,033	238,325	275,083	323,304	375,109
營業毛利淨額	18,574	21,599	67,591	125,736	195,764
營業費用	18,127	17,896	19,366	22,525	25,337
營業利益	448	3,704	48,226	103,211	170,427
EBITDA	29,288	31,433	78,226	129,211	196,427
業外收入及支出	4,076	2,755	89,032	107,195	70,795
稅前純益	4,523	6,459	137,258	210,406	241,223
所得稅	1,216	1,393	22,702	42,081	48,245
稅後純益	3,340	4,519	110,363	155,080	171,996
稅後 EPS(元)	0.42	0.57	13.92	19.55	21.69
完全稀釋 EPS**	0.42	0.57	13.92	19.55	21.69

註 1：稅後純益係指本期淨利歸屬於母公司業主。

註 2：稅後 EPS 以股本 793.08【最新股本】億元計算

註 3：完全稀釋 EPS 以股本 793.08 億元計算

比率分析

(百萬元)	2024	2025	2026F	2027F	2028F
成長力分析(%)					
營業收入淨額	-0.06%	0.12%	31.85%	31.03%	27.13%
營業毛利淨額	-3.39%	16.29%	212.93%	86.02%	55.69%
營業利益	-77.44%	727.55%	1202.14%	114.02%	65.13%
稅後純益	-47.07%	35.28%	2342.41%	40.52%	10.91%
獲利能力分析(%)					
毛利率	7.15%	8.31%	19.72%	28.00%	34.29%
EBITDA(%)	11.28%	12.09%	22.83%	28.77%	34.41%
營益率	0.17%	1.42%	14.07%	22.98%	29.85%
稅後純益率	1.29%	1.74%	32.20%	34.54%	30.13%
總資產報酬率	0.54%	0.74%	14.11%	15.47%	14.51%
股東權益報酬率	0.92%	1.22%	22.99%	25.99%	23.94%
償債能力檢視					
負債比率(%)	41.28%	39.11%	38.65%	40.48%	39.39%
負債/淨值比(%)	70.29%	64.23%	63.00%	68.00%	65.00%
流動比率(%)	131.87%	132.02%	135.00%	149.00%	169.00%
其他比率分析					
存貨天數	77.42	78.87	78.00	78.00	78.00
應收帳款天數	56.38	61.93	59.80	58.30	58.80

資料來源：CMoney、群益

季度損益表

(百萬元)	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	1Q28F
營業收入淨額	65,716	64,177	64,467	68,596	83,647	92,752	97,698	99,746	107,665	115,337	126,293	128,637
營業成本	60,038	58,856	58,502	60,408	67,822	72,732	74,121	73,946	78,595	82,638	88,125	88,380
營業毛利淨額	5,683	5,321	5,970	8,187	15,808	20,020	23,576	25,799	29,070	32,699	38,168	40,256
營業費用	4,380	4,285	4,641	4,445	4,451	5,194	5,276	5,187	5,383	5,767	6,188	5,917
營業利益	1,302	1,036	1,329	3,742	11,357	14,826	18,301	20,612	23,687	26,932	31,980	34,339
業外收入及支出	-5,496	2,959	4,515	12,338	19,407	30,006	27,281	26,234	27,107	28,316	25,538	17,800
稅前純益	-4,194	3,995	5,844	16,080	30,764	44,832	45,582	46,846	50,795	55,248	57,518	52,139
所得稅	11	521	541	1,405	3,214	8,966	9,116	9,369	10,159	11,050	11,504	10,428
稅後純益	-4,116	3,256	4,925	14,255	26,753	34,538	34,816	35,033	37,703	40,672	41,672	37,554
最新股本	79,308	79,308	79,308	79,308	79,308	79,308	79,308	79,308	79,308	79,308	79,308	79,308
稅後EPS(元)	-0.52	0.41	0.62	1.80	3.37	4.35	4.39	4.42	4.75	5.13	5.25	4.74
獲利能力(%)												
毛利率(%)	8.65%	8.29%	9.26%	11.93%	18.90%	21.58%	24.13%	25.86%	27.00%	28.35%	30.22%	31.29%
營業利益率(%)	1.98%	1.61%	2.06%	5.45%	13.58%	15.98%	18.73%	20.66%	22.00%	23.35%	25.32%	26.69%
稅後純益率(%)	-6.26%	5.07%	7.64%	20.78%	31.98%	37.24%	35.64%	35.12%	35.02%	35.26%	33.00%	29.19%
QoQ(%)												
營業收入淨額	0.25%	-2.34%	0.45%	6.41%	21.94%	10.89%	5.33%	2.10%	7.94%	7.13%	9.50%	1.86%
營業利益	3534.12%	-20.41%	28.26%	181.51%	203.52%	30.54%	23.44%	12.63%	14.92%	13.70%	18.74%	7.38%
稅前純益	N.A	N.A	46.27%	175.16%	91.32%	45.73%	1.67%	2.77%	8.43%	8.77%	4.11%	-9.35%
稅後純益	N.A	N.A	51.27%	189.41%	87.68%	29.10%	0.81%	0.62%	7.62%	7.87%	2.46%	-9.88%
YoY(%)												
營業收入淨額	-1.04%	-3.38%	-5.40%	4.64%	27.29%	44.52%	51.55%	45.41%	28.71%	24.35%	29.27%	28.96%
營業利益	103.45%	4.25%	N.A	10342.71%	772.18%	1330.57%	1276.82%	450.86%	108.56%	81.66%	74.74%	66.60%
稅前純益	N.A	714.70%	682.17%	1875.42%	N.A	1022.10%	680.00%	191.33%	65.11%	23.23%	26.19%	11.30%
稅後純益	N.A	4851.25%	1551.81%	3045.97%	N.A	960.71%	606.88%	145.77%	40.93%	17.76%	19.69%	7.20%

註1：稅後EPS以股本793.08億元計算

註2：自2013年開始，稅後純益係指本期淨利歸屬於母公司業主。

2026 年 10 月 6 日

【投資評等說明】

評等	定義
強力買進(Strong Buy)	首次評等潛在上漲空間≥35%
買進(Buy)	15%≤首次評等潛在上漲空間<35%
區間操作(Trading Buy)	5%≤首次評等潛在上漲空間<15%
中立(Neutral)	無法由基本面給予投資評等 預期近期股價將處於盤整 建議降低持股

【投資評等調整機制】

投資評等係「首次」給予特定個股投資評等時，其前一日收盤價相較 3 個月目標價之潛在上漲空間計算而得。個股投資評等分為四個等級，定義如上。爾後的投資評等係依循「首次評等」，直到停止推薦。

「強力買進」、「買進」及「區間操作」均有 upside 目標價。差別僅在於，首次評等時潛在報酬率不同。「中立」則無目標價。

一旦我們給予特定個股「強力買進」、「買進」或「區間操作」之投資評等，就是責任的開始，爾後將透過各式研究報告作定期性、持續性追蹤基本面及股價變化，直到停止推薦。

停止推薦情境：

1. 達目標價。
2. 雖未達原訂目標價，但檢視基本面、訊息面、籌碼面等多方訊息，研判股價上漲空間已然有限，將適時出具降評報告。
3. 推薦後股價不漲反跌。

【免責聲明】

本研究報告僅提供予特定人之客戶作為參考資料「非經同意不得轉載」。我們並不確保此資訊的完整性與正確性，投資人應了解，報告中有關未來預測之陳述可能不會實現，因而不應被依賴。而且此報告並非根據特定投資目的或依預定對象之財務狀況所撰寫出來的，因此，此研究報告的目的，既非對投資人於買賣證券、選擇權、期貨或其他證券相關之衍生性商品提供詢價服務，亦非作為進行交易的要約。投資人應注意到相關證券之價值及收益，可能會有無預警地上升或下降，產生投資回報金額可能比原始投資來得少的情形。

個股報告

2026 年 10 月 6 日

研究員：陳執中 tc.chen@capital.com.tw

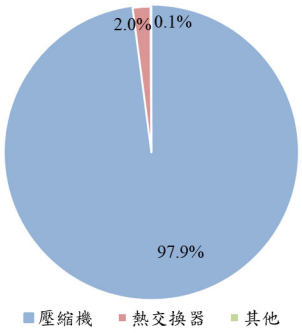
前日收盤價	84.00 元
目標價	
3 個月	元
12 個月	元

近期報告日期、評等及前日股價

公司基本資訊

目前股本(百萬元)	317
市值(億元)	23.65
目前每股淨值(元)	51.9
外資持股比(%)	
投信持股比(%)	
董監持股比(%)	
融資餘額(張)	
現金股息配發率(%)	

產品組合



湧盛(2249 TT)　競拍報告

汽車空調壓縮機製造大廠。

結論：湧盛(2249 TT) 2026 年營運動能下滑，預估 2027 年有望回升。公司將於 10/22/2026 掛牌上櫃，10/06/2026~10/08/2026 為競價拍賣期間。最低投標價格為 55 元，10/05/2026 興櫃收盤價 84 元。考量市場資金集中於 AI 相關族群，汽車零組件廠商目前的市場評價相對較低，建議競拍價格為 75 元~94 元(以 2027 年 EPS 及 PE 8~10 倍評價)。

公司概況：湧盛為台灣最大的汽車空調壓縮機製造商。汽車空調壓縮機佔營收比重約 97%~98%，為主要核心產品；熱交換器與其他零組件佔營收約 2%，包含水箱、冷凝器、中冷器及離合器等。市場定位與全球布局方面，湧盛市場定位聚焦於中價位 AM 件，兼具接近原廠件(OE)的穩定品質與副廠件的價格競爭力。公司具備外部結構逆向工程與模組化開發能力，能避開原廠專利限制。公司外銷比重高達 97%以上，主要市場為歐美地區(合計佔 80%以上)。

2026 年營運偏弱，預估 2027 年營運動能回升：2026 年方面，1H26 營運下滑主要原因包含：(1)歐洲客戶下單模式由長單改變為滾動式下單，影響短期營收認列節奏。(2)受到關稅影響，部分美國客戶提前建立高庫存，且 2025 年歐洲特定機種拉貨量大導致營收基期較高。(3)受到地緣政治、戰爭等因素影響，國際油價上漲壓抑終端車市購車意願及零組件需求。而受惠於歐洲客戶下單模式趨於穩定，美國客戶庫存調整周期告一段落，預估 2H26 營運動能回升。預估 2026 年營收 23.71 億元，YoY-10.04%，獲利能力方面，預估毛利率表現平穩，營業利益 YoY-17.47%。因業外、稅率影響相對有限，預估稅後純益 2.68 億元，YoY-7.57%，EPS 8.45 元。2027 年方面，除了既有業務恢復增長外，美國客戶委託開發逾 40 項新機種，預計於 2026 年底至 2027 年陸續通過認證並進入量產階段。而公司積極投入新建廠房與自動化設備，皆有助於提升其營運體質。預估 2027 年營收 26.60 億元，YoY+12.17%，稅後純益 2.98 億元，YoY+11.47%，EPS 9.42 元。

(百萬元)	2026F	2027F	2028F	1Q26	2Q26	3Q26	4Q26	1Q27	2Q27	3Q27
營業收入淨額	2,371	2,660	2,893	707	457	409	798	777	519	470
營業毛利淨額	533	603	659	157	103	89	184	174	118	103
營業利益	331	379	418	113	57	46	115	125	66	55
稅後純益	268	298	328	102	34	38	93	101	45	46
稅後 EPS(元)	8.45	9.42	10.37	3.23	1.08	1.21	2.93	3.18	1.42	1.44
毛利率(%)	22.48%	22.68%	22.78%	22.18%	22.58%	21.87%	23.01%	22.34%	22.72%	22.01%
營業利益率(%)	13.95%	14.23%	14.45%	16.05%	12.45%	11.15%	14.38%	16.06%	12.76%	11.71%
稅後純益率(%)	11.29%	11.22%	11.35%	14.48%	7.47%	9.35%	11.64%	12.97%	8.65%	9.73%
營業收入 YoY (%)	-10.04%	12.17%	8.76%	-20.57%	-18.45%			9.90%	13.49%	14.82%
稅後純益 YoY (%)	-7.57%	11.47%	10.09%	-22.24%	104.01%			-1.54%	31.42%	19.38%

註：稅後純益係指本期淨利歸屬於母公司業主；稅後 EPS 以股本 3.17 億元計算。

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湧盛(2249 TT) ESG：

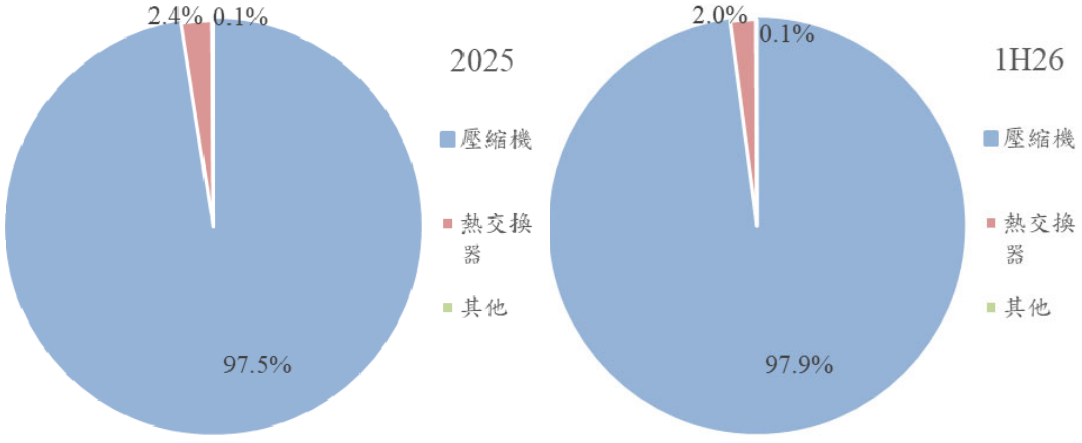
湧盛(2249 TT)在 ESG 永續發展上建立了全面的政策與實踐機制。環境永續(E)方面，公司工廠皆符合 ISO 9001 品質管理與 ISO 14001 環境管理系統認證，並通過 IATF 16949 汽車品質管理體系換證評鑑。湧盛配合主管機關完成溫室氣體盤查，以 2024 年為基準年，設定於 2030 年達成總排放量或單位產品排放量減量 25%的具體目標。產品端則將冷媒升級為全球暖化潛勢極低(GWP<1)的 R1234yf 環保冷媒，以降低碳成本風險並符合歐盟 CBAM 等國際減碳規範，同時積極布局渦捲式與電動壓縮機等綠色產品線，並於營運總部大樓規劃太陽能發電系統、綠建築標章建材、建物擴張網隔熱及 UBIKE 綠色交通停放空間。社會責任(S)方面，湧盛定期編列專款支持公益慈善，並於總部規劃友善職場環境，提供員工完整的人才專業培訓。公司治理(G)方面，公司遴選涵蓋財務、法律、學術及國際市場等多元專業背景的董事成員，定期安排公司治理與 ESG 培訓課程，由董事會嚴格監督內控與風險管理，並落實資訊透明化，將重要治理資訊充分揭露於網站。

湧盛(2249 TT)公司概況：

湧盛(2249 TT)成立於 2008 年，總部位於台灣高雄前鎮科技產業園區，並於日本東京設有子公司。公司員工規模約 183 人，為台灣最大的汽車空調壓縮機製造商。

- 產品組成方面：(1)汽車空調壓縮機佔營收比重約 97%~98%，為主要核心產品。壓縮機主要分為兩大類，傳統的皮帶式壓縮機、電動車使用的電動壓縮機。傳統皮帶式壓縮機依內部結構分為固定排量(渦捲式、旋葉式、斜盤及搖盤)、變排量(斜盤、搖盤)。公司主力自製產品為變排量斜盤式(開發達 366 款)，並有固定排量斜盤式(37 款)、旋葉式(11 款)及量產測試中的渦捲式平台。新能源車電動壓縮機涵蓋油電混合(HEV)與純電動(BEV)車款，已建置相關產線，並積極向車廠與非車用領域延伸(如電池熱管理、資料中心與機櫃溫控散熱)。(2)熱交換器與其他零組件佔營收約 2%，包含水箱、冷凝器、中冷器及離合器等。湧盛累積開發產品機種超過 3,000 項，因對應歐、美、日系各大主流車款，能滿足全球客戶一站式購足的需求。

圖一、湧盛產品比重

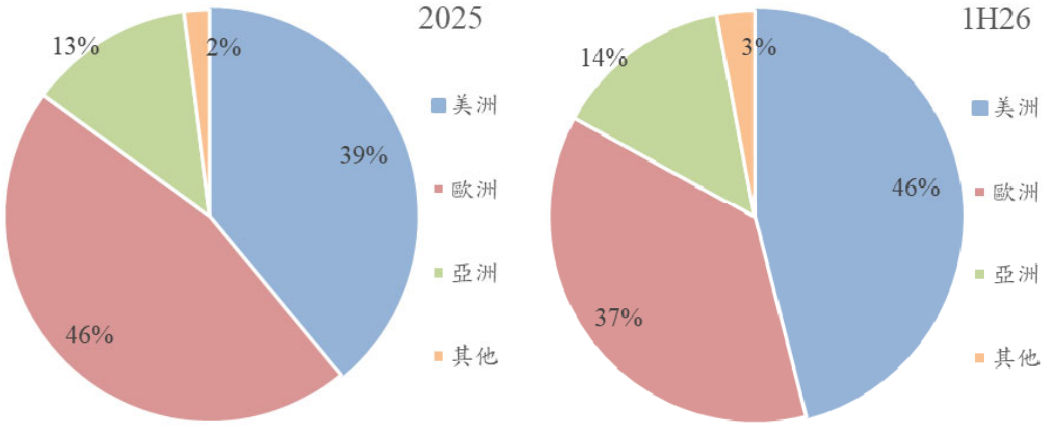


資料來源：湧盛

- 市場定位與全球布局方面，湧盛市場定位聚焦於中價位 AM 件，兼具接近原廠件(OE)的穩定品質與副廠件的價格競爭力。公司具備外部結構逆向工程與模組化開發能力，能避開原廠專利限制。公司外銷比重高達 97%以上，主要市場為歐美地區(合計佔 80%以上)。湧盛旗下自有品牌「SIGMA JP」已於歐美亞多國註冊商標；日本地區則透過子公司 SINLIN 深耕當地通路，並成功切入日本賓士總代理(Yanase)、BMW 及 DENSO 等 OES/AM 供應鏈。

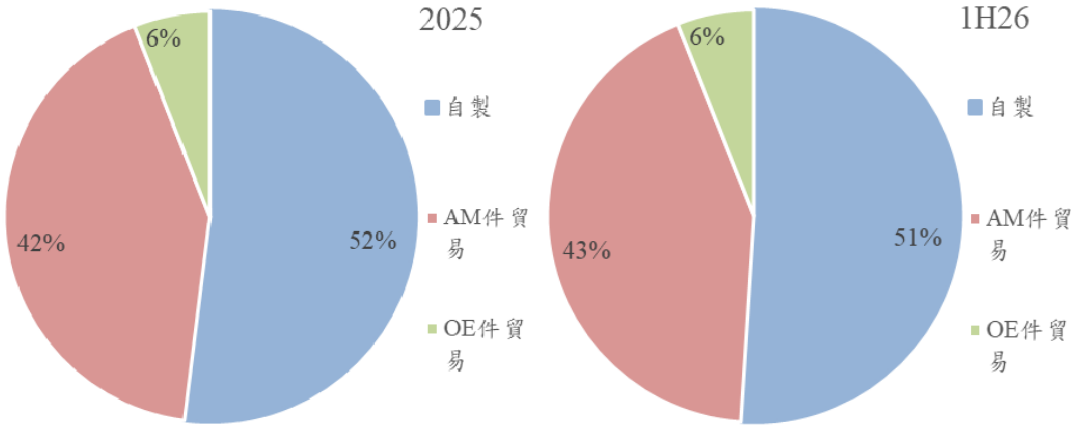
- 核心競爭優勢方面，湧盛延攬具備日本原廠經驗的資深工程師，自主研發產品已逾 400 項。公司積極將貿易外購件轉為自製，目前自製 AM 件比重已超過 51%，有效優化產品毛利率與成本結構。湧盛為 AM 市場中少數導入高自動化機械手臂產線的製造商，並全面通過 ISO 9001、ISO 14001 及 IATF 16949 等國際車廠品質與環境管理認證。因應全球減碳政策與歐盟 CBAM 規範，產品已升級導入極低全球暖化潛勢(GWP<1)之 R1234yf 環保冷媒，並加速電動車用壓縮機的布局。

圖二、湧盛區域別銷售比重



資料來源：湧盛

圖三、湧盛壓縮機產品別銷售比重



資料來源：湧盛

湧盛於北美市場具有關稅優勢：

美國政府於 05/27/2026 正式公告實施台美投資 MOU 中所載明的對台非半導

體 232 關稅優惠措施，並回溯至 05/01/2026 起開始生效。汽車零組件方面，相關輸美產品過去總稅率為 27.5%，已直接降為 15%。而湧盛主要競爭對手中國汽車壓縮機供應商，面臨的關稅落於 25%~50%：中國出口之汽車壓縮機價格落在 88 至 92 美元區間且符合關稅豁免條件者能適用 25%關稅；若單價落在 92 美元以上或 88 美元以下，則面臨 50%的關稅稅率。相較之下，湧盛於北美市場具有關稅及價格競爭優勢。

預估 2026 年營收 23.71 億元，YoY-10.04%，稅後純益 2.68 億元，YoY-7.57%，EPS 8.45 元：

2026 年方面，1H26 營運下滑，2H26 開始恢復穩定。1H26 營運下滑主要原因包含：(1)歐洲客戶下單模式由長單改變為滾動式下單，影響短期營收認列節奏。(2)受到關稅影響，部分美國客戶提前建立高庫存，且 2025 年歐洲特定機種拉貨量大導致營收基期較高。(3)受到地緣政治、戰爭等因素影響，國際油價上漲壓抑終端車市購車意願及零組件需求。而受惠於歐洲客戶下單模式趨於穩定，美國客戶庫存調整周期告一段落，預估 2H26 營運動能回升。預估 2026 年營收 23.71 億元，YoY-10.04%，獲利能力方面，預估毛利率表現平穩，營業利益 YoY-17.47%。因業外、稅率影響相對有限，預估稅後純益 2.68 億元，YoY-7.57%，EPS 8.45 元。

預估 2027 年營運動能回升：

2027 年方面，除了既有業務恢復增長外，美國客戶委託開發逾 40 項新機種，預計於 2026 年底至 2027 年陸續通過認證並進入量產階段。而公司積極投入新建廠房與自動化設備，皆有助於提升其營運體質。預估 2027 年營收 26.60 億元，YoY+12.17%，稅後純益 2.98 億元，YoY+11.47%，EPS 9.42 元。

投資建議：

投資建議方面，湧盛(2249 TT) 2026 年營運動能下滑，預估 2027 年有望回升。公司將於 10/22/2026 掛牌上櫃，10/06/2026~10/08/2026 為競價拍賣期間。最低投標價格為 55 元，10/05/2026 興櫃收盤價 84 元。考量市場資金集中於 AI 相關族群，汽車零組件廠商目前的市場評價相對較低，建議競拍價格為 75 元~94 元(以 2027 年 EPS 及 PE 8~10 倍評價)。

【投資評等說明】

評等	定義
強力買進(Strong Buy)	首次評等潛在上漲空間≥35%
買進(Buy)	15%≤首次評等潛在上漲空間<35%
區間操作(Trading Buy)	5%≤首次評等潛在上漲空間<15%
中立(Neutral)	無法由基本面給予投資評等 預期近期股價將處於盤整 建議降低持股

【投資評等調整機制】

投資評等係「首次」給予特定個股投資評等時，其前一日收盤價相較 3 個月目標價之潛在上漲空間計算而得。個股投資評等分為四個等級，定義如上。爾後的投資評等係依循「首次評等」，直到停止推薦。

「強力買進」、「買進」及「區間操作」均有 upside 目標價。差別僅在於，首次評等時潛在報酬率不同。「中立」則無目標價。

一旦我們給予特定個股「強力買進」、「買進」或「區間操作」之投資評等，就是責任的開始，爾後將透過各式研究報告作定期性、持續性追蹤基本面及股價變化，直到停止推薦。

停止推薦情境：

1. 達目標價。
2. 雖未達原訂目標價，但檢視基本面、訊息面、籌碼面等多方訊息，研判股價上漲空間已然有限，將適時出具降評報告。
3. 推薦後股價不漲反跌，亦將出具降評報告。

【免責聲明】

本研究報告僅提供予特定人之客戶作為參考資料「非經同意不得轉載」。我們並不確保此資訊的完整性與正確性，投資人應了解，報告中有關未來預測之陳述可能不會實現，因而不應被依賴。而且此報告並非根據特定投資目的或依預定對象之財務狀況所撰寫出來的，因此，此研究報告的目的，既非對投資人於買賣證券、選擇權、期貨或其他證券相關之衍生性商品提供詢價服務，亦非作為進行交易的要約。投資人應注意到相關證券之價值及收益，可能會有無預警地上升或下降，產生投資回報金額可能比原始投資來得少的情形。

2026 年 10 月 6 日

研究員：陳俐妍 ly.c@capital.com.tw

前日收盤價	8,600.00 元
目標價	
3 個月	10,290.00 元
12 個月	10,290.00 元

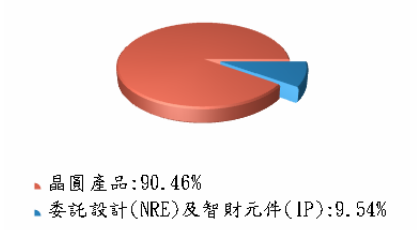
近期報告日期、評等及前日股價

08/03/2026	Trading Buy	3,805.00
06/16/2026	Buy	4,815.00
06/08/2026	Trading Buy	4,390.00
05/22/2026	Buy	5,065.00
05/05/2026	Buy	4,685.00

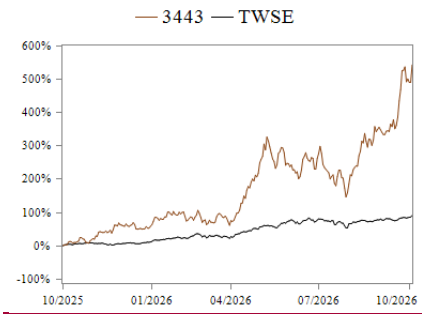
公司基本資訊

目前股本(百萬元)	1,340
市值(億元)	11,524
目前每股淨值(元)	100.49
外資持股比(%)	27.34
投信持股比(%)	11.41
董監持股比(%)	34.84
融資餘額(張)	4,871
現金股息配發率(%)	71.10

產品組合



股價相對大盤走勢



創意(3443 TT)　Buy

成長動能強，建議 Buy。

投資建議：創意客戶自研 N3 CPU 在 2H26 放量後，2027 年為完整年度貢獻。且因應 Agentic AI 與深度推論架構，客戶新平台將 CPU 與 TPU 的配置比從過往 1:4 提升至 1:2，推升 CPU 的需求總量。預估該 CPU 出貨量將由 2026 年的 125 萬顆大幅升至 2027 年約 400 萬顆。另一成長動能為北美車用客戶，預計將採用台積電 N3E 先進製程，預估北美車用專案將於 2H27 加入，顯著優化產品獲利結構，此外新的 CoWoS 訂單也增加。雖然 CSP 訂單和 Turnkey 增加，將使毛利率有壓，銷管研費因公司的圖靈中心營運，營業利益率將維持穩定，預估創意 2026 和 2027 年 EPS 分別為 55.12 元和 158.33 元。考量創意受惠 AI ASIC 訂單增加，除拿到原本客戶下一代訂單，也增加新客戶，2027~2028 年成長動能佳，預期獲利成長性提升，因此對創意投資建議為 Buy，目標價 10,290 元(2027 年 PER65X)。

預估 3Q26 稅後 EPS 14.38 元：創意自結 09/2026 營收 71.45 億元，MoM+22.34%，YoY+97.78%，創歷史新高。3Q26 營收 187.75 億元，QoQ+35.10%，產品營收比重方面，NRE 占營收 11%、Turnkey 則占 89%。3Q26 NRE 營收為 21.03 億元，QoQ-8.96%，Turnkey 營收為 166.71 億元，QoQ+43.89%。由於 Turnkey 占比高，因此預期毛利率將較 2Q26 下滑近 2 個百分點，但營業費用增加有限，營業利益率有望和 2Q26 持平，預估創意 3Q26 稅後 EPS 14.38 元。

AI ASIC 分工深化：CSP 積極投入 AI ASIC 自研，其研發重心並非放在所有晶片規格的開發，而是聚焦最具差異化的系統架構設計，包括 AI 模型、軟硬體協同設計、系統互連架構及整體資料中心效率等核心技術。以 Google 為例，發展 TPU 架構、AI 軟體堆疊等，目的在於提升整體 AI 基礎設施的運算效率，而非自行完成所有衍生晶片規格的設計。因此，隨著 AI ASIC 產品線愈趨多元，專業分工模式預期將更加明確，使創意持續受惠於 SKU 設計需求的增加，而非受到 CSP 自行研發趨勢的排擠。

(百萬元)	2026F	2027F	2028F	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F
營業收入淨額	67,037	188,639	370,913	8,613	12,400	11,448	13,897	18,775	22,918	25,680	37,307	59,347
營業毛利淨額	14,165	36,444	68,451	2,086	2,361	3,113	2,994	3,672	4,386	4,886	7,112	11,620
營業利益	8,420	25,088	44,469	1,055	1,257	1,824	1,676	2,264	2,655	2,948	4,668	8,327
稅後純益	7,386	21,216	36,966	867	1,160	1,646	1,555	1,927	2,257	2,536	3,889	7,050
稅後 EPS(元)	55.12	158.33	275.86	6.47	8.65	12.29	11.61	14.38	16.85	18.92	29.02	52.61
毛利率(%)	21.13%	19.32%	18.45%	24.22%	19.04%	27.20%	21.55%	19.56%	19.14%	19.03%	19.06%	19.58%
營業利益率(%)	12.56%	13.30%	11.99%	12.25%	10.13%	15.93%	12.06%	12.06%	11.59%	11.48%	12.51%	14.03%
稅後純益率(%)	11.02%	11.25%	9.97%	10.06%	9.35%	14.38%	11.19%	10.27%	9.85%	9.87%	10.42%	11.88%
營業收入 YoY/QoQ(%)	96.35%	181.39%	96.63%	41.09%	43.96%	-7.68%	21.39%	35.10%	22.07%	12.05%	45.28%	59.08%
稅後純益 YoY/QoQ(%)	95.95%	187.24%	74.23%	10.90%	33.77%	41.96%	-5.53%	23.92%	17.13%	12.32%	53.35%	81.30%

註：稅後純益係指本期淨利歸屬於母公司業主；稅後 EPS 以股本 13.40 億元計算。

創意電子 ESG 政策：

創意自成立以來，除專注在 IC 設計服務本業中追求最大成就外，同時對於客戶、股東、政府機關、社會、供應商及本公司員工等利害關係人建立良好的互動及平等對待，並積極落實企業社會公民責任，以實現永續經營之理念。

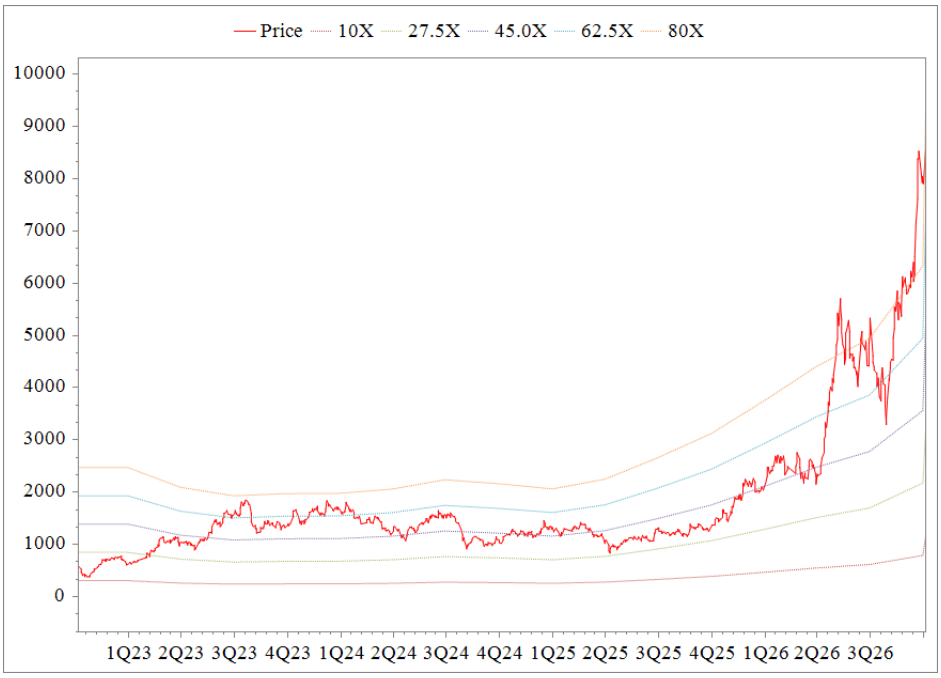
公司執行方針

誠信正直方面，創意提倡誠信經營及道德從業行為，並落實平衡各利害關係人的利益，驅動產業的正向循環，共同成為社會向上提升的力量。

強化環保方面，創意致力於環境永續，堅持綠色設計與綠色供應鏈，追求資源最佳效率，以達成減廢與污染防治的目標。

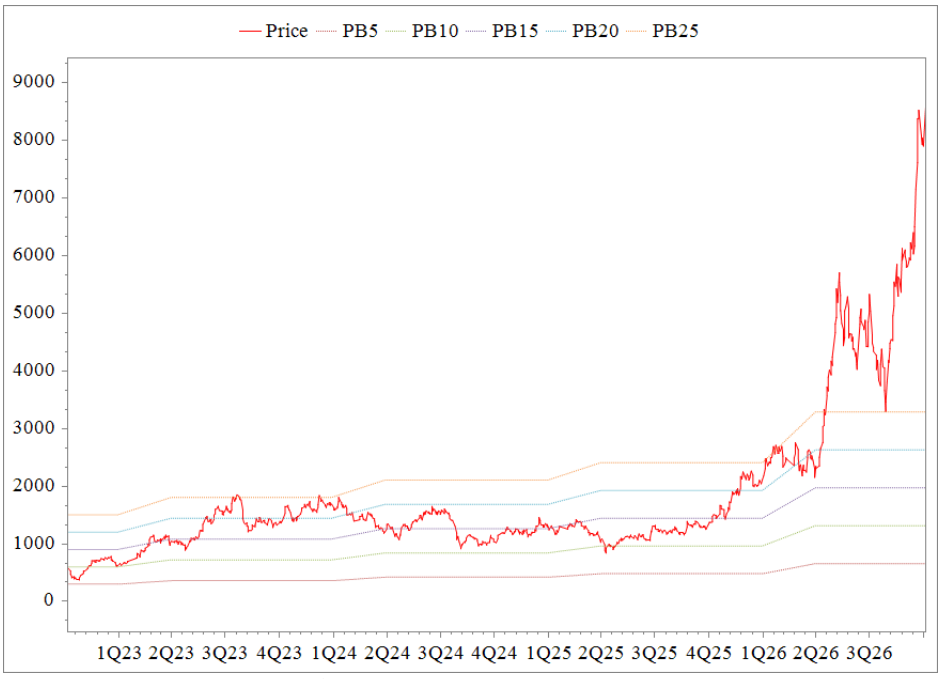
社會關懷方面，創意結合公司資源與同仁自發性的愛心及志工服務，以金錢、物資及人力等方式投入『環境保育』及『偏鄉、弱勢對象』為兩大範疇，期能推動永續環境、並能提供弱勢團體必要的生活協助及急難救助，以建立良好互助社會。

Forward PE Band



資料來源：CMoney，群益預估彙整

Forward PB Band



資料來源：CMoney，群益預估彙整

資產負債表

(百萬元)	2024	2025	2026F	2027F	2028F
資產總計	26,309	28,443	41,541	61,979	95,938
流動資產	24,480	25,737	38,658	61,627	91,766
現金及約當現金	10,427	6,650	13,146	18,945	29,365
應收帳款與票據	2,007	3,417	3,541	13,709	1,922
存貨	2,794	10,148	14,803	24,242	20,206
採權益法之投資	--	--	0	0	0
不動產、廠房設備	942	1,347	2,218	2,276	2,315
負債總計	**15,015**	**15,530**	**23,923**	**26,494**	**30,188**
流動負債	14,669	15,033	23,574	26,414	29,009
應付帳款及票據	1,691	4,126	6,441	5,129	5,185
非流動負債	346	497	349	80	1,179
權益總計	11,294	12,913	17,619	35,485	65,750
普通股股本	**1,340**	**1,340**	**1,340**	**1,340**	**1,340**
保留盈餘	9,924	11,556	16,262	34,127	64,393
母公司業主權益	11,294	12,913	17,618	35,484	65,749
負債及權益總計	**26,309**	**28,443**	**41,541**	**61,979**	**95,938**

現金流量表

(百萬元)	2024	2025	2026F	2027F	2028F
營業活動現金	6,242	-1,143	9,599	14,422	27,791
稅前純益	4,062	4,443	8,633	25,156	44,537
折舊及攤銷	623	677	1,164	1,282	1,436
營運資金變動	2,041	-6,328	-2,464	-20,919	15,878
其他營運現金	-484	65	2,265	8,904	-34,061
投資活動現金	-1,522	-398	-222	1,833	-4,583
資本支出淨額	-471	-838	-470	-554	-583
長期投資變動	-850	688	-446	-202	-217
其他投資現金	-201	-248	694	2,590	-3,782
籌資活動現金	-1,962	-2,227	-2,880	-10,456	-12,788
長借/公司債變動	0	0	-147	-1,271	2,101
現金增資	0	0	0	0	0
發放現金股利	-1,876	-2,144	-2,680	-3,350	-6,701
其他籌資現金	-86	-82	-52	-5,834	-8,189
淨現金流量	2,790	-3,778	6,496	5,799	10,420
期初現金	7,638	10,427	6,650	13,146	18,945
期末現金	10,427	6,650	13,146	18,945	29,365

損益表

(百萬元)	2024	2025	2026F	2027F	2028F
營業收入淨額	25,044	34,141	67,037	188,639	370,913
營業成本	16,937	25,687	52,872	152,196	302,462
營業毛利淨額	8,108	8,454	14,165	36,444	68,451
營業費用	4,305	4,104	5,745	11,356	23,982
營業利益	3,803	4,350	8,420	25,088	44,469
EBITDA	4,536	4,942	9,399	25,922	45,103
業外收入及支出	260	93	213	68	68
稅前純益	4,062	4,443	8,633	25,156	44,537
所得稅	612	673	1,247	3,940	7,571
稅後純益	3,451	3,770	7,386	21,216	36,966
稅後 EPS(元)	25.75	28.13	55.12	158.33	275.86
完全稀釋 EPS**	25.75	28.13	55.12	158.33	275.86

註 1：稅後純益係指本期淨利歸屬於母公司業主。

註 2：稅後 EPS 以股本 13.40【最新股本】億元計算

註 3：完全稀釋 EPS 以股本 13.40 億元計算

比率分析

(百萬元)	2024	2025	2026F	2027F	2028F
成長力分析(%)					
營業收入淨額	-4.56%	36.32%	96.35%	181.39%	96.63%
營業毛利淨額	1.65%	4.27%	67.56%	157.28%	87.83%
營業利益	-4.18%	14.38%	93.58%	197.95%	77.25%
稅後純益	-1.63%	9.24%	95.95%	187.24%	74.23%
獲利能力分析(%)					
毛利率	32.37%	24.76%	21.13%	19.32%	18.45%
EBITDA(%)	18.11%	14.48%	14.02%	13.74%	12.16%
營益率	15.18%	12.74%	12.56%	13.30%	11.99%
稅後純益率	13.78%	11.04%	11.02%	11.25%	9.97%
總資產報酬率	13.12%	13.25%	17.78%	34.23%	38.53%
股東權益報酬率	30.55%	29.19%	41.92%	59.79%	56.22%
償債能力檢視					
負債比率(%)	57.07%	54.60%	57.59%	42.75%	31.47%
負債/淨值比(%)	132.94%	120.27%	135.78%	74.66%	45.91%
流動比率(%)	166.89%	171.20%	181.84%	233.31%	328.31%
其他比率分析					
存貨天數	82.38	91.95	86.12	46.82	26.82
應收帳款天數	29.13	29.00	18.94	16.69	7.69

資料來源：CMoney、群益

季度損益表

(百萬元)	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	1Q28F
營業收入淨額	6,105	8,613	12,400	11,448	13,897	18,775	22,918	25,680	37,307	59,347	66,305	60,904
營業成本	4,071	6,527	10,038	8,335	10,903	15,102	18,533	20,794	30,196	47,726	53,480	49,863
營業毛利淨額	2,034	2,086	2,361	3,113	2,994	3,672	4,386	4,886	7,112	11,620	12,825	11,041
營業費用	1,090	1,031	1,105	1,289	1,318	1,408	1,730	1,939	2,444	3,294	3,680	4,141
營業利益	944	1,055	1,257	1,824	1,676	2,264	2,655	2,948	4,668	8,327	9,145	6,899
業外收入及支出	-15	-40	99	101	80	17	16	18	17	17	16	18
稅前純益	929	1,015	1,356	1,925	1,756	2,281	2,672	2,966	4,685	8,343	9,162	6,917
所得稅	147	148	196	278	201	354	414	430	796	1,293	1,420	1,176
稅後純益	782	867	1,160	1,646	1,555	1,927	2,257	2,536	3,889	7,050	7,742	5,741
最新股本	1,340	1,340	1,340	1,340	1,340	1,340	1,340	1,340	1,340	1,340	1,340	1,340
稅後EPS(元)	5.83	6.47	8.65	12.29	11.61	14.38	16.85	18.92	29.02	52.61	57.77	42.85
獲利能力(%)												
毛利率(%)	33.32%	24.22%	19.04%	27.20%	21.55%	19.56%	19.14%	19.03%	19.06%	19.58%	19.34%	18.13%
營業利益率(%)	15.46%	12.25%	10.13%	15.93%	12.06%	12.06%	11.59%	11.48%	12.51%	14.03%	13.79%	11.33%
稅後純益率(%)	12.80%	10.06%	9.35%	14.38%	11.19%	10.27%	9.85%	9.87%	10.42%	11.88%	11.68%	9.43%
QoQ(%)												
營業收入淨額	-13.09%	41.09%	43.96%	-7.68%	21.39%	35.10%	22.07%	12.05%	45.28%	59.08%	11.73%	-8.15%
營業利益	-13.79%	11.78%	19.12%	45.18%	-8.10%	35.07%	17.27%	11.01%	58.37%	78.38%	9.83%	-24.56%
稅前純益	-18.70%	9.28%	33.55%	41.96%	-8.77%	29.89%	17.13%	11.01%	57.97%	78.08%	9.81%	-24.50%
稅後純益	-18.69%	10.90%	33.77%	41.96%	-5.53%	23.92%	17.13%	12.32%	53.35%	81.30%	9.81%	-25.84%
YoY(%)												
營業收入淨額	-9.16%	30.28%	105.88%	62.99%	127.64%	117.98%	84.83%	124.32%	168.46%	216.10%	189.31%	137.16%
營業利益	-8.39%	-11.41%	43.32%	66.65%	77.64%	114.66%	111.31%	61.58%	178.46%	267.75%	244.43%	134.06%
稅前純益	-15.79%	-15.67%	38.27%	68.45%	89.02%	124.67%	97.05%	54.09%	166.81%	265.80%	242.93%	133.24%
稅後純益	-13.86%	-16.13%	36.93%	71.24%	98.97%	122.33%	94.67%	54.03%	150.03%	265.80%	242.93%	126.42%

註1：稅後EPS以股本13.40億元計算

註2：稅後純益係指本期淨利歸屬於母公司業主。

【投資評等說明】

評等	定義
強力買進(Strong Buy)	首次評等潛在上漲空間≥35%
買進(Buy)	15%≤首次評等潛在上漲空間<35%
區間操作(Trading Buy)	5%≤首次評等潛在上漲空間<15%
中立(Neutral)	無法由基本面給予投資評等 預期近期股價將處於盤整 建議降低持股

【投資評等調整機制】

投資評等係「首次」給予特定個股投資評等時，其前一日收盤價相較 3 個月目標價之潛在上漲空間計算而得。個股投資評等分為四個等級，定義如上。爾後的投資評等係依循「首次評等」，直到停止推薦。

「強力買進」、「買進」及「區間操作」均有 upside 目標價。差別僅在於，首次評等時潛在報酬率不同。「中立」則無目標價。

一旦我們給予特定個股「強力買進」、「買進」或「區間操作」之投資評等，就是責任的開始，爾後將透過各式研究報告作定期性、持續性追蹤基本面及股價變化，直到停止推薦。

停止推薦情境：

1. 達目標價。
2. 雖未達原訂目標價，但檢視基本面、訊息面、籌碼面等多方訊息，研判股價上漲空間已然有限，將適時出具降評報告。
3. 推薦後股價不漲反跌。

【免責聲明】

本研究報告僅提供予特定人之客戶作為參考資料「非經同意不得轉載」。我們並不確保此資訊的完整性與正確性，投資人應了解，報告中有關未來預測之陳述可能不會實現，因而不應被依賴。而且此報告並非根據特定投資目的或依預定對象之財務狀況所撰寫出來的，因此，此研究報告的目的，既非對投資人於買賣證券、選擇權、期貨或其他證券相關之衍生性商品提供詢價服務，亦非作為進行交易的要約。投資人應注意到相關證券之價值及收益，可能會有無預警地上升或下降，產生投資回報金額可能比原始投資來得少的情形。

2026/10/06

CCL

聯茂 (6213 TT)

重新覆蓋

買進(Buy)

漲完價，再升級

- **上游供貨瓶頸帶動漲價，2Q26 獲利跳增**
- **漲價效益 2H26 持續發酵，毛利率續升**
- **伺服器平台換代，2027 年轉向規格升級**

2026 年主軸為上游供貨瓶頸帶動漲價： 聯茂 2Q26 營收 115.3 億元(QoQ+26.1%, YoY+29.9%)，創單季新高；毛利率 22.02%(QoQ +10.6ppts, YoY +6.4ppts)，EPS 為 3.52 元。獲利顯著提升，主因 1Q26 上游玻纖布、銅箔成本急漲，報價僅調漲約 10%且至 3 月才陸續生效，毛利率一度降至 11.42%，形成低基期；2Q26 再全面調漲約 30%，一次補足 1Q26 未能轉嫁的成本，上半年累計漲幅約 40%。

漲價效益在 2H26 持續發酵： 8 月自結營收 50.44 億元(YoY+97.9%)，稅後淨利 9.24 億元(YoY+852.6%)，EPS 為 2.54 元。我們推估 8 月毛利率約 32%，明顯高於市場對 3Q26 毛利率 26.7%的共識預期。公司預期 3Q26、4Q26 全產品線將各再調漲 20%。然而受限上游 E-glass 供給不足(銅箔亦轉趨吃緊)，1Q26~3Q26 稼動率維持在 70~75%，且年底前無新產能開出，營收成長幾乎全數來自漲價。我們預估 3Q26 營收 156.17 億元(QoQ +35.4%)，毛利率 31.8%，EPS 為 7.68 元。

2027 年成長主軸由漲價轉向規格升級： General server 占聯茂營收逾 50%，為規格升級的主軸。兩大客戶 AMD 與 Intel 新平台將於 1Q27 放量，由 PCIe 5.0 升級至 PCIe 6.0，將帶動伺服器 PCB 板材由 M6 升級至 M7，層數亦由 18L 提升至 24L，用量與單價同步成長。我們預估 M7/M8 營收占比將由 2026 年的 8.6%提升至 2027 年的 19.5%，帶動 2027 年毛利率達 35.9%(YoY +8.2ppts)。

聯茂亦積極擴充無錫廠與泰國廠，預計 2026/2027/2028/2029 年底月產能分別達 570 萬/630 萬/825 萬/915 萬張。整體而言，我們認為聯茂在(1)M7 以上占比提升、(2)稼動率回升至 80%以上、(3)產能規模擴大三項動能下，即使漲價趨緩仍可維持成長。我們預估 2026/2027/2028 年 EPS 分別為 23.28/52.49/69.40 元。考量聯茂產品組偏向中低階，長期由產品組合支撐的獲利基礎不如同業，故本益比應較同業折價。我們給予目標價 945 元，對應 2027 年 EPS 18 倍本益比，建議買進。

風險因子：PCIe 6 滲透率不如預期、CCL 價格回跌。

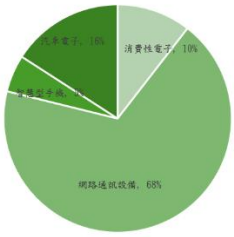
臺灣 ESG 永續指數成分股

潛在漲幅	25.83%
前日收盤價	751.00 元
目標價	945.00 元
前次目標價	-
前次評等	-

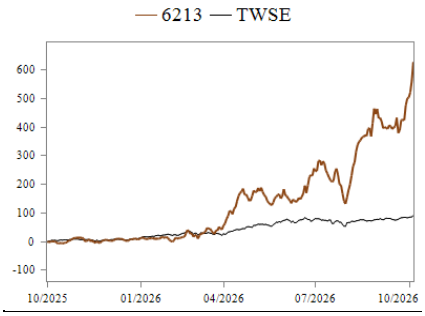
公司基本資訊

目前股本(百萬元)	3,636
市值(億元)	2,731
目前每股淨值(元)	62.14
外資持股比(%)	21.27
董監持股比(%)	19.95

營業比重(2025 年)



個股與大盤走勢(%)

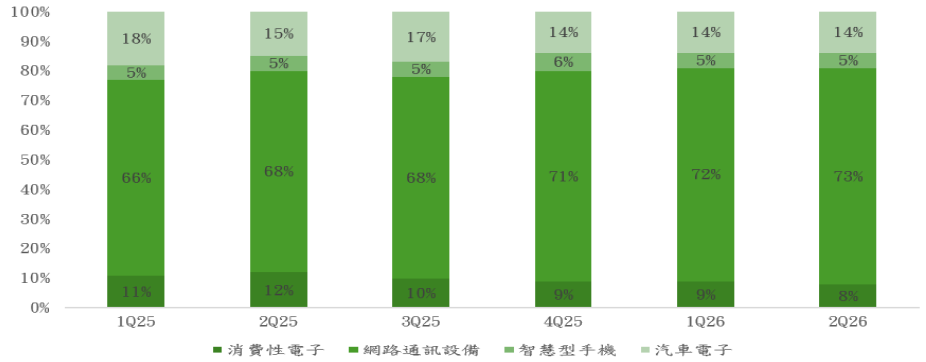


王翊安
wade.wang@ctbcsis.com

(百萬元)	2022	2023	2024	2025	2026F	2027F
營業收入淨額	29,130	25,079	29,378	33,098	56,564	93,053
營業毛利淨額	3,940	3,106	3,690	4,785	15,656	33,421
營業利益	1,896	995	1,351	2,383	12,191	27,931
稅後純益	1,855	677	822	1,510	8,462	19,080
公告基本 EPS(元)	4.94	1.86	2.26	4.16	--	--
調整後 EPS(元)	5.11	1.86	2.26	4.16	23.28	52.49
營業收入 YoY(%)	-10.44%	-13.91%	17.14%	12.66%	70.90%	64.51%
稅後純益 YoY(%)	-41.01%	-63.53%	21.45%	83.77%	460.31%	125.48%
本益比	13.48	43.71	33.82	27.06	32.27	14.31
本淨比	1.20	1.52	1.34	1.92	11.47	9.25
股息(元)	3.00	1.50	1.80	3.00	16.30	36.74
殖利率(%)	4.33%	1.30%	2.05%	0.83%	2.17%	4.89%
ROE(%)	8.96%	3.42%	4.10%	7.21%	37.55%	71.55%

公司簡介：聯茂成立於 1997 年，總部位於新竹縣新埔鎮，主要從事多層印刷電路基材及銅箔基板(CCL)的製造、加工與買賣，為台灣主要 CCL 廠之一。生產基地主要位於中國江西、無錫及泰國，其中江西廠可生產全部等級，高階產品主要於此生產；無錫新廠為未來擴產主力，泰國廠則對應低軌衛星應用。產品依介電性能分級，M4 以下為低階產品，往上依序為 M6、M7、M8、M9，等級越高，規格與單價越高；目前出貨以 M6 以下為主，1H26 M4 以下營收占比約 70%，M7 以上佔比低個位數。主要應用為伺服器(以 general server 為主)、網通與低軌衛星。

圖表一、聯茂營收產品組合趨勢

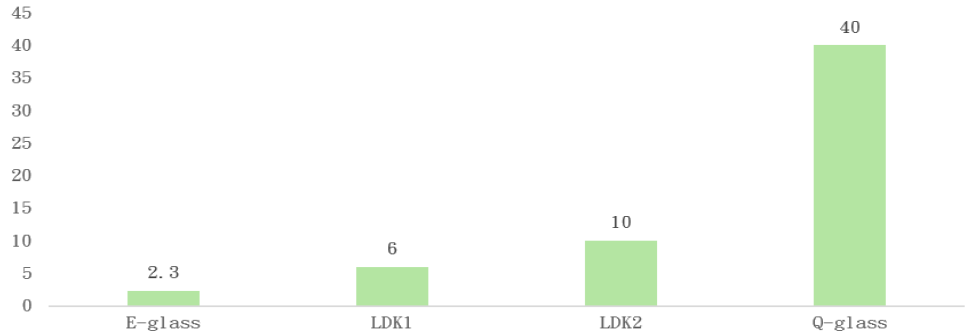


資料來源：Bloomberg、中信投顧彙整

E-glass 缺口延續，稼動率回升帶動 2027 年再創新高：目前聯茂主要使用的玻纖布 E-glass 報價已達 2.3 美元/平方公尺(見圖表二)，預計將逐月調漲至 1Q27，主因為：(1)AI server PCB 規格升級帶動用量倍增、(2)中游電子紗原料緊缺，德、日廠產品交期已排至 2029 年、(3)Toyota 自動織布機嚴重緊缺，交期達 2 年以上，目前亦排至 2029 年。整體而言，目前造成 CCL 緊缺的主因為玻纖布，其中最缺的是用於製作 M6 以下 CCL 的 E-glass；且因 Low-DK 布單價較 E-glass 高約 3 倍以上，原本生產 E-glass 的廠商多已轉作 Low-DK 布，在相同投資金額下，新建窯爐與織布機位幾乎全數用於高階 Low-DK 布，導致 E-glass 缺口持續擴大。

聯茂產品線以 M4 以下為主，故 1Q26~3Q26 稼動率在需求強勁下仍受上游缺料限制，僅維持在 70~75%，且年底前無新產能開出，今年的大幅成長主要來自反映上游原物料漲價。4Q26~1Q27 期間 E-glass 廠中國巨石、泰山玻纖新產能開出後，稼動率可望回升至 80%附近，將帶動聯茂營收規模與毛利率同步提升，預估 2027 年營收達 930.53 億元(YoY +64.5%)，毛利率達 35.9%。

圖表二、各種類玻纖布報價(USD/平方公尺)



資料來源：中信投顧整理

圖表三、各等級板材規格

等級	損耗分類	Df	玻纖布	銅箔	樹脂
M	Standard Loss	>0.015	E	Standard	環氧樹脂
M2	Mid Loss	0.015~0.009	E	Low Profile (LP)	環氧樹脂
M4	Low Loss	0.009~0.006	E	VLP	環氧樹脂
M6	Very Low Loss	0.006~0.005	E	HVLP1~2	PPO
M7	Ultra Low Loss	0.005~0.0015	Low-Dk1	HVLP3	PPO
M8	Super Ultra Low Loss	0.0015~0.0010	Low-Dk1/Dk2	HVLP3/4	PPO
M9	Extreme Low Loss	0.0010~0.0005	Low-Dk2/Q	HVLP4/5	PPO+HC

資料來源：中信投顧整理

General Server 平台換代，PCIe 6 滲透率為規格升級關鍵： General server 為聯茂規格升級的主軸，2026 年占整體營收逾 50%。兩大客戶 AMD 與 Intel 近期皆將進行 2~3 年一次的平台大改版，AMD SP7/8(Venice)與 Intel OakStream 均將由 PCIe 5.0 升級至 PCIe 6.0，板材由以 M6 為主轉為以 M7 為主，板層數亦由 18~22 層提升至 20~24 層(見圖表四)，此規格升級將使聯茂受惠於量價齊揚。以伺服器平台占比而言，AMD 與 Intel 在聯茂營收中的比重約為 7:3。參考主要客戶 AMD 新平台的放量時程，SP7(high-end)將於 4Q26 推出，SP8(mainstream)將於 1H27 推出，兩者皆支援 PCIe 6.0；預計 AMD 與 Intel 新平台均將於 4Q26 開始出貨、1Q27 開始放量，將推升 PCIe 6.0 在 general server 中的滲透率。參考主流研調機構對 general server 出貨量的預估，2027 年年增幅約 6~11%，成長幅度遠低於 AI server，因此我們認為帶動聯茂規格提升的主要變數為 PCIe 6.0 滲透率，而非出貨量成長。

圖表四、Inetl/AMD 伺服器平台規格表

Intel	Platform	Purley		Whitley	Eagle Stream		Birch Stream	Oak Stream
	CPU	Skylake	Cascade Lake	Ice lake	Sapphire Rapids	Emerald Rapids	Granite Rapids	Diamond Rapids
	Nano Process	14 nm	14 nm+	10 nm	Intel 7	Intel 7	Intel 3	Intel 18A
	PCIe Gen	PCIe 3.0	PCIe 3.0	PCIe 4.0	PCIe 5.0	PCIe 5.0	PCIe 5.0	PCIe 6.0
	MP Time	2017 Q3	2019 Q3	2021 Q1	2023 H1	2023 H2	2024	2026
	CCL Material	Mid Loss	Mid Loss	Low Loss	Very Low Loss	Very Low Loss	VLL/ Ultra Low Loss	Ultra Low Loss
	Layer count	8 to 12	8 to 12	12 to 16	16 to 20	16 to 20	18 to 22	20 to 24
AMD	Architecture	Zen	Zen2	Zen3	Zen4		Zen5	Zen6
	CPU	Naples	Rome	Milan	Genoa	Bergamo	Turin	Venice
	Nano Process	14 nm (Global Foundries)	7 nm (TSMC)	7 nm (TSMC)	5 nm (TSMC)	5 nm (TSMC)	4 nm / 3 nm (TSMC)	2 nm (TSMC)
	PCIe Gen	PCIe 3.0	PCIe 4.0	PCIe 4.0	PCIe 5.0	PCIe 5.0	PCIe 5.0	PCIe 6.0
	MP Time	2017 Q3	2019 Q3	2020 Q4	2022 Q4	2023	2024	2026
	CCL Material	Mid Loss	Low Loss	Low Loss	Very Low Loss	Very Low Loss	VLL/ Ultra Low Loss	Ultra Low Loss
	Layer count	8 to 12	12 to 16	12 to 16	16 to 20	16 to 20	18 to 22	20 to 24

資料來源：公司

圖表五、各板材單價及毛利率比較

等級	主要應用	單價(台幣/張)	對應毛利率	2026營收佔比	2027營收佔比
M2	消費性	500~700	24%(整合計算)	66%	57%
M4	PC/Server DB	1000~1200			
M6	X86 Server	1400~1600	30%	26%	23%
M7	OAM/UBB HDI/Next Gen X86 Server	2000~2500	35%	6%	15%
M8	Compute Tray/Switch Tray	6000~8500	40%	3%	5%
M9	Midplane	10000~13000	60%	0%	1%
Note		截至2Q26參考價	截至2Q26，後續考慮漲價		

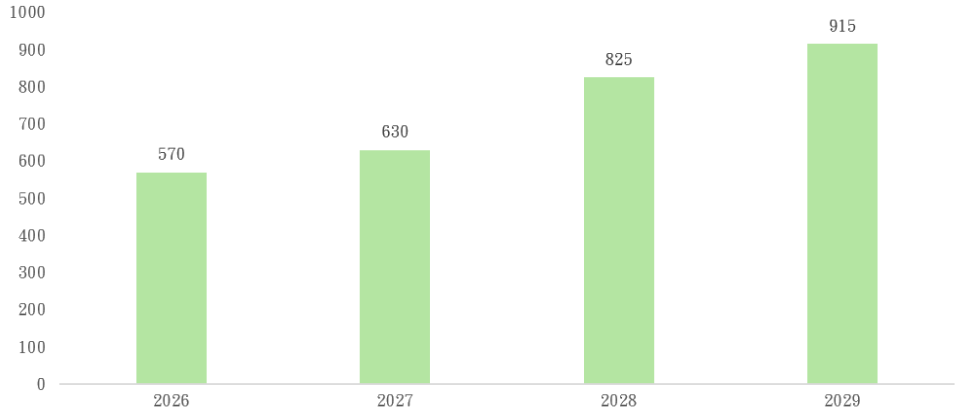
資料來源：中信投顧整理

此外，近期 Meta Muse 等 Agentic AI 應用陸續出現，雖然 Agentic AI 的模型推論仍由 GPU/ASIC 執行，但任務編排(orchestration)、工具執行(tool execution)及每位用戶的運行環境(sandbox)主要仰賴 CPU 處理，將帶動 CSP 對 general server 的需求回溫，並且也是驅動漲價的主要因素之一。我們認為聯茂亦將是 Agentic AI、CPU 等題材的主要受惠者之一，預期將為 2027/2028 年 general server 出貨量帶來進一步上行空間。整體而言，我們預估 2027 年 PCIe 6.0 在 general server 平台的滲透率將達 20%，帶動聯茂 M7 營收占比由 2026 年的 6%提升至 2027 年的 15%，M7 營收金額年增 312%，為 2027 年的主要成長動能。

除了 general server 外，低軌衛星(LEO)也是聯茂 2027 的一大增量，LEO 產品主要在泰國廠製造，使用 M4~M7 CCL，其中以 M4 為主，預計 1Q27~2Q27 開始出貨。

在 CCL 供不應求下，聯茂積極擴充泰國廠與無錫廠：預計 2026 年底泰國一廠開出 30 萬張月產能，月產能達 570 萬張；2H27 無錫廠新增 60 萬張，2027 年底達 630 萬張；1H28 泰國廠新增 60 萬張，2H28 無錫廠再新增 135 萬張，2028 年底達 825 萬張；2029 年再新增 90 萬張，年底達 915 萬張(見圖表六)。

圖表六、聯茂各年底 CCL 月產能(萬張/月)



資料來源：公司、中信投顧整理

圖表七、聯茂各廠區產能規劃

資料來源：公司

目前產能擴張的最大瓶頸在於上膠機交期。聯茂的上膠機主要向亞泰金屬採購，而亞泰金屬於 2027/2028 年均有擴充月產能的計畫，預計聯茂將於 2027 年底有一批上膠機到貨，產能應可順利開出。其中，2028 年新增產能較 2026/2027 年多，主因公司未來將供應美系大型 CSP 的 server、switch 用 CCL，預計 2H27 可取得驗證結果，故我們預期 2028 年在產能大量開出下仍可維持高稼動率，並且公司整體產品報價將有望維持 2027 年的高水準。

整體而言，我們認為聯茂在(1)M7 以上占比快速提升、(2)稼動率隨上游瓶頸緩解回升至 80%以上、(3)產能規模擴大三項動能下，即使 2027 年後有漲價趨緩的風險，獲利仍可維持成長。我們預估 2026/2027/2028 年 EPS 分別為 23.28/52.49/69.40 元。考量聯茂產品組合較台光電、台燿偏向中低階，短期受惠 M6 以下漲價的獲利彈性較大，但長期由產品組合支撐的獲利基礎不如同業，故本益比應較同業折價，但短期內聯茂仍是族群中受惠 CCL 漲價幅度最大的公司，參考過往兩年的 Forward PE 區間落在 10~26 倍之間，我們認為 18 倍的估值為合理評價。

圖表八、季度財報與預估差異

損益表	2Q26					3Q26(F)				
(單位：百萬元)	實際數	中信預估	差異	市場共識	差異	調整後	調整前	差異	市場共識	差異
營業收入	11,530	-	-	11,530	0.0%	15,617	-	-	14,491	7.8%
營業毛利	2,539	-	-	2,539	0.0%	4,973	-	-	3,868	28.6%
營業利益	1,848	-	-	1,848	0.0%	4,036	-	-	2,945	37.0%
稅前純益	1,834	-	-	1,834	0.0%	3,989	-	-	3,007	32.7%
稅後淨利	1,279	-	-	1,279	0.0%	2,792	-	-	2,023	38.0%
EPS(元)	3.52	-	-	3.52	0.0%	7.68	-	-	5.46	40.7%
財務比率(%/百分點)										
毛利率	22.0%	-	-	22.0%	0.0	31.8%	-	-	26.7%	5.1
營益率	16.0%	-	-	16.0%	0.0	25.8%	-	-	20.3%	5.5
稅後淨利率	11.1%	-	-	11.1%	0.0	17.9%	-	-	14.0%	3.9

資料來源：Cmoney、Bloomberg，中信投顧整理

圖表九、年度財報與預估差異

損益表	2026(F)					2027(F)				
(單位：百萬元)	調整後	調整前	差異	市場共識	差異	調整後	調整前	差異	市場共識	差異
營業收入	56,564	-	-	50,496	12.0%	93,053	-	-	75,813	22.7%
營業毛利	15,656	-	-	11,927	31.3%	33,421	-	-	21,558	55.0%
營業利益	12,191	-	-	8,582	42.1%	27,931	-	-	16,726	67.0%
稅前純益	12,129	-	-	8,581	41.3%	27,652	-	-	16,601	66.6%
稅後淨利	8,462	-	-	5,819	45.4%	19,080	-	-	11,340	68.3%
EPS(元)	23.28	-	-	16.56	40.6%	52.49	-	-	32.49	61.6%
財務比率(%/百分點)										
毛利率	27.7%	-	-	23.6%	4.1	35.9%	-	-	28.4%	7.5
營益率	21.6%	-	-	17.0%	4.6	30.0%	-	-	22.1%	8.0
稅後淨利率	15.0%	-	-	11.5%	3.4	20.5%	-	-	15.0%	5.5

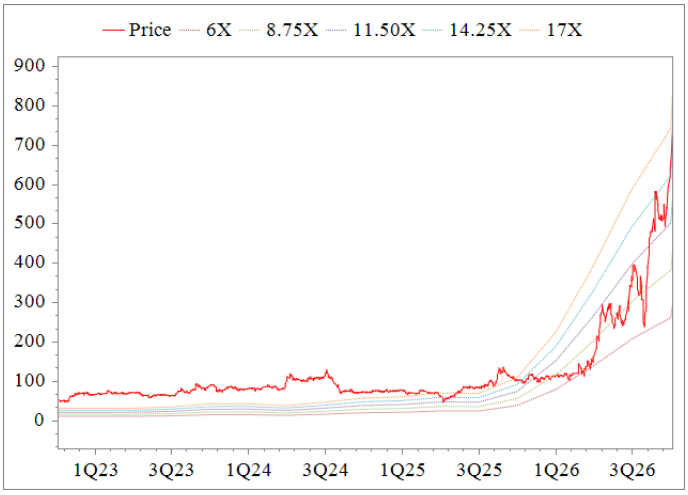
資料來源：Cmoney、Bloomberg，中信投顧整理

2026/10/06

(百萬元)	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F
營業收入	9,143	11,530	15,617	20,274	22,032	22,596	23,592	24,833
營業成本	8,099	8,991	10,644	13,174	14,210	14,575	15,028	15,819
營業毛利	1,044	2,539	4,973	7,100	7,821	8,022	8,564	9,015
營業費用	620	691	937	1,216	1,300	1,333	1,392	1,465
營業利益	424	1,848	4,036	5,883	6,521	6,689	7,172	7,549
營業外收支淨額	59	-14	-47	-61	-66	-68	-71	-75
稅前純益	483	1,834	3,989	5,823	6,455	6,621	7,101	7,475
所得稅	169	554	1,197	1,747	2,001	2,052	2,201	2,317
稅後純益	315	1,279	2,792	4,076	4,454	4,568	4,900	5,158
調整後 EPS(元)	0.87	3.52	7.68	11.21	12.25	12.57	13.48	14.19
毛利率(%)	11.42%	22.02%	31.84%	35.02%	35.50%	35.50%	36.30%	36.30%
營業利益率(%)	4.64%	16.03%	25.84%	29.02%	29.60%	29.60%	30.40%	30.40%
稅後純益率(%)	3.44%	11.10%	17.88%	20.10%	20.22%	20.22%	20.77%	20.77%
營業收入								
QoQ(%)	2.54%	26.11%	35.45%	29.82%	8.67%	2.56%	4.40%	5.26%
YoY(%)	20.62%	29.88%	102.18%	127.37%	140.96%	95.98%	51.06%	22.49%
營業利益								
QoQ(%)	-32.38%	335.67%	118.38%	45.78%	10.84%	2.56%	7.23%	5.26%
YoY(%)	-18.65%	134.51%	805.11%	837.85%	1,437.33%	261.92%	77.71%	28.32%
稅後純益								
QoQ(%)	-26.56%	306.76%	118.25%	45.97%	9.28%	2.56%	7.26%	5.26%
YoY(%)	-6.61%	204.29%	759.95%	851.68%	1,316.15%	257.08%	75.48%	26.54%

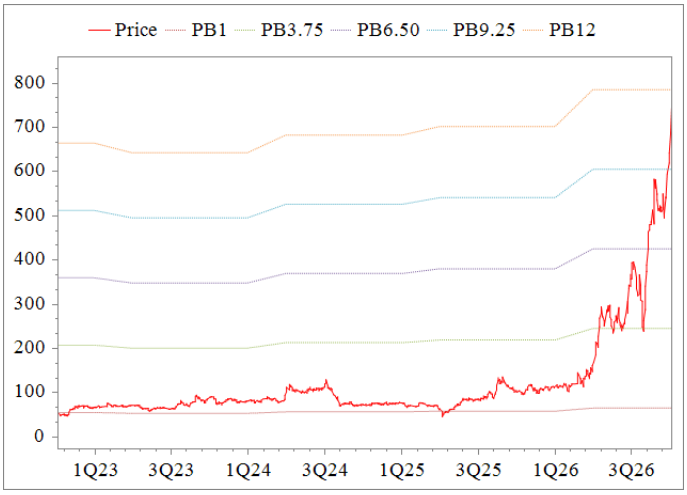
*調整後 EPS 為根據調整後加權平均股數計算

PER Band



資料來源：中信、CMoney

PBR Band



資料來源：CMoney

資產負債表

(百萬元)	2022	2023	2024	2025	2Q26
資產總計	33,384	33,380	34,844	38,213	43,290
流動資產	21,469	21,306	22,259	25,972	30,804
現金及約當現金	5,214	5,594	4,583	4,989	2,822
應收帳款與票據	12,119	11,498	12,790	15,009	18,622
存貨	2,731	3,014	3,763	4,890	7,261
採權益法之投資	48	42	37	30	28
不動產、廠房設備	6,557	7,776	9,649	9,082	8,797
負債總計	13,273	13,940	14,198	16,946	20,696
流動負債	10,883	10,540	11,385	13,561	17,802
應付帳款及票據	5,926	5,991	6,286	8,848	10,662
非流動負債	2,390	3,400	2,813	3,386	2,894
權益總計	20,111	19,440	20,646	21,267	22,594
普通股股本	3,630	3,630	3,630	3,632	3,636
保留盈餘	7,561	7,149	7,431	8,290	8,794
母公司業主權益	20,111	19,440	20,646	21,267	22,594
負債及權益總計	33,384	33,380	34,844	38,213	43,290

損益表

(百萬元)	2023	2024	2025	2026F	2027F
營業收入淨額	25,079	29,378	33,098	56,564	93,053
營業成本	21,973	25,688	28,314	40,908	59,632
營業毛利淨額	3,106	3,690	4,785	15,656	33,421
營業費用	2,111	2,339	2,402	3,464	5,490
營業利益	995	1,351	2,383	12,191	27,931
EBITDA	2,405	2,912	4,068	13,931	29,732
業外收入及支出	157	-33	-16	-63	-279
稅前純益	1,152	1,317	2,367	12,129	27,652
所得稅	476	496	856	3,667	8,572
稅後純益	677	822	1,510	8,462	19,080
公告基本 EPS(元)	1.86	2.26	4.16	- -	- -
調整後 EPS(元)	1.86	2.26	4.16	23.28	52.49

註：稅後純益係指本期淨利歸屬於母公司業主。

現金流量表

(百萬元)	2022	2023	2024	2025	2Q26
營業活動現金	6,429	2,097	925	2,269	-455
稅前純益	2,315	1,152	1,317	2,367	1,834
折舊及攤銷	1,161	1,163	1,474	1,586	412
營運資金變動	2,382	404	-1,747	-784	-2,758
其他營運現金	571	-622	-119	-899	57
投資活動現金	-3,221	-1,248	-1,696	-1,482	-860
資本支出淨額	-3,149	-1,124	-1,709	-1,518	-680
長期投資變動	-44	-82	-471	58	-246
其他投資現金	-28	-43	484	-23	66
籌資活動現金	-1,395	-171	-389	-535	206
長借/公司債變動	1,679	955	164	-470	-539
現金增資	0	0	0	0	0
發放現金股利	-1,915	-1,089	-544	-653	0
其他籌資現金	-1,159	-37	-8	588	746
淨現金流量	791	380	-1,011	406	-740
期初現金	4,423	5,214	5,594	4,583	3,563
期末現金	5,214	5,594	4,583	4,989	2,823

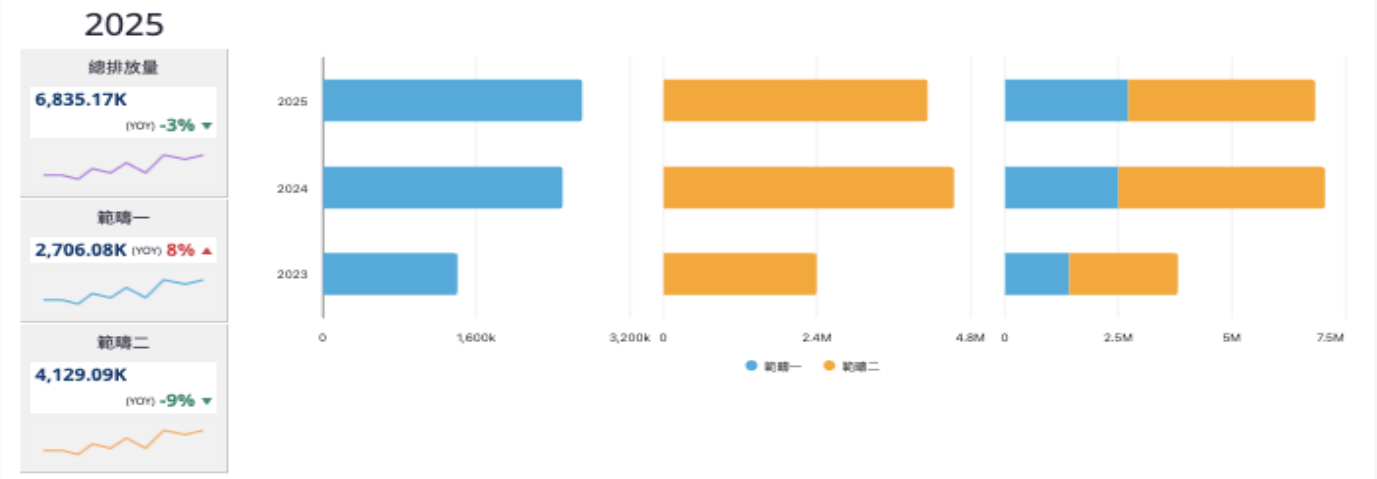
資料來源：中信、CMoney

比率分析

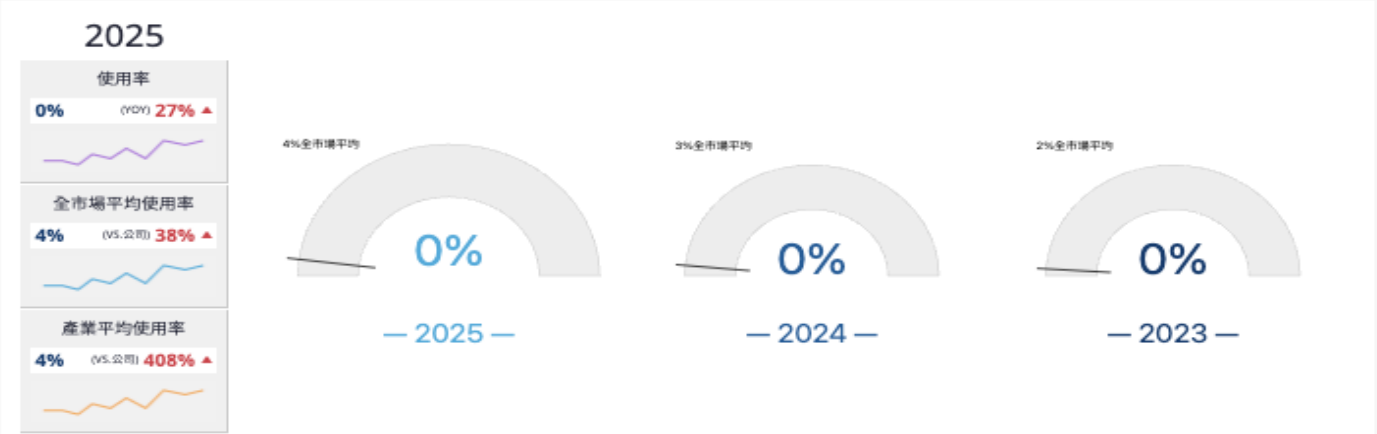
(百萬元)	2023	2024	2025	2026F	2027F
成長力分析(%)					
營業收入淨額	-13.91%	17.14%	12.66%	70.90%	64.51%
營業毛利淨額	-21.17%	18.81%	29.66%	227.22%	113.47%
營業利益	-47.54%	35.76%	76.41%	411.66%	129.10%
稅後純益	-63.53%	21.45%	83.77%	460.31%	125.48%
獲利能力分析(%)					
毛利率	12.38%	12.56%	14.46%	27.68%	35.92%
EBITDA(%)	9.59%	9.91%	12.29%	24.63%	31.95%
營益率	3.97%	4.60%	7.20%	21.55%	30.02%
稅後純益率	2.70%	2.80%	4.56%	14.96%	20.50%
ROE(%)	3.42%	4.10%	7.21%	37.55%	71.55%

ESG 資訊—環境保護(Environmental)

溫室氣體(範疇一/二)排放量

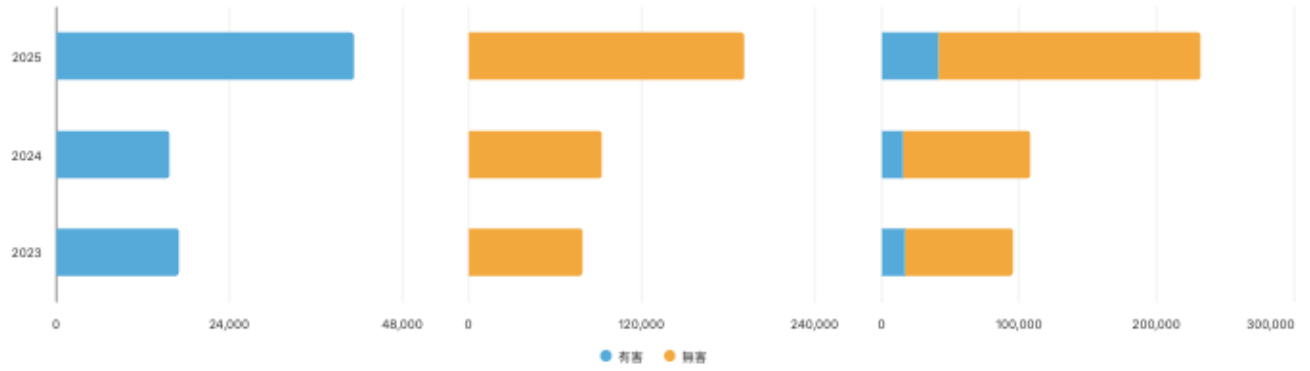


再生能源使用率



用水量

廢棄物量



2025

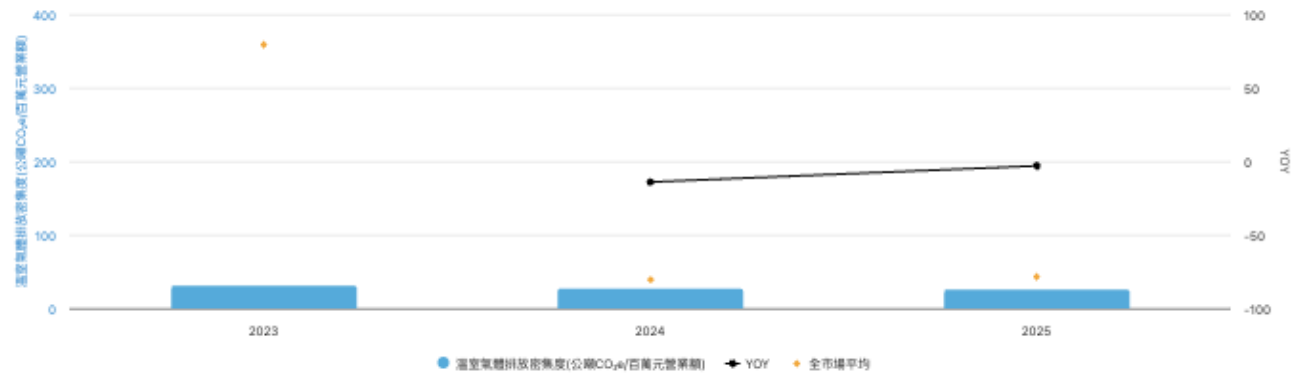
有害	無害	總量
41.26K (YOY) 162% ▲	191.02K (YOY) 106% ▲	232.29K (YOY) 106% ▲

密集度

2025

溫室氣體排放密集度	用水密集度	廢棄物密集度
26.298	113.886	0.894

溫室氣體排放密集度(公噸CO_2e/百萬元營業額)

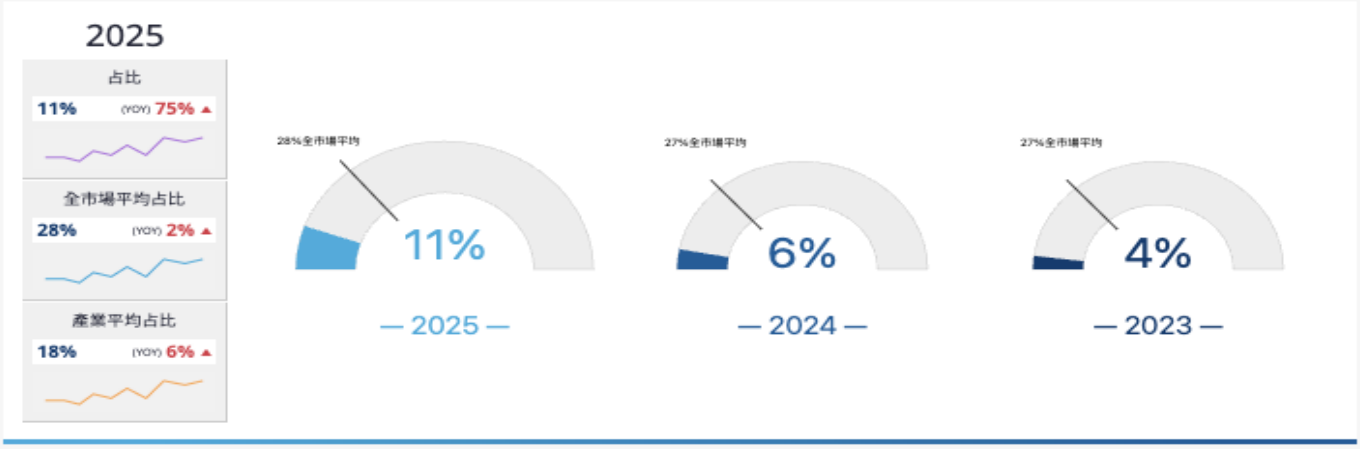


2025

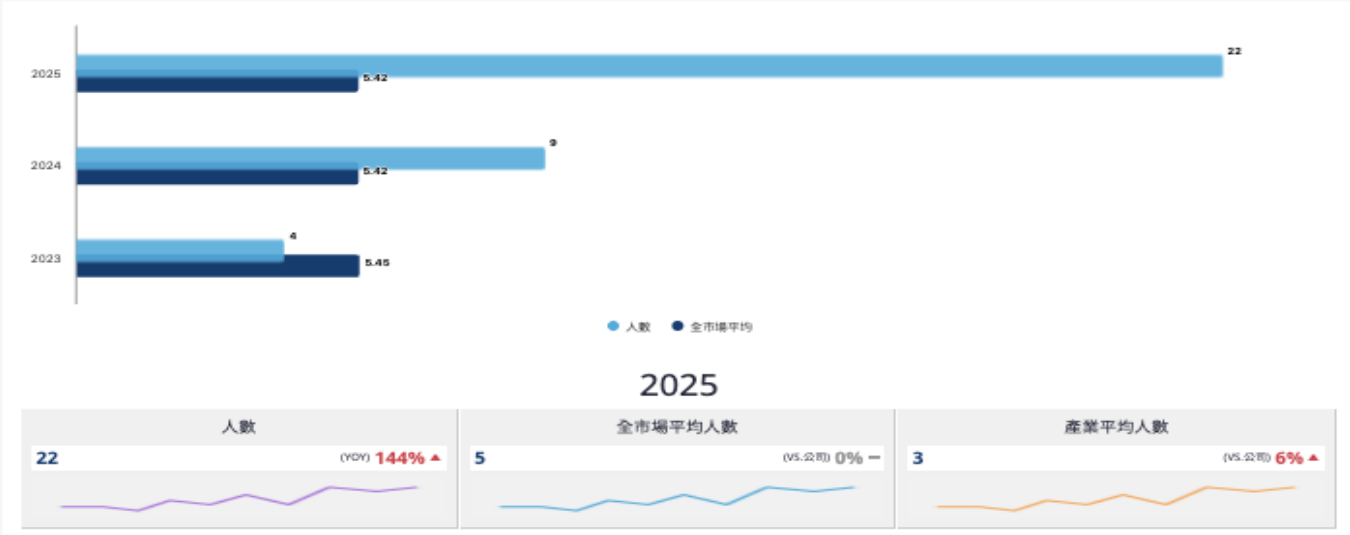
溫室氣體排放密集度(公噸CO_2e/百萬元營業額)	全市場平均溫室氣體排放密集度(公噸CO_2e/百萬元營業額)	產業平均溫室氣體排放密集度(公噸CO_2e/百萬元營業額)
0.03K (YOY) -3% ▼	0.04K (YOY) 10% ▲	0.02K (YOY) -6% ▼

ESG 資訊—社會責任(Social)

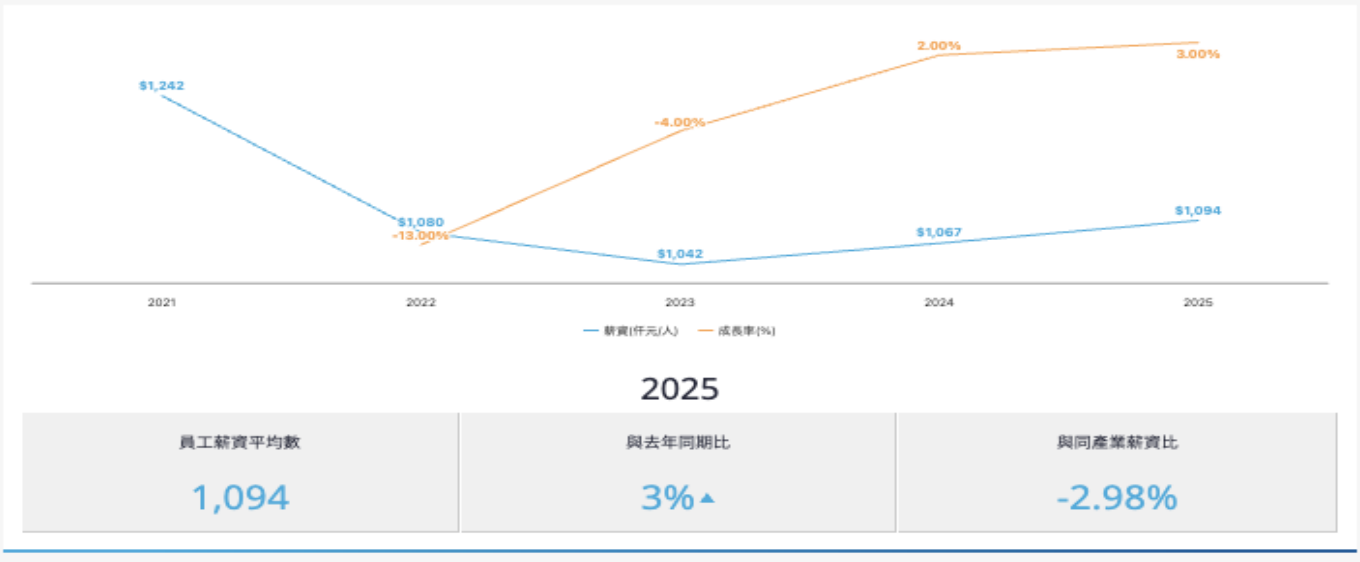
管理職女性主管占比



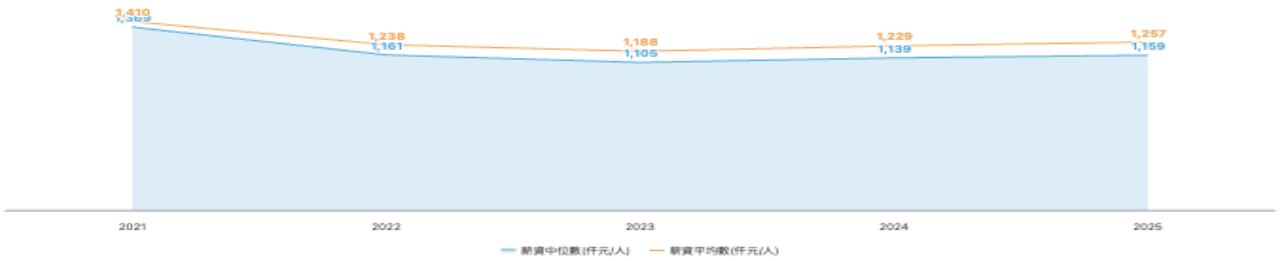
職業災害人數



員工薪資年度趨勢



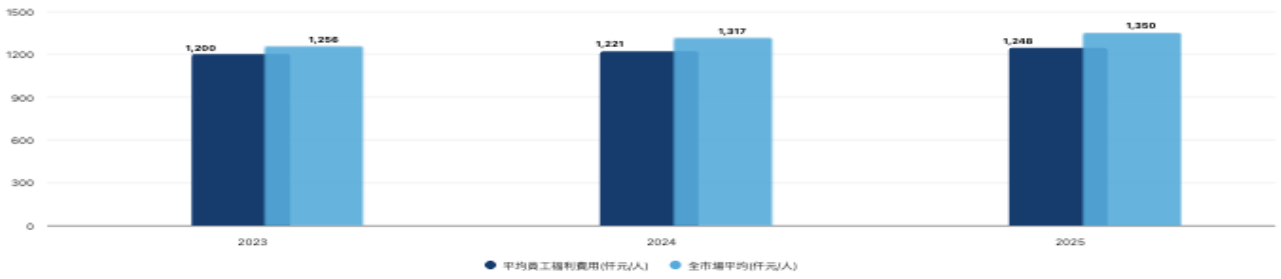
非擔任主管職務之全時員工薪資



2025

平均數	中位數	差異%
1,257 (YOY) 2% ▲	1,159 (YOY) 2% ▲	0% (YOY) 0% —

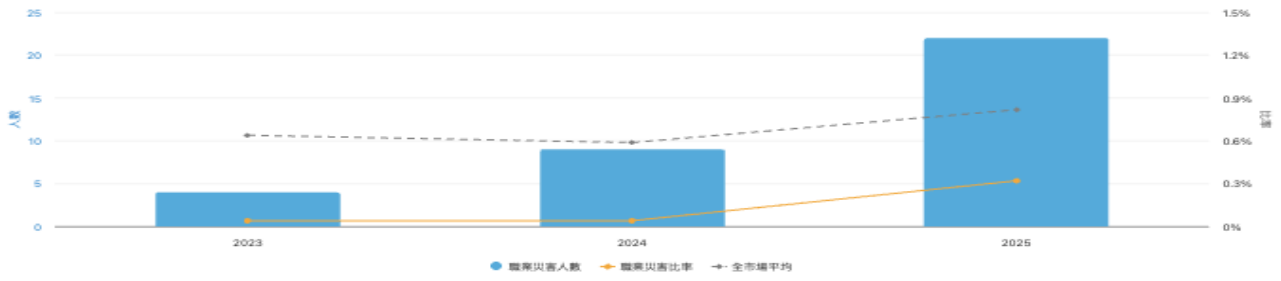
平均員工福利費用



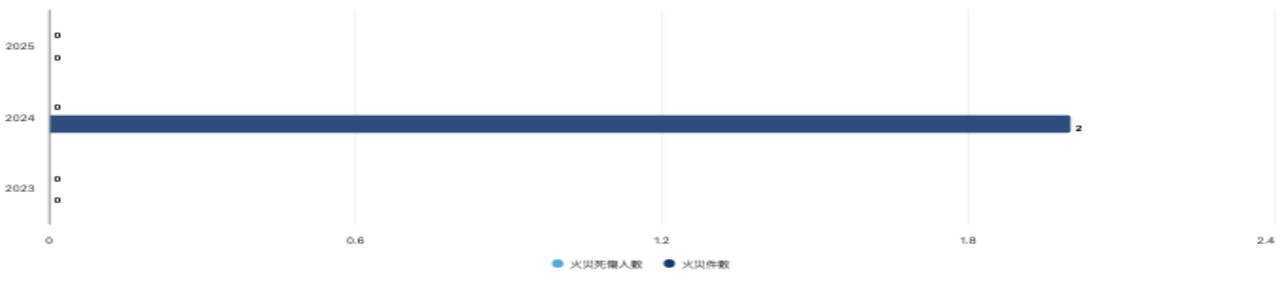
2025

平均福利	全市場平均福利	產業平均福利
1,248 (YOY) 2% ▲	1,350 (YOY) 3% ▲	1,302 (YOY) 3% ▲

職業災害人數與比率



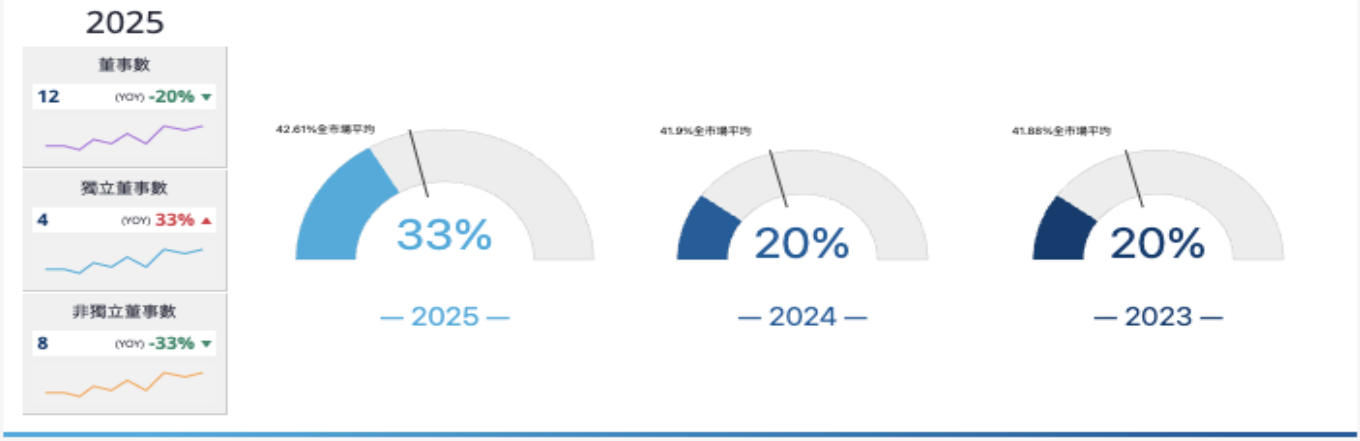
火災件數與死傷人數



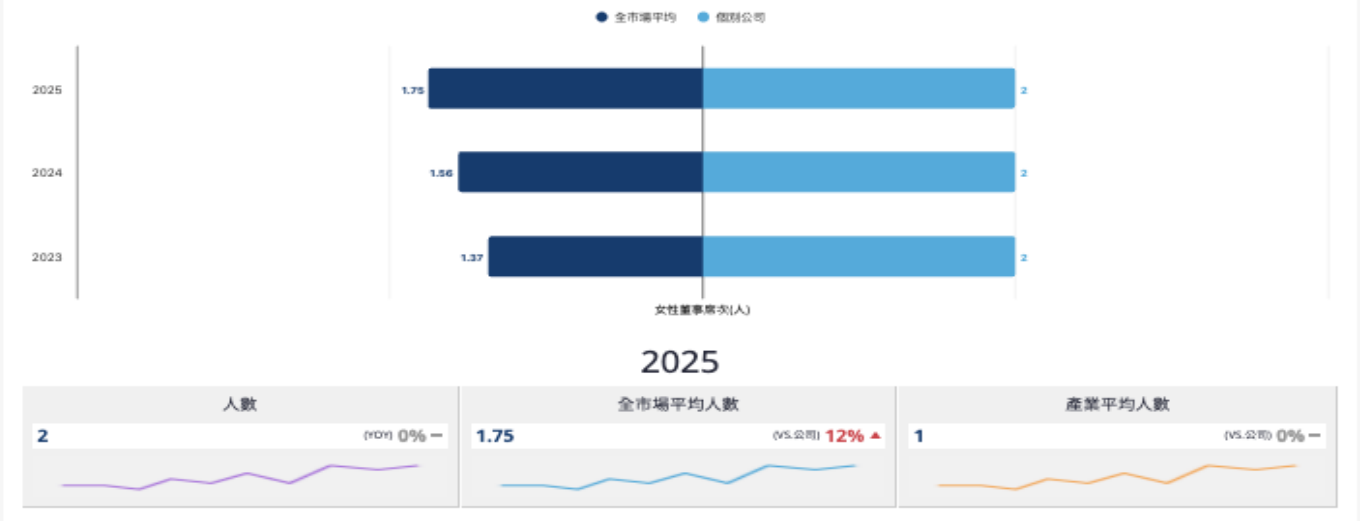
2026/10/06

ESG 資訊—公司治理(Governance)

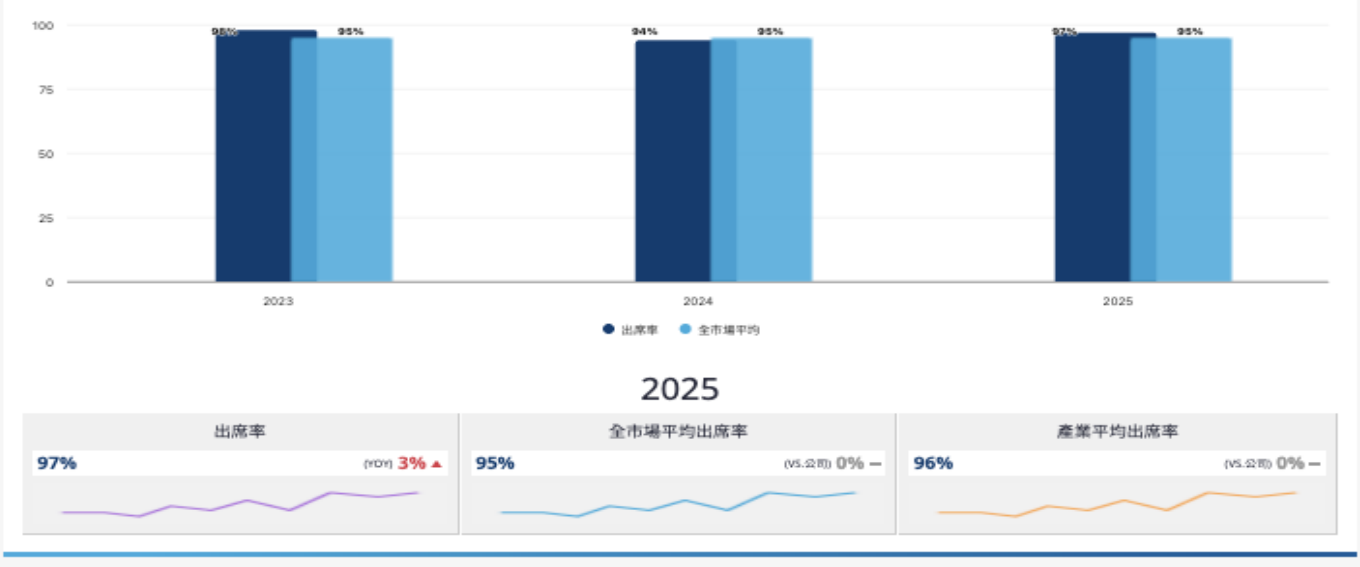
獨立董事vs.董事席次占比



女性董事席次



董事之董事會出席率



2026/10/06

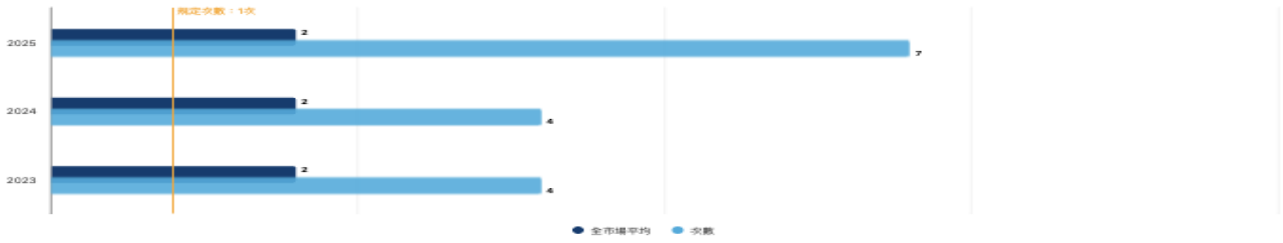
董事進修符合進修要點比率

內圈:個別公司　外圈:全市場平均

2025

進修率	全市場平均進修率	產業平均進修率
100% (YOY) 0% —	94% (VS.公司) 1% ▲	95% (VS.公司) 1% ▲

法說會次數



2025

次數	全市場平均次數	產業平均次數
7 (YOY) 75% ▲	2 (VS.公司) 0% —	2 (VS.公司) 0% —

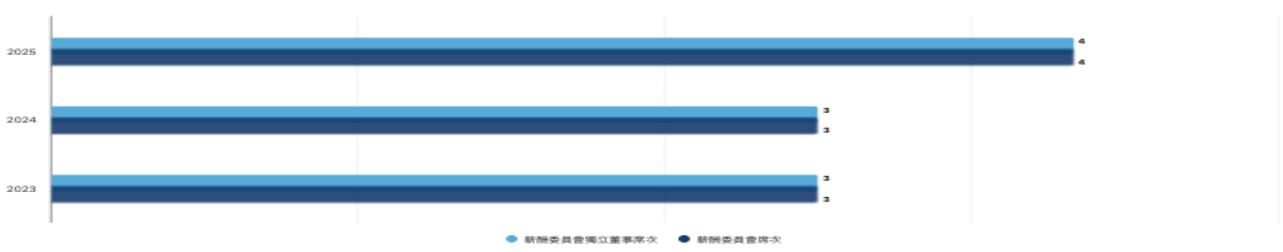
薪酬委員會出席率

內圈:個別公司出席率　外圈:全市場平均出席率

2025

進修率	全市場平均進修率	產業平均進修率
100% (YOY) 0% —	97% (VS.公司) 1% —	97% (VS.公司) 1% —

薪酬委員會席次



資料來源：證交所

2026/10/06

投資評等說明

個股評等	Rating	定義
買進	Buy (B)	相較於市場指數之回報潛力極具吸引力
增加持股	Overweight (OW)	相較於市場指數之回報潛力略有吸引力
中立	Neutral (N)	相較於市場指數之回報潛力相似
降低持股	Underweight (UW)	相較於市場指數之回報潛力稍低
賣出	Sell (S)	相較於市場指數之回報潛力較低
未評等	Not Rated (NR)	

免責聲明

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中國信託證券投資顧問股份有限公司

台北市南港區經貿二路 188 號 5 樓及 18 樓 / (02) 5569-1988

營業執照字號：114 金管投顧新字第 014 號